MAYO CLINIC
LABORATORIES

Rochester
2026 standardTest Catalog

Laboratory Reference Edition

Sorted By Test Name

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Definition of Specimen "Minimum Volume"

Defines the amount of specimen required to perform an assay once, including instrument and container dead space. Submitting the minimum specimen volume makes it impossible to repeat the test or perform confirmatory or perform reflex testing. In some situations, a minimum specimen volume may result in a QNS (quantity not sufficient) result, requiring a second specimen to be collected.

PURPOSE

This document provides policy guidance on topics relevant to laboratory testing performed for Mayo Clinic Laboratories (MCL).

MAYO CLINIC LABORATORIES

Mayo Clinic Laboratories is a reference laboratory operating in partnership with Mayo Clinic's Department of Laboratory Medicine and Pathology (DLMP) and other laboratory partners. Mayo Clinic Laboratories specializes in providing esoteric laboratory testing services to customers across the United States and around the world.

POLICY STATEMENTS

For a statement of qualifications for assistance with meeting regulatory and accreditation compliance requirements for the referral laboratory qualification process, refer to Framework for Quality.

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Abnormal Results

Abnormal results can be defined as one of three different classifications: abnormal, critical, and semi-urgent. A team of laboratory directors and Mayo Clinic physicians determines the critical values for analytical tests. These decision limits are built into the Laboratory Information System (LIS). The call center is notified of possible critical values 24 hours a day, seven days a week by representatives in our performing laboratories. Clients are required to provide contact information to us to facilitate accurate and timely notification of critical values/critical results.

Critical Value/Critical Results

A critical value/critical result is defined as a value/result that represents a pathophysiological state at such variance with normal (expected) values to be life-threatening, unless something is done promptly and for which some corrective action could be taken.

Semi-Urgent Results

A semi-urgent result is defined by MCL as a result or finding that can be unexpected or ambiguous, and does not pose an immediate health threat, but has near-term severe health consequences if not acknowledged and/or treated.

Abnormal Results

Most laboratory tests have established reference values which represent results that are typically seen in a group of healthy individuals. Results outside these reference values may be considered abnormal; however, not all "abnormal" results are considered "critical values" or "semi-urgent results."

Action Taken when Critical Results are Obtained

In addition to the standard results reporting (eg, interface, results available within the ordering application), MCL staff contact the client-provided contact number within 60 minutes following laboratory release of the critical test result(s). If contact is not made within the 60-minute period, continued attempts will be made until the designated party is reached and the result is conveyed.

To complement our standard reporting mechanisms, MCL offers electronic notification of both critical values and semi-urgent results through Message Center, or a customized notification rule can be created. An additional comment may also appear in the final report.

For information regarding the Mayo Clinic Laboratories Critical Values/Semi-Urgent Results List, contact Customer Service at 800-533-1710.

Animal Specimens

Animal specimens are not accepted for laboratory testing.

Assessments

Mayo Clinic Laboratories participates in a wide variety of proficiency testing to ensure the accuracy of test results, including interlaboratory and independent state, national, and international programs. In addition, we participate in external assessments that include, but are not limited to, regulatory inspections and accreditation assessments. Also see Proficiency Testing.

Business Continuity and Contingency Planning

In the event of a local, regional, or national disaster, Mayo Clinic and MCL performing sites have comprehensive contingency plans in place at each location to ensure the impact on laboratory practice is proactively addressed. Test standardization between the performing sites and medical practice locations throughout the country ensure that patient care will not be compromised.

Cancellation of Tests

Cancellations received prior to test setup will be honored at no charge. Requests received following test setup cannot be honored. A report will be issued automatically and charged appropriately.

Chain of Custody

Chain of custody is a documented record of a specimen's disposition that identifies the individuals who collected, handled, and analyzed the associated specimen. This documentation is required when test results may be used in a court of law. MCL has developed packaging and shipping materials that meet legal chain-of-custody requirements. This service is only offered for drug testing.

Compliance Policies

Mayo Clinic Laboratories is committed to compliance with applicable laws and regulations such as the Clinical Laboratory Improvement Amendments (CLIA). Regulatory agencies that oversee our compliance include, but are not limited to, the Centers for Medicare and Medicaid Services (CMS), the US Food and Drug Administration (FDA), and the Department of Transportation (DOT). MCL develops, implements, and maintains policies, processes, and procedures throughout the organization that are designed to meet relevant requirements. We expect that clients utilizing our services will ensure their compliance with patient confidentiality, diagnosis coding, anti-kickback statutes, professional courtesy, CPT-4 coding, CLIA-approved proficiency testing, and other similar regulatory requirements. Also see Health Insurance Portability Accountability Act (HIPAA) Compliance and Reportable Disease Notification.

Confidentiality/Disclosure of Results

Mayo Clinic Laboratories is committed to maintaining confidentiality of patient information. To ensure HIPAA and College of American Pathologists (CAP) compliance for appropriate release of patient results, MCL has adopted the following policies:

Phone Inquiry Policy

One of the following unique identifiers will be required:

- MCL order ID number for specimen;
- Client account number (assigned by MCL) along with patient name;
- Client accession ID number; or
- Identification by individual that he or she is, in fact, "referring healthcare professional" identified on requisition form by MCL client.

Under federal regulation, MCL is authorized to release results to healthcare professionals responsible for the individual patient's care. Patients or patients' personal representatives requesting results can do so via the following link: Contact Us - Mayo Clinic Laboratories.

Environmental Specimens

Environmental specimens are not accepted for laboratory testing.

Extracted Specimens

Extracted nucleic acid will be accepted for clinical testing, provided it is from an acceptable specimen source for the ordered test as listed in the Test Catalog, and if the extraction was performed in a CLIA-certified laboratory, or a laboratory meeting equivalent requirements as determined by CAP or the CMS.

Health Insurance Portability Accountability Act (HIPAA) Compliance

Mayo Clinic Laboratories is fully committed to compliance with all privacy, security, and electronic transaction code requirements of HIPAA. All services provided by MCL that involve joint efforts will be performed in a manner that enables clients to be HIPAA and CAP compliant.

Informed Consent Certification

Submission of an order for any tests contained in the Test Catalog constitutes certification to Mayo Clinic Laboratories by the ordering healthcare professional that: 1) the ordering healthcare professional has obtained "Informed Consent" of subject patient as required by any applicable state or federal laws with respect to each test ordered; and 2) the ordering healthcare professional has obtained from subject patient authorization permitting MCL to report results of each test ordered directly to ordering healthcare professional.

On occasion, a specimen may be forwarded to an outside reference laboratory. The laws of the state where the reference laboratory is located may require written informed consent for certain tests. MCL will request that the ordering healthcare professional pursue and provide such consent. Test results may be delayed or denied if consent is not provided.

Non-Biologic Specimens

Due to the inherent exposure risk of non-biologic specimens, their containers, and the implied relationship to criminal, forensic, and medico-legal cases, MCL does not accept nor refer non-biologic specimen types. Example specimens include unknown solids and liquids in the forms of pills, powder, intravenous fluids, or syringe contents.

Patient Safety Goals/Specimen Identification Policy

The Joint Commission and CAP patient safety goals for improving patient safety through accurate patient identification require the use of two patient identifiers when providing laboratory services. As such, we have developed the following specimen identification policy.

A minimum of two patient-specific identifiers are used to verify the correct patient is matched with the correct specimen and the correct order for testing services. When a specimen is received at MCL, the patient's first and last name, birth date, medical record number, and client accession number are verified by comparing the labels on the specimen tube or container with the electronic order and any paperwork (batch sheet or form) that accompanies the specimen to be tested. When discrepancies are identified, Customer Service will notify the client to verify discrepant information to assure MCL is performing the correct testing for the correct patient. Specimens are considered mislabeled when there is a mismatch between the person-specific identifiers on the specimen and information accompanying the specimen (eg, computer system, requisition form, additional paperwork). When insufficient or inconsistent identification is submitted, MCL will recommend that a new specimen be obtained.

In addition, Anatomic Pathology consultation services require the client pathology report. The pathology report is used to match patient name, patient age and/or birth date, and pathology case number. Since tissue blocks and slides have insufficient space to print the patient name on the block, the pathology report provides MCL another mechanism to confirm the patient identification with the client order and labels on tissue blocks and slides.

Parallel Testing

Parallel testing may be appropriate in some cases to re-establish patient baseline results when converting to a new methodology at Mayo Clinic Laboratories. Contact your Hospital Account Executive (800-533-1710) for further information. Specific planning is required prior to implementation.

Proficiency Testing

Mayo Clinic Laboratories are CAP-accredited, CLIA-licensed laboratories that voluntarily participate in many diverse external and internal proficiency testing programs. It is MCL's expectation that clients utilizing our services will adhere to CLIA requirements for proficiency testing (42 CFR 493.801). This includes a prohibition of discussion about proficiency testing samples or results and sending proficiency testing materials to MCL during the active survey period.

Mayo Clinic Laboratories' proficiency testing includes participation in CMS-approved programs. MCL also performs alternative assessment using independent state, national, and international programs when proficiency testing is not available. Additionally, MCL conducts comparability studies to ensure the accuracy and reliability of patient testing when necessary. We comply with the regulations set forth by the Clinical Laboratory Improvement Amendments (CLIA-88), Occupational Safety and Health Administration (OSHA), or CMS.

Radioactive Specimens

Specimens from patients receiving radioactive tracers or material should be labeled as such. All incoming shipments arriving at MCL are routed through a detection process in receiving to determine if the samples have any levels of radioactivity. If radioactive levels are detected, the samples are handled via an internal process that ensures we do not impact patient care and the safety of our staff. This radioactivity may invalidate the results of radioimmunoassays (RIA).

Record Retention

All test requisitions and patient test results are retained, at a minimum, for the retention period required to comply with and adhere to CAP, CLIA, and New York State (NYS) requirements. A copy of the original report can be reconstructed including reference values, interpretive comments, flags, and footnotes with the source system as the DLMP LIS.

Referral of Tests to Another Laboratory

Mayo Clinic Laboratories forwards tests to other laboratories as a service to its clients. This service should in no way represent an endorsement of such test or referral laboratory, or warrant any specific performance for such test.

Mayo Clinic Laboratories has established collaborative relationships with more than 140 laboratories within the United States. Outside vendors are selected based on certifications, service, turnaround time, methodology, reference values, and price. Laboratory qualifications are reviewed by our internal operations and quality teams. A recommendation is forwarded to our medical director for final consideration. Once selected, each laboratory is monitored to ensure that turnaround time is prompt and certain customer service criteria are met. The referral laboratories must requalify every two years or as their test offerings change.

Reflex Testing

Mayo Clinic Laboratories identifies tests that reflex when medically appropriate. In many cases, MCL also offers these reflex tests as individual tests. Clients should familiarize themselves with the test offerings and decide whether to order a test that includes reflex testing or an individual test and only order test offerings based on medical necessity. Clients who order a test that reflexes can request to receive the Daily Charge Report, which indicates what additional testing has been performed. Clients who wish to receive the Daily Charge Report should contact their Regional Service Representative or Customer Service (800-533-1710).

Reportable Disease Notification

Mayo Clinic Laboratories strives to comply with laboratory reporting requirements for each state health department regarding reportable disease conditions. We report by mail, fax, or electronically, depending upon the specific state health department regulations. Clients shall be responsible for compliance with any state-specific statutes concerning reportable conditions, including, but not limited to, birth defects registries or chromosomal abnormality registries. This may also include providing patient address/demographic information. MCL's reporting does not replace the client or healthcare professional's responsibility to report as per specific state statues.

Request for Ordering Healthcare Professional Name and Number

While providing esoteric reference testing, there are times when we need to contact the ordering healthcare professional directly. The following are 2 examples:

- When necessary for the performance of a test, the ordering healthcare professional's name and phone number are requested as part of "Necessary Information" in the Test Catalog. This information is needed to allow timely consultations or seek clarification of requested services. If this information is not provided at the time of specimen receipt, you will be contacted to obtain the information. By providing this information up front, delays in patient care are avoided.
- In some situations, additional information from the ordering healthcare professional is necessary to clarify or interpret a test result. At that time, MCL will request the ordering healthcare professional's name and phone number so staff can consult with them.

Your rapid assistance in supplying us with the ordering healthcare professional's name and phone number is appreciated when we are required to call. By working together, patients are provided with the highest quality testing services in the shortest possible time.

Special Handling of Time-Sensitive Specimens

Test information, including days tests are performed as well as when the report will be available, is included under each test listing in the Test Catalog on mayocliniclabs.com. Unique circumstances may arise with a patient resulting in an ordering healthcare professional's request that the specimen or results receive special handling. There are several options available. These options can only be initiated by contacting Customer Service (800-533-1710) and providing patient demographic information.

There is a nominal charge associated with any special handling.

- **Expedite:** If you would like us to expedite the specimen to the performing laboratory, you can contact Customer Service and request that your specimen be expedited. Relay the following information: facility name, account number, patient name or MCL accession number, shipping information (eg, courier service, FedEx), date to be sent, and test to be performed. Place specimen in a separate Mayo Clinic Laboratories temperature-appropriate bag. Write "Expedite" in large print on the outside of the bag. Once the shipment is received in our receiving area, we will deliver the specimen to the performing laboratory for the next scheduled analytic run. We will not set up a special run to accommodate an expedite request.
- **STAT:** In rare, preapproved circumstances, STAT testing may be required for patients who need immediate treatment. These cases typically necessitate a special analytic run to turn results around as quickly as possible. To arrange STAT testing, have the pathologist, healthcare professional, or laboratory director call Customer Service. They will be connected with one of our medical/laboratory directors to consult about the patient's case. Once mutually agreed upon that there is a need for a STAT, arrangements will be made to assign resources to run the testing on a STAT basis when the specimen is received.

Specimen Rejection

All tests are unique in their testing requirements. To avoid specimen rejection or delayed turnaround times, check the "Specimen Required" and "Reject Due To" fields within each test. You will be notified of rejected or problem specimens upon receipt.

Review the following conditions prior to submitting a specimen:

- Full 24 hours for timed urine collection
- pH of urine
- Lack of hemolysis/lipemia/icterus
- Specimen type (plasma, serum, whole blood, etc)
- Specimen volume
- Patient information requested
- Proper identification of patient/specimen
- Specimen container (metal-free, separation gel, appropriate preservative, etc)
- Transport medium
- Temperature (ambient, frozen, refrigerated)

Specimen Volume

The "Specimen Required" section of each test includes the preferred volume, and the "Specimen Minimum Volume" is also provided. Preferred volume has been established to optimize testing and allows the laboratory to quickly process specimen containers, present containers to instruments, perform test, and repeat testing if necessary. Many of our testing processes are fully automated; as a result, this volume allows hands-free testing and our quickest turnaround time (TAT). Since patient values are frequently abnormal, repeat testing, dilutions, or other specimen manipulations often are required to obtain a reliable, reportable result. The preferred specimen requirements allow expeditious testing and reporting.

When venipuncture is technically difficult or the patient is at risk of complications from blood loss (eg, pediatric or intensive care patients), smaller volumes may be necessary. "Specimen Minimum Volume" is the amount of sample necessary to provide a clinically relevant result as determined by the testing laboratory.

When patient conditions do not mandate reduced collection volumes, we ask that our clients submit preferred volume to facilitate rapid, cost-effective, reliable test results. Submitting less than preferred volume may negatively impact quality of care by slowing TAT, increasing the hands-on personnel time (and therefore, cost) required to perform test.

Mayo Clinic Laboratories makes every possible effort to successfully test your patient's specimen. If you have concerns about submitting a specimen for testing, contact Customer Service (800-533-1710). Our staff will discuss the test and specimen you have available. While in some cases, specimens are inadequate for desired test, in other cases, testing can be performed using alternative techniques.

Supplies

Supplies for MCL testing, including shipping boxes, specimen vials, special specimen collection containers, etc, are fee inclusive unless indicated otherwise. Supplies can be requested using one of the following methods: Use the online ordering functionality available using the Supplies option in "My Dashboard" on mayocliniclabs.com, or contact Customer Service (800-533-1710).

Test Classifications

Analytical tests offered by MCL are classified according to the FDA labeling of the test kit, or reagents and their usage. Where appropriate, analytical test listings contain a statement regarding these classifications, test development, and performance characteristics.

Test Result Call-Backs

Results will be telephoned to a client when requested by the client (in the ordering application or by a phone call to Customer Service from the client).

Turnaround Time

Mayo Clinic Laboratories' extensive test menu reflects the needs of our own healthcare practice. We are committed to providing the most expedient TAT possible to improve diagnosis and treatment. Laboratory services are considered part of the patient care continuum, wherein the needs of the patient are paramount.

Turnaround time is defined as the time from which the sample is received at MCL to the time when the test result is available. TAT is monitored continuously by each performing laboratory site within the Mayo Clinic DLMP. For the most up-to-date information on TAT for individual tests, visit mayocliniclabs.com or contact Customer Service (800-533-1710).

Unlisted Tests

All available test offerings are not listed in the test catalog. Although we do not usually accept referred tests of a more routine type, special arrangements may be made to provide your laboratory with temporary support during times of special need, such as sustained instrumentation failure. For information about unlisted tests, contact Customer Service (800-533-1710).

DHVD
8822

1,25-Dihydroxyvitamin D, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, preferred but not required

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82652

SFUNG
604094

1,3-Beta-D-Glucan (Fungitell), Serum

Specimen Requirements:

Container/Tube: Serum gel (red top tube is **not acceptable**)

Specimen Volume: 1 mL

Collection Instructions:

1. Avoid exposure of specimen to atmosphere to prevent environmental contamination of the sample.

2. Centrifuge and send entire specimen in original collection tube. **Do not aliquot or open tube**.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Refrigerated (preferred)	14 days	SERUM GEL TUBE
	Frozen	30 days	SERUM GEL TUBE

CPT Code Information:

87449

DOCS
46919

11-Deoxycorticosterone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Morning (8 a.m.) specimen is preferred.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days
	Ambient	7 days

CPT Code Information:

82633

DCORT 46920 11-Deoxycortisol, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Morning (8 a.m.) specimen is preferred.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82634

F11DX 75673 11-Desoxycortisol

Specimen Requirements:

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 45 minutes of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send frozen.

Specimen Stability Information: Frozen 2 years

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

82634

THCMX 62744 11-nor-Delta-9-Tetrahydrocannabinol-9-Carboxylic Acid (Carboxy-THC) Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain-of-Custody Meconium Kit (T653) includes the specimen containers, seals, and documentation required.

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Additional Information: Specimens that arrive with a broken seal do not meet the chain of custody requirements.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Refrigerated	21 days
	Ambient	14 days

CPT Code Information:

G0480

80349 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

THCM 84284 11-nor-Delta-9-Tetrahydrocannabinol-9-Carboxylic Acid (Carboxy-THC) Confirmation, Meconium

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Refrigerated	21 days
	Ambient	14 days

CPT Code Information:

G0480

80349 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

1433E 624066 14-3-3 eta, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Refrigerated (preferred)	14 days
	Frozen	100 days
	Ambient	48 hours

CPT Code Information:

83520

17OHP
81151

17-Hydroxypregnenolone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Refrigerated	28 days

CPT Code Information:

84143

OHPG
9231

17-Hydroxyprogesterone, Serum

Specimen Requirements:

Collection Container/Tube: Red top **(serum gel/SST are not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

83498

FHC18 18-Hydroxycorticosterone, Serum
75675

Specimen Requirements:

Preferred: Red top

Acceptable: SST

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Serum must be separated from cells within 45 minutes of venipuncture. Centrifuge and send 3 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 1 mL (NOTE: Minimum volume does not allow for repeat analysis.)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Ambient	24 hours

	Refrigerated	24 hours

CPT Code Information:

82542

GLIOF 35272 1p/19q Deletion in Gliomas, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Preferred: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded (FFPE) tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Acceptable: Slides

Collection Instructions: Six consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Four consecutive, unstained, 5-micron-thick sections placed on positively charged slides and 1 hematoxylin and eosin-stained slide

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291- DNA probe, each (first probe set), Interpretation and report

88271x2- DNA probe, each; each additional probe set (if appropriate)

88271x1- DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2- DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3- DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274- w/modifier 52- Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274- Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

JGLIF 615002 1p/19q Glioma Deletions, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Provide fixation method used.

Additional Information: Blocks prepared with alternative fixation methods **will not** be accepted.

Acceptable

Specimen Type: Tissue slides

Slides: 1 hematoxylin and eosin-stained and 3 unstained

Collection Instructions:

1. Submit 1 slide stained with hematoxylin and eosin and 3 consecutive, unstained, positively charged, unbaked, slides with 4 to 5 micron-thick sections of the tumor tissue.

2. Provide fixation method used.

Additional Information: Slides cut from blocks prepared with alternative fixation methods **will not** be accepted.

Specimen Minimum Volume: Tissue slides: 2 consecutive unstained and 1 hematoxylin and eosin stained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377 x 2 (2 probe sets)

23BPT 606357 2,3-Dinor 11 Beta-Prostaglandin F2 Alpha, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Patients taking aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) may have decreased concentrations of prostaglandin F2 alpha. If medically feasible, for 2 weeks before specimen collection, patient should **not** take aspirin and for 72 hours before specimen collection, patient should **not** take NSAIDs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine refrigerate for 24 hours.

2. No preservative preferred.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84150

23BPR
606356

2,3-Dinor 11 Beta-Prostaglandin F2 Alpha, Random, Urine

Specimen Requirements:

Patient Preparation: Patients taking aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) may have decreased concentrations of prostaglandin F2 alpha. If medically feasible, for 2 weeks before specimen collection, patient should **not** take aspirin and for 72 hours before specimen collection, patient **should not** take NSAIDs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions: Collect a random urine specimen.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84150

82570

2HGA
614603

2-Hydroxyglutaric Acid Chiral Analysis, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tube, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Pediatric: If the collection volume appears insufficient, submit as much specimen as possible in a single container; the laboratory will determine if volume is sufficient for test.

Collection Instructions:

1. Collect a random urine specimen (first morning void preferred)

2. No preservative

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	416 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

83921 x 2

2OHGP 608030 2-Hydroxyglutaric Aciduria Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see .How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

21DOC
89477

21-Deoxycortisol, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Morning (8 a.m.) specimen is preferred.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days
	Ambient	14 days

CPT Code Information:

82542

21OH
607788

21-Hydroxylase Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

83516

CYPZ
37445

21-Hydroxylase Gene, CYP21A2, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days (preferred)/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Preferred:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) <24 hours/Ambient <24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks is required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated (preferred) <24 hours/Ambient <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours)

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid or CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405-CYP21A2 (cytochrome P450, family 21, subfamily A, polypeptide2) (eg, steroid 21-hydroxylase isoform, congenital adrenal hyperplasia), full gene sequence

81402-CYP21A2 (cytochrome P450, family 21, subfamily A, polypeptide2) (eg, congenital adrenal hyperplasia, 21-hydroxylase deficiency), common variants (eg, IVS2-13G, P30L, I172N, exon 6 mutation cluster [I235N, V236E, M238K], V281L, L307FfsX6, Q318X, R356W, P453S, G110VfsX21, 30-kb deletion variant)

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88235-Tissue culture for amniotic fluid (if appropriate)

88240-Cryopreservation (if appropriate)

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells (if appropriate)

DD22F 22q11.2 Deletion/Duplication, FISH, Varies

35246

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Whole blood

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

4. Cord blood is acceptable

Additional Information:

1. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

2. Specimen cannot be frozen.

Acceptable:

Specimen Type: Fixed cell pellet

Container/Tube: Sterile container with a 3:1 fixative (methanol:glacial acetic acid)

Specimen Volume: Entire specimen

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 to 25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

3. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 12 mL.

4. If ordering with CHRAF / Chromosome Analysis, Amniotic Fluid, submit a minimum of 12 mL.

5. If ordering with both CMAP and CHRAF, submit a minimum of 26 mL.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

4. Specimen cannot be frozen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 25 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium (such as CVS Media (RPMI) and Small Dish (T095).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

4. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 12 mg.

5. If ordering with CHRCV / Chromosome Analysis, Chorionic Villus Sampling, submit a minimum of 12 mg.

6. If ordering with both CMAP and CHRCV, then submit a minimum of 26 mg.

Additional Information:

1. Label each container with the specimen type, patient's name, and laboratory control number.

2. Specimen cannot be frozen.

Specimen Type: Products of conception or stillbirth

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, sterile RPMI transport media, or normal saline

Specimen Volume: 1 cm(3) of placenta (including 20 mg of chorionic villi) **and** a 1 cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions:

1. Do not handle specimen with hands.

2. If a fetus cannot be specifically identified, collect 50-mg villus material or tissue that appears to be of fetal origin.

Additional Information:

1. **Do not** send the entire fetus.

2. If a fetus is sent, please provide a parental release form or complete the Final Disposition of Fetal/Stillborn Remains form and send it with the fetus. (A copy of this form can be found in Special Instructions.) Please note that completion of the form requires a parent's signature.

3. Specimen cannot be frozen.

4. While fresher specimens prepared as described above are preferred, we can attempt analysis on specimens that have been in less-than-ideal conditions.

5. Label each container with the specimen type, patient's name, and laboratory control number.

Specimen Type: Autopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, sterile RPMI transport media, or normal saline.

Specimen Volume: 1 cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. **Do not** use alcohol or iodine preparations.

4. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Additional Information:

1. Label each container with the specimen type, patient's name, and laboratory control number.

2. Specimen cannot be frozen.

Specimen Type: Skin biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, sterile RPMI transport media, or normal saline.

Specimen Volume: 1 cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. **Do not** use alcohol or iodine preparations.

4. A local anesthetic may be used.

5. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Additional Information:

1. Label each container with the specimen type, patient's name, and laboratory control number.

2. Specimen cannot be frozen.

Specimen Minimum Volume: Amniotic fluid: 5 mL; Autopsy, skin biopsy: 4 mm; Whole blood: 2 mL; Chorionic villi: 5 mg; Fixed cell pellet: 1 pellet; Products of conception: 1 cm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88273 w/modifier 52-Chromosomal in situ hybridization, less than 10 cells (if appropriate)

88273-Chromosomal in situ hybridization, 10-30 cells (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

25HDN 25-Hydroxyvitamin D2 and D3, Serum

83670

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within two hours of collection, centrifuge and aliquot serum

into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82306

2425D 63416 25-Hydroxyvitamin D:24,25-Dihydroxyvitamin D Ratio, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82306

82542

3-Hydroxy-3-Methylglutaryl Coenzyme-A (HMG-CoA) Reductase, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82397

3MT
65157

3-Methoxytyramine, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Tricyclic antidepressants, labetalol, and sotalol medications may elevate levels of catecholamines producing results that cannot be interpreted. If clinically feasible, it is optimal to discontinue these medications at least 1 week before collection. Levodopa (Sinemet) medication will cause false-positive results. For advice on assessing the risk of removing patients from these medications and alternatives, consider consultation with a specialist in endocrinology or hypertension.

Supplies: Urine Tubes, 10 mL (T068)

Submission Container/Tube: Plastic urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Complete 24-hour urine collections are preferred, especially for patients with episodic hypertension; ideally the collection should begin at the onset of a "spell."

2. Add 10 g (pediatric: 3 g) of boric acid or 25 mL (pediatric: 15 mL) of 50% acetic acid as preservative **at start of collection**.

3.Collect urine for 24 hours.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

82542

3MGAP 608034 3-Methylglutaconic Aciduria Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not

perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood**

specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

5NTD
622835

5'-Nucleotidase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

83915

FLUC
82741

5-Flucytosine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum for a peak level should be collected 1 to 2 hours after oral dose, 1 hour after intramuscular dose, or 30 minutes after intravenous infusion. Trough specimens should be collected immediately prior to next scheduled dose.

2. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

HIAA
9248

5-Hydroxyindoleacetic Acid, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. Some medications could interfere with test results. If medically feasible, for 48 hours before specimen collection, patient should not take the following medications:

-Acetaminophen (Tylenol or generic versions)

-Tryptophan containing supplements

2. For 48 hours prior to, as well as during, the urine collection, the patient should limit the following to one serving per day:

-Fruits

-Vegetables

-Nuts

-Caffeinated beverages or foods

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative at start of collection. Use 15 mL of 50% acetic acid for children younger than 5 years.

2. Collect a 24-hour urine specimen.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	56 days
	Frozen	365 days

CPT Code Information:

83497

HIAAP 619735 5-Hydroxyindoleacetic Acid, Plasma

Specimen Requirements:

Patient Preparation:

1. Some medications could interfere with test results. If medically feasible, for 48 hours before and during specimen collection, patient should not take the following medications:

-Acetaminophen (Tylenol or generic versions)

-Tryptophan containing supplements.

2. For 24 hours prior to and during collection, the patient should limit the following to one serving per day:

-Fruits

-Vegetables

-Caffeinated beverages or foods

3. For 24 hours prior to and during collection, patients **should not** eat nuts, especially walnuts. Plasma 5-hydroxyindoleacetic acid levels revert to baseline levels when walnuts are ingested after other foods.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge at 4 degrees C.

2. Aliquot plasma into plastic vial.

3. Send plasma frozen.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	60 days
	Refrigerated	72 hours

CPT Code Information:

83497

HIAAR
616090

5-Hydroxyindoleacetic Acid, Random, Urine

Specimen Requirements:

Patient Preparation:

1. Some medications could interfere with test results. If medically feasible, for 48 hours before specimen collection, patient should not take the following medications:

-Acetaminophen (Tylenol or generic versions)

-Tryptophan containing supplements

2. For 48 hours prior to the urine collection, the patient should limit the following to one serving per day:

-Fruits

-Vegetables

-Nuts

-Caffeinated beverages or foods

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Store and send refrigerate

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	56 days
	Frozen	365 days

CPT Code Information:

83497

MAMMX 62732 6-Monoacetylmorphine (6-MAM) Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain of Custody Meconium Kit (T653)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation required

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions:

1. Collect entire random meconium specimen in the container provided. Seal and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

2. Send specimen frozen. When refrigerated, a significant percentage of 6-monoacetylmorphine will convert to morphine in less than 24 hours.

Additional Information: Specimen that arrives with a broken seal does not meet the chain-of-custody requirements.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen	14 days

CPT Code Information:

G0480

80356 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

6MAMM 6-Monoacetylmorphine (6-MAM), Confirmation, Meconium
89659

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions:

1. Collect entire random meconium (newborn's first bowel movements) specimen.

2. Send specimen frozen.

Additional Information: When refrigerated, a significant percentage of 6-monoacetylmorphine (MAM) will convert to morphine in less than 24 hours.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen	14 days

CPT Code Information:

G0480

80356 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

6MAMU 6-Monoacetylmorphine Confirmation, Random, Urine
89605

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container Tube: Plastic urine container

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not** accepted for this test.

Specimen Minimum Volume: 2.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80356 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

6MAMX 6-Monoacetylmorphine, Chain of Custody, Random, Urine
62708

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing specimen containers, seals, and documentation is required.

Specimen Volume: 10 mL

Collection Instructions: Collect a random specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Specimen Minimum Volume: 2.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80356 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

F68KD
91494

68kD (hsp-70) antibodies by Line Blot

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Frozen (preferred)	365 days
	Refrigerated	5 days
	Ambient	48 hours

CPT Code Information:

84182

7AC4 7AC4, Bile Acid Synthesis, Serum

607699

Specimen Requirements:

Patient Preparation:

1. **Fasting: 12 hours, required**; Fasting morning specimen is preferred.

2. Patient should **not use**:

a. Statins for 5 days before specimen collection

b. Bile acid sequestrants for 24 hours before specimen collection

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send specimen frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	72 hours
	Ambient	24 hours

CPT Code Information:

82542

A1R 113437 A1 Antigen Subtype, Blood

Specimen Requirements:

Container/Tube: Pink top (EDTA)

Specimen Volume: 6 mL

Pediatric Volume: 2 mL blood in 6 mL pink-top (EDTA) tube

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	72 hours

CPT Code Information:

86905

RTQPC 620307 Abnormal Prion Protein, Real-Time Quaking Induced Conversion, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

CJDE / Creutzfeldt-Jakob Disease Evaluation, Spinal Fluid

RPDE / Rapidly Progressive Dementia Evaluation, Spinal Fluid

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Frozen (preferred)	28 days	BlueTop SARSTEDT
	Refrigerated	14 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

CPT Code Information:

0584U

ABONR 113498 ABO/Rh Newborn, Blood

Specimen Requirements:

Container/Tube: Pink top (EDTA Micro tube)

Specimen Volume: 0.5 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	10 days
	Ambient	4 days

CPT Code Information:

86900

86901

ABOMR 113490 ABO/Rh, Blood

Specimen Requirements:

Container/Tube: Pink top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Blood for this test cannot be shared with other tests. Submit specimen for this test in its own vial.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	10 days
	Ambient	4 days

CPT Code Information:

86900-ABO

86901-Rh

ACAC
82757

Acacia, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ACARP
64717

Acanthamoeba species, Molecular Detection, PCR, Ocular

Specimen Requirements:

The preferred specimen for this test is corneal scraping or biopsy.

Submit only 1 of the following specimens:

Specimen Type: Fresh tissue

Sources: Ocular

Container/Tube: Sterile container

Specimen Volume: 5-10 mm(3)

Collection Instructions: Submit tissue in a sterile container with 1 mL of sterile saline, minimal essential media (MEM), or viral transport media.

Preferred Paraffin-Embedded Tissue Block:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Ocular

Container/Tube: Tissue block

Collection Instructions: Submit a FFPE tissue block to be cut and returned.

Acceptable Paraffin-Embedded Tissue Block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue (FFPE) section

Sources: Ocular

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Type: Scrapings

Sources: Eye, ocular, cornea

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect corneal scrapings using a scalpel or other sharp device to remove the outer layer of cells from the eye.

2. Swish the collection device in 1 mL of sterile saline, MEM, or viral transport media.

3. Remove the scalpel blade or sharp device from the collection container before submitting to the lab.

4. Specimens containing scalpel blades will be canceled.

Specimen Type: Swabs

Sources: Eye, ocular, cornea

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Swab must be placed into viral transport media and submitted with the specimen.

2. Specimens received without swabs will be canceled.

Additional Information: Swabs are not the preferred specimen for this test and may yield false-negative results.

Specimen Type: Contact lenses

Container/Tube: Sterile container

Specimen Volume: Entire collection

Collection Instructions:

1. Place entire contact lens in a sterile container with 1 mL sterile saline, viral transport media, or MEM.

2. Right and left lenses must be submitted individually using multiple sterile containers or in the original contact lens case. A **separate order must be created for each lens being tested.**

3. Indicate Right or Left in the specimen source.

Specimen Type: Contact lens cases without lenses

Container/Tube: Sterile container

Specimen Volume: 1 mL solution or entire case

Additional Information:

1. Depending on the type of case submitted, it may be necessary to test right and left chambers individually. **A separate order must be created for each chamber being tested.**

2. Indicate Right or Left in the specimen source.

Specimen Minimum Volume: Scrapings: 0.5 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

ACAR
82850

Acarus siro, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ACMA 37030 Acetaminophen, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	24 hours

CPT Code Information:

80143

FACES 75388 Acetoacetate, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between**

multiple orders.***

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collections Instructions:

1. Centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

82010

FCDU3 75778 Acetyl Fentanyl, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

ARBI 8338 Acetylcholine Receptor (Muscle AChR) Binding Antibody, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is

recommended prior to initiation of immunosuppressant medication.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86041

ACMFS 610029 Acetylcholine Receptor Modulating Antibody, Flow Cytometry Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

MGLE / Myasthenia Gravis/Lambert-Eaton Myasthenic Syndrome Evaluation, Serum

MGMR / Myasthenia Gravis Evaluation with Muscle-Specific Kinase (MuSK) Reflex, Serum

PAVAL / Paraneoplastic, Autoantibody Evaluation, Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86043

ACHE_9287 Acetylcholinesterase, Amniotic Fluid

Specimen Requirements:

Container/Tube: Amniotic fluid container

Specimen Volume: 1 mL

Collection Instructions: Specimen must be collected between 14 to 21 weeks gestation; 14 to 18 weeks is preferred.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Amniotic Fld	Refrigerated (preferred)	365 days	
	Frozen	365 days	
	Ambient	14 days	

CPT Code Information:

82013

ASCL1 71355 Achaete-Scute Homolog 1 (ASCL1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GAAW 606267 Acid Alpha-Glucosidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

ASMW
606264

Acid Sphingomyelinase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

SAFB
8213

Acid-Fast Smear for Mycobacterium, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Information: Saliva is **not acceptable**.

Additional Information: If a mycobacterial culture is also requested, collect 1.5 mL.

Specimen Type: Bone marrow

Container/Tube: Sterile container or green top (lithium heparin)

Specimen Volume: Entire collection

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Type: Gastric washing

Container/Tube: Sterile container

Specimen Volume: 10 mL

Collection Instructions: Neutralize specimen within 4 hours of collection with 100 mg of sodium carbonate per 5 to 10 mL of gastric wash.

Specimen Type: Respiratory

Sources: Bronchoalveolar lavage fluid, bronchial washing, sputum

Container/Tube: Sterile container

Specimen Volume: 3 mL

Collection Instructions: Collect 3 respiratory specimens for acid-fast smears and culture in patients with clinical and chest X-ray findings compatible with tuberculosis. These 3 specimens should be collected at 8- to 24-hour intervals (24 hours when possible) and should include at least 1 **first-morning** specimen.

Specimen Type: Feces

Supplies: Stool Collection Kit, Random (T635)

Container/Tube: Sterile container

Specimen Volume: 5 to10 g

Specimen Type: Tissue

Container/Tube: Sterile container

Specimen Volume: 5 to10 mm

Collection Instructions: Collect a fresh, unfixed tissue specimen. Fixed tissue is **not acceptable**.

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions: Collect a random urine specimen.

Acceptable:

Fresh tissue or body fluid are preferred over a swab specimen. Recovery of mycobacteria from swabs is generally very low yield. Only submit a swab specimen if tissue biopsy or fluid aspiration is not possible.

Specimen Type: Swab

Supplies: BD E-swab (T853)

Sources: Surgically collected wound, tissue, or body fluid

Container/Tube: Sterile, screw-capped tube containing Liquid Amies Medium with flocked swab (eg, E-swab)

Specimen Volume: Flocked swab in 1 mL of Liquid Amies Medium

Collection Instructions:

1. Before collecting specimen, wipe away any excessive amount of secretion and discharge, if appropriate.

2. Obtain secretions or fluid from source with sterile flocked swab. **Paranasal sinus collections must use a nasopharyngeal flocked swab.**

3. Place flocked swab in sterile, screw-capped tube containing 1 mL of Liquid Amies Medium.

4. If smear and culture are requested or both a bacterial culture and mycobacterial culture are requested, collect a second swab to maximize test sensitivity. Submit each swab in a separate sterile, screw-capped tube with 1 mL of Liquid Amies Medium.

Additional Information: Swabs from the following sources are **not acceptable**: respiratory fluid (eg, sputum), nasal, sinus, outer ear, mouth, throat, or scalp. Surgically collected inner ear swabs **are** acceptable.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87206

87176-Tissue processing (if appropriate)

87015-Mycobacteria culture, concentration (if appropriate)

SMACN 70551 Actin, Smooth Muscle (SMActin) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

ACT 8221 Actinomyces Culture, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Abscesses, intrauterine devices, percutaneous transtracheal aspirates, sterile body fluids, suprapubic aspirations, wounds

Supplies: Anaerobe Transport Tube (T588)

Container/Tube: Anaerobe transport tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Specimens should be collected by needle and syringe aspiration or surgical drainage.

2. Send ambient.

Specimen Type: Spinal fluid

Container/Tube: Sterile container

Specimen Volume: Entire contents of vial number 2 (minimum volume 0.5 mL)

Collection Instructions: Always send vial number 2, if possible. If not, note which vial was sent. If single vial specimen for multiple laboratories, note “single vial specimen" on card accompanying specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	72 hours

CPT Code Information:

87075-Actinomyces culture

87076-Id MALDI-TOF Mass Spec Anaerobe (if appropriate)

87153-Anaerobe identification by sequencing (if appropriate)

87176-Tissue processing (if appropriate)

APMSC 602182 Activated Partial Thromboplastin Time (APTT) Mix 1:1, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85732

APTSC 602172 Activated Partial Thromboplastin Time (APTT), Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile or as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85730

APTTP 40935 Activated Partial Thromboplastin Time, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge, remove plasma and centrifuge plasma again.

2. Aliquot plasma into plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

Additional Information: Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen (preferred)	30 days
	Ambient	4 hours

CPT Code Information:

85730

APCRV
81967

Activated Protein C Resistance V (APCRV), Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot into a separate plastic vial, leaving 0.25 mL in the bottom of the centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85307

APCRR 60547 Activated Protein C Resistance V, with Reflex to Factor V Leiden, Blood and Plasma

Specimen Requirements:

Blood and plasma are required.

Patient Preparation: Fasting: 8 hours, preferred but not required

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD) or light-blue top (3.2% sodium citrate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Within 4 hours of collection, centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma at -20 degrees C, or ideally, at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL; Whole blood: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days
Whole blood	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

85307

COGMF 113528 Acute Myeloid Leukemia (AML), Children's Oncology Group Enrollment Testing, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88275, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271 x 2, 88275-FISH Probe, Analysis; each additional probe set (if appropriate)

Specimen Requirements:

Container/Tube: Green top (sodium heparin) or yellow top (ACD solution B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

88377 (if 1 probe set)

88377 x 2 (if 2 probe sets)

88377 x 3 (if 3 probe sets)

88377 x 4 (if 4 probe sets)

88377 x 5 (if 5 probe sets)

88377 x 6 (if 6 probe sets)

88377 x 7 (if 7 probe sets)

AMLMF 614202 Acute Myeloid Leukemia (AML), Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. **It is preferable to send the first aspirate from the bone marrow collection.**

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275x1, 88291x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

Acute Porphyria Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal

testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81405

81406 x 2

81479

81479 (if appropriate for government payers)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

APLDF 622913 Acute Promyelocytic Leukemia-PML::RARA fusion, Expedited Diagnostic FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275x1, 88291x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

AHEP 56105 Acute Viral Hepatitis Profile, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.7 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot 2 mL serum into a plastic vial labeled as SST Serum, and ship frozen

(preferred).

Specimen Minimum Volume: 1.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

80074 (if all 4 initial tests are performed)

86709 (if all 4 are not performed)

86705 (if all 4 are not performed)

87340 (if all 4 are not performed)

86803 (if all 4 are not performed)

87522 (if appropriate)

87341 (if appropriate)

FACYS 75396 Acyclovir, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Lavender top or pink top (EDTA) (Plasma gel/PST are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collections Instructions:

1. Centrifuge and aliquot 1 mL of plasma into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Frozen	120 days
	Ambient	30 days

CPT Code Information:

80299

ACRN 82413 Acylcarnitines, Quantitative, Plasma

Specimen Requirements:

Patient Preparation: Collect specimen just before scheduled meal or feeding.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.1 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: 0.04 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	64 days
	Ambient	8 days

CPT Code Information:

82017

ACRNS 60644 Acylcarnitines, Quantitative, Serum

Specimen Requirements:

Patient Preparation: Collect specimen just before scheduled meal or feeding.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.04 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	60 days
	Refrigerated	21 days
	Ambient	72 hours

CPT Code Information:

82017

AGU20 Acylglycines, Quantitative, Random, Urine
608909

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Pediatric: If insufficient collection volume, submit as much specimen as possible in a single container; the laboratory will determine if volume is sufficient for testing.

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	416 days
	Refrigerated	9 days

CPT Code Information:

82542

ABADL
620311

Adalimumab Antibody, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ADALP / Adalimumab Quantitative Antibody, Serum.

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before the next dose of drug administration (trough specimen).

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

83520

INTAD
620312

Adalimumab Panel, Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

ADALP
620309

Adalimumab Quantitative with Antibody, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.4 mL Serum

Collection Instructions:

1. Draw blood immediately before the next dose of drug administration (trough specimen).

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80145

83520

ADALX Adalimumab Quantitative with Reflex to Antibody, Serum
64863

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before the next dose of drug administration (trough specimen).

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80145

83520 (if appropriate)

QNADL Adalimumab Quantitative, Serum (Non-Orderable)
620310

Specimen Requirements:

Only orderable as part of profile. For more information see ADALP / Adalimumab Quantitative with Antibody, Serum.

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.4 mL Serum

Collection Instructions:

1. Draw blood immediately before the next dose of drug administration (trough specimen).

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80145

ADMAB 623136 ADAMTS 13 Antibody, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL plasma

Collection Instructions:

1. Specimen must be collected prior to replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a separate plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

5. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, below -40 degrees C.

Specimen Stability Information: Frozen 21 months

Additional Information:

1. Double-centrifuged specimen is critical for accurate results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

83520

ADAMS 620816 ADAMTS13 Activity Assay, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Polypropylene plastic vials

Specimen Volume: 2 mL Platelet- poor plasma in 2 plastic vials each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1 mL per aliquot) into 2 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Specimen Stability Information: Frozen 2 years

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85397

ADAMP 620815 ADAMTS13 Activity with Reflex Inhibitor Profile, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Polypropylene plastic vials

Specimen Volume: 2 mL Platelet- poor plasma in 2 plastic vials each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1 mL per aliquot) into 2 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Specimen Stability Information: Frozen 2 years

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85397

85335 (if appropriate)

ADMB 620817 ADAMTS13 Inhibitor Bethesda Titer, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see ADAMP / ADAMTS13 Activity with Reflex Inhibitor Profile, Plasma.

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Polypropylene plastic vials

Specimen Volume: 2 mL Platelet-poor plasma in 2 plastic vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1 mL per aliquot) into 2 separate plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Specimen Stability Information: Frozen 2 years

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85335

ADMI
621149

ADAMTS13 Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ADAMP / ADAMTS13 Activity with Reflex Inhibitor Profile, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

AADAM 620818 ADAMTS13 Profile Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see ADAMP / ADAMTS13 Activity with Reflex Inhibitor Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

APBCS 615861 Adaptor Protein 3 Beta2 (AP3B2) Antibody, Cell-Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

APBCC 615860 Adaptor Protein 3 Beta2 (AP3B2) Antibody, Cell-Binding Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a part of a profile or reflex. For more information see:

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

APBTS 616109 Adaptor Protein 3 Beta2 (AP3B2) Antibody, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

APBTC
616110

Adaptor Protein 3 Beta2 (AP3B2) Antibody, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Preferred: Vial number 1

Acceptable: Any vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

ADMPU
615908

Addiction Medicine Profile with Reflex, 22 Drug Classes, High Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine container

Specimen Volume: 30 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 30 mL in 1 plastic bottle.

3. No preservative.

Additional Information:

1. No specimen substitutions.

2. Submitting less than 30 mL may compromise the ability to perform all necessary testing.

3. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 20 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

80307

G0482 (appropriate to use when 0-3 reflexes are required)

G0483 (appropriate to use when 4 reflexes are required)

80347 (if appropriate for select payers)

80364 (if appropriate for select payers)

80326 (if appropriate for select payers)

80323 (if appropriate for select payers)

[Clinical Toxicology CPT Code Client Guidance]

ADSTM
62206

Additional Flow Stimulant (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

86353

AGSTM
62208

Additional Flow Stimulant, LPAGF (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

86353

MGSTM
62207

Additional Flow Stimulant, LPMGF (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

86353

XSRM 607838 Additional Sample for Reflex Oligoclonal Banding, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MSP3 / Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot serum in a plastic vial.

2. Label specimen as serum.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days

VID2 45455 Additional Testing Virus Ident (Bill Only)

Specimen Requirements:

This test is for Billing Purposes Only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87253

FADCF
75666

Adenosine Deaminase in CSF

Specimen Requirements:

Collection Container/Tube: Leak-proof container

Submission Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge cerebrospinal fluid specimen at ambient temperature.

2. Transfer 0.5 mL fluid (supernatant) into a sterile vial and freeze.

3. Send frozen. **Specimen must remain frozen** until received at performing lab.

4. Indicate source on requisition.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

84311

FADFL
75665

Adenosine Deaminase in Peritoneal Fluid

Specimen Requirements:

Specimen Type: Peritoneal fluid (ascites, abdominal fluid, paracentesis fluid

Collection Container/Tube: Leak-proof container

Submission Container/Tube: Plastic vial

Specimen volume: 0.5 mL

Collection Instructions:

1. Centrifuge specimen at ambient temperature.

2. Transfer 0.5 mL into a plastic vial and freeze

3. Send frozen. **Specimen must remain frozen** until received at the performing lab.

4. Indicate source on requisition.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Peritoneal	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

84311

FADDP 75667 Adenosine Deaminase, Pericardial Fluid

Specimen Requirements:

Specimen Type: Pericardial fluid

Collection Container/Tube: Leak-proof container

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge pericardial fluid specimen at ambient temperature.

2. Transfer 0.5 mL of fluid (supernatant) into a plastic vial and freeze.

3. Send frozen. **Specimen must remain frozen** until received at performing lab.

4. Indicate source on requisition.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

84311

FADPF
75664

Adenosine Deaminase, Pleural Fluid

Specimen Requirements:

Specimen Type: Pleural Fluid

Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1.Collect pleural fluid in a leak proof container.

2. Centrifuge specimen at ambient temperature.

3. Transfer 0.5 mL into a plastic vial and freeze.

3. Ship frozen.

Note: Specimen must remain frozen until received at performing lab.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Pleural Fluid	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

84311

FADBC Adenosine Deaminase, RBC
75553

Specimen Requirements:

Container/Tube: Lavender top (EDTA), pink top (K2 EDTA), or green top (sodium or lithium heparin)

Specimen Volume: 1 mL

Collection Instructions:

1. Send whole blood in original tube.

2. Send refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	15 days
	Ambient	15 days

CPT Code Information:

84311

ADVQU Adenovirus DNA Detection and Quantification, Plasma
622150

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL (plasma)

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot plasma into plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

87799

ADV
70352

Adenovirus Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LADV
89074

Adenovirus, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Pleural, peritoneal, ascites, pericardial, or amniotic

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or washing, sputum, or tracheal aspirate

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Stool

Supplies: Stool Collection Kit, Random (T635)

Container/Tube: Sterile container

Specimen Volume: 1 g

Specimen Type: Swab

Sources: Nasal, throat, respiratory, genital, or ocular

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT, M4, M5, Bartels, or Jiangsu) and E-Swab or Culturette

Specimen Volume: Entire specimen

Collection Instructions: Place swab back into a multimicrobe media.

Specimen Type: Tissue

Supplies: M4-RT (T605)

Container/Tube: Sterile container containing 1 mL to 2 mL of sterile saline or multimicrobe medium (M4-RT, M4, M5, Bartels, or Jiangsu)

Specimen Volume: Entire collection

Collection Instructions: Collect fresh tissue specimen.

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Collect a random urine specimen.

Specimen Minimum Volume: Body fluid and respiratory specimens: 0.5 mL
Spinal fluid and urine: 0.3 mL
Stool: 0.5 g
Swab or Tissue: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

AK5CS 623401 Adenylate Kinase 5 (AK5) IgG, Cell Based Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

AK5CC 623400 Adenylate Kinase 5 (AK5) IgG, Cell Based Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

AK5TS 623486 Adenylate Kinase 5 (AK5) IgG, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AK5TC 623485 Adenylate Kinase 5 (AK5) IgG, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AK5IS 623399 Adenylate Kinase 5 (AK5) IgG, Tissue Immunofluorescence, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

AK5IC 623398 Adenylate Kinase 5 (AK5) IgG, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

AK1
607455

Adenylate Kinase Enzyme Activity, Blood

Specimen Requirements:

Collection Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA) or yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

AKC
608421

Adenylate Kinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

ACC
604986

Adrenal Mass Panel, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic urine tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for a full 24 hours (required) and record the total volume.

2. Do not add preservatives. **Specimens containing preservatives will be canceled.**

3. Entire 24 hour collection must be mixed well prior to aliquoting into a 5 mL plastic tube.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	90 days
	Refrigerated	14 days

CPT Code Information:

0015M

ACTHI 70351 Adrenocorticotropic Hormone (ACTH) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

ACTH 8411 Adrenocorticotropic Hormone, Plasma

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: Ice-cooled, lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Morning (7 a.m.-10 a.m.) specimen is desirable.

2. Collect with a pre-chilled lavender top (EDTA) tube and transport to the laboratory on ice.

3. Within 2 hours of collection centrifuge at refrigerated temperature and immediately separate plasma from cells.

4. Immediately freeze plasma.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	28 days
	Refrigerated	3 hours
	Ambient	2 hours

CPT Code Information:

82024

AMLFA 622380 Adult Acute Myeloid Leukemia Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x6, 88275x3, 88291x1-FISH Probe, Analysis, Interpretation; 3 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

ADLTX
62710

Adulterants Survey, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-custody kit containing the specimen containers, seals, and documentation required.

Specimen Volume: 1.5 mL

Collection Instructions: Collect a random specimen without preservative in the provided container, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 1.5 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

ADULT
29345

Adulterants Survey, Random, Urine

Specimen Requirements:

Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 1.5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information: Submitting less than 1.5 mL may compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

ISAE
45246

Aerobe Identification by Sequencing (Bill Only)

Specimen Requirements:

This test is for billing purposes only. This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87153

AERMC
604916

Aeromonas Culture, Feces

Specimen Requirements:

Patient Preparation: Patient should **not** use barium or bismuth for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Specimen Type: Preserved feces

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-Aeromonas Culture, Feces-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

ALAGP 618030 Alagille Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are

inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test

(including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81407

81479

81479 (if appropriate for government payers)

ALT
8362

Alanine Aminotransferase (ALT) (GPT), Serum

Specimen Requirements:

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

84460

ALB24
606718

Albumin, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube ,5 mL (T914)

Container/Tube: Plastic urine tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. Mix well before taking 4-mL aliquot.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

82043

ALBFL
606622

Albumin, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

82042

RALB
603287

Albumin, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

82043

82570

ALBR
609731

Albumin, Random, Urine

Specimen Requirements:

Patient Preparation: Heavy exercise should be avoided prior to collection.

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

82043

82570

RALB1
606730

Albumin, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-ALBR / Albumin, Random, Urine

-RALB / Albumin, Random, Urine.

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

82043

ALB
8436

Albumin, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	150 days
	Frozen	120 days

CPT Code Information:

82040

ALBS1 Albumin, Serum (Non-Orderable)

610525

Specimen Requirements:

Only orderable as part of profile. For more information see: SFIG / Cerebrospinal Fluid (CSF) IgG Index Profile, Serum and Spinal Fluid.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	120 days
	Ambient	7 days

CPT Code Information:

82040

ALBSF
68001

Albumin, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Submit CSF from collection vial number 2 (preferred, not required).

2. Label specimen as CSF.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82042

A_CR
606732

Albumin/Creatinine Ratio (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ALBR / Albumin, Random, Urine

RALB / Albumin, Random, Urine.

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

FALBU Albuterol, Serum/Plasma

90309

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Submit only one of the following specimens:

Serum

Specimen Type: Serum

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a plain, red-top tube(s). **Serum gel tube is not acceptable.**

2. Centrifuge and send 3 mL of serum refrigerated in a plastic, preservative-free vial.

Note: Label specimen appropriately (serum).

Plasma

Specimen Type: Plasma

Container/Tube: Lavender top or pink top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in an EDTA (lavender top or pink top) tube(s). **Plasma gel tube is not acceptable.**

2. Centrifuge and send 3 mL of EDTA plasma refrigerated in a plastic, preservative-free vial.

Note: Label specimen appropriately (plasma).

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	30 days
	Frozen	365 days
	Ambient	30 days

CPT Code Information:

80299

ALS
606872

Aldolase, Serum

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

82085

ALDNA
15150

Aldosterone with Sodium, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) should be discontinued for 4 to 6 weeks before testing. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube 5mL (T914)

Container/Tube: 2 Plastic, 5-mL tubes

Specimen Volume: 10 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative **at start of collection**. Use 15 mL of 50% acetic acid for children under the age of 5 years. This preservative is intended to achieve a pH of between approximately 2 and 4.

2. Collect urine for a full 24 hours (required) and record the total volume.

3. Place 5 mL of well mixed, 24-hour urine in plastic, 5-mL tube and label as Aldosterone.

4. Place 5 mL of well mixed, 24-hour urine in plastic, 5-mL tube and label as Sodium.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

82088-Aldosterone

84300-Sodium

ALDU
8556

Aldosterone, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) should be discontinued for 4 to 6 weeks before specimen collection. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Urine tubes, 10 mL (T068)

Container/Tube: Plastic, urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative **at start of collection**. Use 15 mL of 50% acetic acid for children under the age of 5 years. This preservative is intended to achieve a pH of between approximately 2 and 4.

2. Collect urine for a full 24 hours (required) and record the total volume.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82088

APIVC 65425 Aldosterone, Inferior Vena Cava, Plasma

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) **should be** discontinued for 4 to 6 weeks before specimen collection. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL Plasma

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma:1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	28 days
	Ambient	4 days

CPT Code Information:

82088

AIVC
6503

Aldosterone, Inferior Vena Cava, Serum

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) should be discontinued for 4 to 6 weeks before specimen collection. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.8 mL Serum

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

82088

APLAV 65427 Aldosterone, Left Adrenal Vein, Plasma

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) **should be** discontinued for 4 to 6 weeks before specimen collection. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL Plasma

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	28 days

	Ambient	4 days

CPT Code Information:

82088

ALAV 6349 Aldosterone, Left Adrenal Vein, Serum

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) should be discontinued for 4 to 6 weeks before specimen collection. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.8 mL Serum

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

82088

PALD
65424

Aldosterone, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	28 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

82088

APRAV
65426

Aldosterone, Right Adrenal Vein, Plasma

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) **should be** discontinued for 4 to 6 weeks before specimen collection. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL Plasma

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	28 days
	Ambient	4 days

CPT Code Information:

82088

ARAV 6348 Aldosterone, Right Adrenal Vein, Serum

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) should be discontinued for 4 to 6 weeks before specimen collection. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.8 mL Serum

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for

approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

82088

ALDS
8557

Aldosterone, Serum

Specimen Requirements:

Patient Preparation: Spironolactone (Aldactone) should be discontinued for 4 to 6 weeks before testing. The plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL Serum

Collection Instructions:

1. See Renin-Aldosterone Studies for detailed instructions.

2. The recommended collection time is 8 a.m., after the patient is active for approximately 2 hours. Try to collect the specimen as close to that time as possible and no later than 10 a.m.

3. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

82088

FALPE
57945

Alfalfa (Medicago sativa) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

ALKT
619694

ALK Mutation Analysis, Next-Generation Sequencing,

Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the following slides:

1 slide stained with hematoxylin and eosin

AND

10 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred), or a minimum of 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual
81479

ALPI
622349

Alkaline Phosphatase Isoenzymes, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ALKP / Alkaline Phosphatase, Total and Isoenzymes, Serum.

Patient Preparation:

Fasting: 8 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red top tubes, immediately aliquot into a plastic vial.

3. For serum gel tubes, serum may sit on gel refrigerated but must be aliquoted within 7 days.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84080

ALP
8340

Alkaline Phosphatase, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	60 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84075

ALKP
622157

Alkaline Phosphatase, Total and Isoenzymes, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red top tubes, immediately aliquot into a plastic vial.

3. For serum gel tubes, serum may sit on gel refrigerated but must be aliquoted within 7 days.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84075

84080

FABP2
57698

Allergic Bronchopulmonary Aspergillosis Panel II

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86331

86001

86003

82785

ALLOI
88888

Allo-isoleucine, Blood Spot

Specimen Requirements:

Supplies:

-Card-Blood Spot Collection (Filter Paper) (T493)

-Card-Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Blood Spot Collection (Filter Paper)

Acceptable: Local newborn screening card, Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper, Munktell filter paper, Postmortem Screening Card

Specimen Volume: 2 Blood spots

Collection Instructions:

1. **Do not use** device or capillary tube containing EDTA to collect specimen.

2. At least 1 spot should be complete and unpunched.

3. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

4. Include type of feeding information on the collection card.

5. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

6. Do not expose specimen to heat or direct sunlight.

7. Do not stack wet specimens.

8. Keep specimen dry.

Additional Information:

1. For collection instructions see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	59 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Refrigerated	59 days	FILTER PAPER

CPT Code Information:

82136

FALFG 57519 Almond Food IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

ALM
82882

Almond, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ALPS
82449

Alpha Beta Double-Negative T Cells for Autoimmune Lymphoproliferative Syndrome, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x2

86359

ALDEF
607710

Alpha Defensin, Lateral Flow Assay, Synovial Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red-top tube without preservative or clot activators

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Synovial Fluid	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

83518

AFSH
71768

Alpha Follicle Stimulating Hormone Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

WAGDR Alpha Globin Cluster Locus Deletion/Duplication, Blood
621361

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: Blood: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81269

AGDD 620985 Alpha Globin Cluster Locus Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be

attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81269

88235-Tissue culture for amniotic fluid (if appropriate)

88240-Cryopreservation (if appropriate)

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells (if appropriate)

WASEQ Alpha Globin Gene Sequencing, Varies
61362

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix whole blood.

2. Send whole blood specimen in the original tube. **Do not aliquot**

Specimen Stability Information: Refrigerate 30 days(preferred)/Ambient 14 days

Specimen Type: Extracted DNA from whole blood

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from whole blood within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood: 1 mL; Extracted DNA: 50 mcL at 50 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81259-HBA1/HBA2; full sequence

FALG
57663

Alpha Lactalbumin IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FA1GP 57736 Alpha-1-Acid Glycoprotein

Specimen Requirements:

Patient Preparation:

Fasting: Overnight, preferred but not required

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

82985

A1AFS 604982 Alpha-1-Antitrypsin Clearance, Feces and Serum

Specimen Requirements:

Both feces and serum are required. Blood **must** be drawn during the stool collection period.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial

Specimen Type: Feces

Supplies: Stool Containers - 24, 48, 72 Hour Kit (T291)

Container/Tube: Stool container

Specimen Volume: Entire collection

Collection Instructions:

1. Collect a 24-hour fecal collection.

2. If no specimen is obtained within 24 hours, extend collection time to 48 to 72 hours. Document duration.

Specimen Minimum Volume: Homogenized feces: 1 mL; Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days
Serum	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

82103 x 2

AATRP 70350 Alpha-1-Antitrypsin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

A1APP 26953 Alpha-1-Antitrypsin Phenotype, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.25 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82103

82104

A1ALC
61767

Alpha-1-Antitrypsin Proteotype S/Z, LC-MS/MS, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.25 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82103

82542

82104 (if appropriate)

A1AF 182 Alpha-1-Antitrypsin, Random, Feces

Specimen Requirements:

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions: Collect a random fecal specimen.

Specimen Minimum Volume: Homogenized stool: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

82103

AAT 8161 Alpha-1-Antitrypsin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82103

A124
610363

Alpha-1-Microglobulin, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No added preservative preferred.

3. Mix well before taking 4-mL aliquot.

Additional Information: For multiple collections see Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

83883

RA1U
610364

Alpha-1-Microglobulin, Random, Urine

Specimen Requirements:

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83883

82570

A2PI
602169

Alpha-2 Plasmin Inhibitor, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85410

A2M
9270

Alpha-2-Macroglobulin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83883

AAGTS 623083 Alpha-3 Beta-4 Ganglionic Neuronal Acetylcholine Receptor Binding Antibody, Flow Cytometry Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AAGFS 622854 Alpha-3 Beta-4 Ganglionic Neuronal Acetylcholine Receptor Binding Antibody, Flow Cytometry, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:
-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum
-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

	Ambient	72 hours

CPT Code Information:

86255

AAMY
82866

Alpha-Amylase, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For one allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

ALFP
70353

Alpha-Fetoprotein (AFP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

L3AFP 88878 Alpha-Fetoprotein (AFP) L3% and Total, Hepatocellular Carcinoma Tumor Marker, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum Gel

Acceptable: Red Top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	5 days

CPT Code Information:

82107

AFP
8162

Alpha-Fetoprotein (AFP) Tumor Marker, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

82105

AFPPT
61534

Alpha-Fetoprotein (AFP), Peritoneal Fluid

Specimen Requirements:

Sources: Peritoneal, abdominal, ascites, paracentesis (peritoneal washings are **not acceptable**)

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 2 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Peritoneal	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86316

MAFP1 113382 Alpha-Fetoprotein (AFP), Single Marker Screen, Maternal, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Do not collect specimen after amniocentesis as this could affect results.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Additional Information:

1. Collect blood between 15 weeks, 0 days and 22 weeks, 6 days.

2. Initial or repeat testing is determined in the laboratory at the time of report and will be reported accordingly. To be considered a repeat test for the patient, the testing must be within the same pregnancy and trimester, with interpretable results for the same test, and both tests are performed at Mayo Clinic.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

82105

AFPSF
8876

Alpha-Fetoprotein (AFP), Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Vial number 1

Acceptable: Any vial

Specimen Volume: 1 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

86316

AFPA
9950

Alpha-Fetoprotein, Amniotic Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 0.75 mL

Collection Instructions: Do not centrifuge.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Amniotic Fld	Refrigerated (preferred)	7 days	
	Ambient	7 days	

CPT Code Information:

82106

82013 (if appropriate)

FUCW 8814 Alpha-Fucosidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

82657

AGAW 606261 Alpha-Galactosidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

AGAS 8784 Alpha-Galactosidase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	24 hours

CPT Code Information:

82657

WASQR 47958 Alpha-Globin Gene Sequencing, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-REVE2 / Erythrocytosis Evaluation, Blood

-HBEL1 / Hemoglobin Electrophoresis Evaluation, Blood

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-MEV1 / Methemoglobinemia Evaluation, Blood

-THEV1 / Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	14 days

CPT Code Information:

81259-HBA1/HBA2; full sequence

IDUAW
606276

Alpha-L-Iduronidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

ALFA
82897

Alpha-Lactalbumin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

MANN 62511 Alpha-Mannosidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

82657

APGH
9003

Alpha-Subunit Pituitary Tumor Marker, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	7 days

CPT Code Information:

82397

ASYN
70635

Alpha-Synuclein Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

ASYNC 622855 Alpha-Synuclein Protein Aggregates, Spinal Fluid

Specimen Requirements:

Supplies: Sterile Specimen Tube, 6 mL (T485)

Container/Tube: Sterile tube

Specimen Volume: 1 mL

Collection Instructions:

1. Perform lumbar puncture and discard the first 1 to 2 mL of cerebrospinal fluid (CSF).

2. Collect CSF directly into a sterile tube.

3. Inspect specimen for visible discoloration. Specimen must be clear and colorless to perform testing. **Do not centrifuge**.

4. Freeze sample upright prior to placing in transport container.

5. Collection instructions can also be found on Spinal Fluid Specimen Collection Instructions for Alpha-Synuclein Protein Aggregates.

Note: Polypropylene tubes are recommended for transport.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	60 days
	Refrigerated	14 days

CPT Code Information:

0393U

ABCRS 70636 Alpha/Beta Crystallin IHC, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

ALPRT 70593 Alport (Collagen IV Alpha 5 and Alpha 2)

Immunofluorescent Stain, Renal Biopsy

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Unstained slides (unfixed)

Source: Kidney tissue or skin tissue

Slides: 1 Slide

Collection Instructions: Submit 1 frozen tissue unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections should be 4-microns thick, **centered on the slide,** and shipped on dry ice.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Acceptable:

Specimen Type: Unfixed tissue block (O.C.T)

Source: Kidney tissue or skin tissue

Specimen Volume: Entire specimen

Collection Instructions:

1. Embed in O.C.T. compound.

2. Freeze specimen and ship on dry ice.

Acceptable:

Specimen Type: Wet tissue

Source: Kidney tissue or skin tissue

Supplies: Renal Biopsy Kit (T231)

Container/Tube: Renal Biopsy Kit, Zeus/Michel's

Specimen Volume: Entire specimen

Collection Instructions:

1. Collect specimens according to the instructions in Renal Biopsy Preparation Instructions.

2. If standard immunoglobulin and complement immunofluorescence has already been performed, ship the residual frozen tissue (must contain glomeruli) on dry ice.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Frozen	

CPT Code Information:

88346

ALPGP 618044 Alport Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient 4 days (preferred)/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81407

81408 x 2

81479

81479 (if appropriate for government payers)

ALTN
82910

Alternaria tenuis, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AH50
88676

Alternative Complement Pathway, Functional, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. Do **not** collect a specimen for at least 48 hours following plasma exchange.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

ALU
8828

Aluminum, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Within 4 hours of completion of 24-hour collection, refrigerate specimen.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82108

ALWB
622056

Aluminum, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Greiner Bio-One VACUETTE TUBE 6 mL NH Trace Elements Sodium Heparin tube (T819)

-Metal Free Specimen Vial (T173)

Container/Tube: Greiner Bio-One VACUETTE TUBE 6 mL NH Trace Elements Sodium Heparin tube

Specimen Volume: 1 mL

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	14 days

CPT Code Information:

82108

AL
8373

Aluminum, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been

administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free B-D Tube (No Additive), 6 mL (T184)

-Metal Free Specimen Vial (T173)

Container/Tube: 6-mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 1.2 mL Serum

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

82108

ALUCR 610838 Aluminum/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82570

82108

ALCU
610839

Aluminum/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ALUCR / Aluminum/Creatinine Ratio, Random, Urine.

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert.

Submission Container/Tube: Plastic urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82108

FOXOF 35281 Alveolar Rhabdomyosarcoma (ARMS), 13q14 (FOXO1 or FKHR) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

TFE3F 35319 Alveolar Soft Part Sarcoma (ASPS)/Renal Cell Carcinoma (RCC), Xp11.23 (TFE3), FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Preferred: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded (FFPE) tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Acceptable: Slides

Collection Instructions: Four consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Two consecutive, unstained, 5 micron-thick sections placed on positively charged slides and 1 hematoxylin and eosin-stained slide

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

ADEVL
607273

Alzheimer Disease Evaluation, Spinal Fluid

Specimen Requirements:

Supplies:

Alzheimer's Disease Evaluation (ADEVL) Collection Kit (T836)

CSF AD Biomarker Tubes (T833; also included in T836)

Container/Tube:

Preferred: Sarstedt CSF False Bottom Tube 63.614.625 (2.5 mL)

Acceptable: Sarstedt 72.703.600 (1.5 mL) or Sarstedt 72.694.600 (2 mL)

Specimen Volume: 1.5 to 2.5 mL

Collection Instructions:

1. Perform lumbar puncture and discard the first 1 to 2 mL of cerebrospinal fluid (CSF).

2. Collect CSF directly into one of the listed collection tubes until the tube is at least 50% full.*

3. Send CSF specimen in original collection tube. Do **not aliquot**.

Note: Polystyrene collection tubes are not acceptable. Exposure of CSF to polystyrene tubes may result in falsely low Abeta42 concentrations. For more information see Cautions.

*The Alzheimer's Association consensus protocol for handling of CSF for clinical measurements of Abeta42 and tau recommends using the drip method for CSF collection and directly collecting into a low-bind polypropylene tube. Although some clinicians prefer the syringe pull method due to speed of collection, the drip method reduces the risk of Abeta42 binding to the plastic of any syringe used.

4. Collection instructions can also be found on Spinal Fluid Specimen Collection Instructions for Alzheimer Disease Evaluation (T967).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Refrigerated (preferred)	14 days	BlueTop SARSTEDT
	Frozen	60 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

CPT Code Information:

AB42P-82234

PTAUP-84393

TTAUP-84394

FAMAN 91132 Amantadine, Serum or Plasma

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Serum

Collection Container/Tube: Red top (serum gel/SST are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 3 mL serum into a plastic vial.

2. Send refrigerate.

Specimen Type: Plasma

Collection Container/Tube: Green top (heparin) tube (plasma gel/PST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Plasma

Collection Instructions:

1. Centrifuge and aliquot 3 mL plasma into a plastic vial.

2. Send refrigerate.

Additional Information:

1. Trough levels are the most reproducible.

2. If storing specimens for more than 3 days, refrigerate or frozen storage is optimal.

Specimen Minimum Volume: Serum/Plasma: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	21 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

80299

PAMIK 37032 Amikacin, Peak, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80150

RAMIK 37033 Amikacin, Random, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80150

TAMIK
37031

Amikacin, Trough, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80150

AAMSD
60200

Amino Acids, Maple Syrup Urine Disease Panel, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, required; infants should have specimen collected before next feeding (2-3 hours without total parenteral nutrition if possible)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect specimen and place on wet ice. Note: Thrombin-activated tubes should not be used for collection.

2. Centrifuge immediately or within 4 hours of collection if specimen is kept at refrigerated temperature.

3. Being careful to ensure that no buffy coat is transferred, aliquot plasma into a plastic vial and freeze.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	14 days

CPT Code Information:

82136

AAQP
9265

Amino Acids, Quantitative, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, required; infants should have specimen collected before next feeding (2-3 hours without total parenteral nutrition if possible)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL plasma

Collection Instructions:

1. Collect specimen and place on wet ice. Note: Thrombin-activated tubes **should not be used** for collection.

2. Centrifuge immediately or within 4 hours of collection if the specimen is kept at refrigerated temperature.

3. Being careful to ensure that no buffy coat is transferred, aliquot plasma into a plastic vial and freeze.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	14 days

CPT Code Information:

82139

AAPD
60475

Amino Acids, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Specimen Volume: 2 mL

Collection Instructions: Collect a random urine specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	70 days
	Refrigerated	14 days

CPT Code Information:

82139

AACSF
81934

Amino Acids, Quantitative, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.2 mL

Collection Instructions: Collect specimen from second collection vial.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen	14 days

CPT Code Information:

82139

AAUCD
60202

Amino Acids, Urea Cycle Disorders Panel, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, required; infants should have specimen collected before next feeding (2-3 hours without total parenteral nutrition if possible)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect specimen and place on wet ice. Note: Thrombin-activated tubes **should not be used** for collection.

2. Centrifuge immediately or within 4 hours of collection if the specimen is kept at refrigerated temperature.

3. Being careful to ensure that no buffy coat is transferred, aliquot plasma into a plastic vial and freeze.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	14 days

CPT Code Information:

82136

AACYL
621427

Aminoacylase-1 Deficiency, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Pediatric: If insufficient collection volume, submit as much specimen as possible in a single container; the laboratory will determine if volume is sufficient for testing.

Collection Instructions:

1. Collect a random urine specimen (first morning void preferred).

2. No preservative.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	90 days

	Refrigerated	60 days

CPT Code Information:

82542

AIHL 609784 Aminoglycoside-Induced Hearing Loss, Targeted Variant Testing, Droplet Digital PCR, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies.

Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture

fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81401

ALADW 31895 Aminolevulinic Acid Dehydratase, Washed Erythrocytes

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as ethanol suppresses aminolevulinic acid dehydratase activity, leading to false-positive results.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. **Record volume of packed cells and the total volume of the specimen.**

4. Discard supernatant plasma.

5. Wash packed erythrocytes 2 times by resuspension of at least an equal amount of cold 0.9% saline, mix, and centrifuge for 5 minutes at 2000 rpm, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	7 days
	Refrigerated	14 days
	Ambient	4 days

CPT Code Information:

82657

ALAD
88924

Aminolevulinic Acid Dehydratase, Whole Blood

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as ethanol suppresses aminolevulinic acid dehydratase activity, leading to false-positive results.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Specimen Volume: 4 mL; full tube

Collection Instructions: Refrigerate specimen as soon as possible.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	7 days
	Ambient	4 days

CPT Code Information:

82657

ALAUR 61547 Aminolevulinic Acid, Urine

Specimen Requirements:

Patient Preparation: Patient **must not** consume any alcohol for at least 24 hours before specimen collection.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	28 days
	Frozen	45 days

CPT Code Information:

82135

AMIO 9247 Amiodarone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood no sooner than 12 hours (trough value) after last dose or immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

80151

AMTRP 63506 Amitriptyline and Nortriptyline, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

NH3V
35130

Ammonia, Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plain, plastic screw-top tube

Specimen Volume: 0.5 mL or more

Collection Instructions:

1. Specimens should be put on ice immediately after collection.

2. Centrifuge at refrigerated temperature (4 degrees C).

3. Aliquot plasma into plastic screw-top tube. Keep on ice.

4. Freeze plasma within 2 hours of collection.

Specimen Minimum Volume: See Specimen Collection

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	7 days
	Refrigerated	2 hours

CPT Code Information:

82140

AMMO
606643

Ammonium, 24 Hour, Urine

Specimen Requirements:

Supplies:

-Sarstedt 5 mL Aliquot Tube (T914)

-Diazolidinyl Urea (Germall) 5.0 mL (T822)

Container/Tube: Plastic vial

Specimen Volume: 4 mL

Collection Instructions:

1. Add 5 mL of diazolidinyl urea (Germall) as preservative at start of collection or refrigerate specimen during and after collection.

2. Collect urine for 24 hours.

3. Aliquot urine into plastic vial.

4. Specimens with pH >8 may indicate bacterial contamination and testing will be cancelled. Do not attempt to adjust pH as it will adversely affect results.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82140

RAMBO Ammonium, Random, Urine

606709

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82140

AMOBS Amobarbital, Serum

8325

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before the next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

AMOXY
82663

Amoxicillin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AMPAS
623598

Amphetamine Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	15 days

CPT Code Information:

G0480

80324 (if appropriate for select payers)

80359 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

AMPMX 62712 Amphetamine-Type Stimulants Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain of Custody Meconium Kit (T653) includes the specimen containers, seals, and documentation required.

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Additional Information:

1. Specimen that arrives with a broken seal does not meet the chain of custody requirements.

2. The laboratory recommends sending chain-of-custody specimens by overnight shipment.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

G0480

80324 (if appropriate for select payers)

80359 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

AMPHM
84371

Amphetamine-Type Stimulants Confirmation, Meconium

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Meconium	Frozen (preferred)	28 days	
	Ambient	28 days	
	Refrigerated	28 days	

CPT Code Information:

80324

80359

G0480 (if appropriate)

AMPHX 62711 Amphetamines Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation required

Specimen Volume: 5 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 5 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

G0480

80325 (if appropriate for select payers)

80359 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

AMPHU 8257 Amphetamines Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

G0480

80325 (if appropriate for select payers(

80359 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FCDU5 75780 Amphetamines, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

APHTS 618165

Amphiphysin Antibody Titer Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-PAVAL / Paraneoplastic, Autoantibody Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-SPPS / Stiff-Person Spectrum Disorders Evaluation, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Serum

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

APHTC 618164 Amphiphysin Antibody Titer Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-SPPC / Stiff-Person Spectrum Disorders Evaluation, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

FAMBS
75881

Amphotericin B Susceptibility Testing

Specimen Requirements:

Complete and submit with specimen: Fungus Testing Lab Request form.

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Isolate

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Provide the name of organism/species, source, and collection date with the specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999 - Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 - mould-MIC microdilution or agar dilution (if appropriate)

87186 - yeast-MIC microdilution or agar dilution (if appropriate)

AMP
82664

Ampicillin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AMBF
606595

Amylase, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial fluid

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

82150

AMISO
604930

Amylase, Isoenzymes, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82150 x 2

AMLPC 60078 Amylase, Pancreatic Cyst Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plain vial

Specimen Volume: 1 mL

Additional Information: A minimum of 0.5 mL is required for testing; specimens less than 0.5 mL may be rejected.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Pancreatic Cyst Fluid	Frozen (preferred)	30 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

82150

Amylase, Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	30 days
	Ambient	7 days

CPT Code Information:

82150

AAH
70349

Amyloid A (Hepatic) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FAMBT 75904 Amyloid Beta-Protein

Specimen Requirements:

Collection Container/Tube: Lavender (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. 3 mL of EDTA plasma should be collected and separated as soon as possible.

2. Freeze EDTA plasma immediately after separation.

3. Ship frozen

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	30 days

CPT Code Information:

83519

APPI 70357 Amyloid Precursor Protein (APP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

AMPIP 620247 Amyloid Protein Identification, Paraffin, Mass Spectrometry

Specimen Requirements:

Specimen Type: Formalin-fixed or B5-fixed paraffin-embedded tissue block

Collection Instructions:

1. Do not send fixed tissue slides for testing. Testing can only be done on paraffin-embedded tissue blocks.

2. If Congo red stain has already been performed, send Congo red stained slide along with the tissue block.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
AMYLOID	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88313

82542 (if appropriate)

88380 (if appropriate)

TTRX 83674 Amyloidosis, Transthyretin-Associated Familial, Reflex, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	4 days
	Ambient	4 days

CPT Code Information:

82542

81404 (if appropriate)

ANAID
45010

Anaerobe Ident (Bill Only)

Specimen Requirements:

This test is for Billing Purposes Only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87076

ISAN
45255

Anaerobe Identification by Sequencing (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87153

BATTA
80931

Anaerobe Suscep Battery (Bill Only)

Specimen Requirements:

This test is for Billing Purposes Only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87186

SANA
45337

Anaerobe Suscep per Agent (Bill Only)

Specimen Requirements:

This test is for Billing Purposes Only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87181

AHPV
607829

Anal Pap Human Papillomavirus (HPV) with Genotyping, High-Risk Types, PCR, Varies

Specimen Requirements:

Submit one of the following specimens:

Preferred

Specimen Type: Anus or rectum

Supplies: ThinPrep Media with Spatula and Brush Kit (T434)

Collection Container/Tube: Brush

Submission Container/Tube: ThinPrep

Specimen Volume: A minimum of 20 mL or entire collection

Collection Instructions:

1. Visualize the anal opening by retracting the buttocks.

2. Insert brush past internal anal sphincter until it abuts the distal rectal wall (4-5cm).

3. Rotate the brush 360 degrees and maintain lateral pressure on brush against the walls of the anus. The brush should bow slightly due to the pressure.

4. While rotating, slowly withdraw the brush.

5. Rotate at least 10 times while withdrawing the brush. This should take 20 to 30 seconds.

6. Immediately place brush in the ThinPrep solution and swish at least 20 to 30 seconds.

7. Dispose of the brush.

8. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

9. Bag ThinPrep specimens individually as they tend to leak during transport.

10. Place labels on the vial and on the biohazard bag.

Acceptable

Specimen Type: Anus or rectum

Supplies: PreservCyt Vial (T536)

Collection Container/Tube: Dacron/polyester swab

Submission Container/Tube: PreservCyt solution vial

Specimen Volume: A minimum of 20 mL or entire collection

Collection Instructions:

1. Visualize the anal opening by retracting the buttocks.

2. Insert Dacron or polyester swab past internal anal sphincter until it abuts the distal

rectal wall (4-5cm).

3. Rotate the swab 360 degrees and maintain lateral pressure on swab against the walls of the anus. The swab should bow slightly due to the pressure.

4. While rotating, slowly withdraw the swab.

5. Rotate at least 10 times while withdrawing the swab. This should take 20 to 30 seconds.

6. Immediately place swab in the ThinPrep solution. Swish at least 20 to 30 seconds in the ThinPrep solution.

7. Dispose of the swab.

8. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

9. Bag ThinPrep specimens individually as they tend to leak during transport.

10. Place labels on the vial and on the biohazard bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	21 days

CPT Code Information:

87626

ATPCO 614402 Anal ThinPrep Cytology with Human Papillomavirus (HPV) Co-Test, Varies

Specimen Requirements:

Submit one of the following specimens:

Preferred

Specimen Type: Anus or rectum

Supplies: ThinPrep Media with Spatula and Brush Kit (T434)

Collection Container/Tube: Brush

Submission Container/Tube: ThinPrep

Specimen Volume: A minimum of 20 mL or entire collection

Collection Instructions:

1. Visualize the anal opening by retracting the buttocks.

2. Insert brush past internal anal sphincter until it abuts the distal rectal wall (4-5cm).

3. Rotate the brush 360 degrees and maintain lateral pressure on brush against the walls of the anus. The brush should bow slightly due to the pressure.

4. While rotating, slowly withdraw the brush.

5. Rotate at least 10 times while withdrawing the brush. This should take 20 to 30 seconds.

6. Immediately place brush in the ThinPrep solution and swish at least 20 to 30 seconds.

7. Dispose of the brush.

8. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

9. Bag ThinPrep specimens individually as they tend to leak during transport.

10. Place labels on the vial and on the biohazard bag.

Acceptable

Specimen Type: Anus or rectum

Supplies: PreservCyt Vial (T536)

Collection Container/Tube: Dacron/polyester swab

Submission Container/Tube: PreservCyt solution vial

Specimen Volume: A minimum of 20 mL or entire collection

Collection Instructions:

1. Visualize the anal opening by retracting the buttocks.

2. Insert Dacron or polyester swab past internal anal sphincter until it abuts the distal rectal wall (4-5cm).

3. Rotate the swab 360 degrees and maintain lateral pressure on swab against the walls of the anus. The swab should bow slightly due to the pressure.

4. While rotating, slowly withdraw the swab.

5. Rotate at least 10 times while withdrawing the swab. This should take 20 to 30

seconds.

6. Immediately place swab in the ThinPrep solution. Swish at least 20 to 30 seconds in the ThinPrep solution.

7. Dispose of the swab.

8. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

9. Bag ThinPrep specimens individually as they tend to leak during transport.

10. Place labels on the vial and on the biohazard bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient	21 days	THIN PREP

CPT Code Information:

88112

87626

ANAP
81157

Anaplasma phagocytophilum (Human Granulocytic Ehrlichiosis) Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86666

ALKOT 620405 Anaplastic Lymphoma Kinase (ALK [OTI1A4]) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

ALKLC 615713

Anaplastic Lymphoma Kinase for Lung Cancer, Immunohistochemistry

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-Primary

88341-If additional IHC

ANPAT
70318

Anatomic Pathology Consultation, Wet Tissue

Specimen Requirements:

Submit 1 of the following specimens:

Specimen Type: Lung biopsy

Supplies:

-Pathology Packaging Kit (T554)

-Formalin, 10% Buffered Neutral for Tissue (T118)

-Michel's Transport Media for Immunofluorescent Testing on Tissue (T321)

Specimen Volume: Biopsy specimens as delineated below

Collection Instructions:

For testing to be performed, both of the following must be submitted:

a. For light microscopy:

Preferred: A portion of lung tissue in 10% neutral buffered formalin. Specimens submitted in formalin needs to be fully submerged with the optimal approximate specimen volume of 10:1 or higher.

Acceptable: A representative hematoxylin and eosin (H and E)-stained slide

b. For immunofluorescence, a portion of the unfixed lung tissue in Michel's transport media

Specimen Type: Cardiac biopsy

Supplies:

-Pathology Packaging Kit (T554)

-Formalin, 10% Buffered Neutral for Tissue (T118)

-Gluta (Trumps) (T130)

Specimen Volume: Biopsy specimens as delineated below

Collection Instructions:

1. Collect 4 to 6 biopsy specimens and submit all in 10% neutral buffered formalin. Specimens submitted in formalin needs to be fully submerged with the optimal approximate specimen volume of 10:1 or higher. If formalin biopsy specimens are not available, light microscopy slides would also be acceptable.

2. If electron microscopy (EM) is also requested, submit at least 1 biopsy into 2.5% to 3% buffered glutaraldehyde preservative (Trumps) along with the formalin biopsy specimens.

Additional Information: If the question is drug toxicity, storage disease, or if the patient is younger than 25 years, send at least 1 biopsy specimen in 2.5% to 3% buffered glutaraldehyde (Trumps) in case EM is needed for diagnosis.

Specimen Type: Cardiac explant

Supplies: Pathology Packaging Kit (T554)

Specimen Volume: Entire explant specimen

Collection Instructions:

1. Fix entire explant specimen in formalin for a minimum of 24 hours.

2. Following fixation, lightly wrap specimen in formalin-soaked paper towel, place in a biohazard bag with a small amount of formalin, seal, place in another biohazard bag, and seal.

Specimen Type: Enucleated eye

Supplies:

-Pathology Packaging Kit (T554)

-Formalin - 10% Histo Prep, 45 mL (T117)

Specimen Volume: Entire eye specimen

Collection Instructions:

1. Immediately after removal, place specimen in approximately 300 mL of 10% neutral buffered formalin and fix for 48 hours.

2. Enucleated eye should not be opened or punctured.

3. After the 48-hour fixation, the specimen should be transferred from the larger, 300 mL container to a smaller container with fixative (Formalin - 10% Histo Prep, 45 mL) for shipping.

Specimen Type: Iris, conjunctiva, cornea, and other small biopsies from eye

Supplies:

-Pathology Packaging Kit (T554)

-Formalin - 10% Histo Prep, 45 mL (T117)

Specimen Volume: Entire biopsy specimen

Collection Instructions:

1. Place iris, conjunctiva, cornea, and other small biopsies on a paper mount (a piece of filter paper or other porous paper). This helps to keep the specimen from curling when it is fixed.

2. Immediately, gently place paper-mounted specimen into 10% formalin (Formalin - 10% Histo Prep, 45 mL).

3. In cases where margins are important, placing sutures at the margins of resection with accompanying explanatory draft will help orientation of the specimen in the lab.

Specimen Type: Vitreous fluid/Aqueous humor

Supplies:

-Pathology Packaging Kit (T554)

-Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic screw-top vial

Specimen Volume: Entire specimen

Collection Instructions:

1. Place the liquid (vitreous or aqueous) concentrate (preferred) or washing (diluted, acceptable) into a small tube with a screw-top cap.

2. Refrigerate the specimen immediately (**do not freeze**).

Specimen Stability Information: Refrigerated

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

88300 (if appropriate)

88302 (if appropriate)

88304 (if appropriate)

88305 (if appropriate)

88307 (if appropriate)

88309 (if appropriate)

ANCH
82345

Anchovy, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AREC
70358

Androgen Receptor Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FAGES
75656

Androstanediol Glucuronide (Endocrine Sciences)

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Stability Information: Refrigerated (preferred) 6 days/Ambient 6 days/Frozen 3 years

Specimen Minimum Volume: Serum: 0.5 mL, Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	6 days

	Ambient	6 days
	Frozen	

CPT Code Information:

82154

ANST 9709 Androstenedione, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82157

AECDP 621547 Angioedema and Complement Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the required volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

'Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test

may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

MASF 35859 Angiosarcoma, MYC (8q24) Amplification, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue,

lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 4 unstained

Collection Instructions: Submit 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

FACEC 57824 Angiotensin Converting Enzyme, CSF

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Transfer 1 mL of spinal fluid to a sterile vial.

2. Send frozen.

Note: Gadolinium contrast agents have been reported to inhibit ACE activity. Therefore, CSF containing gadolinium-based contrast agents should not be submitted to the laboratory for evaluation.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	180 days
	Refrigerated	7 days

CPT Code Information:

82164

ACE 603622 Angiotensin Converting Enzyme, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	24 hours

CPT Code Information:

82164

FANG1 Angiotensin I, Plasma

75911

Specimen Requirements:

Patient preparation: Patient should be on a normal sodium diet, 110 mEq of sodium. Patient should be in a recumbent posture for at least 30 minutes prior to collection of specimen. Diuretics, mineralocorticoids, glucocorticoids, estrogens, oral contraceptives, and ACTH medications and sodium, potassium, and posture all affect Angiotensin levels.

Specimen Type: Plasma

Collection Container/Tube: EDTA

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a lavender-top (EDTA) tube(s)

2. Centrifuge and aliquot 3 mL of plasma into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	30 days

CPT Code Information:

83520

ANISP
82857

Anisakis, Parasite, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ANSE
82487

Anise, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FANSE 57520 Annatto Seed (Bixa orellana) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

ANNEX 70355 Annexin-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FABAA 75662 Anti-bestrophin Autoantibodies

Specimen Requirements:

Submit only one of the following specimens:

Serum:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 5 mL serum refrigerated in a plastic vial.

Plasma:

Draw blood in a lavender-top (EDTA) tube(s). Spin down and send 5 mL EDTA plasma refrigerated in a plastic vial.

Complete and submit with specimen:

1. Completed OHSU Ocular request form

2. Clinical history

3. Referring physician information (name & phone number)

NOTE: Without this information, testing cannot be completed.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	7 days

CPT Code Information:

84182

AC1Q 621511 Anti-C1q Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83520

FACN1
75620

Anti-cN-1A Ab (NT5c1A) IBM

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Within 1 hour of specimen collection, centrifuge and aliquot 2 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.5 mL, Note: this volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516

COL7
616879

Anti-Collagen type VII, IgG Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

83516

ADNAS
80204

Anti-DNase B Titer, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

86215

FAEAB 91854 Anti-Enterocyte Antibodies

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 2 mL of serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	

CPT Code Information:

88346

88350 x 2

AGNTS 43434

Anti-Glial/Neuronal Nuclear Antibody-Type 1 (AGNA-1) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-PAVAL / Paraneoplastic, Autoantibody Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AGNTC 43443 Anti-Glial/Neuronal Nuclear Antibody-Type 1 (AGNA-1) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 2 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

FANTI 57892 Anti-IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Ambient (preferred)	7 days
	Frozen	28 days

	Refrigerated	7 days

CPT Code Information:

83516

FAMDA Anti-MDA-5 Ab (CADM-140)

75622

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.3 mL Note: This volume does NOT allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516

FFMI2 Anti-Mi-2 Ab

75591

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516

FCLNE
91321

Anti-Phosphatidylcholine Ab

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Allow specimen to clot for 30 minutes at room temperature.

2. Centrifuge and aliquot 3 mL of serum into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	14 days

CPT Code Information:

83520/x3

FPHET 91322 Anti-Phosphatidylethanolamine Panel

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Allow blood to clot for 30 minutes.

1. Centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	14 days

CPT Code Information:

83520 x 3

FAPMA 75623 Anti-PM/Scl-100 Ab

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.3 mL Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

86235

FARWB 57647 Anti-retinal autoantibodies follow up, WB

Specimen Requirements:

Note: This test should only be ordered as follow up to previous Anti-retinal autoantibodies performed at Oregon Health Sciences University (OHSU) Ocular Immunology Laboratory.

Submit only one of the following specimens:

Serum:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 5 mL serum refrigerated in a plastic vial.

Plasma:

Draw blood in a lavender-top (EDTA) tube(s). Spin down and send 5 mL EDTA plasma refrigerated in a plastic vial.

Complete and submit with specimen:

1. Completed OHSU Ocular request form
2. Clinical history
3. Referring physician information (name & phone number)

-NOTE: Without this information, testing cannot be completed.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	7 days

CPT Code Information:

84182

FAS1A Anti-SAE1 Ab, IgG

75634

Specimen Requirements:

Collection Container/Tube: Serum gel or red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.5 mL Note: This volume does NOT allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516

FFRFT 75624 Anti-Synthetase Profile

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 1 mL; Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516 x 4

86235

FATHO 75619 Anti-Th/To Ab

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.3 mL Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516

FAT1G
75632

Anti-TIF-1gamma Antibody

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.5 mL Note: This volume does NOT allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

86235

FAU3R
75631

Anti-U3 RNP Antibodies (Fibrillarin)

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.3 mL Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

86235

ABIDR
113389

Antibody Identification, Blood and Serum

Specimen Requirements:

Both blood and serum are required.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Specimen Type: Plasma/Blood

Collection Container/Tube: 6-mL Pink-top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume:

3 mL Plasma

3 mL Red blood cells (RBCs)

Collection Instructions:

1. Centrifuge and aliquot plasma into a plastic vial.

2. Label specimen as EDTA plasma or EDTA RBCs as appropriate.

3. Send both tubes.

Specimen Type: Serum/Blood

Collection Container/Tube: 10-mL Red top

Submission Container/Tube: Plastic vial

Specimen Volume:

5 mL Serum

5 mL RBCs

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum or clotted RBCs as appropriate.

3. Send both tubes.

Specimen Minimum Volume: Blood: 6 mL in EDTA; Pediatric: 2 mL Serum

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

86870

86860-(if appropriate)

86886-(if appropriate)

86880 x 3 (if appropriate)

86905-(if appropriate)

86978-(if appropriate)

ABYSR 113387 Antibody Screen with Reflexed Antibody Identification, Blood

Specimen Requirements:

Container/Tube: Pink top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	4 days
	Ambient	4 days

CPT Code Information:

86850

ABTIR 113390 Antibody Titer, Blood and Serum

Specimen Requirements:

Blood cells, plasma, and serum are required.

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Specimen Type: Plasma/Blood

Collection Container/Tube: 6-mL Pink top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume:

3 mL Plasma

3 mL Red blood cells (RBCs)

Collection Instructions:

1. Centrifuge and aliquot plasma into plastic vial.

2. Label specimens as EDTA plasma or EDTA RBCs as appropriate.

3. Send both tubes.

Specimen Type: Serum/Blood

Collection Container/Tube: 10-mL Red top

Submission Container/Tube: Plastic vial

Specimen Volume:

5 mL Serum

5 mL RBCs

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Label specimens as serum or clotted RBCs as appropriate.

3. Send both tubes.

Specimen Minimum Volume: Blood: 6 mL EDTA
Pediatric: 2 mL Serum

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

86886

86870 (if appropriate-per panel tested)

86860 (if appropriate)

86880 x 3 (if appropriate)

86905- (if appropriate)

86978 (if appropriate)

81403 (if appropriate)-Internal only

ENAE
89035

Antibody to Extractable Nuclear Antigen Evaluation, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235 x 6

FASQN
57740

Antidepressant Drug Profile, Quantitative, Urine

Specimen Requirements:

Collection Container/Tube: Plastic preservative-free urine container

Specimen Volume: 3 mL

Collection Instructions:

1. Collect 3 mL from a random urine collection in a preservative free plastic urine container.

2. Send refrigerated.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

80333

80337

80369

FADDS
57772

Antidepressant Drug Screen, Qualitative

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Plasma

Collection Container/Tube: Gray top (NaF/oxalate), green top (heparin), or lavender top (EDTA) (plasma gel/PST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Plasma

Collection Instructions:

1. Centrifuge and aliquot 3 mL plasma into a plastic vial.

2. Send refrigerate.

Specimen Type: Serum

Collection Container/Tube: Red top (serum gel/SST are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 3 mL serum into a plastic vial.

2. Send refrigerate.

Specimen Type: Urine

Submission Container/Tube: Plastic preservative-free container

Specimen Volume: 3 mL

Collection Instructions:

1. Collect 3 mL from a random urine collection in a preservative-free plastic urine container.

2. Send refrigerated.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80333

80337

80369

AHPRU
618553

Antihypertension Panel, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

G0481

80377 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

MMLYP 81602 Antimicrobial Susceptibility Panel, Yeast, Varies

Specimen Requirements:

Specimen Type: Organism in pure culture

Supplies: Infectious Container, Large (T146)

Container/Tube: Sabouraud's dextrose agar slant

Specimen Volume: Infecting yeast isolate

Collection Instructions: Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87186

MMLRG 81601 Antimicrobial Susceptibility, Acid-Fast Bacilli, Rapidly Growing, Varies

Specimen Requirements:

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook 7H10 agar slant or other appropriate media

Specimen Volume: Pure isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87186

MMLSG 34805 Antimicrobial Susceptibility, Acid-Fast Bacilli, Slowly Growing, Varies

Specimen Requirements:

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook 7H10 agar slant or other appropriate media

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87186

ZMMLS 8073 Antimicrobial Susceptibility, Aerobic Bacteria, Varies

Specimen Requirements:

Preferred:

Specimen Type: Bacterial isolate swab

Supplies:

E-Swab (T853)

Infectious Container, Large (T146)

Container/Tube:

Preferred: E-Swab collection and transport system

Acceptable: Flocked swab and 1-mL liquid Amies transport medium in 12 x 80 mm tube

Collection Instruction:

1. Perform isolation of bacterial isolate.

2. Utilize the flocked swab to obtain an adequate sample of pure cultured isolate. **Do not submit mixed cultures.**

3. Place swab into the transport system containing 1-mL liquid Amies transport medium.

4. If needed, break off end of swab and close the transport tube.

5. Place the transport system into the secondary infectious container for shipment.

6. Each isolate must be submitted under a separate order.

Note: For the following organisms, submit an agar slant or other appropriate media to ensure viability upon arrival to the laboratory; *Neisseria gonorrhoeae*, *Campylobacter* species, *Helicobacter pylori*, and any other fastidious organism.

Acceptable:

Specimen Type: Pure culture of organism from source cultured

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Collection Instructions:

1. Perform isolation of bacterial isolate.

2. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures.**

3. Place the agar slant or other appropriate media into the secondary infectious container for shipment.

4. Each isolate must be submitted under a separate order.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

87077-Ident by MALDI-TOF mass spec (if appropriate)

87185-Beta lactamase (if appropriate)

87186-Antimicrobial Susceptibility, Aerobic Bacteria, MIC-per organism for routine battery (if appropriate)

87181-Susceptibility per drug and per organism for drugs not in routine battery (if appropriate)

87150-H pylori + Clarithro Resistance PCR (if appropriate)

87150-mecA PCR (if appropriate)

MMLSA 56031 Antimicrobial Susceptibility, Anaerobic Bacteria, Minimal Inhibitory Concentration, Varies

Specimen Requirements:

Specimen Type: Organism in pure culture

Supplies:

-Anaerobic Transport Tube (T588)

-Infectious Container, Large (T146)

Container/Tube:

Preferred: Anaerobic transport tube

Acceptable: Thioglycollate broth or any other suitable anaerobic transport system. **Agar plates are NOT acceptable.**

Collection Instructions:

1. Perform isolation of infection bacteria.

2. Organism must be in pure culture and actively growing. **Do not submit mixed cultures.**

3. Place specimen in a large infectious container and label as an etiologic agent/infectious substance if appropriate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87186-Antimicrobial Susceptibility, Anaerobic Bacteria, MIC

87076-Organism Ref for ID, Anaerobic Bact (if appropriate)

87076-Anaerobe Ident (if appropriate)

87076-Id MALDI-TOF Mass Spec Anaerobe (if appropriate)

87153-Anaerobe ident by sequencing (if appropriate)

87150-Identification by PCR (if appropriate)

87185-Beta Lactamase (if appropriate)

87181-Anaerobe Susceptibility per Agent (if appropriate)

87150-mecA PCR (if appropriate)

TB1LN
35994

Antimicrobial Susceptibility, Mycobacterium tuberculosis Complex, First Line, Varies

Specimen Requirements:

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook 7H10 agar slant

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87188 x 3-Antimicrobial Susceptibility, *Mycobacterium tuberculosis* Complex, Broth Method

MMLNS
82019

Antimicrobial Susceptibility, Nocardia species and other Aerobic Actinomycetes, Varies

Specimen Requirements:

Specimen Type: Organism in pure culture

Supplies: Infectious Container, Large (T146)

Container/Tube:

Preferred: Middlebrook 7H10 agar slant without antimicrobials

Acceptable: Sabouraud's dextrose agar slant or similar media without antimicrobials (eg, 7H11 agar slant, LJ, MGIT [7H9] broth media)

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87186

AMH1 608824 Antimullerian Hormone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	7 days

CPT Code Information:

82166

AN2TS 43432 Antineuronal Nuclear Antibody Type 2 (ANNA-2) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AN2TC 43441 Antineuronal Nuclear Antibody Type 2 (ANNA-2) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days

	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AN3TS 43433 Antineuronal Nuclear Antibody Type 3 (ANNA-3) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AN3TC 43442 Antineuronal Nuclear Antibody Type 3 (ANNA-3) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AN1TS 43431 Antineuronal Nuclear Antibody-Type 1 (ANNA-1) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

AN1TC 43440 Antineuronal Nuclear Antibody-Type 1 (ANNA-1) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

VASC 83012 Antineutrophil Cytoplasmic Antibodies Vasculitis Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516 x 2

86036 x 2 (if appropriate)

ANA2
9026

Antinuclear Antibodies (ANA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86038

NAIFA 65161 Antinuclear Antibodies, HEp-2 Substrate, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	28 days

CPT Code Information:

86039

RAIFA 622521 Antinuclear Antibodies, HEp-2 Substrate, IgG, with Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86039

IM_04 623464 Antinuclear Antibodies, HEp-2 Substrate, with Reflex, Comment (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see RFLXP / Antinuclear Antibodies, HEp-2, Reflex Panel.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	28 days

RFLXP 624112 Antinuclear Antibodies, HEp-2, Reflex Panel (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see RAIFA / Antinuclear Antibodies, HEp-2 Substrate, IgG, with Reflex, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86225

86235 x5

ASO
80205

Antistrep-O Titer, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collections Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

86060

ATTF
9030

Antithrombin Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

3. Heparin treatment may lower plasma antithrombin.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85300

AATTF 614506 Antithrombin Activity, with Reflex to Antithrombin Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or ideally, at or below -40 degrees C.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

3. Heparin treatment may lower plasma antithrombin.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85300 - AATTF

85301 – ATTI (if appropriate)

ATTI
9031

Antithrombin Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, at-40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85301

GNANT 619005 Antithrombin Deficiency, SERPINC1 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

AATTA 621379 Antithrombin Summary Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see AATTF / Antithrombin Activity, with Reflex to Antithrombin Antigen, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

APIXA
65848

Apixaban, Anti-Xa, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen should be collected 2 to 4 hours (peak) after a dose or just prior (trough) to the next dose for apixaban concentrations.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Plasma Na Cit	Frozen	42 days

CPT Code Information: 80299

FAPIX
75395

Apixaban, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Lavender top or pink top (EDTA) (Plasma gel/PST are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collections Instructions:

1. Centrifuge and aliquot 2 mL of plasma into a plastic vial.

2. Send refrigerated.

Specimen Stability Information: Ambient 29 days/Refrigerated 29 days/Frozen (-20 degrees C) 2 years

Specimen Minimum Volume: Plasma: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	29 days
	Ambient	29 days
	Frozen	

CPT Code Information:

80299

APOL1
605251

APOL1 Genotype, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: Blood: 0.4 mL
Saliva/DNA: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

APOAB 607593 Apolipoprotein A1 and B, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	8 days
	Frozen	60 days
	Ambient	24 hours

CPT Code Information:

82172 x 2

APOA1 607591 Apolipoprotein A1, Serum

Specimen Requirements:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge and aliquot within 2 hours of collection.

2. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	8 days
	Frozen	60 days
	Ambient	24 hours

CPT Code Information:

82172

APOLB
624438

Apolipoprotein B, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	8 days
	Frozen	60 days
	Ambient	24 hours

CPT Code Information:

82172

APOEG Apolipoprotein E Genotyping, Blood
35358

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81401-*APOE (apolipoprotein E)* (eg, hyperlipoproteinemia type III, cardiovascular disease, Alzheimer disease), common variants (eg, *2, *3, *4)

FAPLG
57629

Apple IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

APPL
82712

Apple, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

APR
82835

Apricot, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ARBOP
83267

Arbovirus Antibody Panel, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86651 x 2-California virus (La Crosse) encephalitis antibody, IgG and IgM

86652 x 2-Eastern equine encephalitis antibody, IgG and IgM

86653 x 2-St. Louis encephalitis antibody, IgG and IgM

86654 x 2-Western equine encephalitis antibody, IgG and IgM

ABOPC 83897 Arbovirus Antibody Panel, IgG and IgM, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Vial number 2

Acceptable: Any vial

Specimen Volume: 0.7 mL

Collection Instructions: Submit specimen collected in vial 2, if possible. If not, note which vial from which the aliquot was obtained.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86651 x 2-California Virus (La Crosse) Encephalitis Antibody Panel, IgG and IgM, Spinal Fluid

86652 x 2-Eastern Equine Encephalitis Antibody Panel, IgG and IgM, Spinal Fluid

86653 x 2-St. Louis Encephalitis Antibody Panel, IgG and IgM, Spinal Fluid

86654 x 2-Western Equine Encephalitis Antibody Panel, IgG and IgM, Spinal Fluid

ARGIN 70359 Arginase-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FARI
57112

Aripiprazole

Specimen Requirements:

Submit only 1 of the following specimens

Specimen Type: Serum

Collection Container/Tube: Red top (serum gel/SST are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 2 mL serum into a plastic vial.

2. Send refrigerate.

Additional Information: Trough levels are the most reproducible.

Specimen Type: Plasma

Collection Container/Tube: Green top (sodium heparin) (plasma gel/PST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Centrifuge and aliquot 2 mL plasma into a plastic vial.

2. Send refrigerate.

Additional Information: Trough levels are the most reproducible.

Specimen Minimum Volume: Serum/Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	25 days
	Ambient	72 hours

CPT Code Information:

80299

ARVGG
617127

Arrhythmogenic Cardiomyopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will

be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81439

Arsenic Occupational Exposure, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

ASUOE / Arsenic Occupational Exposure with Reflex, Random, Urine

HMUOE / Heavy Metal Occupational Exposure, with Reflex, Random, Urine

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82175

ASUOE 608895 Arsenic Occupational Exposure, with Reflex, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before specimen collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

3. For industrial exposure monitoring, the recommended timing for specimen collection is at the end of the work week.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic vial or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 6 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82175

82570

SPASU 609383 Arsenic Speciation, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. For the 48-hour period prior to start of collection, as well as during the collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection..

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: For multiple collections see Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82175

SPAS 607691 Arsenic Speciation, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Urine Tubes, 10 mL (T068)

-Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic vial or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82175

ASU24 48537 Arsenic with Reflex, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. For 48-hour period prior to start of collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert.

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert.

Specimen Volume: 6 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82175

ASB
8645

Arsenic, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, a specimen **should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original collection tube. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82175

ASHA 8651 Arsenic, Hair

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Head, beard, mustache, chest, pubic

Specimen Volume: 0.2 g

Collection Instructions: Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Hair	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

82175

ASNA 89848 Arsenic, Nails

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Fingernails or toenails

Specimen Volume: 0.2 g

Collection Instructions:

1. Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

2. Clippings should be taken from all 10 fingernails or toenails.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Nail	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

82175

ASCU 608900 Arsenic/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

ASUCR / Arsenic/Creatinine Ratio, with Reflex, Random, Urine

HMUCR / Heavy Metal/Creatinine Ratio, with Reflex, Random, Urine

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82175

ASUCR 608905 Arsenic/Creatinine, Ratio, with Reflex, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before specimen collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 6 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82175

82570

FART
57913

Artichoke (Cynara scolymus) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

ARSU
8777

Arylsulfatase A, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL tube

Specimen Volume: 6 mL

Collection Instructions:

1. Collect a 24-hour urine specimen.

2. No added preservative.

3. Refrigerate specimen during collection.

Additional Information: For multiple collections see Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	14 days

CPT Code Information:

84311

ARSAB
623012

Arylsulfatase A, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Card-Blood Spot Collection (Filter Paper)

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing EDTA (preferred) or heparin and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

2. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry and freeze (with desiccant if necessary).

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Frozen (preferred)	28 days	FILTER PAPER

	Refrigerated	14 days	FILTER PAPER

CPT Code Information:

82657

ARSAW 8779 Arylsulfatase A, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

82657

ARSBB 616834 Arylsulfatase B, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

83864 (if appropriate)

ARSBW
616835

Arylsulfatase B, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

ASCRI
82764

Ascaris, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

VITC
42362

Ascorbic Acid (Vitamin C), Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required; infants should have specimen collected before next feeding

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin) (plasma gel/PST are not acceptable)

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Immediately place specimen on wet ice and process within 2 hours of collection.

2. Centrifuge at 4 degrees C, aliquot plasma into amber vial to protect from light.

3. Freeze plasma immediately, ideally at or below -60 degrees C, protected from light.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Frozen	14 days	LIGHT PROTECTED

CPT Code Information:

82180

ASPAR
82478

Asparagus, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AST
8360

Aspartate Aminotransferase (AST) (GOT), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84450

FASPE 57947 Aspen (Populus tremuloides) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

ASPAG
84356

Aspergillus (Galactomannan) Antigen, Serum

Specimen Requirements:

Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Avoid exposure of specimen to atmosphere to prevent sample contamination from environment.

2. Centrifuge and send specimen in original serum gel collection tube. **Do not aliquot serum or open tube.**

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Refrigerated (preferred)	14 days	SERUM GEL TUBE
	Frozen	14 days	SERUM GEL TUBE

CPT Code Information:

87305

FASAB
75571

Aspergillus Antibodies, Quantitative, DID

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge immediately after coagulation (30 minutes) to prevent hemolysis.

2. Aliquot 1 mL of serum into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

86606 x 3

ASPBA
61009

Aspergillus Antigen, Bronchoalveolar Lavage

Specimen Requirements:

Container/Tube: Sterile, leak-proof container

Note: Specimen trap collection containers (with suction catheters attached) will be rejected due to high-risk of leakage and contamination upon opening. Avoid use of these for bronchoalveolar lavage specimens.

Specimen Volume: 2 mL

Additional Information:

1. If specimen transfer into an acceptable sterile container is necessary, perform specimen transfer in a biosafety cabinet.

2. For staff safety, the following double bagging of samples must be followed:

-a. Place closed specimen container in an individual, sealed 4 x 6 Biohazard Bag (T877).

-b. Place Biohazard bag in a Refrigerate Specimen Bag (T229).

-c. Multiple sealed Biohazard bags can go into a single transfer bag.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Lavage	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87305

FAFE
57910

Aspergillus flavus IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

ASP
82911

Aspergillus fumigatus, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SASP 9678 Aspergillus fumigatus, IgG Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86001

FASPG
75681

Aspergillus IgG Precipitins Panel

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86331 x 6

ASPG
86324

Aspergillus niger, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ADMA
607697

Asymmetric Dimethylarginine, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of plasma into a plastic vial.

2. Send specimen frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	90 days
	Ambient	7 days

	Refrigerated	7 days

CPT Code Information:

82542

ATRX
70360

ATRX Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

AHUGP
618016

Atypical Hemolytic Uremic Syndrome (aHUS)/Thrombotic Microangiopathy (TMA) /Complement 3 Glomerulopathy (C3G) Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal

essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404
81479
81479 (if appropriate for government payers)

AHUSD 64881 Atypical Hemolytic Uremic Syndrome Complement Panel, Serum and Plasma

Specimen Requirements:

Both serum and plasma are required for this test.

Patient Preparation:

1. Fasting: 12 hours, preferred but not required

2. Do **not** collect specimens for at least 48 hours following plasma exchange.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Specimen Type: Serum

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.75 mL Serum total in **3 separate plastic vials** (see Collection Instructions)

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C.

3. Aliquot serum into **3 separate plastic vials**:

a. Aliquot 0.5 mL into a plastic vial labeled COM3

b. Aliquot 0.5 mL into a plastic vial labeled C3HUS/C4HUS/FBCA/FHCA

c. Aliquot 0.75 mL into a plastic vial labeled AH503

4. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Type: Plasma

Collection Container/Tube:

Preferred: Lavender top (K2 EDTA)

Acceptable: Lavender top (K3 EDTA), light-blue top (sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma total in **2 separate plastic vials**, each containing 0.75 mL

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Centrifuge between 1000 and 2000 x *g* for 10 minutes at 4 degrees C.

3. Aliquot plasma into **2 separate plastic vials**, each containing 0.75 mL.

4. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the plasma is aliquoted and frozen.

Specimen Minimum Volume: Serum: 1.5 mL; Plasma: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days
Serum	Frozen	14 days

CPT Code Information:

86160 x 6

86161

86162

AHLP
619372

AudioloGene Hearing Loss Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will **not** be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81430

81431

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

AUPU
82855

Aureobasidium pullulans, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

APIN
82803

Australian Pine, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ALDG2 Autoimmune Liver Disease Panel, Serum
619516

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86381

86039

86015

86015-Titer (if appropriate)

ALPSG 619746 Autoimmune Lymphoproliferative Syndrome (ALPS) Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume

must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FARP 75446 Autoimmune Retinopathy Panel by Immunoblot (ARP)

Specimen Requirements:

Submit the following required information with specimen:

1. Completed OHSU Ocular request form

2. Clinical history

3. Referring physician information (name & phone number)

-NOTE: Without this information, testing cannot be completed.

Submit only 1 of the following specimens:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send refrigerate.

Specimen Type: Plasma

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen volume: 5 mL Plasma

Collection Instructions:

1. Centrifuge and aliquot plasma into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	7 days

CPT Code Information:

84182 x 8

AUTOG 620092 Autoinflammatory Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FAVCG Avocado IgG
57690

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

AVOC Avocado, IgE, Serum
82812

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigered (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AIAES 606974 Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

	Ambient	72 hours

CPT Code Information:

86255 x 10

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN3TS (if appropriate)

86255 APBCS (if appropriate)

86256 APBTS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 IG5TS (if appropriate)

84182 MP1BS (if appropriate)

86256 PC2TS (if appropriate)

FAZAT
91934

Azathioprine (Imuran) as 6-Mercaptopurine

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL of sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

CTNBT 616499 B-Catenin Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2)

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin stained and 10 unstained
Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.
Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81479

COGBF 113530 B-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), Children's Oncology Group Enrollment Testing, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88275, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271 x 2, 88275-FISH Probe, Analysis; each additional probe set (if appropriate)

BALAF 609537 B-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), FISH, Adult, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x4,88275 x2, 88291 - FISH Probe, Analysis, Interpretation; 2 probe sets

88271 x2, 88275 - FISH Probe, Analysis; each additional probe set (if appropriate)

88271 - FISH Probe (if appropriate)

BALMF 614215 B-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x2, 88275 x1, 88291 x1- FISH Probe, Analysis, Interpretation; 1 probe sets

88271 x2, 88275 x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

88271 x1 -FISH Probe; coverage for sets containing 3 probes (if appropriate)

BCELL B-Cell and Antibody Deficiency Gene Panel, Varies

620106

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination

studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

B-Cell CD40 Expression by Flow Cytometry, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

88184

BALLM
608251

B-Cell Lymphoblastic Leukemia Monitoring, Measurable/Minimal Residual Disease Detection, Flow Cytometry, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as whole blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA), green top (sodium heparin)

Specimen Volume: 6 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 2 mL
Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	72 hours

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker
88185 x 9-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)
88188-Flow Cytometry Interpretation, 9 to 15 Markers

BLBLF 609451 B-Cell Lymphoblastic Leukemia/Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. T**esting may be canceled** if sufficient tumor tissue is not presen**t**.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained for each probe set

Collection Instructions:

1. If individual probe sets are chosen: Submit 1 slide stained with hematoxylin and eosin and, for each probe set ordered, submit 2 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

2. If a complete B-lymphoblastic leukemia/lymphoma (BLBL) panel is ordered: Submit 1 slide stained with hematoxylin and eosin and 20 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained per probe set requested

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

BLYM 65878 B-Cell Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. T**esting may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. For each probe set ordered, 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If ordering MYC, 4 unstained slides are necessary; the break-apart MYC and the MYC/IGH D-FISH probe sets are performed simultaneously.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377 (if 1 probe set)

88377 x 2 (if 2 probe sets)

88377 x 3 (if 3 probe sets)

88377 x 4 (if 4 probe sets)

88377 x 5 (if 5 probe sets)

88377 x 6 (if 6 probe sets)

JLYMF
614564

B-Cell Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods will not be accepted; provide fixation method used.

Additional Information: Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained slide and 1 unstained slide for each probe set plus an additional unstained slide.

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. For each probe set ordered, submit 1 consecutive, unstained, 4 to 5 micron-thick sections placed on positively charged slides, plus 1 additional unstained slide.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained slide and 1 unstained slide for each probe set

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377 (if 1 probe set)

88377 x 2 (if 2 probe sets)

88377 x 3 (if 3 probe sets)

88377 x 4 (if 4 probe sets)

BLPMF
614227

B-Cell Lymphoma, Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88275, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271 x 2, 88275-FISH Probe, Analysis; each additional probe set (if appropriate)

BCMA
623359

B-Cell Maturation Antigen (BCMA), Technical Component only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

IABCS
88800

B-Cell Phenotyping Profile for Immunodeficiency and Immune Competence Assessment, Blood

Specimen Requirements:

Two separate EDTA whole blood specimens are required: 1 refrigerate and 1 ambient transport temperature.

For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Type: Whole blood for TBBS / Quantitative Lymphocyte Subsets: T, B, and Natural Killer (NK) Cells, Blood

Container/Tube: 4 mL Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as TBBS.

3. Ship ambient.

Specimen Stability Information: Ambient <52 hours

Specimen Type: Whole blood for IABC / B-Cell Phenotyping Screen for Immunodeficiency and Immune Competence Assessment, Blood

Container/Tube: Lavender top (EDTA)

Specimen Volume:

< or =14 years: 4 mL

>14 years: 10 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen IABC.

3. Ship refrigerate.

Specimen Stability Information: Refrigerated <48 hours

Specimen Minimum Volume: TBBS: 1 mL; IABC: > 14 years: 5 mL; < or = 14 years: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Varies	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86355-B cells, total count

86357-Natural killer (NK) cells, total count

86359-T cells, total count

86360-Absolute CD4/CD8 count with ratio

86356 x7 - Mononuclear cell antigen, quantitative

PHLFD 622410 B-Lymphoblastic Leukemia/Lymphoma with BCR::ABL1-like Features Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x12, 88275x6, 88291 x1-FISH Probe, Analysis, Interpretation; 6 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

BNP
83873

B-Type Natriuretic Peptide, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot plasma into plastic vial. Freeze immediately or within 7 hours of collection.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	365 days

CPT Code Information:

83880

BABG
81128

Babesia microti IgG Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86753

BABPB
618303

Babesia species, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred).

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798 x2

87469

87999 (if appropriate for government payers)

FBACS
75397

Baclofen, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collections Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	120 days
	Ambient	14 days

CPT Code Information:

80369

MSCSF
620941

Bacteria, Virus, Fungus, and Parasite Metagenomic Sequencing, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 2 mL

Collection Instructions:

1. Masks should be worn by those collecting and processing specimens for this assay.

2. Handle all vials under sterile technique when open to the air.

3. A separate collection vial of CSF is preferred.

4. Submit specimen from collection vial 2, 3, or 4, as specimens from vial 1 are not acceptable.

5. Indicate on the label which vial is being submitted.

6. **Do not centrifuge or heat inactivate**.

7. **Shunt fluid and ventricular fluid specimens are not acceptable**.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	21 days
	Refrigerated	7 days

CPT Code Information:

0480U

URNS 60515 Bacterial Culture, Aerobic, with Antimicrobial Susceptibilities, Urine

Specimen Requirements:

Supplies: Boric Acid Urine Preservative Tube, 4 mL (T996)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Commercially available transport system intended for the collection and transport of urine samples for culture and sensitivity testing of bacteria (4

mL of lyophilized maintenance formula; BD Vacutainer C and S Preservative and BD Vacutainer C and S Boric Acid Sodium Borate/Formate)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Transfer 4 mL into the boric acid transport tube.

3. Send ambient in transport tube. **Do not aliquot from transport tube.**

Additional Information: Aliquots from transport tube and frozen specimens are **not acceptable.**

Specimen Stability Information: Ambient 72 hours

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Varies	

CPT Code Information:

87086-Bacterial Culture, Aerobic, Urine

87077-Identification commercial kit (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria Identification (if appropriate)

87077-Additional Identification procedure (if appropriate)

87077-Identification Staphylococcus (if appropriate)

87077-Identification Streptococcus (if appropriate)

87147 x 1-3-Serologic agglut method 1 ident (if appropriate)

87147-Serologic agglut method 2 ident (if appropriate)

87147 x 4-Serologic agglut method 3 ident (if appropriate)

87147 x 2-6-Serologic Agglut Method 4 Ident (if appropriate)

87153-Aerobe ident by sequencing (if appropriate)

87150-Identification by PCR (if appropriate)

87185-Beta lactamase (if appropriate)

87186-Antimicrobial Susceptibility, Aerobic Bacteria, MIC-per organism for routine battery (if appropriate)

87181-Susceptibility per drug and per organism for drugs not in routine battery (if appropriate)

87150-mec A PCR (if appropriate)

87150 x 5 Carbapenem resistance genes (if appropriate)

ANAES 60519 Bacterial Culture, Anaerobic with Antimicrobial Susceptibilities, Varies

Specimen Requirements:

Supplies: Anaerobic Transport Tube (T588)

Acceptable Sources: Deep tissues, sterile body fluids, abscesses, percutaneous transtracheal aspirates, suprapubic aspirations, or wounds

Collection Instructions: Specimen should be obtained by using a needle and syringe from a source not normally colonized by anaerobes.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	72 hours

CPT Code Information:

87075-Bacterial Culture, Anaerobic

87076-Anaerobe Ident (if appropriate)

87076-Id MALDI-TOF Mass Spec Anaerobe (if appropriate)

87153-Anaerobe Ident by Sequencing (if appropriate)

87150-Identification by PCR (if appropriate)

87176-Tissue Processing (if appropriate)

87185-Beta Lactamase (if appropriate)

87186-Antimicrobial Susceptibility, Anaerobic Bacteria, MIC (if appropriate)

87181-Anaerobe Susceptibility per agent (if appropriate)

87150-mecA PCR (if appropriate)

ANAE 84292 Bacterial Culture, Anaerobic, Varies

Specimen Requirements:

Supplies: Anaerobe Transport Tube (T588)

Specimen Types: Deep tissues, sterile body fluids, abscesses, percutaneous transtracheal aspirates, suprapubic aspirations, or wounds

Collection Instructions: Specimen should be obtained by using a needle and syringe from a source not normally colonized by anaerobes.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	72 hours

CPT Code Information:

87075-Bacterial Culture, Anaerobic

87076-Id MALDI-TOF Mass Spec Anaerobe (if appropriate)

87153-Anaerobe Ident by Sequencing (if appropriate)

87176-Tissue Processing (if appropriate)

87150-Identification by PCR (if appropriate)

CFRCS 60563 Bacterial Culture, Cystic Fibrosis with Antimicrobial Susceptibilities, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Sputum, expectorated or induced

Patient Preparation: Have the patient brush their teeth or gargle with water immediately before specimen collection. This reduces the number of contaminating

oropharyngeal bacteria.

Container/Tube: Sterile container

Specimen Volume: Entire collection

Acceptable:

Specimen Type: Bronchial aspirate or washing, sinus aspirate, bronchoalveolar lavage, endotracheal, or tracheal

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Throat swab

Supplies:

Culturette (BBL Culture Swab) (T092)

BD E-Swab (T853)

Container/Tube: Culture transport swab (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium), or ESwab

Specimen Volume: Entire collection

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	48 hours

CPT Code Information:

87070-Bacterial, Culture, cystic fibrosis, respiratory

87077-Identification commercial kit (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria Identification (if appropriate)

87077-Additional Identification procedure (if appropriate)

87077-Identification Staphylococcus (if appropriate)

87077-Identification Streptococcus (if appropriate)

87147 x 1-3-Serologic agglut method 1 ident (if appropriate)

87147-Serologic agglut method 2 ident (if appropriate)

87147 x 4-Serologic agglut method 3 ident (if appropriate)

87147 x 2-6-Serologic Agglut Method 4 Ident (if appropriate)

87153-Aerobe ident by sequencing (if appropriate)

87150-Identification by PCR (if appropriate)

87185-Beta lactamase (if appropriate)

87186-Antimicrobial Susceptibility, Aerobic Bacteria, MIC-per organism for routine battery (if appropriate)

87181-Susceptibility per drug and per organism for drugs not in routine battery (if appropriate)

87150-mec A PCR (if appropriate)

CFRC 89653 Bacterial Culture, Cystic Fibrosis, Respiratory

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Sputum, expectorated or induced

Patient Preparation: Have the patient brush their teeth or gargle with water immediately prior to specimen collection. This reduces the number of contaminating oropharyngeal bacteria.

Container/Tube: Sterile container

Specimen Volume: Entire collection

Acceptable:

Specimen Type: Bronchial aspirate or washing, sinus aspirate, bronchoalveolar lavage, endotracheal, or tracheal

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Throat swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

Container/Tube: Culture transport swab (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium), or ESwab

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	48 hours

CPT Code Information:

87070-Bacteria, culture, cystic fibrosis, respiratory

87077-Identification commercial kit (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria Identification (if appropriate)

87077-Additional Identification procedure (if appropriate)

87077-Identification Staphylococcus (if appropriate)

87077-Identification Streptococcus (if appropriate)

87147 x 1-3-Serologic agglut method 1 ident (if appropriate)

87147-Serologic agglut method 2 ident (if appropriate)

87147 x 4-Serologic agglut method 3 ident (if appropriate)

87147 x 2-6-Serologic Agglut Method 4 Ident (if appropriate)

87153-Aerobe Ident by sequencing (if appropriate)

87150-Identification by PCR (if appropriate)

BTWGS Bacterial Typing, Whole Genome Sequencing, Varies
65162

Specimen Requirements:

Aerobic Bacteria

Preferred:

Specimen Type: Bacterial isolate swab

Supplies:

E-swab (T853)

Infectious Container, Large (T146)

Container/Tube:

Preferred: E-Swab collection and transport system

Acceptable: Flocked swab and 1-mL liquid Amies transport medium in 12 x 80 mm tube.

Collection Instructions:

1. Perform isolation of infecting bacteria (must be *Acinetobacter baumannii*, *Campylobacter jejuni/coli*, *Enterobacter cloacae*, *Enterococcus faecalis*, *Enterococcus faecium*, *Escherichia coli*, *Klebsiella pneumoniae*, *Legionella pneumophila*, *Pseudomonas aeruginosa*, *Serratia marcescens*, *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Staphylococcus lugdunensis*, *Streptococcus agalactiae*, or *Streptococcus pyogenes)*.

2. Utilize the flocked swab to obtain an adequate sample of pure cultured isolate. **Do not submit mixed cultures**.

3. Place swab into the transport system containing 1-mL liquid Amies transport medium.

4. If needed, break off end of swab and close the transport tube.

5. Place the transport system into the secondary infectious container for shipment.

6. Each isolate must be submitted under a separate order.

Acceptable:

Specimen Type: Pure culture of organism from source cultured

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Collection Instructions:

1. Perform isolation of bacterial isolate.

2. Organism must be in pure culture, actively growing. **Do not submit mixed cultures**.

3. Place the agar slant or other appropriate media into the secondary infectious container for shipment

4. Each isolate must be submitted under a separate order.

Anaerobic Bacteria

Supplies:

Anaerobe Transport Tube (T588)

Infectious Container, Large (T146)

Container/Tube:

Preferred: Anaerobic transport tube

Acceptable: Thioglycollate broth or any other suitable anaerobic transport system

Specimen Volume: Isolates in separate transport tubes in pure culture

Collection Instructions:

1. Isolate the bacteria (must be *Clostridioides difficile* or *Cutibacterium* [*Propionibacterium*] *acnes*).

2. **Do not** submit growth directly from a CHROMagar plate; subculture to anaerobic media to obtain pure isolate and confirm identification prior to submission.

3. Bacterial isolate must be in pure culture, actively growing. **Do not submit mixed cultures.**

4. Each isolate must be submitted under a separate order.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

0010U-Bacterial Typing, Whole Genome Seq

87900-Bioinformatics Reanalysis (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87153-Aerobe ident by sequencing (if appropriate)

87077-Additional identification procedure (if appropriate)

87076-Id MALDI-TOF mass spec anaerobe (if appropriate)

87076-Anaerobe Ident (if appropriate)

87153-Anaerobe ident by sequencing (if appropriate)

BVRNA 621435 Bacterial Vaginosis, Nucleic Acid Amplification, Vaginal

Specimen Requirements:

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. **Specimen must be collected using the Aptima Multitest Swab Collection Kit.**

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place swab into transport tube provided in collection kit. Snap off swab at score line so swab fits into closed tube.

4. Cap tube securely, and label tube with patient's entire name and collection date and time.

5. Maintain swab container between 2 and 30 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection. If longer storage is needed, store frozen between -20 and -70 degrees C for up to 60 days.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

Vaginal	Refrigerated (preferred)	30 days	APTIMA VIAL
	Frozen	60 days	APTIMA VIAL
	Ambient	30 days	APTIMA VIAL

CPT Code Information:

81513

BAHG
82711

Bahia Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BYST
82759

Baker's Yeast, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BCYP
82722

Bald Cypress, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BAMB
82879

Bamboo Shoot, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBANG
57635

Banana IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

BANA
82746

Banana, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BAP1
71481

BAP1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

BAP1Z 614581 BAP1-Tumor Predisposition Syndrome, BAP1 Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

BARBX 62713 Barbiturates Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit (T282) containing the specimen containers, seals, and documentation required

Specimen Volume: 10 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 10 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80345 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

BARBU 80372 Barbiturates Confirmation, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80345 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FBCFS 75839 Barbiturates Confirmation, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

FDA1S / Drugs of Abuse (10 panel) and Alcohol Screen, Serum

FD10S / Drugs of Abuse Screen, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

80345

G0480 (if appropriate)

FCDUB 75786 Barbiturates, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

FBARS 57742 Barium, Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a EDTA metal-free royal blue-top tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL metal-free EDTA plasma refrigerated in a plastic vial.

Serum

Draw blood in a metal-free royal blue-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL metal-free serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

83018

BGRS
82785

Barley Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBARG
57578

Barley IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

BRLY
82687

Barley, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

FBART
91439

Bartonella Antibody Panel, IFA CSF

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Submit 1 mL of spinal fluid (CSF).

2. Send refrigerate.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86611 x 4

BART
81575

Bartonella Antibody Panel, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86611 x 4

BARTB Bartonella, Molecular Detection, PCR, Blood

89983

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Bartonella* species DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred).

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

87801

BARRP Bartonella, Molecular Detection, PCR, Varies

84440

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Bartonella* species DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Fresh tissue or biopsy

Sources: Heart valve, liver, lymph node, spleen, or skin tissue papule/lesion/nodule

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3) - approximately the size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue only, do not add fluid to tissue.

3. Refrigerate or freeze specimen.

Specimen Stability Information: Refrigerated (preferred) 7 days/ Frozen 7 days

Preferred Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Heart valve, liver, lymph node, spleen, or skin tissue papule/lesion/nodule

Supplies: Tissue Block Container (T553)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable: Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Heart valve, liver, lymph node, spleen, or skin tissue papule/lesion/nodule

Container/Tube: Sterile container for each individual cut section (scroll)

Collection Instructions:

1. Perform microtomy and prepare five separate 10-micron sections.

2. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Fluid

Sources: Cerebrospinal (CSF) or ocular (eg, vitreous humor) fluid

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Specimen Stability Information: Refrigerated (preferred) 7 days/Frozen 7 days

Collection Instructions: For CSF, submit specimen from collection vial 2.

Specimen Type: Synovial fluid

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red clot tube (no anticoagulant), or sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send synovial fluid specimen in original tube (preferred).

Specimen Stability Information: Refrigerated (preferred) 7 days /Frozen 7 days

Specimen Minimum Volume: Fresh tissue or biopsy: 5 mm(3); Paraffin-embedded tissue block: two

10-micron sections; Fluid: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87801

RBART 618100 Bartter Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

81406

81407

81479

81479 (if appropriate for government payers)

BMAMA Basic Metabolic Panel, Serum
113630

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	24 hours

CPT Code Information:

84132

84295

82435

82374

84520

82565

82310

82947

FBSLG
57660

Basil IgG

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red Top

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days

	Ambient	7 days

CPT Code Information:

86001

BASL
82489

Basil, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBBLE
57546

Bass Black (Sea Bass) (Centropristis striata) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

BAYL
82601

Bay Leaf, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBWME
57583

Bayberry/Wax Myrtle (Myrica spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

BCL2
70362

BCL-2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

BCL6
70363

BCL-6 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

BCOR
605260

BCOR Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

BCRFX
65248

BCR::ABL1 Qualitative Diagnostic Assay with Reflex to BCR::ABL1 p190 Quantitative Assay or BCR::ABL1 p210 Quantitative Assay, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as whole blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	72 hours	PURPLE OR PINK TOP/EDTA

	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81206

81207

81208

BA190 83336 BCR::ABL1, p190, mRNA Detection, Reverse Transcription-PCR (RT-PCR), Quantitative, Monitoring Assay, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top **(**EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81207

B190R 48391 BCR::ABL1, p190, mRNA Detection, Reverse Transcription-PCR (RT-PCR), Quantitative, Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see BCRFX / *BCR*::*ABL1* Qualitative Diagnostic Assay with Reflex to *BCR*::*ABL1* p190 Quantitative Assay or *BCR*::*ABL1* p210 Quantitative Assay, Varies.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Label specimen as blood.

3. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Type: Bone marrow

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	72 hours	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81207

P210M 621761 BCR::ABL1, p210, mRNA Detection, Reverse Transcription-PCR (RT-PCR), Quantitative, Monitoring Chronic Myeloid Leukemia (CML), Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Bone Marrow	Refrigerated (preferred)	72 hours	PURPLE TOP/EDTA
	Ambient	72 hours	PURPLE TOP/EDTA

CPT Code Information:

81206-BCR::ABL1 (t[9;22]) (eg, chronic myelogenous leukemia) translocation analysis; major breakpoint, qualitative or quantitative

BCRAB 89007 BCR::ABL1, p210, mRNA Detection, Reverse Transcription-PCR (RT-PCR), Quantitative, Monitoring Chronic Myeloid Leukemia (CML), Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 4 mL; Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	72 hours	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81206

B210R 48390 BCR::ABL1, p210, mRNA Detection, Reverse Transcription-PCR (RT-PCR), Quantitative, Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see BCRFX / *BCR::ABL1* Qualitative Diagnostic Assay with Reflex to *BCR::ABL1* p190 Quantitative Assay or *BCR::ABL1* p210 Quantitative Assay, Varies

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Label specimen as blood.

3. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	72 hours	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81206

BADX
89006

BCR::ABL1, Qualitative, Diagnostic Assay, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

Varies	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81206

81207

81208

BARQ
621751

BCR::ABL1, Rare Fusion Monitoring, Quantitative, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81208

BADX: 81206, 81207, 81208 (if appropriate)

BA190: 81207 (if appropriate)

BCRAB: 81206 (if appropriate)

BAKDM 89609 BCR::ABL1, Tyrosine Kinase Inhibitor Resistance, Kinase Domain Mutation Screen, Sanger Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Label specimen as blood.

3. Send whole blood specimen in original tube. **Do not aliquot.**

Acceptable:

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81170-*ABL1* (ABL proto-oncogene 1, non-receptor tyrosine kinase)(eg, acquired imatinib tyrosine kinase inhibitor resistance), gene analysis, variants in the kinase domain

FBEBE
57521

Bean Black (Phaseolus spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.
2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FBGSG 57522 Bean Green/String IgG

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.
2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

Bean Kidney IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FBLME
57523

Bean Lima (Phaseolus limensis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.
2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FBNWG 57655 Bean Navy/White IgG

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red Top

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

BWRS 35376 Beckwith-Wiedemann Syndrome/Russell-Silver Syndrome, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen can be tested only after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks is required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81401-H19 (imprinted maternally expressed transcript [non-protein coding]) (eg, Beckwith-Wiedemann syndrome), methylation analysis

81401-KCNQ1OT1 (KCNQ1 overlapping transcript 1 [non-protein coding]) (eg, Beckwith-Wiedemann syndrome) methylation analysis

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Tissue culture for amniotic fluid (if appropriate)

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells (if appropriate)

BECH
82669

Beech, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBEFG
57626

Beef IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

BEEF
82697

Beef, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL; For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBTRG
57689

Beet Root IgG

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red Top

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

BEETS
82618

Beets (Beetroot), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL; For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPHEN
91136

Benzene as Phenol, Occupational Exposure, Urine

Specimen Requirements:

Collect 10 mL random urine without preservatives. Ship refrigerated in a plastic container.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

82570/Creatinine

84600/Volatiles

FBEN 90294 Benzene, Occupational Exposure, Blood

Specimen Requirements:

Draw blood in a green-top (sodium heparin) tube(s) and send 20 mL in two tubes of sodium heparin whole blood refrigerated. Blood should be drawn at end of shift. Tubes should be filled to prevent loss of volatile compound into headspace.

Specimen Minimum Volume: 2.5 mL (in two tubes)

Transport Temperature:

Specimen Type	Temperature	Time
WB Sodium Heparin	Refrigerated (preferred)	14 days
	Frozen	365 days

CPT Code Information:

84600

FBCQ 75822 Benzodiazepine Confirmation, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For further information see FSHPU/Sedative Hypnotic Panel, Urine

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80346

G0480 (if appropriate)

BNZX 608279 Benzodiazepines Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation required

Specimen Volume: 5 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 5 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

G0480

80347 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

BNZU
608255

Benzodiazepines Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

G0480

80347 (if appropriate for select payers)

80339 (if appropriate for select payers)

80368 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FCDU2
75777

Benzodiazepines, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

FBENZ
90092

Benztropine (Cogentin), Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

BBEET
82838

Berlin Beetle, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BERG
82892

Bermuda Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBERY
91092

Beryllium, Blood

Specimen Requirements:

Draw blood in a metal free, royal blue-top with EDTA tube(s). Send 2 mL of EDTA whole blood refrigerated.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA - Metal Free (ERB)	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

83018

BETV2
609436

BET v2 (Profilin), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86008

WBSEQ 62128 Beta Globin Gene Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix whole blood.

2. Send whole blood specimen in the original tube. **Do not aliquot.**

Specimen Stability Information: Refrigerate 30 days(preferred)/Ambient 14 days

Specimen Type: Extracted DNA from whole blood

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from whole blood within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods

are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood: 1 mL; Extracted DNA: 50 mcL at 50 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81364-HBB (hemoglobin, beta) full sequence

FBLGG 57667 Beta Lactoglobulin IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

AB2GP
86180

Beta-2 Glycoprotein 1 Antibodies, IgA, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86146

B2GMG
62926

Beta-2 Glycoprotein 1 Antibodies, IgG and IgM, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86146 x 2

GB2GP Beta-2 Glycoprotein 1 Antibodies, IgG, Serum
86182

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86146

MB2GP Beta-2 Glycoprotein 1 Antibodies, IgM, Serum
86181

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86146

B2MU
602026

Beta-2 Microglobulin, Random, Urine

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, urine tube

Specimen Volume: 1 mL

Collection Instructions:

1. Patient should empty bladder.

2. Have patient drink at least 0.5 liters of water.

3. Within 1 hour, collect a random urine specimen.

4. Add 1M sodium hydroxide (NaOH) as preservative to the collection. This preservative is intended to achieve an approximate pH of between 6 and 8.

Specimen Minimum Volume: 0.5mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	14 days
	Refrigerated	48 hours

CPT Code Information:

82232

B2MC
60546

Beta-2-Microglobulin (Beta-2-M), Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Send spinal fluid from collection vial 1.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82232

B2M
9234

Beta-2-Microglobulin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82232

BAMY 70634 Beta-Amyloid Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

Beta-Amyloid Ratio (1-42/1-40), Spinal Fluid

Specimen Requirements:

Container/Tube:

Preferred: Sarstedt 1.5 mL tube (Ref. 72.703.600), Collect at least 750 mcL of cerebrospinal fluid (CSF) or (> or =50% full).

Acceptable:

-CSF AD Biomarker Tube (T833; Sarstedt CSF False Bottom Tube 63.614.625 [2.5 mL])

-Sarstedt 72.694.600 (2 mL)

-Sarstedt 10 mL tube Ref. 62.610.018 (Collect at least 5 mL of CSF [> or =50% full])

All tubes should be filled to between 50% to 100% of the total empty container volume.

Specimen Volume: 0.75 to1.5 mL or at least 50% container volume

Collection Instructions:

1. This test is intended for use in adult patients aged 55 years and older. **Do not collect CSF for this test if the patient is younger than 55 years**.

2. Perform lumbar puncture and discard the first 1 to 2 mL of CSF. Note: CSF collection should be performed by the gravity drip method.

3. Discard any CSF visibly contaminated with blood.

4. Collect CSF directly into one of the listed collection tubes until the tube is at least 50% full. Specimens received with less than the required fill volume may be rejected.

5. If transporting frozen, freeze sample upright prior to placing in transport container.

6. If transporting refrigerate, tubes should be transported upright.

Note: Polystyrene collection tubes are not acceptable. Exposure of CSF to polystyrene tubes may result in falsely low beta-amyloid concentrations. For more information see Cautions.

7. Collection instructions can also be found on Spinal Fluid Specimen Collection Instructions for B-Amyloid (1-42/1-40) (T979)

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

CSF	Frozen (preferred)	90 days
	Refrigerated	14 days
	Ambient	48 hours

CPT Code Information:

0358U

BCATN Beta-Catenin Immunostain, Technical Component Only
70361

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CTX Beta-CrossLaps, Serum
83175

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen prior to 10 a.m.

2. Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	72 hours

CPT Code Information:

82523

BGA
8486

Beta-Galactosidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

82657

WBGDR 620975 Beta-Globin Gene Cluster Deletion/Duplication, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-HBEL1 / Hemoglobin Electrophoresis Evaluation, Blood

-MEV1 / Methemoglobinemia Evaluation, Blood

-REVE2 / Erythrocytosis Evaluation, Blood

-THEV1 / Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	

CPT Code Information:

81363-HBB (hemoglobin, beta, beta-globin) (eg, beta thalassemia), duplication/deletion analysis

WBGDD 620979 Beta-Globin Gene Cluster, Deletion/Duplication, Varies

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in the original tube. **Do not aliquot.**

Specimen Stability Information: Refrigerated (preferred)/Ambient

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81363-HBB (hemoglobin, beta, beta-globin) (e.g. beta thalassemia), duplication/deletion analysis

WBSQR 47959 Beta-Globin Gene Sequencing, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-REVE2 / Erythrocytosis Evaluation, Blood

-HBEL1 / Hemoglobin Electrophoresis Evaluation, Blood

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-MEV1 / Methemoglobinemia Evaluation, Blood

-THEV1 / Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	14 days

CPT Code Information:

81364-HBB (hemoglobin, beta) full sequence

GBAW
606273

Beta-Glucosidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in whole blood original tube. **Do not aliquot**.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82963

GUSBB
618288

Beta-Glucuronidase, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell

filter paper, or blood collected in tubes containing EDTA and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours.

Specimen Minimum Volume:

Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

GUSBW Beta-Glucuronidase, Leukocytes
618289

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

BHCG Beta-Human Chorionic Gonadotropin, Quantitative, Serum
61718

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

84702

BHSF 8877 Beta-Human Chorionic Gonadotropin, Quantitative, Spinal Fluid

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient should not take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen from collection vial number 1.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

84702

BHYD 9251 Beta-Hydroxybutyrate, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

82010

BLAC 82896 Beta-Lactoglobulin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

BTP
623088

Beta-Trace Protein, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Collection Container/Tube:

Preferred: Sterile container, syringe (with needle removed), test tube, or microtube

Acceptable: Plain cotton swab, pledget, gauze, or facial tissue

Specimen Volume: 0.5 mL

Collection Instructions:

1. If submitting a syringe, remove the needle. Add cap to end of syringe.

2. If direct collection is not feasible, specimen may be collected using a plain cotton swab, pledget, gauze or facial tissue.

a. For gauze or facial tissue: circle area on the gauze where specimen was collected.

b. For swab, pledget, gauze or facial tissue: place in a small container (plain test tube or sterile container).

3. **Do not** collect specimen with a culture swab.

4. **Do not** add any additional liquid other than source to the swab or gauze.

5. **Do not** collect or send swab or gauze specimens in containers with additional liquids or additives.

Additional Information:

1. Although results may be obtainable on smaller specimens (perhaps as little as 0.05 mL, depending on the protein concentrations and percentage of spinal fluid in the specimen), reliable results are best obtained with an adequate specimen volume.

2.Specimens collected with additives such as microbiology media (eg, Stuart or Amies liquid medium) or TransFix/EDTA (used for analyses in flow cytometry) yield uninterpretable results and will be rejected.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigered (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

83883

HCO3
876

Bicarbonate, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tube must be centrifuged within 2 hours of collection.

2. Red-top tube must be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated	24 hours

CPT Code Information:

82374

FBIUR
75383

Bicarbonate, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect 1 mL from a random urine collection in a preservative free plastic urine container.

2. Send refrigerated.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days
	Ambient	48 hours

CPT Code Information:

82374

BAPS
62538

Bile Acid Profile, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 to 14 hours, required

Supplies:

Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	90 days
	Ambient	90 days
	Frozen	90 days

CPT Code Information:

82542

BAIPD
41445

Bile Acids for Peroxisomal Disorders, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 to 14 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	90 days
	Ambient	90 days
	Frozen	90 days

CPT Code Information:

82542

BAMRP 619919 Bile Acids Malabsorption Panel, Serum and Feces

Specimen Requirements:

Both feces and serum specimens are required.

It is preferred that both specimen types be collected on the same day, but up to 3 days apart is acceptable.

Patient Preparation:

1. For serum collection: **Fasting: 12 hours, required**; fasting morning specimen preferred

2. The patient should **not use**:

a. Antibiotics for 7 days before specimen collection

b. Statins for 5 days before specimen collection

c. Laxatives (particularly mineral oil and castor oil) for 3 days before or during specimen collection

d. Bile acid sequestrants for 24 hours before specimen collection

e. Synthetic fat substitutes (eg, Olestra) or fat-blocking nutritional supplements for 24 hours before specimen collection.

3. If the patient has used barium, it is recommended to wait at least 48 hours before collecting the feces specimen.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send specimen frozen.

Specimen Type: Feces

Supplies: Stool Container, Small (Random), 4 oz (T288)

Container/Tube: Fecal container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a loose, unpreserved, random fecal specimen.

2. Freeze immediately.

Additional Information:

Both feces and serum specimens are required.

It is preferred that both specimen types be collected on the same day, but up to 3 days apart is acceptable.

Patient Preparation:

1. For serum collection: **Fasting: 12 hours, required**; fasting morning specimen preferred

2. The patient should **not use**:

a. Antibiotics for 7 days before specimen collection

b. Statins for 5 days before specimen collection

c. Laxatives (particularly mineral oil and castor oil) for 3 days before or during specimen collection

d. Bile acid sequestrants for 24 hours before specimen collection

e. Synthetic fat substitutes (eg, Olestra) or fat-blocking nutritional supplements for 24 hours before specimen collection.

3. If the patient has used barium, it is recommended to wait at least 48 hours before collecting the feces specimen.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send specimen frozen.

Specimen Type: Feces

Supplies: Stool Container, Small (Random), 4 oz (T288)

Container/Tube: Fecal container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a loose, unpreserved, random fecal specimen.

2. Freeze immediately.

Additional Information:

1. The patient should store specimen frozen immediately after completion.

2. If additional tests are ordered, aliquot and separate stool sample prior to freezing to allow 1 container per test.

Specimen Minimum Volume: Feces: See Specimen Required; Homogenized feces: 1 mL; Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen	30 days
Serum	Frozen (preferred)	90 days
	Refrigerated	72 hours
	Ambient	24 hours

CPT Code Information:

82542 x2

BA48F Bile Acids, Bowel Dysfunction, 48 Hour, Feces

607368

Specimen Requirements:

Patient Preparation:

1. The patient should be on a fat-controlled diet (100-150 g fat per day) for 3 days before and during the collection period.

2. Patient should **not** use:

a. Antibiotics for 7 days before or during specimen collection

b. Statins for 5 days before or during specimen collection

c. Laxatives (particularly mineral oil and castor oil) for 3 days before or during specimen collection.

d. Bile acid sequestrants for 24 hours before or during specimen collection

e. Synthetic fat substitutes (eg, Olestra) or fat-blocking nutritional supplements for 24 hours before or during the specimen collection

3. If the patient has used barium, it is recommended to wait at least 48 hours before collecting the specimen.

Supplies: Stool Containers - 24, 48, 72 Hour Kit (T291)

Collection Container/Tube: Stool container that complies with shipping requirements

Specimen Volume: Entire 48-hour collection

Collection Instructions:

1. Do not use other containers.

2. Freeze immediately.

3. All containers must be sent together.

4. The entire collection must contain at least 5 g of feces.

5. The number of containers sent should be indicated on the labels (1 of 4, for example).

Additional Information:

1. The patient may store specimen at refrigerate temperature during the collection period, but it must be frozen immediately after completion.

2. If additional tests are ordered, aliquot and separate sample prior to freezing to allow 1 container per test.

Specimen Minimum Volume: 5 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen	30 days

CPT Code Information:

82542

BAFS
62234

Bile Acids, Fractionated and Total, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 to 14 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	90 days
	Ambient	90 days
	Frozen	90 days

CPT Code Information:

82542

BILEA
84689

Bile Acids, Total, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required; Infants and pregnant patients do not need to fast.

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and serum aliquoted into plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

82239

FBAC
75012

Bile Acids, Urine

Specimen Requirements:

Patient Information: It is preferable for patients to be off Urso or Actigall for 5 days before sample collection

Collection Container: Sterile urine container

Specimen Volume: 5 to 25 mL

Collection Instructions:

1. Collect 5 to 25 mL random urine specimen.

2.. No preservative.

2. Send frozen.

Specimen Stability Information: Frozen >10 years

Additional Information:

1. Submit Clinical history/Preliminary diagnosis with specimen.

2. If possible, send both urine and serum (ZW166 / Bile Acids Serum [referral lab code 9001004]). Urine is analyzed for all patients, if the urine result shows evidence of a metabolic abnormality, serum will be tested. Urine and serum must be ordered separately as they are 2 separate tests with separate charges.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	
	Ambient	48 hours
	Refrigerated	48 hours

CPT Code Information:

83789

BILAO 71917 Biliary Tract Malignancy, FISH, Varies

Specimen Requirements:

Supplies: PreservCyt Vial (T536)

Specimen Type: Bile duct brushing, bile duct aspirate, hepatobiliary brushing, or hepatobiliary aspirate (fine-needle aspiration is **not acceptable**)

Container/Tube: Separate ThinPrep vial, containing 20 mL PreservCyt or CytoLyt solution for each specimen

Specimen Volume: Entire collection

Collection Instructions:

1. If performing local cytology in addition to fluorescence in situ hybridization testing, aliquot half of the specimen into another ThinPrep vial **before** processing the specimen.

2. Submission of residual specimen (after processing other testing) may compromise the sensitivity of the test.

3. Label each specimen with specific source (eg, right hepatic duct or common bile duct).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88377

FBILM 70587 Biliary Tract Malignancy-Cytology, FISH, Varies

Specimen Requirements:

Supplies:

PreservCyt Vial (T536)

CytoLyt Solution (T564)

Specimen Type: Bile duct brushing, bile duct aspirate, hepatobiliary brushing, or hepatobiliary aspirate

Container/Tube: Separate ThinPrep vial containing 20 mL PreservCyt or CytoLyt solution for each specimen

Specimen Volume: Entire collection

Collection Instructions: Label with site specimen was collected from (eg, right hepatic duct or common bile duct).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88112

88377 (if appropriate)

BFBL
606895

Bilirubin, Body Fluid

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial fluid

Acceptable Source: Other body fluid, write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Opaque, amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into an amber vial to protect from light.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Body Fluid	Frozen (preferred)	70 days	LIGHT PROTECTED
	Refrigerated	14 days	LIGHT PROTECTED

CPT Code Information:

82247

BILID
81787

Bilirubin, Direct, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, aliquot serum into an amber vial within 6 hours. Protect from light.

3. For red top tubes, aliquot serum into an amber vial immediately. Protect from light.

4. Refrigerate or freeze serum.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	24 hours	LIGHT PROTECTED
	Frozen	30 days	LIGHT PROTECTED
	Ambient	6 hours	LIGHT PROTECTED

CPT Code Information:

82248

BILI3
8452

Bilirubin, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into an amber vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	24 hours	LIGHT PROTECTED
	Frozen	30 days	LIGHT PROTECTED
	Ambient	6 hours	LIGHT PROTECTED

CPT Code Information:

82247-Bilirubin, total

82248-Bilirubin, direct

BILIT
81785

Bilirubin, Total, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into an amber vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	30 days	LIGHT PROTECTED
	Ambient	24 hours	LIGHT PROTECTED

CPT Code Information:

82247

BIOTN
606867

Biotin, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

84591

BTD
622553

Biotinidase Deficiency, BTD Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report.

Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

BIOTS
88205

Biotinidase, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 4 Hours, recommended but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	21 days
	Refrigerated	5 days

CPT Code Information:

82261

FBFPI
57925

Bird Fancier's Precipitin Panel I

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86331 x10

BHDZ 614584 Birt-Hogg-Dube Syndrome, FLCN, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

BIWB
64274

Bismuth, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Collection Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381)

Specimen Volume: 0.25 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

BIVAL
614120

Bivalirudin, Ecarin, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen should be collected 2 hours after initiation of continuous infusion of bivalirudin.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

5. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, below -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	42 days

CPT Code Information:

80299

PBKQN
614567

BK Virus DNA Detection and Quantification, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87799

UBKQN BK Virus DNA Detection and Quantification, Random, Urine
614568

Specimen Requirements:

Supplies: COBAS PCR - Urine Sample Kit (T903)

Container/Tube: cobas PCR urine tube

Specimen Volume: 4.3 mL

Collection Instructions:

1. Collect random urine into a sterile, plastic, preservative-free container.

2. Transfer 4.3 mL of urine into the cobas PCR urine sample tube using the disposable pipette provided within 24 hours of collection. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

3. Transport and store urine specimen transport container at 2 to 30 degrees C (refrigerate is preferred temperature).

Additional Information: cobas PCR media contains guanidine hydrochloride. Do not allow these tubes to come in direct contact with sodium hypochlorite (bleach) or other highly reactive reagents such as acids and bases. These mixtures can release noxious gas.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Refrigerated (preferred)	84 days	COBAS PCR URINE
	Ambient	84 days	COBAS PCR URINE

CPT Code Information:

87799

BLPEP
82814

Black/White Pepper, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BLACK
82361

Blackberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SBL 8237 Blastomyces Antibody Immunodiffusion, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86612

CBL
81541

Blastomyces Antibody Immunodiffusion, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Submit specimen from collection vial number 2 (preferred), 3, or 4.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86612

BLAST
35793

Blastomyces Antibody, Enzyme Immunoassay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86612

86612 (if appropriate)

ALBLD
603305

Bleeding Diathesis Profile, Limited, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). Treatment with heparin causes false-positive results of in vitro coagulation testing for lupus anticoagulant. Warfarin treatment may impair ability to detect the more subtle varieties of lupus-like anticoagulants.

If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 6 mL platelet-poor plasma in six (6) 13-mm x 75-mm, 5-mL, polypropylene vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Within 4 hours of collection, centrifuge the specimen, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 6 separate 13-mm x 75-mm, 5-mL, polypropylene vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately (within 2 hours of processing) freeze plasma at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 4 Plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85610-PTSC

85730-APTSC

85670-TTSC

85384-CLFIB

85379-DIMER

85390-26-ALBLI

85240-F8A

85250-F_9

85291-FXIII

85246-VWAG

85397-VWACT

85130-Chromogenic factor VIII (if appropriate)

85130-Chromogenic factor IX (if appropriate)

85210-Factor II (if appropriate)

85220-Factor V (if appropriate)

85230-Factor VII (if appropriate)

85245-Ristocetin cofactor (if appropriate)

85247-von Willebrand factor multimer (if appropriate)

85260-Factor X (if appropriate)

85270-Factor XI (if appropriate)

85280-Factor XII (if appropriate)

85300-Antithrombin activity (if appropriate)

85301-Antithrombin antigen (if appropriate)

85335-Bethesda units (if appropriate)

85335-Factor II inhibitor screen (if appropriate)

85335-Factor V inhibitor screen (if appropriate)

85335-Factor VII Inhibitor screen (if appropriate)

85335-Factor VIII inhibitor screen (if appropriate)

85335-Factor IX inhibitor screen (if appropriate)

85335-Factor X inhibitor screen (if appropriate)

85335-Factor XI inhibitor screen (if appropriate)

85366-Soluble fibrin monomer (if appropriate)

85385-PT-Fibrinogen (if appropriate)

85410-Alpha-2 plasmin inhibitor (if appropriate)

85415-PAI-1 Ag (if appropriate)

85420-Plasminogen Activity (if appropriate)

85597-Platelet neutralization for lupus inhibitor (if appropriate)

85598-Hex LA (if appropriate)

85611-PT mix 1:1 (if appropriate)

85613-DRVVT (if appropriate)

85613-DRVVT mix (if appropriate)

85613-DRVVT confirm (if appropriate)

85635-Reptilase time (if appropriate)

85732-APTT mix 1:1 (if appropriate)

GNBLC 619257 Bleeding Disorders, Comprehensive Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information:

1. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

GNBLF 619243 Bleeding Disorders, Focused Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81238

81407

81408

81479

81479 (if appropriate for government payers)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

BTROP Blomia tropicalis, IgE, Serum

82374

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BUN
81793

Blood Urea Nitrogen (BUN), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

84520

BWOR
82840

Blood Worm, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MUSS
82548

Blue Mussel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBLUG
57658

Blueberry IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

BLUE
82359

Blueberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BOB1
70365

BOB-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

BAP
82985

Bone Alkaline Phosphatase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

84080

BHISI
70314

Bone Histomorphometry, Consultant Interpretation, Slides Only

Specimen Requirements:

Specimen Type: Bone

Source: Anterior iliac crest

Slides: 2

Submission Container/Tube: Plastic slide holder

Specimen Volume: A minimum of 1 Goldner Trichrome-stained slide and 1 hematoxylin and eosin-stained slide are required. One tetracycline slide should be submitted as applicable.

Collection Information: For complete instructions see Bone Histomorphometry Specimen Preparation (T579).

Additional Information: Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

88321

BHISC 70312 Bone Histomorphometry, Gross Microscopic Exam

Specimen Requirements:

Supplies: Metal Free Specimen Vial (T173)

Specimen Type: Bone

Preferred: Anterior iliac crest

Container/Tube: Metal-free specimen vial

Specimen Volume: Entire specimen

Collection Instructions:

1. Fix specimen in 70% ethanol.

2. Quantitation of bone turnover requires 2 time-spaced tetracycline labels.

3. **The use of metal-free containers is required** to avoid aluminum or iron contamination.

4. For complete instructions see Bone Histomorphometry Specimen Preparation (T579).

Additional Information: Consultation with a Mayo Clinic Laboratories pathologist or endocrinologist/nephrologist is recommended for first-time users of this service. Written instructions are available upon request.

Specimen Minimum Volume: 5 mm bone

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88346

88307

88313

BMAPC 113350 Bone Marrow Aspirate (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

85097 GC

BMBPC 113351 Bone Marrow Biopsy (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305

BMCPC 113352 Bone Marrow Clot (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305

BPRPV 618299 Bordetella pertussis and Bordetella parapertussis, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction (PCR) requires the specimen to be processed in an environment in which contamination of the specimen by *Bordetella pertussis* or *Bordetella parapertussis* DNA is unlikely.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Nasopharyngeal swab

Supplies: Culture Swab - Liquid Stuarts/Single Swab (NP Swab) (T515)

Container/Tube: Rayon swab with an aluminum shaft placed in transport medium such as a green-top nasopharyngeal swab (rayon mini-tip) with Stuart's media (no charcoal), or Stuart's media with charcoal, or Amies media with or without charcoal (Transwab Nasopharyngeal with Charcoal System)

Additional Information:

1. Swab transport containers without charcoal must contain a pledget saturated with either Stuart's or Amies liquid media. Clear semi-solid/solid media is gel and will be rejected.

2. Other swab or media types may be inhibitory to PCR testing and will be rejected.

Acceptable:

Specimen Type: Nasopharyngeal (not throat) aspirate/wash or nasal aspirate/wash

Container/Tube: Sterile container with a screw top cap (no transport media)

Specimen Volume: Entire collection

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

87798 x 2

87798 (if appropriate for government payers)

BORDG 64780 Bordetella pertussis Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86615

BOAC 9723 Boron, Serum/Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Submit only 1 of the following specimens:

Specimen Type: Serum

Supplies: NMS Acid Washed "Precleaned" Vial, 10 mL (T619)

Collection Container/Tube: Plastic Royal Blue top tube (Trace metal-free; No additive)

Submission Container/Tube: Acid washed plastic vial

Specimen Volume: 2 mL serum

Collection Instructions:

1. Draw blood in a trace metal free royal blue-top, no additive tube(s). **Serum gel tube is**

not acceptable.

2. Centrifuge and aliquot 2 mL serum into an acid washed plastic screw capped vial.

3. Label specimen as serum.

4. Ship refrigerate.

Specimen Type: Plasma

Supplies: NMS Acid Washed "Precleaned" Vial, 10 mL (T619)

Collection Container/Tube: Royal blue top tube (Trace metal-free; EDTA)

Submission Container/Tube: Acid washed plastic vial

Specimen volume: 2 mL plasma

Collection Instructions:

1. Draw blood in a trace metal free royal blue-top EDTA tube(s). **Plasma gel tube is not acceptable**.

2. Centrifuge and aliquot 2 mL plasma into an acid washed plastic screw capped vial.

3. Label specimen as plasma.

4. Ship refrigerate.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	30 days	Acid Washed Plastic (MML Supply T619)
	Ambient	30 days	Acid Washed Plastic (MML Supply T619)
	Frozen	30 days	Acid Washed Plastic (MML Supply T619)

CPT Code Information:

83018

BMIYC 64969 Borrelia miyamotoi Detection PCR, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Submit aliquot from collection vial 2.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87478

BMIPB
618298

Borrelia miyamotoi Detection, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred).

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87478

BOT
82715

Botrytis cinerea, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BOV
82135

Bovine Serum Albumin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

BXMPL
82876

Box Elder/Maple, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BRACH
70366

Brachyury Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

BRFKT 616786 BRAF and KIT Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the following slides:

1 slide stained with hematoxylin and eosin

AND

10 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred), or a minimum of 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81406

81479

81479 (if appropriate for government payers)

MBRAF 616925 BRAF V600 Somatic Mutation Analysis, Bone Marrow

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81210

TBRAF 616940 BRAF V600 Somatic Mutation Analysis, Tumor, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block. Blocks prepared with alternative fixation methods will not be accepted; provide fixation method used

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 4 or more unstained

Collection Instructions:

1. A minimum of 10 percent tumor is required.

2. Submit 1 hematoxylin and eosin-stained slide and 4 to 10 unstained, nonbaked slides with 7 to 10-micron thick sections of the tumor tissue with an aggregate of 25 mm squared. 10 microns is preferred.

3. 5-micron sections will be accepted if a 50 mm squared aggregate is cut

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained slide and 1 unstained slide for each probe set

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81210

BRAFD 608305

BRAF V600E/V600K Somatic Mutation Analysis, Tumor

Specimen Requirements:

Preferred: Submit **both** of the following specimens.

Acceptable: Submit **at least** one of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 5 unstained

Collection Instructions:

Submit the following slides:

1 slide stained with hematoxylin and eosin

AND

5 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue

Note: The total amount of required tumor nuclei can be obtained by scraping up to 5 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81210

88381-Microdissection, manual

MSUSC 618716 Branched-Chain Amino Acids, Self-Collect, Blood Spot

Specimen Requirements:

Supplies: Blood Spot Collection-Self Collect (T858)

Container/Tube: Blood Spot Self Collection Card

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Order test each time the patient is to collect a dried blood specimen at home and mail the specimen directly to Mayo Clinic Laboratories.

2. Order should be placed a minimum of 3 days prior to desired date of collection.

3. Enter patient's address information for each order created, including street address (**post-office [PO] boxes are not acceptable delivery locations**), city, state abbreviation, ZIP (postal) code, country, and home phone number.

4. For each order, the Blood Spot Collection-Self Collect kit will be mailed directly to the patient for self-collection (**delivery to a PO box will not occur**).

5. For more information on how to collect blood spots, see the following:

-How to Collect a Dried Blood Spot Sample via fingerstick.

-Blood Spot Collection Instructions-Fingerstick

-Blood Spot Collection Instructions-Fingerstick-Spanish

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	59 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Refrigerated	59 days	FILTER PAPER

CPT Code Information:

0381U

FBNC1
75583

Brazil Nut Component rBer e 1

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

86008

BRAZX
618852

Brazil Nut Component, IgE, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see BRAZR / Brazil Nut, IgE, with Reflex to Brazil Nut Component, IgE, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

BRAZR 618851 Brazil Nut, IgE with Reflex to Brazil Nut Component, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BRAZ 82899 Brazil Nut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BRCAT 616495 BRCA1/2 Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the following slides:

1 slide stained with hematoxylin and eosin

AND

10 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred), or a minimum of 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381 - Microdissection, manual

81163

HBOCZ 614570 BRCA1/BRCA2 Genes, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81162

C2729 606583 Breast Carcinoma-Associated Antigen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial. **Specimens that have not been aliquoted will be canceled**.

2. Send refrigerated.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	4 days

CPT Code Information:

86300

BRG1
71537

BRG1 (SMARCA4) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

BRIVA
620767

Brivaracetam, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (K2 EDTA)

Acceptable: Lavender top (K3 EDTA), green top (sodium or lithium heparin), light-blue top (sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. **For sustained-release formulations only**, draw blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

BRBPS 65058 Broad Range Bacterial PCR and Sequencing, Varies

Specimen Requirements:

Fresh tissue is generally preferred over formalin-fixed, paraffin-embedded tissue, but either can be tested.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Fresh tissue or biopsy

Sources: Normally sterile tissue such as bone, lymph node, joint, heart valve, brain, viscera, organ, lung, prostate

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3)-approximately the size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue only, do not add fluid (eg, saline, broth, formalin, formaldehyde, acetone) to tissue. Body fluid accompanying tissue is acceptable.

3. Freeze specimen.

Specimen Stability Information: Frozen (preferred) 21 days/Refrigerated 21 days

Acceptable:

Preferred: Paraffin-embedded tissue block

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded (FFPE) tissue block

Source: Normally sterile or deep tissues such as bone, lymph node, joint, heart valve, brain, viscera, organ, lung, prostate

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable: Paraffin-embedded tissue block

Specimen Type: Section (scrolls) of FFPE tissue block

Source: Normally sterile or deep tissues such as bone, lymph node, joint, heart valve, brain, viscera, organ, lung, prostate

Container/Tube: Sterile container for each individual cut section (scroll)

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Fluid

Sources: Normally sterile body fluids such as vitreous humor, pleural, abdominal, peritoneal, ascites, pericardial, pelvic, prostatic

Container/Tube: Screw-capped, sterile container

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect fresh fluid specimen.

2. Freeze specimen.

Specimen Stability Information: Frozen (preferred) 21 days/Refrigerated 21 days

Specimen Type: Spinal fluid

Container/Tube: Screw-capped, sterile container

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect fresh spinal fluid (CSF) specimen using sterile technique.

2. Submit specimen from collection vial 2 or higher, specimens in vial 1 are not acceptable.

3. Indicate on the label which vial is being submitted.

4. CSF collected by lumbar puncture is preferred. CSF collected via shunt and ventricular fluid are also acceptable; label tube with applicable collection information if submitting nonlumbar puncture collected CSF.

Specimen Stability Information: Frozen (preferred) 21 days/Refrigerated 21 days

Specimen Type: Synovial fluid

Container/Tube:

Preferred: Red top or sterile container

Acceptable: Lavender top (EDTA), pink top (EDTA), royal blue top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 0.5 mL

Collection Instructions: Send specimen in original tube (preferred).

Specimen Stability Information: Frozen (preferred) 21 days/Refrigerated 21 days

Specimen Minimum Volume: Fluid: 0.2 mL; Fresh tissue or biopsy: 5 mm(3); Paraffin-embedded tissue block: two 10-micron sections

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87801-Broad Range Bacterial PCR and Sequencing

87798-Specimen Identification by PCR (if appropriate)

87798-Ident by Next Generation Sequencing (if appropriate)

87483-Meningitis Encephalitis Panel, PCR (if appropriate)

87999-Joint Infection Panel, PCR, Synovial Fluid (if appropriate)

BROC
82817

Broccoli, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BROM
82919

Brome Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBROM
75695

Bromine - Total, Blood

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Whole blood

Collection Container/Tube: Royal blue top (Trace metal-free EDTA)

Specimen Volume: 2 mL

Collection Instructions:

1. Do not use disinfectants containing iodine, eg, Betadine, during venipuncture.

2. Send whole blood specimen in original tube.

3. Send refrigerated.

Note: Avoid exposure to gadolinium or iodine-based contrast media for 96 hours prior to specimen collection.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

Whole Blood EDTA - Metal Free (ERB)	Refrigerated (preferred)	30 days	METAL FREE
	Frozen	365 days	METAL FREE
	Ambient	30 days	METAL FREE

CPT Code Information:

82542

BRCMG 607346 Brucella Antibody Screen, IgM and IgG, ELISA, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a sterile, plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86622 x 2-*Brucella* antibody, IgG and IgM

86622-*Brucella* total antibody, agglutination (if appropriate)

BRUTA 8112 Brucella Total Antibody Confirmation, Agglutination, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86622

SCN5A
617449

Brugada Syndrome, SCN5A Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions.

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81407

BSPR
82480

Brussels Sprouts, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BTKSG 619760 Bruton Tyrosine Kinase, BTK Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information:: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2**.** A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

BTK
89011

Bruton Tyrosine Kinase, Protein Expression, Flow Cytometry, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

88184

BUCW
82727

Buckwheat, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BDRP
82791

Budgerigar Droppings, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BFTH
82779

Budgerigar Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

BPAB 606816 Bullous Pemphigoid, BP180 and BP230, IgG Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission container/tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

83516 x 2

FMARC 75307 Bupivacaine (Marcaine)

Specimen Requirements:

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 1 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	180 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

80299

BUPMX 65215 Buprenorphine and Norbuprenorphine, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing specimen containers, seals, and required documentation.

Specimen Volume: 5 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

G0480

80348 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

BUPM 66200 Buprenorphine and Norbuprenorphine, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

G0480

80348 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

BUPR
63222

Buprenorphine Screen with Reflex, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 5 mL in 1 plastic vial.

3. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 5 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

BUPS
63119

Buprenorphine Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 5 mL in 1 plastic vial.

3. No preservative.

Additional Information:

1. **No specimen substitutions.**

2. STAT requests are **not accepted** for this test.

3. Submitting less than 5 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

FBUMT 75387 Bupropion and Metabolite, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial (preservative free)

Specimen Volume: 1 mL Serum

Collections Instructions:

1. Centrifuge and aliquot 1 mL of serum into a preservative-free plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	30 days

CPT Code Information:

80338

FBUS
91115

Buspirone (Buspar)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

BUAUC
83188

Busulfan, Intravenous Dose, Area Under the Curve, Plasma

Specimen Requirements:

Four plasma specimens with different collection times (keep all specimens under 1 order) are required.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin) (Plasma gel/PST are **not**

acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. The first specimen should be collected immediately after completion of the first intravenous infusion of 0.8 mg/kg busulfan.

2. Additional specimens should also be collected at 1 hour, 2 hours, and 4 hours after completion of infusion.

3. Label each specimen with exact time of collection.

4. Busulfan degrades quickly at ambient temperature. Specimens must be kept in wet ice slurry or refrigerated at 4 degrees C. Specimens must be centrifuged within 2 hours after collection. Separate the plasma and transfer to individual 5-mL plastic vials, labeled with exact time of collection. Immediately freeze at -20 degrees C.

Additional Information:

This test should only be ordered when the following criteria are met:

-Busulfan dosing protocol must be intravenous administration of 0.8 mg/kg doses every 6 hours over 4 days, for a total of 16 doses

Specimens must be collected as described below:

-1 specimen collected immediately after completion of the first 2-hour IV infusion of busulfan

-1 specimen collected 1 hour after the infusion is completed

-1 specimen collected 2 hours after the infusion is completed

-1 specimen collected 4 hours after the infusion is completed and prior to the next infusion of busulfan

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Frozen (preferred)	28 days
	Refrigerated	72 hours

CPT Code Information:

80299 x 4

BTBAS
622885

Butabarbital, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before the next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

BUTAS
8427

Butalbital, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before the next scheduled dose.

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

CCR4
623468

C-C Motif Chemokine Receptor 4 (CCR4), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

FCPEU
75894

C-Peptide, 24-Hour Urine

Specimen Requirements:

Patient Preparation: It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to specimen collection.

Collection Container/Tube: Clean, plastic 24-hour urine collection container.

Submission Container/Tube: Plastic urine container

Specimen Volume: 10 mL

Collection Instructions:

1. Instruct the patient to void at 8 a.m. and discard the specimen.

2. Collect all urine including the final specimen voided at the end of the 24-hour collection period (ie, 8 a.m. the next morning) in a 24-hour urine collection container. pH should be between 4 and 7.

3. Refrigerate specimen during the 24-hour collection.

4. Aliquot and send 10 mL urine in a plastic urine container frozen.

5. **24-Hour collection volume is required**.

Specimen Minimum Volume: 1 mL; Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Urine 24hr	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

84681

CPR
8804

C-Peptide, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, required

2. For 12 hours before specimen collection, patient **should not take** multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

84681

CRPRO 70409 C-Reactive Protein (CRP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CRP
9731

C-Reactive Protein (CRP), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tube should be centrifuged within 2 hours of collection.

2. Red-top tube should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	365 days

	Ambient	14 days

CPT Code Information:

86140

HSCRP 606909 C-Reactive Protein, High Sensitivity, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

86141

CXCR3 623466 C-X-C Motif Chemokine Receptor 3 (CXCR3), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

C1ES
8198

C1 Esterase Inhibitor Antigen, Serum

Specimen Requirements:

Patient Preparations:

Fasting: 12 hours, preferred but not required.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Centrifuge and aliquot serum into plastic vial.

3. Freeze specimen within 30 minutes.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	72 hours

CPT Code Information:

83883

C1INF 616906 C1 Esterase Inhibitor, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	28 days

CPT Code Information:

83520

C1QFX C1q Complement, Functional, Serum
83374

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Frozen	14 days

CPT Code Information:

86161

C2FXN 32137 C2 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	21 days

CPT Code Information:

86161

C2
81835

C2 Complement, Functional, with Reflex, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	21 days

CPT Code Information:

86161

86160 x 2 (if appropriate)

C3 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

FC3AR
75729

C3a Level By RIA

Specimen Requirements:

Container/Tube: Lavender top (K2 EDTA) (Plasma gel tube/PST are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions:

1. Mix well.

2. Within 30 minutes of collection, centrifuge at 1600 (+/-200) x *g* (rcf) at 4 degrees C for 15 minutes.

3. Aliquot 1.5 mL of plasma into a plastic vial.

4. Freeze immediately at -20 degrees C for up to 2 weeks.

5. Send frozen.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

86160

C4U
88829

C4 Acylcarnitine, Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: If clinically feasible, discontinue L-carnitine supplementation at least 72 hours before specimen collection**.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10 mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Freeze specimen immediately.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	7 days
	Refrigerated	24 hours

CPT Code Information:

82017

C4FX
83391

C4 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL of serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

FC4AL
75726

C4 Level by RIA

Specimen Requirements:

Container/Tube: Lavender top (K2 EDTA) (Plasma gel tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions:

1. Mix well.

2. Within 30 minutes of collection, centrifuge at 1600 (+/-200) x *g* (rcf) at 4 degrees C for 15 minutes.

3. Aliquot 1.5 mL of plasma into a plastic vial.

4. Freeze plasma immediately at -20 degrees C for up to 4 weeks.

5. Send frozen.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	28 days

CPT Code Information:

86160

C5AG
9266

C5 Complement, Antigen, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	60 days
	Refrigerated	28 days
	Ambient	7 days

CPT Code Information:

86160

C5FX
83392

C5 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL of serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

C5DCU
88831

C5-DC Acylcarnitine, Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: If clinically feasible, discontinue L-carnitine supplementation at least 72 hours before specimen collection.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Freeze specimen immediately.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Frozen (preferred)	7 days
	Refrigerated	24 hours

CPT Code Information:

82017

C5OHU
88830

C5-OH Acylcarnitine, Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: If clinically feasible, discontinue L-carnitine supplementation at least 72 hours before specimen collection.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Freeze specimen immediately.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	7 days
	Refrigerated	24 hours

CPT Code Information:

82017

C6FX
83393

C6 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL of serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. After specimen has clotted on wet ice, centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

C7FX
81064

C7 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

C8FX
81065

C8 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

C9FX
81066

C9 Complement, Functional, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86161

C9ORF 35377 C9orf72 Hexanucleotide Repeat, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

FCABB
57672

Cabbage IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CABB
86327

Cabbage, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

COCOA
60112

Cacao/Cocoa, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CDOMB
89539

Cadmium for Occupational Monitoring, Blood

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, a specimen **should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original collection tube. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82300

CDUOE
608896

Cadmium Occupational Exposure, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect urine a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82300

82570

CDOU 608892 Cadmium Occupational Exposure, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

CDUOE / Cadmium Occupational Exposure, Random, Urine

HMUOE / Heavy Metal Occupational Exposure, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82300

CDU 8678 Cadmium, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium or iodine containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect urine for 24 hours and record the total volume.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days

	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82300

CDB 8682 Cadmium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original collection tube. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82300

CDUCR 608906 Cadmium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect urine a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82300

82570

CDCU 608902 Cadmium/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information, see:

CDUCR / Cadmium/Creatinine Ratio, Random, Urine

HMUCR / Heavy Metal/Creatinine Ratio, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82300

CAFF
8754

Caffeine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, serum may sit on the gel or be aliquoted into a plastic vial.

3. For red top tubes, aliquot serum into a plastic vial immediately after centrifugation.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Ambient	72 hours

	Refrigerated	72 hours

CPT Code Information:

80155

CALCI 70368 Calcitonin (CALCI) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CATLN 61527 Calcitonin, Fine-Needle Aspiration Biopsy Needle Wash, Lymph Node

Specimen Requirements:

Patient Preparation: For 12 hours before this procedure do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. Needle wash specimens for analysis should be collected in conjunction with cytology specimens.

2. Have saline available prior to start of procedure. Saline is the only acceptable solution for needle washings.

3. After each fine-needle aspiration biopsy (FNAB) has been collected and the material in the needle has been expelled onto a slide for cytologic analysis, attach the used FNAB needle to an empty syringe.

4. Withdraw between 0.10 mL and 0.25 mL of saline up through the needle until the saline starts to fill the hub of the needle or end of the syringe.

5. Expel this fluid back through the needle into a separate tube. This is the needle washing used for analysis.

6. Repeat steps 2 through 4 for each needle pass of the same biopsied site and empty into the same tube, accumulating a total of 0.5 mL to 1.5 mL of fluid to send to the laboratory. (If more than 1 site is biopsied, see Additional Information)

7. Inspect specimen for visible blood or tissue contamination:

a. If bloody, centrifuge specimen and transfer supernatant to a new plastic aliquot tube (5-mL standard tube) to send to laboratory. The supernatant, not the cellular material, is used for analysis.

b. If specimen is clear, centrifugation is not necessary.

8. Freeze within 2 to 4 hours of collection.

Additional Information:

1. If more than 1 site is biopsied, each washing material should be submitted in a separate tube and under a different order number.

2. A minimum of 0.5 mL is required for testing; however, the total collection volume should not exceed 1.5 mL. Sample volumes outside these parameters may be rejected.

3. Do not send saline control. This test has been validated to rule-out saline matrix effect.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fine Needle Wash	Frozen (preferred)	7 days
	Refrigerated	4 hours

CPT Code Information:

82308

CATN 9160 Calcitonin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient should not take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. After collection, immediately place specimen on ice.

2. Refrigerate specimen during centrifugation and immediately transfer serum to a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	24 hours
	Ambient	8 hours

CPT Code Information:

82308

CALU
610595

Calcium, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Patient **should not** use a laxative during the 24-hour collection period.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

82340

CAI
8378

Calcium, Ionized, Serum

Specimen Requirements:

Container/Tube: Serum gel or serum gel microtainer

Specimen Volume: Full tube

Collection Instructions:

1. Allow blood to clot for 30 minutes.

2. Serum gel tube/microtainer must be centrifuged within 1 hour of collection. Centrifuge with stopper in place for 7 minutes at 3000 rpm to ensure that the gel barrier separates the serum and cells.

3. **Keep specimen anaerobic; do not aliquot.**

4**. Test cannot be combined with other testing. Separate specimens must be submitted when multiple tests are ordered.**

Specimen Minimum Volume: 1.75 mL in a 3.5 mL (50% full) in serum gel tube or 1 full serum gel microtainer

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Refrigerated	7 days	SERUM GEL TUBE

CPT Code Information:

82330

CALC5 Calcium, Random, Urine (Non-Orderable)

610591

Specimen Requirements:

Only orderable as part of a profile. For more information see CACR3 / Calcium/Creatinine Ratio, Random, Urine

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5-mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

82310

CA
601514

Calcium, Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. If drawing for more than total calcium, send first tube drawn.

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	240 days
	Refrigerated	21 days

CPT Code Information:

82310

CACR3
610594

Calcium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5-mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

82310

82570

CCTR 610592 Calcium/Creatinine Ratio, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CACR3 / Calcium/Creatinine Ratio, Random, Urine

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5-mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CALD
70369

Caldesmon Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CAMTA 603417 Calmodulin-Binding Transcription Activator 1 (CAMTA1), Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CALPN 70370 Calponin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CALPR
63016

Calprotectin, Feces

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz Random (T288)

Submission Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a fresh random fecal specimen, no preservatives.

2. If specimen is sent refrigerate, send immediately after collection.

3. If specimen cannot be sent immediately, freeze specimen, and send frozen.

Additional Information:

1. **Separate specimens must be submitted when multiple tests are ordered**, with the exception of ELASF / Pancreatic Elastase, Feces. **If only a single specimen is collected, it must be split prior to transport**.

2. Testing cannot be added on to a previously collected specimen.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	7 days
	Refrigerated	72 hours

CPT Code Information:

83993

CALFM 618654 CALR Mutation Analysis, Myeloproliferative Neoplasm (MPN), Bone Marrow

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81219

CALX 36997 CALR Mutation Analysis, Myeloproliferative Neoplasm (MPN), Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNR / Myeloproliferative Neoplasm, *JAK2* V617F with Reflex to *CALR* and *MPL*, Varies.

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not** aliquot.

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred)7 days/Refrigerate 7 days

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)7 days/Refrigerate 7 days

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA from blood or bone marrow.

2. Provide volume and concentration of the DNA on the label.

Specimen Stability Information: Frozen (preferred)/Refrigerate/Ambient

Specimen Minimum Volume: Whole blood/Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81219-*CALR* (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9

CALR
62912

CALR Mutation Analysis, Myeloproliferative Neoplasm (MPN), Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Additional Information: To ensure minimum volume and concentration of DNA is

met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood/bone marrow: 1 mL; Extracted DNA: 50 mcl at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81219-*CALR* (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9

CALJM 606806 CALR Variant Analysis, Myeloproliferative Neoplasm, Reflex, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNJM / Myeloproliferative Neoplasm, *JAK2* V617F with Reflex to *CALR* and *MPL*, Bone Marrow.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Note: Extracted DNA from bone marrow is **not** acceptable.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81219-*CALR* (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9

CALRC 71486 Calreticulin ex9mut Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: Formalin-fixed, paraffin-embedded (FFPE) tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CALNN 70371 Calretinin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CAMPC 606218 Campylobacter Culture, Feces

Specimen Requirements:

Patient Preparation: Patient **should not** use barium or bismuth for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-Campylobacter Culture, Stool-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

CFTH 82778 Canary Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CAGR 82829 Canary Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CA25
9289

Cancer Antigen 125 (CA 125), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	5 days
	Frozen	168 days

CPT Code Information:

86304

CA153 81607 Cancer Antigen 15-3 (CA 15-3), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

86300

FCARP 75447 Cancer-Associated Retinopathy Panel (CARP) by Immunoblot and IHC

Specimen Requirements:

Submit only one of the following specimens:

Serum:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 5 mL serum refrigerated in a plastic vial.

Plasma:

Draw blood in a lavender-top (EDTA) tube(s). Spin down and send 5 mL EDTA plasma refrigerated in a plastic vial.

Complete and submit with specimen:

1. Completed OHSU Ocular request form

2. Clinical history

3. Referring physician information (name & phone number)

-NOTE: Without this information, testing cannot be completed.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	7 days

CPT Code Information:

84182 x 8

CDAB
82690

Candida albicans (Monilia), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCANG 75605 Candida albicans IgG

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Container/Tube: Red

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 0.5 mL of serum Frozen in a sterile, screw top tube.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	28 days
	Ambient	7 days

CPT Code Information:

86001

CAURS 607883 Candida auris Surveillance, Molecular Detection, PCR,

Varies

Specimen Requirements:

Supplies: E-swab (T853)

Source: Axilla and groin composite

Container/Tube: ESwabs in liquid Amies medium

Specimen Volume: Swab

Collection Instructions:

1. Swab transport containers without charcoal must contain a pledget saturated with either Stuart's or Amies liquid media.

2. Swab used for this test cannot be shared with fungal culture. When fungal culture is ordered with this test, send separate swabs for each.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87481

CVRNA 620888 Candida Vaginitis, Nucleic Acid Amplification RNA, Vaginal

Specimen Requirements:

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Multitest Swab Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place swab into transport tube provided in collection kit. Snap off swab at score line so swab fits into closed tube.

4. Cap tube securely, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for up to 60 days.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Vaginal	Refrigerated (preferred)	30 days	APTIMA VIAL
	Frozen	60 days	APTIMA VIAL
	Ambient	30 days	APTIMA VIAL

CPT Code Information:

87481

FCBDS 75527 Cannabidiol, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (Serum gel/SST are not acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collections Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Stability Information: Ambient 14 days/Refrigerated 30 days/Frozen (-20 degrees C) 2 years

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	30 days
	Ambient	14 days
	Frozen	

CPT Code Information:

80299

FMARI 75172 Cannabinoid Analysis, Whole Blood

Specimen Requirements:

Draw blood in a purple-top (EDTA), green-top (sodium-heparin) or grey-top (NaF/oxalate) tube(s). Send 7 mL of whole blood refrigerated in a plastic vial.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307

80349 (if appropriate)

G0480 (if appropriate)

FCDU7 75782 Cannabinoids, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

CWAY
82493

Caraway, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CARBR
610048

Carbamazepine Hypersensitivity Pharmacogenomics, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81381 x 2

CARTF Carbamazepine Profile, Serum

37037

Specimen Requirements:

Container/Tube: Red top

Specimen Volume: 2 mL

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Draw blood 12 hours (trough value) after last dose.

2. Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

80156-Carbamazepine, total

80157-Carbamazepine, free

80161-Carbamazepine-10,11-Epoxide

CARFT
37039

Carbamazepine, Free and Total, Serum

Specimen Requirements:

Container/Tube: Red top (serum gel/SST are not acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

80156

80157

CARF
37038

Carbamazepine, Free, Serum

Specimen Requirements:

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

80157

CARTA
37035

Carbamazepine, Total, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Specimen Volume: 0.5 mL

Submission Container/Tube: Plastic vial

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours

of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

80156

CARBG 37036 Carbamazepine-10,11-Epoxide, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Draw blood 12 hours (trough value) after last dose.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

80156-Carbamazepine, Tot, S

80161-Carbamazepine-10,11-Epoxide

Carbapenem Resistance Genes, Molecular Detection, PCR, Rectal Swab

Specimen Requirements:

Specimen Type: Rectal/perirectal swab

Supplies: Culturette (BBL Culture Swab) (T092)

Container/Tube: Culture transport swab

Specimen Volume: Entire collection

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Ambient	5 days

CPT Code Information:

87798

CARBI
620159

Carbapenem Resistance Genes, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Klebsiella pneumoniae* carbapenemase (KPC), New Delhi metallo-beta-lactamase (NDM), Verona integron-encoded metallo-beta-lactamase (VIM), oxacillin-hydrolyzing beta-lactamase (OXA-48), and imipenemase-type metallo-beta-lactamase (IMP) DNA is not likely.

Supplies: Infectious Container, Large (T146)

Container/Tube: Slant

Specimen Volume: Isolate

Collection Instructions:

1. Perform isolation of infecting bacteria.

2. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures.**

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87183

CARNB 35953 Carbapenemase Detection-Carba NP Test (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87185

CARNP 62606 Carbapenemase Detection-Carba NP Test, Varies

Specimen Requirements:

Preferred:

Specimen Type: Pure culture of Enterobacterales or *Pseudomonas aeruginosa* from source cultured

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Collection Instructions:

1. Perform isolation of bacterial isolate.

2. Submit Enterobacterales or *Pseudomonas aeruginosa* isolate in pure culture actively

growing. **Do not submit mixed cultures**.

3. Place the slant into the secondary infectious container for shipment.

4. Each isolate must be submitted under a separate order.

Acceptable:

Specimen Type: Enterobacterales or *Pseudomonas aeruginosa* isolate swab

Supplies:

-E-Swab (T853)

-Infectious Container, Large (T146)

Container/Tube: E-Swab collection and transport system

Collection Instructions:

1. Perform isolation of bacterial isolate.

2. Utilize the E-swab to obtain an adequate sample of Enterobacterales or *Pseudomonas aeruginosa* isolate in pure culture. **Do not submit mixed cultures.**

3. Place swab into the transport system containing 1-mL liquid Amies transport medium.

4. If needed, break off end of swab and close the transport tube.

5. Place the transport system into the secondary infectious container for shipment.

6. Each isolate must be submitted under a separate order.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87185

199PC Carbohydrate Antigen 19-9 (CA 19-9), Pancreatic Cyst Fluid
89508

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Source: Pancreatic cyst or pancreatic pseudocyst

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 1 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Pancreatic Cyst Fluid	Frozen (preferred)	90 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

86301

199PT 61530 Carbohydrate Antigen 19-9 (CA 19-9), Peritoneal Fluid

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Sources: Peritoneal, abdominal, ascites, paracentesis fluid (peritoneal washings are **not acceptable**)

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 2 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Peritoneal	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86301

PF199 Carbohydrate Antigen 19-9 (CA 19-9), Pleural Fluid
60230

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Source: Pleural fluid, thoracentesis fluid

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 2 mL

Specimen Minimum Volume: 0.5 mL (Samples <0.5 mL may be rejected)

Transport Temperature:

Specimen Type	Temperature	Time
Pleural Fluid	Frozen (preferred)	90 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

86301

CA19 Carbohydrate Antigen 19-9 (CA 19-9), Serum
9288

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient should not take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Within 2 hours of collection, serum gel tubes should be centrifuged.

2. Within 2 hours of collection, red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	8 hours

CPT Code Information:

86301

CDG
89891

Carbohydrate Deficient Transferrin for Congenital Disorders of Glycosylation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.05 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	45 days
	Refrigerated	28 days
	Ambient	7 days

CPT Code Information:

82373

CDTA
82425

Carbohydrate Deficient Transferrin, Adult, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.05 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	45 days
	Refrigerated	28 days
	Ambient	7 days

CPT Code Information:

82373

CHOU
9255

Carbohydrate, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect an early-morning (preferred) random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	42 days

CPT Code Information:

84379

FCMEB 75880 Carbon Monoxide Exposure, Blood

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Send refrigerated.

Specimen Minimum Volume: 1.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	28 days
	Ambient	28 days

CPT Code Information:

82375

CAIX 606251 Carbonic Anhydrase IX (CA-IX) Immunostain, Technical

Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CEAPC
89509

Carcinoembryonic Antigen (CEA), Pancreatic Cyst Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 1 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Pancreatic Cyst Fluid	Frozen (preferred)	90 days
	Refrigerated	72 hours
	Ambient	24 hours

CPT Code Information:

82378

CEAPT
61528

Carcinoembryonic Antigen (CEA), Peritoneal Fluid

Specimen Requirements:

Sources: Peritoneal, abdominal, ascites, paracentesis fluid (peritoneal washings are **not acce**ptable)

Container/Tube: Plain, plastic, screw top tube

Specimen Volume: 2 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Peritoneal	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

82378

PFCEA
83742

Carcinoembryonic Antigen (CEA), Pleural Fluid

Specimen Requirements:

Sources: Pleural fluid, thoracentesis fluid

Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 2 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Pleural Fluid	Frozen (preferred)	90 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

82378

CEA
8521

Carcinoembryonic Antigen (CEA), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, serum may sit on gel refrigerated but must be aliquoted within 7 days.

3. For red top tubes, aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

82378

CEASF 90695 Carcinoembryonic Antigen (CEA), Spinal Fluid

Specimen Requirements:

Collection Container/Tube: Sterile vial

Submission Container/Tube: 13 x 75-mm tube

Specimen Volume: 0.5 mL

Collection Instructions: Submit specimen from collection vial number 1.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

82378

MCEA 70506 Carcinoembryonic Antigen, Monoclonal Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PCEAI
70535

Carcinoembryonic Antigen, Polyclonal Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

CRMP1 Cardiovascular Risk Marker Panel, Serum
616727

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required unless directed by the ordering provider

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

80061-Lipid panel (includes: HDL [CPT Code 83718], total cholesterol [CPT Code 82465], and triglycerides [CPT Code 84478])

83695-Lipoprotein (a)

86141-C-reactive protein; high sensitivity (hsCRP)

82172-Apolipoprotein B

FCRDE Carmine Dye/Red Dye Cochineal (Dactylopius coccus) IgE (Red # 4)
57524

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

CARN 8802 Carnitine, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA), green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial

Specimen Minimum Volume: Plasma: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	60 days
	Refrigerated	21 days
	Ambient	7 days

CPT Code Information:

82379

CARNU
81123

Carnitine, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 1.5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	365 days
	Refrigerated	72 hours

CPT Code Information:

82379

CARNS
60449

Carnitine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	60 days
	Refrigerated	21 days
	Ambient	7 days

CPT Code Information:

82379

CAROB
82368

Carob, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCATB
75927

Carotene, Beta

Specimen Requirements:

Patient Preparation:

1. **Fasting: 12 hours, required**

a. Blood should be collected before breakfast in the morning and prior to receiving any medication

b. Infants younger than 6 months, collect specimen before next feeding.

2. Patient **must not** consume any alcohol for 24 hours before specimen collection.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.7 mL

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 0.7 mL of serum into an amber vial to protect from light.

2. Ship ambient.

Specimen Minimum Volume: 0.4 mL
Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Ambient (preferred)	14 days	LIGHT PROTECTED
	Frozen	14 days	LIGHT PROTECTED
	Refrigerated	14 days	LIGHT PROTECTED

CPT Code Information:

82380

CSFP 608336 Carrier Screen, Focused Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81220

81329

81479

81257

81363

81222

81479 (if appropriate for government payers)

FCRTG 57630 Carrot IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CROT
82742

Carrot, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCASG
57555

Casein IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CASE
82895

Casein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

FCCA3 Cashew Component rAna o 3
75556

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

86008

CASHX Cashew Component, IgE, Serum (Non-Orderable)
618846

Specimen Requirements:

Only orderable as a reflex. For more information see CASHR / Cashew, IgE, with Reflex to Cashew Component, IgE, Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

FCASH
57687

Cashew IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CASHR 618845 Cashew, IgE with Reflex to Cashew Component, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CASH 82881 Cashew, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCFS
75883

Caspofungin Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Complete and submit with specimen:

1. Fungus Testing Lab request form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999-Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 - mould - MIC microdilution or agar dilution (if appropriate)

87186 - yeast - MIC microdilution or agar dilution (if appropriate)

CASRG 618058 CASR Full Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight..

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that

all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

CAT
82665

Cat Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CATPF
616093

Cat Epithelium, IgE, with Reflex to Cat Epithelium

Components, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

COMTQ
610049

Catechol-O-Methyltransferase (COMT) Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

0032U

CATU 9276 Catecholamine Fractionation, Free, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. If medically feasible, discontinue drugs that release or hinder metabolism of epinephrine, norepinephrine, or dopamine for at least 1 week prior to specimen collection (see Cautions for details). If this is not possible for medical reasons, contact the laboratory to discuss whether a shorter drug-withdrawal period may be acceptable.

2. Unless the reason for testing is drug monitoring, the patient should stop any epinephrine, norepinephrine, or dopamine injections or infusions for at least 12 hours prior to specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative at start of collection. Use 15 mL of 50% acetic acid for children younger than 5 years. This preservative is intended to achieve a pH of between approximately 2 and 4.

2. Collect urine for 24 hours.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82384

CATP 8532 Catecholamine Fractionation, Free, Plasma

Specimen Requirements:

Patient Preparation:

1. Discontinue drugs that release epinephrine, norepinephrine, or dopamine, or hinder their metabolism, for at least one week before obtaining the specimen (see Cautions for details). If this is not possible for medical reasons, contact the laboratory and discuss whether a shorter drug withdrawal period may be possible in a particular case.

2. Unless the purpose of the measurement is drug monitoring, discontinue any epinephrine, norepinephrine, or dopamine injections or infusions for at least 12 hours before specimen collection.

3. Patient **must not eat, use tobacco, or drink caffeinated beverages** for at least 4 hours before the specimen is collected.

Supplies:

Catecholamine Tubes-EDTA (T066) (contain sodium metabisulfite, may come as 10-mL or 6-mL tubes, and **have a 6-month expiration time**)

Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: 10-mL Catecholamine tubes containing EDTA-sodium metabisulfite solution

Acceptable: 6-mL Catecholamine tubes containing EDTA-sodium metabisulfite solution

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL plasma

Collection Instructions:

Note: If the collection instructions are not followed, falsely elevated test results are highly likely.

1. **Drawing from an indwelling intravenous (IV) line/catheter/butterfly is required.**

2. Calm the patient by giving complete instructions and reassurance regarding the procedure.

3. Insert an indwelling IV catheter. Flush with 3 mL of sodium chloride (NaCl) using positive pressure.

4. Have the patient rest for 30 minutes in the supine position in a quiet room.

5. At the end of the 30 minutes, withdraw and discard a minimum of 2 mL of blood to remove the saline out of the catheter.

6. If provocative sampling (eg, standing specimen) is required, perform provocative maneuver immediately after obtaining supine specimen. Obtain standing specimen immediately.

7. For each specimen, draw 10 mL of blood into the chilled 10 mL catecholamine tube containing EDTA-sodium metabisulfite solution. A 6 mL pink top EDTA-metabisulfite tube is an acceptable substitute.

8. Specimens **must** remain at refrigerated temperature at all times.

9. Separate plasma in a refrigerated centrifuge within 30 minutes of collection.

10. Freeze specimen immediately. Transport frozen.

Specimen Minimum Volume: Plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA Meta	Frozen	28 days

CPT Code Information:

82384

CPVTG
617211 Catecholaminergic Polymorphic Ventricular Tachycardia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days /Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81408

81479

81479 (if appropriate for government payers)

FCATE
57554

Catfish (Siluriformes spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FCLPF
75676

Cathartic Laxatives Profile, Stool

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Stool

Container/Tube: Acid-washed or trace metal-free plastic container, MCL supply T656

Specimen Volume: 10 g

Collection Instructions: Collect 10 g of stool with no preservative. Send specimen in an acid-washed or trace metal-free plastic container, MCL supply T656. Send specimen refrigerated.

Specimen Minimum Volume: 10 mL stool liquid or 10 g stool solid

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

83735

84100

CTSK
607887

Cathepsin K Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered;

sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FCAFG
57680

Cauliflower IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 0.5 mL

Submission Container/tube: Plastic vial

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days

	Ambient	7 days

CPT Code Information:

86001

CALFL 82617 Cauliflower, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

IN16Q 610266 CBFB::MYH11 Inversion(16), Quantitative Detection and Minimal Disease Risk Monitoring, qRT-PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: Blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	72 hours

CPT Code Information:

81401-*CBFB-MYH11*

CD10
70373

CD10 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD103
70372

CD103 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD11C
70412

CD11c Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD123
70413

CD123 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD13
70374

CD13 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD138
70414

CD138 (Syndecan) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD14I
70375

CD14 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD15
70376

CD15 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD163
70415

CD163 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

C19BM 603205 CD19 Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostains Technical Only Envelope

Preferred: 2 Unstained positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick; B5 fixed, decalcified, paraffin-embedded bone marrow tissue.

Acceptable: B5 fixed, decalcified, paraffin-embedded bone marrow tissue block.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD19I
70377

CD19 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD1A
70378

CD1a Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD2B
603203

CD2 Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostains Technical Only Envelope

Preferred: 2 Unstained positively charged glass slides (25- x 75- x 1-mm) per test

ordered; sections 4-microns thick; B5 fixed, decalcified, paraffin-embedded bone marrow tissue.

Acceptable: B5 fixed, decalcified, paraffin-embedded bone marrow tissue block.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD2
70384

CD2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CEE20 CD20 Cell Expression Evaluation, Varies
65660

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 10 mL

Collection Instructions:

1. Label specimen as blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Minimum Volume: Blood: 3 mL; Bone marrow aspirate: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 3-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker

88187-Flow Cytometry Interpretation, 2 to 8 Markers

CD20I
70379

CD20 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time

TECHONLY	Ambient (preferred)
	Refrigerated

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD20B
89584

CD20 on B Cells, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information:

1. **Secondary aliquot tubes will be rejected.**

2. **Testing will be canceled** if the specimen is not received ambient.

3. For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	4 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86355

86356

CD200
619517

CD200 (OX2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD21
70380

CD21 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD22I
70381

CD22 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD23
70382

CD23 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD25
70383

CD25 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD273
601986

CD273 (PD-L2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD279
70417

CD279 (PD-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD3I 70391 CD3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD30 70385 CD30 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD303
607891

CD303 (BDCA-2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD31
70386

CD31 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD33 CD33 Immunostain, Technical Component Only
70387

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD34I CD34 Immunostain, Technical Component Only
70388

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD35
70389

CD35 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD38
70390

CD38 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TCD4 CD4 Count for Immune Monitoring, Blood
84348

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86359-T cells, total count

86360-Absolute CD4/CD8 count with ratio

CD4NY CD4 Count for Monitoring, New York, Blood
28334

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86359-T cells, total count

86360-Absolute CD4/CD8 count with ratio

CD4I
70393

CD4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD4RT
89504

CD4 T-Cell Recent Thymic Emigrants, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356

CD43
70392

CD43 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD45I
70348

CD45 Leukocyte Common Antigen (LCA) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CEE49 CD49d Cell Expression Evaluation, Varies

65658

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 10 mL

Collection Instructions:

1. Label specimen as blood.

2. Send specimen in original tube. **Do not aliquot**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Minimum Volume: Blood: 3 mL; Bone marrow aspirate: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 3-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker

88187-Flow Cytometry Interpretation, 2 to 8 Markers

CD5
70396

CD5 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CEE52
65659

CD52 Cell Expression Evaluation, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 10 mL

Collection Instructions:

1. Label specimen as blood.

2. Send specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Minimum Volume: Blood: 3 mL; Bone marrow aspirate: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 3-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker

88187-Flow Cytometry Interpretation, 2 to 8 Markers

CD56
70394

CD56 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD57
70395

CD57 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD61
70397

CD61 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CDKPB 603209 CD68 (KP1) Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CDKP1 70486 CD68 (KP1) Immunostain, Tissue, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PGM1 70536

CD68 (PG-M1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD7
70399

CD7 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD71
70398

CD71 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

CD79
70418

CD79a Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CD8I
70400

CD8 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GLICP 89369 CD8 T-Cell Immune Competence Panel, Global, Whole Blood

Specimen Requirements:

Two separate whole blood specimens, EDTA and (possibly 2 tubes) sodium heparin, are required.

For serial monitoring, it is recommended that specimens are collected at the same time of day.

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Specimen Type: EDTA whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as TBBS / Quantitative Lymphocyte Subsets: T, B, and Natural Killer (NK) Cells, Blood

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Type: Sodium heparin whole blood

Container/Tube: Green top (sodium heparin)

Specimen Volume: 15 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as GLIC / CD8 T-Cell Immune Competence, Global, Blood

Specimen Minimum Volume: Sodium heparin whole blood: 10 mL
EDTA whole blood: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP
Whole Blood EDTA	Ambient	52 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86355-B cells, total count

86357-Natural killer (NK) cells, total count

86359-T cells, total count

86360-Absolute CD4/CD8 count with ratio

86356 x 2

GLIC
89317

CD8 T-Cell Immune Competence, Global, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 15 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86356 x 2

CD99
70508

CD99 (MIC-2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CDKZ
35385

CDKN1C Gene, Full Gene Analysis, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes This does not include cultured chorionic villi.

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

CDX2
70401

CDX2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FRCE
57952

Cedar Red (Juniperus virginiana) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

CEDR
82482

Cedar, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CELY
82766

Celery, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CELI
88906

Celiac Associated HLA-DQ Alpha 1 and DQ Beta 1 DNA Typing, Blood

Specimen Requirements:

Container/Tube: Yellow top (ACD Solution A or B)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood ACD-B	Refrigerated (preferred)
	Ambient

CPT Code Information:

81376 x 2-HLA Class II typing, low resolution (eg, antigen equivalents); one locus (eg, HLA-DRB1/3/4/5, -DQB1, -DQA1, -DPB1, or -DPA1), each

CDCOM 89201 Celiac Disease Comprehensive Cascade, Serum and Whole Blood

Specimen Requirements:

Both whole blood and serum are required.

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: No additional anticoagulants are acceptable.

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not aliquot**.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Whole blood: 3 mL; Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days
Whole Blood ACD-B	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

81376 x 2

82784

86258 (if appropriate)

86364 (if appropriate)

86231 (if appropriate)

CDGF 89200 Celiac Disease Gluten-Free Cascade, Serum and Whole Blood

Specimen Requirements:

Both whole blood and serum are required.

Specimen Type: Whole Blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: No additional anticoagulants are acceptable.

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not aliquot**.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: Blood: 3 mL
Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days
Whole Blood ACD-B	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

81376 x 2

82784 (if appropriate)

86258 (if appropriate)

86364 (if appropriate)

86231 (if appropriate)

CDSP
89199

Celiac Disease Serology Cascade, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days

CPT Code Information:

82784

86258 (if appropriate)

86364 (if appropriate)

86231 (if appropriate)

BFCC 608873 Cell Count and Differential, Body Fluid

Specimen Requirements:

For Local Accounts Only

Sources: Synovial, pleural, peritoneal, pericardial fluid

Container/Tube:

Preferred: Body fluid container

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 1 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Ambient (preferred)	24 hours
	Refrigerated	24 hours

CPT Code Information:

89051-Cell count with differential

BRAFB 65100

Cell-Free DNA BRAF V600, Blood

Specimen Requirements:

Supplies: Streck Black/Tan Top Tube Kit (T715)

Container/Tube: Streck Cell-Free DNA blood collection kit

Specimen Volume: Two 10-mL Streck Cell-Free DNA blood collection tubes

Additional Information: Only blood collected in Streck Cell-Free DNA tubes will be accepted for analysis. Whole blood will be processed to produce platelet-poor plasma before cfDNA isolation.

Specimen Minimum Volume: One 10 mL Streck cell-free DNA blood collection tube

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81210

T790M
113410

Cell-Free DNA EGFR T790M Mutation Analysis, Blood

Specimen Requirements:

Supplies: Streck Black/Tan Top Tube Kit (T715)

Specimen Volume: Two, 10-mL Streck cell-free DNA (cfDNA) blood collection tubes

Additional Information:

1. Only blood collected in Streck cfDNA tubes will be accepted for analysis.

2. Whole blood will be processed to produce platelet-poor plasma before cfDNA isolation.

Specimen Minimum Volume: One 10 mL Streck tube

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	7 days	Streck Black/Tan top
	Refrigerated	7 days	Streck Black/Tan top

CPT Code Information:

81235

KRASD 68003 Cell-Free DNA KRAS 12, 13, 61,146, Blood

Specimen Requirements:

Supplies: Streck Black/Tan Top Tube Kit (T715)

Container/Tube: Streck Cell-Free DNA blood collection kit

Specimen Volume: Two 10-mL Streck Cell-Free DNA blood collection tubes

Additional Information: Only blood collected in Streck Cell-Free DNA BCT tubes will be accepted for analysis. Whole blood will be processed to produce platelet-poor plasma before cfDNA isolation.

Specimen Minimum Volume: One 10 mL Streck tube

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	7 days	Streck Black/Tan top
	Refrigerated	7 days	Streck Black/Tan top

CPT Code Information:

81275

81276

CNSA 70644 Central Nervous System Consultation, Autopsy, Varies

Specimen Requirements:

Hematoxylin-and-eosin stained sections are also acceptable, but must be accompanied by paraffin blocks and/or remaining wet tissue.

In cases submitted as part of a research protocol, contact 507-284-3887 for further guidance.

Supplies: Central Nervous System Consult Kit (T633)

Sources: Brain (and spinal cord when indicated)

Container/Tube: Plastic container

Specimen Volume: 1 cm(3) cube

Collection Instructions:

1. Take a small (1 cm[3]) cube of brain from 1 of the frontal lobes (typically, right inferior frontal).

2. Wrap in aluminum foil and place in plastic container.

3. Label container with identifying information (ie, patient name, date of birth, autopsy number, and date of collection).

4. Freeze and store in a -70 degrees C freezer.

5. Place the remaining brain in 10% formalin if non-Creutzfeldt-Jakob disease (CJD), and 15% formalin for suspected cases of CJD, suspended by a thread under the basilar artery and fixed for 7 to 10 days.

Additional Information: Upon completion of consultation, the brain tissue will be stored 10 years in the Mayo Clinic Tissue Registry. The Mayo Clinic policy precludes our evaluation of cases under litigation that involve non-Mayo Clinic patients.

Specimen Minimum Volume: Entire collection

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

88037 (if appropriate)

88036 (if appropriate)

CMA 9278 Centromere Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

CEAC
82387

Cephalosporium acremonium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

Ceramide Trihexosides and Sulfatides, Random, Urine

Specimen Requirements:

Patient Preparation: Prior to urine collection, patient **should not** use baby wipes or wipes containing soaps as these may interfere with testing.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL urine tube

Specimen Volume: 2 mL

Collection Instructions: Collect a **first-morning**, random urine specimen.

Specimen Stability Information: Refrigerated (preferred) 45 days/Ambient 45 days/Frozen 19 months

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	45 days
	Ambient	45 days
	Frozen	

CPT Code Information:

83789

SFINC 2762 Cerebrospinal Fluid (CSF) IgG Index, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SFIG / Cerebrospinal Fluid IgG Index Profile, Serum and Spinal Fluid.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Preferred: Collection vial number 2

Specimen Volume: 1 mL

Collection Instructions: Label specimen as SFINC.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82042

82784

SFIG 610783 Cerebrospinal Fluid IgG Index Profile, Serum and Spinal Fluid

Specimen Requirements:

Both serum and spinal fluid are required.

-Spinal fluid must be obtained within 7 days of serum collection.

-Two individual serum samples are required.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: 2 Plastic vials

Specimen Volume: 2 mL Serum in 2 separate plastic vials (see Collection Instructions)

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into 2 plastic vials, each containing 1 mL.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Preferred: Collection vial number 2

Specimen Volume: 1 mL

Collection Instructions: Label specimen as SFINC.

Specimen Minimum Volume: Serum: 1 mL in 2 plastic vials, each vial containing 0.5 mL; Spinal fluid: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82040

82042

82784 x 2

CTXWB
113444

Cerebrotendinous Xanthomatosis, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin), yellow top (ACD B)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood in original vial. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	72 hours
	Ambient	48 hours

CPT Code Information:

82542

CTXBS 65630 Cerebrotendinous Xanthomatosis, Blood Spot

Specimen Requirements:

Supplies:

-Card-Blood Spot Collection (Filter Paper) (T493)

-Card-Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Blood Spot Collection Card (Filter Paper)

Acceptable: Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper, Munktell filter paper, Postmortem Screening Card or collected with EDTA, sodium heparin, lithium heparin, or ACD B-containing devices

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Let blood dry completely on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

2. At least 1 spot should be complete (ie, unpunched).

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	10 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Ambient	10 days	FILTER PAPER

CPT Code Information:

82542

CTXP
65631

Cerebrotendinous Xanthomatosis, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin), yellow top (ACD B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions:

1. Centrifuge at 4 degrees C if possible

2. Aliquot plasma into a plastic vial. **Do not disturb or transfer the buffy coat layer**.

3. Send frozen.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	65 days

CPT Code Information:

82542

FCZAC
75563

Certolizumab and Anti-Certolizumab Antibody, DoseASSURE CTZ

Specimen Requirements:

Patient Preparation: High serum biotin concentrations in patients taking biotin supplements may cause an interference in this assay. Patients may be advised to stop biotin consumption at least 72 hours prior to sample collection.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Allow a minimum clotting time of 30 to 60 minutes.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

3. Send frozen.

To avoid delays in turnaround time when requesting multiple tests, **submit separate frozen specimens for each test requested**.

Specimen Minimum Volume: Serum: 0.6 mL, Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

80299

82397

CERS
624326

Ceruloplasmin, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial (false-bottom vials are **not acceptable**)

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

82390

CHGL
82384

Chaetomium globosum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FAMCE Cheese American IgE
57914

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FCCGG Cheese Cheddar IgG
57573

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FSCE
57936

Cheese Swiss IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

CCHZ
82752

Cheese, Cheddar, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MCHZ
82751

Cheese, Mold, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL; For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CHER
82798

Cherry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CTRE
82607

Chestnut Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CNUT
82870

Chestnut, Sweet, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CHXP
82494

Chick Pea, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CDROP
82142

Chicken Droppings, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CHCK
82713

Chicken Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

FCHXG 57625 Chicken IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CSPR 82351 Chicken Serum Proteins, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CHIC
82703

Chicken, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CHIKG 63868 Chikungunya IgG, Antibody, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CHIKV / Chikungunya IgM and IgG, Antibody, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86790

CHIKV 64173 Chikungunya IgM and IgG, Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86790 x2

CHIKM
63867

Chikungunya IgM, Antibody, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CHIKV / Chikungunya IgM and IgG, Antibody, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86790

CHIKI
37102

Chikungunya Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CHIKV / Chikungunya IgM and IgG, Antibody, Serum.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CHIKS
603833

Chikungunya Virus, PCR, Molecular Detection, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Sterile container

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot the serum into a sterile container.

Additional Information: Serum specimens not aliquoted from the serum gel collection tube into a sterile container **will be rejected**.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

CHIKC 603832 Chikungunya Virus, PCR, Molecular Detection, Spinal Fluid

Specimen Requirements:

Collection Container/Tube:

Preferred: Vial number 2

Acceptable: Any vial number

Submission Container/Tube: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge or heat inactivate.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

CHIMU 62983 Chimerism Transplant No Cell Sort, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Only 1 tube is required.

2. Invert several times to mix blood.

3. Label specimen as blood.

4. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Whole Blood: 3 mL
Bone marrow: 1 mL
Lesser volumes may be acceptable, depending on white cell count.
Call 800-533-1710 or 507-266-5700 with questions.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	7 days	PURPLE OR PINK TOP/EDTA
	Refrigerated	7 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81267-Chimerism (engraftment) analysis, post hematopoietic stem cell transplantation specimen, includes comparison to previously performed baseline analyses, without cell selection

Chimerism Transplant Sorted Cells, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Only 1 tube is required.

2. Invert several times to mix blood.

3. Label specimen as blood.

4. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Whole blood: 3 mL
Bone marrow: See Specimen Required
Lesser volumes may be acceptable, depending on white cell count.

Call 800-533-1710 or 507-266-5700 with questions.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	4 days	PURPLE OR PINK TOP/EDTA
	Refrigerated	4 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81268-Chimerism (engraftment) analysis, post hematopoietic stem cell transplantation specimen, includes comparison to previously performed baseline analyses; with cell selection (eg, CD3, CD33), each cell type (if appropriate)

CHIDB
83182

Chimerism-Donor, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Label specimen as blood.

3. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not** aliquot.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA from blood or bone marrow

2. Indicate volume and concentration of the DNA

Specimen Type: Buccal swab

Supplies: Buccal Swab Kit (T543)

Container/Tube: Buccal smear collection kit

Specimen Volume: 2 Cyto-Pak brushes-1 per cheek

Collection Instructions:

1. Patient should rinse out mouth vigorously with mouthwash for approximately 15 seconds.

2. Remove Cyto-Pak brush from container only touching "stick" end. Save container.

3. Using **medium** pressure, rotate brush several times on inside of cheek.

4. Return brush to container and cap.

5. Repeat steps 2 through 4 on other cheek using second brush.

6. It is important that patient's buccal cells are not contaminated with cells from any other source. Do not touch bristles. Do not brush too vigorously. If blood appears, discard brush and restart collection process.

7. Label each container with patient's name and order number or hospital/clinic number.

Additional Information: It is important that the cells do not dry out during shipping. Ensure that container is tightly sealed.

Specimen Minimum Volume: Whole blood: 3 mL
Bone marrow/Buccal swab: See Specimen Required
Extracted DNA from blood or bone marrow: 50 microliters at 20 ng/microliter
Lesser volumes may be acceptable, depending on white cell count.

Call 800-533-1710 or 507-266-5700 with questions.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81265-Comparative analysis using short tandem repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells)

CHRGB Chimerism-Recipient Germline (Pretransplant), Varies
83186

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Label specimen as blood.

3. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot**.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA from blood or bone marrow

2. Indicate volume and concentration of the DNA

Specimen Type: Buccal swab

Supplies: Buccal Swab Kit (T543)

Container/Tube: Buccal smear collection kit

Specimen Volume: 2 Cyto-Pak brushes-1 per cheek

Collection Instructions:

1. Patient should rinse out mouth vigorously with mouthwash for approximately 15 seconds.

2. Remove Cyto-Pak brush from container only touching "stick" end. Save container.

3. Using **medium** pressure, rotate brush several times on inside of cheek.

4. Return brush to container and cap.

5. Repeat steps 2 through 4 on other cheek using second brush.

6. It is important that patient's buccal cells are not contaminated with cells from any other source. Do not touch bristles. Do not brush too vigorously. If blood appears, discard brush and restart collection process.

7. Label each container with patient's name and order number or hospital/clinic number.

Additional Information: It is important that the cells do not dry out during shipping. Ensure that container is tightly sealed.

Specimen Minimum Volume: Whole blood: 3 mL
Bone marrow/buccal swab: See Specimen Required
Extracted DNA from blood or bone marrow: 50 microliters at 20 ng/microliter
Lesser volumes may be acceptable, depending on white cell count.
Call 800-533-1710 or 507-266-5700 with questions.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells)

CHLG
619389

Chlamydia IgG, Immunofluorescence, Serum

Specimen Requirements:

Specimen Required

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86631 x 2

CHLAP 619387 Chlamydia IgM and IgG Panel, Immunofluorescence, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86632 x 2

86631 x 2

CHLM 619388 Chlamydia IgM, Immunofluorescence, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86632 x 2

MCTGC 43721 Chlamydia trachomatis and Neisseria gonorrhoeae, Miscellaneous Sites, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Ocular (corneal/conjunctiva)

Supplies:

-Aptima Unisex Swab Collection Kit (T583)

-Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab or Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using Aptima Multitest Swab or Aptima Unisex Swab. **Specimens must be collected using either one of these Aptima swabs.**

Note: The white swab provided within the collection kit is a cleaning swab and should

not be used for collection. Discard the white cleaning swab.

2. Place collection swab in transport tube provided with the collection kit.

3. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and time.

5. Maintain specimen at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 60 days of collection.

Specimen Type: Peritoneal fluid (pelvic wash, cul-de-sac fluid)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Transfer 1 mL of peritoneal fluid directly into the Aptima specimen transfer tube within 24 hours of collection.

2. Cap tube securely and label tube with patient's entire name and collection date and time.

3. Maintain specimen at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 30 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87494

CGRNA 61553 Chlamydia trachomatis and Neisseria gonorrhoeae, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Endocervix/Cervix

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Unisex Swab Collection Kit.**

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

3. Discard the cleaning swab.

4. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

5. Place blue swab into transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Multitest Swab Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urethra (Males Only)

Supplies: Swab, Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Unisex Swab Collection Kit.

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate blue swab gently at least 1 full rotation using sufficient pressure to ensure swab comes into contact with all urethral surfaces. Allow blue swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection, transfer 2 mL of urine into the urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: ThinPrep Specimen (Endocervix/Cervix)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect ThinPrep sample as per normal collection process.

2. ThinPrep specimen must be aliquoted (as outlined below) **before** it is processed for Papanicolaou testing (Pap smear).

3. Vortex ThinPrep/PreservCyt vial 3 to 10 seconds.

4. Within 1 minute of vortexing:

a. Transfer 1 mL of specimen into the Aptima Specimen Transport Tube using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug).

b. Process only 1 ThinPrep and transfer tube set at a time.

c. Recap Aptima specimen transfer tube tightly and gently invert 3 times to mix.

5. Label Aptima transfer tube with appropriate label.

6. Use remainder of ThinPrep specimen for Pap testing.

7. Maintain specimen transport tube at 2 to 8 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection, or within 14 days if stored at 15 to 30 degrees C. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Oropharynx/Pharynx/Throat

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Multitest Swab Collection Kit.

2. Swab site using Aptima Multitest Swab (pink shaft).

3. Place pink swab in transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Type: Rectal/Anal

Supplies: Swab, Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Multitest Swab Collection Kit.

2. Insert swab into rectum about 3 to 5 cm past anal margin and gently rotate swab for 10 seconds.

3. Place collection swab in transport tube provided in collection kit.

4. Snap off swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87494

SCCGR 621934 Chlamydia trachomatis and Neisseria gonorrhoeae, Self-Collect, Amplified RNA, Rectal

Specimen Requirements:

Specimen Type: Rectal/anal

Supplies: Aptima Rectal Swab Self-Collection Kit (T1000)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1000 or available separately)**.**

2. Provide patient with the Aptima Rectal Swab Self-Collection Kit or Aptima Multitest Swab and Rectal Swab Patient Collection Guide.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL

	Ambient	APTIMA VIAL
	Frozen	APTIMA VIAL

CPT Code Information:

87494

SCCGT 621931 Chlamydia trachomatis and Neisseria gonorrhoeae, Self-Collect, Amplified RNA, Throat

Specimen Requirements:

Specimen Type: Oropharynx/pharynx/throat

Supplies: Aptima Throat Swab Self-Collection Kit (T1002)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. **Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab** (provided in T1002 or available separately)**.**

2. Provide patient with the Aptima Throat Swab Self-Collection Kit or Aptima Multitest Swab and Throat Swab Patient Collection Guide.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87494

SCCGV 621937 Chlamydia trachomatis and Neisseria gonorrhoeae, Self-Collect, Amplified RNA, Vaginal

Specimen Requirements:

Specimen Type: Vaginal

Supplies: Aptima Vaginal Swab Self-Collection Kit (T1001)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1001 or available separately)**.**

2. Provide patient with the Aptima Vaginal Swab Self-Collection Kit or Aptima Multitest Swab and Vaginal Swab Patient Collection Guide.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for up to 12 months.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87494

CTCON 623154 Chlamydia trachomatis Confirmation, Nucleic Acid Amplification, Varies

Specimen Requirements:

If testing is being performed on a previously tested specimen that resulted as positive, submit that specimen.

If testing is being performed on a newly collected specimen, submit only 1 of the following specimens:

Specimen Type: Ocular (corneal/conjunctiva)

Supplies:

-Aptima Unisex Swab Collection Kit (T583)

-Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab or Aptima Collection Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using Aptima Multitest Swab or Aptima Collection Unisex Swab. Specimens must be collected using either of these options.

Note: The white swab provided within the collection kit is a cleaning swab and should not be used for collection. Discard the white cleaning swab.

2. Place collection swab in transport tube provided in collection kit.

3. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimen must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerated temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection. transfer 2 mL of urine into the urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport tube at 2 to 30 degrees C (refrigerated temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Oropharynx/Pharynx/Throat

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Collection Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Swab site using Aptima Collection Multitest Swab (pink shaft).

3. Place pink swab in transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Rectal/Anal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab into rectum about 3 to 5 cm past anal margin and gently rotate swab for 10 seconds.

3. Place collection swab in transport tube provided in collection kit.

4. Snap off swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL

	Frozen	APTIMA VIAL

CPT Code Information:

87491

MCRNA 61554 Chlamydia trachomatis, Miscellaneous Sites, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Ocular (corneal/conjunctiva)

Supplies:

-Aptima Unisex Swab Collection Kit (T583)

-Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab or Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using Aptima Multitest Swab or Aptima Unisex Swab. **Specimens must be collected using either of these Aptima swab**s.

Note: The white swab provided within the collection kit is a cleaning swab and should not be used for collection. Discard the white cleaning swab.

2. Place collection swab in transport tube provided in collection kit.

3. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred); transport within 60 days of collection.

Pediatric Specimen Type: Nasopharyngeal (NP) (**only accepted for patients aged 6 months or younger**)

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using NP Swab or Aptima Unisex Swab. **Specimens must be collected using either one of these swabs**.

Note: The white swab provided within the Aptima Unisex Swab Collection kit is a cleaning swab and should not be used during this collection process. This white swab should be discarded.

2. Place NP swab in Aptima Specimen Transfer Tube.

3. If using the Aptima Unisex Swab collection kit, place the unisex swab in the Aptima Collection kit tube. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred); transport within 60 days.

Specimen Type: Peritoneal fluid (pelvic wash, cul-de-sac fluid)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Transfer 1mL of peritoneal fluid directly into the Aptima specimen transfer tube within 24 hours of collection.

2. Cap tube securely and label tube with patient's entire name and collection date and time.

3. Maintain Aptima specimen transfer tube at 2 to 30 degrees C (refrigerate temperature is preferred); transport within 30 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL

	Frozen	APTIMA VIAL

CPT Code Information:

87491

CTRNA 61551 Chlamydia trachomatis, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Endocervix/Cervix

Supplies: Swab, Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Unisex Swab Collection Kit.

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

3. Discard the cleaning swab.

4. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

5. Place blue swab into transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Vaginal

Supplies: Swab, Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name, and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urethra (Males Only)

Supplies: Swab, Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Unisex Swab Collection Kit.

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate blue swab gently at least 1 full rotation using sufficient pressure to ensure swab comes into contact with all urethral surfaces. Allow blue swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date, and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection, transfer 2 mL of urine into the urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: ThinPrep Specimen (Endocervix/Cervix)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: ThinPrep collection kit

Specimen Volume: 1 mL

Collection Instructions:

1. Collect ThinPrep sample as per normal collection process.

2. **ThinPrep specimen must be aliquoted (as outlined below) before it is processed/tested for Papanicolaou test (Pap smear)**.

3. Vortex ThinPrep/PreservCyt vial 3 to 10 seconds.

4. Within 1 minute of vortexing:

a. Transfer 1 mL of specimen into the Aptima Specimen Transport Tube using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug).

b. Process only 1 ThinPrep and transfer tube set at a time.

c. Recap Aptima specimen transfer tube tightly and gently invert 3 times to mix.

4. Label Aptima specimen transfer tube with appropriate label.

5. Use remainder of ThinPrep specimen for Papanicolaou (Pap) testing.

6. Maintain specimen transport tube at 2 to 8 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection, or within 14 days if stored at 15 to 30 degrees C. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Oropharynx/Pharynx/Throat

Supplies: Swab, Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Swab site using Aptima Collection Multitest Swab (pink shaft).

3. Place pink collection swab in transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Type: Rectal/anal

Supplies: Swab, Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab into rectum about 3 to 5 cm past anal margin and gently rotate swab for 10 seconds.

3. Place collection swab in transport tube provided in collection kit.

4. Snap off swab at score line so swab fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C and transport within 60 days of collection.

Specimen Minimum Volume: Urine: 2 mL; Endocervical/Cervix in PreservCyt or Swab specimens: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87491

SCCTV 621938 Chlamydia trachomatis, Self-Collect, Amplified RNA, Vaginal (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SCCGV / *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Vaginal.

Specimen Type: Vaginal

Supplies: Aptima Vaginal Swab Self-Collection Kit (T1001)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1001 or available separately)**.**

2. Provide patient with the Aptima Vaginal Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided with the kit and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for up to 12 months.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

SCCTR 621935 Chlamydia trachomatis, Self-Collect, Nucleic Acid Amplification, Rectal

Specimen Requirements:

Only orderable as part of a profile. For more information see SCCGR / *Chlamydia trachomatis* and *Neisseria gonorrhoeae,* Self-Collect, Amplified RNA, Rectal

Specimen Type: Rectal/anal

Supplies: Aptima Rectal Swab Self-Collection Kit (T1000)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1000 or available separately**.**

2. Provide patient with the Aptima Rectal Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then

return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

SCCTT 621932 Chlamydia trachomatis, Self-Collect, Nucleic Acid Amplification, Throat (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SCCGT / *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Throat.

Specimen Type: Oropharynx/pharynx/throat

Supplies: Aptima Throat Swab Self-Collection Kit (T1002)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. **Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab** (provided in T1002 or available separately)**.**

2. Provide patient with the Aptima Throat Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

CDP 8610 Chlordiazepoxide and Metabolites, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Red top (serum get/SST are **not acceptable)**

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into an amber vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum Red	Refrigerated (preferred)	28 days	LIGHT PROTECTED
	Ambient	28 days	LIGHT PROTECTED
	Frozen	28 days	LIGHT PROTECTED

CPT Code Information:

80299

CLU 614058 Chloride, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-Hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82436

RCHLU 610607 Chloride, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82436

CL
8460

Chloride, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

82435

FCHPZ
57719

Chlorpromazine (Thorazine)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 1 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

FCHCG 57644 Chocolate/Cacao IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and send 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FCCK1 Cholecystokinin (CCK)
75914

Specimen Requirements:

Patient Preparation:

1. Fasting: 10 to 12 hours

2. Antacid medications and medications that affect intestinal motility should be discontinued, if possible, for at least 48 hours prior to collection of specimen.

Supplies: GI Preservative Tube (T125)

Collection Container/Tube: Special tube containing GI preservative

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL GI plasma

Collection Instructions:

1. Collect 10 mL blood in special GI preservative tube.

2. Centrifuge immediately and aliquot 3 mL plasma into a plastic vial.

3. Freeze immediately and ship frozen.

Specimen Minimum Volume: GI plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
GI Plasma	Frozen	180 days

CPT Code Information:

83519

CHLGP 608018 Cholestasis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

CHLBF
606915

Cholesterol, Body Fluid

Specimen Requirements:

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

-Pericardial fluid

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

84311-Spectrophotometry, analyte not specified (cholesterol)

HDCH 8429 Cholesterol, High-Density Lipoprotein (HDL), Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required unless directed by the ordering provider.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tube must be centrifuged within 2 hours of collection.

2. Red-top tube must be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

83718

CLDL1 617023 Cholesterol, Low-Density Lipoprotein (LDL), Calculated, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see LPSC1 / Lipid Panel, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CHOL
8320

Cholesterol, Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

82465

CHLE
606888

Cholesteryl Esters, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial. Send refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days
	Ambient	24 hours

CPT Code Information:

84311

FCNAB 57737 Chromatin (Nucleosomal) Antibody

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

86235

CRCOF 606424 Chromium and Cobalt, Synovial Fluid

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Specimen Volume: 1 mL

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: Cobalt and chromium are present in the black rubber plunger seals found in most disposable syringes. As a result, synovial fluid **should not be** collected in these devices as contamination may occur.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Synovial Fluid	Refrigerated (preferred)	90 days
	Ambient	90 days
	Frozen	90 days

CPT Code Information:

83018

82495

CRUO 65719 Chromium Occupational Exposure, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5-mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82495

82570

CRU 8593 Chromium, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube:

Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82495

CRWB
65601

Chromium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Metal Free EDTA 3 mL Tube (T989)

Container/Tube: 3 mL Royal blue-top BD vacutainer with EDTA blood collection tube (T989) (BD catalog no. 367777)

Specimen Volume: 1.0 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82495

CRS 8638 Chromium, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free B-D Tube (No Additive), 6 mL (T184)

-Metal Free Specimen Vial (T173)

Collection Container/Tube: 6 mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 1.6 mL Serum

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

82495

CRSY
606353

Chromium, Synovial Fluid

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: Chromium is present in the black rubber plunger seals found in most disposable syringes. As a result, synovial fluid **should not be** collected in these devices as contamination may occur.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Synovial Fluid	Refrigerated (preferred)	90 days

	Ambient	90 days
	Frozen	90 days

CPT Code Information:

82495

CRCRU 607758 Chromium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic vial or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82495

82570

Chromogenic Factor IX Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85130

CHF8
610419

Chromogenic Factor VIII Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85130

CH8BI 606844 Chromogenic Factor VIII Inhibitor Bethesda Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time

Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

CHF8P 610420 Chromogenic Factor VIII Inhibitor Bethesda Profile, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 2 mL Platelet-poor plasma in two (2) 13-mm x 75-mm, 5-mL, polypropylene vials, each containing 1 mL

Collection Instructions:

1. Specimen **must be** collected prior to factor replacement therapy.

2. If collecting sample through a port/line, be sure to waste the appropriate amount prior to collection.

3. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

4. Within 4 hours of collection, Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

5. Aliquot plasma (1 mL per aliquot) into 2 separate 13-mm x 75-mm, 5-mL, polypropylene vials, leaving 0.25 mL in the bottom of centrifuged vial.

6. Immediately (within 2 hours of processing) freeze plasma at -20 degrees C or ideally, -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

CHF8-85130

CH8B-85335

CH8BI-85390-26

CH8B 606843 Chromogenic Factor VIII Inhibitor Bethesda Titer, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CHF8P / Chromogenic Factor VIII Inhibitor Bethesda Profile, Plasma.

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 2 mL Platelet-poor plasma in two (2) 13-mm x 75-mm, 5-mL, polypropylene vials, each containing 1 mL

Collection Instructions:

1. Specimen **must be** collected prior to factor replacement therapy.

2. If collecting sample through a port/line, be sure to waste the appropriate amount prior to collection.

3. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

4. Within 4 hours of collection, centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

5. Aliquot plasma (1 mL per aliquot) into 2 separate 13-mm x 75-mm, 5-mL, polypropylene vials, leaving 0.25 mL in the bottom of centrifuged vial.

6. Immediately (within 2 hours of processing) freeze plasma at -20 degrees C or, ideally -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

CGAK
34641

Chromogranin A, Serum

Specimen Requirements:

Patient Preparation: For at least 2 weeks before specimen collection, patient should stop taking proton pump inhibitor medications.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Information: Centrifuge and aliquot serum into plastic vial. **Do not submit in original tube**.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Ambient	48 hours
	Refrigerated	48 hours

CPT Code Information:

86316

CHRO 70402 Chromogranin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CMAFF 35263 Chromosomal Microarray (CMA) Familial Testing, FISH

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube: Green top (sodium heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Other anticoagulants are not recommended and are harmful to the viability of the cells.

Additional Information: Provide the name of the child (originally tested family member) on the request form. If testing was performed outside of Mayo Clinic Laboratories, consultation with the laboratory is required prior to ordering this test.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88273 w/modifier 52-Chromosomal in situ hybridization, less than 10 cells (if appropriate)

88273-Chromosomal in situ hybridization, 10-30 cells (if appropriate)

88274 w/modifier 52 Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

CMAPC 63042 Chromosomal Microarray, Autopsy, Products of Conception, or Stillbirth

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Products of conception or stillbirth

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's solution, Ringer's solution, or normal saline

Specimen Volume: 1 cm(3) of placenta (including 50-mg chorionic villi) **and** 1 cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions:

1. Attempt to identify and send only fetal tissue for analysis.

2. If a fetus cannot be specifically identified, collect 50-mg villus material or tissue that appears to be of fetal origin.

3. If multiple specimen types are sent, send each specimen in a separate container. Multiple specimens received (eg, placenta and fetal thigh) will be ordered under 1 test. All specimens will be processed separately.

Additional Information:

1. **Do not** send entire fetus.

2. **Do not** send the container from a dilation and curettage (D and C) procedure.

3. **Do not** send other containers with large volumes of tissue.

4. While fresher specimens prepared as described above are preferred, we can attempt analysis on specimens that have been in less-than-ideal conditions.

Specimen Type: Autopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's solution, Ringer's solution, or normal saline

Specimen Volume: 1 cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. Do not use alcohol or iodine preparations.

4. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 to 30 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. Results will be reported and telephoned or faxed if requested.

Specimen Type: Chorionic villus

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 50 mg

Collection Instructions:

1. Collect chorionic villus specimen (CVS) by transabdominal or transcervical method.

2. Transfer CVS to a Petri dish containing transport medium (such as CVS Media [RPMI] and Small Dish).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of villi and remove any blood clots and maternal decidua.

Acceptable

Specimen Type: Cultured cells

Container/Tube: T25 flasks with culture media

Specimen Volume: 2 T25 flasks

Specimen Type: Tissue

Supplies: Hank's Solution (T132)

Container/Tube: In sterile Hank's solution

Specimen Minimum Volume: Chorionic villus: 10 mg; Muscle-fascia: 1 cm(3); Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		
	Refrigerated		

CPT Code Information:

81229

CMAMT 62667 Chromosomal Microarray, Autopsy/Products of Conception/Stillbirth, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block containing fetal or placental (including chorionic villi) tissue.

Specimen Type: Slides

Specimen Volume: 6 Consecutive, unstained, 5-micron-thick sections placed on positively charged slides and 1 hematoxylin and eosin-stained slide.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Formalin-fixed, paraffin-embedded tissue block; 5 Consecutive, unstained slides and 1 hematoxylin and eosin-stained slide

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81229

CMACB 35247 Chromosomal Microarray, Congenital, Blood

Specimen Requirements:

This test requires 2 blood specimens: 1 sodium heparin and 1 EDTA.

Submit only 1 of the following specimen types:

Specimen Type: Whole blood

Container/Tube: Green top (sodium heparin) **and** lavender top (EDTA)

Specimen Volume: 3-mL EDTA tube **and** 4-mL sodium heparin tube

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimens in original tubes. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Cord blood

Container/Tube: Green top (sodium heparin) **and** lavender top (EDTA)

Specimen Volume: 3-mL EDTA tube **and** 4-mL sodium heparin tube

Note: The lab will attempt testing on a minimum of 1-mL whole blood, EDTA preferred.

Collection Instructions:

1. Invert several times to mix blood.

2. Send cord blood specimens in original tubes. **Do not aliquot.**

3. Label specimen as cord blood.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81229

CMAH
35899

Chromosomal Microarray, Hematologic Disorders, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Bone marrow

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA)

Specimen Volume: 1-2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Type: Whole blood

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA)

Specimen Volume: 7-10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as whole blood.

Specimen Minimum Volume: Blood: 2 mL; Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81277

CMAP 35898 Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 30 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium (such as CVS Media [RPMI] and Small Dish).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 to 30 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. Results will be reported and telephoned or faxed if requested.

Specimen Minimum Volume: Amniotic fluid: 12 mL; Chorionic villi: 12 mg; If ordering in conjunction with other testing: With PADF: 14 mL or 14 mg; with CHRAF: 24 mL; with CHRCV: 24 mg; with PADF and CHRAF/CHRCV: 26 mL or 26 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		
	Ambient		

CPT Code Information:

81229

CMAPT 35901 Chromosomal Microarray, Tumor, Formalin-Fixed Paraffin-Embedded

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Container/Tube: Formalin-fixed, paraffin-embedded tumor tissue block

Specimen Type: Slides

Specimen Volume: 10 Consecutive, unstained, 5-micron-thick sections placed on positively charged slides and 1 hematoxylin and eosin-stained slide

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81277

CMAT 35900 Chromosomal Microarray, Tumor, Fresh or Frozen

Specimen Requirements:

Submit only 1 of the following specimens:

Supplies: Hank's Solution (T132)

Specimen Type: Tumor biopsy

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5-3 cm(3) or larger

Specimen Type: Lymph node

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline.

Specimen Volume: 1 cm(3)

Specimen Type: Skin biopsy

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline.

Specimen Volume: 4-mm diameter

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. **Do not use alcohol or iodine preparations**.

4. A local anesthetic may be used.

5. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Minimum Volume: Tumor biopsy: 3 cm(3); Lymph node: 1 cm(3); Skin biopsy: 4 mm diameter

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81277

CHRAF 35243 Chromosome Analysis, Amniotic Fluid

Specimen Requirements:

Specimen Type: Amniotic fluid

Submission Container/Tube: Centrifuge tube

Specimen Volume: 20 to 25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted.

2. Discard the first 2 mL of amniotic fluid.

3. If ordering with PADF / Prenatal Aneuploidy Detection, FISH, submit a minimum of 14 mL.

4. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 24 mL.

5. If ordering with both PADF and CMAP, then submit a minimum of 26 mL.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. If the specimen does not grow in culture, the client will be notified within 7 days of receipt.

3. Bloody specimens are undesirable.

Specimen Type: Fetal body fluid

Container/Tube: Sterile tube

Specimen Volume: Entire specimen

Additional Information:

1. If the specimen does not grow in culture, the client will be notified within 7 days of receipt.

2. Clearly indicate on tube and paperwork that specimen is fetal body fluid.

Specimen Minimum Volume: The following are the minimum volumes when only this test is ordered:
Amniotic fluid: 12 mL
Fetal body fluid: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Amniotic Fld	Refrigerated (preferred)		
	Ambient		

CPT Code Information:

88235, 88291-Tissue culture for amniotic fluid or chorionic villus cells, Interpretation and report

88269 w/modifier 52-Chromosome analysis, in situ for amniotic fluid cells, <6 colonies, 1 karyotype with banding (if appropriate)

88269-Chromosome analysis, in situ for amniotic fluid cells, 6 or greater colonies, 1 karyotype with banding (if appropriate)

88267, 88285-Chromosome analysis, amniotic fluid or chorionic villus, greater than 15 cells, 1 karyotype with banding (if appropriate)

88267 w/modifier 52-Chromosome analysis, amniotic fluid or chorionic villus, <15 cells, 1 karyotype with banding (if appropriate)

CHRCV 35251 Chromosome Analysis, Chorionic Villus Sampling

Specimen Requirements:

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20-30 mg

Collection Instructions:

1. Collect chorionic villus specimen (CVS) by the transabdominal or transcervical method.

2. Transfer the CVS to a Petri dish containing transport medium (Such as CVS Media [RPMI] and Small Dish [T095]).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

4. If ordering with PADF / Prenatal Aneuploidy Detection, FISH, submit a minimum of 14 mg.

5. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 24 mg.

6. If ordering with both PADF and CMAP, then submit a minimum of 26 mg.

Specimen Minimum Volume: The following is the minimum volume when only this test is ordered:
12 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Tissue	Refrigerated (preferred)		
	Ambient		

CPT Code Information:

88235, 88291-Tissue culture for amniotic fluid or chorionic villus cells, Interpretation and report

88267 w/modifier 52-Chromosome analysis, amniotic fluid or chorionic villus, <15 cells, 1 karyotype with banding (if appropriate)

88267-Chromosome analysis, amniotic fluid or chorionic villus, 15 cells, 1 karyotype with banding (if appropriate)

88267, 88285-Chromosome analysis, amniotic fluid or chorionic villus, <15 cells, 1 karyotype with banding (if appropriate)

CHRCB 35248 Chromosome Analysis, Congenital Disorders, Blood

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Green top (sodium heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Other anticoagulants are not recommended and are harmful to the viability of the cells.

3. Label specimen as whole blood.

Specimen Type: Cord blood

Container/Tube: Green top (sodium heparin)

Specimen Volume: As much as possible

Collection Instructions:

1. Invert several times to mix blood.

2. Other anticoagulants are not recommended and are harmful to the viability of the cells.

3. Label specimen as cord blood.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88230, 88291- Tissue culture for Lymphocytes, Interpretation and report

88262 w/modifier 52-Chromosome analysis less than15 cells (if appropriate)

88262-Chromosome analysis with 15 to 20 cells (if appropriate)

88262, 88285-Chromosome analysis with greater than 20 cells (if appropriate)

88280-Chromosome analysis, greater than 2 karyotypes (if appropriate)

88283-Additional specialized banding technique (if appropriate)

CHRHB 35308 Chromosome Analysis, Hematologic Disorders, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not** aliquot.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole blood	Ambient (preferred)
	Refrigerated

CPT Code Information:

88237, 88291-Tissue culture for neoplastic disorders; bone marrow, blood, Interpretation and report

88264 w/ modifier 52-Chromosome analysis with less than 20 cells (if appropriate)

88264-Chromosome analysis with 20 to 25 cells (if appropriate)

88264,88285-Chromosome analysis with greater than 25 cells (if appropriate)

88283-Additional specialized banding technique (if appropriate)

CHRBM 35245 Chromosome Analysis, Hematologic Disorders, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88237, 88291-Tissue culture for neoplastic disorders; bone marrow, blood, Interpretation and report

88264 w/ modifier 52-Chromosome analysis with less than 20 cells (if appropriate)

88264-Chromosome analysis with 20 to 25 cells (if appropriate)

88264, 88285-Chromosome analysis with greater than 25 cells (if appropriate)

88283-Additional specialized banding technique (if appropriate)

COGBL 113532 Chromosome Analysis, Hematologic Disorders, Children's Oncology Group Enrollment Testing, Blood

Specimen Requirements:

Specimen Type: Blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Invert several times to mix blood.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88237, 88291-Tissue culture for neoplastic disorders; bone marrow, blood, Interpretation and report

88264 w/ modifier 52-Chromosome analysis with less than 20 cells (if appropriate)

88264-Chromosome analysis with 20 to 25 cells (if appropriate)

88264, 88285-Chromosome analysis with greater than 25 cells (if appropriate)

88283-Additional specialized banding technique (if appropriate)

COGBM 113531 Chromosome Analysis, Hematologic Disorders, Children's Oncology Group Enrollment Testing, Bone Marrow

Specimen Requirements:

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88237, 88291-Tissue culture for neoplastic disorders; bone marrow, blood, Interpretation and report

88264 w/ modifier 52-Chromosome analysis with less than 20 cells (if appropriate)

88264-Chromosome analysis with 20 to 25 cells (if appropriate)

88264, 88285-Chromosome analysis with greater than 25 cells (if appropriate)

88283-Additional specialized banding technique (if appropriate)

CHFXH Chromosome Analysis, Hematologic Disorders, Fixed Cells
64922

Specimen Requirements:

Specimen Volume: 2 mL

Additional Information: Advise Express Mail or equivalent if not on courier service.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88291- Interpretation and report

88264 w/ modifier 52-Chromosome analysis with less than 20 cells (if appropriate)

88264-Chromosome analysis with 20 to 25 cells (if appropriate)

88264, 88285-Chromosome analysis with greater than 25 cells (if appropriate)

88283-Additional specialized banding technique (if appropriate)

CHRTI
35250

Chromosome Analysis, Skin Biopsy

Specimen Requirements:

Specimen Type: Skin biopsy

Container/Tube: Sterile container with sterile RPMI transport media, Ringer's solution, or normal saline-RPMI transport media (T095-Petri dish is not needed for this test).

Specimen Volume: 4 mm diameter

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. Do not use alcohol or iodine preparations.

4. A local anesthetic may be used.

5. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Minimum Volume: 4 mm punch biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88233, 88291- Tissue culture for skin/biopsy, Interpretation and report

88262 w/modifier 52-Chromosome analysis less than 15 cells(if appropriate)

88262-Chromosome analysis with 15 to 120 cells (if appropriate)

88262, 88285-Chromosome analysis with greater than 20 cells (if appropriate)

88280-Chromosome analysis, greater than 2 karyotypes (if appropriate)

88283-Additional specialized banding technique (if appropriate)

EOSFD 622440 Chronic Eosinophilia Panel, Diagnostic FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole Blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x12, 88275x6, 88291x1-FISH Probe, Analysis, Interpretation; 6 probe sets

88271x2, 88275x1–FISH Probe, Analysis; each additional probe set (if appropriate)

EOSMF 614254 Chronic Eosinophilia, Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2-3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x2, 88275 x1, 88291 x1-FISH Probe, Analysis, Interpretation; 1 probe sets

88271 x2, 88275 x1-FISH Probe, Analysis; each additional probe set (if appropriate)

88271 x1-FISH Probe; coverage for sets containing 3 probes (if appropriate)

CHSBP 9023 Chronic Hepatitis B Monitoring Profile, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection**,** patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Transfer serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86707

87340

87350

87341 (if appropriate)

CHBVS 615268 Chronic Hepatitis B Screen, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

Specimen Minimum Volume: Serum: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

87340

G0499-(if appropriate)

87350 (if appropriate)

87341 (if appropriate)

86707 (if appropriate)

CIDP
616443

Chronic Inflammatory Demyelinating Polyradiculoneuropathy/Nodopathy Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended before initiation of immunosuppressant medication or intravenous immunoglobulin treatment.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 2

CIDPI 616444 Chronic Inflammatory Demyelinating Polyradiculoneuropathy/Nodopathy Interpretation, Serum (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CLLMD 608972 Chronic Lymphocytic Leukemia (CLL) Monitoring Measurable/Minimal Residual Disease Detection, Flow Cytometry, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 9-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88188-Flow Cytometry Interpretation, 9 to 15 markers

CLLDF
610713

Chronic Lymphocytic Leukemia, Diagnostic FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Specimen Minimum Volume: Whole blood: 2 mL; Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x12, 88275x6, 88291-FISH Probe, Analysis, Interpretation; 6 probe sets

88271x2, 88275-FISH Probe, Analysis; each additional probe set (if appropriate)

CLLMF 610724 Chronic Lymphocytic Leukemia, Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Specimen Minimum Volume: Whole blood: 2 mL; Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275-FISH Probe, Analysis; each additional probe set (if appropriate)

Chronic Viral Hepatitis (Unknown Type), Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: 1.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

86704

86706

86803

87340

87341 (if appropriate)

87522 (if appropriate)

CHUB
82822

Chub Mackerel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CSU
606921

Chyluria Screen, Random, Urine

Specimen Requirements:

Patient Preparation: Before specimen collection, the patient **should not** ingest foods or beverages rich in vitamin C or take vitamin C supplements.

Supplies: Urine Container, 60 mL (T313)

Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 15 mL

Collection Instructions: Collect a **first-morning**, random urine collection.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	10 days
	Refrigerated	10 days
	Ambient	4 hours

CPT Code Information:

82664-Electrophoretic technique, not elsewhere specified (Chylomicrons and lipoproteins)

84311-Spectrophotometry, analyte not specified (Cholesterol)

84478-Triglycerides

FCHYS 57806 Chymotrypsin, Stool

Specimen Requirements:

Specimen Type: Random stool

Container/Tube: Sterile leak proof stool container

Specimen Volume: 1 g

Collection Instructions: Collect 1 gm random stool in sterile leak proof container, ship refrigerate.

Note:

Dietary restrictions: Patients receiving pancreatic enzymes should discontinue taking the enzymes at least 5 days before the collection of the stool sample.

Specimen Minimum Volume: 0.5 gram

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

84311

FCING 57676 Cinnamon IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CINN
82624

Cinnamon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CINPC
618275

CINtec IHC Multiplex, PC (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

CPT Code Information:

88344

FCICP
75925

Circulating Immune Complex Panel

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Draw blood in plain, red-top tube(s) or serum gel tube(s).

2. Allow complete clotting of red blood cells (up to 1 hour)

3. Within 30 minutes of clotting completion, centrifuge and aliquot 2 mL into a plastic vial.

3. Freeze immediately.

4. Ship frozen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86332 x 2

FCQBA
57301

Circulating Immune Complex, C1q Binding

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Allow complete clotting of red blood cells (up to 1 hour).

2. Within 30 minutes after clotting is complete, centrifuge and aliquot serum into a plastic vial.

3. Freeze immediately.

4. Send frozen.

Note: Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86332

FCIC3
75772

Circulating Immune Complex, C3 fragments

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Allow complete clotting of red blood cells (up to 1 hour).

2. Within 30 minutes after clotting is complete, centrifuge and aliquot serum into a plastic vial.

3. Freeze immediately.

4. Send frozen

Note: Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	30 days

CPT Code Information:

86332

FCIC
91497

Circulating Immune Complexes (CIC)

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	14 days

CPT Code Information:

86332

CITR2
606637

Citrate Concentration, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CITRA / Citrate Excretion, Random, Urine.

Patient Preparation: Any drug that causes alkalemia or acidemia may be expected to alter citrate excretion and should be avoided, if possible. The patient must avoid laxative use for24 hours prior to collection.

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

82507

CITR
606710

Citrate Excretion, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Any drug that causes alkalemia or acidemia may be expected to alter citrate excretion and should be avoided, if possible. The patient must avoid laxative use for 24 hour collection period.

Supplies:

-Diazolidinyl Urea (Germall) 5.0 mL (T822)

-Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic tube

Specimen Volume: 4 mL

Collection Instructions:

1. Add 5 mL of diazolidinyl urea (Germall) as preservative at start of collection **or** refrigerate specimen during and after collection.

2. Collect urine for 24 hours.

3. Mix well before taking 4-mL aliquot.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

82507

CITRA
606715

Citrate Excretion, Random, Urine

Specimen Requirements:

Patient Preparation: Any drug that causes alkalemia or acidemia may be expected to alter citrate excretion and should be avoided, if possible. The patient must avoid laxative use for 24 hours prior to collection.

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

82507

82570

RAT10
606642

Citrate/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CITRA / Citrate Excretion, Random, Urine.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

82507

CLAD
82912

Cladosporium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CLAM
82884

Clam, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CLD18 620665 Claudin 18 (CLDN18) (43-14A), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 50 tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

CLAUD 70403 Claudin-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

Claudin-4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FCLCN
75262

CLCN1 DNA Sequencing Test

Specimen Requirements:

Note: Collection date and informed consent are required.

Specimen Type: Whole blood

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 8 mL

Collection Instructions:

1. Invert several times to mix.

2. Send ambient.

Specimen Minimum Volume: 6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

81406

CLIR
605136

CLIR Supplemental Report

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CLOBZ
65483

Clobazam and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect blood immediately before next scheduled dose.

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial.

Additional Information: Trough specimens are recommended as therapeutic ranges are based on specimens collected at trough (ie, immediately before the next dose).

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

G0480

80339 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

CLOM
80902

Clomipramine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

CZPS
65044

Clonazepam and 7-Aminoclonazepam, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection, the specimen must be centrifuged and the serum aliquoted into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80299

FCLMS
75524

Clonazepam and Metabolite, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Serum

Collection Container/Tube: Red top

Submission Container/Tube: Plastic, preservative-free vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

80299

FCLON 91107 Clonidine (Catapres)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 5 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

Clostridioides difficile Culture with Antimicrobial Susceptibilities, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Patient Preparation: Patient should **not** use antacids, barium, bismuth, antidiarrheal medication, zinc oxide paste, Vagisil cream or oily laxatives prior to specimen collection.

Preferred:

Specimen Type: Preserved feces

Supplies:

-Culture and Sensitivity Stool Transport Vial (T058)

-Stool Collection Kit, Random (T635)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S [T058])

Specimen Volume: Representative portion of feces; 5 mL

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

Specimen Stability Information: Ambient (preferred) 96 hours/Refrigerated 96 hours/Frozen 7 days

Additional Information: Only diarrheal (ie., unformed) feces should be tested. Testing formed feces for *Clostridioides difficile* is not clinically indicated.

Acceptable:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Stool container

Specimen Volume: Representative portion of stool

Collection Instructions: Collect fresh stool and submit representative sample in stool container.

Specimen Stability Information: Ambient (preferred) 72 hours/Frozen 7 days

Additional Information: Only diarrheal (i.e., unformed) stool should be tested. Testing formed stool for *Clostridioides difficile* is not clinically indicated.

Specimen Type: Fresh tissue or biopsy

Sources: Colon

Supplies: Anaerobe Transport Tube (T588)

Specimen Volume: Entire collection, 1 to 2 cm(3)

Collection Instructions: Aseptically collect 1 to 2 cm(3) piece of tissue whenever possible. In general, a larger piece of tissue is preferred. Submit in an anaerobic transport tube.

Specimen Stability Information: Ambient 72 hours

Specimen Minimum Volume: Stool: 1 gram or 5 mL; Tissue: 5 mm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87081-*C. difficile* Culture

87076-Anaerobe Ident (if appropriate)

87076-Id MALDI-TOF Mass Spec Anaerobe (if appropriate)

87153-Anaerobe Ident by Sequencing (if appropriate)

87181-Anaerobe Susceptibility per agent (if appropriate)

87181 x 3-Antimicrobial Susceptibility, Anaerobic Bacteria, MIC (if appropriate)

CDIF 64354 Clostridioides difficile Culture, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Patient Preparation: Patient should **not** use antacids, barium, bismuth, antidiarrheal medication, zinc oxide paste, Vagisil cream or oily laxatives prior to specimen collection.

Preferred:

Specimen Type: Preserved feces

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of feces; 5 mL

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

Specimen Stability Information: Ambient (preferred) 96 hours/Refrigerated 96 hours/Frozen 7 days

Additional Information: Only diarrheal (ie, unformed) feces should be tested. Testing formed feces for *Clostridioides difficile* is generally not clinically indicated.

Acceptable:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Stool container

Specimen Volume: Representative portion of feces

Collection Instructions: Collect fresh feces and submit representative sample in stool container.

Specimen Stability Information: Ambient (preferred) 72 hours/Frozen 7 days

Additional Information: Only diarrheal (ie, unformed) feces should be tested. Testing formed feces for *Clostridioides difficile* is generally not clinically indicated.

Specimen Type: Fresh tissue or biopsy

Sources: Colon

Supplies: Anaerobe Transport Tube (T588)

Specimen Volume: Entire collection, 1 to 2 cm(3)

Collection Instructions: Aseptically collect 1 to 2 cm(3) piece of tissue whenever possible. In general, a larger piece of tissue is preferred. Submit in an anaerobic transport tube.

Specimen Stability Information: Ambient 72 hours

Specimen Minimum Volume: Stool: 1 gram or 5 mL; Tissue: 5 mm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87081-*C. difficile* Culture

87076-Anaerobe Ident (if appropriate)

87076-Id MALDI-TOF Mass Spec Anaerobe (if appropriate)

87153-Anaerobe Ident by Sequencing (if appropriate)

EIACD 622780 Clostridioides difficile Toxin, EIA, Feces

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Preserved feces

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of nonnutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of feces; 5 mL

Collection Instructions:

1. Collect fresh fecal specimen and submit in container with transport medium.

2. Within 2 hours of collection place feces in preservative.

Specimen Stability Information: Ambient (preferred) <7 days/Refrigerated <7 days

Acceptable:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: Representative portion of feces

Collection Instructions: Collect fresh fecal specimen and submit representative sample in fecal container.

Specimen Stability Information: Refrigerated (preferred) <7 days/Frozen <7 days

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies	7 days

CPT Code Information:

87324

CDPCR Clostridioides difficile Toxin, PCR, Feces

618554

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Clostridioides difficile* toxin DNA is unlikely.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Preserved feces

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of nonnutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of feces; 5 mL

Collection Instructions:

1. Collect fresh fecal specimen and submit in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

Specimen Stability Information: Ambient (preferred) <5 days/Refrigerated <5 days

Acceptable:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz Random (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: Representative portion of feces

Collection Instructions: Collect fresh fecal specimen and submit representative sample in fecal container.

Specimen Stability Information: Refrigerated (preferred) <5 days/Frozen <5 days

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies	5 days

CPT Code Information:

87493

CLOV
82490

Clove, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CLZ
42366

Clozapine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Additional Information: Therapeutic range (trough level) applies to specimens collected immediately prior to next dose.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80159

CLUS
70404

Clusterin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information:

Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FCMVQ
91734

CMV by PCR

Specimen Requirements:

1 mL amniotic fluid shipped frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Amniotic Fld	Frozen	180 days

CPT Code Information:

87497

FDMZ
57859

CNBP DNA Test (DM2)

Specimen Requirements:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA)

Specimen volume: 8 mL

Collection instructions:

1. Draw blood in a lavender top (EDTA) tube(s).

2. Invert several times to mix.

3. Send ambient.

Note: **Collection date is required.**

Specimen Minimum Volume: 6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

81187

CDS1
65565

CNS Demyelinating Disease Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended before initiation of immunosuppressant medication.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

	Ambient	72 hours

CPT Code Information:

86053

86363

86053-Titer (if appropriate)

86363-Titer (if appropriate)

F_2 9121 Coagulation Factor II Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Patient Preparation: Patient **must not** be receiving coumadin (warfarin) or heparin therapy. (If not possible for medical reasons, note on request.)

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

5. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85210

F_9
9065

Coagulation Factor IX Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or ideally, at or below -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85250

FACTV 9054 Coagulation Factor V Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Specimen must be collected prior to initiation of anticoagulants and thrombolytic therapy.

2. Patient **must not** be receiving warfarin, heparin, direct thrombin inhibitors (argatroban, dabigatran), or direct factor Xa inhibitors (apixaban, rivaroxaban, and edoxaban).

a. If medically feasible, for 4 to 6 hours before specimen collection, do not administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, do not administer subcutaneous heparin or warfarin.

3. It is best to collect the specimen pretransfusion if possible. If patient has been recently transfused, wait at least 48 hours after transfusion to collect the specimen.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a separate plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85220

F_7
9055

Coagulation Factor VII Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Specimen **must be** collected prior to initiation of anticoagulants and thrombolytic therapy.

2. Patient **must not** be receiving warfarin, heparin, direct thrombin inhibitors (argatroban,

dabigatran), or direct factor Xa inhibitors (apixaban, rivaroxaban, and edoxaban).

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

3. It is best to collect the specimen pretransfusion if possible. If patient has been recently transfused, wait at least 48 hours after transfusion to collect the specimen.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a separate plastic vial, leaving 0.25 mL in the bottom of the centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

F8A 9070 Coagulation Factor VIII Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Within 4 hours of specimen collection, centrifuge,, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a 13-mm x 75-mm, 5-mL, polypropylene vial leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately (within 2 hours of processing) freeze plasma at -20 degrees C or ideally, at or below -40 degrees C.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85240

F8IS 7289 Coagulation Factor VIII Inhibitor Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

8INHE / Factor VIII Inhibitor Evaluation, Plasma

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

APROL / Prolonged Clot Time Profile, Plasma

AVWPR / von Willebrand Disease Profile, Plasma

For more information see Coagulation Guidelines for Specimen Handling and Processing.

Specimen Minimum Volume: Platelet-poor plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

F_10 9066 Coagulation Factor X Activity Assay, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Within 4 hours of collection, centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a separate plastic vial, leaving 0.25 mL in the bottom of the centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85260

FXCH
89042

Coagulation Factor X Chromogenic Activity Assay, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a separate plastic vial leaving 0.25 mL in the bottom of the centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

3. If priority specimen, mark request form, give reason, and request a call-back.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85260

F_11 9067 Coagulation Factor XI Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or ideally, at or below -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85270

F_12 9069 Coagulation Factor XII Activity Assay, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or ideally, at or below -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85280

13ACT 623415 Coagulation Factor XIII (13) Activity Assay, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. It is best to perform this study pre-plasma transfusion if possible. If patient has been recently transfused, wait at least 48 hours after transfusion before collecting the specimen.

3. Specimen should be collected prior to factor replacement therapy.

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	180 days

CPT Code Information:

85290

CMMPP 606103 Cobalamin, Methionine, and Methylmalonic Acid Pathways, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Immediately place specimen on ice.

2. Within 4 hours of collection, centrifuge and aliquot plasma into plastic vial. **Note:** If blood cannot be placed on wet ice immediately, centrifuge and aliquot plasma into a plastic vial within 1 hour of collection.

3. A refrigerated centrifuge is not required if the above time restrictions are met.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Refrigerated (preferred)	28 days
	Frozen	309 days
	Ambient	28 days

CPT Code Information:

83090

83918

82136

82542 (if appropriate for government payers)

CMMPS 606111 Cobalamin, Methionine, and Methylmalonic Acid Pathways, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 4 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	309 days
	Ambient	28 days

CPT Code Information:

83090

83918

82136

82542 (if appropriate for government payers)

COUO 607762 Cobalt Occupational Exposure, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic 10 mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. At the end of the work week, collect a random urine specimen at the end of the employee's work shift.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83018

82570

COU
80083

Cobalt, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

COWB
60355

Cobalt, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Metal Free EDTA 3 mL Tube (T989)

Container/Tube: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (T989) (BD catalog no. 367777)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

COS
80084

Cobalt, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Metal Free Specimen Vial (T173)

Collection Container/Tube: Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 1.6 mL Serum

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

83018

COSY
606352

Cobalt, Synovial Fluid

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (T989) (BD catalog no. 367777)

Acceptable: Royal blue-top BD vacutainer with EDTA blood collection tube (6mL) (BD Catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: Cobalt is present in the black rubber plunger seals found in most disposable syringes. As a result, synovial fluid **should not be** collected in these devices as contamination may occur.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Synovial Fluid	Refrigerated (preferred)	90 days
	Ambient	90 days
	Frozen	90 days

CPT Code Information:

83018

COBRU 607760 Cobalt/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83018

82570

COKMX 62720 Cocaine and Metabolite Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain of Custody Meconium Kit (T653) includes the specimen containers, seals, and documentation required.

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Additional Information:

1. Specimen that arrives with a broken seal does not meet the chain of custody requirements.

2. The laboratory recommends sending chain-of-custody specimens by overnight shipment.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	21 days
	Refrigerated	21 days
	Ambient	72 hours

CPT Code Information:

G0480

80353 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

COKEX 62719 Cocaine and Metabolite Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation required.

Specimen Volume: 10 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 10 mL will compromise the ability to

perform all necessary testing.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80353 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

COKEU 9286 Cocaine and Metabolite Confirmation, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not** accepted for this test.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80353 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

COKEM 84140 Cocaine and Metabolites Confirmation, Meconium

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	21 days
	Refrigerated	21 days
	Ambient	72 hours

CPT Code Information:

G0480

80353 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

COKEP 624104 Cocaine and Metabolites, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Gray top (sodium fluoride/potassium oxalate plasma) (PST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Within 2 hours of collection, centrifuge at 4 degrees C and aliquot plasma into a plastic vial.

2. Freeze the specimen.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	28 days
	Refrigerated	6 days

CPT Code Information:

G0480

80353 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FCDU6 75781 Cocaine and Metabolites, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

RSCOC 35928 Coccidioides Antibody Reflex, Complement Fixation and Immunodiffusion, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see COXIS / *Coccidioides* Antibody Screen with Reflex, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86635 x 3

COXIS
62079

Coccidioides Antibody Screen with Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86635

86635 x3 (if appropriate)

SCOC 8295 Coccidioides Antibody, Complement Fixation and Immunodiffusion, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86635 x 3

CCOC 81542 Coccidioides Antibody, Complement Fixation and

Immunodiffusion, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 2 mL

Collection Instructions: Submit specimen from collection vial number 2 (preferred), vials 3, or 4 are acceptable.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86635 x 3

CIMT
62204

Coccidioides immitis/posadasii, Molecular Detection, PCR, Paraffin, Tissue

Specimen Requirements:

Preferred Paraffin-embedded tissue block:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Body tissue

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Acceptable Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Body tissue

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87798

CIMRP 88804 Coccidioides immitis/posadasii, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction (PCR) requires the specimen to be processed in an environment in which contamination of the specimen by *Coccidioides* species DNA is unlikely.

Preferred Specimens: Body fluid, cerebrospinal fluid (CSF), ocular fluid, respiratory (eg, bronchoalveolar lavage [BAL], bronchial washing, sputum), fresh tissue, or bone

Acceptable Specimens: If no fresh specimen is available, digested respiratory specimens treated with N-acetyl-L-cysteine-sodium hydroxide (NALC/NaOH) are acceptable (eg, BAL, bronchial washing, respiratory fluid, sputum, or tracheal secretion)

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Body, ocular, or CSF

Container/Tube: Sterile container

Specimen Volume: 1 mL

Additional Information: Only fresh, non-NALC/NaOH-digested body fluid is acceptable.

Specimen Type: Respiratory

Sources: BAL, bronchial washing, or sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL if only PCR ordered or 3 mL if PCR ordered with smear and culture

Specimen Type: Tissue

Sources: Fresh tissue or bone

Container/Tube: Sterile container

Specimen Volume: 5 to 10 mm

Collection Instructions: Keep moist with sterile water or sterile saline

Additional Information: Only fresh, non-NALC/NaOH-digested tissue is acceptable.

Acceptable

Specimen Type: NALC/NaOH-digested respiratory specimens

Sources: BAL, bronchial washing, respiratory fluid, sputum, or tracheal secretion

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Minimum Volume: Body fluid or nondigested respiratory specimen: 0.5 mL; Fresh tissue or

bone: 5 mm; NALC-NaOH-digested specimen: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

FCKTF 57966 Cockatiel Feathers IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

CBUR 82802 Cocklebur, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCAIG
75561

Cockroach American (Periplaneta americana) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

COCR
82693

Cockroach, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCOCN
57668

Coconut IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CCNT
82739

Coconut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CODES
623449

Codeine and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80361 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

COD
82889

Codfish, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

Q10
87853

Coenzyme Q10, Reduced and Total, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (lithium or sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Immediately after collection, place specimen on wet ice. Maintain on wet ice until processing.

2. Within 3 hours of collection, centrifuge, aliquot plasma into a plastic vial, and freeze immediately.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Heparin	Frozen	7 days

CPT Code Information:

82542

TQ10
63148

Coenzyme Q10, Total, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (lithium or sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Immediately after collection, place specimen on wet ice. Maintain on wet ice until processing.

2. Within 3 hours of collection, centrifuge, aliquot plasma into a plastic vial, and freeze immediately.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Heparin	Frozen (preferred)	60 days
	Refrigerated	7 days
	Ambient	4 days

CPT Code Information:

82542

FCOFE 57525 Coffee (Coffea spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

CATTR
614034

Cold Agglutinin Titer, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL

Pediatric Volume: 1 mL

Collections Instructions:

1. Use a warm pack to keep specimen at 37 degrees C prior to and after collecting.

2. Allow specimen to clot at 37 degrees C.

3. Centrifuge at 37 degrees C and aliquot serum into plastic vial immediately after blood clots or within one hour of collection.

4. Do not refrigerate prior to separation of serum from red cells.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	72 hours

CPT Code Information:

86157

COLIV
70408

Collagen IV Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25 x 75 x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FFTYC
91496

Collagen Type II antibody

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in plain, red-top tube(s) or serum-gel tube(s).

2. Centrifuge and aliquot serum into a plastic vial.

3. Send refrigerated.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

83520

MSCG 615298 Collagenofibrotic Glomerulopathy Confirmation, Mass Spectrometry

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Formalin-fixed, paraffin-embedded kidney tissue block

Collection Instructions: Do not send fixed tissue slides. Testing can only be done on paraffin-embedded tissue blocks.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

82542

88380

CRMTS
43436

Collapsin Response-Mediator Protein-5 (CRMP-5) Neuronal IgG Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-PAVAL / Paraneoplastic, Autoantibody Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

-PVLE / Paraneoplastic Vision Loss Evaluation, Serum

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

CRMTC
43445

Collapsin Response-Mediator Protein-5 (CRMP-5) Neuronal IgG Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

CRMWS 83107 Collapsin Response-Mediator Protein-5-IgG, Western Blot, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

CRMWC 21747 Collapsin Response-Mediator Protein-5-IgG, Western Blot, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 3 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

COMID 621533 Combined Humoral and Cell-Mediated Immunodeficiency Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When

applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions.

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

CMITO 617103 Combined Mitochondrial Full Genome and Nuclear Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777) 5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81460

81440

81465

FHEXN
75955

Comment (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see FHEX / Hexagonal Phase Phospholipid (HPP).

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	28 days

FHEXP
75956

Comment (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see FHEX / Hexagonal Phase Phospholipid (HPP).

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	28 days

CMIL
82833

Common Millet, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL

For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

REED
82902

Common Reed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CO4D
70407

Complement 4d Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

C1Q
8851

Complement C1q, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

86160

C341Q
621739

Complement C3 and C4 with Anti-C1q Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	21 days

CPT Code Information:

86160 x2

83520

C3
8174

Complement C3, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86160

C4
8171

Complement C4, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86160

COM
8167

Complement, Total, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into 5 mL plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Frozen	28 days

CPT Code Information:

86162

CBC
9109

Complete Blood Cell Count (CBC) with Differential, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	48 hours
	Ambient	24 hours

CPT Code Information:

85025

85007 (if appropriate)

85060 (if appropriate)

CACMG
617141

Comprehensive Arrhythmia and Cardiomyopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81439

CARGG 617169 Comprehensive Arrhythmia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81413

CCMGG 617183 Comprehensive Cardiomyopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81439

CVHBG 617225 Comprehensive Cerebrovascular Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405 x5

81406 x3

81408

81479

81479 (if appropriate for government payers)

DWPAN 617545 Comprehensive Distal Weakness Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

FCDSU 75775 Comprehensive Drug Screen, Umbilical Cord Tissue

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Umbilical cord tissue

Container/Tube: Plastic, preservative-free container

Specimen Volume: 10 g

Collection Instructions:

1. Collect at least 6 inches of umbilical cord (approximately the length of an adult hand).
2. Drain and discard any blood.
3. Rinse the exterior of the cord segment with normal saline or sterile water.
4. Pat the cord dry and place in a preservative-free plastic container for transport.
5. Ship refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

EPPAN 616550 Comprehensive Epilepsy With or Without Encephalopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.
2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81419

CAORG 617155 Comprehensive Marfan, Loeys-Dietz, Ehlers-Danlos, and Aortopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to

determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81410

CMAMA 113631 Comprehensive Metabolic Panel, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged and protected from light within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into an amber vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	24 hours

CPT Code Information:

KS-84132

NAS-84295

CL-82435

HCO3-82374

BUN-84520

CRTS1-82565

CA-82310

GLURA-82947

TP-84155

ALB-82040

AST-84450

ALP-84075

ALT-84460

BILIT-82247

NGSMC 622766 Comprehensive Myeloid Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD-B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow with anticoagulant.

2. Send bone marrow in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood with anticoagulant.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Minimum Volume: Whole blood: 2 mL; Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	5 days

CPT Code Information:

81455

NEPHP
618086

Comprehensive Nephrology Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81401 x 2

81404 x 12

81405 x 8

81406 x 22

81407 x 13

81408 x 5

81479

81479 (if appropriate for government payers)

MUPAN 617662 Comprehensive Neuromuscular Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic

donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81443

PEPAN 617688 Comprehensive Peripheral Neuropathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81448

TBT
80667

Concentration, Mycobacteria (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87015

FFCAH
75305

Congenital Adrenal Hyperplasia (CAH) Pediatric Profile 6, Comprehensive Screen

Specimen Requirements:

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3.5 mL Serum

Collection Instructions:

1. Within 45 minutes of collection, centrifuge and aliquot serum into a plastic vial.

2. Send frozen.

Additional Information: To avoid delays in turnaround time when requesting multiple

tests, **submit separate frozen specimens for each test requested**.

Specimen Minimum Volume: Serum: 2 mL Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	90 days

CPT Code Information:

82633 - DOC

82634 - 11-Desoxycortisol

82157 - Androstenedione

82533 - Cortisol

82626 - DHEA

84143 - 17-OH-Pregnenolone

84144 - Progesterone

83498 - 17-OH-Progesterone

84403 - Testosterone

CAH21 87815 Congenital Adrenal Hyperplasia (CAH) Profile for 21-Hydroxylase Deficiency, Serum

Specimen Requirements:

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions:

1. Morning (8 a.m.) and afternoon (4 p.m.) specimens are preferred.

2. Include time of collection.

3. Centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are collected, send a separate order for each specimen.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82157

82533

83498

CAH2T
42202

Congenital Adrenal Hyperplasia Newborn Screen, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Local newborn screening card, Whatman 903 filter paper, PerkinElmer 226 filter paper, Munktell filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Do not use device or capillary tube containing EDTA to collect specimen.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	90 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Refrigerated	90 days	FILTER PAPER

CPT Code Information:

82542

CCMVS 620659 Congenital Cytomegalovirus (cCMV), Molecular Detection, PCR, Saliva

Specimen Requirements:

Supplies: BD UVT with mini tip flocked swab (T971)

Collection Container/Tube: Flocked swab

Submission Container/Tube: Sterile, screw-capped, tube containing viral transport media with flocked swab containing saliva. See Collection Instructions for acceptable media.

Specimen Volume: 1 Swab in viral transport media

Collection Instructions:

1. Collect a saliva specimen using a flocked swab per swab manufacturer instructions.

2. Place in a sterile, screw capped tube containing BD Universal Viral Transport Medium (UVT) (1 mL or 3 mL), Copan Universal Transport Medium (UTM) (1 mL or 3 mL), Remel M4-RT, Remel M4, or Remel M6.

Specimen Minimum Volume: Saliva swab submitted in minimum volume of 0.3 mL of viral transport media.

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Frozen	7 days

CPT Code Information:

87496

CCMVU 620658 Congenital Cytomegalovirus (cCMV), Molecular Detection, PCR, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic sterile container, 5 mL

Specimen Volume: 0.2 mL

Collection Instruction:

1. Collect a random urine from a urine bag, container, or catheter.

2. Transfer urine to sterile container.

3. No preservative

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	72 hours

CPT Code Information:

87496

CDGGP 608010 Congenital Disorders of Glycosylation Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 14 days

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin (Eagle's minimum essential medium with 1% penicillin and streptomycin).

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblast

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 (formerly Ahlstrom 226) filter paper, or blood spot collection card

Specimen Volume: 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

2. For collection instructions, see Blood Spot Collection Instructions.

3. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

4. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

SALIVA:

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior

to collection.

Preferred: 1 OGD-675 Saliva Collection Container

Acceptable: 2 ORAcollect OCD-100 Swabs

Transport: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information:

1. Saliva is acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from saliva, it is possible that additional specimen may be required to complete testing.

Specimen Minimum Volume: Blood: 1 mL; Blood spots: 2 spots; Skin biopsy, cultured fibroblasts, or saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

CDGN
65485

Congenital Disorders of N-Glycosylation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.15 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	45 days
	Ambient	7 days

CPT Code Information:

83789

NCDA 619075 Congenital Dyserythropoietic Anemia Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

GNFIB 619159 Congenital Fibrinogen Disorders, FGA, FGB, and FGG Genes, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with whole blood testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

CHDGG 617197 Congenital Heart Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional information:

1. If amniotic fluid or nonconfluent cultures are received, CULAF / Culture for Genetic Testing, Amniotic Fluid will be added at an additional charge.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated

Additional Information:

1. If nonconfluent cultures are received, CULFB / Fibroblast Culture for Biochemical or Molecular Testing will be added at an additional charge.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information: All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

81405 x 3

81406 x 6

81407 x 3

81408

81479

81479 (if appropriate for government payers)

81265-Maternal cell contamination (if appropriate)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88235-Amniotic Fluid culture (if appropriate)

88240-Cryopreservation (if appropriate)

CLADP 608019 Congenital Lactic Acidosis Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

81460

81465

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

CTDC
83631

Connective Tissue Diseases Cascade, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86038

86200

83516-Centromere (if appropriate)

83516-Ribosome (if appropriate)

86225-ds-DNA AB IgG, Serum (if appropriate)

86235 x 6-RNP, Sm, SS-B, SS-A, Jo 1, and Scl 70 (if appropriate)

SLSUR 622224 Consult Slide Surcharge (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

CRHPC 113329 Consult, w/Comp Rvw of His (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88325

CSPPC
113327

Consult, w/Slide Prep (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88323

CUPPC
113328

Consult, w/USS Prof (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88323-26

CONCS
616442

Contactin-1 IgG Cell Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-CIDP / Chronic Inflammatory Demyelinating Polyradiculoneuropathy/Nodopathy Evaluation, Serum

-DMNES / Peripheral Nervous System Demyelinating Neuropathy, Autoimmune Evaluation, Serum

Patient Preparation: For optimal antibody detection, specimen collection is recommended to occur prior to starting immunosuppressant medication or intravenous immunoglobulin treatment (IVIg).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

CSMEU 615907 Controlled Substance Monitoring Enhanced Profile with Reflex, 21 Drug Classes, High Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine container

Specimen Volume: 30 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 30 mL in 1 plastic bottle.

3. No preservative.

Additional Information:

1. No specimen substitutions.

2. Submitting less than 30 mL may compromise the ability to perform all necessary testing.

3. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 20 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

80307

G0482

80347 (if appropriate for select payers)

80364 (if appropriate for select payers)

80326 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

CSMHU 615293 Controlled Substance Monitoring Hybrid Drug Profile, 20 Drug Classes, High-Resolution Mass Spectrometry and

Immunoassay Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 10 mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 5 mL in 1 plastic bottle.

3. No preservative

Additional Information:

1. No specimen substitutions.

2. Submitting less than 5 mL may compromise the ability to perform all necessary testing.

3. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

80307

G0482

80347 (if appropriate for select payers)

80364 (if appropriate for select payers)

80326 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

CSMOF 621483 Controlled Substance Monitoring Panel, 11 Drug Classes, Screen Only, Immunoassay, Oral Fluid

Specimen Requirements:

Patient Preparation:

1. If the patient has recently taken an oral medication or used an inhaled medication, wait 2 hours before collecting the specimen.

2. Patient must empty mouth of any gum, food, or tobacco prior to oral fluid collection. If patient's mouth is not empty immediately before collection, have them rinse their mouth with water (up to 4 oz) and wait 10 minutes before collection. The patient may discard or drink the water after rinsing.

Supplies: Quantisal Oral Fluid Collection Device (T980)

Note: Check expiration date on Quantisal packaging

Container/Tube: Quantisal collection device

Specimen Volume: 1 mL

Collection Instructions:

1. Review detailed instructions before specimen collection.

-Patient video: Oral fluid specimen collection for controlled substance monitoring.

-Practitioner video: Oral fluid specimen collection for controlled substance monitoring

-Printable instructions: Oral Fluid Specimen Collection Instructions for Controlled Substance Monitoring.

2. Peel open package and remove collector using its handle. Do not touch the collection pad with fingers before or after specimen collection. To expedite the collection process, move tongue side to side to accumulate saliva in mouth before starting. Keep the tip of the device pointed down.

3. Position collector under tongue and close mouth. Keep head down to allow gravity to help with saliva collection.

Important: Do not chew on pad, talk, or remove collector from mouth until indicator turns BLUE, or until 10 minutes has passed, whichever occurs first.

4. Hold transport tube in an upright position and uncap by pushing up with thumbs. Do not stand tube on table. Do not spill or empty the liquid from tube.

5. Insert collector into the uncapped transport tube and replace cap. Do not place collector back in mouth after it has been placed in the transport tube.

6. **Snap cap firmly for transport**. Place center of specimen seal on top of tube and press down both sides.

7. Complete paperwork and send sample to laboratory.

8. Send in original tube. **Do not aliquot.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fluid	Ambient (preferred)	7 days
	Frozen	28 days
	Refrigerated	7 days

CPT Code Information:

80307

CSMPU 610271 Controlled Substance Monitoring Profile, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine container

Specimen Volume: 30 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 30 mL in 1 plastic bottle.

3. No preservative.

Additional Information:

1. No specimen substitutions.

2. Submitting less than 30 mL may compromise the ability to perform all necessary testing.

3. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 20 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

80307

G0482

80347 (if appropriate for select payers)

80364 (if appropriate for select payers)

80326 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

CSMTU 615292 Controlled Substance Monitoring Targeted Profile, 17 Drug Classes, Mass Spectrometry, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 10 mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 5 mL in 1 plastic bottle.

3. No preservative

Additional Information:

1. No specimen substitutions.

2. Submitting less than 5 mL may compromise the ability to perform all necessary testing.

3. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

G0482

80347 (if appropriate for select payers)

80364 (if appropriate for select payers)

80326 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

CPAPD 70329 Conventional Smear-Diagnostic, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, or sexual intercourse for 24 hours prior to specimen collection.

Container/Tube: Slide

Specimen Volume: Circular scrape of cervical os

Collection Instructions:

1. Specimen containers must be labeled with a minimum of 2 unique identifiers (patient's name, and medical record number or date of birth). Containers should also be labeled with specimen source, and date collected.

2. Glass slides may be labeled with a single unique identifier, but 2 identifiers are preferred. If multiple slides are submitted, each slide must have proper identification. Glass slides should be identified with the patient's name and a second patient identifier that is also on the accompanying paperwork (ie, medical record number or date of birth)

3. Fix slides immediately in 95% alcohol or treat with commercially available spray fixative.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		SLIDE
	Refrigerated		SLIDE

CPT Code Information:

P3000

88164

88141-CVSPC (if appropriate)

CPAPS 70331 Conventional Smear-Screen, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, or sexual intercourse for 24 hours prior to specimen collection.

Container/Tube: Slide

Specimen Volume: Circular scrape of cervical os

Collection Instructions:

1. Specimen containers must be labeled with a minimum of 2 unique identifiers (patient's name, and medical record number or date of birth). Containers should also be labeled with specimen source, and date collected.

2. Glass slides may be labeled with a single unique identifier, but 2 identifiers are preferred. If multiple slides are submitted, each slide must have proper identification. Glass slides should be identified with the patient's name and a second patient identifier that is also on the accompanying paperwork (ie, medical record number or date of birth)

3. Fix slides immediately in 95% alcohol or treat with commercially available spray fixative.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		SLIDE
	Refrigerated		SLIDE

CPT Code Information:

P3000

88164

88141-CVSPC (if appropriate)

CPAVP 603599 Copeptin proAVP, Plasma

Specimen Requirements:

Patient Preparation: For water-deprivation testing, for at least 8 hours, the patient should fast and thirst (no liquids, including water, are allowed).

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic screw-top vial

Specimen Volume: 0.5 mL Plasma

Collection Information: Centrifuge and aliquot plasma into a plastic vial. Do **not** submit in original tube.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84588

CUU 8590 Copper, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of barium are known to interfere with this test. If barium-containing contrast media has been administered, **the specimen should not be collected for at least 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82525

CUT
8687

Copper, Liver Tissue

Specimen Requirements:

Supplies: Metal Free Specimen Vial (T173)

Container/Tube:

Preferred: Metal-free specimen vial

Acceptable: Paraffin block with no more than 1 or 2 cuts previously made

Specimen Volume: 2 mg

Collection Instructions: Two mg of liver tissue is required. This is typically a piece of tissue from a 22-gauge needle biopsy at least 2 cm long. If an 18-gauge needle is used, the tissue must be at least 1 cm in length.

Specimen Stability Information:

Fresh or formalin-fixed liver tissue specimens: Frozen (-30 to -10 degrees C) at least 20 years

Paraffin-embedded (block) liver tissue specimens: Ambient (16 to 24 degrees C) at least

12 1/2 years

Additional Information: Paraffin blocks will be returned 7 days after analysis is complete.

Specimen Minimum Volume: Needle biopsy: See Specimen Required; 2 mm x 2 mm (punch): 0.3 mg by dry weight

Transport Temperature:

Specimen Type	Temperature	Time
Liver Tissue	Refrigerated (preferred)	
	Ambient	
	Frozen	

CPT Code Information:

82525

FCOPP 75391 Copper, RBCs

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: RBCs

Collection Container/Tube: Royal Blue top tube (Trace metal-free; EDTA)

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a royal blue top (trace metal free; EDTA) tube(s). Centrifuge and separate plasma within two hours of collection. Leave RBCs in the original collection container and replace stopper. Send 1 mL RBC's refrigerated.

NOTE: Tubes containing Heparin based anticoagulants are not acceptable.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

RBCS	Refrigerated	14 days	METAL FREE

CPT Code Information:

82525

CUS1
616155

Copper, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium, iodine, and barium are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium-, iodine-, and barium containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: 6-mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.8 mL serum

Collection Instructions:

1. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully **pour** specimen into metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not ream** the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

82525

CURCU 615257 Copper/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of barium are known to interfere with this test. If barium-containing contrast media has been administered, **the specimen should not be collected for at least 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82525

82570

CURC 615258 Copper/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CURCU /

Copper/Creatinine Ratio, Random, Urine.

Patient Preparation: High concentrations of barium are known to interfere with most metal tests. If barium-containing contrast media has been administered, the specimen should not be collected for at least 96 hours.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82525

CORI
82476

Coriander, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCORG 57526 Corn IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge down and aliquot 0.5 mL of serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FCOR4
57569

Corn IgG4

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CRNP
82718

Corn Pollen, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspac

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CORN
82705

Corn-Food, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CORTC
88221

Corticosterone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Morning (8 a.m.) specimen is preferred.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82528

SALCC 623604 Cortisol and Cortisone, Saliva

Specimen Requirements:

If multiple specimens are collected, submit each Salivette under a separate order number.

Patient Preparation:

1. Patient **should not** brush teeth, eat, drink, or take any oral medication for at least 60 minutes before specimen collection.

2. Have patient rinse mouth thoroughly with water for 1 to 5 seconds. Wait 10 minutes before collecting the specimen to avoid contamination of the saliva by an interfering substance.

Supplies: Cortisol, Saliva Collection Kit (T514)

Container/Tube: Sarstedt Salivette

Specimen Volume: 1.5 mL

Collection Instructions:

1. Provide patient with a Saliva Collection Kit (Salivette) containing the Cortisol and Cortisone - Saliva Collection Instructions and ask them to follow the instructions as written.

2. If performing a late-night salivary cortisol (LNSC) test, specimen should be collected immediately prior to going to bed. If patient is undergoing assessment for adrenal insufficiency, specimen should be collected immediately after waking up. The patient should follow their normal sleeping schedule. Record collection time on the Cortisol and Cortisone - Saliva Collection Instructions sheet.

3. Instruct patient to return Cortisol and Cortisone - Saliva Collection Instructions with the appropriately labeled Salivette to the laboratory.

Additional Information: Reference intervals are available for LNSC testing (immediately before going to bed) and immediately after waking for assessment of adrenal insufficiency.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Saliva	Refrigerated (preferred)	28 days
	Frozen	60 days
	Ambient	14 days

CPT Code Information:

82533 - Cortisol

82542 - Cortisone

CORTO
65484

Cortisol, Free and Total, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.85 mL

Collection Instructions:

1. Morning (8 a.m.) specimens are preferred. The 8 a.m. cortisol can be referred to as the a.m. cortisol and can be collected anywhere between 6 a.m. and 10:30 a.m. in the morning.

2. Centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are collected, send separate order for each specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82530

82533

CORTU 8546 Cortisol, Free, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10-mL (T068)

Submission Container/Tube: Plastic urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Add 10 g of boric acid as preservative at the start of the collection.

2. Collect urine for 24 hours and record the total volume.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82530

CORTF 65423 Cortisol, Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.25 mL

Collection Instructions:

1. Morning (8 a.m.) specimens are preferred. The 8 a.m. cortisol can be referred to as the a.m. cortisol and can be collected any time between 6 a.m. and 10:30 a.m. in the morning.

2. Centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are collected, send a separate order for each specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82530

CIVC 6347 Cortisol, Inferior Vena Cava, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Specimen Volume: 1.5 mL

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Morning (8 a.m.) and afternoon (4 p.m.) specimens are preferred.

2. Include time of collection.

3. Centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are collected, send a separate order for each specimen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82533

CLAV 6346 Cortisol, Left Adrenal Vein, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Specimen Volume: 1.5 mL Serum

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Morning (8 a.m.) and afternoon (4 p.m.) specimens are preferred.

2. Include time of collection

3. Centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are collected, send a separate order for each specimen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82533

CINP 9369 Cortisol, Mass Spectrometry, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Specimen Volume: 0.6 mL Serum

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Morning (8 a.m.) and afternoon (4 p.m.) specimens are preferred.

2. Include time of collection.

3. Centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are collected, send a separate order for each specimen.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82533

CRAV 6345 Cortisol, Right Adrenal Vein, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Specimen Volume: 1.5 mL Serum

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Morning (8 a.m.) and afternoon (4 p.m.) specimens are preferred.

2. Include time of collection.

3. Centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are collected, send a separate order for each specimen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82533

CORT 8545 Cortisol, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions:

1. Morning (8 a.m.) and afternoon (4 p.m.) specimens are preferred.

2.Within 2 hours of collection, centrifuge the specimen.

3. For red-top tubes aliquot the serum into a plastic vial after centrifugation.

Additional Information:

1. Include time of collection.

2. **If multiple specimens are collected, send separate order for each specimen**.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

82533

COCOU Cortisol/Cortisone, Free, 24 Hour, Urine

82948

Specimen Requirements:

Supplies: Urine Tubes, 10-mL (T068)

Submission Container/Tube: Plastic, urine tube (T068)

Specimen Volume: 5 mL

Collection Instructions:

1. Add 10 g of boric acid as preservative at the start of the collection.

2. Collect urine for a full 24 hours (required) and record the total volume.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

82530-Cortisol; free

82542

COTT
82859

Cotton Fiber, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CSED
82804

Cottonseed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CTWD
82748

Cottonwood, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

COW
82873

Cow Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CBBRP
62248

Coxiella burnetii (Q Fever), Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen

to be processed in an environment in which contamination of the specimen by *Coxiella burnetii* DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred).

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

CBSRP 62194 Coxiella burnetii (Q Fever), Molecular Detection, PCR, Serum

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Coxiella burnetii* DNA is unlikely.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a sterile vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

87798

CBRP 62193 Coxiella burnetii (Q Fever), Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Coxiella burnetii* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Fresh tissue or biopsy

Sources: Lung, bone, liver, heart valve, aorta, or endocardium

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3) - approximately the size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue only, do not add fluid to tissue

3. Refrigerate or freeze specimen.

Specimen Stability Information: Refrigerated (preferred) <7 days/ Frozen <7 days

Preferred Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Lung, bone, liver, heart valve, aorta, or endocardium

Supplies: Tissue Block Container (T553)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Lung, bone, liver, heart valve, aorta, or endocardium

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare 5 separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: Fresh tissue or biopsy: 5 mm(3)
Paraffin-embedded tissue block: two 10-micron sections

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87798

FCRAB 57674 Crab IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CRAB
82745

Crab, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CRANB
86307

Cranberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CRAY
82343

Crayfish, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CRDPP
606130

Creatine Disorders Panel, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin), yellow top (ACD)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into plastic vial. Send plasma frozen.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	14 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

82540

82565

82542

CRDPU Creatine Disorders Panel, Random, Urine

88697

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10 mL urine tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Immediately freeze urine specimen.

3. If possible, **do not** send other tests ordered on same vial of urine. In doing so, the other tests may have increased turnaround time due to the strict frozen criteria of this assay.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	29 days

CPT Code Information:

82540

82570

82542

CRDPS Creatine Disorders Panel, Serum

606131

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial. Send serum frozen.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

82540

82565

82542

CK
8336

Creatine Kinase (CK), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

82550

CRCL
615813

Creatinine Clearance, Serum and 24-Hour Urine

Specimen Requirements:

Both serum and urine are required. Serum must be collected no earlier than 72 hours before start of urine collection and no later than 72 hours after urine collection is completed.

Specimen Type: Serum

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot serum into plastic vial.

2. Label specimen as serum.

Specimen Type: Urine

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. Label specimen as urine.

Additional Information: For multiple collections see Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens

Specimen Minimum Volume: Serum: 0.5 mL
Urine: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	
	Frozen	
Urine	Refrigerated (preferred)	
	Ambient	
	Frozen	

CPT Code Information:

82575

CRTS1 48216 Creatinine with Estimated Glomerular Filtration Rate (eGFR), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

82565

CTU
610601

Creatinine, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information:

1. This test does not require the use of a chemical preservative; if a chemical preservative is used, it must be added to the specimen within 4 hours of completion of 24-hour collection.

2. See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

82570

CRT24 Creatinine, 24 Hour, Urine (Non-Orderable)

610597

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-NMH24 / *N*-Methylhistamine, 24 Hour, Urine

-RB24 / Retinol-Binding Protein, 24 Hour, Urine

-A124 / Alpha-1-Microglobulin, 24 Hour, Urine

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

CRT2F Creatinine, 24 Hour, Urine (Non-Orderable)

618161

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-TLTE4 / Leukotriene E4, 24 Hour, Urine

-MCM24 / Mast Cell Mediators, 24 Hour, Urine

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	30 days
	Refrigerated	30 days
	Ambient	14 days

CRBF Creatinine, Body Fluid

606601

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

-Peritoneal dialysate (dialysis fluid)

-Pericardial

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

82570

RCTUR 610603 Creatinine, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 6 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

82570

CRETR
610598

Creatinine, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see orderable test ID.

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days

	Ambient	14 days

CPT Code Information:

82570

CRTFR 618560 Creatinine, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see RLTE4 / Leukotriene E4, Random, Urine.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	30 days
	Refrigerated	30 days
	Ambient	14 days

CPT Code Information:

82570

CRE2 614427 Creatinine, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-ALBR / Albumin, Random, Urine

-RALB / Albumin, Random, Urine.

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

82570

CJDEI 620375 Creutzfeldt-Jakob Disease Evaluation Interpretation, Spinal Fluid (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Frozen (preferred)	28 days	BlueTop SARSTEDT
	Refrigerated	14 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

CJDE 620374 Creutzfeldt-Jakob Disease Evaluation, Spinal Fluid

Specimen Requirements:

Supplies: CJD/RPD Evaluation Kit (T966)

Container/Tube:

Preferred: 2 Sarstedt CSF False Bottom Tubes 63.614.625 (2.5 mL)

Acceptable: Sarstedt 72.703.600 (1.5 mL) or Sarstedt 72.694.600 (2 mL)

Specimen Volume: 2 tubes, each containing 1.5 mL to 2.5 mL

Collection Instructions:

1. Perform lumbar puncture and discard the first 1 to 2 mL of cerebrospinal fluid (CSF).

2. Collect two tubes of CSF directly into an acceptable collection tube until the tube is at least 50% full.

3. Send CSF specimen in original collection tube. **Do not aliquot.**

4. Collection instructions can also be found on Spinal Fluid Specimen Collection Instructions for Creutzfeldt-Jakob Disease and Rapidly Progressive Dementia Evaluations (T974).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Frozen (preferred)	28 days	BlueTop SARSTEDT
	Refrigerated	14 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

CPT Code Information:

0584U

84393

84394

CRGSP 83659 Cryoglobulin and Cryofibrinogen Panel, Serum and Plasma

Specimen Requirements:

Both plasma and serum are required.

Patient Preparation: Fasting: 12 hours, preferred but not required

Specimen Type: Plasma

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Tube must remain at 37 degrees C.

2. Centrifuge at 37 degrees C. **Do not use a refrigerated centrifuge.** If absolutely necessary, ambient temperature is acceptable. It is very important that the specimen remains at 37 degrees C until after separation of plasma from red blood cells.

3. Place plasma into an appropriately labeled plastic vial.

Specimen Type: Serum

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL Serum

Collection Instructions:

1. Tube must remain at 37 degrees C.

2. Allow blood to clot at 37 degrees C.

3. Centrifuge at 37 degrees C. **Do not use a refrigerated centrifuge.** If absolutely necessary, ambient temperature is acceptable. It is very important that the specimen remains at 37 degrees C until after separation of serum from red blood cells.

4. Place serum into an appropriately labeled plastic vial.

Additional Information: Analysis cannot be performed with less than 3 mL of serum. Smaller volumes are insufficient to detect clinically important trace (mixed) cryoglobulins. Less than 3 mL will require collection and submission of a new specimen.

Specimen Minimum Volume: Plasma: 0.5 mL; Serum: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	
	Frozen	
Serum Red	Refrigerated (preferred)	
	Frozen	

CPT Code Information:

82585

82595

86334-Immunofixation (if appropriate)

CRY_S 80988 Cryoglobulin, Serum

Specimen Requirements:

Patient Preparation: Fasting 12 hours, preferred but not required

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL Serum

Collection Instructions:

1. Tube must remain at 37 degrees C.

2. Allow blood to clot at 37 degrees C.

3. Centrifuge at 37 degrees C. **Do not use a refrigerated centrifuge.** If absolutely necessary, ambient temperature is acceptable. It is very important that the specimen remains at 37 degrees C until after separation of serum from red blood cells.

4. Place serum into an appropriately labeled plastic vial.

Additional Information: Analysis cannot be performed with less than 3 mL of serum. Smaller volumes are insufficient to detect clinically important trace (mixed) cryoglobulins. Less than 3 mL will require collection and submission of a new specimen.

Specimen Minimum Volume: Serum: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	
	Frozen	

CPT Code Information:

82595

86334-(if appropriate)

SLFA
62075

Cryptococcus Antigen Screen with Titer, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899

CLFA
62074

Cryptococcus Antigen Screen with Titer, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen from collection vial 2 (preferred), 3, or 4.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899-*Cryptococcus* screen

87899-*Cryptococcus* titer (if appropriate)

PLFA 42396 Cryptococcus Antigen Screen, Lateral Flow Assay, Pleural Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Collect pleural fluid in a sterile vial.

Specimen Minimum Volume: 0.25

Transport Temperature:

Specimen Type	Temperature	Time
Pleural Fluid	Refrigerated (preferred)	21 days
	Frozen	30 days

CPT Code Information:

87899-*Cryptococcus* Ag Screen, LFA, PF

87899-*Cryptococcus* Ag Titer, LFA, PF (if appropriate)

ULFA 604095 Cryptococcus Antigen Screen, Lateral Flow Assay, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899

87899 (if appropriate)

PLFAT
48431

Cryptococcus Antigen Titer, Lateral Flow Assay, Pleural Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see PLFA / *Cryptococcus* Antigen Screen, Lateral Flow Assay, Pleural Fluid.

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Collect pleural fluid in a sterile vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Pleural Fluid	Refrigerated (preferred)	21 days
	Frozen	30 days

CPT Code Information:

87899

SLFAT
62077

Cryptococcus Antigen Titer, Lateral Flow Assay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899

CLFAT 62076 Cryptococcus Antigen Titer, Lateral Flow Assay, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Submit specimen from collection vial 2 (preferred), 3, or 4.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899

ULFAT
604369

Cryptococcus Antigen Titer, Lateral Flow Assay, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see ULFA / *Cryptococcus* Antigen Screen, Lateral Flow Assay, Random, Urine.

Supplies: Urine Tubes, 10 mL (T068)

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899

CRYPS
80335

Cryptosporidium Antigen, Feces

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Preserved feces

Supplies:

-Formalin 10% Buffered Neutral 15 mL (T466)

-Stool Collection Kit, Random (T635)

Container/Tube:

Preferred: Stool container with 10% buffered formalin preservative

Acceptable: SAF (sodium acetate formalin)

Specimen Volume: 5 grams

Specimen Stability Information: Ambient (preferred) 60 days

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Stool container

Specimen Volume: 5 grams

Specimen Stability Information: Frozen 60 days

Specimen Minimum Volume: 1 gram

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies	

CPT Code Information:

87328

CRSBF 610646 Crystal Identification, Body Fluid

Specimen Requirements:

Specimen Required

Specimen Type: Synovial Fluid, Prosthetic Joint Fluid

Container/Tube: Lavender top (EDTA)

Acceptable: Green top (heparin)

Specimen Volume: 2 mL

Specimen Type: Bile Fluid

Container/Tube: Red top

Specimen Volume: 2 mL

Specimen Type: Bronchoalveolar Lavage (BAL)

Container/Tube: Body fluid container

Specimen Volume: 2 mL

Specimen Type: Pleural Fluid, Peritoneal Fluid, Pericardial fluid

Container/Tube:

Preferred: Body fluid container

Acceptable: Lavender top (EDTA) or Green top (heparin)

Specimen Volume: 2 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	
	Ambient	24 hours
	Frozen	

CPT Code Information:

89060

CSTB
616515

CSTB Gene, Repeat Expansion Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient 4 days (preferred)/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81188

FCUIP
57590

CU (Chronic Urticaria) Index Panel

Specimen Requirements:

Patient preparation: Patients taking calcineurin inhibitors should stop medication 72 hours prior to draw. Patients taking prednisone should be off their medication for 2 weeks prior to draw.

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 3 mL

Collection Instructions: Draw blood in Red-top tube (SST is acceptable). Separate from cells within 2 hours of draw. Send 3 mL of serum refrigerate in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	5 days

CPT Code Information:

84443

86343

86376

86800

FCUIX
57549

CU Index

Specimen Requirements:

Patient Preparation: Patients taking calcineurin inhibitors should stop medication 72 hours prior to draw. Patients taking prednisone should be off their medication for 2 weeks prior to draw.

Specimen Type: Serum

Collection Container/Tube: Red or SST

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Draw 5 mL blood in a serum separator tube (SST) (plain, red-top tube is acceptable).

2. Separate from cells within 2 hours of draw. Send 2 mL of serum ambient in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

86343

FCUKG
57651

Cucumber IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

CUKE
82861

Cucumber, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

OATC
82916

Cultivated Oat, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CRYE
82918

Cultivated Rye, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WHTC
82915

Cultivated Wheat, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CULAF
35244

Culture for Genetic Testing, Amniotic Fluid

Specimen Requirements:

Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted.

Container/Tube: Amniotic fluid container

Specimen Volume: 5 to 10 mL

Collection Instructions:

1. Discard the first 2 mL of amniotic fluid. If the culture will be performed in conjunction with other cytogenetic testing, such as CHRAF / Chromosome Analysis, Amniotic Fluid or CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, a total of 25 to 30 mL will be needed.

4. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

5. Bloody specimens are undesirable.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Amniotic Fld	Refrigerated (preferred)		
	Ambient		

CPT Code Information:

88235

88240

FUNID
8223

Culture Referred for Identification, Fungus

Specimen Requirements:

Specimen Type: Yeast or filamentous fungus in pure culture

Supplies: Infectious Container, Large (T146)

Container/Tube:

Preferred: Sabouraud dextrose agar slant

Acceptable: Inhibitory mold agar slant

Specimen Volume: Isolated mold or yeast

Collection Instructions: Yeast or filamentous fungus must be in pure culture, actively growing. **Do not submit mixed cultures.**

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87107-Culture, fungi, definitive identification

87106-Culture, fungi, definitive identification, each organism; yeast (if appropriate)

87106-Id MALDI-TOF Mass Spec Yeast (if appropriate)

87107-Id MALDI-TOF Mass Spec Fungi (if appropriate)

87107-Culture, fungi, definitive identification, each organism; mold (if appropriate)

87107-Fungal identification Panel A (if appropriate)

87107-Fungal identification Panel B (if appropriate)

87153-D2 fungal sequencing Identification (if appropriate)

87153-ITS Fungal Sequencing (if appropriate)

TBIDS 64714 Culture Referred for Identification, Mycobacterium and Nocardia with Antimicrobial Susceptibility Testing, Varies

Specimen Requirements:

Specimen Type: Organism in pure culture

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook (7H10 or 7H11) or Lowenstein-Jensen medium slant or in broth (eg, Mycobacteria Growth Indicator Tube [7H9] broth)

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

Culture Referred for Identification, *Mycobacterium*

87118-Identificaiton of mycobacteria

87158-Identification of mycobacteria by other methods (if appropriate)

87118 -Id MALDI-TOF Mass Spec AFB (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87186-Susceptibility Rapid Grower (if appropriate)

87186-Susceptibility Slow Grower (if appropriate)

87186-Susceptibility Nocardia species (if appropriate)

87188 x 3-Antimicrobial Susceptibility, Mycobacterium tuberculosis Complex, Broth Method (if appropriate)

87186-Suscetpbility, Mtb Cx, 2nd Line (if appropriate)

87188-Susceptibility, Mycobacterium tuberculosis Complex, Pyrazinamide (if appropriate)

87153-Mtb PZA Confirmation, pncA Sequencing (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

CTBID 80278 Culture Referred for Identification, Mycobacterium and Nocardia, Varies

Specimen Requirements:

Specimen Type: Mycobacterium species or aerobic actinomycetes organism in pure culture

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook (7H10 or 7H11) or Lowenstein-Jensen medium slant or in broth (eg, Mycobacteria Growth Indicator Tube [7H9] broth)

Specimen Volume:

Solid media: Visible growth of isolate

Isolate in broth media: > or =3 mL

Collection Instructions: Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

Additional Information: A minimum volume of 3 mL is recommended to perform all initial testing, this may include: stains, sub-culture media, nucleic acid probes, and any additional testing that may be required to determine the identification. If the broth sample volume is less than 3 mL, initial testing may be limited, and increased turnaround time is likely.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

Culture Referred for Identification, *Mycobacterium*

87118-Identification of mycobacteria

87158-Identification of mycobacteria by other methods (if appropriate)

87118 -Id MALDI-TOF Mass Spec AFB (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

CURL
82852

Curvularia lunata, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCURV
57898

Curvularia spicifera/Bipolaris IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

CGPH 605198 Custom Gene Panel, Hereditary, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For instructions about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

CPT codes are based on the gene content of the custom gene panel. Refer to the Custom Gene Ordering Tool for custom gene panel specific CPT code information.

81165 (if appropriate)

81166 (if appropriate)

81167 (if appropriate)

81162 (if appropriate)

81201 (if appropriate)

81216 (if appropriate)

81218 (if appropriate)

81223 (if appropriate)

81249 (if appropriate)

81252 (if appropriate)

81286 (if appropriate)

81292 (if appropriate)

81295 (if appropriate)

81298 (if appropriate)

81307 (if appropriate)

81317 (if appropriate)

81319 (if appropriate)

81321 (if appropriate)

81329 (if appropriate)

81351 (if appropriate)

81403 (if appropriate)

81404 (if appropriate)

81405 (if appropriate)

81406 (if appropriate)

81407 (if appropriate)

81408 (if appropriate)

81430 (if appropriate)

81431 (if appropriate)

81432 (if appropriate)

81435 (if appropriate)

81437 (if appropriate)

81440 (if appropriate)

81441 (if appropriate)

81443 (if appropriate)

81448 (if appropriate)

81479 (if appropriate)

81189 (if appropriate)

81419 (if appropriate)

CIB 607602 Cutaneous Direct Immunofluorescence Assay, Varies

Specimen Requirements:

Each biopsy requires **separate** specimen vials. If 2 or more biopsies are received in 1 specimen vial, expect testing delay.

Note: Multiple biopsies can be ordered together. Tests performed on each site will be billed accordingly.

Transport Medium Method

Specimen Type: Tissue

Supplies: Michel's Transport Media for Immunofluorescent Testing on Tissue (T321)

Sources: Skin or 1 of the following mucosae: oral (oropharyngeal), nasal, genital, esophageal, conjunctival, laryngeal, or epiglottis

Container/Tube: Screw-capped container or vial containing transport medium (Michel's, also called Zeus media)

Specimen Volume: 2- to 8-mm punch specimen, intact or bisected; excisional biopsy specimen intact or bisected

Collection Instructions:

1. Collect biopsy of skin or mucosa avoiding old lesions (including facial), ulcers, erosions, or bullae. Refer to Recommended Biopsy Site Selection Based on Disease State below.

2. Immediately place specimen into a labeled vial of transport medium and seal tightly.

Acceptable

Snap-Frozen Method

Specimen Type: Tissue

Sources: Skin or 1 of the following mucosae: oral (oropharyngeal), nasal, genital, esophageal, conjunctival, laryngeal, or epiglottis

Container/Tube: Plastic vial

Specimen Volume: 2- to 8-mm punch specimen, intact or bisected; excisional biopsy specimen, intact or bisected

Collection Instructions:

1. Collect biopsy of skin or mucosa avoiding old lesions (including facial), ulcers, erosions, or bullae. Refer to Recommended Biopsy Site Selection Based on Disease State below.

2. Immediately place specimen into liquid nitrogen and allow to freeze thoroughly (do not allow specimen to desiccate). If liquid nitrogen is not available, specimen may be frozen by placing it on a small square of aluminum foil on a block of dry ice. Liquid nitrogen is preferred.

3. Immediately wrap specimen carefully in aluminum foil. At no time should the specimen be allowed to thaw.

4. Place the wrapped specimen into the prelabeled plastic vial and seal tightly. Ship frozen.

Recommended Biopsy Site Selection Based on Disease State

1. Pemphigus and pemphigoid groups (including linear IgA bullous dermatosis and chronic bullous disease of childhood)**:** Biopsy erythematous perilesional skin or mucosa. Avoid erosions, ulcers, and bullae while obtaining tissue adjacent to active lesions. Label as perilesional skin.

2. Dermatitis herpetiformis: Biopsy normal-appearing skin, 0.5-1 cm away from lesion. Label as perilesional skin.

3. Lupus erythematosus: Involved areas of skin such as erythematous or active borders are preferred biopsy sites to confirm the diagnosis of lupus erythematosus, either discoid or systemic. Label as involved skin. Avoid ulcers, old lesions, and facial lesions, if possible. Uninvolved, nonexposed skin is the preferred site to detect a lupus band as may be found in systemic lupus erythematosus. Should unexposed skin be desired, buttock or medial thigh is suggested. Label as uninvolved, nonexposed skin.

4. Mixed connective tissue disease: Biopsy as for lupus erythematosus except when sclerodermoid features are present. For sclerodermoid features, biopsy inflamed area. Label as involved or uninvolved, exposed or nonexposed skin.

5. Vasculitis and urticaria: The erythematous or active border of a new lesion is

preferred. Avoid old lesions and ulcers. Label as involved skin. If appropriate, skin lesion is not present, diagnosis may sometimes be made from uninvolved skin.

6. Porphyria: Biopsy involved skin. Avoid old lesions and ulcers. Label as involved skin.

7. Lichen planus and lichenoid reactions: Biopsy involved skin. Avoid old lesions and ulcers. Label as involved skin.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	30 days
	Frozen	120 days
	Refrigerated	30 days

CPT Code Information:

Per biopsy:

88346

88350 x 5

CIFST 622323 Cutaneous Immunofluorescence Antibodies Titer, IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see CIFS / Cutaneous Immunofluorescence Antibodies, IgG and IgG4, Serum.

Collection Container/Tube:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

88350

CIFA 610627 Cutaneous Immunofluorescence Antibodies, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

88346

CIFS 8052 Cutaneous Immunofluorescence Antibodies, IgG and IgG4, Serum

Specimen Requirements:

Collection Container/Tube:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

88346

88350

88350 (if appropriate)

CXC13
113163

CXCL13 Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained

slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FCXCL 75827 CXCL9

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1.Within 8 hours of collection, centrifuge and aliquot plasma into a plastic vial. Freeze immediately.

2.Send frozen.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	180 days

CPT Code Information:

83520

CXLPL 64759 CXCR4 Mutation Analysis, Somatic, Lymphoplasmacytic Lymphoma/Waldenstrom Macroglobulinemia, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 10 days /Refrigerated 10 days

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable

Specimen Type: Tissue slide

Slides: 20 Unstained slides

Container/ Tube: Transport in plastic slide holders

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection

2. Indicate specimen source.

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 50 mcL at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81479

FCYNB
75370

Cyanide, Blood Test

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between**

multiple orders.***

Container/Tube: Gray top (potassium oxalate/sodium fluoride)

Specimen volume: 2 mL

Collection instructions: Collect 2 mL whole blood in potassium oxalate/sodium fluoride Gray top tube, send frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood NaFl-KOx	Frozen (preferred)	90 days
	Refrigerated	7 days

CPT Code Information:

82600

CARU 609739 Cyclic Adenosine Monophosphate (cAMP), Urinary Excretion, Serum and Urine

Specimen Requirements:

Both serum and urine are required. Serum must be collected at the time of the urine collection.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection serum gel tubes should be centrifuged.

2. Within 2 hours of collection red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial.

3. Label specimen as serum.

Specimen Type: Urine

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Label specimen as urine.

Specimen Minimum Volume: Serum: 0.5 mL
Urine: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82030

82570

82565

CCP
84182

Cyclic Citrullinated Peptide Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86200

CYC1
70411

Cyclin D1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FFLEX 90085 Cyclobenzaprine (Flexeril)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Specimen Type: Plasma

Container/Tube: Green Top

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a green-top (sodium heparin) tube, **plasma gel tube is not acceptable.** Spin down and send 2 mL of sodium heparin plasma refrigerated in a plastic vial.

Serum

Specimen Type: Serum

Container/Tube: Red

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a plain, red-top tube, **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

80369

CYCL
81506

Cyclospora Stain, Feces

Specimen Requirements:

Patient Preparation: Patient should avoid use of antidiarrheal medication (eg, loperamide [Imodium-AD] or Pepto-Bismol).

The presence of barium will interfere with this test.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Preserved stool

Supplies:

-ECOFIX Stool Transport Vial (Kit) (T219)

-Formalin-Meridian 10% Buffered Neutral (T466)

Container/Tube:

Preferred: ECOFIX Stool Transport Vial (Kit)

Acceptable: 10% Buffered Formalin Stool Transport (Kit), Sodium Acetate Formalin (SAF)

Specimen Volume: 10 g

Specimen Stability Information: Ambient 21 days (preferred)/Refrigerated 21 days

Acceptable

Specimen Type: Unpreserved stool

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 5 g

Specimen Stability Information: Refrigerated 3 days (preferred)

Specimen Minimum Volume: Preserved stool: 1 g
Unpreserved stool: 2 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies	

CPT Code Information:

87015

87207

CYSPR 35143 Cyclosporine, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Collect specimen immediately before a scheduled dose.

2. Do not centrifuge.

3. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Therapeutic range applies to trough specimens collected immediately prior to a.m. dose.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80158

CYCPK 42427 Cyclosporine, Peak, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Do not centrifuge.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: No definitive therapeutic or toxic ranges have been established for this peak testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80158

CSTCE 614154 Cystatin C with Estimated Glomerular Filtration Rate (eGFR), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82610

CFSMN 608349 Cystic Fibrosis and Spinal Muscular Atrophy Carrier Screen Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81220

81329

81222

81479 (if appropriate for government payers)

CFTRN 619774 Cystic Fibrosis Transmembrane Conductance Regulator, CFTR, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the

cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a

minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order

MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81223

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

CFMP 605197 Cystic Fibrosis, CFTR Gene, Variant Panel, Varies

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	

	Refrigerated

CPT Code Information:

81220

81222

CKDGP 618072 Cystic Kidney Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Prenatal Specimens:

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional information:

1. If amniotic fluid or nonconfluent cultures are received, CULAF / Culture for Genetic Testing, Amniotic Fluid will be added at an additional charge.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated

Additional Information:

1. If nonconfluent cultures are received, CULFB / Fibroblast Culture for Biochemical or Molecular Testing will be added at an additional charge.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information: All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid/CVS: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

81405

81406 x 6

81407 x 4

81408 x 3

81479

81265-Maternal cell contamination (if appropriate)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88235-Amniotic Fluid culture (if appropriate)

81479 (if appropriate for government payers)

CYSTG
616730

Cysticercosis (Taenia solium), IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

86682

FCAEC
75587

Cysticercus Antibody (IgG), ELISA, CSF

Specimen Requirements:

Specimen Type: Spinal fluid

Collection Container/Tube: Sterile container

Submission Container/Tube: Sterile, plastic screw-cap vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect spinal fluid (CSF) in a sterile container.

2. Aliquot 1 mL CSF into a sterile, plastic screw-cap vial.

2. Refrigerate specimen.

3. Send refrigerate.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86682

CYSGP
608027

Cystinuria Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic

donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell [or liver] transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

CYSQN 8376 Cystinuria Profile, Quantitative, 24 Hour, Urine

Specimen Requirements:

-Urine Tubes, 10 mL (T068)

-Diazolidinyl Urea (Germall) 5.0 mL (T822)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect before intravenous pyelogram.

2. Add 5 g of diazolidinyl urea (Germall) as preservative **at start of collection**. If preservative is not available, refrigerate during collection.

3. Collect urine for 24 hours.

4. Mix well before taking 5-mL aliquot.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	70 days
	Refrigerated	14 days

CPT Code Information:

82136

CYSR 81067 Cystinuria Profile, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Specimen Volume: 2 mL

Collection Instructions: Collect a random urine specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	70 days
	Refrigerated	14 days

CPT Code Information:

82136

METR1
65609

Cytochrome b5 Reductase Enzyme Activity, Blood

Specimen Requirements:

Collection Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	18 days

CPT Code Information:

82657

1A2Q
610041

Cytochrome P450 1A2 Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor or a liver transplant will interfere with testing. For information about testing patients who have received a hematopoietic stem cell or liver transplant call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

2B6Q
610042

Cytochrome P450 2B6 Genotype, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor or a liver transplant will interfere with testing. For information about testing patients who have received a hematopoietic stem cell or liver transplant call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested

volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

2C19R
610043

Cytochrome P450 2C19 Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell or liver transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not

perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81225

2C9QT 610044 Cytochrome P450 2C9 Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 9 days/Refrigerated 30 days

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube: 2-mL screw top tube

Specimen Volume: 100 mcL (microliters)

Collection Instructions:

1. The preferred volume is 100 mcL at a concentration of 50 ng/mcL.

2. Provide concentration of DNA and volume on tube.

Specimen Stability Information: Frozen (preferred)/Ambient/Refrigerated

Specimen Minimum Volume: Blood: 0.4 mL
Saliva, extracted DNA: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81227

2D6Q
610045

Cytochrome P450 2D6 Comprehensive Cascade, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 9 days/Refrigerated 30 days

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube: 2 mL screw top tube

Specimen Volume: 100 mcL (microliters)

Collection Instructions:

1. The preferred volume is 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Specimen Minimum Volume: Whole blood: 1 mL
Saliva: 1 swab

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

0070U

0071U (if appropriate)

0076U (if appropriate)

3A4Q
610046

Cytochrome P450 3A4 Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor or a liver transplant will interfere with testing. For more information about testing patients who have received a hematopoietic stem cell or liver transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81230-CYP3A4

3A5Q
610047

Cytochrome P450 3A5 Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor or liver transplant will interfere with testing. For more information about testing patients who have received a hematopoietic stem cell or liver transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81231-CYP3A5

Cytokine Panel 13, CSF

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Transfer 1 mL CSF to a sterile plastic container.

2. Send frozen.

Note: Critical frozen. **Additional specimens must be submitted** when multiple tests are ordered.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen	30 days

CPT Code Information:

83520 x 12

83529

LCYP
621646 Cytokine Panel, 13 Analytes, Quantitative, Serum

Specimen Requirements:

Collection Container/Tube: Serum gel or red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. As soon as possible and within 2 hours of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send frozen.

Note: Critical frozen. **Separate specimens must be submitted** when multiple tests are ordered.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	30 days

CPT Code Information:

83520 x 12

83529

CYPAN
610259

Cytokine Panel, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender-top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Centrifuge at 4 degrees C, 1500 x *g* for 10 minutes.

3. Aliquot plasma into plastic vial.

4. Within 2 hours of collection, freeze specimen.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

83520 x 10

83529

CTFNA Cytology Fine-Needle Aspiration, Varies

70334

Specimen Requirements:

Specimen Type: Slide

Container/Tube: Plastic slide container

Specimen Volume: Smear

Collection Instructions:

1. Smears should be immediately fixed in 95% ethanol or sprayed with commercially available fixative. Smears that have been air-dried or Diff-Quik stained may also be accepted.

2. Label containers with a minimum of 2 unique identifiers (eg, patient name and medical record number or date of birth), specimen source, and date of collection. Label each glass slide in pencil with a minimum of 2 unique identifiers. If multiple slides are submitted, each slide must have proper identification.

Specimen Type: Fluid

Container/Tube: 60-mL (2 oz) jar with screw cap, 50-mL disposable centrifuge tube with screw cap, or 15-mL test tube with screw cap

Specimen Volume: Any amount

Collection Instructions:

1. Preferred method is no fixative added to fluid prior to processing and the **specimen must be received and processed by the Cytology Laboratory within 1 hour of collection**.

2. If not possible to submit within 1 hour, specimen should be refrigerated no longer than 62 hours. Additional acceptable fixatives are specimens with equal volume of 50%, 70%, 80%, or 95% ethanol, PreservCyt solution, CytoRich Red, or CytoLyt.

3. Label containers with a minimum of 2 unique identifiers (eg, patient name and medical record number or date of birth), specimen source, and date of collection.

Specimen Type: Tissue

Container/Tube: 50-mL disposable centrifuge tube with screw cap or 60-mL (2 oz) jar with screw cap containing 10% neutral-buffered formalin

Specimen Volume: Any amount

Collection Instructions:

1. Tissue fragments must be submitted in 10% neutral-buffered formalin.

2. Label containers with a minimum of 2 unique identifiers (eg, patient name and medical record number or date of birth), specimen source, and date of collection.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88173

88305-CBKPC (if appropriate)

CFNPC
113344

Cytology FNA (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88173

CYTNG
70330

Cytology Non-Gynecologic, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Contact the testing lab for specific instructions.

Specimen Type: Non-gynecologic cytology specimen

Supplies: CytoLyt Solution (T564)

Container/Tube:

Preferred: CytoLyt solution

Acceptable:

1. PreservCyt solution in prefilled vial

2. Specimens with equal volume of 50%, 70%, 80%, or 95% ethanol

3. Specimens fixed in carbowax, CytoSpin collection fluid, or CytoRich red

Specimen Volume: A minimum of 20 mL or entire collection

Collection Instructions:

1. Specimen containers **must be labeled** with a minimum of 2 unique identifiers (patient's name and medical record number).

2. Indicate the specimen source and source location on the label.

Specimen Type: Spinal fluid

Collection Container/Tube: Sterile vial

Submission Container/Tube: Container with equal volume of 50%, 70%, 80%, or 95% ethanol

Specimen Volume: A minimum of 1 mL

Collection Instructions: Specimen containers **must be labeled** with a minimum of 2 unique identifiers (patient's name and clinic number).

Specimen Type: Smear

Container/Tube: Plastic slide container

Specimen Volume: Glass slide

Collection Instructions:

1. Slides should be immediately fixed in 95% ethanol or sprayed with commercially available fixative. Smears that have been air-dried or Diff-Quik stained may also be

accepted.

2. Label containers with a minimum of 2 unique identifiers (eg, patient name and medical record number or date of birth), specimen source, and date of collection. Label each glass slide in pencil with a minimum of 2 unique identifiers. If multiple slides are submitted, each slide must have proper identification.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88104-NDSPC (if appropriate)

88108-NCSPC (if appropriate)

88112-NTPPC (if appropriate)

88161-CSOPC (if appropriate)

88162-CSAPC (if appropriate)

88305-CBKPC (if appropriate)

CTPPC
113340

Cytology Touch Prep (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88333

CTAPC
113341

Cytology Touch Prep Additional (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88334

CMVG
800030

Cytomegalovirus (CMV) Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86644

CMVP 800029 Cytomegalovirus (CMV) Antibodies, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86644-CMV, IgG

86645-CMV, IgM

CMVM 800031 Cytomegalovirus (CMV) Antibodies, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86645

CMVC8 88826 Cytomegalovirus (CMV) CD8 T-Cell Immune Competence, Quantitative Assessment by Flow Cytometry, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 20 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86356 x 6

86359

86352

CMVQN
601954

Cytomegalovirus (CMV) DNA Detection and Quantification by Real-Time PCR, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87497

CMVI
70406

Cytomegalovirus (CMV) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CMVLR 621770 Cytomegalovirus (CMV) Molecular Detection, PCR, Lower Respiratory

Specimen Requirements:

Specimen Type: Lower respiratory

Source: Bronchial washing, bronchoalveolar lavage, fluid/washings from lung, sputum, tracheal secretions, tracheal aspirates

Container/Tube:

Preferred: Sterile, screwcap, 5-mL aliquot tube

Acceptable: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Do not centrifuge.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87496

CMVPV 618968 Cytomegalovirus (CMV) Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Spinal, pleural, peritoneal, ascites, pericardial, amniotic, or ocular

Container/Tube:

Preferred: Sterile, screwcap, 5-mL aliquot tube

Acceptable: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Upper respiratory tract fluid

Sources: Nasopharyngeal aspirate or washing

Container/Tube:

Preferred: Sterile, screwcap, 5-mL aliquot tube

Acceptable: Sterile container

Specimen Volume: 1.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Swab

Sources: Dermal, eye, nasal, saliva, throat, or genital

Supplies:

-Culturette (BBL Culture Swab) (T092)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT, M4, M5, Bartels, or Jiangsu) and ESwab or Culturette

Collection Instructions: Place swab back into multimicrobe media.

Specimen Type: Tissue

Sources : Brain, colon, kidney, liver, lung, etc.

Supplies: M4-RT (T605)

Container/Tube: Sterile container containing 1 mL to 2 mL of sterile saline or multimicrobe medium (M4-RT, M4, M5, Bartels, or Jiangsu)

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue.

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Collect a random urine specimen.

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA)

Specimen Volume: 0.5 mL

Collection Instructions: Send bone marrow in original tube. **Do not aliquot.**

Specimen Minimum Volume: Ocular Fluid, Spinal Fluid, or Urine: 0.3 mL; Body Fluid (pleural, peritoneal, ascites, pericardial): See Specimen Required; Upper respiratory tract specimens: (Nasopharyngeal aspirate or washing): 1 mL; Tissue: 2 x 2-mm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87496

LCMV 800145 Cytomegalovirus (CMV), Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Spinal, pleural, peritoneal, ascites, pericardial, amniotic, or ocular

Container/Tube:

Preferred: Sterile screw cap 5-mL aliquot tube

Acceptable: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Genital swab

Sources: Cervix, vagina, urethra, anal/rectal, or other genital sources

Supplies:

-Culturette (BBL Culture Swab) (T092)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT) and ESwabs

Collection Instructions: Place swab back into multimicrobe media (M4-RT, M4, or M5)

Specimen Type: Swab

Sources: Dermal, eye, nasal, saliva, or throat

Supplies:

-Culturette (BBL Culture Swab) (T092)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT) and ESwabs

Collection Instructions: Place swab back into multimicrobe media (M4-RT, M4, or M5)

Specimen Type: Tissue

Sources: Brain, colon, kidney, liver, lung, etc.

Supplies: M4-RT (T605)

Container/Tube: Sterile container containing 1 mL to 2 mL of sterile saline or multimicrobe medium (M4-RT, M4, or M5)

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue in multimicrobe media (M4-RT) or a sterile container with 1 to 2 mL sterile saline

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Collect a random urine specimen.

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA)

Specimen Volume: 0.5 mL

Collection Instructions: Send bone marrow in original tube. **Do not aliquot.**

Specimen Minimum Volume: Body fluid, ocular fluid, spinal fluid, or urine: 0.3 mL; Respiratory specimens: 1 mL; Tissue: 2 x 2-mm Biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87496

CMVGS 621835 Cytomegalovirus Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	180 days
	Ambient	7 days

CPT Code Information:

86644

CMVAB 621833 Cytomegalovirus Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86645-Cytomegalovirus IgM

86644-Cytomegalovirus IgG

CMVMS 621834 Cytomegalovirus Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86645

FCYTG 75445 Cytomegalovirus IgG Avidity

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1.Centrifuge and aliquot serum into a plastic vial

2.Send refrigerated.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86644

ANCA2 610030 Cytoplasmic Neutrophil Antibodies, Inflammatory Bowel Disease Panel, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see IBDP2 / Inflammatory Bowel Disease Serology Panel, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86036

ANCA
9441

Cytoplasmic Neutrophil Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86036 x2

86037-Titer (if appropriate)

DDITT 40936 D-Dimer, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL plasma

Collection Instructions:

1. Centrifuge, remove plasma, and centrifuge plasma again.

2. Aliquot plasma into plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

Additional Information: Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen (preferred)	90 days
	Ambient	4 hours

CPT Code Information:

85379

DIMER 602174 D-Dimer, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile or as a reflex. For more information see:

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

ALUPP / Lupus Anticoagulant Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85379

DLAC
8878

D-Lactate, Plasma

Specimen Requirements:

Supplies: Sodium Fluoride/Potassium Oxalate Tube, 2 mL (T275)

Collection Container/Tube:

Preferred: Sodium fluoride/potassium oxalate tube

Acceptable: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Note: If collecting in sodium heparin tubes, centrifugation must occur within one hour of collection.

Specimen Minimum Volume: Plasma: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma NaFl-KOx	Frozen (preferred)	91 days
	Ambient	91 days
	Refrigerated	91 days

CPT Code Information:

83605

DLAU
8873

D-Lactate, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic urine tube

Specimen Volume: 0.50 mL

Collection Instructions:

1. Collect a timed or random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	90 days
	Ambient	90 days
	Refrigerated	90 days

CPT Code Information:

83605

DAGR
31768

Dairy and Grain Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

DAND
82694

Dandelion, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DATE
82358

Date, Fruit, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DATRE
82481

Date, Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DCALP
622511

Decalcification (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88311

DOCK8
608112

Dedicator of Cytokinesis 8 (DOCK8) Deficiency, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x 4

DHEA_
81405

Dehydroepiandrosterone (DHEA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Additional Information: Requests for this test cannot be added to a previously received specimen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	28 days
	Refrigerated	21 days
	Ambient	6 hours

CPT Code Information:

82626

DHES1 Dehydroepiandrosterone Sulfate, Serum
113595

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82627

THCCR 616333 Delta 9-Carboxy-Tetrahydrocannabinol (THC-COOH) Confirmation and Creatinine Ratio, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 10 mL in a plastic container.

3. No preservative.

Additional Information:

1. No specimen substitutions.

2. Submitting less than 10 mL may compromise the ability to perform all necessary testing.

3. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

82570 (if appropriate for select payers)

80349 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

THCX 62743 Delta-8 and Delta-9-Carboxy-Tetrahydrocannabinol (THC) Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and required documentation.

Specimen Volume: 5 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80349 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

THCU 8898 Delta-8 and Delta-9-Carboxy-Tetrahydrocannabinol (THC) Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80349 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

THCCU 616334 Delta-9-Carboxy-Tetrahydrocannabinol Confirmation and Creatinine Ratio, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see THCCR / Delta 9-Carboxy-Tetrahydrocannabinol (THC-COOH) Confirmation and Creatinine Ratio, Random, Urine

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80349 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

DLL3
603332

Delta-Like 3 Protein (SP347), Semi-Quantitative Immunohistochemistry, Manual, Tissue

Specimen Requirements:

A minimum of 100 tumors cells are required for interpretation of DLL3 expression.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

DMS2 Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

92114

Specimen Requirements:

Patient Preparation:

1. For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin treatment.

2. For 24 hours before specimen collection, patient **should not** receive general anesthetic or take muscle-relaxant drugs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 21

86341

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86255 AK5CS (if appropriate)

86256 AK5TS (if appropriate)

86256 AMPIS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN2TS (if appropriate)

86256 AN3TS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

86256 IG5TS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86255 GL5CS (if appropriate)

86256 GL5TS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86256 NMDIS (if appropriate)

86256 PC2TS (if appropriate)

84182 PCTBS (if appropriate)

86256 PCTTS (if appropriate)

86256 PDETS (if appropriate)

86255 T46CS (if appropriate)

86256 T46TS (if appropriate)

DMC2 92115 Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 23

86341

84182 AGNBC (if appropriate)

86256 AGNTC (if appropriate)

86255 AK5CC (if appropriate)

86256 AK5TC (if appropriate)

86255 AINCC (if appropriate)

86256 AMPIC (if appropriate)

86256 APHTC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 AN2TC (if appropriate)

86256 AN3TC (if appropriate)

86256 CRMTC (if appropriate)

84182 CRMWC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

86256 IG5TC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 GL5CC (if appropriate)

86256 GL5TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86255 NFHCC (if appropriate)

86256 NIFTC (if appropriate)

86255 NFLCC (if appropriate)

86256 NMDIC (if appropriate)

86256 PC2TC (if appropriate)

84182 PCTBC (if appropriate)

86256 PCTTC (if appropriate)

86256 PDETC (if appropriate)

86255 T46CC (if appropriate)

86256 T46TC (if appropriate)

DCME 609795 Dendritic Cell and Monocyte Enumeration, Blood

Specimen Requirements:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not open tube**. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	36 hours	GREEN TOP/HEP

CPT Code Information:

86356 x 3

DENGM 83865 Dengue Virus Antibody, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

IgM-86790

IgG-86790

DENVP 62869 Dengue Virus Antibody/Antigen Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86790 x 3

DNSAG Dengue Virus NS1 Antigen, Serum
36781

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86790

DENGS Dengue Virus, Molecular Detection, PCR, Serum
606372

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Sterile container

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot the serum into a sterile container.

Additional Information: Serum specimens not aliquoted from the serum gel collection tube into a sterile container **will be rejected**.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

DENGC
606371

Dengue Virus, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Collection Container/Tube:

Preferred: Vial number 2

Acceptable: Any vial number

Submission Container/Tube: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge or heat inactivate.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days

	Frozen	7 days

CPT Code Information:

87798

DRPL 35402 Dentatorubral-Pallidoluysian Atrophy (DRPLA) Gene Analysis, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81177-ATN1 (ataxin 2) (eg, denatatorubral-pallidolyuysian atrophy) gene analysis, evaluation to detect abnormal (eg, expanded) alleles

FDCU 58048 Deoxypyridinoline Crosslinks, Urine

Specimen Requirements:

Container/Tube: Plastic, 5-mL urine tube

Specimen Volume: 3.5 mL

Collection Instructions:

1. Collect a first morning urine.

2. No preservative.

3. Mix the specimen well and aliquot 3.5 mL of urine to a plastic urine tube.

4. Send frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	90 days
	Refrigerated	7 days

CPT Code Information:

82523

DEXT 70596 Dermatopathology Consultation, Wet Tissue

Specimen Requirements:

Supplies: Dermatopathology Media (T101)

Sources: Skin or oral mucosa

Container/Tube: Screw-capped container or vial containing 10% formalin

Specimen Volume: Entire specimen

Collection Instructions: For scalp biopsies: when the differential diagnosis includes a scarring alopecia, 2 separate 4-mm punch biopsies are recommended (1

for vertical and 1 for horizontal sections). If a single scalp biopsy is received with the clinical diagnosis of a scarring alopecia, the specimen will be processed with horizontal sections.

Additional Information: STAT requests are not accepted for this test.

Specimen Minimum Volume: Size needed depends on diagnosis and size of lesion.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient	

CPT Code Information:

88302 (if appropriate)

88304 (if appropriate)

88305 (if appropriate)

DMIC 82828 Dermatophagoides microceras, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DCP 61844 Des-Gamma-Carboxy Prothrombin, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

83951

DESPR 37123 Desipramine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

DESMN
70421

Desmin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

DSGAB 606818 Desmoglein 1 (DSG1) and Desmoglein 3 (DSG3), IgG Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

83516 x 2

DESG3 70420 Desmoglein 3 (DSG3) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

DEXA
609439

Dexamethasone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood between 7:30 a.m. and 9:00 a.m. the morning following an evening dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial and freeze immediately. **Do not send in original tube.**

3. Send frozen.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Refrigerated	14 days
	Ambient	72 hours

CPT Code Information:

80299

FDXAP 57720 Dexedrine (Dextroamphetamine)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 5 mL sodium heparin plasma refrigerated in a plastic vial.

Serum
Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

80324

FCDU8 75783 Dextro/Levo Methorphan, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

FDM
90117

Dextromethorphan (DM), Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

80362

FDAIP
75937

Diabetes Autoimmune Profile

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions:

1. Within 24 hours of collection, centrifuge and aliquot 2.5 mL of serum into a plastic vial.

2. Freeze within 24 hours and send frozen.

Specimen Minimum Volume: 1 mL (Note: This volume does not allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	7 days

CPT Code Information:

86337

86341x3

DIA
8629

Diazepam and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days

	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

DICET 619667 DICER1 Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block
Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide
Slides: 1 Hematoxylin and eosin stained and 10 unstained
Collection Instructions:

Submit the following slides:

1 slide stained with hematoxylin and eosin

AND

10 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.
Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.
Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)
Slides: 1 to 3 Slides
Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.
Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.
Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81479

FDGTX 75374 Digitoxin, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

80299

FRDIG
82130

Digoxin, Free, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood 6 to 8 hours after last dose of digoxin.

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days

CPT Code Information:

80163

DIG
8674

Digoxin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood 6 to 8 hours after the last dose of digoxin.

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days

	Frozen	180 days

CPT Code Information:

80162

DPYDQ
610052

Dihydropyrimidine Dehydrogenase Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81232

DPYDZ 610051 Dihydropyrimidine Dehydrogenase, DPYD Full Gene Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant or a liver transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell or liver transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Gray top (sodium fluoride/potassium oxalate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days (preferred)/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not

perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81232

DHRP 62765 Dihydrorhodamine Flow Cytometric Phorbol Myristate Acetate Test, Blood

Specimen Requirements:

Two whole-blood sodium heparin specimens are required, one from the testing patient and the other from an unrelated healthy donor as a control.

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Patient:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Normal Control:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a control specimen from the unrelated healthy donor within an hour of the patient's specimen collection time.

2. Label clearly as **Normal Control** and the corresponding patient information.

3. Send the whole blood specimen in the original tube. **Do not aliquot.**

4. Rubber band patient specimen and normal control vials together.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86352

DHR
62764

Dihydrorhodamine Flow Cytometric Test, Blood

Specimen Requirements:

Two whole-blood sodium heparin specimens are required, one from the testing patient and the other from an unrelated healthy donor as a control.

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Patient:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Normal Control:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a control specimen from the unrelated healthy donor within an hour of the patient's specimen collection time.

2. Clearly label as **Normal Control** on the outermost label.

3. Send the whole blood specimen in the original tube. **Do not aliquot.**

4. Rubber band patient specimen and control vial together.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86352 x2

DHTS
81479

Dihydrotestosterone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	90 days
	Ambient	28 days

CPT Code Information:

82642

DCLNG
617239

Dilated Cardiomyopathy and Left Ventricular Noncompaction Cardiomyopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that

additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81439

DILL Dill, IgE, Serum

82602

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FDILT
91118

Diltiazem (Cardizem, Dilacor)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

DRVI3 602181 Dilute Russell's Viper Venom Time (DRVVT) Confirmation Ratio, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see DRVI1 / Dilute Russell's Viper Venom Time (DRVVT), with Reflex, Plasma.

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot into a separate plastic vial, leaving 0.25 mL in the bottom of the centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Specimen Minimum Volume: Platelet poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Plasma Na Cit	Frozen	14 days

CPT Code Information:

85613

DRV3 602178 Dilute Russell's Viper Venom Time (DRVVT) Confirmation, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85613

DRVI4 603310 Dilute Russell's Viper Venom Time (DRVVT) Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as a part of a profile. For more information see DRVI1 / Dilute Russell's Viper Venom Time (DRVVT), with Reflex, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

DRVI2 602180

Dilute Russell's Viper Venom Time (DRVVT) Mix Ratio,
Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see DRVI1 / Dilute Russell's Viper Venom Time (DRVVT), with Reflex, Plasma.

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial and centrifuge plasma again.

3. Aliquot into a separate plastic vial, leaving 0.25 mL in the bottom of the centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Specimen Minimum Volume: Platelet poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85613

DRV2 602177 Dilute Russell's Viper Venom Time (DRVVT) Mix, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85613

DRV1 602176 Dilute Russell's Viper Venom Time (DRVVT), Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile or as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85613-DRVVT

85613-DRVVT mix (if appropriate)

85613-DRVVT confirmation (if appropriate)

DRVI1 602179 Dilute Russell's Viper Venom Time (DRVVT), with Reflex, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot into a separate plastic vial, leaving 0.25 mL in the bottom of the centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet poor plasma 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85613

85613 (if appropriate)

85613 (if appropriate)

DIPGS
36664

Diphtheria Toxoid IgG Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86317

DTABS
36670

Diphtheria/Tetanus Antibody Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86317 x 2

DCTR 113489 Direct Antiglobulin Test (Polyspecific), Blood

Specimen Requirements:

Container/Tube: Pink top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 3 mL; Neonates: EDTA Micro tube 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient (preferred)	48 hours
	Refrigerated	48 hours

CPT Code Information:

86880

DSAC 608236 Disaccharidase Activity Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue

Source: Intestinal biopsy

Supplies: 15 mL Tissue Tube-DSAC (T993)

Container/Tube: 15 mL tissue tube

Specimen Volume: 5 mg

Collection Instructions:

1. Place specimen in tissue tube.

2. Specimen **should not** be placed on gauze, filter paper, or swabs/wooden sticks and **should not** have any saline, water, support, or embedding material added.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Frozen	28 days

CPT Code Information:

82657

ADICI 603182 Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

ADIC 603306 Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation: Patient **should not** be receiving warfarin or heparin therapy. If so, note on request.

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 5 mL Platelet-poor plasma in five (5) 13-mm x 75-mm, 5-mL, polypropylene, plastic vials , each containing 1 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma (1 mL per aliquot) into 5 separate 13-mm x 75-mm, 5-mL, polypropylene, plastic vials , leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally -40 degrees C or below.

Additional Information: Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85610-PTSC

85730-APTSC

85670-TTSC

85379-DIMER

85384-CLFIB

85390-26-ADICI

85210-Factor II (if appropriate)

85220-Factor V (if appropriate)

85230-Factor VII (if appropriate)

85240-Coagulation factor VIII assay (if appropriate)

85250-Factor IX (if appropriate)

85260-Factor X (if appropriate)

85270-Factor XI (if appropriate)

85280-Factor XII (if appropriate)

85366-Soluble fibrin monomer (if appropriate)

85385-PT-Fibrinogen (if appropriate)

85597-Platelet neutralization for lupus inhibitor (if appropriate)

85598-Hex LA (if appropriate)

85611-PT mix 1:1 (if appropriate)

85613-DRVVT (if appropriate)

85613-DRVVT mix (if appropriate)

85613-DRVVT confirm (if appropriate)

85635-Reptilase time (if appropriate)

85732-APTT mix 1:1 (if appropriate)

FDISU
75938

Diuretic Profile, Urine

Specimen Requirements:

Specimen Type: Urine

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 10 mL

Collection Instructions:

1. Collect 10 mL random urine without preservative.

2. Send specimen refrigerated.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80377

DMDZ
617532

DMD Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81408

FDM1 DMPK DNA Test (DM1)
91592

Specimen Requirements:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA)

Specimen volume: 8 mL

Pediatric (0-3 years) Volume: 2 mL (1 mL minimum)

Collection instructions:

1. Draw blood in a lavender-top (EDTA) tube.

2. Invert several times to mix.

3. Send ambient

Note: **Collection date is required.**

Specimen Minimum Volume: 6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

81234

DNJB9 DNAJB9 Immunostain, Technical Component Only
71739

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FDKYE Dock Yellow (Rumex crispus) IgE
57528

Specimen Requirements:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

DOGD
60108

Dog Dander, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DOGPF 616091 Dog Dander, IgE, with Reflex to Dog Dander Components, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FDFEN 75547 Dog Fennel (Anthemis cotula) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

DOG1
70422

DOG-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FDLS 58007 Donath Landsteiner

Specimen Requirements:

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. Maintain specimen at 37 degrees C until after centrifugation.

2. Centrifuge and aliquot 3 mL of serum into a plastic vial.

3. Send refrigerated.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Ambient	72 hours

CPT Code Information:

86940

86941

ADNAP 620812 Double-Stranded DNA (dsDNA) Antibodies Panel, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	21 days

CPT Code Information:

ADNA1-86225

CRTHS-86255

ADNA1 Double-Stranded DNA (dsDNA) Antibodies, IgG, Serum
620809

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86225

CRTHS 623426 Double-Stranded DNA (dsDNA) IgG Antibodies, Crithidia luciliae Immunofluorescence Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	21 days

CPT Code Information:

86255

CRTHT 623427 Double-Stranded DNA (dsDNA) IgG Antibodies, Crithidia luciliae Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see CRTHS / Double-Stranded DNA (dsDNA) IgG Antibodies, *Crithidia luciliae* Immunofluorescence Screen, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	21 days

CPT Code Information:

86256

DXPIN 63507 Doxepin and Nordoxepin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

CDA7X 62716 Drug Abuse Survey with Confirmation, Panel 9, Chain of Custody, Random, Urine

Specimen Requirements:

Container/Tube: Chain-of-Custody Kit (T282) containing specimen containers, seals, and documentation required

Specimen Volume: 30 mL

Collection Instructions: Collect specimen in the provided container, seal, and submit it with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 30 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

CDAU7 81410 Drug Abuse Survey with Confirmation, Panel 9, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 30 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 30 mL in 1 plastic bottle.

3. No preservative.

Additional Information:

1. No specimen substitutions.

2. Submitting less than 30 mL will compromise the ability to perform all necessary testing.

3. STAT requests are **not accepted** for this test

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

PNRCH 65061 Drug Immunoassay Panel, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

-CSMPU / Controlled Substance Monitoring Panel, Random, Urine

-ADMPU / Addiction Medicine Profile with Reflex, 22 Drug Classes, High Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine

-CSMEU / Controlled Substance Monitoring Enhanced Profile with Reflex, 21 Drug Classes, High Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

PDSUX 62741 Drug Screen, Prescription/Over the Counter, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain-of-Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation required.

Specimen Volume: 30 mL

Collection Instructions: Collect a random urine specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 30 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	3 hours

CPT Code Information:

80307

PDSU 88760 Drug Screen, Prescription/Over the Counter, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	3 hours

CPT Code Information:

80307

DSS
8421

Drug Screen, Prescription/Over the Counter, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.75 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days

	Frozen	14 days
	Ambient	3 hours

CPT Code Information:

80307

FDA1S 75525 Drugs of Abuse (10 Panel) and Alcohol Screen, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions:

1. Collect specimen using alcohol-free skin preparation.

2. Promptly centrifuge and aliquot serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307x2

DSM4X 62721 Drugs of Abuse Screen 4, Chain of Custody, Meconium

Specimen Requirements:

Container/Tube: Chain of Custody Meconium Kit (T653) includes the specimen

containers, seals, and documentation required.

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Additional Information:

1. Specimen that arrives with a broken seal does not meet the chain-of-custody requirements.

2. The laboratory recommends sending chain-of-custody specimens by overnight shipment.

Specimen Minimum Volume: 0.45 g (approximately 0.5 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	21 days
	Refrigerated	21 days
	Ambient	72 hours

CPT Code Information:

80307

80324 (if appropriate)

80359 (if appropriate)

80353 (if appropriate)

80361 (if appropriate)

80365 (if appropriate)

80349 (if appropriate)

DSM5X 62722 Drugs of Abuse Screen 5, Chain of Custody, Meconium

Specimen Requirements:

Container/Tube: Chain of Custody Meconium Kit (T653) includes the specimen containers, seals, and documentation required

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen

Additional Information:

1. Specimen that arrives with a broken seal does not meet the chain-of-custody requirements.

2. The laboratory recommends sending chain-of-custody specimens by overnight shipment.

Specimen Minimum Volume: 0.45 g (approximately 0.5 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	21 days
	Refrigerated	21 days
	Ambient	72 hours

CPT Code Information:

80307

80324 (if appropriate)

80359 (if appropriate)

80353 (if appropriate)

80361 (if appropriate)

80365 (if appropriate)

80349 (if appropriate)

83992 (if appropriate)

DASM4 Drugs of Abuse Screen, Meconium 4
60553

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Specimen Minimum Volume: 0.45 g (approximately 0.5 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	21 days
	Refrigerated	21 days
	Ambient	72 hours

CPT Code Information:

80307

80324 (if appropriate)

80359 (if appropriate)

80353 (if appropriate)

80361 (if appropriate)

80365 (if appropriate)

80349 (if appropriate)

DASM5 Drugs of Abuse Screen, Meconium 5

60250

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Specimen Minimum Volume: 0.45 g (approximately 0.5 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	21 days
	Refrigerated	21 days
	Ambient	72 hours

CPT Code Information:

80307

80324 (if appropriate)

80359 (if appropriate)

80353 (if appropriate)

80361 (if appropriate)

80365 (if appropriate)

80349 (if appropriate)

83992 (if appropriate)

FD10S Drugs of Abuse Screen, Serum
75352

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot 4 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 1.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

DBMD Duchenne/Becker Muscular Dystrophy, DMD Gene, Large
58125

Deletion/Duplication Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal

specimen.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year /Ambient/ Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81161-DMD (dystrophin) (eg, Duchenne/Becker muscular dystrophy) deletion analysis and duplication analysis, if performed

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88235-Tissue culture for amniotic fluid (if appropriate)

88240-Cryopreservation (if appropriate)

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells (if appropriate)

DUCK
82708

Duck Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FDME
57926

Duck Meat IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

DYS2 92121 Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 8

86256 AAGTS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86255 APBCS (if appropriate)

86256 APBTS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

84182 MP1BS (if appropriate)

86256 PC2TS (if appropriate)

ECADB 603211 E-Cadherin Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time

TECHONLY	Ambient (preferred)
	Refrigerated

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

ECAD
70423

E-Cadherin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

EOIBD
620120

Early Onset Monogenic Inflammatory Bowel Disease (IBD) Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1.Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2.A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions.

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

ESYC
82721

Eastern Sycamore, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ECHNG
616731

Echinococcus Antibody, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

86682

ECMP 618696 Eculizumab Monitoring Panel, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. Suggest discontinuing natalizumab at least 4 weeks prior to testing for eculizumab quantitation in serum. Patient should consult the healthcare provider who prescribed this drug to determine if discontinuation is an option. If not, ok to proceed with testing while taking natalizumab.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: 2 Plastic vials

Specimen Volume: 2 mL Serum in **2 separate plastic vials** (see Collection Instructions)

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

3. Centrifuge at 4 degrees C.

4. Aliquot serum into two separate plastic vials, each containing 1 mL.

5. Within 30 minutes of centrifugation, freeze specimen. Serum must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 1.25 mL total in 2 separate plastic vials containing 0.5 mL (ECULI) and 0.75 mL (EAH50)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days
Serum Red	Frozen	14 days

CPT Code Information:

80299

86161

ECULI
65676

Eculizumab, Serum

Specimen Requirements:

Patient Preparation: Suggest discontinuing natalizumab at least 4 weeks prior to testing for eculizumab quantitation in serum. Patient should consult the healthcare professional who prescribed this drug to determine if discontinuation is an option. If not, ok to proceed with testing while taking natalizumab**.**

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Immediately after specimen collection, place the tube on wet ice.

3. After specimen has clotted on wet ice, centrifuge at 4 degrees C and aliquot serum into a plastic vial.

4. Freeze specimen within 30 minutes of centrifugation. Specimen must be placed on dry ice if not frozen immediately.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

80299

EGFRS 614665 EGFR Gene, Targeted Mutation Analysis, 51 Mutation Panel, Tumor

Specimen Requirements:

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded (FFPE) tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Source: FFPE tissue, cell block (prepared from fine-needle aspirate, or pleural fluid)

Slides: 1 Hematoxylin and eosin-stained and 5 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

5 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears; smears stained with Pap stain are not acceptable)

Slides: 2 Slides

Collection Instructions: Submit 2 slides stained with Diff Quik and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.
2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Formalin-fixed, paraffin-embedded tissue block or Slides: see Specimen Required
Cytology smear stained with Diff Quik: >1000 cells

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81235-EGFR (epidermal growth factor receptor) (eg, non-small cell lung cancer) gene analysis, common variants (eg, exon 19 LREA deletion, L858R, T790M, G719A, G719S, L861Q)

88381-Microdissection, manual

EGGPF 610705 Egg Comprehensive Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL; if needed, 0.5 mL for every 5 additional allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 3

86008 x 2

EGWTP 610706 Egg White Component Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.75 mL; if needed, 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 1

86008 x 2

FEGWH Egg White IgG

57584

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FEWG4 Egg White IgG4
57529

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

EGG Egg White, IgE, Serum
82872

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FEWHG Egg Whole IgG
57530

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FEGYK
57582

Egg Yolk IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

YOLK
82753

Egg Yolk, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

EGGP
82477

Eggplant, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

EDSGG
617253

Ehlers-Danlos Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81408 x2

81479

81479 (if appropriate for government payers)

EHRCP 81480 Ehrlichia Antibody Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86666 x 2

EHRC 81478 Ehrlichia chaffeensis (HME) Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86666

EPCRB 618301 Ehrlichia/Anaplasma, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred).

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87468

87484

87798 x 2

87999 (if appropriate for government payers)

EHBAP 608396 Ehrlichia/Babesia Antibody Panel, Immunofluorescence, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86666 x 2

86753

FPCEL 75901 Elastase, Pancreatic, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 10 to 12 hours, recommended

2. Medications that affect pancreatic activity should be discontinued, if possible, for at least 48 hours prior to specimen collection.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. As soon as possible after collection, centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Freeze immediately.

3. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

83520

ELDR
82392

Elder, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume:

0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ELPSR
113632

Electrolyte Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	24 hours

CPT Code Information:

80051-Electrolyte Panel (if all 4 are performed)

82435-Chloride (if all 4 are not performed)

84295- Sodium (if all 4 are not performed)

84132-Potassium (if all 4 are not performed)

82374-Bicarbonate (if all 4 are not performed)

EM
70316

Electron Microscopy, Varies

Specimen Requirements:

Specimen Type: Fixed wet tissue

Supplies: Electron Microscopy Kit (T660)

Container/Tube: Electron Microscopy Kit or leak-proof container

Specimen Volume: Entire specimen

Collection Instructions: Collect specimen according to the instructions in Electron Microscopy Procedures of Handling Specimens for Electron Microscopy. **Do not place on ice, dry ice, or freeze**.

Additional Information:

1. PATHC / Pathology Consultation may be added if deemed necessary by the reviewing pathologist.

2. **Liver/gastrointestinal and hair shaft specimens are not acceptable**. Testing will be canceled if one of these specimen types is received.

For neuronal ceroid lipofuscinosis (NCL) testing only

Specimen Type: Whole blood

Container/Tube: Green top (sodium heparin) or yellow top (ACD solution B)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Additional Information: If test indication is for NCL, whole blood may be submitted in lieu of fixed wet tissue. This is only applicable for a presumptive diagnosis of NCL; **whole blood specimens submitted for any other reason will be rejected**.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
EM	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88348

ELM
82672

Elm, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

EMR
113366

EM, Renal Biopsy (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88348

EMICZ
610074

Emicizumab, Modified One Stage Assay Factor VIII, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation: For at least 12 hours before specimen collection, the patient should not be given infusions of factor VIII concentrates.

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	42 days

CPT Code Information:

80299

FENC 90087 Encainide (Enkaid), ODE and MODE

Specimen Requirements:

Submit only 1 of the following:

Plasma

Draw blood in a green-top (sodium heparin) tube(s). Plasma gel tube is not acceptable. Spin down and send 1 mL of sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s). Serum gel tube is not acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

ENS2 92116 Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin (IVIg) treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 24

86341

0432U

83516

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86255 AK5CS (if appropriate)

86256 AK5TS (if appropriate)

86256 AMPIS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN2TS (if appropriate)

86256 AN3TS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

86256-GABIS (if appropriate)

86256 IG5TS (if appropriate)

86256 K11TS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86255 GL5CS (if appropriate)

86256 GL5TS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86256 NMDIS (if appropriate)

84182 PC1BS (if appropriate)

86256 PC1TS (if appropriate)

86256 PC2TS (if appropriate)

84182 PCTBS (if appropriate)

86256 PCTTS (if appropriate)

86256 PDETS (if appropriate)

86255 SP7CS (if appropriate)

86256 SP7TS (if appropriate)

86255 T46CS (if appropriate)

86256 T46TS (if appropriate)

ENC2 92117 Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 26

86341

0432U

83516

84182 AGNBC (if appropriate)

86256 AGNTC (if appropriate)

86255 AK5CC (if appropriate)

86256 AK5TC (if appropriate)

86255 AINCC (if appropriate)

86256 AMPIC (if appropriate)

86256 APHTC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 AN2TC (if appropriate)

86256 AN3TC (if appropriate)

86256 CRMTC (if appropriate)

84182 CRMWC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

86256 IG5TC (if appropriate)

86256 K11TC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 GL5CC (if appropriate)

86256 GL5TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86255 NFHCC (if appropriate)

86256 NIFTC (if appropriate)

86255 NFLCC (if appropriate)

86256 NMDIC (if appropriate)

84182 PC1BC (if appropriate)

86256 PC1TC (if appropriate)

86256 PC2TC (if appropriate)

84182 PCTBC (if appropriate)

86256 PCTTC (if appropriate)

86256 PDETC (if appropriate)

86255 SP7CC (if appropriate)

86256 SP7TC (if appropriate)

86255 T46CC (if appropriate)

86256 T46TC (if appropriate)

MPS1R 622360 Endogenous Mucopolysaccharidosis Type I (IDUA [Alpha-L-Iduronidase]) Biomarker Reflex, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell filter paper, local newborn screening card, or postmortem screening card.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Completely fill at least 2 circles on filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	30 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	13 days	FILTER PAPER

CPT Code Information:

82542

MPS1B 622354 Endogenous Mucopolysaccharidosis Type I (IDUA [Alpha-L-Iduronidase]) Biomarker, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell filter paper, local newborn screening card, or postmortem screening card.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Completely fill at least 2 circles on filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish

Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	30 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	13 days	FILTER PAPER

CPT Code Information:

82542

MPS2R 622363 Endogenous Mucopolysaccharidosis Type II (I2S [Iduronate-2-Sulfatase]) Biomarker Reflex, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell filter paper, local newborn screening card, or postmortem screening card.

Specimen Volume: 2 Blood spots

Collection Instructions

1. An alternative blood collection option for patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Completely fill at least 2 circles on filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	30 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	13 days	FILTER PAPER

CPT Code Information:

82542

MPS2B 622357 Endogenous Mucopolysaccharidosis Type II (I2S [Iduronate-2-Sulfatase]) Biomarker, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell filter paper, local newborn screening card, or postmortem screening card.

Specimen Volume: 2 Blood spots

Collection Instructions

1. An alternative blood collection option for patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Completely fill at least 2 circles on filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	30 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	13 days	FILTER PAPER

CPT Code Information:

82542

ESTUF 35851 Endometrial Stromal Tumors (EST), 7p15 (JAZF1), 6p21.32 (PHF1), 17p13.3 (YWHAE) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 3 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2

88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

EMA
9360

Endomysial Antibodies, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

86231

86231-titer (if appropriate)

EMAT
65091

Endomysial Antibodies, IgA, Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see EMA / Endomysial Antibodies, IgA, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

86231

EMAIG
608880

Endomysial Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

86231-screen

86231-titer (if appropriate)

FENDI
75913

Endothelin I

Specimen Requirements:

Patient Preparation:

Patient should NOT be on any ACTH, Corticosteroids, or hypertension medications, if possible, for at least 48 hours prior to collection of specimen.

Submit only one of the following:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: 12x75 mm screw capped vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a plain, red-top or serum-gel tube(s).

2. Centrifuge and immediately aliquot 3 mL of serum into a plastic vial.

3. Send frozen.

Specimen Type: Plasma

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: 12x75 mm screw capped vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a lavender top (EDTA) tube(s).

2. Centrifuge and immediately aliquot 3 mL plasma into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	180 days
	Refrigerated	7 days

CPT Code Information:

83520

EGPL
82704

English Plantain, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ELF
623424

Enhanced Liver Fibrosis (ELF), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial. **Specimens that have not been aliquoted will be rejected.**

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days
	Ambient	48 hours

CPT Code Information:

81517

EHOLG
616732

Entamoeba histolytica Antibody, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

86753

FEHAG 75350 Entamoeba histolytica Antigen, EIA

Specimen Requirements:

Preferred Specimen Type: Unpreserved stool

Supplies: Sterile stool container

Container/Tube: Sterile stool container

Specimen Volume: 2 g

Specimen Stability Information: Frozen

Collection Instructions: Collect 2 grams of fresh unpreserved stool in sterile container. Send specimen frozen.

NOTE: Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: 1 gram

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	7 days
	Refrigerated	48 hours

CPT Code Information:

87337

STL
8098

Enteric Pathogens Culture, Feces

Specimen Requirements:

Patient Preparation: Patient **should not** use barium or bismuth for 7 to 10 days before specimen collection.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Fecal Swab

Supplies: Copan Fecal Swab (T995)

Container/Tube: Culture transport swab (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Remove swab from packaging.

2. Collect a small amount of stool by inserting all the tip of the flocked swab into the stool sample and rotate.

3. Within 2 hours of collection transfer swab into the tube with preservative medium and visually check to ensure MAX fill line is not exceeded.

4. Mix stool specimen against side of tube to evenly disperse.

5. Break swab off into the tube.

6. Place screw cap in tube and tighten.

7. Shake vial until sample homogenous.

8. Place vial in a sealed plastic bag.

Specimen Stability Information: Refrigerated 3 days (preferred)/Ambient 2 days

Acceptable:

Specimen Type: Preserved Feces

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Stability Information: Ambient 4 days (preferred)/Refrigerated 4 days

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies (preferred)	
	Ambient	
	Refrigerated	

CPT Code Information:

87045-Enteric Pathogens Culture, Stool-with isolation and preliminary examination

87046 x 3-Stool Culture Aerobic Bacteria, each

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147 x 3-Serologic Agglut Method 1 Ident (if appropriate)

87147 x 4-Serologic Agglut Method 3 Ident (if appropriate)

87150 Carbapenem resistance genes (if appropriate)

ENTP 89893 Enterovirus, Molecular Detection, PCR, Plasma

Specimen Requirements:

Submit a raw clinical sample (not a culture isolate) for enterovirus polymerase chain reaction .

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Screw-capped, sterile container

Specimen Volume: 1 mL

Collection Instructions: Centrifuge blood collection tube and aliquot plasma into plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87498

LENT 80066 Enterovirus, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit a raw clinical sample (**not a culture isolate**) for enterovirus testing.

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Pericardial, peritoneal

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Submit specimen from collection vial 2.

2. Do not centrifuge.

Specimen Type: Swab

Sources: Dermal, eye, rectal, genital, nasopharyngeal, oropharyngeal, throat, nasal, or urethral

Supplies:

-Culturette (BBL Culture Swab) (T092)

-E-Swab (T853)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT, M4, M5, Bartels, or Jiangsu) and E-Swab or Culturette

Specimen Volume: Entire specimen

Collection Instructions:

1. Rectal swab must have no visible fecal matter

2. Place swab back into multimicrobe media.

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or

washing, pleural fluid, sputum, or tracheal aspirate

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Collection Instructions: Do not centrifuge.

Specimen Minimum Volume: Body and respiratory fluids: 0.5 mL; Spinal fluid: 0.3 mL; Swab: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87498

EDN
618211

Eosinophil Derived Neurotoxin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Within 12 hours of collection, centrifuge and aliquot serum into a plastic vial. Serum cannot sit on either gel or cells for longer than 12 hours.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	21 days

CPT Code Information:

83520

EOSU1 Eosinophils, Random, Urine

610552

Specimen Requirements:

Container/Tube: Unstained cytospin slide

Specimen Volume: 1 slide

Collection Instructions:

1. Collect a random urine specimen.

2. Centrifuge 10 mL of a random urine collection.

3. Pour off supernatant. Assess urine sediment for presence of white blood cells (WBC). If no WBC are seen, the eosinophil test cannot be performed. If WBC are present, prepare slides using the centrifuged sediment.

4. The unstained slide should be prepared within 2 hours of collection. Only an unstained slide will be accepted. **Any other specimen, including the raw urine specimen, will be rejected**.

5. Centrifuge urine in cytospin centrifuge at speed of 750 rpm for 5 minutes.

6. Slides need to be labeled with patient name and date of birth

7. Guidelines for slide preparation. Use the sediment to prepare slides as follows:

a. 150 mcL of sample for 1 to 3 WBC/high power field (hpf)

b. 100 mcL of sample for moderate to normal sediment (<50 cell/hpf)

c. 50 mcL of sample for heavier sediment (>50 cells/hpf)

d. 25 mcL for packed fields of sediment

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Ambient (preferred)	14 days	SLIDE
	Frozen	14 days	SLIDE
	Refrigerated	14 days	SLIDE

CPT Code Information:

85999

FEPHD 90109 Ephedrine, Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top sodium heparin tube(s), **plasma gel tube is not acceptable.** Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

If specimen is not light protected foil wrap specimen to protect from light.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 3 mL of serum refrigerated in a plastic vial.

If specimen is not light protected foil wrap specimen to protect from light.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	14 days	LIGHT PROTECTED
	Frozen	180 days	LIGHT PROTECTED
	Ambient	72 hours	LIGHT PROTECTED

CPT Code Information:

80324

G0480 (if appropriate)

EPUR 82854 Epicoccum purpurascens, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SPBX
70599

Epidermal Nerve Fiber Density Consultation, Varies

Specimen Requirements:

Preferred:

Specimen Type: Skin punch biopsy

Supplies: Epidermal Nerve Fiber Density Kit (T696)

Source: Distal leg, mid-thigh, dorsal foot, or lower abdomen

Container/Tube: Epidermal Nerve Fiber Density Kit (**required**, no substitutions accepted)

Collection Instructions:

1. The standard biopsy for evaluating distal small fiber sensory neuropathy includes two 3-mm skin punch biopsies from the **same side of the body**.

2. Prepare specimen per instructions on the Epidermal Nerve Fiber Density Preparation Instructions (T703).

Acceptable:

Specimen Type: PGP 9.5-reacted slides

Additional Information:

1. At least one slide reacted with PGP 9.5, using a PGP 9.5 protocol for visualizing epidermal nerve fibers, is required.

2. At least one hematoxylin and eosin-stained slide and one Congo red-stained slide are optional.

Additional Information: Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: PGP 9.5-reacted slides **and** tissue block

Note: Visualization of epidermal nerve fibers cannot be done on paraffin blocks.

Additional Information:

1. At least one slide reacted with PGP 9.5, using a PGP 9.5 protocol for visualizing epidermal nerve fibers, is required.

2. Tissue block may **only** be used to create hematoxylin and eosin-stained slides and Congo red-stained slides.

Additional Information: Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88305 (if appropriate)

88313 (if appropriate)

88321 (if appropriate)

88323 (if appropriate)

88323-26 (if appropriate)

88325 (if appropriate)

88348 (if appropriate)

88356 (if appropriate)

88342 (if appropriate)

FEPI
57960

Epidermophyton floccosum IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

EPS2
92118

Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin (IVIg) treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 21

86341

0432U

83516

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86255 AK5CS (if appropriate)

86256 AK5TS (if appropriate)

86256 AMPIS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN2TS (if appropriate)

86256 AN3TS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

86256 K11TS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86255 GL5CS (if appropriate)

86256 GL5TS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86256 NMDIS (if appropriate)

86256 PC2TS (if appropriate)

84182 PCTBS (if appropriate)

86256 PCTTS (if appropriate)

86256 PDETS (if appropriate)

86255 T46CS (if appropriate)

86256 T46TS (if appropriate)

EPC2 92119 Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 22

86341

0432U

83516

84182 AGNBC (if appropriate)

86256 AGNTC (if appropriate)

86255 AK5CC (if appropriate)

86256 AK5TC (if appropriate)

86256 AMPIC (if appropriate)

86256 APHTC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 AN2TC (if appropriate)

86256 AN3TC (if appropriate)

86256 CRMTC (if appropriate)

84182 CRMWC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

86256 K11TC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 GL5CC (if appropriate)

86256 GL5TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86256 NMDIC (if appropriate)

86256 PC2TC (if appropriate)

84182 PCTBC (if appropriate)

86256 PCTTC (if appropriate)

86256 PDETC (if appropriate)

86255 T46CC (if appropriate)

86256 T46TC (if appropriate)

EPIP1 Epithelia Panel # 1, Serum

81709

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

EPIP2
81881

Epithelia Panel # 2, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

EMAI
70424

Epithelial Membrane Antigen (EMA) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

EBNA2 71487 Epstein Barr Nuclear Antigen 2 (EBNA2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

EBLPD
619788

Epstein Barr Virus (EBV) Susceptibility and Lymphoproliferative Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

LMP1I 70502 Epstein Barr Virus Latency Membrane Protein 1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SEBV 800033 Epstein-Barr Virus (EBV) Antibody Profile, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86664-EBNA

86665 x 2-VCA, IgG and IgM

EBVE 800748 Epstein-Barr Virus (EBV), IgG Antibody to Early Antigen, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86663

LEBV
800244

Epstein-Barr Virus (EBV), Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Spinal, peritoneal, ascites, pericardial, pleural, thoracentesis, amniotic, or ocular

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Sterile, screw cap, 5-mL aliquot tube

Acceptable: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Respiratory fluid

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or washing, sputum, or tracheal aspirate

Supplies: Sarstedt Aliquot Tube, 5 mL

Container/Tube:

Preferred: Sterile, screw cap, 5-mL aliquot tube

Acceptable: Sterile container

Specimen Volume: 1.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Swab

Sources: Eye and upper respiratory (nasal, throat)

Supplies:

-Culturette (BBL Culture Swab) (T092)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT) and Eswabs

Collection Instructions: Place swab back into multimicrobe media (M4-RT, M4 or M5)

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA)

Specimen Volume: 0.5 mL

Additional Information: Clotted specimens will be rejected.

Specimen Type: Tissue

Sources: Brain, colon, kidney, liver, lung, cornea, etc.

Supplies: M4-RT (T605)

Container/Tube:

Preferred: Multimicrobe medium (M4-RT)

Acceptable: Sterile container containing 1 to 2 mL of sterile saline or multimicrobe medium (M4-RT, M4 or M5)

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue

Specimen Minimum Volume: Fluids: 0.3 mL; Respiratory Specimens: 1 mL; Tissue: 2 x 2-mm biopsy; Swab: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

EBVPV 618306 Epstein-Barr Virus (EBV), Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Fluid

Sources: Spinal fluid, sterile body fluids (peritoneal fluid/ascites, pericardial fluid, pleural fluid/thoracentesis), amniotic, or ocular

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Preferred: Sterile screwcap 5-mL plastic vial

Acceptable: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Fluid

Sources: Respiratory; bronchial washing, bronchoalveolar lavage, nasopharyngeal

aspirate or washing, sputum, or tracheal aspirate

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Sterile screwcap 5-mL plastic vial

Acceptable: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Swab

Sources: Eye and upper respiratory (nasal, throat)

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-M4-RT(T605)

Container/Tube: Multimicrobe media (M4-RT, M4, M5, Bartels, or Jiangsu) and E-Swab or Culturette

Collection Instructions: Place swab back into multimicrobe media.

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) only

Specimen Volume: 0.5 mL

Additional Information: Clotted specimens will be rejected.

Specimen Type: Fresh tissue

Supplies: M4-RT (T605)

Container/Tube:

Preferred: Sterile container containing multimicrobe medium (M4-RT, M4, M5, Bartels, or Jiangsu)

Acceptable: Sterile container containing 1-2 mL of sterile saline

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue. Fixed tissue is **not** acceptable.

Specimen Minimum Volume: Ocular Fluid, Spinal Fluid: 0.3 mL
Sterile body fluids (peritoneal fluid/ascites, pericardial fluid, pleural fluid/thoracentesis): See Specimen Required
Respiratory Specimens: 1 mL
Tissue: 2 x 2-mm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

EBVAB
621373

Epstein-Barr Virus Antibody Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	48 hours

CPT Code Information:

86664-EBNA

86665 x 2-VCA, IgG and IgM

EBVQN 615297 Epstein-Barr Virus DNA Detection and Quantification, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87799

EAEBV 621374 Epstein-Barr Virus Early Antigen, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	48 hours

CPT Code Information:

86663

ERG
70426

ERG Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

REVE2 618830 Erythrocytosis Evaluation, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD solution B)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

83020-26

83020

83021

83789

83068 (if appropriate)

82664 (if appropriate)

88184 (if appropriate)

REVEI 608426 Erythrocytosis Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

REVE0 Erythrocytosis Summary Interpretation (Non-Orderable)
608094

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	

EPO Erythropoietin, Serum
80173

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions:

1. Morning collection, 7:30 a.m. to 12 p.m. is preferred due to diurnal variation. For more information see Cautions.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

82668

E157C Escherichia coli O157:H7 Culture, Feces
606219

Specimen Requirements:

Patient Preparation: Patient **should not** use barium or bismuth for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-*Escherichia coli* O157:H7 Culture, Stool-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147-Serologic Agglut Method 2 Ident (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

ESR1T 616500 ESR1 Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

NOTE: Submit tissue from either local recurrence or metastatic disease collected after

endocrine therapy has been administered (see Clinical Information for more details).

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381 - Microdissection, manual

81479

FFES 91215 Estradiol Free, Serum (includes Estradiol and SHBG)

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel tube

Submission Container/tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Within 45 minutes of collection, centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Freeze immediately after separation and send frozen.

Additional Information: To avoid delays when requesting multiple tests on frozen specimens, submit separate specimens for each test requested.

Specimen Minimum Volume: Serum: 1 mL; Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	180 days
	Ambient	48 hours
	Refrigerated	48 hours

CPT Code Information:

82681-Free Estradiol

84270-Sex Hormone binding globulin

82670-Estradiol

ESTS 8575 Estradiol, Rapid, Immunoassay, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume:

0.5 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Frozen (preferred)	180 days	
	Refrigerated	5 days	
	Ambient	24 hours	

CPT Code Information:

82670

EEST
81816

Estradiol, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (SST/serum gel tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Additional Information: For more information see Steroid Pathways.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82670

UE3
81711

Estriol, Unconjugated, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	14 days	
	Frozen	90 days	

CPT Code Information:

82677

ESTR
70427

Estrogen Receptor Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

ERPR 70589 Estrogen/Progesterone Receptor, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Breast carcinoma

Preferred: A paraffin-embedded tissue block containing in-situ, invasive or metastatic breast carcinoma tissue that has been fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total of 6 to 72 hours and shipped at ambient temperature

Acceptable: 3 unstained sections, containing carcinoma, on charged slides cut at 4 microns <1 month ago and shipped at ambient temperature

Collection Instructions: Submit paraffin-embedded carcinoma tissue

Specimen Type: Non-breast carcinoma

Preferred: A paraffin-embedded tissue block containing carcinoma tissue that has been fixed in 10% neutral buffered formalin and shipped at ambient temperature

Acceptable: 3 unstained sections, containing carcinoma, on charged slides cut at 4 microns <1 month ago and shipped at ambient temperature

Collection Instructions: Submit paraffin-embedded carcinoma tissue

Additional Information:

1. According to the American Society of Clinical Oncology (ASCO)/College of American Pathologists (CAP) guidelines, estrogen/progesterone receptor protein immunohistochemical test results are only valid for nondecalcified, paraffin-embedded specimens fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total time of 6 to 72 hours. Delay to fixation, under- or overfixation may affect these results.

2. Paraffin blocks will be returned with final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360 x 2

ESTF 84230 Estrogens, Estrone (E1) and Estradiol (E2), Fractionated, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST is **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum in plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82671-Estrogens, fractionated

82670-Estradiol (If appropriate)

82679-Estrone (If appropriate)

E1
81418

Estrone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Additional Information: For more information see Steroid Pathways.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82679

ETX
8769

Ethosuximide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, serum may sit on the gel or be aliquoted into a plastic vial.

3. For red top tubes, aliquot serum into a plastic vial immediately after centrifugation.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

80168

ETGX 63418 Ethyl Glucuronide Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-custody kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 5 mL

Collection Instructions: Collect specimen in the provided container, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 5 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80307

G0480

80321 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

ETGC
63421

Ethyl Glucuronide Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days

	Frozen	28 days

CPT Code Information:

G0480

80321 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

ETGR 63419 Ethyl Glucuronide Screen with Reflex, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5 mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80307

ETGSR 616033 Ethyl Glucuronide Screen with Reflex, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

-ADMPU / Addiction Medicine Profile with Reflex, 22 Drug Classes, High Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine

-CSMEU / Controlled Substance Monitoring Enhanced Profile with Reflex, 21 Drug Classes, High Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine

Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80307

ETGS 63420 Ethyl Glucuronide Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5 mL tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80307

ETGL
8749

Ethylene Glycol, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST tubes are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

G0480

80320 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

EOXD
82767

Ethylene Oxide, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ETV6F
63433

ETV6 (12p13.2) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the

success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2

88291 -DNA probe, each (first probe set), Interpretation and report

88271x2 -DNA probe, each; each additional probe set (if appropriate)

88271x1 -DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2 -DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3 -DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52 -Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274 -Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275 -Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

EUCL
82758

Eucalyptus, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FEUCT
75738

Euglobulin Clot Lysis Time

Specimen Requirements:

Specimen Type: Citrated plasma

Collection Container/Type: Light-blue top (sodium citrate)

Specimen Volume: 2 mL

Collection Instructions: Collect blood in 3/2% sodium citrate light blue top tube. Centrifuge within 30 minutes after collection, freeze immediately. Send 2 mL platelet-poor plasma in plastic vial frozen.

Note:

1. Prohibit exercise prior to drawing sample.

2. To avoid release of plasminogen activator, do not massage vein vigorously, pump fist excessively or leave tourniquet in place for a prolonged period.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	21 days

CPT Code Information:

85360

EMAY 82846 Euroglyphus maynei, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

EHOR 82662 European Hornet, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

EVROL
35146

Everolimus, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood immediately before a scheduled dose.

2. Do not centrifuge.

3. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Therapeutic range applies to trough specimens collected immediately prior to a.m. dose.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood EDTA	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

80169

EWSF 35268 Ewing Sarcoma, 22q12 (EWSR1) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc.).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of

the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2

88291 - NA probe, each (first probe set), Interpretation and report

88271x2 -DNA probe, each; each additional probe set (if appropriate)

88271x1 -DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2 -DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3 -DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52 -Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274 -Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275 -Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

EXT2 614548 Exostosin 2 (EXT2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FEBGP 75552 Exotic Bird Panel IgG

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

86001 x4

PANGP 621965 Expanded Pancreatitis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the

cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81223

81404 x2

81405 x2

81479

81479 (if appropriate for government payers)

ESBLS 607828 Extended-Spectrum Beta-Lactamase-Producing Gram-Negative Bacteria Surveillance Culture, Feces

Specimen Requirements:

Patient Preparation: Do not use barium or bismuth before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Specimen Type: Preserved feces

Container/Tube: Cary-Blair or modified Cary-Blair transport system is required.

Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of nonnutritive transport medium containing phenol red as a pH indicator. Submit sample in original Cary Blair medium container (not an aliquot of Cary Blair medium).

Specimen Volume: Representative portion of feces; 1 gram or 5 mL

Collection Instructions:

1. Collect fresh fecal specimen and place 1 gram or 5 mL in container with transport medium.

2. Place feces in preservative within 2 hours of collection.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87081-ESBL GNB surveillance culture

87077-Ident by MALDI-TOF mass spec (if appropriate)

87184-ESBL SUS test charge (if appropriate)

F2ISO
607698

F2-Isoprostanes, Random, Urine

Specimen Requirements:

Patient Preparation: Patient should not take nonsteroidal anti-inflammatory drugs within the 72 hours or aspirin within the 2 weeks prior to specimen collection.

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82542

GLA
622568

Fabry Disease, GLA Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405-GLA (galactosidase, alpha) (eg, Fabry disease), full gene sequence

88233-Tissue culture, skin, or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FA13R
75852

Factor 13 1:1 Mix (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information, see FA13Q / Factor 13, Qualitative, with Reflex to Factor 13 1:1 Mix

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85291

FA13Q
75851

Factor 13, Qualitative, with Reflex to Factor 13 1:1 Mix

Specimen Requirements:

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Platelet-poor plasma

Collection Instructions:

1. Centrifuge immediately and aliquot 2 mL platelet-poor plasma in a plastic vial.

2. Send frozen.

Additional Information: Critical frozen. **Separate specimens must be submitted** when multiple tests are ordered.

Specimen Minimum Volume: Platelet-poor plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85291-Factor 13, Qualitative

85291-Factor 13, 1:1 Mix (if appropriate)

FC13A
70428

Factor 13a Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections

4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FACR8 70430 Factor 8 Related Antigen Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time

TECHONLY	Ambient (preferred)
	Refrigerated

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

AFH 615249 Factor H Autoantibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. After sample has clotted on wet ice, centrifuge at 4 degrees C and aliquot serum into a plastic vial.

3. Freeze specimen within 30 minutes of centrifugation. Sample must be placed on dry ice if not frozen immediately.

Additional Information: If the specimen is to be shared with AHUSD / Atypical Hemolytic Uremic Syndrome Complement Panel, Serum and Plasma, **only serum collected in a red-top tube is acceptable.**

NOTE: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen..

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	28 days

	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

83520

2INHE 607427 Factor II Inhibitor Evaluation, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in 3 plastic vials, each containing 1 mL

Collection Instructions:

1. **Specimen must be collected prior to factor replacement therapy**.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 2 Plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

85210

85335 (if appropriate)

85335 (if appropriate)

85390 (if appropriate)

2AINH 607445 Factor II Inhibitor Profile, Professional Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

2INHT 607438 Factor II Inhibitor Profile, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see 2INHE / Factor II Inhibitor Evaluation, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

F2_IS
7806

Factor II Inhibitor Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

2INHE / Factor II Inhibitor Evaluation, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

APROL / Prolonged Clot Time Profile, Plasma

For information see Coagulation Guidelines for Specimen Handling and Processing.

Specimen Minimum Volume: Platelet-poor plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

9INHE
607425

Factor IX Inhibitor Evaluation, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in 3 plastic vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 2 Plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

85250

85335 (if appropriate)

85335 (if appropriate)

85390 (if appropriate)

9INHT 607436 Factor IX Inhibitor Profile Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see 9INHE / Factor IX Inhibitor Evaluation, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

9AINH 607443 Factor IX Inhibitor Profile, Professional Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

F9_IS 7802

Factor IX Inhibitor Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

9INHE / Factor IX Inhibitor Evaluation, Plasma

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

APROL / Prolonged Clot Time Profile, Plasma

For more information see Coagulation Guidelines for Specimen Handling and Processing.

Specimen Minimum Volume: Platelet poor plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

5BETH
607433

Factor V Bethesda Units, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ALUPP / Lupus Anticoagulant Profile, Plasma

-ALBLD / Bleeding Diathesis Profile, Limited, Plasma

-APROL / Prolonged Clot Time Profile, Plasma

-5INHE / Factor V Inhibitor Evaluation, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

5INHE 607426 Factor V Inhibitor Evaluation, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in 3 plastic vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet

contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 2 Plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

85220

85335 (if appropriate)

85335 (if appropriate)

85390 (if appropriate)

5AINH 607444 Factor V Inhibitor Profile, Professional Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

5INHT 607437 Factor V Inhibitor Profile, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see 5INHE / Factor V Inhibitor Evaluation, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

F5_IS 7808 Factor V Inhibitor Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

5INHE / Factor V Inhibitor Evaluation, Plasma

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

APROL / Prolonged Clot Time Profile, Plasma

For more information see Coagulation Guidelines for Specimen Handling and Processing.

Specimen Minimum Volume: Platelet-poor plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

F5DNA 81419 Factor V Leiden (R506Q) Mutation, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD solution B), light-blue top (sodium citrate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not** aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

81241

GNF7 619089 Factor VII Deficiency, F7 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Whole blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

7INHE 607428 Factor VII Inhibitor Evaluation, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in 3 plastic vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 2 Plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

85230

85335 (if appropriate)

85335 (if appropriate)

85390 (if appropriate)

7AINH 607446 Factor VII Inhibitor Profile, Professional Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

7INHT 607439 Factor VII Inhibitor Profile, Technical Interpretation (Non-

Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see 7INHE / Factor VII Inhibitor Evaluation, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

F7_IS
7810

Factor VII Inhibitor Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

7INHE / Factor VII Inhibitor Evaluation, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

APROL / Prolonged Clot Time Profile, Plasma

For more information see Coagulation Guidelines for Specimen Handling and Processing.

Specimen Minimum Volume: Platelet-poor plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

8INHE
607424

Factor VIII Inhibitor Evaluation, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in 3 plastic vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 2 Plastic vials, each containing 1

mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-Factor VIII Tech Interp

85240-Factor VIII activity assay

85335-Bethesda titer (if appropriate)

85335-Factor VIII inhibitor screen (if appropriate)

85390-Factor VIII Professional Interp (if appropriate)

8AINH 607442 Factor VIII Inhibitor Profile, Professional Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

8INHT 607435 Factor VIII Inhibitor Profile, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see 8INHE / Factor VIII Inhibitor Evaluation, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time

Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

10INE
607429

Factor X Inhibitor Evaluation, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in 3 plastic vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results, as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 2 Plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

85260

85335 (if appropriate)

85335 (if appropriate)

85390 (if appropriate)

10AIH 607447 Factor X Inhibitor Profile, Professional Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

10INT 607440 Factor X Inhibitor Profile, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see 10INE / Factor X Inhibitor Evaluation, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

10_IS 7812 Factor X Inhibitor Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

10INE / Factor X Inhibitor Evaluation, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

APROL / Prolonged Clot Time Profile, Plasma

For more information see Coagulation Guidelines for Specimen Handling and Processing .

Specimen Minimum Volume: Platelet poor plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

11INE 607430 Factor XI Inhibitor Evaluation, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in 3 plastic vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 2 Plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

85270

85335 (if appropriate)

85335 (if appropriate)

85390 (if appropriate)

11AIH 607448 Factor XI Inhibitor Profile, Professional Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26

11INT 607441 Factor XI Inhibitor Profile, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see 11INE / Factor XI Inhibitor Evaluation, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

11_IS 7804

Factor XI Inhibitor Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

11INE / Factor XI Inhibitor Evaluation, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

APROL / Prolonged Clot Time Profile, Plasma

For more information see Coagulation Guidelines for Specimen Handling and Processing .

Specimen Minimum Volume: Platelet-poor plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

FXIII
9068

Factor XIII (13), Screen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ALBLD / Bleeding Diathesis Profile, Limited, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85291

GNF13
619145

Factor XIII Deficiency, F13A1 and F13B Genes, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with whole blood testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 4 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

FRW
82684

False Ragweed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMTT
63032

Familial Variant, Targeted Testing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimen types:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A

maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: Whatman FTA Classic paper, PerkinElmer 226 filter paper, or Blood Spot Collection Card

Specimen Volume: 2 to 5 Blood spots on collection card

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

2. For collection instructions, see Blood Spot Collection Instructions

3. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

4. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:
The applicable CPT code will be applied on a case-by-case basis.

Gene	CPT Code (as appropriate)
Any gene not listed below	81403
APC	81202
AR	81174
BRCA1	81215
BRCA2	81217

CACNA1A	81186
CFTR	81221
CSTB	81190
FXN	81289
G6PD	81248
GJB2	81253
HBA1	81258
HBA2	81258
HBB	81362
MECP2	81303
MLH1	81293
MSH2	81296
MSH6	81299
PALB2	81308
PMP22	81326
PMS2	81318
PTEN	81322
SMN1	81337
TP53	81353

81265-Maternal cell contamination (if appropriate)

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88235-Tissue culture for amniotic fluid (if appropriate)

88240-Cryopreservation (if appropriate)

CMPRE 616069 Family Member Comparator Specimen for Exome Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CMPRG 614611 Family Member Comparator Specimen for Genome Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens for each family member.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80 mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient <24 hours/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

FASC
70431

Fascin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered;

sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

INTC1
617028

Fasting (8 Hours or more) (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

FATF
607701

Fat, Feces

Specimen Requirements:

Patient Preparation:

1. For 3 days prior to and during the collection period:

a. Patient should be on a fat-controlled diet (100-150 g fat per day).

b. No laxatives (particularly mineral oil and castor oil).

c. No synthetic fat substitutes (eg, Olestra) or fat-blocking nutritional supplements.

2. The use of diaper rash ointments will falsely elevate test results. Discontinue use during collection period.

3. Barium interferes with test procedure; a waiting period of 48 hours before stool collection analysis is recommended.

Supplies: Stool Containers - 24, 48, 72 Hour Kit (T291)

Container/Tube: Stool container complies with shipping requirements, do not use other containers.

Specimen Volume:

Preferred: Entire 48-hour or 72-hour collection

Acceptable: Entire 24-hour or a minimum of 5 g from a random collection

Collection Instructions:

1. All containers must be sent together.

2. The entire collection must contain at least 5 g of feces.

3. For a random collection, a minimum of 5 g (do not send entire collection) is required.

4. The number of containers sent should be indicated on the labels (1 of 4, for example).

Additional Information:

1. Patient can store sample at refrigerate temperature during collection period.

2. A separate order and collection should take place if stool, chloride, magnesium, osmolality, phosphorus, pH, or any microbiology testing is desired.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	180 days
	Refrigerated	180 days

CPT Code Information:

82710

HFAOP 608028 Fatty Acid Oxidation Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24

hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FAO
81927

Fatty Acid Oxidation Probe Assay, Fibroblast Culture

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Cultured fibroblasts

Container/Tube: T-75 or T-25 flask

Specimen Volume: 1 Full T-75 flask or 2 full T-25 flasks

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin. Tubes can be supplied upon request (Eagle's minimum essential medium with 1% penicillin and streptomycin).

Specimen Volume: 4-mm Punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Varies	

CPT Code Information:

82017

PFAPC
60466

Fatty Acid Profile, Comprehensive (C8-C26), Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting:

a. **For nutritional assessment: 12 hours, required**

b. For patients with a suspected fatty acid oxidation disorder, prolonged fasting is contraindicated. Collect as close to the patient's next scheduled meal/feeding as possible.

2. Patient **must not** consume any alcohol for 24 hours before the specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	72 hours

CPT Code Information:

82725

FAPCP 82042 Fatty Acid Profile, Comprehensive (C8-C26), Serum

Specimen Requirements:

Patient Preparation:

1. Fasting:

a. **For nutritional assessment: 12 hours, required**

b. For patients with a suspected fatty acid oxidation disorder, prolonged fasting is

contraindicated. Collect as close to the patient's next scheduled meal/feeding as possible.

2. Patient **must not** consume any alcohol for 24 hours before the specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	92 days
	Refrigerated	72 hours

CPT Code Information:

82725

PFAPE
60464

Fatty Acid Profile, Essential, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting:

a. **For nutritional assessment: 12 hours, required**

b. For patients with a suspected fatty acid oxidation disorder, prolonged fasting is contraindicated. Collect as close to the patient's next scheduled meal/feeding as possible.

2. Patient **must not** consume any alcohol for 24 hours before the specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	72 hours

CPT Code Information:

82725

FAPEP
82426

Fatty Acid Profile, Essential, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting:

a. **For nutritional assessment: 12 hours, required**

b. For patients with a suspected fatty acid oxidation disorder, prolonged fasting is contraindicated. Collect as close to the patient's next scheduled meal/feeding as possible.

2. Patient **must not** consume any alcohol for 24 hours before the specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	92 days
	Refrigerated	72 hours

CPT Code Information:

82725

FAPM
81939

Fatty Acid Profile, Mitochondrial (C8-C18), Serum

Specimen Requirements:

Patient Preparation:

1. Fasting:

a. **For nutritional assessment: 12 hours, required**

b. For patients with a suspected fatty acid oxidation disorder, prolonged fasting is contraindicated. Collect as close to the patient's next scheduled meal/feeding as possible.

2. Patient **must not** consume any alcohol for 24 hours before the specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Frozen (preferred)	92 days
	Refrigerated	72 hours

CPT Code Information:

82725

POXP
60468

Fatty Acid Profile, Peroxisomal (C22-C26), Plasma

Specimen Requirements:

Patient Preparation:

1. **Fasting: 12 hours, required**; Infants and small children should have specimen collected just before next feeding/meal

2. Patient **must not** consume any alcohol for 24 hours before the specimen is collected.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL plasma

Collection Instructions: Centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: Plasma: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	15 days

CPT Code Information:

82726

POX
81369

Fatty Acid Profile, Peroxisomal (C22-C26), Serum

Specimen Requirements:

Patient Preparation:

1. **Fasting: 12 hours, required**; Infants and small children should have specimen collected just before next feeding/meal

2. Patient **must not** consume any alcohol for 24 hours before the specimen is collected.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	92 days
	Refrigerated	15 days

CPT Code Information:

82726

MFBNG 617365 FBN1 Full Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81408

FETH2
81880

Feather Panel # 2, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOBT
607700

Fecal Occult Blood, Colorectal Cancer Screen, Qualitative, Immunochemical, Feces

Specimen Requirements:

Supplies: Fecal Occult Blood Test Kit (T1009)

Container/Tube: Fecal Occult Blood Test Kit

Specimen Volume: Specimen must fill the grooved portion of the sample probe

Collection Instructions:

1. Collect a random stool specimen.

2. See Fecal Occult Blood Test Kit package insert for instructions.

3. Specimen must be collected in specific sample vial within 4 hours of defecation.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Fecal	Refrigerated (preferred)	30 days	FOBT
	Ambient	15 days	FOBT

CPT Code Information:

82274

G0328 (if appropriate for government payers)

FELBA
80782

Felbamate (Felbatol), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge the specimen.

3. For red-top tubes, immediately aliquot serum into a plastic vial.

4. For serum-gel tubes, aliquot serum into a plastic vial within 24 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80167

FENR
63061

Fentanyl Screen with Reflex, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

FENS
63060

Fentanyl Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

FENTX
62726

Fentanyl with Metabolite Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of Custody Kit (T282) containing the specimen containers,

seals, and documentation required

Specimen Volume: 5 mL

Collection Instructions: Collect urine specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 5 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 2.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

G0480

80354 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FENTU 89655 Fentanyl with Metabolite Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this procedure.

3. Submitting less than 5 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

G0480

80354 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FENTS
89654

Fentanyl, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.3 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	28 days
	Refrigerated	14 days
	Ambient	72 hours

CPT Code Information:

G0480

80354 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FEEP
82143

Ferret Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FERR1
619953

Ferritin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days
	Ambient	24 hours

CPT Code Information:

82728

FMB
88841

Fetomaternal Bleed, Flow Cytometry, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. **Fill evacuated tube as completely as possible.**

2. **Do not centrifuge.**

3. Invert several times to mix blood.

4. Send whole blood specimen in original tube. **Do not aliquot** as aliquoting into or out of a sample tube can adversely affect test results.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Refrigerated (preferred)	5 days	
	Ambient	5 days	

CPT Code Information:

88184-Flow cytometry, cell surface, cytoplasmic

FMBNY 30320 Fetomaternal Bleed, New York, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Fill evacuated tube as completely as possible.

2. Do not centrifuge.

3. Invert several times to mix blood.

4. Send specimen in original tube. **Do not aliquot** as aliquoting into or out of a sample tube can adversely affect test results.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Refrigerated (preferred)	5 days	
	Ambient	5 days	

CPT Code Information:

88184-Flow cytometry; cell surface cytoplasmic

FGF1F 58124 FGFR1 (8p11.2) Amplification, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc.).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

FGFR2 FGFR2 (10q26.1) Rearrangement, FISH, Tissue

63432

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc.).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291 - DNA probe, each (first probe set), Interpretation and report

88271x2 - DNA probe, each; each additional probe set (if appropriate)

88271x1 - DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2 - DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3 - DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52 - Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274 - Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275 - Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

TFH 619658 FH Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)
-Minimum amount of tumor area: tissue 36 mm(2)
-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.
-Tissue fixation: 10% neutral buffered formalin, not decalcified
-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will

not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81405

MSFGN 620248 Fibrillary Glomerulonephritis Confirmation, Mass Spectrometry, Paraffin Tissue

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Formalin-fixed or B5-fixed, paraffin-embedded tissue block

Collection Instructions:

1. Do not send fixed tissue slides. Testing can only be done on paraffin-embedded tissue blocks.

2. Attach the green pathology address label included in the kit to the outside of the transport container.

Transport Temperature:

Specimen Type	Temperature	Time
AMYLOID	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

82542

88380

FIBAG 606887

Fibrinogen Antigen, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Light-blue top (3.2% sodium citrate at 9:1 ratio)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL plasma

Collection Instructions:

1. Centrifuge and aliquot plasma into plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

85385

CLFIB
602173

Fibrinogen, Clauss, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile or as a reflex. For more information, see:

AATHR / Thrombophilia Profile, Plasma and Whole Blood

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

ALUPP / Lupus Anticoagulant Profile, Plasma

APROL / Prolonged Clot Time Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85384

FIBTP
40937

Fibrinogen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, at -40 degrees C or below.

Additional Information: Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen (preferred)	14 days
	Ambient	24 hours

CPT Code Information:

85384

CULFB
35257

Fibroblast Culture for Biochemical or Molecular Testing, Chorionic Villi/Products of Conception/Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Autopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 4-mm diameter

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. Do not use alcohol or iodine preparations.

4. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15 mL tube containing 15 mL of transport media

Specimen Volume: 20-30 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer the chorionic villi specimen to a Petri dish containing transport medium.

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Type: Products of conception or stillbirth

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 1 cm(3) of placenta (including 20 mg of chorionic villi) **and** a 1 cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions: If a fetus cannot be specifically identified, collect 50 mg villus material or tissue that appears to be of fetal origin.

Additional Information: Do not send entire fetus.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Full T-25 flasks

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin. Tubes of culture media can be supplied upon request (Eagle's minimum essential medium with 1% penicillin and streptomycin).

Specimen Volume: 4-mm punch

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88233

88240

FGFRC Fibroblast Growth Factor Receptor 1, Immunostain, 71483

Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

23FBG
620249

Fibroblast Growth Factor-23 (FGF23), In Situ Hybridization

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

5 Unstained, positively charged glass slides; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-Primary

PRKAF
64777

Fibrolamellar Carcinoma, 19p13.1 (PRKACA) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Container/Tube: Formalin-fixed, paraffin-embedded tumor tissue block

Specimen Type: Slides

Slides/Slide Count: Three consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Two consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Transport Temperature:

Specimen Type	Temperature	Time

Tissue	Ambient (preferred)
	Refrigerated

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274-w/modifier 52 -Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

MSFNG 615305 Fibronectin Glomerulopathy Confirmation, Mass Spectrometry

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Formalin-fixed, paraffin-embedded kidney tissue block

Collection Instructions: Do not send fixed tissue slides. Testing can only be done on paraffin-embedded tissue blocks.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

82542

88380

FIBRO 38292 FibroTest-ActiTest, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 3 mL

Collection Instructions:

1. Centrifuge and aliquot serum into an amber vial within 2 hours of collection.

2. Centrifuged serum must be light protected within 4 hours of collection. It is acceptable to draw the blood and then protect it from light after centrifugation as long as it is within 4 hours of collection.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	14 days	LIGHT PROTECTED
	Ambient	24 hours	LIGHT PROTECTED

CPT Code Information:

81596

3FBLN 609271 Fibulin 3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FIGE 57916 Fig (Ficus carica) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FACT 61620 Filamentous-Actin (F-actin) Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

FFAG4 57875 Filaria Antibody (IgG4)

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86682

FIL
9232

Filaria, Blood

Specimen Requirements:

Container/Tube: Light-blue top (3.2% sodium citrate)

Specimen Volume: 2.7 mL

Collection Instructions: Certain microfilariae have a nocturnal or diurnal periodicity, therefore the blood collection should be timed to correspond with the release of microfilariae in the peripheral circulation. For the agents of lymphatic filariasis (*Wuchereria bancrofti* and the *Brugia* species), blood should be collected between 10 p.m. and 2 a.m., whereas for detection of *Loa loa,* blood should be collected between 10 a.m. and 2 p.m.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood Na Cit	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

87015

87210

FINCH
82146

Finch Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FANT
82698

Fire Ant, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FBSH
82735

Firebush (Kochia), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FFSPG
57927

Fish and Shellfish Panel IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 12

FLEC
9243

Flecainide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80181

FLI1
70432

FLI-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FLNDR
57895

Flounder (Bothidae/Pleuronectidae Fam) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FCINS 88467 Flow Cytometry Interpretation, 16 or More Markers (Bill Only)

Specimen Requirements:

Only to be ordered by pathology consultant.

This test is for billing purposes only.

CPT Code Information:

88189

FLT 19739 FLT3 Mutation Analysis, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collections Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot**.

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred) 7 days/Refrigerate 7 days

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 2 mL

Collections Instructions:

1. Invert several times to mix bone marrow.

2. Send specimens in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 7 days/Refrigerate 7 days

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection

2. Label specimen as extracted DNA and source of specimen

2. Provide volume and concentration of DNA on the label.

Specimen Stability Information: Frozen (preferred)/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, Bone marrow: 1 mL; Extracted DNA: 50 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies	7 days

CPT Code Information:

81245-FLT3 (fms-related tyrosine kinase) (eg, acute myeloid leukemia), gene analysis, internal tandem duplication (ITD) variants (ie, exons 14, 15)

81246-FLT3 (fms-related tyrosine kinase 3) (eg, acute myeloid leukemia), gene analysis; tyrosine kinase domain (TKD) variants (eg, D835, I836)

FLUCS 75885 Fluconazole (Diflucan) Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Complete and submit with specimen: Fungus Testing Lab request form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as an etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999 - Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen source.)

87188 - mould - MIC microdilution or agar dilution (if appropriate)

87186 - yeast - MIC microdilution or agar dilution (if appropriate)

FFLRO 91795 Flunitrazepam Confirmation, Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable**. Spin down and send 5 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	48 hours

CPT Code Information:

80346

FL
8641

Fluoride, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial. Glass tubes **are not** acceptable.

Specimen Minimum Volume: Plasma: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Heparin	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

82735

PROLX
80458

Fluphenazine (Prolixin), Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. Centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

FFLUR
90091

Flurazepam (Dalmane) and Desalkylflurazepam

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable**. Spin down and send 2 mL sodium heparin plasma frozen in plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 2 mL of serum frozen in plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen	180 days

CPT Code Information:

80346

FFVOX 57731 Fluvoxamine (Luvox)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top sodium heparin tube(s), plasma gel tube is not acceptable. Spin down and send 3mL of plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain, red-top tube(s), serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80332

IAPC 113345 FNA Immediate Adequacy (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88172

IAAPC 113346 FNA Immediate Adequacy Add'l (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88177

RFSGS 618114

Focal Segmental Glomerulosclerosis (FSGS) and Nephrotic Syndrome Gene Panel, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81408 x 2

81405 x 2

81406 x 4

81407 x 4

81479

81479 (if appropriate for government payers)

Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81406 x 2

81407

81479 (if appropriate for government payers)

PGXQP 610057 Focused Pharmacogenomics Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81418

0071U (if appropriate)

0072U (if appropriate)

0073U (if appropriate)

0074U (if appropriate)

0075U (if appropriate)

0076U (if appropriate)

AFOLR 620731 Folate Receptor Alpha (FOLR1), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Source: Ovarian tumor

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively-charged" glass slides with 4-microns thick formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

FOL 9198 Folate, Serum

Specimen Requirements:

Patient preparation:

1. Fasting: 8 hours, required

2. **Do not order** on patients who have recently received methotrexate or other folic acid antagonists.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume:0.6 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes aliquot the serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

82746

FSHB 70433 Follicle Stimulating Hormone, Beta Subunit (Beta FSH) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FSH
602753

Follicle-Stimulating Hormone (FSH), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	7 days	
	Frozen	180 days	
	Ambient	24 hours	

CPT Code Information:

83001

FDP1 Food Panel #2, Serum
86207

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FFPG4 Food Panel IgG4 (532)
58090

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 8

FFPII
57850

Food Panel II IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 X 19

FOOD6
81874

Food Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOOD2 Food-Fruit Panel, Serum

81869

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOOD4
81872

Food-Grain Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOOD8
81876

Food-Nut Panel # 1, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOOD1 Food-Nut Panel # 2, Serum
81868

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOOD7 Food-Seafood Panel, Serum
81875

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FRMH
82869

Formaldehyde, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FORAC
75402

Formic Acid, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a plastic vial.

2. Send refrigerated.

Specimen Stability Information: Refrigerate (preferred) 30 days/Ambient 7 days/Frozen 15 months

Specimen Minimum Volume: Serum: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	30 days
	Ambient	7 days
	Frozen	

CPT Code Information:

83921

BFOS 603419 FosB, Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

XL2T 616501 FOXL2 Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381 - Microdissection, manual

81479

FOXP1
70435

FOXP1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FOXP3 FOXP3 Immunostain, Technical Component Only

70436

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FMIL Foxtail Millet, IgE, Serum

82832

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FXS
35428

Fragile X Syndrome, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See

Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal

testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Direct testing of an uncultured specimen may be attempted for this test. Contact the laboratory at 800-533-1710 if direct testing is desired.).

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks is required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies

on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen can be tested from a direct extraction or culture: Contact the laboratory at 800-533-1710 if a direct testing is desired.

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81243

88233 (if appropriate)

88240 (if appropriate)

88235 (if appropriate)

81265 (if appropriate)

81244 (if appropriate)

FUFXS 35427 Fragile X, Follow-up Analysis (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see FXS / Fragile X Syndrome, Molecular Analysis, Varies.

No additional specimen is required. Lab will utilize specimen they already have in the lab for this test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81244

TULG 605952 Francisella tularensis Antibody, IgG, ELISA, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see TULAB / *Francisella tularensis* Antibody, IgM and IgG, ELISA, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

86668

TULAB 605950 Francisella tularensis Antibody, IgM and IgG, ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

86668 x 2

TULI 605953 Francisella tularensis Antibody, IgM and IgG, Technical Interpretation, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see TULAB / *Francisella tularensis* Antibody, IgM and IgG, ELISA, Serum.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

TULM 605951 Francisella tularensis Antibody, IgM, ELISA, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see TULAB / *Francisella tularensis* Antibody, IgM and IgG, ELISA, Serum.

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

86668

NEFA
606892

Free Fatty Acids, Total, Serum

Specimen Requirements:

Patient Preparation:

1. **Fasting: 12 hours, required**

2. Patient must not consume any alcohol for 24 hours before the specimen is collected.

3. Patient should not be receiving therapeutic heparin.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 45 minutes of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Immediately freeze specimen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	7 days

CPT Code Information:

82725

FRTUP 62583 Free Thyroxine Index (FTI), Serum

Specimen Requirements:

Patient Preparation:

1. For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

2. If patient is receiving treatment with lipid-lowering agents containing dextrothyroxine (D-T4), discontinue for 4 to 6 weeks prior to specimen collection.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84479-Thyroxine binding capacity

84436-Thyroxine total

FLARP 64718 Free-Living Amebae, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Spinal fluid

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send vial number 2.

Specimen Type: Tissue: Fresh

Sources: Brain, skin, lung

Container/Tube: Sterile container

Specimen Volume: 5 to 10 mm

Collection Instructions: Submit tissue in a sterile container with 1 mL of sterile saline or minimal essential media (MEM).

Preferred: Paraffin-embedded tissue block:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Brain, skin, lung

Container/Tube: Tissue block

Collection Instructions: Submit a FFPE tissue block to be cut and returned.

Acceptable: Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Brain, skin, lung

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Minimum Volume: Spinal Fluid: 0.3 mL; Tissue: 5 mm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798 x 3

87999 (if appropriate for government payers)

FFRWB 60477 Friedreich Ataxia, Frataxin, Quantitative, Blood

Specimen Requirements:

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Frozen (preferred)	70 days
	Ambient	70 days
	Refrigerated	70 days

CPT Code Information:

83520

AFXN 609751 Friedreich Ataxia, Repeat Expansion Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days /Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks is required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81284

81265-Maternal Cell Contamination (if appropriate)

88233-Fibroblast Culture (if appropriate)

88235-Amniotic Fluid Culture (if appropriate)

88240-Cryopreservation (if appropriate)

PCIFS
113333

Frozen Section, 1st Block (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88331

PCAFS
113334

Frozen Section, Additional Blocks (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88332

FRUCT
81610

Fructosamine, Serum

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days
	Ambient	72 hours

CPT Code Information:

82985

FROS2 Fructose, Qualitative, Semen

92187

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection.

Submit only 1 of the following specimens:

Specimen Type: Semen

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic container

Specimen Volume: Total ejaculate

Collection Instructions: Do not dilute specimen. Freeze specimen at -20 degrees C.

Specimen Type: Seminal plasma

Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. After semen collection, wait 30 to 40 minutes until the semen is liquefied, then centrifuge the semen for 10 minutes at maximum centrifuge speed.

2. Remove top 3/4 of specimen with a pipet and place in a plastic vial. Freeze specimen at -20 degrees C.

3. Discard remainder of centrifuged specimen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Frozen	

CPT Code Information:

82757

FFPG
57932

Fruit Panel IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x16

IHCFH
606334

Fumarate Hydratase Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FBL
50022

Fungal Culture, Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: SPS (sodium polyanethol sulfonate)

Specimen Volume: 5 mL

Pediatric Volume: 3 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. SPS tubes are acceptable but not preferred.

Note: SPS tubes must be clearly labeled as SPS. If label is obscured, sample may be cancelled, as ACD (yellow top) is not an acceptable tube type.

Specimen Minimum Volume: Adults: 1 mL; Pediatrics: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

87103-Blood

87106-Id MALDI-TOF Mass Spec Yeast (if appropriate)

87107-Id MALDI-TOF Mass Spec Fungi (if appropriate)

87107-Fungal identification panel A (if appropriate)

87107-Fungal identification panel B (if appropriate)

87153-D2 fungal sequencing identification (if appropriate)

87153- ITS Fungal Sequencing (if appropriate)

FDERM 87283 Fungal Culture, Dermal

Specimen Requirements:

Note:

-Aseptic techniques should be used when collecting specimens to minimize contamination.

-For optimal recovery of organisms, sufficient clinical material should be collected.

Specimen Type: Hair

Container/Tube: Dry sterile container or specimen collection envelope

Specimen Volume: 10 to 12

Collection Instructions: Using forceps collect affected hairs with base of the shaft intact.

Specimen Type: Nails

Container/Tube: Dry sterile container or specimen collection envelope

Specimen Volume: Entire collection

Collection Instructions:

1. Wipe the nail with 70% alcohol using gauze (not cotton).

2. Clip away a generous portion of the affected area.

3. Collect material or debris from under the nail.

Specimen Type: Skin

Container/Tube: Dry sterile container or specimen collection envelope

Specimen Volume: Entire specimen

Collection Instructions:

1. Cleanse the affected area with 70% alcohol.

2. Gently scrape the surface of the skin at the active margin of the lesion, being careful to not draw blood.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	7 days

CPT Code Information:

87101-Fungal culture, dermal

87106-Id MALDI-TOF Mass Spec Yeast (if appropriate)

87107-Id MALDI-TOF Mass Spec Fungi (if appropriate)

87107-Fungal identification panel A (if appropriate)

87107-Fungal identification panel B (if appropriate)

87153-D2 fungal sequencing identification (if appropriate)

87153-ITS Fungal Sequencing (if appropriate)

FGEN 84389 Fungal Culture, Routine

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Fresh tissue

Container/Tube: Sterile container

Specimen Volume: Pea size

Collection Instructions: Tissue should be placed in small amount of sterile saline or sterile water.

Acceptable:

Specimen Type: Bone marrow

Container/Tube: Sterile container, SPS/Isolator system, or green top (lithium or sodium heparin)

Specimen Volume: Entire collection

Specimen Type: Respiratory specimen

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions: Collect a random urine specimen.

Fresh tissue or body fluid are preferred over a swab specimen. Recovery of mycobacteria from swabs is generally very low yield. Only submit a swab specimen if tissue biopsy or fluid aspiration is not possible.

Specimen Type: Swab

Supplies: BD E-Swab (T853)

Sources: Dermal, ear, mouth, ocular, throat, or wound

Container/Tube: Sterile, screw-capped tube containing Liquid Amies Medium with flocked swab (eg, E-Swab)

Specimen Volume: 1 mL in swab container with swab

Collection Instructions:

1. Before collecting specimen, wipe away any excessive amount of secretion and discharge, if appropriate.

2. Obtain secretions or fluid from source with sterile flocked swab. **Paranasal sinus collections must use a nasopharyngeal flocked swab.**

3. Place flocked swab in sterile, screw-capped tube containing 1 mL of Liquid Amies Medium.

4. If smear and culture are requested or both a bacterial culture and fungal culture are requested, collect a second swab to maximize test sensitivity. Submit each swab in a separate sterile, screw-capped tube with 1 mL of Liquid Amies Medium.

Specimen Minimum Volume: Bone marrow or body fluid: 1 mL; Cerebrospinal fluid: 0.5 mL;

Respiratory specimen: 1.5 mL; All other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87102-Fungal culture, routine

87106-Id MALDI-TOF Mass Spec Yeast (if appropriate)

87107-Id MALDI-TOF Mass Spec Fungi (if appropriate)

87107-Fungal identification panel A (if appropriate)

87107-Fungal identification panel B (if appropriate)

87153-D2 fungal sequencing identification (if appropriate)

87153-ITS Fungal Sequencing (if appropriate)

87176-Tissue processing (if appropriate)

FVAG 5184 Fungal Culture, Vaginal

Specimen Requirements:

Specimen Type: Swab

Source: Vaginal secretions

Container/Tube: Culture transport swab (noncharcoal)

Collection Instructions:

1. Before collecting specimen, wipe away any excessive amount of secretion and discharge.

2. Obtain secretions from the mucosal membrane of the vaginal vault with a sterile swab.

3. If smear and culture are requested or both a bacterial culture and fungal culture are requested, collect a second swab to maximize test sensitivity.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87102-Fungal culture, vaginal

87106-Id MALDI-TOF Mass Spec Yeast (if appropriate)

87107-Id MALDI-TOF Mass Spec Fungi (if appropriate)

87107-Fungal identification panel A (if appropriate)

87107-Fungal identification panel B (if appropriate)

87153-D2 fungal sequencing identification (if appropriate)

87153-ITS Fungal Sequencing (if appropriate)

FUNA 45196 Fungal Ident Panel A (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87107

FUNB 45205 Fungal Ident Panel B (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87107

D2F
45079

Fungal Sequencing Identification (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87153

FS
84390

Fungal Smear, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Fresh tissue

Container/Tube: Sterile container

Specimen Volume: Pea size

Collection Instructions: Tissue should be placed in small amount of sterile saline or sterile water.

Acceptable:

Specimen Type: Bone marrow

Container/Tube: Sterile container, SPS/Isolator system, or green top (lithium or sodium heparin)

Specimen Volume: Entire collection

Specimen Type: Respiratory specimen

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions: Collect a random urine specimen.

Fresh tissue or body fluid are preferred over a swab specimen. Recovery of mycobacteria from swabs is generally very low yield. Only submit a swab specimen if tissue biopsy or fluid aspiration is not possible.

Specimen Type: Swab

Sources: Dermal, ear, mouth, ocular, throat, or wound

Supplies: BD E-swab (T853)

Container/Tube: Sterile, screw-capped tube containing Liquid Amies Medium with flocked swab (eg, E-Swab)

Specimen Volume: Flocked swab in 1 mL of Liquid Amies Medium

Collection Instructions:

1. Before collecting specimen, wipe away any excessive amount of secretion and discharge, if appropriate.

2. Obtain secretions or fluid from source with sterile flocked swab.

3. Place flocked swab in sterile, screw-capped tube containing 1 mL of Liquid Amies Medium.

4. If smear and culture are requested or both a bacterial culture and fungal culture are requested, collect a second swab to maximize test sensitivity. Submit each swab in a separate sterile, screw-capped tube with 1 mL of Liquid Amies Medium.

Specimen Minimum Volume: Bone marrow, body fluid, or respiratory specimen: 0.2 mL; Any other specimen type: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87206

87176-Tissue processing (if appropriate)

FUNBL 57873 Fungitell, BAL

Specimen Requirements:

Specimen Type: Bronchoalveolar lavage

Container/Tube: Sterile container

Specimen Volume: 1 to 3 mL

Collection Instructions:

1. Collect 1 to 3 mL bronchoalveolar lavage (BAL) in a sterile container.

2. Send frozen

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Lavage	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

87449

FUNBW 57872 Fungitell, bronch wash

Specimen Requirements:

Collect 1-3 mL Bronchial Wash in a sterile screw top tube, ship frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bronchial Washing	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	4 days

CPT Code Information:

87449

FUNSF 57871 Fungitell, CSF

Specimen Requirements:

Container/Tube: Sterile, leak-proof container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 1 to 3 mL of spinal fluid (CSF) in a sterile, leak-proof container.

2. Send 2 mL refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

87449

FFURO
91119

Furosemide (Lasix)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 1 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

FUSI
70434

FUS Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: Formalin-fixed, paraffin-embedded tissue block

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FUSM
82750

Fusarium moniliforme, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

FCDX1
75841

Gabapentin Confirmation (Qualitative), Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80171

GABA
80826

Gabapentin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST is **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80171

FCDUE 75840 Gabapentin, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

FGABA 51115 Gabapentin, Urine

Specimen Requirements:

Collect 10 mL random urine without preservatives. Ship refrigerated in a plastic container.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307

GDU 89301 Gadolinium, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 0.35 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

GDS
89299

Gadolinium, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered **a specimen should not be collected for at least 96 hours.**

Supplies: Metal Free Specimen Vial (T173)

Collection Container/Tube: Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.35 mL Serum

Collection Instructions:

1. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into Mayo metal-free, screw-capped vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipette into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

83018

GDUCR 615338 Gadolinium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, a specimen should not be collected for 96 hours..

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5.35 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83018

82570

GDCU
615339

Gadolinium/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see GDUCR / Gadolinium/Creatinine Ratio, Random, Urine.

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5.35 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

GATOL 62440 Galactitol, Quantitative, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82542

GALCW 606270 Galactocerebrosidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

GALK
8628

Galactokinase, Blood

Specimen Requirements:

Multiple whole blood tests for galactosemia can be performed on 1 specimen. Prioritize order of testing when submitting specimens. For a list of tests that can be ordered together see Galactosemia-Related Test List.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin) or yellow top (ACD)

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	10 days
	Ambient	72 hours

CPT Code Information:

82759

KM55
621066

Galactose Deficient IgA1 (KM55) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

GALP
83638

Galactose, Quantitative, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial

Specimen Minimum Volume: Plasma: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Frozen (preferred)	365 days
	Ambient	20 days
	Refrigerated	20 days

CPT Code Information:

82760

GALTP
80341

Galactose-1-Phosphate Uridyltransferase Biochemical Phenotyping, Erythrocytes

Specimen Requirements:

Multiple whole blood tests for galactosemia can be performed on 1 specimen. Prioritize order of testing when submitting specimens. For a list of tests that can be ordered together see Galactosemia-Related Test List.

Patient Preparation

Fasting: 4 hours, preferred not required

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	28 days
	Ambient	14 days

CPT Code Information:

82664

82775

GALT
8333

Galactose-1-Phosphate Uridyltransferase, Blood

Specimen Requirements:

Multiple whole blood tests for galactosemia can be performed on 1 specimen. Prioritize order of testing when submitting specimens. For a list of tests that can be ordered together see Galactosemia-Related Test List.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin) or yellow top (ACD)

Specimen Volume: 5 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	28 days
	Ambient	14 days

CPT Code Information:

82775

GAL1B
622838

Galactose-1-Phosphate, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, Revvity 226 filter paper, or blood

collected in tubes containing EDTA and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 91 days/Refrigerate 91 days/Frozen 91 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin) or yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 48 hours/Frozen 14 days

Specimen Minimum Volume:

Blood Spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

84378

GAL1P
80337

Galactose-1-Phosphate, Erythrocytes

Specimen Requirements:

Multiple whole blood tests for galactosemia can be performed on 1 specimen. Prioritize order of testing when submitting specimens. For a list of tests that can be ordered together, see Galactosemia-Related Test List.

Patient Preparation: Specimens collected following a meal can exhibit postprandial elevations. For infants, collect a specimen immediately prior to feeding to avoid this.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	72 hours

CPT Code Information:

84378

APGAL
609738

Galactose-Alpha-1,3-Galactose (Alpha-Gal) Mammalian Meat Allergy Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 4

86008

ALGAL 609737 Galactose-Alpha-1,3-Galactose (Alpha-Gal), IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

GCT
84360

Galactosemia Reflex, Blood

Specimen Requirements:

Multiple whole blood tests for galactosemia can be performed on one specimen. Prioritize order of testing when submitting specimens. For a list of tests that can be ordered together, see Galactosemia-Related Test List.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin) or yellow top (ACD)

Specimen Volume: 5 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	28 days
	Ambient	14 days

CPT Code Information:

82775

81406 (if appropriate)

GALZ
608016

Galactosemia, GALT Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

4. Multiple whole blood tests for galactosemia can be performed on one specimen. Prioritize order of testing when submitting specimens. For a list of tests that can be

ordered together, see Galactosemia-Related Test List.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a

minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight..

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

GALMP 606343 Galactosemia, GALT Gene, Variant Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or

equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81401

GAL1
606832

GALAD Score, Serum (Non-Orderable

Specimen Requirements:

Only orderable as part of a profile. For more information see HCCGS / Hepatocellular Carcinoma Risk Panel with GALAD Score, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days

	Refrigerated	5 days

GALN3 Galectin-3 Immunostain, Technical Component Only
70438

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GAL3 Galectin-3, Serum
86202

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	365 days
	Refrigerated	24 hours

CPT Code Information:

82777

GBACS 620232 Gamma-Amino Butyric Acid Type A (GABA-A) Receptor Antibody, Cell-Binding Assay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

GBACC 620231

Gamma-Amino Butyric Acid Type A (GABA-A) Receptor Antibody, Cell-Binding Assay, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 1 mL

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

WGSEQ
62982

Gamma-Globin Full Gene Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix whole blood.

2. Send whole blood specimen in the original tube. **Do not aliquot.**

Specimen Stability Information: Refrigerate 30 days(preferred)/Ambient 14 days

Specimen Type: Extracted DNA from whole blood

Container/Tube: 1.5 to 2 mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from whole blood within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood: 1 mL; Extracted DNA: 50 mcL at 50 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479-Unlisted molecular

GGT
8677

Gamma-Glutamyltransferase (GGT), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

82977

FGHSP 58034 Gamma-Hydroxybutyric Acid (GHB), Serum/Plasma

Specimen Requirements:

Submit only 1 of the following specimens:

Serum

Specimen Type: Serum

Container/Tube: red-top tube(s)

Specimen Volume: 5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 5 mL of serum refrigerated in a plastic vial.

Min Vol: 1.2 mL

Plasma

Specimen Type: Plasma

Container/Tube: green-top (sodium heparin)

Specimen Volume: 5 mL

Collection Instructions: Draw blood in a green-top (sodium heparin) tube(s), plasma gel

tube is not acceptable. Spin down and send 5 mL of sodium heparin plasma refrigerated in a plastic vial.

Min Vol: 1.2 mL

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307

FGHSU 58036 Gamma-Hydroxybutyric Acid (GHB), Urine

Specimen Requirements:

Collect 10 mL random urine without preservatives. Ship refrigerated in a plastic container.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307

FGANP 75518 Ganciclovir, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: EDTA (lavender top) or pink top

Submission Container/Tube: Plastic preservative-free vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in EDTA (lavender top) or pink top tube(s). Plasma gel tube is not acceptable. Centrifuge and send 1 mL of plasma refrigerated in a preservative-free plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Frozen	120 days
	Ambient	30 days

CPT Code Information:

80299

FGAGM 58017 Ganglioside (Asialo-GM1, GM1, GM2, GD1a, GD1b, and GQ1b) Antibodies

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Refrigerated (preferred)	14 days

	Frozen	365 days

CPT Code Information:

83516 x 6

GAES 621108 Ganglioside Antibodies Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended to occur prior to initiation of immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516 x5

83520 x4 (if appropriate)

FGGAG 57812 Ganglioside GD1a Antibody (IgG)

Specimen Requirements:

Patient Preparation:

Fasting: Overnight, preferred but not required

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collections Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

83520

GQ1ES 621107 Ganglioside GQ1b Antibody, IgG, ELISA, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended to occur prior to initiation of immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516

FGARG
57634

Garlic IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

GARL
82760

Garlic, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPEPA
89123456

Gastric Pepsin Assay

Specimen Requirements:

Specimen Type: Body fluid

Sources: Tracheal or bronchial secretions

Container/Tube: Standard transport tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a minimum of 1 mL tracheal or bronchial fluid.

2. Transfer to a tightly capped plastic tube and freeze immediately.

3. Ship frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen	30 days

CPT Code Information:

83986

84157

82657

GASTN 70439 Gastrin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

GAST 8512 Gastrin, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required

2. For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

3. For 1 week before specimen collection, if medically feasible, patient **should not** take proton pump inhibitors (omeprazole, lansoprazole, dexlansoprazole, esomeprazole, pantoprazole, and rabeprazole).

4. For at least 2 weeks before specimen collection, patient **should not** take or receive drugs that interfere with gastrointestinal motility (eg, opioids).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. If multiple specimens are collected, submit each vial under a separate order.

2. Label specimens with corresponding collection time.

3. Within 2 hours of collection, centrifuge at refrigerated temperature. Refrigerated centrifugation is preferred but not required.

4. Immediately aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	24 hours

CPT Code Information:

82941

Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin (IVIg) treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 8

86256 AAGTS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86255 APBCS (if appropriate)

86256 APBTS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

84182 MP1BS (if appropriate)

86256 PC2TS (if appropriate)

GIP
63169

Gastrointestinal Pathogen Panel, PCR, Feces

Specimen Requirements:

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube:

Preferred: Specific modified Cary-Blair transport system; see Additional Information for acceptable collection media

Acceptable: Approved Cary-Blair transport system (15 mL of non-nutritive transport medium containing phenol red as a pH indicator)

Specimen Volume: Representative portion of feces

Collection Instructions:

1. Collect fresh fecal specimen and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Submit preserved feces in original container. **Do not aliquot**.

4. **If unpreserved specimens are received, testing will be canceled**.

Additional Information:

If collection media other than those listed is utilized, testing may be canceled. Media listed have been verified for use by Mayo Clinic Laboratories.

Modified Cary-Blair media:

Preferred: Culture and Sensitivity Stool Transport Vial (T058)

Acceptable: Meridian Para-Pak C and S, Cardinal Health Culture and Sensitivity Stool transport Vial

Cary Blair media: Remel Cary-Blair, Remel; Protocol Cary-Blair

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87507

GATA3 70440 GATA Binding Protein 3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

GATA5 619802 GATA-Binding Protein 2, GATA2, Full Gene Analysis, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Blood: 1 mL; Skin biopsy or cultured fibroblasts: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

GATAB 603213 GATA-Binding Protein 3 Immunostain, Technical Component Only, Bone Marrow,

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred: 2 Unstained positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick; B5 fixed, decalcified, paraffin-embedded bone marrow tissue.

Acceptable: B5 fixed, decalcified, paraffin-embedded bone marrow tissue block.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GBA
622596

Gaucher Disease, GBA1 Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

GCDF
70441

GCDFP-15 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

FGPE
57919

Gelatin Porcine IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

GELA
86326

Gelatin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GBETH
607434

General Factor Bethesda Units, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ALBLD / Bleeding Diathesis Profile, Limited, Plasma

-APROL / Prolonged Clot Time Profile, Plasma

-2INHE / Factor II Inhibitor Evaluation, Plasma

-7INHE / Factor VII Inhibitor Evaluation, Plasma

-10INE / Factor X Inhibitor Evaluation, Plasma

-11INE / Factor XI Inhibitor Evaluation, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85335

GENPA
37042

Gentamicin, Peak, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80170

GENRA 37044 Gentamicin, Random, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80170

GENTA 37043 Gentamicin, Trough, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80170

GERB
82545

Gerbil Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GCTF 35270 Germ Cell Tumor (GCT), Isochromosome 12p, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

GCET
70442

Germinal Center B-cell Expressed Transcript 1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

FGHTP
75898

Ghrelin Total (Plasma)

Specimen Requirements:

Patient Preparation:

1. Fasting: 10 to 12 hours

2. Patient should not be on any medications or supplements that may influence cholecystokinin (CCK), glucose, growth hormone, insulin and/or somatostatin levels, if possible, for at least 48 hours prior to specimen collection.

Supplies: GI Preservative (T125)

Collection Container/Tube: GI preservative plasma tube

Submission Container/Tube: Plastic vial

Specimen Volume: 3 to 5 mL plasma

Collection Instructions:

1. Draw 10 mL of blood in a GI preservative plasma tube.

2. As soon as possible, centrifuge specimen in a refrigerated centrifuge.

3. Aliquot 3 to 5 mL plasma into a plastic vial and freeze at -20 degrees C.

4. Ship frozen.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
GI Plasma	Frozen	180 days

CPT Code Information:

83519

GRW
82685

Giant Ragweed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GIAR
80231

Giardia Antigen, Feces

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Preserved feces

Supplies:

-Formalin 10% Buffered Neutral 15 mL (T466)

-Stool Collection Kit, Random (T635)

Container/Tube:

Preferred: Fecal container with 10% buffered formalin preservative

Acceptable: SAF (sodium acetate formalin)

Specimen Volume: 5 grams

Specimen Stability Information: Ambient (preferred) 60 days

Acceptable:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: 5 grams

Specimen Stability Information: Frozen 60 days

Specimen Minimum Volume: 2 grams

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies	

CPT Code Information:

87329

GING
82488

Ginger, IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DGLDN 89031 Gliadin (Deamidated) Antibodies Evaluation, IgG and IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86258 x 2

DAGL 89029 Gliadin (Deamidated) Antibody, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86258

DGGL 89030 Gliadin (Deamidated) Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86258

GFAPP 623425 Glial Fibrillary Acidic Protein (GFAP), Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL plasma

Collection Information: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
EDTA Plasma	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

83520

GFATS 605133 Glial Fibrillary Acidic Protein Alpha Subunit Antibody,

Immunofluorescence Titer Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see PNEFS / Neuroimmunology Antibody Follow-up, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

GFATC
605134

Glial Fibrillary Acidic Protein Alpha Subunit Antibody, Immunofluorescence Titer Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information, see:

DMC2 / Dementia Autoimmune Evaluation, Spinal Fluid

ENC2 / Encephalopathy Autoimmune Evaluation, Spinal Fluid

EPC2 / Epilepsy Autoimmune Evaluation, Spinal Fluid

MAC1 / Autoimmune Myelopathy Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 2 mL

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

GFAP
70443

Glial Fibrillary Acidic Protein Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

FGLIP 91097 Glipizide (Glucotrol)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

LGBWB 602351 Globotriaosylsphingosine, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin) and yellow top (ACD B)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood in original vial. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	72 hours
	Ambient	48 hours

CPT Code Information:

82542

LGBBS
113521

Globotriaosylsphingosine, Blood Spot

Specimen Requirements:

Supplies:

-Card-Blood Spot Collection (Filter Paper) (T493)

-Card Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Blood Spot Collection Card (Filter Paper)

Acceptable: Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper, Munktell filter paper, Postmortem Screening Card or collected with EDTA, sodium heparin, lithium heparin, or ACD B-containing devices

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry completely on filter paper at ambient temperature in a horizontal

position for a minimum of 3 hours.

3. At least 1 spot should be complete,(ie, unpunched).

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	10 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Ambient	10 days	FILTER PAPER

CPT Code Information:

82542

LGB3S 65532 Globotriaosylsphingosine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	48 hours

CPT Code Information:

82542

GBM
8106

Glomerular Basement Membrane Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

GLUCG
70445

Glucagon Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GLP 9358 Glucagon, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Pre-chill lavender top (EDTA) tube at 4 degrees C before collecting the specimen.
2. Draw blood into the pre-chilled tube and process as follows:
   a. Chill filled tube in wet ice for 10 minutes.
   b. Centrifuge in a refrigerated centrifuge or in a pre-chilled centrifuge carrier.
   c. Immediately after centrifugation, aliquot plasma into a plastic vial and freeze.

Specimen Minimum Volume: Plasma: 0.45 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	90 days

CPT Code Information:

82943

GPSYW 113430 Glucopsychosine, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin) or yellow top (ACD B)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood in original vial. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	72 hours
	Ambient	48 hours

CPT Code Information:

82542

GPSY 62236

Glucopsychosine, Blood Spot

Specimen Requirements:

Supplies:

-Card-Blood Spot Collection Card (Filter Paper) (T493)

-Card-Postmortem Screening (Filter Paper) (T525)

Collection Container/Tube:

Preferred: Blood Spot Collection (Filter Paper)

Acceptable: Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper, Munktell filter paper, Postmortem Screening card, or collected with sodium heparin, lithium heparin, ACD-B or EDTA containing devices

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Let blood dry completely on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

2. At least 1 spot should be complete, (ie, unpunched)

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	10 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Ambient	10 days	FILTER PAPER

CPT Code Information:

82542

GPSYP 65632 Glucopsychosine, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin), yellow top (ACD B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL plasma

Collection Instructions:

1. Centrifuge at 4 degrees C, if possible

2. Aliquot plasma into plastic vial. **Do not disturb or transfer the buffy coat layer**.

3. Send frozen

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	65 days

CPT Code Information:

82542

G6PDC 608417 Glucose 6 Phosphate Dehydrogenase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82955

G6PD1 607460 Glucose 6-Phosphate Dehydrogenase Enzyme Activity, Blood

Specimen Requirements:

Collection Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA) or yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not** aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82955

GPI1 607463 Glucose Phosphate Isomerase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not** aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

84087

GPIC 608419 Glucose Phosphate Isomerase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

84087

GLBF 606609 Glucose, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

-Peritoneal dialysate (dialysis fluid)

-Pericardial

-Amniotic fluid+

-Synovial fluid

- Pancreatic cyst

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

82945

GLURA Glucose, Random, Serum

89115

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

82947

GLUR1
609796

Glucose, Random, Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 20 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	72 hours
	Ambient	2 hours

CPT Code Information:

82945

GLSF
152

Glucose, Spinal Fluid

Specimen Requirements:

Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge to remove any cellular material.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

82945

G6PDZ
610053

Glucose-6-Phosphate Dehydrogenase (G6PD) Full Gene Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81249

HEX4 64174 Glucotetrasaccharides, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL urine tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	87 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

82542

82570

GLUT
70446

GLUT-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FGADA 75968 Glutamic Acid Decarboxylase (GAD) Autoantibody

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.4 mL
Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

86341

GD65S 81596 Glutamic Acid Decarboxylase (GAD65) Antibody Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

-SPPS / Stiff-Person Spectrum Disorders Evaluation, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Serum

-PNEFS / Neuroimmunology Antibody Follow-up, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86341

GD65C 84221 Glutamic Acid Decarboxylase (GAD65) Antibody Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

-SPPC / Stiff-Person Spectrum Disorders Evaluation, including Progressive

Encephalomyelitis with Rigidity and Myoclonus, Spinal Fluid

-PNEFC / Neuroimmunology Antibody Follow-up, Spinal Fluid

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86341

GLUTS 70450 Glutamine Synthetase Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GA2P 608029 Glutaric Aciduria Type II Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information**Specimen Stability Information**: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A

maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight..

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture

fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

GSH
608409

Glutathione, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82978

FGLUT
57559

Gluten IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

GLT
82894

Gluten, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GLYCS 606972 Glycine Receptor Alpha1 IgG, Cell Binding Assay, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

0431U

GLYCC 606973 Glycine Receptor Alpha1 IgG, Cell Binding Assay, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 2 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

0431U

GSDGP 608012 Glycogen Storage Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a ski or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FGLMA GlycoMark
91742

Specimen Requirements:

Submit only 1 of the following specimens:

Serum

Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 1 mL of serum ambient in a plastic vial.

Plasma

Draw blood in a lavender-top (EDTA) tube(s). Spin down and send 1 mL of EDTA plasma ambient in a plastic vial.

Specimen Minimum Volume: 0.75 mL Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	7 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

84378

GLYCF 70448 Glycophorin A (CD235a) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GPNMB 623081 Glycoprotein Non-Metastatic Melanoma Protein B

(GPNMB), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GDOM
82847

Glycyphagus domesticus, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GLYP3 Glypican-3 Immunostain, Technical Component Only
70447

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GOAT
82783

Goat Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GMILK
82550

Goat's Milk, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GLDR
82717

Goldenrod, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FGAGA
75564

Golimumab and Anti-Golimumab Antibody, DoseASSURE GOL

Specimen Requirements:

Specimen Type: Serum

Container/Tube: SST or Red

Specimen Volume: 3 mL

Collection Instructions: Draw blood in a serum gel tube(s), plain red-top tube(s) is acceptable. **Serum must be separated from cells within 45 minutes of venipuncture**. Spin down and send 3 mL of serum frozen in a plastic vial.

To avoid delays in turnaround time when requesting multiple tests, **please submit separate frozen specimens for each test requested**.

NOTE: High serum biotin concentrations in patients taking biotin supplements may cause an interference in this

assay. Patients may be advised to stop biotin consumption at least 72 hours prior to sample collection.

Specimen Minimum Volume: 1 mL (Note: This volume does not allow for repeat testing.)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	7 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

80299

82397

FGNDR 75915 Gonadotropin Releasing Hormone (Gn-RH, Luteinizing Hormone-Releasing Hormone LH-RH)

Specimen Requirements:

Patient Preparation: Patient should not be on any steroid, ACTH, gonadotropin, or estrogen medications, if possible, for at 48 hours prior to collection of specimen.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge as soon as possible and aliquot 3 mL serum into a plastic vial.

2. Freeze specimen immediately after separation.

3. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	7 days

CPT Code Information:

83727

GOOS
82714

Goose Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GP210 GP210 Antibody, IgG, Serum
620726

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

LAGGN Granulocyte Antibody Screen, Serum
621630

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Additional Information: Only a specimen collected before a transfusion reaction is acceptable.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	30 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86021 x2

GRANB 70449 Granzyme B Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

FGRPG 57653 Grape IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

GRAP
82800

Grape, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GRFR
82836

Grapefruit, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GRAS1
81706

Grass Panel # 1, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GRAS2
81707

Grass Panel # 2, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GRAS3
81708

Grass Panel # 3, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GAB1
70437

GRB2-Associated Binding Protein 1 (GAB1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

GRFE
82365

Greek Fennel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GPEA
82887

Green Pea, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GPEP Green Pepper, IgE, Serum
82623

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GSTB Green String Bean, IgE, Serum
82610

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ALDR
82671

Grey Alder, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GDF15 64637 Growth Differentiation Factor 15, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood and centrifuge immediately.

2. Aliquot plasma into plastic vial.

3. Do not expose specimen to heat or direct sunlight.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	28 days

CPT Code Information:

83520

GRH 70444 Growth Hormone Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HGH
8688

Growth Hormone, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.6 mL

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. If multiple specimens are collected, submit each vial under a separate order.

3. Label specimens appropriately with the corresponding collection times.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

83003

FGHRH 75910 Growth Hormone-Releasing Hormone (GH-RH, also known as GHRF or GRF)

Specimen Requirements:

Patient preparation: Patient should not be on any medications that may influence pituitary secretion.

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: SST

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge as soon as possible and aliquot 3 mL of serum into a plastic vial.

2. Freeze specimen immediately after separation.

3. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	180 days

CPT Code Information:

83520

GGUM
82479

Guar Gum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GUIN
82706

Guinea Pig Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

GUM
82367

Gum Arabic, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FGUMX Gum Xanthan IgE
57974

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

GUSAB Guselkumab Antibodies, Serum (Non-Orderable)
623122

Specimen Requirements:

Only orderable as part of profile. For more information see GUSAP / Guselkumab Quantitation with Antibodies, Serum.

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83520

GUSAP 623123 Guselkumab Quantitation with Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.1 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

83520

GUS
622837

Guselkumab, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

FHACK 57951 Hackberry (Celtis occidentalis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FHADE 57556 Haddock (Melanogrammus aeglefinus) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

HIBSG
614598

Haemophilus influenzae Type B Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

86684

HAKE
82348

Hake, Fish, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FHALG
57637

Halibut IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

HALI
82633

Halibut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HALO
80339

Haloperidol, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80173

HEPI
82780

Hamster Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HANTG 624187 Hantavirus IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see HANGM / Hantavirus IgM and IgG, ELISA, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86790

HANGM 624185 Hantavirus IgM and IgG, ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86790 x 2

HANTI 624188 Hantavirus IgM and IgG, Technical Interpretation, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see HANGM / Hantavirus IgM and IgG, ELISA, Serum.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	30 days
	Ambient	7 days

HANTM 624186 Hantavirus IgM, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see HANGM / Hantavirus IgM and IgG, ELISA, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86790

HAPT
9168

Haptoglobin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume:1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83010

FHZCP 75565 Hazelnut Component Panel

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

86008 x 4

FHCC1 75576 Hazelnut Component rCor a 1

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

86008

NUTHX 618848 Hazelnut-Food Components, IgE, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see NUTHR / Hazelnut-Food, IgE with Reflex to Hazelnut-Food Components, IgE, Serum.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

86008 x 3

NUTHR 618847 Hazelnut-Food, IgE with Reflex to Hazelnut-Food Components, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

NUTH 82743 Hazelnut-Food, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HAZ
82670

Hazelnut-Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HRTVS
620057

Heartland Virus, RNA, Molecular Detection, PCR, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Sterile container

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a sterile container.

Additional Information: Serum specimens not aliquoted from the serum gel collection tube into a sterile container **will be rejected**.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	24 hours

CPT Code Information:

87798

HRTVC
620056

Heartland Virus, RNA, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Send specimen from collection vial 2.

2. Do not centrifuge or heat inactivate.

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	24 hours

CPT Code Information:

87798

HMUOE 608889 Heavy Metal Occupational Exposure, with Reflex, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 6 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

82175

82300

83825

83655

82570

HMUCR 608899 Heavy Metal/Creatinine Ratio, with Reflex, Random, Urine

Specimen Requirements:

1. For 48 hours before collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 6 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

82175

82300

83825

83655

82570

HMDB 39183 Heavy Metals Screen with Demographics, Blood

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original collection tube. **Do not aliquot**.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

82175

82300

83655

83825

HMU24 Heavy Metals Screen, with Reflex, 24 Hour, Urine

48538

Specimen Requirements:

1. For the 48-hour period prior to the start of collection, as well as during the collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic aliquot container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 6 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

82175

82300

83825

83655

HMHA
45479

Heavy Metals, Hair

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Head (preferred), beard, mustache, chest, pubic

Specimen Volume: 0.2 g

Collection Instructions: Prepare and transport specimen per the instructions in kit or see Collecting Hair and Nails for Metals Testing.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Hair	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

82175

83655

83825

HMNA
31070

Heavy Metals, Nails

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Fingernails or toenails

Specimen Volume: 0.2 g

Collection Instructions:

1. Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

2. Clippings should be taken from all 10 fingernails or toenails.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Nail	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

82175

83655

83825

HEG1
615261

HEG1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

HPYL
70466

Helicobacter pylori (H pylori) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HPSAF 800272 Helicobacter pylori Antigen, Feces

Specimen Requirements:

Collection Container/Tube: Stool container

Submission Container/Tube: Plastic container

Specimen Minimum Volume: 5 g

Collection Instructions: Mix stool well.

Specimen Minimum Volume: 5 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	60 days
	Refrigerated	72 hours

CPT Code Information:

87338

UBT 81590 Helicobacter pylori Breath Test

Specimen Requirements:

Patient Preparation:
1. **Do not administer this test** if this list of instructions is not followed, as test results may be inaccurate:

a. **Do not administer this test if patient is allergic to citric acid.** Note: Product contains aspartame**.**

b. For 2 weeks prior to testing, patients **should not** take proton-pump inhibitors (eg, Prilosec, Prevacid, Aciphex, Protonix, and Nexium), histamine 2-receptor antagonists (H2RA), (eg Pepcid, Tagamet, Axid, or Zantac), or bismuth compounds (eg, Pepto-Bismol).

c**.** For 4 weeks prior to testing, patients **should not** take antibiotics**.**

2. Carafate (sucralfate) does not interfere with this test.

3. Use of antacids does not affect the accuracy of this test**.**

Fasting: 1 hour, required

Supplies: H pylori Breath Kit - Meridian BreathID (T906; fees apply)

Collection Instructions:

1. Do not collect if patient is younger than 3 years.

2. Follow instructions included with kit.

3. Mixing the (13)C-Urea Tablet

a. Dissolve the Citrica and the (13)C-enriched urea tablet in 150 to 200 mL (5.1 to 6.8 oz.) of tap water in the provided drinking cup.

b. Close the lid firmly using both hands. Place fingers over lid and shake thoroughly for a few minutes, until the Citrica Powder and the urea tablet are completely dissolved.

Note: Tiny particles may remain visible after thorough mixing. However, if more substantial particulate matter is still present after five minutes of mixing, discard the solution and repeat the procedure with a new kit.

Specimen Minimum Volume: Bag of "breath" must be full

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Breath	Ambient	14 days	BREATH TEST BAG

CPT Code Information:

83013

HELIS 62769 Helicobacter pylori Culture with Antimicrobial Susceptibilities, Varies

Specimen Requirements:

Preferred:

Specimen Type: Fresh tissue or biopsy

Sources: Gastric or duodenum

Container/Tube: Sterile container

Specimen Volume: 0.5 x 0.2 x 0.2-cm sized piece of tissue

Collection Instructions: Acquire biopsied tissue; moisten with sterile saline

Acceptable:

Specimen Type: Fluid

Sources: Gastric brushings, gastric aspirate

Container/Tube: Sterile container

Specimen Volume: Entire collection or 0.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	48 hours

CPT Code Information:

87081-*Helicobacter pylori* culture

87077-Bacteria identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87176-Tissue processing (if appropriate)

87181-Susceptibility per drug and per organism for drugs not in routine battery (if appropriate)

87186-Antimicrobial Susceptibility, Aerobic Bacteria, MIC-per organism for routine battery (if appropriate)

87150-H pylori + Clarithro Resistance PCR (if appropriate)

HPCR1 607597 Helicobacter pylori with Clarithromycin Resistance Prediction, Molecular Detection, PCR (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

87513

HPFRP 607594 Helicobacter pylori with Clarithromycin Resistance Prediction, Molecular Detection, PCR, Feces

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Helicobacter pylori* DNA is unlikely.

Patient Preparation:

1. For 4 weeks prior to testing, patient **should not** take antibiotics.

2. For 2 weeks prior to testing, patient **should not** take proton pump inhibitors or bismuth compounds.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Specimen Type: Preserved feces

Submission Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of nonnutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect fresh fecal specimen and submit 1 gram or 5 mL in container with transport medium.

2. Place feces in preservative within 2 hours of collection.

3. Place vial in a sealed plastic bag and send ambient or refrigerated. **Specimens sent frozen will be rejected.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Fecal	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

87513

HPRP 607595 Helicobacter pylori with Clarithromycin Resistance Prediction, Molecular Detection, PCR, Tissue

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Helicobacter pylori* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Fresh tissue or biopsy

Sources: Stomach (or duodenum)

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm (3) approximate size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue in a sterile container (without adding anything).

3. Refrigerate or freeze the specimen.

Specimen Stability Information: Refrigerated (preferred) 7 days/Frozen 7 days

Preferred:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded (FFPE) tissue block

Sources: Stomach (or duodenum)

Container/Tube: Tissue block

Collection Instructions: Submit FFPE tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable:

Specimen Type: Formalin-fixed, paraffin-embedded (FFPE) tissue scroll

Sources: Stomach (or duodenum)

Container/Tube: Sterile container for each individual cut section (scroll)

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: Fresh tissue or biopsy: 5 mm(3)
Formalin-fixed paraffin-embedded tissue block: One block
Formalin-fixed paraffin-embedded tissue scroll: Two 10-micron sections

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87513

HPCRP 607596 Helicobacter pylori with Clarithromycin Resistance Prediction, Molecular Detection, PCR, Varies

Specimen Requirements:

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Isolate

Collection Instructions:

1. Perform isolation of *Helicobacter pylori* in culture.

2. *H pylori* isolate must be submitted in pure culture. **Do not submit mixed cultures.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

87513

HELM 82749 Helminthosporium halodes, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FHSSE 57532 Helminthosporium sativum/Drechslera IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

HOLDC 35848 Hematologic Disorders, Chromosome Hold, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin), lavender top (EDTA)

Specimen Volume: 1 to 2 mL

Collection Instructions: Invert several times to mix bone marrow.

Acceptable:

Specimen Type: Blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin), lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Invert several times to mix blood.

Specimen Minimum Volume: Blood: 2 mL
Bone marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

See Individual Components

EXHR 65114 Hematologic Disorders, DNA and RNA Extract and Hold, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: Blood: 8 mL
Bone Marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

EXHD Hematologic Disorders, DNA Extract and Hold, Varies

64779

Specimen Requirements:

Specimen must arrive within 7 days of collection. Collect and package specimen as close to shipping time as possible.

Submit only 1 of the following specimens:

Specimen Type: Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: Blood, Bone Marrow: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	7 days	PURPLE OR PINK TOP/EDTA
	Refrigerated	7 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

Hematologic Disorders, Fluorescence In Situ Hybridization (FISH) Hold, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Standard	

CPT Code Information:

See individual reflex tests

HLLFH 34854 Hematologic Disorders, Leukemia/Lymphoma; Flow Hold, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or Green top (sodium heparin)

Specimen Volume: 10 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as blood.

3. Include 1 to 2 unstained blood smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

4. Include 1 to 2 unstained bone marrow smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Note: A fresh (less than 4 days post-collection), unfixed, nonembedded bone marrow core biopsy, bone or bone lesion is acceptable as an equivalent source for bone marrow aspirate for this test **only in the event of a dry tap** during the bone marrow harvesting procedure. Indicate "dry tap" in performing lab notes or paperwork when submitting this specimen type.

Specimen Type: Fluid

Sources: Serous effusions, pleural, pericardial, or abdominal (peritoneal fluid)

Container/Tube: Body fluid container

Specimen Volume: 20 mL

Collection Instructions:

1. If possible, fluids should be anticoagulated with heparin (1 U/mL of fluid).

2. Label specimen with fluid type.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Additional Information: The volume of fluid necessary to phenotype the lymphocytes or blasts in serous effusions depends upon the cell count in the specimen. Usually, 20 mL of pleural or peritoneal fluid is sufficient. Smaller volumes can be used if there is a high cell count.

Specimen Type: Tissue

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: 5 mm(3) or larger biopsy

Collection Instructions:

1. Send intact specimen (**do not mince**)

2. Specimen cannot be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. An original cytospin preparation (preferably unstained) should be included with the spinal fluid specimen so correlative morphologic evaluation can occur.

2. Label specimen as spinal fluid.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Additional Information: The volume of spinal fluid necessary to phenotype the lymphocytes or blasts depends upon the cell count in the specimen. A cell count should be determined and submitted with the specimen. Usually, 1 to 1.5 mL of spinal fluid is sufficient. Smaller volumes can be used if there is a high cell count. If cell count is less than 10 cells/mcL, a larger volume of spinal fluid may be required. When cell counts drop below 5 cells/mcL, the immunophenotypic analysis may not be successful.

Specimen Minimum Volume: Blood: 3 mL
Bone Marrow: 0.5 mL
Fluid: 5 mL
Tissue: 1 mm(3) or larger biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

P53CA 62402 Hematologic Neoplasms, TP53 Somatic Mutation, DNA Sequencing Exons 4-9, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Acceptable

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA), yellow top (ACD), or green top (heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Additional Information: To ensure minimum volume and concentration of DNA are

met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Stabilize fresh tissue in tissue culture medium or freeze immediately after collection.

Specimen Stability Information: Refrigerated (preferred) 24 hours/ Frozen 10 days

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81352-TP53 (tumor protein 53) (eg, tumor samples), full gene sequence or targeted sequence analysis of >5 exons

HEMMF 614265 Hematologic Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2-3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275x1, 88291x1 - FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

HFWET 616820 Hematopathology Consult, MCL

Specimen Requirements:

Multiple specimens are required to perform testing.

Submit each of the following:

Specimen Type: Bone marrow core biopsy unprocessed

Container/Tube: 10% formalin solution

Collection Instructions:

1. If bone marrow units are sparse or absent or aspirate is a dry tap, make 3 biopsy touch prep slides.

2. Place biopsy core in 10% formalin immediately after collection.

3. Place Parafilm around the container to prevent leaking and exposure to formalin fumes during transport.

Specimen Type: Bone marrow clot unprocessed

Container/Tube: 10% formalin solution

Collection Instructions:

1. Place 0.5 mL bone marrow aspirate in clot tube.

2. After clot has formed, place clot in 10% formalin.

3. Place Parafilm around the container to prevent leaking and exposure to formalin fumes during transport.

Specimen Type: Whole blood

Container/Tube: Transport in plastic slide holders.

Collection Instructions:

1. Prepare 2 good quality smears of even thickness from whole blood EDTA within 8 hours of collection.

2. Submit unstained and unfixed slides.

3. Place slides in a plastic slide holder and place parafilm around the slide holder. Place slides in a separate bag apart from any formalin-fixed clot or core biopsy specimens during transport. If using slide carriers, make sure they have not previously been used to carry fixed slides.

Specimen Type: 2 slides from bone marrow aspirate and 1 slide from roll preparation on bone marrow biopsy (or 3 slides from a roll preparation on bone marrow biopsy)

Container/Tube: Transport in plastic slide holders.

Collection Instructions:

1. Prepare fresh prep slides of good quality (push and squash) made at the time of sample collection of even thickness from aspirate and roll prep slides from bone biopsy. Select 1 push, 1 squash, 1 roll of best quality, and send slides unfixed and unstained.

2. If bone marrow units are sparse or absent or aspirate is a dry tap, make 3 bone biopsy roll prep slides and send all 3 slides unfixed and unstained.

3. Air dry slides.

4. Place slides in a plastic slide holder and place parafilm around the slide holder. Place slides in a separate bag apart from any formalin-fixed clot or core biopsy specimens during transport. If using slide carriers, make sure they have not previously been used to carry fixed slides.

Specimen Type: Bone marrow aspirate in anticoagulant for possible ancillary testing

Container/Tube: Lavender top (EDTA), yellow top (ACD), and green top (sodium heparin)

Specimen Volume:

3 mL in EDTA

6 mL in ACD

6 mL in sodium heparin

Collection Instructions:

1. Aspirate per standard bone marrow collection procedure.

2. Send bone marrow specimens in original tubes. **Do not aliquot.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

85060 (if appropriate)

85097 (if appropriate)

88305 (if appropriate)

88311 (if appropriate)

88313 (if appropriate)

HPCUT 71743 Hematopathology Consultation, Client Embed

Specimen Requirements:

All specimens are required to perform testing.

Additional Information:

All specimens and paperwork must be labeled with:

-Two patient identifiers (patient name, date of birth, medical record number, case number, or pathology ID

-Specimen type

Information on collecting, packaging, and shipping specimens, is available:

-Bone Marrow Core Biopsy, Clot, and Aspirate Collection Guideline

-Assistance with Bone Marrow Collection

Specimen Type: Bone marrow aspirate

Slides: 5

Preferred: Freshly prepared slides made at the time of specimen collection

Acceptable: Slides made from EDTA bone marrow specimen, within 2 hours of collection

Submission Container/Tube: Plastic slide holder

Collection Instructions:

1. Prepare slides of bone marrow aspirate.

2. If bone marrow units are sparse, or absent or aspirate is a dry tap, make biopsy touch prep slides.

3. Air dry slides. **Do not place** on hot plate to dry.

4. Place Parafilm around the slide carriers holding unstained slides to prevent exposure to formalin fumes during transport. Place slides in a separate bag apart from any formalin-fixed clot or core biopsy specimens. If using slide carriers, make sure they have not been used to carry fixed slides previously.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) and yellow top (ACD solution A or B)

Specimen Volume: 2 x 3 mL in EDTA and 2 x 6 mL in ACD solution A or B

Collection Instructions:

1. Aspirate per standard bone marrow collection procedure.

2. Send bone marrow specimens in original tubes. **Do not aliquot**.

Specimen Type: Bone marrow aspirate clot

Container/Tube: Bone marrow clot embedded in paraffin block

Collection Instructions: Process and embed clot in paraffin

Specimen Type: Bone marrow core biopsy

Container/Tube: Fixed biopsy core embedded in paraffin block

Collection Instructions: Process and embed core in paraffin.

Specimen Type: Blood

Slides: 2

Preferred: 2 Freshly prepared fingerstick slides

Acceptable: 2 Slides made from whole blood in EDTA, within 8 hours of collection

Submission Container/Tube: Plastic slide holder

Collection Instructions:

1. Prepare 2 smears of even thickness.

2. Place Parafilm around the slide carriers holding unstained slides to prevent exposure to formalin fumes during transport. Place slides in a separate bag apart from any formalin-fixed clot or core biopsy specimens. If using slide carriers, make sure they have not been used to carry fixed slides previously.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

88321 (if appropriate)

88323 (if appropriate)

88325 (if appropriate)

HPWET 70343 Hematopathology Consultation, MCL Embed

Specimen Requirements:

All specimens are required to perform testing.

Supplies: Bone Marrow Collection Kit (T793)

Additional Information:

All specimens and paperwork must be labeled with:

-Two patient identifiers (patient name, date of birth, medical record number, case number, or pathology ID)

-Specimen type

Information on collecting, packaging, and shipping specimens is available:

-Bone Marrow Core Biopsy, Clot, and Aspirate Collection Guideline

-Assistance with Bone Marrow Collection

Specimen Type: Bone marrow aspirate

Slides: 5

Preferred: Freshly prepared slides made at the time of specimen collection

Acceptable: Slides made from EDTA bone marrow specimen, within 2 hours of collection

Submission Container/Tube: Plastic slide holder

Specimen Volume: 5 Total slides: 2 direct smears and 3 unit prep slides; unfixed and unstained, per unilateral collection

Collection Instructions:

1. Prepare slides of bone marrow aspirate.

2. If bone marrow units are sparse or absent or aspirate is a dry tap, make biopsy touch prep slides.

3. Air dry slides. **Do not place** on hot plate to dry.

4. Place Parafilm around the slide carriers holding unstained slides to prevent exposure to formalin fumes during transport. Place slides in a separate bag apart from any formalin-fixed clot or core biopsy specimens. If using slide carriers, make sure they have not been used to carry fixed slides previously.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) and yellow top (ACD solution A or B)

Specimen Volume: 2 x 3 mL in EDTA and 2 x 6 mL in ACD solution A or B

Collection Instructions:

1. Aspirate per standard bone marrow collection procedure.

2. Send bone marrow specimens in original tubes. **Do not aliquot**.

Specimen Type: Bone marrow aspirate clot

Container/Tube: Bone marrow aspirate clot in 10% formalin

Specimen Volume: 0.5 mL

Collection Instructions:

1. Place 0.5 mL bone marrow aspirate in clot tube.

2. After clot has formed, place clot in 10% formalin.

3. Place Parafilm around the container to prevent exposure.

Specimen Type: Bone marrow core biopsy

Container/Tube: Fixed biopsy core in 10% formalin

Collection Instructions:

1. If bone marrow units are sparse or absent, or aspirate is a dry tap, make biopsy touch prep slides.

2. Place biopsy core in 10% formalin immediately after collection.

3. Fix in 10% formalin for 1 to 2 hours.

4. Place Parafilm around the 10% formalin container to prevent exposure.

Specimen Type: Blood

Slides: 2

Preferred: 2 Freshly prepared fingerstick slides

Acceptable: 2 Slides made from whole blood in EDTA, within 8 hours of collection

Submission Container/Tube: Plastic slide holder

Specimen Volume: 2 Unstained and unfixed slides

Collection Instructions:

1. Prepare 2 smears of even thickness.

2. Place Parafilm around the slide carriers holding unstained slides to prevent exposure to formalin fumes during transport. Place slides in a separate bag apart from any formalin-fixed clot or core biopsy specimens. If using slide carriers, make sure they have not been used to carry fixed slides previously.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

85007 (if appropriate)

85060 (if appropriate)

85097 (if appropriate)

88305 (if appropriate)

88311 (if appropriate)

HMEP 616524 Hemiplegic Migraine With or Without Epilepsy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal

essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing . An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblast

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated <24 hours

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filtration Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient 1 year of age or older is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

2. For collection instructions, see Blood Spot Collection Instructions

3. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

4. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81185

81405

81406 x 2

81407

81408

81479 (if appropriate for government payers)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate) 88240-Cryopreservation (if appropriate)

HEMB 70454 Hemoglobin (Hb) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HBA1C
82080

Hemoglobin A1c, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send specimen in original tube. Do **not** aliquot.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	24 hours

CPT Code Information:

83036

HBEL1
608083

Hemoglobin Electrophoresis Evaluation, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD solution B)

Specimen Volume: 10 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL (this volume will limit reflex testing possibilities); 3 mL if multiplex ligation-dependent probe amplification is needed

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

83020

83021

82664 (if appropriate)

83068 (if appropriate)

83789 (if appropriate)

88184 (if appropriate)

83020-26 (if appropriate)

HBELI 608088 Hemoglobin Electrophoresis Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

HBEL0 608091 Hemoglobin Electrophoresis Summary Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	

CPT Code Information:

83020-26

HPFH 8270 Hemoglobin F Distribution, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-HBEL1 / Hemoglobin Electrophoresis Evaluation, Blood

-MEV1 / Methemoglobinemia Evaluation, Blood

-REVE2 / Erythrocytosis Evaluation, Blood

-THEV1 / Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	14 days

CPT Code Information:

88184

UNHB 9095 Hemoglobin Stability, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile or as a reflex. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-HBEL1 / Hemoglobin Electrophoresis Evaluation, Blood

-THEV1 / Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

-REVE2 / Erythrocytosis Evaluation, Blood

-MEV1 / Methemoglobinemia Evaluation, Blood

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

83068

HGBCE
65039

Hemoglobin Variant, A2 and F Quantitation, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Submit fresh specimen.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	10 days

CPT Code Information:

83020

HGB
801417

Hemoglobin, Blood

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	48 hours
	Ambient	24 hours

CPT Code Information:

85018

THEVI
608425

Hemoglobinopathy Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

83020-26

HAEV1 Hemolytic Anemia Evaluation, Blood
607494

Specimen Requirements:

The following specimens are required for testing:

2 Patient whole blood EDTA specimens

2 Patient whole blood ACD specimens

1 Shipping control whole blood EDTA specimen (collected from a healthy individual)

2 Well-made peripheral blood smears (Wright stained or fixed in absolute methanol)

For complete instructions, refer to the Specimen Collection and Labeling Instructions for Osmotic Fragility Testing of Erythrocytes.

Shipping Control Specimen:

Specimen Type: Whole blood (non-patient)

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a shipping control specimen from a healthy person unrelated to the patient at the same time (or within 4 hours) as the patient specimen.

Note: The shipping control specimen can be collected from a phlebotomist, volunteer or another heathy patient.

2. Clearly handwrite "CONTROL" on the outermost label of the tube.

3. Refrigerate (or place on cold gel pack/small amount of wet ice) specimen immediately after collection.

4. Send shipping control in the original tube. **Do not aliquot**.

5. Keep the shipping control and the patient specimens **together**, either rubber banded or in a bag.

Additional Information:

1. The shipping control and patient specimens **must be handled identically** from the time of collection through receipt in the testing laboratory.

2. **If the shipping control is not sent with the patient specimen, test cancellation is likely.**

3. The shipping control specimen evaluates whether the patient result has been compromised by handling conditions such as temperature, motion, or other transportation interferences, as temperature and handling extremes can adversely impact the integrity of the specimen.

Patient Specimen:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) and yellow top (ACD)

Specimen Volume: Two 4-mL EDTA tubes AND two 6-mL ACD tubes

Collection Instructions:

1. Collect and label patient specimens.

2. Refrigerate (or place on cold gel pack/small amount of wet ice) specimen immediately after collection.

3. Send whole blood specimens in the original tubes. **Do not aliquot**.

4. Keep the shipping control and the patient specimens together, either rubber banded or in a bag.

Additional Information: The shipping control and patient specimens must be handled identically from the time of collection through receipt in the testing laboratory.

Patient Specimen:

Specimen Type: Slides

Container/Tube: Blood smears

Specimen Volume: 2 Well-made peripheral blood smears

Collection Instructions:

1. Prepare 2 peripheral blood smears from the EDTA tube collected from the patient.

2. Either stain the smear with Wright stain or fix the smear with absolute methanol prior to shipping.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: EDTA whole blood patient/Shipping control: 3 mL; ACD whole blood patient: 5 mL; Slides: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Control	Refrigerated	72 hours	PURPLE OR PINK TOP/EDTA
Whole Blood ACD-B	Refrigerated	72 hours	
Whole Blood EDTA	Refrigerated	72 hours	
Whole Blood Slide	Refrigerated		CARTRIDGE

CPT Code Information:

83020-26-Hemolytic Anemia Interpretation

82657-Hexokinase, B

82955-G6PD Enzyme Activity, B

83020-Hemoglobin electrophoresis

83021-High-Performance Liquid Chromatography (HPLC)

83068-Hemoglobin Stability

84087-Glucose phosphate isomerase, B

84220-Pyruvate Kinase Enzyme Activity, B

82657-Adenylate Kinase, B

82657-Phosphofructokinase, B

82657-Phosphoglycerate Kinase, B

82657-Trisephosphate Isomerase, B

85060-Morphology review

85557-Osmotic fragility

88184-Band 3 Fluorescence Staining, RBC

83915-Pyrimidine 5' Nucleotidase

82978-Glutathione, B

85547 (if appropriate)

83789 (if appropriate)

82664 (if appropriate)

88184 (if appropriate)

HAEVI 608427 Hemolytic Anemia Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	72 hours

CPT Code Information:

83020-26

HAEV0 608090 Hemolytic Anemia Summary Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	

F8INV 66205 Hemophilia A F8 Gene, Intron 1 and 22 Inversion Mutation Analysis, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD) or blue top (3.2% sodium citrate)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable if approved by the laboratory. See Additional Information.

Additional Information:

1. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

2. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are performed to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal blood specimen under a separate order number.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Frozen	7 days
	Refrigerated	7 days

CPT Code Information:

81403

F8INP 66206 Hemophilia A F8 Gene, Intron 1 and 22 Inversion Mutation Analysis, Prenatal

Specimen Requirements:

Results will be reported and telephoned or faxed if requested.

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 5-10 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted.

2. Discard the first 2 mL of amniotic fluid. If the culture will be performed in conjunction with chromosome analysis and alpha-fetoprotein, a total of approximately 25 mL to 30 mL will be needed for the combined studies.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated

Additional Information:

1. Place the tubes in a Styrofoam container.

2. Fill remaining space with packing material.

3. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

4. Bloody specimens are undesirable.

5. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

6. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

7. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20-30 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer the chorionic villi specimen to a Petri dish containing transport medium.

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Stability Information: Refrigerated (preferred) <24 hours/Ambient

Additional Information:

1. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

2. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks approximately 90% confluent

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated

Additional Information: **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Amniotic fluid: See Specimen Required
Chorionic villi: 5 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		

CPT Code Information:

81403

F81P 88806 Hemophilia A F8 Gene, Intron 1 Inversion Known Mutation Analysis, Prenatal

Specimen Requirements:

Results will be reported and telephoned or faxed, if requested.

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 5-10 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted.

2. Discard the first 2 mL of amniotic fluid. If the culture will be performed in conjunction with chromosome analysis and alpha-fetoprotein, a total of approximately 25 to 30 mL will be needed for the combined studies.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated

Additional Information:

1. Place the tubes in a Styrofoam container.

2. Fill remaining space with packing material.

3. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

4. Bloody specimens are undesirable.

5. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

6. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

7. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20-30 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer the chorionic villi specimen to a Petri dish containing transport medium.

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Stability Information: Refrigerated (preferred) <24 hours/Ambient

Additional Information:

1. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing, Chorionic Villi/Products of Conception/Tissue. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

2. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks approximately 90% confluent

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated

Additional Information: All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Amniotic fluid: See Specimen Required; Chorionic villi: 5 mg

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

F81B 60555 Hemophilia A F8 Gene, Intron 1 Inversion Known Mutation, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD) or blue top (3.2% sodium citrate)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable if approved by the laboratory. See Additional Information.

Additional Information:

1. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

2. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are performed to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal blood specimen under a separate order number.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Frozen	7 days
	Refrigerated	7 days

CPT Code Information:

81403

F822B 60554 Hemophilia A F8 Gene, Intron 22 Inversion Known Mutation, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD) or blue top (3.2% sodium citrate)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable if approved by the laboratory. See Additional Information.

Additional Information:

1. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

2. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are performed to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal blood specimen under a separate order number.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Frozen	7 days
	Refrigerated	7 days

CPT Code Information:

81403

F822P 89454 Hemophilia A F8 Gene, Intron 22 Inversion Mutation Analysis, Prenatal

Specimen Requirements:

Results will be reported and telephoned or faxed if requested.

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 5 to 10 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted.

2. Discard the first 2 mL of amniotic fluid. If the culture will be performed in conjunction with chromosome analysis and alpha-fetoprotein, a total of approximately 25 to 30 mL will be needed for the combined studies.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated

Additional Information:

1. Place the tubes in a Styrofoam container.

2. Fill remaining space with packing material.

3. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

4. Bloody specimens are undesirable.

5. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

6. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

7. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 30 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer the chorionic villi specimen to a Petri dish containing transport medium.

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Stability Information: Refrigerated (preferred) <24 hours/Ambient

Additional Information:

1. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

2. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 flasks approximately 90% confluent

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated

Additional Information: **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Amniotic fluid: See Specimen Required
Chorionic villi: 5 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		

CPT Code Information:

81403

GNHMA 619103 Hemophilia A, F8 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with whole blood testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81407

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

GNHMB 619117 Hemophilia B, F9 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic vill

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81238

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

GNF11 619131 Hemophilia C (Factor XI Deficiency), F11 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Whole blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

HQ 607706 HemoQuant, Feces

Specimen Requirements:

Patient Preparation: Patient should refrain from ingesting red meat and aspirin-containing products (eg, Excedrin, Aspirin) for 3 days prior to specimen collection.

Collection Container/Tube: Hemoquant Specimen Collection (T134)

Submission Container/Tube: Screw-capped tube

Specimen Volume: 1 g

Collection Instructions: Collect random specimen from a single defecation.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	

CPT Code Information:

84126

UHSD2 620554 Hemosiderin, Random, Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 13 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 12 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	2 hours

CPT Code Information:

83070

FWWE
57956

Hemp Western Water (Acnida tamariscina) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

HEPTP
40938

Heparin Anti-Xa, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. For complete instructions see Coagulation Guidelines for Specimen Handling and

Processing.

2. Centrifuge, aliquot plasma, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

Additional Information: Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen (preferred)	14 days
	Ambient	2 hours

CPT Code Information:

85520

HITIG
86533

Heparin-PF4 IgG Antibody, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Stability Information: Frozen (preferred) 2 years/Refrigerate 7 days

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	

	Refrigerated	7 days

CPT Code Information:

86022

HAIGM
48064

Hepatitis A Virus IgM Antibody, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86709

HAVTA
620407

Hepatitis A Virus Total Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg. hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86708

HEPBC 70451 Hepatitis B Core (HBc) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained

slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

HEPBS 70453 Hepatitis B Surface (HBs) Antigen Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HBGCD 83626 Hepatitis B Surface Antigen for Cadaveric or Hemolyzed Specimens, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

87340

87341 (if appropriate)

HBVQN 65555 Hepatitis B Virus (HBV) DNA Detection and Quantification by Real-Time PCR, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87517

HBIM 9015 Hepatitis B Virus Core IgM Antibody, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic tube.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86705

HBCPR 610462 Hepatitis B Virus Core Total Antibodies Prenatal, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86704

HBCSN
610471

Hepatitis B Virus Core Total Antibodies Screen, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86704

G0499 (if appropriate)

HBC
8347

Hepatitis B Virus Core Total Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86704

CORAB 32111 Hepatitis B Virus Core Total Antibodies, with Reflex to Hepatitis B Virus Core IgM Antibody, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86704

86705 (if appropriate)

HBCRQ 621765 Hepatitis B Virus Core-Related Antigen, Quantitative, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

82397

HEAB 80973 Hepatitis B Virus e Antibody, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86707

HEAG 8311 Hepatitis B Virus e Antigen and Hepatitis B Virus e Antibody, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86707

87350

EAG
80510

Hepatitis B Virus e Antigen, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

87350

FHBG
57618

Hepatitis B Virus Genotyping

Specimen Requirements:

Specimen Type: Plasma

Collection Container/Tube: EDTA (lavender or pink top)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1.Centrifuge and aliquot 2 mL plasma into a plastic vial.

2. Send frozen.

Additional Information: This test may be unsuccessful if the HBV Viral load is less than log 3.0 or 1000 IU/mL of plasma.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	42 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

87912

HBPES
615271

Hepatitis B Virus Past Exposure Panel, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86706

86704

87340

87341 (if appropriate)

G0499 (if appropriate)

HBABY 63137 Hepatitis B Virus Perinatal Exposure Follow-up Panel, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements containing biotin (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86706

86704

87340

87341 (if appropriate)

HBABT 87893 Hepatitis B Virus Surface Antibody Monitor, Post-Transplant, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86317

HBABP 610465 Hepatitis B Virus Surface Antibody Prenatal, Qualitative/Quantitative, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86706

HBBSN 610474 Hepatitis B Virus Surface Antibody Screen, Qualitative/Quantitative, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86706

G0499 (if appropriate)

HBAB 8254 Hepatitis B Virus Surface Antibody, Qualitative/Quantitative, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

86706

HBNTP 35936 Hepatitis B Virus Surface Antigen Confirmation, Prenatal, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see HBAGP / Hepatitis B Virus Surface Antigen Prenatal, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

87341

HBAGP Hepatitis B Virus Surface Antigen Prenatal, Serum
86185

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.9 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Frozen (preferred)	90 days	
	Refrigerated	6 days	
	Ambient	72 hours	

CPT Code Information:

87340

87341 (if appropriate)

HBGSN
610468

Hepatitis B Virus Surface Antigen Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.9 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

87340

G0499 (if appropriate)

87341 (if appropriate)

HBAGQ
621764

Hepatitis B Virus Surface Antigen, Quantitative, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

87467

HBAG
9013

Hepatitis B Virus Surface Antigen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.9 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Refrigerated	6 days
	Ambient	72 hours

CPT Code Information:

87340

87341 (if appropriate)

HCVSP 609748 Hepatitis C Virus (HCV) Antibody Screen Prenatal, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.3 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

86803

G0472 (if appropriate for government payers)

87522 Hepatitis C, quantification (if appropriate)

HCSRN 113122 Hepatitis C Virus (HCV) Antibody Screen with Reflex to HCV RNA, PCR, Asymptomatic, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.3 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

86803

G0472 (if appropriate for government payers)

87522 Hepatitis C, quantification (if appropriate)

HCVDX 113121 Hepatitis C Virus (HCV) Antibody with Reflex to HCV RNA, PCR, Symptomatic, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.3 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

86803

87522 (if appropriate)

HCVQN 97291 Hepatitis C Virus (HCV) RNA Detection and Quantification by Real-Time Reverse Transcription-PCR, Serum

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87522

HCVRP 609749 Hepatitis C Virus (HCV) RNA Detection and Quantification, Real-Time Reverse Transcription-PCR, Prenatal, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87522

HCVQG 603602 Hepatitis C Virus (HCV) RNA Quantification with Reflex to HCV Genotype, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	42 days
	Refrigerated	72 hours

CPT Code Information:

87522

87902 (if appropriate)

87902 (if appropriate)

HCVL
63063

Hepatitis C Virus Antibody Confirmation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.4 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

86804

HCCAD 87858 Hepatitis C Virus Antibody Screen, Cadaveric or Hemolyzed Specimens, Asymptomatic, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86803

G0472 (if appropriate for government payers)

86804 (if appropriate)

HCCDD 58127 Hepatitis C Virus Antibody, Cadaveric or Hemolyzed Specimens, Symptomatic, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86803

86804 (if appropriate)

HCVG 81618 Hepatitis C Virus Genotype, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Additional Information:

1. This test requires a minimum hepatitis C virus viral load of 500 IU/mL within the 30 days preceding collection.

2. Serum specimens previously submitted to other laboratories for non-microbiology tests are **not acceptable** for add-on test requests due to possible sample-to-sample carryover from automation used for those tests.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Frozen (preferred)	42 days	ALIQUOT TUBE
	Refrigerated	72 hours	ALIQUOT TUBE

CPT Code Information:

87902

HCVDR 604410 Hepatitis C Virus Genotypic Antiviral Drug Resistance, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial. **Specimens sent in serum gel tubes will be canceled**.

Specimen Minimum Volume: Serum: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Frozen (preferred)	60 days	ALIQUOT TUBE
	Refrigerated	7 days	ALIQUOT TUBE

CPT Code Information:

87900

87902

87999 (if appropriate for government payers)

AHDV 9209 Hepatitis D Virus (HDV) Total Antibodies with Reflex to HDV RNA Detection and Quantitation, PCR, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Frozen (preferred)	30 days
	Refrigerated	5 days

CPT Code Information:

86692

87523 if needed

HDVQU 621375 Hepatitis D Virus RNA Detection and Quantification, Real-Time RT-PCR, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: 1.0 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Frozen (preferred)	35 days	ALIQUOT TUBE
	Refrigerated	5 days	ALIQUOT TUBE

CPT Code Information:

87523

HEVG 86211 Hepatitis E Virus IgG Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	
	Refrigerated	24 hours

CPT Code Information:

86790

HEVML 61903 Hepatitis E Virus IgM Antibody Confirmation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum SST	Frozen (preferred)	
	Refrigerated	7 days

CPT Code Information:

86790

HEVM 86212 Hepatitis E Virus IgM Antibody Screen with Reflex to Confirmation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	
	Refrigerated	24 hours

CPT Code Information:

86790

HEVQU 62929 Hepatitis E Virus RNA Detection and Quantification, Real-Time RT-PCR, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.8 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Frozen (preferred)	35 days	ALIQUOT TUBE
	Refrigerated	5 days	ALIQUOT TUBE

CPT Code Information:

87799

HCCGS 606585 Hepatocellular Carcinoma Risk Panel with GALAD Score, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days

	Refrigerated	5 days

CPT Code Information:

82107

83951

HEPAT 70456 Hepatocyte Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HNF1B 70461 Hepatocyte Nuclear Factor 1Beta Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HSMWB 65695 Hepatosplenomegaly Panel, Blood

Specimen Requirements:

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin), yellow top (ACD B)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	72 hours
	Ambient	48 hours

CPT Code Information:

82542

HSMBS 601519 Hepatosplenomegaly Panel, Blood Spot

Specimen Requirements:

Supplies:

-Card-Blood Spot Collection (Filter Paper) (T493)

-Card-Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Blood Spot Collection card (Filter Paper)

Acceptable: Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper, Munktell filter paper, Postmortem Screening Card, or collected with EDTA, sodium heparin, lithium heparin, or ACD B-containing devices

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood completely dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. At least 1 spot should be complete (ie, unpunched).

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	10 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Ambient	10 days	FILTER PAPER

CPT Code Information:

82542

HSMP 65694 Hepatosplenomegaly Panel, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin), yellow top (ACD B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions:

1. Centrifuge at 4 degrees C, if possible

2. Aliquot plasma into plastic vial. **Do not disturb or transfer the buffy coat layer**.

3. Send frozen

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Plasma	Frozen	65 days

CPT Code Information:

82542

H2BR 65879 HER2 Amplification Associated with Breast Cancer, FISH, Tissue

Specimen Requirements:

Note: In accordance with College of American Pathologists guidelines, place specimens for *HER2* (*ERBB2*) testing in fixative within one hour of biopsy or resection (cold ischemia time). Specimens should remain in 10% neutral buffered formalin for a minimum of 6 hours to a maximum of 72 hours (formalin fixation time). Do not use decalcification solutions with strong acids.(CAP Accreditation Program. CYG.48932 Fixation - HER2 (ERBB2) Breast Predictive Marker Testing. Cytogenetics Checklist. College of American Pathologists. 08/2023)

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377

H2GE 65880 HER2 Amplification Associated with Gastroesophageal Cancer, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2.Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377

H2MT 65881 HER2 Amplification, Miscellaneous Tumor, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377

HER2I 70457 HER2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

HERBN 70913 HER2, Breast, Quantitative Immunohistochemistry, Automated, No Reflex

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: A paraffin-embedded tissue block containing breast cancer tissue that has been fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total of 6 to 72 hours and shipped at ambient temperature

Acceptable: 2 unstained sections, containing breast carcinoma, on charged slides cut at 4 microns less than 1 month ago and shipped at ambient temperature. Tissue on the slides should have been fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total of 6 to 72 hours.

Submission Container/Tube: Pathology Packaging Kit (T554)

Collection Instructions: Submit paraffin-embedded invasive or metastatic breast carcinoma tissue.

Additional Information:

1. When ordering this test, the following questions, as stated on the order form or presented electronically, must be answered:

a. "Was specimen fixed in 10% NB formalin w/in 1 hour? Yes, No, or Unknown"

b. "Was specimen fixed in 10% NB formalin 6-72 hours? Yes, No, or Unknown"

c. "Tissue was decalcified? Yes, No, or Unknown."

d. "Tumor type? Primary invasive breast carcinoma or metastatic breast carcinoma."

e. "Tumor classification? Invasive breast carcinoma, metastatic breast carcinoma, or micro-invasive breast carcinoma."

2. According to the American Society of Clinical Oncology (ASCO)/College of American Pathologists (CAP) guidelines, HER2 protein immunohistochemical test results are only valid for nondecalcified, paraffin-embedded specimens fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total time of 6 to 72 hours. Under- or overfixation may affect these results.

3. HER2 immunohistochemistry testing on intracystic papillary carcinoma and solid papillary carcinoma, without clearly stating invasive carcinoma, is not appropriate and will be canceled without processing.

4. Paraffin blocks will be returned with final report.

Specimen Minimum Volume: Entire block

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88361

HERMB 620800 HER2, Breast, Semi-Quantitative Immunohistochemistry, Manual with HER2 FISH Reflex

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: A paraffin-embedded tissue block containing breast cancer tissue that has been fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total of 6 to 72 hours.

Acceptable: 5 Unstained sections containing breast carcinoma on charged slides cut at 4 microns less than 1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total of 6 to 72 hours.

Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit paraffin-embedded invasive or metastatic breast carcinoma tissue.

Additional Information:

1. According to the American Society of Clinical Oncology (ASCO)/College of American Pathologists (CAP) guidelines, HER2 protein immunohistochemical test results are only valid for nondecalcified, paraffin-embedded specimens fixed in 10% neutral buffered formalin within 1 hour from surgical collection time and for a total time of 6 to 72 hours. Delay to fixation, or under- or over-fixation may affect these results.(1)

2. HER2 immunohistochemistry testing on intracystic papillary carcinoma and solid papillary carcinoma, without clearly stating invasive carcinoma, is not appropriate and will be reported as indeterminate.

3. Paraffin blocks will be returned with final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

HERGM 70911 HER2, Gastric/Esophageal, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Only formalin-fixed, paraffin-embedded (FFPE) gastric or esophageal adenocarcinoma specimens will be accepted. Fixation in 10% neutral-buffered

formalin is preferred. The performance and quality of immunohistochemical stains in 10% neutral-buffered FFPE tissue depends on proper fixation. It is recommended (not required) for surgical specimens to be fixed between 18 and 24 hours and biopsy specimens between 6 and 8 hours.

If being ordered for prognostic purposes:

Specimen Type: Gastric or esophageal adenocarcinoma

Supplies: Pathology Packaging Kit (T554)

Preferred: Paraffin-embedded tissue block containing invasive gastric or esophageal adenocarcinoma tissue

Additional Information: Paraffin blocks will be returned with final report.

Acceptable: Slides

Specimen Volume: 5

Collection Instructions: 5 Unstained sections, containing gastric or esophageal adenocarcinoma, on charged slides cut at 4 microns less than 1 month ago.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

HERGN 70914 HER2, Gastric/Esophageal, Semi-Quantitative Immunohistochemistry, Manual, No Reflex

Specimen Requirements:

Only formalin-fixed, paraffin-embedded (FFPE) gastric or esophageal adenocarcinoma specimens will be accepted. Fixation in 10% neutral-buffered formalin is preferred. The performance and quality of immunohistochemical stains in 10% neutral-buffered FFPE tissue depends on proper fixation. It is recommended (not required) for surgical specimens to be fixed between 18 and 24 hours and biopsy specimens between 6 and 8 hours.

If being ordered for prognostic purposes:

Specimen Type: Gastric or esophageal adenocarcinoma

Supplies: Pathology Packaging Kit (T554)

Preferred: Paraffin-embedded tissue block containing invasive gastric or esophageal adenocarcinoma tissue

Additional Information: Paraffin blocks will be returned with final report.

Acceptable: Slides

Specimen Volume: 5

Collection Instructions: 5 Unstained sections, containing gastric or esophageal adenocarcinoma, on charged slides cut at 4 microns less than 1 month ago.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

BRGYP 614320 Hereditary Breast/Gynecologic Cancer Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81432

COMCP Hereditary Common Cancer Panel, Varies
614319

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81201

81408 x2

81162

81406 x4

81404

81403

81405 x2

81292

81295

81298

81307

81317

81321

81351

81407

81479

81479 (if appropriate for government payers)

CDHZ
614582

Hereditary Diffuse Gastric Cancer Syndrome, CDH1, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal

specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

ENDCP 614578 Hereditary Endocrine Cancer Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81437

NHEM 618991 Hereditary Erythrocytosis Focused Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with whole blood testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred) < 24 hours/Ambient < 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblast

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) < 24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Whole blood: 1 mL; Skin biopsy, Cultured fibroblast: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

81479

81479 (if appropriate for government payers)

NHEP 619019 Hereditary Erythrocytosis Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with whole blood testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred) < 24 hours/Ambient < 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblast

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) < 24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Whole blood: 1 mL; Skin biopsy, Cultured fibroblast: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

81405

81479

81479 (if appropriate for government payers)

XCP 614318 Hereditary Expanded Cancer Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole Blood1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81292

81295

81298

81162

81201

81307

81317

81321

81351

81404 x 4

81405 x 6

81406 x 7

81407 x2

81408 x 2

81479

81479 (if appropriate for government payers)

CRCGP 614571 Hereditary Gastrointestinal Cancer Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81435

HFET
614573

Hereditary Hemochromatosis, HFE Variant Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81256-*HFE (*hemochromatosis*)* (eg, hereditary hemochromatosis) gene analysis, common variants (C282Y and H63D)

NHHA
619033

Hereditary Hemolytic Anemia Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81443

HHTGG 617295 Hereditary Hemorrhagic Telangiectasia and Vascular Malformations Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 x3

81479

81479 (if appropriate for government payers)

LRCCZ 614583 Hereditary Leiomyomatosis and Renal Cell Cancer Syndrome, FH, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume

must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information : Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that

all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

PANCP 614574 Hereditary Pancreatic Cancer Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal

essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81432

HPGLP 614579 Hereditary Paraganglioma/Pheochromocytoma Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81437

PRS8P Hereditary Prostate Cancer Panel, Varies
614576

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information: 1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81432

RENCP 614575 Hereditary Renal Cancer Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405 x 3

81321

81406 x 2

81404 x 2

81351

81407

81479

81479 (if appropriate for government payers)

GNHTC 619341 Hereditary Thrombocytopenia Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are

met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Blood: 1 mL; Cultured fibroblasts/skin biopsy: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

GNADM 619229 Hereditary Thrombotic Thrombocytopenic Purpura, ADAMTS13 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

THYRP 614580 Hereditary Thyroid Cancer Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81437

WILMP
614577

Hereditary Wilms Tumor Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days /Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that

additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee

that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81351

81405

81479

81479 (if appropriate for government payers)

LHSVZ 800315 Herpes Simplex Virus (HSV) and Varicella-Zoster Virus (VZV), Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-M4-RT (T605)

Sources: Genital, dermal, eye, or throat

Container/Tube: Multimicrobe media (M4-RT)

Specimen Volume: Swab

Collection Instructions: Place swab back into multimicrobe media (M4-RT, M4, or M5).

Additional Information: Source information should include the main anatomical source of collection.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Specimen Type: Fluid

Sources: Pleural, peritoneal, ascites, pericardial, amniotic, or ocular

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or washing, sputum, or tracheal aspirate

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Tissue

Supplies: M4-RT (T605)

Sources: Brain, colon, kidney, liver, lung, etc

Container/Tube: Sterile container containing 1 mL to 2 mL of sterile saline or multimicrobe medium (M4-RT, M4, or M5)

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue in multimicrobe media (M4-RT) or a sterile container with 1 to 2 mL sterile saline.

Specimen Minimum Volume: Body Fluid or Ocular Fluid: 0.3 mL; Respiratory: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87529 x2 HSV-1 and HSV-2
87798-VZV
87999 (if appropriate for government payers)

HSVG 84429 Herpes Simplex Virus (HSV) Type 1- and Type 2-Specific Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86695

86696

LHSVB 802067 Herpes Simplex Virus (HSV), Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87529 x 2

87529 (if appropriate for government payers)

HSVPB 618308 Herpes Simplex Virus (HSV), Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87529 x 2

87529 (if appropriate for government payers)

HSVC 63434 Herpes Simplex Virus (HSV), Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 0.2 mL

Collection Instructions:

1. Submit from second spinal fluid collection vial.

2. Transfer into aliquot vial using sterile technique.

3. Do not centrifuge or heat inactivate.

Additional Information:

1. The high sensitivity of amplification by polymerase chain reaction (PCR) requires the specimen to be processed in an environment in which contamination of the specimen by herpes simplex virus DNA is not likely.

2. Specimens that are received with less than the minimum volume required for all testing requested will be canceled.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87529 x 2

87529 (if appropriate for government payers)

LHSV 800143 Herpes Simplex Virus (HSV), Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Pleural, peritoneal, ascites, pericardial, amniotic, or ocular

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Swab

Sources: Genital, dermal, ocular, nasal, throat, or oral

Supplies: M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT)

Specimen Volume: Entire collection

Collection Instructions: Place swab back into multimicrobe media (M4-RT)

Additional Information: Source information should include main anatomical site of collection.

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or washing, sputum, or tracheal aspirate

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Tissue

Sources: Brain, colon, kidney, liver, lung, etc.

Supplies: M4-RT (T605)

Container/Tube: Sterile container containing 1 mL to 2 mL of sterile saline or multimicrobe medium (M4-RT)

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue.

Additional Information: Source information should include main anatomical site of collection.

Specimen Type: Urine (<1-month old infant)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Minimum Volume: Body or Ocular Fluid: 0.4 mL
Respiratory Specimen: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87529 x 2

87529 (if appropriate for government payers)

HSVPV 618307 Herpes Simplex Virus (HSV), Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Pleural, peritoneal, ascites, pericardial, amniotic, or ocular

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Swab

Sources: Genital, dermal, ocular, nasal, throat, or oral

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-M4-RT (T605),

Container/Tube: Multimicrobe media (M4-RT, M4, M5, Bartels, or Jiangsu) and E-Swab or Culturette

Specimen Volume: Entire collection

Collection Instructions: Place swab back into multimicrobe media.

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or washing, sputum, or tracheal aspirate

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Fresh tissue

Supplies: M4-RT (T605)

Container/Tube: Sterile container containing 1 mL to 2 mL of sterile saline or multimicrobe medium (M4-RT, M4, M5, Bartels, or Jiangsu)

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue. Fixed tissue is **not** acceptable.

Specimen Type: Urine (newborns < or =35 days old)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Minimum Volume: Amniotic or ocular fluid: 0.4 mL; Sterile body fluid (Pleural, peritoneal, ascites, pericardial): 0.5 mL; Respiratory Specimen: 1 mL; Swab, tissue, or urine: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Frozen	7 days

CPT Code Information:

87529 x 2

87529 (if appropriate for government payers)

HSVSW 623119 Herpes Simplex Virus (HSV), Molecular Detection, PCR, Varies

Specimen Requirements:

Specimen Type: Swab

Sources: Genital, dermal, ocular, nasal, throat, or oral

Supplies:

-Culturette (BBL Culture Swab) (T092)

-E-Swab (T853)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT) or Universal Transport Medium (UTM) and E-Swab or Culturette

Specimen Volume: Entire collection

Collection Instructions: Place swab back into multimicrobe media (M4-RT, M4, or M5) or Universal Transport Medium (UTM).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

87529 x 2

87529 (if appropriate for government payers)

HRPSV 70467

Herpes Simplex Virus, I and II (HSV I and II) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FHHV6
91311

Herpes Virus 6 (HHV-6) DNA, Qualitative Real-Time PCR

Specimen Requirements:

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Draw blood in lavender-top (EDTA) tube(s) and send 1 mL of whole blood refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	48 hours

CPT Code Information:

87532

FHV7P 75755 Herpesvirus 7 (HHV-7) DNA, Quantitative Real-Time PCR

Specimen Requirements:

Submit only 1 of the following specimens:

Source is required

Whole Blood

Specimen Type: Whole Blood

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 1 mL

Collection Information: Draw blood in a lavender-top (EDTA) tube(s) and send 1 mL whole blood refrigerated (DO NOT FREEZE).

Stability: Ambient 48 hours; Refrigerated 7 days

Serum

Specimen Type: Serum

Container/Tube: Red-top

Submission Container/Tube: 12x75 mm screw-capped vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain red-top tube(s). Spin down and send 1 mL serum in a plastic, screw-capped vial. Send specimen refrigerated.

Stability: Ambient 48 hours; Refrigerated 7 days; Frozen 30 days

Plasma

Collection Container/Tube: Lavender-top (EDTA)

Submission Container/Tube: 12x75 mm screw-capped vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a lavender-top (EDTA) tube(s). Spin down and transfer 1 mL EDTA plasma into a plastic screw-capped vial. Send specimen refrigerated.

Stability: Ambient 48 hours; Refrigerated 7 days; Frozen 30 days

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	48 hours

CPT Code Information:

87799

FH7AB 75964 Herpesvirus 7 Antibody Panel (IgG, IgM), IFA

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86790 x 2

FHV8P Herpesvirus 8 (HHV-8) DNA, Quantitative Real-Time PCR

75756

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send 1 mL EDTA whole blood refrigerated. Frozen whole blood is **not acceptable**.

Specimen Type: Serum

Collection Container/Tube: Red-top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL serum into a plastic vial.

2. Send refrigerated.

Specimen Type: Plasma

Collection Container/Tube: Lavender-top (EDTA) or PPT (white-top) tube

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL plasma

Collection Instructions:

1. Centrifuge and aliquot 1 mL EDTA plasma into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	48 hours

CPT Code Information:

87799

HERR
82823

Herring, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions:

Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HEXLA 617974 Hexagonal Lupus Anticoagulant, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a reflex. For more information see:

-AATHR / Thrombophilia Profile, Plasma and Whole Blood

-ACBL / Bleeding Diathesis Profile, Comprehensive, Plasma

-ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

-ALBLD / Bleeding Diathesis Profile, Limited, Plasma

-ALUPP / Lupus Anticoagulant Profile, Plasma

-APROL / Prolonged Clot Time Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	60 days

CPT Code Information:

85598

FHEX 75954 Hexagonal Phase Phospholipid (HPP)

Specimen Requirements:

Patient Preparation: Ideally, the patient should not be on anticoagulant therapy. Avoid warfarin (Coumadin) therapy for two weeks prior to the test and heparin, direct Xa, and thrombin inhibitor therapies for about three days prior to testing. **Do not** draw from an arm with a heparin lock or heparinized catheter.

Specimen Type: Citrated plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Draw blood in a blue-top tube containing 3.2% buffered sodium citrate.

2. Mix immediately by gentle inversion at least six times to ensure adequate mixing of the anticoagulant with the blood.

3. Centrifuge for 10 minutes and remove 2/3 of the plasma using a plastic transfer pipette, being careful not to disturb the cells.

4. Transfer to a plastic transport tube, cap, and centrifuge again for 10 minutes.

5. Use a second plastic pipette to remove the plasma, staying clear of the platelets at the bottom of the tube.

6. Transfer the plasma into a plastic vial and freeze immediately.

7. Send frozen.

Note: **Critical frozen**. **Separate specimens must be submitted** when multiple tests are ordered.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	28 days

CPT Code Information:

85598

HK1
607461

Hexokinase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

HKC
608420

Hexokinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

NAGW
8775

Hexosaminidase A and Total Hexosaminidase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

83080 x 2

NAGS
8774

Hexosaminidase A and Total Hexosaminidase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Frozen (preferred)	30 days	
	Refrigerated	7 days	

CPT Code Information:

83080 x 2

NAGR
82943

Hexosaminidase A and Total, Leukocytes/Molecular Reflex, Whole Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

83080 x2

81406 (if appropriate)

81479 (if appropriate)

MUGS
80350

Hexosaminidase A, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, required; Infants and small children should have specimen collected before next feeding/meal

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.15 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	5 days

CPT Code Information:

83080

FSHAG 57950 Hickory Shagbark (Carya ovata) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

HMGA 70460 High Mobility Group A2 (HMGA2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HSCHP 623137 High Sensitivity Clonal Hematopoiesis Panel, Next-Generation Sequencing, Blood

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have

received a hematopoietic stem cell transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81479

Hemoglobin Variant, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-HBEL1 / Hemoglobin Electrophoresis Evaluation, Blood

-MEV1 / Methemoglobinemia Evaluation, Blood

-REVE2 / Erythrocytosis Evaluation, Blood

-THEV1 / Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	10 days

CPT Code Information:

83021

H1083
618598

HIK1083 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

HIPA
9756

Hippuric Acid, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Urine

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 2 mL random urine specimen in a plastic preservative-free urine container at end of shift.

2. Send refrigerate.

NOTE: The acidification of the urine with 1.0 mL acetic acid per 100 mL urine will prolong the stability of the analyte.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days
	Ambient	5 days

CPT Code Information:

82570

83921

81002 (if appropriate)

FHSPL 57533 Histamine Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Immediately place specimen on ice.

2. Centrifugation should be performed within 20 minutes of collection at 900 x g for 10 minutes at 4 degrees C.

3. Carefully remove 1 mL of EDTA plasma from the upper part of the tube and place in a plastic vial.

4. Send frozen.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	28 days

CPT Code Information:

83088

FHI24 75835 Histamine, 24-Hour Urine

Specimen Requirements:

Patient Preparation: Avoid direct sunlight. Patient should refrain from taking allergy causing drugs, antihistamines, oral corticosteroids, and substances which block H2 receptors for at least 24 hours prior to specimen collection.

Specimen Type: Urine

Submission Container/Tube: Plastic, 10-mL tube (T068)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours, with 10 mL 6N HCL preservative.

2. Avoid direct sunlight during the 24-hour collection.

3. Send specimen refrigerated in the plastic, 10-mL urine tube (T068).

4. Collection volume and duration are required.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	48 hours

CPT Code Information:

83088

FHSTW 57368 Histamine, Whole Blood

Specimen Requirements:

Specimen Type: Whole Blood

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood in a green-top tube (sodium or lithium heparin).

2. Aliquot 1 mL well mixed whole blood into a plastic vial.

3. Freeze specimen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
WB Heparin	Frozen	180 days

CPT Code Information:

83088

HG34W
604697

Histone 3.3 G34W (H3F3A G34W) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

HISGT
619586

Histone Genes Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4mm x 4mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens..

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells or a minimum of at least 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81445

HK27M Histone H3 K27M Mutant (H3 K27M) Immunostain, 604999

Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HK36M 604699 Histone H3 K36M Mutant (H3F3 K36M) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HISME 72127 Histone H3 Trimethyl K27 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HIBAG 618715 Histoplasma and Blastomyces Antigen, Enzyme Immunoassay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

87449

UHBAG 621248 Histoplasma and Blastomyces Antigen, Enzyme

Immunoassay, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen, with no preservative.

2. **Do not centrifuge** to remove particulates.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

87449

HISER
621213

Histoplasma Antibody Complement Fixation and Immunodiffusion, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86698 x2

HICSF 621216 Histoplasma Antibody Complement Fixation and Immunodiffusion, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen from collection vial number1.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86698 x2

HBRP 60213 Histoplasma capsulatum/Blastomyces species, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Histoplasma* or *Blastomyces* species DNA is not likely.

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Sources: Body, spinal fluid, bone marrow

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Type: Respiratory

Sources: Bronchoalveolar lavage, bronchial washing, sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Type: Tissue or bone

Container/Tube: Sterile container

Specimen Volume: 5-10 mm

Collection Instructions: Collect a fresh tissue or bone specimen.

Acceptable:

Specimen Type: N-acetyl-l-cysteine-sodium hydroxide (NALC/NaOH)-digested respiratory specimens

Sources: Lavage fluid, bronchial washing, gastric washing, respiratory fluid, sputum, or tracheal secretion

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Minimum Volume: Body fluid or respiratory specimen: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

87798 x 2

HIBC
621305

Histoplasma/Blastomyces Panel, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Collection Instructions: Submit specimen from collection vial number 1.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86698 x2

86612

HIVSP
48393

HIV Antigen and Antibody Prenatal Routine Screen, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Transfer plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

87389

G0475 (if appropriate)

HVPRS 617827 HIV Antigen and Antibody Prenatal Routine Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

87389

G0475 (if appropriate)

86701 (if appropriate)

86702 (if appropriate)

87535 (if appropriate)

87538 (if appropriate)

HV1CD
83628

HIV-1 and HIV-2 Antibodies for Cadaveric or Hemolyzed Specimens, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86703

86701

86702

HVPPS 617828 HIV-1 and HIV-2 Antibody Confirmation and Differentiation Prenatal, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

86701

86702

87535 (if appropriate)

87538 (if appropriate)

HVDIP 601758 HIV-1 and HIV-2 Antibody Confirmation and Differentiation, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: Plasma: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

86701

86702

87535 (if appropriate)

87538 (if appropriate)

87536 (if appropriate)

HVDSP 601759 HIV-1 and HIV-2 Antibody Confirmation and Differentiation, Prenatal, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

86701

86702

87535 (if appropriate)

87538 (if appropriate)

87536 (if appropriate)

HIVDI 62421 HIV-1 and HIV-2 Antibody Confirmation and Differentiation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

86701

86702

87535 (if appropriate)

87538 (if appropriate)

HV1CM 60357 HIV-1 and HIV-2 Antibody Screen for Hemolyzed Specimens, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86703

G0432

86701 (if appropriate)

86702 (if appropriate)

HIVDX 48392 HIV-1 and HIV-2 Antigen and Antibody Diagnostic Evaluation, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

87389

86701 (if appropriate)

86702 (if appropriate)

87536 (if appropriate)

HIVDS 617825 HIV-1 and HIV-2 Antigen and Antibody Diagnostic Evaluation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

87389

86701 (if appropriate)

86702 (if appropriate)

87535 (if appropriate)

87538 (if appropriate)

HVCOP 48341 HIV-1 and HIV-2 Antigen and Antibody Routine Screen, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

87389

G0475

86701 (if appropriate)

86702 (if appropriate)

87536 (if appropriate)

HIVSS 617826 HIV-1 and HIV-2 Antigen and Antibody Routine Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	30 days
	Refrigerated	6 days

CPT Code Information:

87389

G0475 (if appropriate)

86701 (if appropriate)

86702 (if appropriate)

87535 (if appropriate)

87538 (if appropriate)

HIVDR 616052 HIV-1 Genotypic Drug Resistance to Reverse Transcriptase, Protease, and Integrase Inhibitors, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. Centrifuge blood collection tube and aliquot plasma into plastic vial per collection tube manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Freeze aliquoted plasma for shipment.

Additional Information: Specimens submitted for HIV-1 genotyping must contain 1000 copies/mL or more of HIV-1 RNA.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	60 days
	Refrigerated	7 days

CPT Code Information:

0219U

HIVQN 113581 HIV-1 RNA Detection and Quantification, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87536

HIQNP 65567 HIV-1 RNA Detection and Quantification, Prenatal, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87536

PSHIV 65819 HIV-1 RNA Patient Source, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

87536

HIQDR 616917 HIV-1 RNA Quantification with Reflex to Genotypic Drug Resistance to Reverse Transcriptase, Protease, and Integrase Inhibitors, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 3.6 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	55 days
	Refrigerated	5 days

CPT Code Information:

87536

0219U (if appropriate)

HPP12 615817 HIV-1/HIV-2 RNA Detection Prenatal, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	40 days
	Refrigerated	6 days

CPT Code Information:

87535

87538

HPS12 615818 HIV-1/HIV-2 RNA Detection Prenatal, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into

plastic vial.

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	40 days
	Refrigerated	6 days

CPT Code Information:

87535

87538

HIP12 615815 HIV-1/HIV-2 RNA Detection, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: Plasma: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	40 days
	Refrigerated	6 days

CPT Code Information:

87535

87538

HIS12 615816 HIV-1/HIV-2 RNA Detection, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	40 days
	Refrigerated	6 days

CPT Code Information:

87535

87538

HV2QU 623478 HIV-2 RNA Detection and Quantification, Real-Time qRT-PCR, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.1 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma EDTA	Frozen (preferred)	60 days	ALIQUOT TUBE

	Refrigerated	6 days	ALIQUOT TUBE

CPT Code Information:

87539

FHLAA 91498 HLA A High Resolution

Specimen Requirements:

Histocompatibility Lab Transplant Testing Requisition **is required.**

Container/Tube: Lavender top (EDTA)**Specimen Volume:** 14 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Specify blood on the label

3. Send ambient.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient	14 days

CPT Code Information:

81380

FHLAB 91499 HLA B High Resolution

Specimen Requirements:

Histocompatibility Lab Transplant Testing Requisition **is required.**

Container/Tube: Lavender top (EDTA)

Specimen Volume: 14 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.
2. Specify blood on the tube label.
3. Send ambient.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient	14 days

CPT Code Information:

81380

FHLAC
91500

HLA C High Resolution

Specimen Requirements:

Histocompatibility Lab Transplant Testing Requisition **is required.**

Container/Tube: Lavender top (EDTA)

Specimen Volume: 14 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.
2. Specify blood on the label.
3. Send ambient.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient	14 days

CPT Code Information:

81380

HL57R
610054

HLA-B*57:01 Genotype, Pharmacogenomics, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant or liver transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell or liver transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee

that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81381

HL58R 610055 HLA-B*5801 Genotype, Allopurinol Hypersensitivity, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant or liver transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell or liver transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies: Saliva Swab Collection Kit (T786)

Specimen Volume: 1 Swab

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81381

LY27B
9648

HLA-B27, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient	4 days

CPT Code Information:

86812

HMB45
70459

HMB45 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HCMM 89047 Homocysteine (Total), Methylmalonic Acid, and Methylcitric Acid, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card (Filter Paper)

Acceptable: Local newborn screening card, Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper, Munktell filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. **Do not use** device or capillary tube containing EDTA or ACD to collect specimen. Sodium heparin is acceptable but must be spotted on card the same day as collected.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle) using blood from a heel or finger stick.

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)		FILTER PAPER
	Frozen		FILTER PAPER
	Refrigerated		FILTER PAPER

CPT Code Information:

83090

83918

HCYSP
80379

Homocysteine, Total, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Immediately place specimen on wet ice.

2. Within 4 hours of collection, centrifuge and aliquot plasma into a plastic vial.

3. If blood cannot be placed on wet ice immediately, then within 1 hour of collection, centrifuge and aliquot plasma into a plastic vial.

4. A refrigerated centrifuge is not required if the above time restrictions are met.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	28 days
	Frozen	309 days
	Ambient	28 days

CPT Code Information:

83090

HCYSS
35836

Homocysteine, Total, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 4 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	309 days
	Ambient	28 days

CPT Code Information:

83090

HVA
9253

Homovanillic Acid, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Administration of L-dopa may falsely increase homovanillic acid results. For 24 hours prior to and during specimen collection, the patient should **not** take L-dopa.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative at start of collection. Use 15 mL of 50% acetic acid for children younger than 5 years. If specimen is refrigerated during collection, preservative may be added up to 12 hours after collection. This preservative is intended to achieve a pH of between approximately 1 and 5.

2. Collect a 24-hour urine specimen.

3. If necessary, adjust urine pH to a level between 1 and 5 by adding 50% acetic acid or hydrochloric acid dropwise and checking the pH.

Additional Information:

1. The sensitivity of this test is greater on a 24-hour specimen than on a random specimen.

2. See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	180 days

CPT Code Information:

83150

HVAR 60275 Homovanillic Acid, Random, Urine

Specimen Requirements:

Patient Preparation: Administration of L-dopa may falsely increase homovanillic acid results. For 24 hours prior to specimen collection, the patient should **not** take L-dopa.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Adjust the urine pH to a level between 1 and 5 by adding 50% acetic acid or hydrochloric acid dropwise and checking the pH.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	180 days

CPT Code Information:

83150

HBV
82551

Honeybee Venom, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

HBEA Hornbeam, IgE, Serum
82484

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HRSPF Horse Dander, IgE with Reflex to Horse Dander Component, IgE, Serum
616095

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HORS
82874

Horse Dander, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HFSF
82608

Horsefly/Stablefly, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FHORS
57934 Horseradish (Armoracia rusticana/A.lapathifolia) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

DF
82905

House Dust Mites/Dermatophagoides farinae, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For one allergen: 0.3 mL
For more than one allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DP
82904

House Dust Mites/Dermatophagoides pteronyssinus, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HD1
81877

House Dust Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

HDG
82906

House Dust/Greer Lab, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HDHS
82903

House Dust/H-S Lab, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FHTL
91491

HTLV I/II DNA, Qualitative Real-Time PCR

Specimen Requirements:

Collection Container/Tube: Lavender-top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Draw whole blood specimen in a lavender-top (EDTA) tube.

2. Send refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	48 hours

CPT Code Information:

87798 x 2

FHAM
57856

Human Anti-mouse Antibody (HAMA)

Specimen Requirements:

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 2 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	90 days

CPT Code Information:

83520

HCG 70455 Human Chorionic Gonadotropin (hCG) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

THCG
80678

Human Chorionic Gonadotropin (hCG), Quantitative, Pregnancy, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	72 hours	
	Frozen	365 days	

CPT Code Information:

84702

HE4
62137

Human Epididymis Protein 4, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing

biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	84 days
	Refrigerated	7 days

CPT Code Information:

86305

HRPV8 70458 Human Herpes Virus, Type 8 (HHV-8) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained

slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

QHV6P 619925 Human Herpesvirus-6 A and B DNA Detection and Quantification, PCR, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube:

Preferred: Plastic vial

Acceptable: Screw-capped, sterile container

Specimen Volume: 1 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	24 hours

CPT Code Information:

87533

QHV6C
619924

Human Herpesvirus-6 A and B DNA Detection and Quantification, PCR, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Aliquot tube

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect specimen from collection vial 2.

2. **Do not centrifuge.**

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	24 hours

CPT Code Information:

87533

1DIS
609354

Human Leukocyte Antigens (HLA) A-B-C Disease Association Typing Low Resolution, Blood

Specimen Requirements:

Container/Tube: Yellow top (ACD solution A or B)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not aliquot**.

Additional Information: Specimen acceptability is based on extracted DNA concentration and not sample age.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81372

DIS 618568 Human Leukocyte Antigens (HLA) Class I and II Disease Association Typing, Low Resolution, Blood

Specimen Requirements:

Container/Tube: Yellow top (ACD solution A or B)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not aliquot**.

Additional Information: Specimen acceptability is based on extracted DNA concentration and not sample age.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81370

81376 x3

2DIS 609356 Human Leukocyte Antigens (HLA)-DR-DQ Disease Association Typing Low Resolution, Blood

Specimen Requirements:

Container/Tube: Yellow top (ACD solution A or B)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not aliquot**.

Additional Information: Specimen acceptability is based on extracted DNA concentration and not sample age.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81375

81376 x3

SCHPV 621927 Human Papillomavirus (HPV) Detection and High-Risk Genotyping, Self-Collect, PCR, Vaginal

Specimen Requirements:

Patient Preparation: For 24 hours prior to specimen self-collection, patients **should avoid** using feminine hygiene products.

Supplies: Evalyn Brush (T990)

Specimen Type: Vaginal

Collection Container/Tube: Evalyn Brush

Submission Container/Tube: ThinPrep/PreservCyt solution vial

Specimen Volume: 3 mL of solution in ThinPrep/PreservCyt vial

Specimen Stability Information:

Evalyn Brush: Ambient 72 hours

ThinPrep/PreservCyt Vial: Ambient (preferred) 42 days/Refrigerated 42 days

Collection Instructions:

1**. Specimen must be collected by the patient in a healthcare setting.**

2. Provide patient with a labeled Evalyn brush and the accompanying self-collection instruction pamphlet.

3. Following patient self-collection, ensure that the Evalyn brush is tightly capped and

appropriately labeled.

4. Place labeled Evalyn brush in a biosafety bag and send to the laboratory.

5. Perform the following steps prior to shipment to Mayo Clinic Laboratories:

a. Label PreservCyt (ThinPrep) vial with appropriate patient information.

b. Uncap PreservCyt vial. Remove pink cap from Evalyn brush.

c. Depress the pink plunger on the Evalyn brush to expose the brush (white bristles).

d. Vigorously plunge the brush, smashing the white brush against the bottom and interior wall of the vial 10 times to maximize sample release. Be careful not to splash.

e. Discard Evalyn brush

f. Tightly recap the PreservCyt vial.

g. Submit the PreservCyt vial for testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87626

HPVP 62995 Human Papillomavirus (HPV) DNA Detection with Genotyping, High Risk Types by PCR with Papanicolaou Smear Reflex, ThinPrep, Varies

Specimen Requirements:

Original ThinPrep/PreservCyt collection vial is required for testing.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume=3 mL).

For optimal interpretation, Pap smears should be collected near the middle of the menstrual cycle. Avoid douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Specimen source is required.

Submit only 1 of the following specimens:

Broom Collection Device:

Specimen Type: Cervical (endocervical or ectocervical)

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep/PreservCyt vial

Specimen Volume: 20 mL of solution in ThinPrep/PreservCyt vial

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the broom collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Endocervical Brush/Spatula Collection Device:

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep/PreservCyt vial

Specimen Volume: 20 mL of solution in ThinPrep/PreservCyt vial

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatula as quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate 1/4 or 1/2 turn in one direction. **Do not over-rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial.

Specimen Minimum Volume: 17 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	
	Refrigerated	42 days	

CPT Code Information:

87626

G0476 (if appropriate)

88142 (if appropriate)

SHPV 62599 Human Papillomavirus (HPV) DNA Detection with Genotyping, High-Risk Types by PCR, SurePath, Varies

Specimen Requirements:

Supplies: Cobas PCR Media Tube w/Cap (T945)

Specimen Type: Cervical (endocervical or ectocervical) or vaginal

Specimen Volume: 3.0 mL

Collection Instructions:

1. Aliquot 3 mL SurePath specimen into Cobas PCR Media Tube w/Cap tube.

2. Bag specimens individually as they have a tendency to leak during transport.

3. Place labels on the vial and on the bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	
	Refrigerated	42 days	

CPT Code Information:

87626

G0476 (if appropriate)

HPV
62598 Human Papillomavirus (HPV) DNA Detection with Genotyping, High-Risk Types by PCR, ThinPrep, Varies

Specimen Requirements:

Specimen Type: Cervical (endocervical or ectocervical)

Container/Tube: ThinPrep/PreservCyt solution vial

Specimen Volume: 3 mL of solution in ThinPrep/PreservCyt vial

Collection Instructions:

1. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

2. Place labels on the vial and on the bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	
	Refrigerated	42 days	

CPT Code Information:

87626

G0476 (if appropriate)

HPVE6 71405 Human Papillomavirus (HPV) High-Risk E6/E7, RNA In Situ Hybridization

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Collection Instructions: Formalin-fixed, paraffin-embedded tissue block; or 5 unstained glass, "positively charged" slides with 4-microns, formalin-fixed, paraffin-embedded tissue

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-Primary

88364-If additional ISH

HPVHL 70464 Human Papillomavirus (HPV) High/Low Risk, In Situ

Hybridization

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Formalin-fixed, paraffin-embedded tissue block

Specimen Volume: Entire block

Specimen Type: Slides

Slides: 6 Unstained glass, positively charged slides with 5 (+ or - 1)-microns formalin-fixed, paraffin-embedded tissue

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-Primary

88364-If additional ISH

HPVLR 70465 Human Papillomavirus (HPV) Low Risk, In Situ Hybridization

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Collection Instructions: Formalin-fixed, paraffin-embedded tissue block; or 5 unstained glass, "positively charged" slides with 4-microns, formalin-fixed, paraffin-embedded tissue

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-Primary

88364-If additional in situ hybridization

DHPVR 623292 Human Papillomavirus (HPV) Type 16 with Type 18, 31, 33, and 35 Reflex, Droplet Digital PCR, Blood

Specimen Requirements:

Supplies: Streck Tan Top Tube Kit (T715)

Container/Tube: Streck Cell-Free DNA blood collection kit

Specimen Volume: Two 10-mL Streck Cell-Free DNA blood collection tubes

Additional Information: Only blood collected in Streck Cell-Free DNA tubes will be accepted.

Specimen Minimum Volume: One 10 mL Streck Cell-Free DNA blood collection tube

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

0661U

DHPVA 622798 Human Papillomavirus (HPV) Type 16, Droplet Digital PCR, Blood

Specimen Requirements:

Supplies: Streck Tan Top Tube Kit (T715)

Container/Tube: Streck Cell-Free DNA blood collection kit

Specimen Volume: Two 10-mL Streck Cell-Free DNA blood collection tubes

Additional Information: Only blood collected in Streck Cell-Free DNA tubes will be accepted for analysis.

Specimen Minimum Volume: One 10 mL Streck Cell-Free DNA blood collection tube

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

0661U

DHPVB 622799 Human Papillomavirus (HPV) Types 18, 31, 33, and 35, Droplet Digital PCR, Blood

Specimen Requirements:

Supplies: Streck Tan Top Tube Kit (T715)

Container/Tube: Streck Cell-Free DNA blood collection kit

Specimen Volume: Two 10-mL Streck Cell-Free DNA blood collection tubes

Additional Information: Only blood collected in Streck Cell-Free DNA tubes will be accepted for analysis.

Specimen Minimum Volume: One 10 mL Streck Cell-Free DNA blood collection tube

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

0660U

VHPV
619401

Human Papillomavirus (HPV) Vaginal Detection with Genotyping for High-Risk Types by PCR

Specimen Requirements:

Patient Preparation: For 24 hours prior to specimen collection, patients **should not use** carbomer-containing feminine hygiene products (eg, KY Jelly [Physician Formula], Vagisil creme regular strength). For additional information about carbomer-containing feminine hygiene products, including additional examples, see Cautions.

Specimen Type: Vaginal

Container/Tube: ThinPrep/PreservCyt solution vial

Specimen Volume: 3 mL of solution in ThinPrep/PreservCyt vial

Collection Instructions:

1. Obtain an adequate sampling from vagina using a sterile brush, broom, or spatula.

2. Rinse the collection device as quickly as possible into the PreservCyt Solution vial.

3. DISCARD THE COLLECTION DEVICE.

4. Place ThinPrep specimens in individual bags, as they tend to leak during transport.

5. Place labels on the vial and on the bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Vaginal	Ambient (preferred)	42 days
	Refrigerated	42 days

CPT Code Information:

87626

G0476 (if appropriate)

FHPL
91178

Human Placental Lactogen (HPL)

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Specimen Type: Serum

Container/Tube: Red top

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge at 3000 rpm for 10 minutes

2. Aliquot 1 mL of serum into a plastic vial and freeze immediately.

3. Send serum frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	90 days

CPT Code Information:

83632

HPL 70462 Human Placental Lactogen Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HTLLC 604935 Human T-Cell Lymphotropic Virus Types 1 and 2 (HTLV-1/-2) Antibody Confirmation, Spinal Fluid

Specimen Requirements:

Collection Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Submit spinal fluid (CSF) specimen from collection vial number 2, 3, or 4 (preferred). If vial number is not provided, then only clear CSF without visible hemolysis or xanthochromia will be accepted for testing.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	30 days
	Refrigerated	14 days

CPT Code Information:

86689

HTLVC 604934 Human T-Cell Lymphotropic Virus Types 1 and 2 (HTLV-1/-2) Antibody Screen with Confirmation, Spinal Fluid

Specimen Requirements:

Collection Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Submit spinal fluid (CSF) specimen from collection vial number 2, 3, or 4 (preferred). If vial number is not provided, then only clear CSF without visible hemolysis or xanthochromia will be accepted for testing.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	30 days
	Refrigerated	14 days

CPT Code Information:

86790

86689 (if appropriate)

HTLVL 83277 Human T-Cell Lymphotropic Virus Types I and II (HTLV-I/-II) Antibody Confirmation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	
	Refrigerated	7 days

CPT Code Information:

86689

HTLVI 9539 Human T-Cell Lymphotropic Virus Types I and II Antibody Screen with Confirmation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge blood collection tube per collection tube manufacturer's instructions (eg, centrifuge within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Refrigerated	7 days

CPT Code Information:

86790

86689 (if appropriate)

FHTGF 75825 Human Transforming Growth Factor beta 1 (TGF-b1)

Specimen Requirements:

Specimen Type: Platelet-free Plasma

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood in a lavender top (EDTA) tube.

2. Centrifuge for 15 minutes at 1000 x g within 30 minutes of collection. Centrifuge plasma again at 3000 x g for 10 minutes for complete platelet removal.

3. Aliquot plasma into a plastic vial and freeze at -20 degrees C or below.

4. Send 1 mL platelet-free plasma frozen in plastic vial.

Additional Information:

For fixed speed centrifuges such as 645e: Within 30 minutes of collection, centrifuge 3 times for 10 minutes at 1600 x g while decanting the plasma each time before the next centrifugation. Aliquot plasma into a plastic vial and freeze at -20 degrees C or below.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

83520

HAD 35452 Huntington Disease, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81271-HTT (huntingtin) (eg, Huntington disease) gene analysis; evaluation to detect abnormal (eg, expanded) alleles

HYDCS
623451

Hydrocodone and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80361 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

HYDCU 62614 Hydrocodone with Metabolite Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STATS are **not** accepted for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

HYDMU 62615 Hydromorphone Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not** accepted for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

HYDMS 623455 Hydromorphone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80361 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

HCQWB 623295 Hydroxychloroquine, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood EDTA	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80220

HCQ
64947

Hydroxychloroquine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

80220

HGEM
62230

Hydroxyglutaric Acids, Glutaric Acid, Ethylmalonic Acid, and Methylsuccinic Acid, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Card-Blood Spot Collection (Filter Paper) from heel or finger stick

Acceptable: PerkinElmer (formerly Ahlstrom 226) filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, available newborn screening card, blood collected in tubes containing sodium heparin or EDTA and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Do not use device or capillary tube containing ACD to collect specimen. Sodium heparin or EDTA are acceptable but must be spotted on card the same day as collected.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not stack wet specimens.

5. Do not expose specimen to heat or direct sunlight.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	123 days	FILTER PAPER
	Frozen	123 days	FILTER PAPER
	Refrigerated	123 days	FILTER PAPER

CPT Code Information:

83918

and Methylsuccinic Acid, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA), green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.1 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.02 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	12 days

CPT Code Information:

83918

HGEMS
62231

Hydroxyglutaric Acids, Glutaric Acid, Ethylmalonic Acid, and Methylsuccinic Acid, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.02 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	4 days

CPT Code Information:

83918

FVIST
90121

Hydroxyzine (Vistaril, Atarax), Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable**. Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

HIESG
619816

Hyper-IgE Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

HCHLG 617267 Hypercholesterolemia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese

Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 x2

81407

81479

81479 (if appropriate for government payers)

HYOX
86213

Hyperoxaluria Panel, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Immediately freeze specimen.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	90 days
	Refrigerated	14 days

CPT Code Information:

82542

FAVI
91509

Hypersensitivity Pneumonitis Avian Panel

Specimen Requirements:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	30 days
	Frozen	365 days

CPT Code Information:

86331x5

BHYPS 623579 Hypersensitivity Pneumonitis Panel, Immunoglobulin G, Serum

Specimen Requirements:

Container/Tube: Sarstedt Aliquot Tube, 5 mL (T914)

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL Serum

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

86001 x 7

HYPTG 617323 Hypertriglyceridemia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. Do not aliquot.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

HCMGG 617281 Hypertrophic Cardiomyopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit of 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See

Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81439

HYPBG 617309 Hypobetalipoproteinemia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline.

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information:: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese

Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

81407

81479

81479 (if appropriate for government payers)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

Hypoglycemic Agent Screen, Serum

Specimen Requirements:

Patient Preparation: Specimen must be collected during an episode of hypoglycemia.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume:1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	7 days

CPT Code Information:

G0480
80377 (if appropriate for select payers)

FIA2A
75939

IA-2 Antibody

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 1 mL

Collection Instruction:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send ambient.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Ambient (preferred)	7 days
	Frozen	30 days
	Refrigerated	7 days

CPT Code Information:

86341

FIAAB
75936

IA2 Autoantibodies

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. With 24 hours of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Freeze immediately after separation and send frozen.

Specimen Minimum Volume: 0.5 mL (Note: Minimum volume does not allow for repeat analysis)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	107 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

86341

FIBUP
57703

Ibuprofen (Motrin, Advil, Nuprin), serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 1 mL sodium heparin plasma refrigerated in plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 1 mL of serum refrigerated in plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80329

ICOSI
113518

ICOS (CD278), Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

RMALM 60778 Id MALDI-TOF Mass Spec AFB (Bill Only)

Specimen Requirements:

**For billing purposes only

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87118

RMALA 62258 Id MALDI-TOF Mass Spec Anaerobe (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87076

RMALD 60029 Ident by MALDI-TOF Mass Spec (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87077

PCRID 64706 Identification by PCR (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87150

RTBSP 60768 Identification Mycobacterium tuberculosis Complex

Speciation, PCR (Bill Only)

Specimen Requirements:

**For billing purposes only

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87150

STAP
45362

Identification Staphylococcus (Bill Only)

Specimen Requirements:

This test is for Billing Purposes Only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87077

STRP
45371

Identification Streptococcus (Bill Only)

Specimen Requirements:

This test is for Billing Purposes Only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87077

COMM 45070 Identification, Commercial Kit (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87077

ISNGS 609732 Identification, Next-Generation Sequencing (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87798

IDHQ 615859 IDH1 (R132) and IDH2 (R140 and R172) Quantitative Detection, Droplet Digital PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD-B) or green top (heparin)

Specimen Volume: 4mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD-B) or green top (heparin)

Specimen Volume: 2mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube with indication of volume and concentration of DNA

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA and source of specimen

2. Indicate volume and concentration of DNA on label. The required volume of DNA is at least 50 mcL at a concentration of 50 ng/mcL

Specimen Stability: Frozen (preferred)/Refrigerated

Specimen Minimum Volume: Whole blood: 4mL; Bone marrow: 2mL; Extracted DNA: 50 mcL at 50 ng/mcL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Ambient	7 days

CPT Code Information:

81120

81121

IDHT
616498

IDH1 and IDH2 Mutation Analyses, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81479

IDH1
70468

IDH1 Mutation (R132H) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IDTRT 619649 IDH1, IDH2, and TERT Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)
-Minimum amount of tumor area: tissue 36 mm(2)
-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.
-Tissue fixation: 10% neutral buffered formalin, not decalcified
-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

0481U

IPFGP 621589 Idiopathic Pulmonary Fibrosis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have

received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2.A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1.Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2.To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

I2SB
618290

Iduronate-2-Sulfatase, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood spot collection card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on the filter paper at room temperature in a horizontal position for 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

83864 (if appropriate)

I2SWB
618291

Iduronate-2-Sulfatase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

IFPCA
113304

IF Additional (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88350

IFA26
603534

IF Additional, Professional Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88350-26

IFTOA
603532

IF Additional, Technical Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88350-TC

IFPCI
113303

IF Initial (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88346

IFI26
603533

IF Initial, Professional Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88346-26

IFTOI
603531

IF Initial, Technical Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88346-TC

IGAI
70470

IgA Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IGAS
87938

IgA Subclasses, Serum

Specimen Requirements:

Patient Preparation: Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

82784

82787 x 2

IGDI
70471

IgD Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FIERA
75754

IgE Receptor Antibody

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Immediately centrifuge and aliquot 1.5 mL serum into a plastic vial.

2. Send frozen

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	7 days
	Ambient	48 hours

CPT Code Information:

88184

88185 x 2

FIGF1 IGF-1, LC/MS
75932

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collections Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

84305

FFIG2 IGF-2
75924

Specimen Requirements:

Collection Container/Tube: Serum gel or red top

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 1 hour of collection, centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Freeze immediately after separation and send frozen.

Specimen Minimum Volume: Serum: 0.3 mL; Note: Minimum volume does not allow for repeat analysis.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

83520

IGGI
70473

IgG Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IGGS
9259

IgG Subclasses, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

82784

82787 x 4

SFIGS
610784

IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SFIG / Cerebrospinal Fluid IgG Index Profile, Serum and Spinal Fluid.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82784

CASF
8271

IgG/Albumin Ratio, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal fluid (CSF)

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Submit CSF from collection vial number 2 (preferred, not required).

2. Label specimen as CSF.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days

	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82042

82784

FG4FI
57851

IgG4 Food Panel I

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 X 10

FGFP2
57904

IgG4 Food Panel II

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 9

FG4FP
57591

IgG4 Food Panel VIII

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 6

IGG4I
70472

IgG4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IGGS4 84250 IgG4, Immunoglobulin Subclasses, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

82787

BCLL 89008

IGH Somatic Hypermutation Analysis, B-Cell Chronic Lymphocytic Leukemia (B-CLL), Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability: Refrigerated (preferred) 7 days/Ambient 7 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability: Ambient (preferred) 7 days/Refrigerate 7 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2-mL screw-top tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen

3. Provide volume and concentration of the DNA

Specimen Stability Information: Frozen (preferred)/Refrigerate/Ambient

Additional Information:

1. Specimens are preferred to be received within 1 year after DNA extraction.

2. DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood/Bone marrow: 1 mL; Extracted DNA: 50 mcL at a concentration of 20 ng/mcL.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81263-*IGH* (immunoglobulin heavy chain locus) (eg, leukemia and lymphoma, B-cell), variable region somatic mutation analysis

IGMI
70474

IgM Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IHPCA
113298

IHC Additional (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88341

IHA26
113300

IHC Additional, Professional Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88341-26

IHPCI
113297

IHC Initial (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342

IHC26
113299

IHC Initial, Professional Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-26

IHMPC 113301 IHC Multiplex (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344

IHM26 113302 IHC Multiplex, Professional Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-26

IHMTO 113211 IHC Multiplex, Tech Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

IDH2 616903 IHD2, R172K Mutation, Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IMIPR 63508 Imipramine and Desipramine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

IMPLF 622937 Immature Platelet Fraction, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	48 hours
	Ambient	24 hours

CPT Code Information:

85055

IFXED 606458 Immunofixation Heavy Chain Type Delta and Epsilon, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

86334

FIFLC 75678 Immunofixation with Free Light Chains, Quantitative, Urine

Specimen Requirements:

Container/Tube: Two plastic aliquot tubes

Specimen Volume: 8 mL in 2 plastic aliquot tubes, each containing 4 mL

Collection Instructions:

1. Collect urine for 24 hours (No preservative).

2. Refrigerate specimen during the 24-hour collection.

3. Mix the collection and then aliquot urine (4 mL per aliquot) into two plastic aliquot tubes

4. **Collection volume and duration are required.**

5. Send refrigerated.

Specimen Minimum Volume: 4 mL total, 2 vials each containing 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	21 days
	Frozen	180 days

CPT Code Information:

84156

86335

83521 x 2

FIXCF 75749 Immunofixation, CSF

Specimen Requirements:

Specimen Type: CSF

Container/Tube: sterile screw cap container

Specimen Volume: 6 mL

Collection Instructions: Collect 6 mL of spinal fluid (CSF) in a sterile screw cap container. Ship frozen.

Specimen Minimum Volume: 4.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	30 days
	Refrigerated	10 days
	Ambient	6 days

CPT Code Information:

86335

FHLCA 75550 Immunoglobulin A (IgA) Heavy and Light Chain (HLC) Pairs, Kappa and Lambda with Ratio

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.75 mL serum

Collection Instructions:

1. Centrifuge immediately after coagulation (30 minutes) to prevent hemolysis.

2. Aliquot 0.75 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83521 x 2

IGA
8157

Immunoglobulin A (IgA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82784

IGD
9272

Immunoglobulin D (IgD), Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

82784

IGE
8159

Immunoglobulin E (IgE), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For only total IgE: 0.3 mL
For total IgE and more than 1 allergen: 0.05 mL x number of allergen-specific tests + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785

FLCS
608250

Immunoglobulin Free Light Chains, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83521 x 2

FHLCG
75551

Immunoglobulin G (IgG) Heavy and Light Chain (HLC) Pairs, Kappa and Lambda with Ratio

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.75 mL serum

Collection Instructions:

1. Centrifuge immediately after coagulation (30 minutes) to prevent hemolysis.

2. Aliquot 0.75 mL of serum into a plastic vial.

3. Send refrigerate

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83521 x 2

SUBIF 70620 Immunoglobulin G (IgG) Subtypes Immunofluorescence, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Unstained slides (unfixed)

Source: Kidney tissue

Slides: 4 Slides

Collection Instructions: 4 frozen tissue unstained positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick, **centered on the slide,** and submitted on dry ice.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Acceptable:

Specimen Type: Unfixed tissue block (O.C.T)

Source: Kidney tissue

Specimen Volume: Entire specimen

Collection Instructions:

1. Embed in O.C.T. compound.

2. Freeze specimen and ship on dry ice.

Acceptable:

Specimen Type: Wet tissue

Source: Kidney tissue

Supplies: Renal Biopsy Kit (T231)

Container/Tube: Renal Biopsy Kit, Zeus/Michel's

Specimen Volume: Entire specimen

Collection Instructions:

1. Collect specimens according to the instructions in Renal Biopsy Preparation Instructions.

2. If standard immunoglobulin and complement immunofluorescence has already been performed, submit the residual frozen tissue (must contain glomeruli) on dry ice.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Frozen	

CPT Code Information:

88346-Primary IF

88350-If additional IF

Immunoglobulin G (IgG), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82784

BCGR
83123

Immunoglobulin Gene Rearrangement, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81261-IGH (Immunoglobulin heavy chain locus) (eg, leukemias and lymphomas B-cell), gene rearrangement analysis to detect abnormal clonal populations; amplified methodology (eg. polymerase chain reaction)

81264-IGK (Immunoglobulin kappa light chain locus) (eg, leukemia and lymphoma, B-cell) gene rearrangement analysis, evaluation to detect abnormal clonal populations

BCGBM Immunoglobulin Gene Rearrangement, PCR, Bone Marrow
31141

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81261-IGH (Immunoglobulin heavy chain locus) (eg, leukemias and lymphomas B-cell), gene rearrangement analysis to detect abnormal clonal populations; amplified methodology (eg. polymerase chain reaction)

81264-IGK (Immunoglobulin kappa light chain locus) (eg, leukemia and lymphoma, B-cell) gene rearrangement analysis, evaluation to detect abnormal clonal populations

BCGRV 31142 Immunoglobulin Gene Rearrangement, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Paraffin-embedded tissue

Container/Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

3. Include pathology report.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

4. Include pathology report.

Specimen Stability Information: Ambient

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Sources: Pleural, peritoneal, vitreous, and spinal fluid (CSF)

Container/Tube: Sterile container

Specimen Volume: At least 5 mL

Collection Instructions:

1. If the volume is large, pellet cells prior to sending.

2. Send less volume at ambient temperature or as a frozen cell pellet.

3. Specify the type of fluid being submitted.

Specimen Stability Information:

Body fluid: Ambient 4 days/Refrigerated/Frozen

Cell pellet: Frozen

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection.

2. Indicate specimen source.

Specimen Stability Information: Frozen

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Body Fluid: 1 mL; Frozen tissue: 50 mg; Extracted DNA: 50 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81261-IGH (Immunoglobulin heavy chain locus) (eg, leukemias and lymphomas B-cell), gene rearrangement analysis to detect abnormal clonal populations; amplified methodology (eg. polymerase chain reaction)

81264-IGK (Immunoglobulin kappa light chain locus) (eg, leukemia and lymphoma, B-cell) gene rearrangement analysis, evaluation to detect abnormal clonal populations

BCGET
802123

Immunoglobulin Gene Rearrangement, Tissue

Specimen Requirements:

Specimen Type: Paraffin-embedded bone marrow aspirate clot or paraffin-embedded tissue

Container/Tube: Paraffin block

Specimen Volume: Minimum of 4 slides, 10 um preferred

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	

	Refrigerated

CPT Code Information:

81261-IGH (Immunoglobulin heavy chain locus) (eg, leukemias and lymphomas B-cell), gene rearrangement analysis to detect abnormal clonal populations; amplified methodology (eg. polymerase chain reaction)

81264-IGK (Immunoglobulin kappa light chain locus) (eg, leukemia and lymphoma, B-cell) gene rearrangement analysis, evaluation to detect abnormal clonal populations

KCSF 65572 Immunoglobulin Kappa Free Light Chain, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal fluid (CSF)

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Label specimen as spinal fluid.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	28 days
	Refrigerated	72 hours
	Ambient	24 hours

CPT Code Information:

83521

KCSFP 607839 Immunoglobulin Kappa Free Light Chain, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MSP3 / Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid.

Specimen Type: Spinal fluid (CSF)

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Submit CSF from collection vial number 4 (preferred); vial number 1, 3, and 2 are also acceptable (in this order).

2. Label specimen as CSF.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	28 days
	Refrigerated	72 hours
	Ambient	24 hours

CPT Code Information:

83521

IGM
8158

Immunoglobulin M (IgM), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days

	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82784

TLCU
87934

Immunoglobulin Total Light Chains, Urine

Specimen Requirements:

Submit only 1 of the following specimens:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Specimen Type: Random urine

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Refrigerate after collection and send refrigerate.

Specimen Type: 24-Hour urine

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No additive needed for preservation.

3. Urine may be kept ambient during the collection period but should be refrigerated within 4 hours of collection completion.

Additional Information: See Urine Preservatives-Collection and Transportation for

24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	20 days
	Ambient	72 hours

CPT Code Information:

83883 x 2

IMMG 8156 Immunoglobulins (IgG, IgA, and IgM), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82784 x 3

Immunoglobulins, CSF Quantitative

Specimen Requirements:

Collection Container/Tube: Leak-proof container

Submission Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge spinal fluid to remove cellular material.

2. Transfer 1 mL into a plastic vial.

3. Send refrigerated.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	30 days
	Frozen	180 days

CPT Code Information:

82784 x3

IEICP
621519

Inborn Errors of Immunity Comprehensive Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions.

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

IMMAU 619858 Inborn Errors of Immunity with Immune Dysregulation and Autoimmunity Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Blood: 1 mL; Skin biopsy or cultured fibroblasts: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MONOS Infectious Mononucleosis, Rapid Test, Serum
9081

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86308

IBDP2 Inflammatory Bowel Disease Serology Panel, Serum
610004

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86671 x 2

86036

IMTF 35277 Inflammatory Myofibroblastic Tumors (IMT), 2p23 (ALK) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

INFXP Infliximab Quantitation with Antibodies to Infliximab, Serum
620151

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not t**ake multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

INFX-80230

INXA -82397

INFXR 63437 Infliximab Quantitation with Reflex to Antibodies to Infliximab, Serum

Specimen Requirements:

Patient Preparation:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80230

82397-(if appropriate)

HPFLU
610411

Influenza Virus Type A and Type B and Respiratory Syncytial Virus (RSV) RNA, Molecular Detection, PCR, Varies

Specimen Requirements:

Preferred:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with viral transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4 or M5), saline, or phosphate buffered saline (PBS). Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchoalveolar lavage fluid

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Acceptable:

Specimen Type: Oropharyngeal (throat) swab, nasal mid-turbinate, or nares/nasal swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-Swab, Sterile Polyester (T507)

Collection Container/Tube:

Preferred: BBL Culture Swab or COPAN Mid-Turbinate Swab

Acceptable: Dacron-tipped swab with plastic handle

Submission Container/Tube: Swab in sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media

Collection Instructions: Swab must be placed into viral transport media (eg, M4-RT, M4, or M5), saline, or PBS. Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchial washings, endotracheal aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Specimen Minimum Volume: Upper respiratory tract swab: See Specimen Required; Lower respiratory specimens: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	14 days

	Refrigerated	72 hours

CPT Code Information:

87631

FLUNP 802208 Influenza Virus Type A and Type B, and Respiratory Syncytial Virus (RSV), Molecular Detection, PCR, Nasopharyngeal Swab

Specimen Requirements:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with viral transport media

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed in viral transport media (for example, M4-RT, M4, or M4 media).

Specimen Minimum Volume: Nasopharyngeal swab submitted in minimum volume of 0.3 mL of viral transport media (eg, M4-RT)

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87631

ATAXP 617506 Inherited Ataxia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

BMFGP Inherited Bone Marrow Failure Gene Panel, Varies
621561

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81441

CMSP 617519 Inherited Congenital Myasthenic Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

EDMDP 617558 Inherited Emery-Dreifuss Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81404

81406 x 2

81479

81479 (if appropriate for government payers)

AFTDP 617493 Inherited Frontotemporal Dementia and Amyotrophic Lateral Sclerosis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81406 x 9

81404 x 3

81405 x 2

81407

81479

81479 (if appropriate for government payers)

LGCMP 617623 Inherited Limb-Girdle Muscular Dystrophy and Congenital Myasthenic Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not

perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

IMSNP 617584 Inherited Motor and Sensory Neuropathy Gene Panel, Varies

Specimen Requirements:

A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81448

MNDP 617649 Inherited Motor Neuron Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

IMNP
617571

Inherited Motor Neuropathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404 x 2

81405

81406 x 2

81479

81479 (if appropriate for government payers)

MDYSP 617636 Inherited Muscular Dystrophy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

PARDP 617675 Inherited Parkinson Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81404 x 3

81405 x 4

81406 x 9

81407

81408

81479

81479 (if appropriate for government payers)

RABMP 617701 Inherited Rhabdomyolysis and Metabolic Myopathy Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

ISNP 617597 Inherited Sensory Neuropathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81448

SMCP 617727 Inherited Skeletal Muscle Channelopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81406 x 3

81479

81479 (if appropriate for government payers)

ISPP 617610 Inherited Spastic Paraplegia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81448

INHAB Inhibin A and B, Tumor Marker, Serum

86336

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

83520-Inhibin B

86336-Inhibin A

INHA
81049

Inhibin A, Tumor Marker, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86336

INHB
88722

Inhibin B, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

83520

INHIB
70476

Inhibin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

INHU
82789

Insulin (Human), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

INSAL
624132

Insulin Analogs, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required for human insulin reference values to apply

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial

Additional Information: Human insulin reference values are not applicable to patients who were not fasting for 8 hours or received pharmaceutical/synthetic insulin analogs.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	14 days

CPT Code Information:

G0480

80375 (if appropriate for select payers)

FINAB
75935

Insulin Autoantibodies (IAA)

Specimen Requirements:

Patient preparation: No radioactive isotopes should be administered 24 hours prior to venipuncture.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 24 hours of collection, centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Freeze immediately after separation and send frozen.

Specimen Minimum Volume: 0.2 mL Note: This volume does not permit repeat analysis.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	
	Refrigerated	72 hours

CPT Code Information:

86337

INSUL
70478

Insulin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

INSFT 62990 Insulin, Free and Total, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, required

2. For 12 hours before specimen collection, **do not** take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.45 mL

Collection Instructions:

1. Avoid hemolysis

2. Label specimens with corresponding collection times.

3. Serum-gel tubes should be centrifuged within 2 hours of collection.

4. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

5. Send specimen refrigerated

Additional Information: If multiple specimens are collected, send separate order for each specimen.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	24 hours

CPT Code Information:

83527-Free Insulin

83525-Total Insulin

INS
8664

Insulin, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, required

2. **For 12 hours before specimen collection do not** take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Avoid hemolysis

2. Label specimens with corresponding collection times.

3. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial.

Additional Information: If multiple specimens are drawn, send separate order for each specimen.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	7 days

CPT Code Information:

83525

IGFPN 623574 Insulin-Like Growth Factor 1 and Insulin-Like Growth Factor Binding Protein 3 Growth Panel, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: 2 Plastic vials

Specimen Volume: 1.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into 2 plastic vials, each containing 0.8 mL of serum.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

83520

84305

IGF1S
623571

Insulin-Like Growth Factor 1, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84305

IGFB3
83300

Insulin-Like Growth Factor-Binding Protein 3, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions: Centrifuge promptly and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

83520

FGBP1 Insulin-like Growth Factor-binding Protein-1 (IGFBP-1)

75670

Specimen Requirements:

Collection container/tube: 5 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down within one hour and send 0.5 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 0.1 mL Note: This volume does not permit repeat analysis.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	
	Refrigerated	48 hours

CPT Code Information:

83520

INSM1 602573 Insulinoma-Associated Protein 1 (INSM1), Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IFG23 607216 Intact Fibroblast Growth Factor 23, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

83520

INI1
70477

Integrase Interactor 1 (INI1/BAF47) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

IEHCG 606587 Interference Evaluation Heterophile, Beta-Human Chorionic Gonadotropin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

84702 x 2

IETG 610069 Interference Evaluation Heterophile, Thyroglobulin Tumor

Marker, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84432

86800

84432

IRF8
620064

Interferon Regulatory Factor 8 (IRF8) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FIFNY
57586

Interferon-gamma (IFN-y) Serum

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube: Red or SST

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	365 days

CPT Code Information:

83520

FIN1A
75908

Interleukin 1-Alpha

Specimen Requirements:

Patient preparation: Patient **should not** be on any corticosteroids, anti-inflammatory medications, or pain killers, if possible, for at least 48 hours prior to specimen collection.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 3 mL of serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	30 days

CPT Code Information:

83520

FIL2M
57826

Interleukin 2, Serum

Specimen Requirements:

Collection Container/Tube: Serum gel or red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. As soon as possible or within 2 hours of collection, centrifuge and aliquot 1 mL of

serum into a plastic vial.

2. Send frozen.

Note: **Separate specimens must be** submitted when multiple tests are ordered.

Cytokine levels may demonstrate diurnal variation. For longitudinal comparison, it is recommended that cytokine levels be determined at the same time of day.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen	365 days

CPT Code Information:

83520

IL28Q
610056

Interleukin 28B (IL28B) Variant (rs12979860), Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Specimen Volume: 100 mcL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1year /Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction

platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81283

IL1B
622327

Interleukin-1 Beta, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Immediately after specimen collection, place tube on wet ice.

2. Centrifuge at 4 degrees C, 1500 x g for 10 minutes.

3. Aliquot plasma into plastic vial.

4. Freeze specimen within 2 hours of collection.

Additional Information:

1. Plasma for this test cannot be shared with other tests. Submit specimen for this test in its own vial.

2. This test cannot be added on to a previously collected specimen.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

83520

SIL2R
622328

Interleukin-2 Receptor Alpha Soluble, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender-top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Immediately after specimen collection, place tube on wet ice.

2. Centrifuge at 4 degrees C, 1500 x g for 10 minutes.

3. Aliquot plasma a into plastic vial.

4. Freeze specimen within 2 hours of collection.

Additional Information:

If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation and is frozen immediately afterward.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

83520

FIL4S
57585

Interleukin-4 (IL-4) Serum

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube: Red or SST

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

83520

IL6DX
618776

Interleukin-6, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

83529

FIL8S
57563

Interleukin-8 (IL-8) Serum

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube: Red or SST

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

83520

IFBA
9335

Intrinsic Factor Blocking Antibody, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, required

2. This test should not be performed on patients who have received a vitamin B12 injection or radiolabeled vitamin B12 injection within the previous 2 weeks.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86340

UIOD
9549

Iodine, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

2. If this test is used in conjunction with the (131)I uptake test, then specimen collection

should begin immediately after the dose of (131)I is given (ie, the patient should void and discard urine just prior to the (131)I dose, and all subsequent urine should be collected for the next 24 hours). The last void should be included in the collection.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic aliquot container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83789

IOD
81574

Iodine, Serum

Specimen Requirements:

Patient Preparation:

1. Disinfectants, such as Betadine that contain iodine should **not** be used during specimen collection..

2. High concentrations of gadolinium and iodine may interfere with inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free B-D Tube (No Additive), 6 mL (T184)

-Metal Free Specimen Vial (T173)

Collection Container/Tube: Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Allow specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into Mayo metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and do not ream the specimen with a wooden stick to assist with serum transfer.

3. See Trace Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83789

IODCU
610708

Iodine/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83789

82570

HEXP 606984 Iohexol, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Heparin	Refrigerated (preferred)	7 days

	Frozen	35 days
	Ambient	7 days

CPT Code Information:

82542

HEXU 606985 Iohexol, Timed Collection, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions: Collect a timed urine specimen. Timing may be variable and patient dependent.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	35 days
	Ambient	7 days

CPT Code Information:

82542

SFEC 621385 Iron and Total Iron-Binding Capacity, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. For 24 hours before collection, patient **should not** take iron-containing supplements.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood before noon (preferred).

2. Within 2 hours of collection, serum gel tubes should be centrifuged.

3. Within 2 hours of collection, red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days

CPT Code Information:

83540-Iron

83550-Iron-binding capacity

FET
8350

Iron, Liver Tissue

Specimen Requirements:

Supplies: Metal Free Specimen Vial (T173)

Container/Tube:

Preferred: Metal-free specimen vial

Acceptable: Paraffin block, with no more than 1 or 2 cuts previously made

Specimen Volume: 2 mg

Collection Instructions: Two mg of liver tissue are required. This is typically a piece of tissue from a 22-gauge needle biopsy at least 2 cm long. If an 18-gauge needle is used, the tissue must be at least 1 cm in length.

Specimen Stability Information:

Fresh or formalin-fixed liver tissue specimens: Frozen (-30 to -10 degrees C) at least 20 years

Paraffin-embedded (block) liver tissue specimens: Ambient (16 to 24 degrees C) at least 12 1/2 years

Additional Information: Paraffin blocks will be returned 7 days after analysis is complete.

Specimen Minimum Volume: Needle biopsy: See Specimen Required; 2 mm x 2 mm (punch): 0.3 mg by dry weight

Transport Temperature:

Specimen Type	Temperature	Time
Liver Tissue	Refrigerated (preferred)	
	Ambient	
	Frozen	

CPT Code Information:

83540

IRON
2502

Iron, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see SFEC / Iron and Total Iron-Binding Capacity, Serum

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. For 24 hours before collection, patient **should not** take iron-containing supplements.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood before noon (preferred).

2. Within 2 hours of collection, serum gel tubes should be centrifuged.

3. Within 2 hours of collection, red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	365 days

CPT Code Information:

83540-Iron

FIVCZ
75575

Isavuconazole (CRESEMBA) LC-MS/MS

Specimen Requirements:

Collection Container/Tube: Red top (serum gel/SST **are not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

80299

FISST
75887

Isavuconazole Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen volume: Organism in pure culture

Complete and submit with specimen:

1.Fungus Testing Lab request form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999-Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 x 4 for mould MIC microdilution or agar dilution (if appropriate)

87186 x 4 for yeast MIC microdilution or agar dilution (if appropriate)

ISPCA
113306

ISH Additional (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88364

ISA26
113308

ISH Additional, Professional Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88364-26

ISTOA
113217

ISH Additional, Tech Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88364-TC

ISPCI
113305

ISH Initial (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365

ISH26
113307

ISH Initial, Professional Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-26

ISTOI
113216

ISH Initial, Tech Only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-TC

ISLET
70479

Islet 1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

ATR
113383

Isoagglutinin Titer, Anti-A, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Pediatric Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Ambient (preferred)	4 days
	Frozen	10 days
	Refrigerated	10 days

CPT Code Information:

86886

BTR 113384 Isoagglutinin Titer, Anti-B, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Pediatric Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Ambient (preferred)	4 days
	Frozen	10 days
	Refrigerated	10 days

CPT Code Information:

86886

IHDI
82773

Isocyanate HDI, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

IMDI
82774

Isocyanate MDI, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ITDT
82775

Isocyanate TDI, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FISON
91349

Isoniazid (INH)

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red-top

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

ISPG
82768

Ispaghula, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FITZS
75882

Itraconazole Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Complete and submit with specimen:

1. Fungus Testing Lab request form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999-Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 - mould - MIC microdilution or agar dilution (if appropriate)

87186 - yeast - MIC microdilution or agar dilution (if appropriate)

ITCON
81247

Itraconazole, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top (serum gel/SST are **not** acceptable)

Acceptable: None

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.18 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	29 days
	Ambient	29 days
	Frozen	29 days

CPT Code Information:

80189

JCHAI
7048

J-Chain Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

JMACK
82819

Jack Mackerel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

JAKXB
89189

JAK2 Exon 12 and Other Non-V617F Mutation Detection, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 8 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	5 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

0027U-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder), exon 12 sequence and exon 13 sequence, if performed

JAKXM 60025 JAK2 Exon 12 and Other Non-V617F Mutation Detection, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: EDTA (lavender top)

Acceptable: ACD (yellow top)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Bone Marrow	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	5 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

0027U-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder), exon 12 sequence and exon 13 sequence, if performed

JAKXR 44178 JAK2 Exon 12-15 Sequencing, Polycythemia Vera Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information, see PVJAK / Polycythemia Vera, *JAK2* V617F with Reflex to *JAK2* Exon 12-15, Sequencing for Erythrocytosis, Varies.

Submit only 1 of the following specimens:

Specimen Type: Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	5 days

CPT Code Information:

0027U-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder), exon 12 sequence and exon 13 sequence, if performed

JAK2B
88715

JAK2 V617F Mutation Detection, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient (preferred)	7 days	PURPLE OR PINK TOP/EDTA
	Refrigerated	7 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81270-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

JAK2M 31155 JAK2 V617F Mutation Detection, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Bone Marrow	Ambient (preferred)	7 days	PURPLE OR PINK TOP/EDTA
	Refrigerated	7 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81270-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

JAKFM 618647 JAK2 V617F Mutation Detection, Bone Marrow

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81270

JAK2V
31156

JAK2 V617F Mutation Detection, Varies

Specimen Requirements:

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube with indication of volume and concentration of the DNA

Specimen Volume: Entire specimen

Collection Instructions: Label specimen as extracted DNA from blood or bone marrow and indicate volume and concentration of the DNA.

Specimen Stability Information: Refrigerated/Ambient

Specimen Minimum Volume: 50 microliter at a concentration of 20 ng/microliter

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81270-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

FJPE
57921

Jalapeno/Chipotle (Capsicum annuum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

JCEDR
82865

Japanese Cedar, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL

For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FJCQP 75758 JC Polyoma Virus DNA, Quantitative Real-Time PCR, Serum/Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA), yellow top (ACD-A), white top (PPT) tube

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Plasma

Collection Instructions:

1. Centrifuge and aliquot plasma into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	48 hours

CPT Code Information:

87799

JCV 70475 JC Virus Detection by In Situ Hybridization

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block (preferred)

OR

4 Unstained glass, positively-charged slides with 5 (+ or -1)-micron formalin-fixed, paraffin-embedded tissue

Additional Information: Broken or cracked unstained slides will be rejected.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-Primary

88364-If additional ISH

LCJC 800170 JC Virus, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Container/Tube:

Preferred: Aliquot tube

Acceptable: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

JCPCR
618305

JC Virus, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Collection Container/Tube:

Preferred: Vial number 2

Acceptable: Any vial number

Submission Container/Tube: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

JO1
80179

Jo 1 Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

JOHN
82900

Johnson Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

JIP
621699

Joint Infection Panel, PCR, Synovial Fluid

Specimen Requirements:

Specimen Type: Synovial fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Do not process or treat sample in any way.

2. Label specimen as synovial fluid.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Synovial Fluid	Refrigerated (preferred)	7 days
	Ambient	4 hours

CPT Code Information:

87627

JUNE
82893

June Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FWJR 57953 Juniper Western (Juniperus occidentalis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

KAIHC 70482 Kappa Light Chain Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KAIPC
113330

KappaLambda IHC (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342

KBAL
623286

Karius Focus, Bronchoalveolar Lavage

Specimen Requirements:

Container/Tube: Sterile, leak-proof freezable container

Specimen Volume: 1.5 mL

Collection Instructions:

1. Collect bronchoalveolar lavage fluid (BALF) specimen according to your institutional guidelines.

2. Within 12 hours of specimen collection, transfer at least 1.5 mL of BALF specimen to a sterile container and freeze.

3. See Karius Focus: BAL Specimen Collection and Preparation Process for complete instructions.

Additional Information:

1. **Specimen cannot be shared with other tests**.

2. **Specimen trap collection containers (with suction catheters attached) will be rejected** due to the high risk of leakage and contamination upon opening. Avoid use of these for BALF specimens.

3. If specimen transfer into an acceptable sterile container is necessary, perform specimen transfer in a biosafety cabinet. Place the container in a separate sealed plastic bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Bronchoalveolar Lavage	Frozen	180 days

CPT Code Information:

87999

KDBAL
623288

Karius Focus, Direct Ship, Bronchoalveolar Lavage

Specimen Requirements:

Supplies: Karius KDBAL Kit (T1015)

Container/Tube: Streck CP tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect bronchoalveolar lavage fluid (BALF) specimen according to your institutional guidelines.

2. Within 12 hours of specimen collection, transfer at least 1 mL of BALF specimen to the Streck CP tube using disposable sterile pipette, taking care to avoid spilling storage reagent in Streck CP tube.

3. Screw the cap tightly and gently invert the tube 8 to 10 times.

4. **Do not freeze the Streck CP tube.**

5. See Shipping Instructions for how to package and ship the specimen to Karius Labs.

Additional Information:

1. **Specimen cannot be shared with other tests**.

2. **Specimen trap collection containers (with suction catheters attached) will be rejected** due to the high risk of leakage and contamination upon opening. Avoid use of these for BALF specimens.

Alternate Collection Instructions for when storage is needed before shipment

1. Collect BALF specimen according to your institutional guidelines.

2. Freeze BALF in the sterile collection tube. **Do not use the Streck CP tube**, as it must not be frozen.

3. When ready to prepare for shipment, thaw the frozen BALF specimen.

4. Label the Streck CP tube with the patient's first and last name, a unique patient identifier (eg, date of birth or medical record number), the collection date and time, **and the thawing date**.

5. Use a disposable pipette to carefully transfer 1 mL of BALF into the labeled Streck CP tube containing storage reagent. Avoid spilling the reagent.

6. Securely tighten the cap and gently invert the tube 8 to 10 times to mix.

7. **Do not freeze the Streck CP tube**.

8. See Shipping Instructions for how to package and ship the specimen to Karius Labs.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Bronchoalveolar Lavage	Ambient	7 days

CPT Code Information:

87999

KDPLA
623289

Karius Spectrum, Direct Ship, Plasma

Specimen Requirements:

Supplies: Karius KDPLA Kit (T1014)

Collection Container/Tube:

Preferred: Plasma Preparation Tube (PPT)

Acceptable: Lavender top (K2 EDTA)

Submission Container/Tube:

Preferred: PPT

Acceptable: Sterile polypropylene tube

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Gently invert tube 8 to 10 times to mix whole blood.

2. Centrifuge specimen as follows:

a. For PPT: Within 6 hours of collection, centrifuge at 1100 x *g* for 10 minutes.

b. For K2 EDTA:

i. Within 24 hours of collection, centrifuge at 1600 x *g* for 10 minutes. For tubes less than 4 mL, refer to tube manufacturer's instructions for centrifugation speed and time.

ii. Aliquot 1 mL of plasma into a sterile polypropylene tube, taking care not to disturb the buffy coat.

3. See Shipping Instructions for how to package and ship the specimen to Karius Labs.

Specimen Minimum Volume: Plasma: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Ambient	4 days

CPT Code Information:

0152U

KPLA
623287

Karius Spectrum, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Plasma preparation tube (PPT)

Acceptable: Lavender top (K2 EDTA)

Submission Container/Tube:

Preferred: PPT

Acceptable: Sterile polypropylene tube

Specimen Volume: 1 mL Plasma

Collection Instructions :

1. Gently invert tube 8 to 10 times to mix whole blood.

2. Centrifuge specimen as follows:

a. For PPT: Within 6 hours of collection, centrifuge at 1100 x *g* for 10 minutes.

b. For K2 EDTA:

i. Within 24 hours of collection, centrifuge at 1600 x *g* for 10 minutes. For tubes less than 4 mL, refer to tube manufacturer's instructions for centrifugation speed and time.

ii. Aliquot 1 mL of plasma into a sterile polypropylene tube, taking care not to disturb the buffy coat.

3. For complete instructions see Karius Spectrum Specimen Collection and Preparation Process.

Specimen Minimum Volume: Plasma: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	180 days
	Ambient	4 days

CPT Code Information:

0152U

KLHCS 622117 Kelch-Like Protein 11 Antibody, Cell Binding Assay, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended before starting immunosuppressant medication or intravenous immunoglobulin (IVIg) treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

0432U

KLHCC 622118 Kelch-Like Protein 11 Antibody, Cell Binding Assay, Spinal Fluid

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended before initiation of immunosuppressant medication, or corticosteroid or intravenous immunoglobulin (IVIg) treatment.

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 1 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

0432U

KRT34 Keratin (34BE12) Immunostain, Technical Component Only

70492

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRTAE 70493 Keratin (AE1/AE3) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRTOS 70495 Keratin (OSCAR) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT19 70490 Keratin 19 (KRT19) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT20 70491 Keratin 20 (KRT20) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time

TECHONLY	Ambient (preferred)
	Refrigerated

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT5 606999 Keratin 5 (KRT5) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT7 70488 Keratin 7 (KRT7) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

K818 Keratin 8 and 18 (KRT8/18), Technical Component only

622524

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FKEMS
75730

Ketamine and Metabolite Screen, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Plasma

Collection Container/Tube: Lavender-top or pink top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in an EDTA (lavender top or pink top) tube(s). **Plasma gel tube is not acceptable.**

2. Centrifuge and send 3 mL of plasma refrigerated in a preservative-free plastic vial.

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	14 days

	Frozen	270 days
	Ambient	14 days

CPT Code Information:

80307-Screen

80357-Confirmation, if appropriate

KETGP 608024 Ketone Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight..

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SKETC 606546 Ketones, Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 20 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	72 hours

CPT Code Information:

81003

KI67
70481

Ki-67 (MIB-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-Primary

88341-If additional IHC

KI67B
70588

Ki-67(MIB-1), Breast, Quantitative Immunohistochemistry, Automated

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Preferred: Formalin-fixed, paraffin-embedded tissue block containing

invasive or metastatic breast carcinoma

Acceptable: 2 Unstained sections, containing invasive or metastatic breast carcinoma, on charged slides cut at 4 microns less than 1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin. Also send one hematoxylin and eosin-stained slide if possible.

Submission Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit paraffin-embedded invasive or metastatic breast carcinoma tissue.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88361

KIBM 71668 Ki-67(MIB-1), Breast, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This is not an orderable test. Order PATHC / Pathology Consultation. The consultant will determine the need for special stains.

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing invasive or metastatic breast carcinoma

Acceptable: 2 unstained sections, containing invasive or metastatic breast carcinoma, on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Submission Container/Tube: Pathology Packaging Kit (T554)

Collection Instructions: Submit paraffin-embedded invasive or metastatic breast carcinoma tissue.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigered	

CPT Code Information:

88360

KINET 71503 Ki-67(MIB-1), Gastrointestinal/Pancreatic Neuroendocrine Tumors, Quantitative Immunohistochemistry, Automated

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing neuroendocrine tumor of the pancreas or gastrointestinal (GI) tract including metastases.

Acceptable: 2 unstained sections, containing neuroendocrine tumor of the pancreas or GI tract including metastases, on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Submission Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit formalin-fixed, paraffin-embedded tissue block.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88361

KINM
71667

Ki-67(MIB-1), Gastrointestinal/Pancreatic Neuroendocrine Tumors, Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This is not an orderable test. Order PATHC / Pathology Consultation. The consultant will determine the need for special stains.

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing neuroendocrine tumor of the pancreas or gastrointestinal (GI) tract including metastases.

Acceptable: 2 unstained sections, containing neuroendocrine tumor of the pancreas or GI tract including metastases, on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Submission Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit formalin-fixed, paraffin-embedded tissue block

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

KI67P
72130

Ki-67(MIB-1), Pulmonary, Quantitative Immunohistochemistry, Automated

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing carcinoid/atypical carcinoid tumor of the lung including metastases.

Acceptable: 2 Unstained sections containing carcinoid/atypical carcinoid tumor of the lung including metastases on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit formalin-fixed, paraffin-embedded tissue block.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88361

KIPM 72131 Ki-67(MIB-1), Pulmonary, Quantitative Immunohistochemistry, Manual (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see KI67P / Ki-67 (MIB-1), Pulmonary, Quantitative Immunohistochemistry, Automated.

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing carcinoid/atypical carcinoid of the lung including metastases.

Acceptable: 2 Unstained sections on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Container/Tube: Pathology Packaging Kit

Collection Instructions:

1. Submit formalin-fixed, paraffin-embedded tissue block.

2. Attach the green pathology address label included in the kit to the outside of the transport container.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

KIMEL
70483

Ki67 + Melan A Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

KIDBN
82619

Kidney Bean (Red), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

KIDST
605761

Kidney Stone Analysis

Specimen Requirements:

Supplies: Stone Analysis Collection Kit (T550)

Sources: Bladder, kidney, prostatic, renal, or urinary

Specimen Volume: Entire dried calculi specimen

Collection Instructions:

1. Have patient collect specimen using the Patient Collection Instructions for Kidney Stones.

2. For detailed instructions on kidney stone preparation and packaging, see Kidney Stone Packaging Instructions.

3. Prepare stone by cleaning any blood or foreign material from the stone with deionized water.

4. Place stone on a clean filter or paper towel and let dry at ambient temperature for a minimum of 24 hours.

5. **Do not place stone directly in a bag.** If specimen is received in a bag, either transfer stone into a screw-capped, plastic container or place bag containing stone in a screw-capped, plastic container.

6. Indicate source of specimen on the outside of the container (eg, left kidney, bladder, right ureter).

7. Repeat steps 2 through 5 for each stone received.

Specimen Stability Information: Ambient (preferred) 2 years/Refrigerated 1 year/Frozen 1 year

Specimen Minimum Volume: Entire stone

Transport Temperature:

Specimen Type	Temperature	Time
Stone	Ambient (preferred)	
	Frozen	365 days
	Refrigerated	365 days

CPT Code Information:

82365

KKBRP 65202 Kingella kingae, Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Kingella kingae* DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send specimen in original tube (preferred).

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

KKRP 65201 Kingella kingae, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Kingella kingae* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Synovial fluid

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red clot tube (no anticoagulant), or sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send specimen in original tube (preferred).

Specimen Stability Information: Refrigerated (preferred) <7 days /Frozen <7 days

Specimen Type: Fresh tissue or biopsy

Sources: Bone, joint, synovium, heart valve, aorta, or endocardium

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3)- approximately the size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue only, do not add fluid to tissue

3. Refrigerate or freeze specimen.

Specimen Stability Information: Refrigerated (preferred) <7 days/ Frozen <7 days

Preferred Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Bone, joint, synovium, heart valve, aorta, or endocardium

Supplies: Tissue Block Container (T553)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Bone, joint, synovium, heart valve, aorta, or endocardium

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: Fluid/fresh tissue or biopsy: See Specimen Required
Paraffin-embedded tissue block: Two 10-micron sections

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87798

KIT
70485

KIT Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KITQ
620901

KIT p.Asp816Val Variant Analysis, Quantitative, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix.

2. Send specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix.

2. Send specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA

2. Provide indication of volume of DNA.

Specimen Stability Information: Frozen (preferred)/ Refrigerate/Ambient

Additional Information: We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied.

Specimen Type: Paraffin-embedded bone marrow aspirate clot

Container/Tube: Paraffin block

Specimen Stability Information: Ambient

Specimen Type: Tissue (FFPE)

Container/Tube: Paraffin block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block.

Specimen Stability Information: Ambient

Additional Information: Decalcified core biopsies are not accepted.

Specimen Minimum Volume: Blood, bone marrow: 1 mL; Extracted DNA: 50 mcL at 10 ng/mcL concentration; Paraffin block: 1 block

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81273

KIWI
82761

Kiwi Fruit, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

XYMF 35307 Known 45,X, Mosaicism Reflex Analysis, FISH, Whole Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

4. Cord blood is acceptable

Additional Information:

1. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

2. Specimen cannot be frozen.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

GALC
622624

Krabbe Disease, GALC Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's

solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test

may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

KRASW 616452 KRAS Somatic Mutation Analysis, Peritoneal Fluid

Specimen Requirements:

Container/Tube: 50-mL Falcon tube

Preferred: Fresh, peritoneal washing; no fixatives added to wash

Specimen Volume: Two 50-mL Falcon tubes

Collection Instructions: Containers must be labeled with two unique patient identifiers.

Specimen Minimum Volume: 100 mL of peritoneal washing

Transport Temperature:

Specimen Type	Temperature	Time
Peritoneal	Refrigerated (preferred)	10 days
	Ambient	5 days

CPT Code Information:

81275-KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene) (eg, carcinoma) gene analysis, variants in codons 12 and 13

81276-KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene) (eg, carcinoma) gene analysis, additional variants

KRASP 610679 KRAS Somatic Mutation Analysis, Tumor

Specimen Requirements:

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides

: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred), or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81275-KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene) (eg, carcinoma) gene analysis, variants in codons 12 and 13

81276-KRAS additional variant(s)

88381-Microdissection, manual

LBCS 620895 Labile Bound Copper, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours**.

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: 6-mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.75 mL serum

Collection Instructions:

1. For complete instructions, see Metals Analysis Specimen Collection and Transport.

2. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

3. Remove the stopper. Carefully pour specimen into metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

4. Freeze sample on dry ice immediately after pouring off the serum.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Frozen	28 days	METAL FREE

CPT Code Information:

82525

LACO
62905

Lacosamide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. For sustained-release formulations ONLY, collect specimen a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80235

LDBF
606612

Lactate Dehydrogenase (LDH), Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial

-Synovial

-Cerebral spinal fluid

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Ambient (preferred)	7 days
	Refrigerated	48 hours

CPT Code Information:

83615

LD
8344

Lactate Dehydrogenase (LDH), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Ambient (preferred)	7 days
	Frozen	30 days
	Refrigerated	48 hours

CPT Code Information:

83615

LAPYP
616609

Lactate Pyruvate Panel, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Green top (Sodium heparin)

Acceptable: Green top (Lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

83605

84210

LACS1 Lactate, Plasma
601685

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Gray top (potassium oxalate/sodium fluoride)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.625 mL Plasma

Collection Instructions:

1. Draw at least 1 mL whole blood in a 2-mL collection tube or at least 2 mL whole blood in a 4-mL collection tube.

2. Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma NaFl-KOx	Refrigerated (preferred)	14 days
	Ambient	8 hours

CPT Code Information:

83605

LASF1 Lactic Acid, Spinal Fluid
601821

Specimen Requirements:

Specimen Type: Spinal fluid

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Centrifuge to remove any cellular material.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	60 days
	Refrigerated	24 hours
	Ambient	3 hours

CPT Code Information:

83605

LCTF
623285

Lactoferrin, Feces

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a fresh random fecal specimen, no preservatives.

2. If specimen is sent refrigerate, send immediately after collection.

3. If specimen cannot be sent immediately, freeze specimen and send frozen.

Additional Information:

1. Separate specimens must be submitted when multiple tests are ordered, with the exception of ELASF / Pancreatic Elastase, Feces and CALPR / Calprotectin, Feces. If only a single specimen is collected, it must be split prior to transport.

2. Testing cannot be added on to a previously collected specimen.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	7 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

83631

LACTO Lactotransferrin Immunostain, Technical Component Only
70625

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LAMQ Lamb's Quarter, IgE, Serum
82682

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LAMB
82699

Lamb, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LAIHC
70499

Lambda Light Chain Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LAMO
80999

Lamotrigine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect blood immediately before next scheduled dose.

2. **Only** for sustained-release formulations, collect blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge.

4. For red-top tubes, immediately aliquot serum into a plastic vial.

5. For serum gel tubes, **the serum must be aliquoted into a plastic vial within 24 hours of collection**.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80175

LANGR
70496

Langerin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LANG
82349

Langust (Lobster), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LATX
82787

Latex, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLDLD
75160

LDL Cholesterol, Direct

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

83721

PBUOE
608898

Lead Occupational Exposure, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic vial or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83655

82570

PBOU
608894

Lead Occupational Exposure, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

PBUOE / Lead Occupational Exposure, Random, Urine

HMUOE / Heavy Metal Occupational Exposure, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBPFD
619722

Lead Profile Occupational Exposure, Blood

Specimen Requirements:

Both EDTA whole blood and heparin whole blood specimens are required.

Patient Preparation:

1. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively couple plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

2. Patient should abstain from alcohol for 24 hours prior to specimen collection.

Specimen Type: EDTA Whole blood

Supplies:

-Metal Free B-D Tube (EDTA), 6 mL (T183)

-Metal Free EDTA 3 mL Tube (T989)

-Microtainer (EDTA) Tube, 0.5 mL (T174)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 36777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183) or BD Microtainer with EDTA (0.5 mL) (T174)

Specimen Volume: 2 mL

Collection Instructions:

1. See Metal Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Refrigerate specimen as soon as possible after collection.

Specimen Type: Heparin Whole blood

Container/Tube:

Preferred: Green top (Sodium heparin)

Acceptable: Dark blue top (metal free heparin), green top (lithium heparin), or lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions: Refrigerate specimen as soon as possible after collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

83655

82542

Lead, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium or iodine containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBDC
113400

Lead, Capillary, with Demographics, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal testing. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Microtainer (EDTA) Tube, 0.5 mL (T174)

Collection Container/Tube: BD Microtainer MAP (microtube for automated process) with EDTA

Specimen Volume: 0.4 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBHA
8495

Lead, Hair

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Head (preferred), beard, mustache, chest, pubic

Specimen Volume: 0.2 g

Collection Instructions: Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time

Hair	Ambient (preferred)
	Frozen
	Refrigerated

CPT Code Information:

83655

PBNA
89857

Lead, Nails

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Fingernails or toenails

Specimen Volume: 0.2 g

Collection Instructions:

1. Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

2. Clippings should be taken from all 10 fingernails or toenails.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Nail	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

83655

PBDV
113401

Lead, Venous, with Demographics, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based

metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 2 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBUCR 608908 Lead/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83655

82570

PBCU
608904

Lead/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

PBUCR / Lead/Creatinine Ratio, Random, Urine

HMUCR / Heavy Metal/Creatinine Ratio, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

LEFLU 60292 Leflunomide Metabolite (Teriflunomide), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Draw blood no sooner than 12 hours (trough value) after last dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Ambient (preferred)	28 days
	Frozen	28 days
	Refrigerated	28 days

CPT Code Information:

80193

LAGU 81268 Legionella Antigen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Excessively bloody or very turbid specimens containing protein, cells, or particulates will be canceled. They can inhibit the function of the test.

4. Centrifuging to remove particulates is not approved.

5. Specimens with any dyes or unnatural color are not acceptable and will be canceled.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	24 hours

CPT Code Information:

87899

LEGI 8204 Legionella Culture, Varies

Specimen Requirements:

Specimen Type: Lower respiratory

Sources: Bronchoalveolar lavage, bronchial aspirate/brushing/lavage/washing, tracheal/endotracheal secretions/aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Fresh tissue or biopsy

Sources: Lung, pleura, heart valve, pericardium

Container/Tube: Sterile container

Specimen Volume: 5 mm(3)

Collection Instructions: Aseptically collect a 1 to 2 cm(3) piece of tissue whenever possible

Specimen Type: Fluid

Sources: Pericardial, pleural, chest, chest tube drainage, thoracentesis, empyema

Container/Tube: Sterile container

Specimen Volume: 2 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	48 hours

CPT Code Information:

87081-*Legionella* culture

87077-Ident by MALDI-TOF mass spec (if appropriate)

87153-Aerobe ident by sequencing (if appropriate)

87176-Tissue processing (if appropriate)

SLEG 8122 Legionella pneumophila (Legionnaires Disease), Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86713

LEGRP 89564 Legionella species, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Legionella* DNA is unlikely.

Specimen Type: Lower respiratory

Sources: Bronchoalveolar lavage, bronchial aspirate/brushing/lavage/washing, tracheal/endotracheal secretions/aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Type: Fresh tissue or biopsy

Sources: Lung, pleura, heart valve, pericardium

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3) - approximately the size of a pencil eraser

Collection Instructions: Aseptically collect a 1 to 2 cm(3) piece of tissue whenever possible

Specimen type: Fluid

Sources: Pericardial, pleural, chest, chest tube drainage, thoracentesis, empyema

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87801

LEIS
86219

Leishmaniasis (Visceral) Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86717

FLEMG
57643

Lemon IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), a serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3 Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

LEM
82678

Lemon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLENG
57685

Lentil IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

LEN
82885

Lentil, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LEPD
82849

Lepidoglyphus destructor, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLEP
91339

Leptin

Specimen Requirements:

Patient Preparation:

Fasting: Recommended

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 45 minutes of collection, centrifuge and aliquot serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.3 mL NOTE: Minimum volume does not allow for repeat analysis.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	
	Refrigerated	14 days
	Ambient	8 days

CPT Code Information:

83520

LEPDT
65183

Leptospira, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Additional Information: If acute and convalescent specimens are being obtained to determine seroconversion, they should be collected 2 or more weeks apart.

Specimen Minimum Volume: Serum 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86720

FLETG 57639 Lettuce IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

LETT 82805 Lettuce, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LLTOT 621505 Leukemia and Lymphoma Immunophenotyping, Technical Only, Tissue

Specimen Requirements:

Submit 1 of the following specimens:

Preferred

Specimen Type: Tissue

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: 5 mm(3) or larger biopsy

Collection Instructions:

1. Place tissue into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen **(do not mince)**.

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Acceptable

Specimen Type: Fine needle aspirate (FNA)

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: Entire collection

Collection Instructions:

1. Collect FNA and transfer entire collection into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen **(do not mince)**.

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Minimum Volume: Tissue: 1 mm(3); Fine needle aspirate: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

Additional CPTs may be added if consultative help is needed with the case, or algorithm dictates Mayo consultant involvement.

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LLTOF 34488 Leukemia and Lymphoma Phenotyping, Technical Only, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 10 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as blood.

3. Include 1 to 2 unstained blood smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

4. Include 1 to 2 unstained bone marrow smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Note: A fresh (less than 4 days post-collection), unfixed, nonembedded bone marrow core biopsy, bone or bone lesion is acceptable as an equivalent source for bone marrow aspirate for this test **only in the event of a dry tap** during the bone marrow harvesting

procedure. Indicate "dry tap" in performing lab notes or paperwork when submitting this specimen type.

Specimen Type: Body fluid

Sources: Serous effusions, pleural, pericardial, or abdominal (peritoneal fluid)

Container/Tube: Body fluid container

Specimen Volume: 20 mL

Collection Instructions:

1. If possible, body fluids other than spinal fluid should be anticoagulated with heparin (1 U/mL of fluid).

2. Label specimen with body fluid type.

Additional Information: The volume of serous effusion necessary to phenotype lymphocytes or blasts depends upon the cell count in the specimen. Usually, 20 mL of pleural or peritoneal fluid is sufficient. Smaller volumes can be used if there is a high cell count.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. An original cytospin preparation (preferably unstained) should be included with the spinal fluid specimen so correlative morphologic evaluation can occur.

2. Label specimen as spinal fluid.

Specimen Stability Information: Refrigerated /Ambient 4 days

Additional Information: The volume of spinal fluid necessary to phenotype the lymphocytes or blasts depends upon the cell count in the specimen. A cell count should be determined and submitted with the specimen. Usually, 1 to 1.5 mL of spinal fluid is sufficient. Smaller volumes can be used if there is a high cell count. If the cell count is less than 10 cells/mcL, a larger volume of spinal fluid may be required. When cell counts drop below 5 cells/mcL, the immunophenotypic analysis may not be successful.

Specimen Minimum Volume: Whole blood: 3 mL; Bone marrow: 0.5 mL; Spinal fluid: 1 mL; Fluid from serous effusions: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

Additional CPTs may be added if consultative help is needed with the case, or algorithm dictates Mayo consultant involvement.

88187-Flow cytometry interpretation, 2 to 8 markers (if appropriate)

88188-Flow cytometry interpretation, 9 to 15 markers (if appropriate)

88189-Flow cytometry interpretation, 16 or more markers (if appropriate)

LLPB 800433 Leukemia/Lymphoma Immunophenotyping, Flow Cytometry, Blood

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Slides: If possible, include 5 to 10 unstained blood smears labeled with **two unique identifiers**

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LLPT 19499 Leukemia/Lymphoma Immunophenotyping, Flow Cytometry, Tissue

Specimen Requirements:

Submit 1 of the following specimens:

Preferred

Specimen Type: Tissue

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: 5 mm(3) or larger biopsy

Collection Instructions:

1. Place tissue into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen (**do not mince**)

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Acceptable

Specimen Type: Fine needle aspirate (FNA)

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: Entire collection

Collection Instructions:

1. Collect FNA and transfer entire collection into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen (**do not mince**)

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Minimum Volume: 1 mm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LCMS 3287 Leukemia/Lymphoma Immunophenotyping, Flow Cytometry, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

3. Include 1 to 2 unstained blood smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

4. Include 1 to 2 unstained bone marrow smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Note: A fresh (less than 4 days post-collection), unfixed, nonembedded bone marrow core biopsy, bone or bone lesion is acceptable as an equivalent source for bone marrow aspirate for this test **only in the event of a dry tap** during the bone marrow harvesting procedure. Indicate "dry tap" in performing lab notes or paperwork when submitting this specimen type.

Specimen Type: Body fluid

Sources: Serous effusions, pleural fluid, pericardial fluid, abdominal (peritoneal) fluid,

bronchoalveolar lavage (BAL)

Container/Tube: Body fluid container

Specimen Volume: 20 mL

Collection Instructions:

1. If possible, body fluids other than spinal fluid should be anticoagulated with heparin (1 U/mL of fluid).

2. Label specimen with body fluid type.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Additional Information: The volume of serous effusions necessary to phenotype lymphocytes or blasts depends upon the cell count in the specimen. Usually, 20 mL of pleural or peritoneal fluid is sufficient. Smaller volumes can be used if there is a high cell count.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. An original cytospin preparation (preferably unstained) should be included with the spinal fluid specimen so correlative morphologic evaluation can occur.

2. Label specimen as spinal fluid.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Additional Information: The volume of spinal fluid necessary to phenotype lymphocytes or blasts depends upon the cell count in the specimen. A cell count should be determined and submitted with the specimen. Usually, 1 to 1.5 mL of spinal fluid is sufficient. Smaller volumes can be used if there is a high cell count. If cell count is less than 10 cells/mcL, a larger volume of spinal fluid may be required. When cell counts drop below 5 cells/mcL, the immunophenotypic analysis may not be successful.

Specimen Minimum Volume: Blood: 3 mL; Bone marrow: 0.5mL; Spinal fluid: 1 mL; Fluid from serous effusions: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LAD1
81155

Leukocyte Adhesion Deficiency Type 1, CD11a/CD18 and CD11b/CD18 Complex Immunophenotyping, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x 3

LECT2
70497

Leukocyte Cell-Derived Chemotaxin 2 (LECT2), Immunostains Without Interpretation

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TLTE4 Leukotriene E4, 24 Hour, Urine

606355

Specimen Requirements:

Patient Preparation: Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Start collection within a few hours of symptom onset and collect urine for 24 hours

2. No preservative preferred.

3. Aliquot urine into a plastic vial and send frozen.

Additional Information: See Urine Preservatives-Collection and Transportation for

24-Hour Urine Specimens for multiple collection.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	28 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

82542

RLTE4 Leukotriene E4, Random, Urine
606354

Specimen Requirements:

Patient Preparation: Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

3. Aliquot urine into a plastic vial and send frozen.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	28 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

82542

82570

FLEVA
75401

Levamisole, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container: Plastic, preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect 1 mL random urine specimen in a plastic preservative-free urine container.

2. Ship refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80375

LEV1P
113309

Level 1 Gross only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88300

LEV2P 113310 Level 2 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88302

LEV3P 113311 Level 3 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88304

LEV4P 113312

Level 4 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305

LV4RP 113313 Level 4 Gross and Microscopic, RB (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305

LEV5P 113314 Level 5 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88307

LEV6P
113315

Level 6 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88309

LEVE
83140

Levetiracetam, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. **For sustained-release formulations only**, draw blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80177

LID
8382

Lidocaine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80176

LMO2
70501

LIM Domain Only 2 (LMO2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LIME
82360

Lime, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ALBLI 603181 Limited Bleeding Diathesis Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

LDPU 615289 Limited Drug Profile, 3 Drug Classes, Immunoassay, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see CSMHU / Controlled Substance Monitoring Hybrid Drug Profile, 20 Drug Classes, High-Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 10 mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 5 mL in 10 plastic tube.

3. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

FLLYS 75928 Limulus Amebocyte Lysate, Endotoxin

Specimen Requirements:

Specimen Type: Aqueous solutions

Container/Tube: Nonpyrogenic, plastic, leak-proof container

Specimen Volume: 5 mL

Collection Instructions:

1.Collect 5 mL aqueous solutions in a nonpyrogenic, plastic, leak-proof container used in patient management.

2. Send frozen.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Frozen	30 days

CPT Code Information:

87999

LIND
82862

Linden, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LINS
86311

Linseed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LPSBF
606615

Lipase, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Sources:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

83690

FLIPR
90347

Lipase, Random Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect random urine without preservative.
2. Ship 2 mL urine in a sterile screw capped plastic container ship ambient.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Ambient (preferred)	5 days
	Frozen	21 days
	Refrigerated	5 days

CPT Code Information:

83690

LPS
8328

Lipase, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tube must be centrifuged within 2 hours of collection.

2. Red-top tube must be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

83690

BFLA1 Lipid Analysis, Body Fluid

614164

Specimen Requirements:

Specimen Type: Body fluid

Preferred Sources:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

-Pericardial

Acceptable Source:

Collection Container/Tube: Sterile container, no additive

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

82664-Electrophoretic technique, not elsewhere specified (Chylomicrons and lipoproteins)

84311-Spectrophotometry, analyte not specified (Cholesterol)

84478-Triglycerides

LPSC1 616696 Lipid Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tube must be centrifuged within 2 hours of collection.

2. Red-top tube must be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days

	Frozen	30 days

CPT Code Information:

80061-Lipid panel (if all 3 performed)

82465-Cholesterol, total (if all 3 are not performed)

84478-Triglycerides (if all 3 are not performed)

83718-Cholesterol, HDL (if all 3 are not performed)

LIPOG 617337 Lipodystrophy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 x2

81408

81479

81479 (if appropriate for government payers)

LPALD 610738 Lipoprotein (a) and Low-Density Lipoprotein Cholesterol, Serum

Specimen Requirements:

Patient Preparation:

1. **Fasting: 8 hours, required**

2. Patient must abstain from alcohol for 24 hours before collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

83700

83701

LMPP
83673

Lipoprotein Metabolism Profile, Serum

Specimen Requirements:

Patient Preparation:

1**. Fasting: 12 hours, required**

2. Patient **must not** consume any alcohol for 24 hours before the specimen is collected.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

80061-Lipid panel (includes: HDL [CPT Code 83718], total cholesterol [CPT Code 82465], and triglycerides [CPT Code 84478] if all performed)

82172-Apolipoprotein B

83700-Lp(a) cholesterol electrophoresis

LIPA1 Lipoprotein(a), Serum

624437

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	20 days
	Frozen	91 days
	Ambient	5 days

CPT Code Information:

83695

LUCSF LiquidHALLMARK CSF

622814

Specimen Requirements:

Supplies: Sterile Specimen Tube, 6 mL (T485)

Container/Tube: Sterile tube

Specimen Volume: 5 mL

Collection Instructions:

1. Perform lumbar puncture and discard the first 1 mL to 2 mL of cerebrospinal fluid (CSF).

2. Collect CSF directly into a sterile tube.

3. Inspect specimen for visible discoloration. **Specimen must be clear and colorless to perform testing.**

4. Freeze sample upright prior to placing in transport container.

5. Send frozen.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen	30 days

CPT Code Information:

81455

LUCHM 622811 LiquidHALLMARK ctDNA and ctRNA

Specimen Requirements:

Specimen Type: Whole blood

Supplies: Streck Tan Top Tube Kit (T715)

Container/Tube: Two 10-mL Streck cell-free DNA (cfDNA) blood collection tubes

Specimen Volume: 20 mL; 10 mL in two Streck tubes

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not centrifuge or aliquot.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Streck	Ambient	4 days	Streck Black/Tan top

CPT Code Information:

0571U

FLISD
75638

Lisdexamfetamine as Metabolite, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions:

1. Send 1 mL random urine specimen in a plastic preservative-free urine container.

2. Ship refrigerate.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	180 days
	Frozen	180 days
	Ambient	7 days

CPT Code Information:

80324

LITH
37046

Lithium, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood 8 to 12 hours after last dose (trough specimen).

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Additional Information: Peak serum concentrations do not correlate with symptoms.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80178

LFABP 70429 Liver Fatty Acid-Binding Protein (L-FABP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LIVPR
113633

Liver Profile, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into an amber vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated	7 days

CPT Code Information:

82247

82248

84450

84460

84075

82040

84155

80076 (if appropriate if all analytes performed)

LKM
80387

Liver/Kidney Microsome Type 1 Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86376

LOB
82744

Lobster, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LQTSG
617351

Long QT Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81406 x2

81407

81479

81479 (if appropriate for government payers)

LORAZ
80459

Lorazepam (Ativan), Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma:

Specimen Type: Plasma

Collection Container/Tube: Green top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a green-top sodium heparin tube(s), **plasma gel tube is not acceptable.** Centrifuge and send 2 mL of plasma refrigerated in a plastic vial.

Serum:

Specimen Type: Serum

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Centrifuge and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80346

G0480 (if appropriate)

LDLD 606898 Low-Density Lipoprotein (LDL) Cholesterol, Beta-Quantification, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial. Send refrigerated.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	60 days

CPT Code Information:

83701

FUSF 35304 Low-Grade Fibromyxoid Sarcoma (LGFMS), 16p11.2 (FUS or TLS) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Preferred: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded (FFPE) tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Acceptable: Slides

Collection Instructions: Four consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Two consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

FLXPN Loxapine, Plasma

75950

Specimen Requirements:

Must submit one specimen per order. Specimens cannot be shared between multiple orders.

Collection Container/Tube: Lavender or pink top (EDTA)

Submission Container/Tube: Plastic screw-capped vial

Specimen Volume: 1 mL plasma

Collection Instructions:

1. Promptly centrifuge and aliquot 1 mL plasma into a plastic screw-capped vial.

2. Send refrigerated.

Specimen Stability Information: Refrigerated (preferred) 30 days/Ambient 30 days/Frozen 15 months

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	

CPT Code Information:

80342

FLSDA
75680

LSD Trace Analysis, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Urine

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions: Collect 1 mL random urine without preservative. Send specimen refrigerated in a plastic, preservative-free urine container.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

80323

LCAF
35282

Lung Cancer, ALK (2p23) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

RETF 35846 Lung Cancer, RET (10q11) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

ROS1F 35845 Lung Cancer, ROS1 (6q22) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

LUPN
82613

Lupin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLUPV
91714

Lupus Anticoagulant Evaluation with Reflex

Specimen Requirements:

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL platelet-poor plasma, in plastic vial

Collection Instructions: Draw blood into a 3.2% sodium citrate (light blue-top) tube.

Centrifuge light blue-top tube 15 minutes at approx. 1500 g within 60 minutes of collection. Using a plastic pipette, remove plasma, taking care to avoid the WBC/platelet buffy layer and place into a plastic vial. Centrifuge a second time and transfer platelet-poor plasma into a new plastic vial. Plasma must be free of platelets (<10,000/mcl).

Freeze immediately and ship on dry ice.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	30 days

CPT Code Information:

85613/ Russell viper venom time (includes venom); diluted

85730/ Thromboplastin time, partial (PTT); plasma

85598/ Platelet neutralization (Hexagonal Phase Confirm)-(if appropriate)

85597/ Platelet neutralization (dRVVT Confirm)-(if appropriate)

85613/dRVVT 1:1 Dilution (if appropriate)

85670/Thrombin Clotting Time (if appropriate)

ALUPO 603465 Lupus Anticoagulant Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

ALUPP 603463 Lupus Anticoagulant Profile, Plasma

Specimen Requirements:

Patient Preparation:

1. If possible, patient should not be receiving anticoagulant treatment (eg, warfarin, heparin): If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, do not administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, do not administer subcutaneous heparin or Coumadin (warfarin).

2. Patient should not be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator: tPA).

3. If patient has been recently transfused, it is best to perform this study pretransfusion, if possible.

Specimen Type: Platelet-poor plasma

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Light-blue top (citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 4 mL in 4 plastic vials each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 4 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: 3 mL in 3 plastic vials each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85610

85730

85613

85390

85130 (if appropriate)

85130 (if appropriate)

85210 (if appropriate)

85220 (if appropriate)

85230 (if appropriate)

85240 (if appropriate)

85245 (if appropriate)

85246 (if appropriate)

85247 (if appropriate)

85250 (if appropriate)

85260 (if appropriate)

85270 (if appropriate)

85280 (if appropriate)

85335 (if appropriate)

85335 (if appropriate)

85335 (if appropriate)

85366 (if appropriate)

85379 (if appropriate)

85384 (if appropriate)

85385 (if appropriate)

85390-26 (if appropriate)

85397 (if appropriate)

85597 (if appropriate)

85598 (if appropriate)

85611 (if appropriate)

85613 (if appropriate)

85613 (if appropriate)

85635 (if appropriate)

85670 (if appropriate)

85732 (if appropriate)

ALUPI 603464 Lupus Anticoagulant Profile, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ALUPP / Lupus Anticoagulant Profile, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

LUTHI 70498 Luteinizing Hormone (LH) Beta Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LHPED 62999 Luteinizing Hormone (LH), Pediatrics, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube. 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.25 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red top tubes, aliquot serum into a plastic vial immediately.

3. For serum gel tubes, serum may sit on gel refrigerated but must be aliquoted into a plastic vial within 4 hours.

Specimen Minimum Volume: 0.13 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

83002

LH
602752

Luteinizing Hormone (LH), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days

	Frozen	180 days
	Ambient	24 hours

CPT Code Information:

83002

SLYME 602732 Lyme Antibody Modified 2-Tier with Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	30 days

CPT Code Information:

86618

86617 x2 (if appropriate)

LNBAB 63502 Lyme Central Nervous System Infection IgG with Antibody Index Reflex, Serum and Spinal Fluid

Specimen Requirements:

Both spinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within 24 hours (maximum) of each other.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A CSF sample of 1.2 mL needs to be collected within 24 hours of the serum specimen, preferably at the same time.

2. Label vial as spinal fluid or CSF.

3. CSF aliquot should be from the second, third, or fourth CSF vial collected during the lumbar puncture.

Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection.

4. Band specimens together.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A serum sample of 1.2 mL needs to be collected within 24 hours of the spinal fluid specimen, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label as serum.

4. Band specimens together.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	11 days
	Frozen	35 days
Serum	Refrigerated (preferred)	11 days
	Frozen	35 days

CPT Code Information:

86618

86618 x2-(if appropriate)

82040-(if appropriate)

82042-(if appropriate)

82784 x2-(if appropriate)

LNBAI 63249 Lyme Central Nervous System Infection IgG, Antibody Index, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see LNBAB / Lyme Central Nervous System Infection IgG with Antibody Index Reflex, Serum and Spinal Fluid.

Both cerebrospinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within 24 hours maximum of each other. CSF specimens with blood contamination will be rejected.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A spinal fluid (CSF) sample of 1.2 mL needs to be collected within 24 hours of the serum specimen, preferably at the same time.

2. Label vial as spinal fluid or CSF.

3. CSF aliquot should be from the second, third, or fourth CSF vial collected during the

lumbar puncture.

Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection.

4. Band specimens together.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A serum sample of 1.2 mL needs to be collected within 24 hours of the spinal fluid specimen, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label as serum.

4. Band specimens together.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	11 days
	Frozen	35 days

CPT Code Information:

86618 x 2

82040

82042

82784 x 2

LYWB
9535

Lyme Disease Antibody, Immunoblot, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.75 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

86617 x 2

LYME
9129

Lyme Disease Serology, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	30 days

CPT Code Information:

86618

86617 x 2-Lyme disease confirmation (if appropriate)

PBORB 87973 Lyme Disease, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87476

87798 x 2

87801 (if appropriate for government payers)

LYMPV 618310 Lyme Disease, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Label specimen as spinal fluid.

Specimen Type: Synovial fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Label specimen as synovial fluid.

Specimen Type: Tissue (fresh only)

Sources: Skin or synovial biopsy

Container/Tube: Sterile container with normal saline

Specimen Volume: Approximately 4 mm(3)

Collection Instructions:

1. Submit only fresh tissue.

2. Skin biopsies:

a. Wash biopsy site with an antiseptic soap. Thoroughly rinse area with sterile water. Do not use alcohol or iodine preparations. A local anesthetic may be used.

b. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

3. Label specimen with source of tissue.

Specimen Minimum Volume: Spinal fluid: 0.3 mL; Synovial fluid: 0.5 mL; Tissue: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87476

87798 x 2

87801 (if appropriate for government payers)

TLYME 608104 Lyme IgM and IgG, Whole Cell Sonicate, ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	30 days

CPT Code Information:

86617 x 2

PM3CX 606164 Lymph3Cx Assay, Primary Mediastinal Large B-cell Lymphoma and Diffuse Large B-cell Lymphoma, mRNA Gene Expression, NanoString, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 6 to 10 unstained

Collection Instructions:

1. Submit 1 slide stained with hematoxylin and eosin.

2. If the formalin-fixed paraffin-embedded (FFPE) biopsy specimen is a core needle biopsy or fine needle aspirate, submit 10 unstained slides.

3. If the FFPE biopsy specimen is a resection, submit 6 unstained slides.

4. Submitted slides should be 4- to 5-micron thick sections placed on positively charged slides.

Additional Information: Paraffin embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

Acceptable:

Specimen Type: Tissue block

Collection Instructions: Submit an FFPE tumor tissue block.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Minimum 60% tumor with or without macrodissection. Minimum required unstained tissue section input: 0.12 mm(3); Slides: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

0120U

88381

LPA3P 62205 Lymphocyte Proliferation to Anti-CD3/Anti-CD28 and Anti-CD3/Interleukin-2 (IL-2), Flow Cytometry, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 20 mL

See table for information on recommended volume based on absolute lymphocyte count

Pediatric Volume:

<3 months: 1 mL

3-24 months: 3 mL

25 months-18 years: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Table. Blood Volume Recommendations Based on Absolute Lymphocyte Count (ALC)

ALC	Blood volume for minimum aCD28 only	Blood volume for minimum of aCD3, aCD28, and IL-2	Blood volume for full assay
<0.5	>15 mL	>28 mL	>50 mL
0.5-1.0	15 mL	28 mL	50 mL
1.1-1.5	6.5 mL	12 mL	24 mL
1.6-2.0	4.5 mL	8.5 mL	16 mL
2.1-3.0	3.5 mL	6.5 mL	12 mL
3.1-4.0	2.5 mL	4.5 mL	8 mL
4.1-5.0	1.8 mL	3.5 mL	6 mL
>5.0	1.5 mL	2.5 mL	5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86353 x 2

86353 (as appropriate)

LPAGF 60592 Lymphocyte Proliferation to Antigens, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 20 mL

See tables for information on recommended volume based on absolute lymphocyte count

Pediatric Volume:

<3 months: 1 mL

3-24 months: 3 mL

25 months-18 years: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Table. Blood Volume Recommendations Based on Absolute Lymphocyte Count (ALC)

Antigen only		
ALC x 10(9)/L	**Blood volume for minimum *Candida albicans* (CA) and tetanus toxoid (TT) Only**	**Blood volume for full assay**
<0.5	>18.5 mL	>40 mL
0.5-1.0	18.5 mL	40 mL
1.1-1.5	8.5 mL	20 mL
1.6-2.0	6.0 mL	12 mL

2.1-3.0	4.5 mL	10 mL
3.1-4.0	3.0 mL	6 mL
4.1-5.0	2.5 mL	5 mL
>5.0	2.0 mL	4 mL
Mitogen and antigen		
ALC x 10(9)/L	**Blood volume for minimum of each assay**	**Blood volume for full assay**
<0.5	>28 mL	>60 mL
0.5-1.0	28 mL	60 mL
1.1-1.5	12 mL	30 mL
1.6-2.0	8.5 mL	20 mL
2.1-3.0	6.5 mL	15 mL
3.1-4.0	4.5 mL	10 mL
4.1-5.0	3.5 mL	8 mL
>5.0	2.5 mL	6 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86353

86353 (if appropriate)

LPMGF 60591 Lymphocyte Proliferation to Mitogens, Blood

Specimen Requirements:
Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 20 mL

See tables for information on recommended volume based on absolute lymphocyte count

Pediatric Volume:

<3 months: 1 mL

3-24 months: 3 mL

25 months-18 years: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Table. Blood Volume Recommendations Based on Absolute Lymphocyte Count (ALC)

Mitogen only			
ALC x 10(9)/L	**Blood volume for minimum phytohemagglutinin (PHA) only**	**Blood volume for minimum PHA and pokeweed mitogen (PWM)**	**Blood volume for full assay**
<0.5	>6.5 mL	>8.5 mL	>22 mL
0.5-1.0	6.5 mL	8.5 mL	22 mL
1.1-1.5	3.0 mL	4.0 mL	10 mL
1.6-2.0	2.0 mL	2.5 mL	7 mL
2.1-3.0	1.5 mL	2.0 mL	6 mL
>3.1	1.0 mL	1.5 mL	4 mL

4.1-5.0	1.0 mL	1.0 mL	3 mL
>5.0	1.0 mL	1.0 mL	2 mL

Mitogen and antigen

ALC x 10(9)/L	Blood volume for minimum of each assay	Blood volume for full assay
<0.5	>28 mL	>60 mL
0.5-1.0	28 mL	60 mL
1.1-1.5	12 mL	30 mL
1.6-2.0	8.5 mL	20 mL
2.1-3.0	6.5 mL	15 mL
3.1-4.0	4.5 mL	10 mL
4.1-5.0	3.5 mL	8 mL
>5.0	2.5 mL	6 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86353

86353 (if appropriate)

LEF1 71356 Lymphoid Enhancer-Binding Factor 1(LEF1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LPLFX 61114 Lymphoplasmacytic Lymphoma/Waldenstrom Macroglobulinemia, MYD88 L265P with Reflex to CXCR4, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 10 days /Refrigerated 10 days

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not

be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection

2. Indicate specimen source.

Specimen Stability Information: Frozen

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 50 mcL at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81305

LYNCP 614572 Lynch Syndrome Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81435

LPCBS 61766 Lysophosphatidylcholines, LC MS/MS, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 Paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA, and then spotted and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800.

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	90 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	28 days	FILTER PAPER

CPT Code Information:

82542

LALB
62954

Lysosomal Acid Lipase, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD) or green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood in original tube. **Do not aliquot**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

82657

LALBS
62955

Lysosomal Acid Lipase, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Collection Container/Tube:

Preferred: Blood spot collection card

Acceptable: PerkinElmer 226 filter paper, Munktell TFN, or Whatman Protein Saver 903 Paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

2. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	28 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	7 days	FILTER PAPER

CPT Code Information:

82657

LDALD 64907 Lysosomal and Peroxisomal Disorders Newborn Screen, Blood Spot

Specimen Requirements:

Patient must be older than 24 hours and younger than 1 week.

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 Paper, local newborn screening card, or blood collected in tubes containing ACD, or EDTA and then spotted and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	90 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	28 days	FILTER PAPER

CPT Code Information:

83789

PLSD
89678

Lysosomal and Peroxisomal Disorders Screen, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA and dried on acceptable filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	90 days	FILTER PAPER

	Frozen	90 days	FILTER PAPER
	Ambient	28 days	FILTER PAPER

CPT Code Information:

83789

LSDS
606771

Lysosomal Disorders Screen, Random, Urine

Specimen Requirements:

Patient Preparation:

1. Do not administer low-molecular weight heparin before specimen collection.

2. Baby wipes or wipes containing soaps or lotions should **not** be used before specimen collection because these may interfere with results.

Supplies: Urine Container, 60 mL (T313)

Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Specimen Volume: 12 mL

Pediatric Volume: 3.5 mL

Collection Instructions:

1. Collect a random urine specimen **(early morning preferred)**.

2. No preservative.

Specimen Minimum Volume: 3.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	15 days
	Frozen	90 days

CPT Code Information:

83789

83864

84377

84275

LSDGP 608011 Lysosomal Storage Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies

on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MURA
607462

Lysozyme (Muramidase), Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

2. Freeze immediately.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

85549

LYSOZ
70503

Lysozyme Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LYSO
82398

Lysozyme, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

MA2ES 605970 Ma2 Antibody by ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516

MA2EC 605971 Ma2 Antibody, ELISA, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 1 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516

MACNT
65405

Macadamia Nut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MACE
82492

Mace, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MACK
82342

Mackerel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMACR 57817 Macroamylase

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

82150

MCRPL 34643 Macroprolactin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

84146 x 2

MAGU
610768

Magnesium, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic urine container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

83735

MGS
8448

Magnesium, Serum

Specimen Requirements:

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Additional Information: If other metal tests are also desired when drawing for a serum magnesium level; the specimen must be drawn in a plain, royal blue-top Vacutainer plastic trace element blood collection tube.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

83735

MAGRU
621396

Magnesium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube or a clean, plastic aliquot container

with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

83735

82570

FMME
57924

Mahi Mahi IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

MOBRD
28339

Mail Out Research, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Varies		

MALI
601988

MAL Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MALP 618475 Malabsorption Evaluation Panel, Feces

Specimen Requirements:

Supplies: Malabsorption Panel (T920)

Container/Tube: Malabsorption kit or 2 small stool containers

Specimen Volume: 18 g split equally between 2 containers

Collection Instructions:

1. Collect a fresh, random fecal specimen, no preservatives.

2. Split specimen equally between 2 small containers.

3. Label one small container with the A1AF and UREDF sample collection labels. Label the other small container with the CALPR, ELASF sample collection label.

4. Freeze immediately

Additional Information:

1. Specimen must be split prior to transport.

2. Testing cannot be added to a previously collected specimen.

3. Specimen cannot be collected from a diaper.

Specimen Minimum Volume: 5 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen	7 days

CPT Code Information:

0430U

LCMAL 87860 Malaria, Molecular Detection, PCR, Varies

Specimen Requirements:

Both blood specimens and slides are required.

Specimen Type: Blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot.**

Specimen Type: Blood films

Container/Tube: Clean, grease-free slides in plastic slide container

Specimen Volume: 2 thin blood films and 2 thick blood films

Collection Instructions:

1. For complete instructions, see Thick and Thin Blood Slide Preparation Instructions.

2. Ideally, blood films should be made directly from uncoagulated blood acquired via fingerstick. However, EDTA anticoagulated blood is also acceptable.

3. Prepare thin blood films as follows:

a. Prepare a thin film with a "feathered edge" that is no more than a single cell thick.

b. Allow the film to thoroughly air dry and then fix by briefly immersing in either absolute or 95% methyl alcohol.

c. Allow to air dry after fixation.

4. Prepare thick blood films as follows:

a. Place a large drop of blood (approximately the size of a dime and preferably from a fingerstick) on a slide.

b. Using a corner of a second slide, spread the drop in a circular motion while applying firm pressure to literally scratch the blood onto the carrier slide. This technique allows the blood to dry quickly and adhere well to the slide. Use approximately 20 circular sweeps with the second slide. The drop of blood should be about the size of a quarter when finished.

c. **Do not fix.** Air dry thoroughly (approximately 45 minutes) before placing in transport container.

Specimen Minimum Volume: Blood: 1 mL
Slides: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87798

MCMF
113355

Malignant Cells Cyto/Heme (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88108

MALT
82834

Malt, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MAMLF 58105 MAML2 (11q21) Rearrangement, Mucoepidermoid Carcinoma (MEC), FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

MGB
70507

Mammaglobin (MGB) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MAND
82352

Mandarin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MNU 8080 Manganese, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 0.8 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Within 4 hours of completion of 24-hour collection, refrigerate specimen.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

83785

MNB 89120 Manganese, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83785

FMNRB Manganese, RBC and Plasma

57196

Specimen Requirements:

Both plasma and RBCs are required for testing.

Specimen Type: Packed RBCs and Plasma

Container/Tube: Royal blue-top (EDTA metal-free tube; metal-free transfer tubes

Specimen Volume: 3 mL each of plasma and packed RBCs

Collection Instructions: Both packed cells and plasma are required. Plasma and red cells must be separated ASAP and shipped separately. Failure to separate may cause rejection due to hemolysis of red cells and contamination of the plasma. Transfer RBCs and plasma into separate plastic, acid-washed, trace metal-free tubes. Ship refrigerated.

NOTE: Specimen must be received at the testing facility within five days of collection.

Specimen Minimum Volume: 0.5 mL each plasma and RBCs

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Metal Free EDTA Plasma	Refrigerated (preferred)	5 days	METAL FREE
	Ambient	72 hours	METAL FREE
RBCS	Refrigerated (preferred)	5 days	METAL FREE
	Ambient	72 hours	METAL FREE

CPT Code Information:

83785

MNS
8413

Manganese, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free B-D Tube (No Additive), 6 mL (T184)

-Metal Free Specimen Vial (T173)

Collection Container/Tube: 6 mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene

vial

Specimen Volume: 1.6 mL Serum

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

83785

MNRCU 614993 Manganese/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82570

83785

MNCU
614994

Manganese/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MNRCU / Manganese/Creatinine Ratio, Random, Urine.

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

83785

MANGO 82811 Mango, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MBLF 614985 Mannan Binding Lectin Complement Pathway, Functional, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. Do **not** collect a specimen for at least 48 hours following plasma exchange.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	14 days

CPT Code Information:

86161

FMPRE 57535 Maple Red (Acer rubrum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

MSUDP Maple Syrup Urine Disease Gene Panel, Varies
608031

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81406 x 3

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

MARE
82141

Mare's Milk, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MFRGG
617379 Marfan, Loeys-Dietz, and Aortopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81410

MARJ
82605

Marjoram, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MSPTC
113288

Mass Spectrometry (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

82542-Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS)

MSMNT
621774

Mass Spectrometry, Membranous Nephropathy, Technical Component (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

82542

MCM24
608378

Mast Cell Mediators, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. Patients taking monoamine oxidase inhibitors (MAOIs) or aminoguanidine may have increased levels of *N*-methylhistamine. If medically feasible, patient **must not** take MAOIs or aminoguanidine for 3 weeks before specimen collection. For more information see Cautions.

2. Patients taking aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) may have decreased concentrations of prostaglandin F2 alpha. If medically feasible, the patient **should not** take aspirin for 2 weeks or NSAIDs for 72 hours before specimen collection.

3. Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 20 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Aliquot urine into plastic tube and send to lab.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	28 days

CPT Code Information:

82542

84150

MCMRU 608379 Mast Cell Mediators, Random, Urine

Specimen Requirements:

Patient Preparation:

1. Patients taking monoamine oxidase inhibitors (MAOI) or aminoguanidine may have increased N-methylhistamine levels. If medically feasible, patient **must not** take MAOI or aminoguanidine for 3 weeks before specimen collection. For more information see Cautions.

2. Patients taking aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) may have decreased concentrations of prostaglandin F2 alpha. If medically feasible, the patient **should not** take aspirin for 2 weeks or NSAIDs for 72 hours before specimen collection.

3. Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 20 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	28 days

CPT Code Information:

82570

84150

82542

MATCC 35479 Maternal Cell Contamination, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Maternal blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Prenatal Specimens:

Submit only 1 of the following specimens:

Specimen Type: Cord blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Additional Information: A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

Specimen Stability Information: Refrigerated (preferred)/Ambient

Specimen Type: Cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Genetic Testing, Tissue.

Specimen Stability Information: Refrigerated

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: Amniotic Fluid: 10 mL
Blood, Cord Blood: 0.5 mL
Chorionic Villus: 5 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		

CPT Code Information:

81265

Fibroblast Culture for Genetic Test

88233-(if appropriate)

88240-(if appropriate)

Amniotic Fluid Culture/Genetic Test

88235-(if appropriate)

88240-(if appropriate)

Each additional specimen

81266

FFMSS 75692 Maternal Serum Screening, Integrated, Specimen #1, PAPP-A, NT

Specimen Requirements:

Specimen # 1 collection must occur between 10 weeks, 0 days and 13 weeks, 6 days gestation. If gestational age is based on Crown-Rump length (CRL), the specimen must be collected when the CRL is between 32.4 - 83.9 mm.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

3. Ship refrigerate.

Note:

Submit with order: Patient's date of birth, current weight, number of fetuses present, patient's race, if the patient was diabetic at the time of conception, if there is a known family history of neural tube defects, if the patient has had a previous pregnancy with a trisomy, if the patient is currently smoking, if the patient is taking valproic acid or carbamazepine (Tegretol), if this is a repeat sample, and the age of the egg donor if in vitro fertilization.

In addition to the above:

If an NT measurement is performed: the date of ultrasound, the CRL measurement, the nuchal translucency (NT) measurement and the name and certification number of the sonographer are required. NT must be measured when the CRL is between 38-83.9mm. The NT measurement must be performed by an ultrasonographer that is certified by the Fetal Medicine Foundation (FMF).

OR

If no NT measurement is performed: a due date or CRL measurement with the date of ultrasound is required.

See the PATIENT HISTORY FOR MATERNAL SERUM TESTING form.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	72 hours

CPT Code Information:

84163

FMSS2
75351

Maternal Serum Screening, Integrated, Specimen #2, Alpha Fetoprotein, hCG, Estriol, and Inhibin A

Specimen Requirements:

Specimen collection must occur between 14 weeks, 0 days and 24 weeks, 6 days gestation. The recommended time for maternal serum screening is 16 to 18 weeks gestation. Acceptable date ranges to collect the second sample is provided in the Integrated-1 report.

Collection Container/Tube: Serum gel or red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. As soon as possible or within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

2. Send refrigerate.

Note: This test requires that a previous first trimester specimen, FFMSS / Maternal Serum Screening, Integrated, Specimen #1, PAPP-A, NT has been performed.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	72 hours

CPT Code Information:

81511

FMT21 MaterniT21 Plus
57852

Specimen Requirements:

NOTE: Completed Sequenom Test Requisition form is required

A Core Option must be marked on TRF under MaterniT 21 PLUS test

If nothing indicated by client, mark option- Core (chr 21, 18, 13, sex)

Preferred evacuated tube: (1)10 mL Streck tube kit (MCL supply number T715).

Absolute minimum collection for analysis: (1) 10 mL in Streck tube

Collection instructions: Draw 1 tube of blood, 10 mL in special Streck tube kit (MCL supply number T715). Ship ambient.

REQUIRED:

1. Specimen MUST be received at MCL within 72 hours of collection.

2. Specimen collected NOT less than 9 weeks of gestation

3. Sequenom collection kit (MCL Supply T715)

4. Completed Sequenom Test Requisition form

5. Maternal Height (inches)

6. Maternal Weight (pounds)

7. Gestational Age (weeks)

8. Gestational Age (days)

9. Gestation (Number of fetuses)

10. Increased risk due to

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Streck	Ambient	7 days	Streck Black/Tan top

CPT Code Information:

81420

MSMRT 64946 Mayo Algorithmic Approach for Stratification of Myeloma and Risk-Adapted Therapy Report, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88182-Flow cytometry, cell cycle or DNA analysis

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow cytometry interpretation, 2 to 8 Markers

NGAML 65089 MayoComplete Acute Myeloid Leukemia, 11-Gene Panel, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5 to 2 mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, one marrow: 1 mL; Extracted DNA: 100 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

NGAMT 65661 MayoComplete Acute Myeloid Leukemia, Therapeutic Gene Mutation Panel (FLT3, IDH1, IDH2, TP53), Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred)/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Blood, Bone marrow: 1 mL
Extracted DNA: 100 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81120

81121

81245

81246

81352

81479 (if appropriate for government payers)

NGBCL 618477 MayoComplete B-Cell Lymphoma, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%

3. Required amount of tissue area is at least 25 mm(2)

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted.

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Indicate volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

MCBPP 619604 MayoComplete Bladder and Prostate Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of

5000 nucleated cells (preferred), or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual
81457

NGCLN 618476 MayoComplete Chronic Lymphoid Neoplasms, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%

3. Required amount of tissue area is at least 25 mm(2)

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not

acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted.

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Indicate volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Forms

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue:

50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

MCCRC 616489 MayoComplete Colorectal Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381 - Microdissection, manual

81457

MCSRC 616492 MayoComplete Comprehensive Sarcoma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 72 mm(2)

-These amounts are cumulative over up to 15 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Sold Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 15 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

15 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (Diff-Quik or Pap stained direct smears or ThinPrep)

Slides: 2 to 6 Slides

Collection Instructions: Submit 2 to 6 slides, stained and coverslipped, with a total of 10,000 nucleated cells (preferred) or 2 slides with at least 2000 nucleated cells per slide (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81457

81456

MCECP 619631 MayoComplete Endometrial Carcinoma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.
Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

MCGST 616490 MayoComplete Gastrointestinal Stromal Tumor (GIST) Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens..

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting,

digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual
81457

MCGYN 619622 MayoComplete Gynecological Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

NGHIS 618480 MayoComplete Histiocytic Neoplasms, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%.

3. Required amount of tissue area is at least 25 mm(2).

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted.

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA and source of specimen

2. Indicate volume and concentration of DNA on label

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies	14 days

CPT Code Information:

81450

MCKCP 619595 MayoComplete Kidney Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

MCLBP 614939 MayoComplete Liquid Biopsy Panel, Next-Generation Sequencing, Cell-Free DNA

Specimen Requirements:

Supplies: Streck Black/Tan Top Tube Kit (T715)

Container/Tube: Streck Cell-Free DNA (cfDNA) blood collection kit

Specimen Volume: Two 10-mL Streck Cell-Free DNA blood collection tubes

Additional Information: Only blood collected in Streck Cell-Free DNA BCT tubes will be accepted for analysis. Whole blood will be processed to produce platelet-poor plasma before cfDNA isolation.

Specimen Minimum Volume: One 10-mL Streck tube

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	7 days	Streck Black/Tan top
	Refrigerated	7 days	Streck Black/Tan top

CPT Code Information:

81463

MCLNM 616487 MayoComplete Lung Cancer Mutations, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

- For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381–Microdissection, manual

81457

MCLNG 616486 MayoComplete Lung Cancer-Targeted Gene Panel with Rearrangement, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 72 mm(2)

-These amounts are cumulative over up to 20 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 20 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 20 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 20 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

20 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 20 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (Diff-Quik, Pap-stained direct smears, or ThinPrep)

Slides: 2 to 3 Slides

Collection Instructions: Submit at least 2 slides stained and coverslipped with a total of 10,000 nucleated cells (preferred) or at least 2000 nucleated cells (minimum) per slide.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will

not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

MCLNR 616488 MayoComplete Lung Rearrangements, Rapid Test, Tumor

Specimen Requirements:

This assay requires at least 10% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 36 mm(2)

-Minimum amount of tumor area: tissue 18 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81449

MCMLN 616491 MayoComplete Melanoma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

OncoHeme Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred)/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Blood, Bone marrow: 1 mL
Extracted DNA: 100 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

NGPCM 618479 MayoComplete Plasma Cell Myeloma, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender or pink top (EDTA) or yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Minimum plasma cell percentage is 5%.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

4. Label specimen as bone marrow.

5. **Fresh specimen is required for this test**, as testing is performed on sorted cells.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded bone marrow clot

Container/Tube: Paraffin block

1. Send 1 hematoxylin and eosin-stained slide in addition to the paraffin block.

2. Minimum plasma cell percentage is 20%.

3. Required amount of tissue area is at least 25 mm(2).

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the specimen is poor or the plasma cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide, bone marrow clot

Slides: 20 unstained slides

Container/Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue and 1 hematoxylin and eosin-stained slide.

2. Minimum amount of plasma cell percentage is 20%.

4. Required amount of tissue area is at least 25 mm(2).

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Bone marrow aspirate: 1 mL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81450

MCFRC 619613 MayoComplete Renal Cell Carcinoma with Fibromyomatous Stroma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Acceptable:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.
Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)
Slides: 1 to 3 Slides
Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.
Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.
Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81479

81406

81407

81404

81479 (if appropriate for government payers)

MCSMP 616493 MayoComplete Sarcoma Mutation Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81457

MCSTP 606162 MayoComplete Solid Tumor Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei. However, 40% tumor is preferred.

-Preferred amount of tumor area: 432 mm(2) tissue on up to 20 unstained slides

-Minimum amount of tumor area: 72 mm(2) tissue on up to 20 unstained slides

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For this test, at least 4.6 mm x 4.6 mm areas on 20 unstained slides is preferred: this is approximately equivalent to 432 mm(2). The minimum acceptable area is 1.9 mm x 1.9 mm on 20 unstained slides: approximately equivalent to 72 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 20 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

20 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 20 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (Diff-Quik, Pap-stained direct smears, or ThinPrep)

Slides: 2 to 6 Slides

Collection Instructions: Submit at least 2 slides, stained and coverslipped, with a total of 10,000 nucleated cells (preferred) or at least 2000 nucleated cells (minimum) per slide.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81459

88381-Microdissection, manual

NGTCL 618478 MayoComplete T-Cell Lymphoma, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%

3. Required amount of tissue area is at least 25 mm(2)

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA and source of specimen

2. Indicate volume and concentration of DNA on label

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

MCRSP 622969 MayoComplete Targeted RNA Sequencing Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 10% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 144 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slides (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81456

JMDMF
616053

MDM2 (12q15) Amplification, Well-Differentiated Liposarcoma/Atypical Lipomatous Tumor, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods are not accepted; provide fixation method used.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 2 consecutive, unstained, positively charged, unbaked slides with 4 to 5-micron-thick sections of the tumor tissue.

Specimen Minimum Volume: Slides: 1 hematoxylin and eosin-stained and 1 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377

MDM2F
63049

MDM2 (12q15) Amplification, Well-Differentiated Liposarcoma/Atypical Lipomatous Tumor, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

MEAD
82890

Meadow Fescue, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MFOX
82914

Meadow Foxtail, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ROPG
34941

Measles (Rubeola) Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Instructions: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765

ROM
80979

Measles (Rubeola) Antibodies, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765

ROGM
62066

Measles (Rubeola) Virus Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765 x2

MEASU 617822 Measles Virus, Molecular Detection, PCR, Random, Urine

Specimen Requirements:

Container/Tube: Sterile tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen during the 6-to-14-day period after rash onset in a sterile urine tube.

2. No preservative.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	7 days

CPT Code Information:

87638

MEASR 617821 Measles Virus, Molecular Detection, PCR, Throat

Specimen Requirements:

Specimen Type: Throat Swab

Supplies: Culturette (BBL Culture Swab) (T092)

Container/Tube: Sterile container with transport media

Specimen Volume: Entire collection

Collection Instructions:

1. Apply swab tip back and forth firmly over mucosal surface of pharynx (throat) to maximize recovery of cells.

2. Place swab in viral transport media (eg, Liquid Stuart, M4-RT, M4, M5, Bartels FlexTrans Transport Media) and refrigerate for transport to the testing laboratory.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87638

MMRV
61853

Measles, Mumps, Rubella, and Varicella (MMRV) Immune Status Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Instructions: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735

86762

86765

86787

FFMTH
75544

Meconium Methadone Screen with Reflex Confirmation

Specimen Requirements:

Container/Tube: Sterile collection container

Specimen Volume: 3 g

Collection Instructions: Collect specimen into the same sterile collection container until 3 g of meconium have been collected or until the first milk stool appears. When at least 3 g of meconium has been collected, tightly screw on the cap of the collection vial and send specimen ambient.

Note: Specimens from different voids may be pooled if necessary.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Ambient (preferred)	14 days
	Frozen	365 days
	Refrigerated	14 days

CPT Code Information:

80307

MCP2Z
616576

MECP2 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblast

Container/Tube: T-25 flask

Specimen Volume: 2 flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filtration Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 (formerly Ahlstrom 226) filter paper or blood spot collection card

Specimen Volume: 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

2. For collection instructions, see Blood Spot Collection Instructions

3. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

4. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Blood: 1 mL; Skin biopsy, cultured fibroblasts, dried blood spots, or saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81302

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

Medium-Chain Acyl-CoA Dehydrogenase Deficiency (MCAD), ACADM Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm (3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MEGR Megrim, IgE, Serum

82347

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MELAI 82724 Melaleuca leucadendron, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MELAN 70504 Melan A (MART-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FMARP 75449 Melanoma Associated Retinopathy MAR Panel by Immunoblot and IHC

Specimen Requirements:

Submit only one of the following specimens:

Serum:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 5 mL serum refrigerated in a plastic vial.

Plasma:

Draw blood in a lavender-top (EDTA) tube(s). Spin down and send 5 mL EDTA plasma refrigerated in a plastic vial.

Complete and submit with specimen:

1. Completed OHSU Ocular request form
2. Clinical history
3. Referring physician information (name & phone number)

-NOTE: Without this information, testing cannot be completed.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	7 days

CPT Code Information:

84182 x 6

MAGE4 622925 Melanoma-Associated Antigen-A4 (MAGE-A4), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most

cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

FMELA
75386

Melatonin, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Plasma

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA)

Submission Container/Tube: Plastic vial

Specimen volume: 3 mL

Collection instructions:

1. Draw blood in EDTA (lavender top) tube(s).

2. Centrifuge and send 1 mL of plasma refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Frozen	365 days
	Ambient	14 days

CPT Code Information:

80299

FMELG
57652

Melons IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MELN
82762

Melons, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

MSMN 620788 Membranous Nephropathy Target Antigen Identification, Mass Spectrometry, Tissue

Specimen Requirements:

Specimen Type: Kidney tissue

Supplies: Pathology Packaging Kit (T554)

Container/Tube: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Do not send fixed tissue slides. Testing can only be done on paraffin-embedded tissue blocks.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

82542

88380

CSFME 67846 Meningitis/Encephalitis Pathogen Panel, PCR, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Frozen specimens are **not** acceptable.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated	7 days

CPT Code Information:

87483

FMEP 90090 Meperidine (Demerol) and Normeperidine, serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s). Plasma gel tube is not acceptable. Spin down and send 3 mL of sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s). Serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80362

FMMM 57766 Mephedrone, MDPV and Methylone, Urine

Specimen Requirements:

Container: Preservative-free plastic urine container

Specimen volume: 10 mL

Collection instructions

1. collect a random urine specimen
2. No preservative

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307-single drug

80371-(if appropriate)

FMERC 91120 Mercaptopurine (6-MP, Purinethol)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

HGUOE
608897

Mercury Occupational Exposure, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

82570

HGOU 608893 Mercury Occupational Exposure, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

HGUOE/ Mercury Occupational Exposure, Random, Urine

HMUOE / Heavy Metal Occupational Exposure, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

HGU 8592 Mercury, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. For the 48-hour period prior to the start of collection, as well as during the collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

HG
8618

Mercury, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Collection Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83825

HGHAR 8498 Mercury, Hair

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Head, beard, mustache, chest, pubic

Specimen Volume: 0.2 g

Collection Instructions: Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Hair	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

83825

HGNA 89856 Mercury, Nails

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Fingernails or toenails

Specimen Volume: 0.2 g

Collection Instructions:

1. Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

2. Clippings should be taken from all 10 fingernails or toenails.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Nail	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

83825

HGUCR 608907 Mercury/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued

insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

82570

HGCU 608903 Mercury/Creatinine Ratio, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

HGUCR / Mercury/Creatinine Ratio, Random, Urine

HMUCR / Heavy Metal/Creatinine Ratio, with Reflex, Random Urine.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

MERKC 71538 Merkel CC (MCPyV) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NF2 622010 Merlin (NF2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

HBME 70452 Mesothelial Cell (HBME-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

MESOF 609714 Mesothelioma, CDKN2A FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291 - DNA probe, each (first probe set), Interpretation and report

88271x2 - DNA probe, each; each additional probe set (if appropriate)

88271x1 - DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2 - DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3 - DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52 - Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274 - Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275 - Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

MESQ
82776

Mesquite, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

METF
58123

MET (7q31), FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%.Testing **may be** canceled if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 3 unstained

Collection Instructions: Submit 3 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

METSP 623179 MET (SP44), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Collection Instructions: Store slides at 2 to 8 degrees C and send within 45 days.

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

GL5CS 623397 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Cell Based Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

GL5CC 623396 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Cell Based Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

GL5TS 623488 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

GL5TC 623487 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

GL5IS 623395 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

GL5IC 623394 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid
-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

MT 619972 Metallothionein Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MHBRP 65129 Metamycoplasma hominis, Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Metamycoplasma hominis* DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred). **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

MHPRP 65134 Metamycoplasma hominis, Molecular Detection, PCR, Plasma

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by

Metamycoplasma hominis DNA is unlikely.

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA)

Submission Container/Tube: Screw-capped, sterile container

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot plasma into a sterile vial within 24 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

MHRP 60756 Metamycoplasma hominis, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction (PCR) requires the specimen to be processed in an environment in which contamination of the specimen by *Mycoplasma hominis* DNA is not likely.

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-M4-RT (T605)

Sources: Vaginal, cervix, urethra, urogenital, chest/mediastinal; bronchus or lung (donor swab); or upper respiratory sources (only infants <3 months: nasopharynx, nose, throat)

Container/Tube:

Preferred: Culture swab transport system (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Acceptable: Swab in transport media: M4, M4-RT, M5, M6, universal transport media, or ESwab

Specimen Volume: 1 swab

Collection Instructions:

Vaginal:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Urethra or cervical:

1. Collect specimen by inserting swab 1 to 3 cm and rotating 360 degrees.

2. Place swab back into swab cylinder.

Wound:

1. Collect specimen by swabbing back and forth over wound surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Specimen Type: Fluid

Sources: Pelvic, peritoneal, amniotic, prostatic secretions, semen, reproductive drainage or fluid, pleural/chest, chest tube, pericardial

Container/Tube: Sterile container

Specimen Volume: 1 to 2 mL

Specimen Type: Respiratory

Sources: sputum, tracheal secretions, bronchial washings, bronchoalveolar lavage, lung; or nasal washings (Note: Nasal washings may only be submitted for infants <3 months of age.)

Container/Tube: Sterile container

Specimen Volume: 1 to 2 mL

Specimen Type: Synovial fluid

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red clot tube (no anticoagulant), or sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send specimen in original tube.

Specimen Type: Urine (first void), kidney/bladder stone, or ureter

Container/Tube: Sterile container

Specimen Volume: 10 mL or entire specimen

Collection instructions: Urine first void: Specimen can be collected at any time during the day. The patient should not have urinated for at least 1 hour prior to specimen collection. The first voided portion is the initial 20 to 30 mL of the urine stream obtained **without** cleaning the external urethra.

Specimen Type: Tissue

Sources: Placenta, products of conception, urogenital, respiratory, bronchus, chest/mediastinal, bone, spine, or joint

Container/Tube: Sterile container

Specimen Volume: 5 mm(3)

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit fresh tissue only, do not add fluid to tissue

3. Refrigerate or freeze specimen.

Specimen Minimum Volume: Fluid: 1 mL
Urine, first void: 2 mL
Swab: 1 swab
Tissue: 5 mm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

META3 Metanephrines with 3-Methoxytyramine, 24 Hour, Urine
65158

Specimen Requirements:

Patient Preparation: Tricyclic antidepressants, labetalol, and sotalol medications may elevate levels of metanephrines producing results that cannot be interpreted. If clinically feasible, it is optimal to discontinue these medications at least 1 week before collection. For advice on assessing the risk of removing patients from these medications and alternatives, consider consultation with a specialist in endocrinology or hypertension.

Supplies: Urine Tubes, 10 mL (T068)

Submission Container/Tube: Plastic urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Complete 24-hour urine collections are preferred, especially for patients with episodic hypertension; ideally the collection should begin at the onset of a "spell."

2. Add 10 g (pediatric: 3 g) of boric acid or 25 mL (pediatric: 15 mL) of 50% acetic acid as preservative **at start of collection**.

3. Collect urine for 24 hours.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

82542

83835

METAF
83006

Metanephrines, Fractionated, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Tricyclic antidepressants, labetalol, and sotalol medications may elevate levels of metanephrines producing results that cannot be interpreted. If clinically feasible, it is optimal to discontinue these medications at least 1 week before collection. For advice on assessing the risk of removing patients from these medications and alternatives, consider consultation with a specialist in endocrinology or hypertension.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Complete 24-hour urine collections are preferred, especially for patients with episodic hypertension; ideally the collection should begin at the onset of a "spell."

2. Add 10 g (pediatric: 3 g) of boric acid or 25 mL (pediatric: 15 mL) of 50% acetic acid as preservative **at start of collection**.

3. Collect urine for 24 hours.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days

	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83835

PMET
81609

Metanephrines, Fractionated, Free, Plasma

Specimen Requirements:

Patient Preparation: Use of an Epi-pen within the last 7 days may produce inaccurate results.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	28 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

83835

FMETN
75389

Metformin, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Plasma

Collection Container/Tube: Lavender top or pink top (EDTA) (Plasma gel tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen volume: 1 mL

Collection instructions:

1. Draw blood in an EDTA (lavender top or pink top) tube.

2. Centrifuge and aliquot 1 mL of plasma into a preservative-free plastic vial.

3. Send refrigerated (preferred)

Specimen Stability Information: Refrigerated (preferred) 30 days/Ambient 30 days/Frozen 2 years

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	

CPT Code Information:

80299

MDNS 36309 Methadone and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.0 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80358 (if appropriate for select payers)

MTDNX
62734

Methadone Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-custody kit containing the specimen containers, seals, and documentation required.

Specimen Volume: 10 mL

Collection Instructions: Collect urine specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 10 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80358 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

MTDNU
83129

Methadone Confirmation, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80358 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FMETH
57996

Methaqualone Confirmation, Urine

Specimen Requirements:

Container/Tube: Plastic, preservative-free urine container

Specimen Volume: 20 mL

Collection Instructions:

1. Collect 20 mL random urine without preservative.

2. Transfer into a preservative-free plastic urine container.

3. Ship refrigerate.

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	10 days
	Frozen	396 days
	Ambient	72 hours

CPT Code Information:

80375

MET 81032 Methemoglobin and Sulfhemoglobin, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: Full tube

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Patient's age is required.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	72 hours

CPT Code Information:

83050-Methemoglobin

83060-Sulfhemoglobin

MEV0 608089 Methemoglobin Summary Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood EDTA	Refrigerated

MEV1 Methemoglobinemia Evaluation, Blood

607495

Specimen Requirements:

The following specimens are required for testing:

Whole blood ACD-B specimen

2 Whole blood EDTA specimens

Container/Tube: Lavender top (EDTA) **and** yellow top (ACD solution B)

Specimen Volume:

EDTA: Two 4 mL tubes

ACD: One 6 mL tube

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: EDTA blood: 3 mL; ACD blood: 2.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	72 hours
Whole Blood EDTA	Refrigerated	72 hours

CPT Code Information:

83020-26-Hemoglobinopathy Interpretation

83020-Hb Variant, A2 and F Quantitation

83021-HPLC Hb Variant

82657-Methemoglobin reductase

83050-Methemoglobin, quantitative

83060-Sulfhemoglobin, quantitative

82664 (if appropriate)

83068 (if appropriate)

83789 (if appropriate)

88184 (if appropriate)

MEVI 608086 Methemoglobinemia Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	72 hours

CPT Code Information:

83020-26

MECPI 621419 Methicillin Resistance Gene, Molecular Detection, PCR, Varies

Specimen Requirements:

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Collection Instructions:

1. Perform isolation of bacteria.

2. Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87641

MRSAP 609735 Methicillin Resistant Staphylococcus aureus, PCR, Nasal

Specimen Requirements:

Specimen Type: Nasal cavity swab

Supplies: BD E-Swab (T853)

Container/Tube:

Preferred: BD Liquid Amies Elution Swab (E-Swab)

Acceptable: Copan Liquid Amies Elution Swab (ESwab); Cepheid nasal sample collection device (dual swab in liquid Stuart media); or Copan swab and transport systems (LQ Stuart or LQ Amies) (scored swabs only)

Specimen Volume: Entire collection

Collection Instructions:

1. Ask the patient to blow their nose prior to collection.

2. Use one swab for both right and left nares.

3. Insert the white Dacron swab tip (do not insert any further in) into the anterior nares.

4. Rotate the swab for 3 seconds against the nasal mucosa. Apply slight pressure on the outside of the nose with your finger to ensure good contact.

5. Repeat the process on the other nostril.

6. Insert the swab back into the carrier. The swabs should go all the way into the transport container. Tightly cap the container.

7. Label the transport container with patient label and send refrigerate.

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Refrigerated (preferred)	5 days
	Ambient	24 hours

CPT Code Information:

87641

MTXSG 62580 Methotrexate Post Glucarpidase, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into an amber vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	14 days	LIGHT PROTECTED
	Ambient	7 days	LIGHT PROTECTED

CPT Code Information:

80204

MTHX
37047

Methotrexate, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Methotrexate is sensitive to fluorescent light; avoid prolonged exposure of specimen to direct light.

2. Within 2 hours of collection, centrifuge the specimen.

3. For serum gel tubes, aliquot serum into a plastic vial within 2 hours of collection. Keep protected from light.

4. For red-top tubes, aliquot serum into a plastic vial immediately. Keep protected from light.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	14 days	LIGHT PROTECTED
	Frozen	28 days	LIGHT PROTECTED
	Ambient	72 hours	LIGHT PROTECTED

CPT Code Information:

80204

FMETX 91822 Methsuximide (Celontin) as Desmethylmethsuximide

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80339

MMAP 31927 Methylmalonic Acid, Quantitative, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Refrigerated (preferred)	48 days
	Ambient	48 days
	Frozen	48 days

CPT Code Information:

83921

MMAS 80289 Methylmalonic Acid, Quantitative, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	48 days
	Ambient	48 days
	Frozen	48 days

CPT Code Information:

83921

MMAU
80290

Methylmalonic Acid, Quantitative, Urine

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 4 mL

Collection Instructions:

1. After fasting at least 8 hours overnight, patient should void first-morning urine as normal. **Do not collect**.

2. Patient should continue to fast and then collect the next (second morning) urine voided.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	21 days

CPT Code Information:

83921

MMAGP 608021 Methylmalonic Aciduria Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies.

Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and

will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required

to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order

MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MPAGP 608022 Methylmalonic Aciduria-Propionic Aciduria Combined Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See

Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight..

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MPHNU 608882 Methylphenidate and Metabolite, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	10 days
	Frozen	28 days

CPT Code Information:

G0480

80360 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

MPHNS
624101

Methylphenidate and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	28 days

CPT Code Information:

G0480

80360 (if appropriate for select payers)

MEX 9245 Mexiletine, Serum

Specimen Requirements:

Patient Preparation: Specimens should only be collected after patient has been receiving mexiletine for at least 3 days. Trough concentrations should be collected just before administration of the next dose.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

MGMTD 621858 MGMT Promoter Methylation Analysis, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 144 mm(2) tissue (4 x 6 mm x 6 mm areas)

-Minimum amount of tumor area: 36 mm(2) tissue (1 x 6 mm x 6 mm area)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: formalin-fixed paraffin-embedded (FFPE), non-decalcified

Submit only 1 of the following specimens:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed non-decalcified, paraffin-embedded tissue block.

Specimen Type: Tissue slide

Slides: 1 Stained and 10 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 10 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: 5 unstained slides at 5-microns thickness

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81287

CERAM
606777

MI-Heart Ceramides, Plasma

Specimen Requirements:

Patient Preparation: Patients should not be receiving Intralipid because it may cause false elevations in measured ceramides.

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge, aliquot at least 1 mL of plasma into a plastic vial and freeze within 8 hours.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	24 hours
	Ambient	8 hours

CPT Code Information:

0119U

MADON
622523

MLCPC
113370

Microbiology Add-On
Microdissection, Laser Capture (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88380-Microdissection; laser capture

MNLCP 621773 Microdissection, Laser Capture, Membranous Nephropathy (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88380

TMSI 609364 Microsatellite Instability, Tumor

Specimen Requirements:

This assay requires at least 40% tumor nuclei for endometrial specimens and at least 20% tumor nuclei for colorectal specimens.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 72 mm(2)

-Minimum amount of tumor area: 18 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: formalin-fixed paraffin-embedded, non-decalcified

Preferred: Submit 2, if available, of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81301

88381-Microdissection, manual

LCMSP 63097 Microsporidia species, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: 5 g

Specimen Type: Preserved feces

Supplies:

-ECOFIX Stool Transport Vial (Kit) (T219)

-Stool Collection Kit, Random (T635)

Container/Tube: ECOFIX preservative

Specimen Volume: 5 g

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Mid-stream, clean-catch, suprapubic aspirates and catheterization collections are acceptable. Submit in a clean, sterile container free from preservatives. The first portion of the voided urine (first void) is also acceptable.

Specimen Minimum Volume: Feces: 1 g
Urine: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Ambient	7 days
	Frozen	7 days

CPT Code Information:

87798

MTBS 81507 Microsporidia Stain, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Duodenal aspirate (small intestinal aspirate, jejunal aspirate, small bowel aspirate)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Additional Information: ECOFIX and 10% formalin are acceptable preservatives.

Specimen Stability Information: Preserved Ambient (preferred) <10 days/Refrigerated <3 days

Specimen Type: Respiratory secretions (bronchoalveolar lavage [BAL], sputum, bronchial wash, pleural fluid)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Stability Information: Refrigerated <3 days (preferred)/Frozen <10 days

Specimen Type: Eye (vitreous fluid, ocular fluid)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Stability: Refrigerated <3 days

Specimen Type: Corneal scraping

Container/Tube: Sterile container

Specimen Volume: 0.5mL in sterile saline

Additional Information: Place scrapings in sterile saline for shipping.

Specimen Stability: Refrigerated <3 days

Specimen Type: Fresh tissue (lung, eye, bladder, rectal, intestinal, colon, skin, muscle, kidney)

Container/Tube: Sterile container

Specimen Volume: 3-mm biopsy in 0.1-mL sterile saline

Specimen Stability: Refrigerated <3 days

Specimen Type: Gallbladder aspirate/Bile aspirate

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Stability: Refrigerated <3 days/Frozen <10 days

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87015-Concentration

87207-Stain

MP1BS 621406 Microtubule-Associated Protein 1B (MAP1B) Antibody, Immunoblot, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum:1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

MP1BC 621405 Microtubule-Associated Protein 1B (MAP1B) Antibody, Immunoblot, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

FMIDZ 90112 Midazolam (Versed), serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80346

G0480 (if appropriate)

FMCG4 57536 Milk Cow IgG4

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MILK
82871

Milk, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PMLK
82827

Milk, Processed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMINT
57885

Mint (Mentha Piperita) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

MIRI
623669

Mirikizumab, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FMIRT
57749

Mirtazapine (Remeron)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80335

ZW153
91213

Misc Center for Human Genetics

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW204 Misc Cincinnati Childrens Hospital Medical Center

91801

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW59 Misc Esoterix Endocrinology

90498

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW179 Misc Medical Neurogenetics, LLC

91403

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo

Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW149 Misc Monogram Biosciences, Inc.

91170

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie,

consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW196 Misc National Jewish Health Mycobacteriology Test

91643

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW146 Misc Prometheus Laboratories Test
90585

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW305 Misc Washington University Neuromuscular Clinical Lab
75411

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW206 Misc. Seattle Children's Hospital Laboratories

91803

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW185 Miscellaneous Ambry Genetics
91516

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW242 Miscellaneous ARUP Testing
57378

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo

Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW127 Miscellaneous Athena Testing

90566

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie,

consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW221 Miscellaneous Baylor Medical Genetics Laboratories

57103

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW155 Miscellaneous BioAgilytix Diagnostics

91272

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

SCT2 Miscellaneous Biochemical Genetics Testing

20521

Specimen Requirements:

Only orderable as part of the miscellaneous test process.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

ZW293 Miscellaneous CBL Path Inc.

75252

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW172 Miscellaneous Center for Genetic Testing at St. Francis
91302

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW10 Miscellaneous Chemistry Testing, Varies
8921

Specimen Requirements:

For specimen requirements, call Mayo Clinic Laboratories at 800-533-1710.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen	30 days

ZW140 Miscellaneous Child Med Ctr Dallas
90579

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW163 Miscellaneous Children's Hospital of Philadelphia (CHOP)

91291

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW299 Miscellaneous Childrens Hospital Los Angeles

76271

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW246 Miscellaneous Childrens Hospital of Colorado Testing

57992

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW239 57180 Miscellaneous Cincinnati Children's Hospital Medical Center-Hematology/Oncology

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW166 Miscellaneous Cincinnati Childs Hosp Center

91294

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW346 Miscellaneous Cleveland Clinic

75850

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW58 Miscellaneous Duke University

90497

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW57 Miscellaneous Esoterix Coagulation

90496

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW69 Miscellaneous Eurofins Viracor

90508

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW168 Miscellaneous GeneDx, Inc. Testing

91296

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW68 Miscellaneous Greenwood Genetic Ctr

90507

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW193 Miscellaneous HNL Genomics

91601

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW143 Miscellaneous IMMCO Diagnostics

90582

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW71 Miscellaneous Inter Science Institute

90510

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW298 Miscellaneous IntrinsicDx

75278

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW175 Miscellaneous Kennedy Krieger Institute

91323

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW76 Miscellaneous LabCorp of America

90515

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW113 Miscellaneous LabCorp-RTP,NC

90552

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW266 Miscellaneous Machaon Diagnostics
58067

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW342 Miscellaneous MDX Health Inc.
75816

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW78 Miscellaneous Medical Coll of WI

90517

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW80 Miscellaneous Medtox Laboratories, Inc.

90519

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW111 Miscellaneous MiraVista Diagnostics

90550

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW1 Miscellaneous MML Referral Test 1

99991

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW2
99992

Miscellaneous MML Referral Test 2

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW3 99993 Miscellaneous MML Referral Test 3

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW227 57163 Miscellaneous National B Virus Resource Laboratory

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW85 Miscellaneous National Jewish Health

90524

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW333 Miscellaneous Nationwide Children's Hospital

75744

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW190 89123468 Miscellaneous Nemours Children's Health-Gastroenterology

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW359 624088 Miscellaneous Neurocode USA, Inc

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW335 Miscellaneous New York Blood Center

75641

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW86 Miscellaneous NMS Testing
90525

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW214 Miscellaneous Ocular Immunology Laboratory OHSU
91861

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW87 Miscellaneous Palo Alto Testing

90526

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW194 Miscellaneous Prevention Genetics Lab

91602

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW83 Miscellaneous Quest Diagnostics

90522

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW51 Miscellaneous Quest Testing

90490

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW131 Miscellaneous Quest/Nichols Testing

90570

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW224 Miscellaneous Revvity Omics

57157

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW91 Miscellaneous RFFIT Testing

90530

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW124 Miscellaneous Sequenom

75456

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

MISCF 35267 Miscellaneous Studies Using Chromosome-Specific Probes, FISH

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20-25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

4. Results will be reported and telephoned or faxed if requested.

Specimen Type: Blood (only accepted for Congenital/Hereditary [nonhematologic] testing)

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

Specimen Type: Chorionic villi (CVS)

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 25 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium (eg, CVS media (RPMI)).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Type: Lymph node

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 1 cm(3)

Specimen Type: Skin biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 1-cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. Do not use alcohol or iodine preparations.

4. A local anesthetic may be used.

5. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Type: Tissue

Preferred: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Additional Information:

1. Paraffin embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable: Slides

Collection Instructions: For each probe set ordered, 4 consecutive, unstained, 5 micron-thick sections placed on positively charged slides. Include 1 hematoxylin and eosin-stained slide for the entire test order.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Tumor

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Minimum Volume: Amniotic fluid: 5 mL; Blood: 2 mL; Chorionic villi: 5 mg; Lymph node: 0.5 cm(3); Solid tumor: 0.5 cm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88273 w/modifier 52-Chromosomal in situ hybridization, less than 10 cells (if appropriate)

88273-Chromosomal in situ hybridization, 10-30 cells (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

ZW102 Miscellaneous U of TX San Ant Test

90541

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW288 Miscellaneous UF Health Pathology Laboratories

75170

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW99 Miscellaneous Univ of AL-Metabolic Disease Lab

90538

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW210 Miscellaneous Univ of IA Molecular Otolaryngology

91857

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW128 Miscellaneous University Hospital Clinical Lab Test

90567

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW234 Miscellaneous University of Alabama at Birmingham

57175

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW186 Miscellaneous University of Chicago Genetics Services

91515

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW187 Miscellaneous University of Iowa Diagnostic Labs

91514

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW61 90500 Miscellaneous University of Minnesota Outreach Laboratory

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW238 57179 Miscellaneous University of Southern California Endocrine Laboratories

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW191 91599 Miscellaneous University of Texas Health Center at Tyler

Microbiology

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW280 58092 Miscellaneous University of Washington Medical Center (Molecular Diagnosis Clinical Microbiology Lab)

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW282 58088 Miscellaneous University of Washington Medical Center (UW Virology Dept of Lab Medicine)

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW278 58078 Miscellaneous University of Washington Medical Center-Clinical Immunology Lab

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW334 75750 Miscellaneous Variantyx, Inc.

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW121 Miscellaneous Versiti Blood Center of Wisconsin

90560

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

IHC 35466 Mismatch Repair (MMR) Protein Immunohistochemistry Only, Tumor

Specimen Requirements:

Tumor tissue is required.

Preferred: Submit both of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88341 MLH1, MLH2, or MLH6 (if appropriate)

88342 PMS2 (if appropriate)

MITF
70509

MiTF Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

AMA
8176

Mitochondrial Antibodies (M2), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86381

DMITO
618558

Mitochondrial DNA Deletion Heteroplasmy, ddPCR, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic

donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80 mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient <24 hours/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Snap frozen nerve tissue biopsy

Collection Instructions: Prepare snap frozen tissue biopsy per surgical procedure

Specimen Volume: 0.25 to 0.5 cm

Specimen Stability Information: Frozen

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will

be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

MITOP 62510 Mitochondrial Full Genome Analysis, Next-Generation Sequencing (NGS), Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A

maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Snap frozen nerve tissue biopsy

Collection Instructions: Prepare snap frozen tissue biopsy per surgical procedure

Specimen Volume: 0.25 to 0.5 cm

Specimen Stability Information: Frozen

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80 mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient <24 hours/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies

on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81460-Whole Mitochondrial Genome

81465-Whole Mitochondrial Genome Large Deletion Analysis

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MMPP 616610 Mitochondrial Metabolites, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Green top (Sodium heparin)

Acceptable: Green top (Lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	7 days

CPT Code Information:

82542

MITAN 621811 Mitotane, Plasma

Specimen Requirements:

Collection Container/Tube: Green top (sodium heparin) (Lithium heparin and PST/plasma gel tubes **are not** acceptable.)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Refrigerated (preferred)	21 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80299

FMITU 75773 Mitragynine (Qualitative), Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 2 mL random urine in a plastic, preservative-free urine container.

2. Send refrigerated.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80323

FCDX2 75842 Mitragynine Confirmation (Qualitative), Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80375

MLHPB 35500 MLH1 Hypermethylation Analysis, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Requirements

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81288

ML1HM
35494

MLH1 Hypermethylation Analysis, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 2, if available, of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81288

88381

BRMLH 35491 MLH1 Hypermethylation and BRAF Mutation Analysis, Tumor

Specimen Requirements:

Submit 1 of the following specimens:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

Slide Review

88381-Microdissection, manual

81210-*BRAF* (*v-raf murine sarcoma viral oncogene homolog B1*) (eg, colon cancer), gene analysis, V600E variant, if appropriate

81288-*MLH1* promoter methylation analysis, if appropriate

MLH1
70510

MLH1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MOC31 MOC-31 Immunostain, Technical Component Only

70505

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

MOLD1
81878

Mold Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOLPS
89270

Molybdenum, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours**.

Supplies:

-Metal Free B-D Tube (No Additive), 6 mL (T184)

-Metal Free Specimen Vial (T173)

Collection Container/Tube: Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.25 mL Serum

Collection Instructions:

1. Allow specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a Mayo metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	14 days	METAL FREE
	Ambient	14 days	METAL FREE
	Frozen	14 days	METAL FREE

CPT Code Information:

83018

QMPTS 620169 **Monoclonal Protein Isotype, Quantitative, Serum (Non-Orderable)**

Specimen Requirements:

Only orderable as part of a profile. For more information see QMPSS / Monoclonal Protein Study, Quantitative, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

0077U

MPQU 616907 Monoclonal Protein Quantitation, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect urine for 24 hours and refrigerate during collection.

2. Aliquot between 30 mL and 50 mL urine into plastic, 60-mL urine bottle.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

84166

0077U (if appropriate)

RMPQU
616909

Monoclonal Protein Quantitation, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect random urine specimen.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60-mL urine bottle and refrigerate.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

82570

84166

0077U (if appropriate)

SMPU 616913 Monoclonal Protein Screen, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect urine for 24 hours and refrigerate specimen during collection.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60-mL urine bottle.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

84166 Electrophoresis, protein (if appropriate)

0077U

RSMPU 616915 Monoclonal Protein Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect random urine specimen.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60 mL urine bottle and refrigerate.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

82570

84166 Electrophoresis, protein (if appropriate)

0077U

MPU 616911 Monoclonal Protein Studies, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect urine for 24 hours and refrigerate during collection.

2. Aliquot between 30 mL and 50 mL urine into plastic, 60-mL urine bottle.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

84166

0077U

RMPU
616912

Monoclonal Protein Studies, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60-mL urine bottle and refrigerate.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

82570

84166

0077U

QMPSS 620919 Monoclonal Protein Study, Quantitative, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum total; 2 separate plastic vials, each containing 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into 2 plastic vials, each containing 1 mL

Specimen Minimum Volume: Serum: 2 Plastic vials, each containing 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

0077U

82784 x 3

MONOF 610018 Monocyte Repartition by CD14/CD16, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	72 hours

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 7-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 markers

MORS
623447

Morphine and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80361 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

MSPP
82845

Mosquito Species, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOTH
82738

Moth, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMNPP
75811

Motor Neuropathy Panel

Specimen Requirements:

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions:

1. As soon as possible or within 2 hours of specimen collection, centrifuge and aliquot 4 mL serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

82784 x 3

83516 x 7

84155

84165

86334

CED
82668

Mountain Cedar, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOUS
82707

Mouse Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOSP
82792

Mouse Serum Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOUP
82795

Mouse Urine Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

JAMOVATSS Movat Pentachrome Stain
102516

Specimen Requirements:

This is not an orderable test. Order CONFL / Pathology Consultation. The consultant will determine the need for special stains.

CPT Code Information:

88313

MDS2 Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum
606192

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86596

86255 x 26

83516

86341

0432U

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86256 AMPIS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN2TS (if appropriate)

86256 AN3TS (if appropriate)

86255 APBCS (if appropriate)

86256 APBTS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

86255 GRFCS (if appropriate)

86256 GRFTS (if appropriate)

86256 IG5TS (if appropriate)

86255 ITPCS (if appropriate)

86256 ITPTS (if appropriate)

86256 K11TS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86255 GL5CS (if appropriate)

86256 GL5TS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86256 NMDIS (if appropriate)

84182 PC1BS (if appropriate)

86256 PC1TS (if appropriate)

86256 PC2TS (if appropriate)

84182 PCTBS (if appropriate)

86256 PCTTS (if appropriate)

86256 PDETS (if appropriate)

86255 SP5CS (if appropriate)

86256 SP5TS (if appropriate)

86255 SP7CS (if appropriate)

86256 SP7TS (if appropriate)

86255 T46CS (if appropriate)

86256 T46TS (if appropriate)

MDC2 606193 Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication, corticosteroids, or intravenous immunoglobulin treatment.

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x28

83516

86341

0432U

84182 AGNBC (if appropriate)

86256 AGNTC (if appropriate)

86255 AINCC (if appropriate)

86256 AMPIC (if appropriate)

86256 APHTC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 AN2TC (if appropriate)

86256 AN3TC (if appropriate)

86255 APBCC (if appropriate)

86256 CRMTC (if appropriate)

84182 CRMWC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

86255 GRFCC (if appropriate)

86256 GRFTC (if appropriate)

86256 IG5TC (if appropriate)

86255 ITPCC (if appropriate)

86256 ITPTC (if appropriate)

86256 K11TC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 GL5CC (if appropriate)

86256 GL5TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86255 NFHCC (if appropriate)

86256 NIFTC (if appropriate)

86255 NFLCC (if appropriate)

86256 NMDIC (if appropriate)

84182 PC1BC (if appropriate)

86256 PC1TC (if appropriate)

86256 PC2TC (if appropriate)

84182 PCTBC (if appropriate)

86256 PCTTC (if appropriate)

86256 PDETC (if appropriate)

86255 SP5CC (if appropriate)

86256 SP5TC (if appropriate)

86255 SP7CC (if appropriate)

86256 SP7TC (if appropriate)

86255 T46CC (if appropriate)

86256 T46TC (if appropriate)

MPLFM MPL Exon 10 Mutation Detection, Bone Marrow

618661

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81339

MPLR 36682 MPL Exon 10 Mutation Detection, Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNR / Myeloproliferative Neoplasm, *JAK2* V617F with Reflex to *CALR* and *MPL*, Varies.

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not** aliquot.

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81339-*MPL* (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence

MPLVS
602599

MPL Exon 10 Mutation Detection, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collections Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 2 mL

Collections Instructions:

1. Invert several times to mix bone marrow.

2. Send specimens in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2- mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or

equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, Bone marrow: 0.5 mL; Extracted DNA: 50 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81339-*MPL* (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence

MPNML 44179 MPL Exon 10 Sequencing, Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNCM / Myeloproliferative Neoplasm, *CALR* with Reflex to *MPL*, Varies.

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5 to 2 mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81339-MPL (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence

Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNJM / Myeloproliferative Neoplasm, *JAK2* V617F with Reflex to *CALR* and *MPL*, Bone Marrow.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send specimen in original tube. Do not aliquot.

3. Label specimen as bone marrow.

Note: Extracted DNA from bone marrow is not acceptable.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81339

FMPVP 75817 Mpox Virus DNA, Qualitative Real-Time PCR

Specimen Requirements:

Specimen Type: Lesion swab

Container/Tube: Viral Culture Media (VCM), Viral Transport Media (VTM), Universal Transport Media (UTM), or equivalent

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab the pustule/lesion vigorously and place the swab in 3 mL Viral Culture Media (VCM), Viral Transport Media (VTM), Universal Transport Media (UTM), or equivalent.

2. Include the following required information with the specimen: Patient race, ethnicity, specimen type (source), and anatomic location specimen was collected.

3. Send frozen.

Additional Information: Each individual specimen submitted should be transported in its own sealed bag. Multiple specimens collected on a single patient should be submitted separately.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

87593 x 2

MSH2
70512

MSH2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

MSH6
70513

MSH6 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

CSMRT 607626 mSMART Plasma Cell Proliferative Disorder, Pre-Analysis Cell Sorting, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see one of the following:

MSMRT / Mayo Algorithmic Approach for Stratification of Myeloma and Risk-Adapted Therapy Report, Bone Marrow

MSMRD / Myeloma Stratification and Risk-Adapted Therapy with Reflex to Minimal Residual Disease, Bone Marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow Cytometry, additional cell surface, cytoplasmic or nuclear marker (each)

MPCDS 606090 mSMART, Plasma Cell Proliferative Disorder, FISH, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MSMRT / Mayo Algorithmic Approach for Stratification of Myeloma and Risk-Adapted Therapy Report, Bone Marrow.

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88274, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271 x 2, 88274-FISH Probe, Analysis; each additional probe set (if appropriate)

MUCN2 Mucin 2, Immunostain, Technical Component Only

605116

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUCN4 Mucin 4, Immunostain, Technical Component Only
601740

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUCN5 Mucin 5AC, Immunostain, Technical Component Only
605118

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUCN6 Mucin 6, Immunostain, Technical Component Only
605120

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MPSQU 606299 Mucopolysaccharides Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL urine tube

Specimen Volume: 2 mL

Pediatric Volume: 1 mL

Collection Instructions: Collect a random urine specimen **(early morning preferred).**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	90 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

83864

82570

MPSER 604905 Mucopolysaccharides Quantitative, Serum

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Pediatric: 0.2 mL

Collection Instructions:

1. Do not collect specimen from a line that may have been used to infuse heparin or has been flushed with heparin.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	14 days

CPT Code Information:

83864

MPS3W 618293

Mucopolysaccharidosis III, Four-Enzyme Panel, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

MPS3B 618292 Mucopolysaccharidosis III, Three-Enzyme Panel, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole blood	Varies

CPT Code Information:

82657

83864 (if appropriate)

MPS4B 618294 Mucopolysaccharidosis IV Enzyme Panel, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete, ie, unpunched.

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

83864 (if appropriate)

MPS4W 618295 Mucopolysaccharidosis IV Enzyme Panel, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

IDUA
622652

Mucopolysaccharidosis Type I, IDUA Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal

essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24

hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 IDUA (iduronidase, alpha-L-) (eg, mucopolysaccharidosis type I), full gene sequence

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

IDS
622666

Mucopolysaccharidosis Type II, IDS Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific

gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will

be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405-*IDS* (iduronate 2-sulfatase) (eg, mucopolysaccharidosis, type II), full gene sequence

88233-Tissue culture, skin, or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MPSWB Mucopolysaccharidosis, Blood
113435

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Do not collect specimen from a line that may have been used to infuse heparin or has been flushed with heparin.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

83864

MPSBS Mucopolysaccharidosis, Blood Spot
65095

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Collection Container:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell

filter paper, local newborn screening card, postmortem screening card, or blood collected in tubes containing ACD or EDTA and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions

1. Do not collect specimen from a line that may have been used to infuse heparin or has been flushed with heparin.

2. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

3. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

4. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

5. Do not expose specimen to heat or direct sunlight.

6. Do not stack wet specimens.

7. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	100 days	FILTER PAPER
	Frozen	100 days	FILTER PAPER
	Refrigerated	100 days	FILTER PAPER

CPT Code Information:

83864

MUC 82675 Mucor, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MUG
82683

Mugwort, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MULB
82864

Mulberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RETZZ
614587

Multiple Endocrine Neoplasia Type 2 Syndrome, RET, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific

gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

MRDMR 616031 Multiple Myeloma Measurable Residual Disease (MRD) by Flow Cytometry, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MSMRD / Myeloma Stratification and Risk-Adapted Therapy with Reflex to Minimal Residual Disease, Bone Marrow

Specimen Type: Redirected bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 9-Flow Cytometry; additional cell surface, cytoplasmic or nuclear marker

88188-Flow Cytometry Interpretation, 9 to 15 Markers

MRDMM 65218 Multiple Myeloma Measurable Residual Disease (MRD), Flow Cytometry, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 9-Flow Cytometry; additional cell surface, cytoplasmic or nuclear marker

88188-Flow Cytometry Interpretation, 9 to 15 Markers

MSP3 607837 Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid

Specimen Requirements:

Both serum and spinal fluid are required. Spinal fluid must be obtained within 1 week of serum collection.

Specimen Type: Spinal fluid (CSF)

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Submit CSF from collection vial number 4 (preferred); vial number 1, 3, 2 are also acceptable (in this order).

2. Label specimen as spinal fluid.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

Specimen Minimum Volume: Serum, Spinal fluid: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	14 days
	Refrigerated	72 hours
	Ambient	24 hours
Serum	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

83521

83916 x2 (if appropriate)

MSDBS Multiple Sulfatase Deficiency, Blood Spot
618296

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient 1 year of age or older is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

83864 (if appropriate)

MSDW
618297

Multiple Sulfatase Deficiency, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

MUM1B 603217 MUM-1/IRF4 Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUM1 70514 MUM-1/IRF4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CMUMP Mumps Virus Antibodies, IgG and IgM, Spinal Fluid
81435

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Submit aliquot from collection vial 1.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days

	Frozen	14 days

CPT Code Information:

86735 x 2

MPPG 34947 Mumps Virus Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735

MMPGM 61854 Mumps Virus Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735-Mumps, IgG

86735-Mumps, IgM

MMPM
80977

Mumps Virus Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735

MUMPR 617823 Mumps Virus, Molecular Detection, PCR, Buccal

Specimen Requirements:

Specimen Type: Buccal Swab

Supplies: Culturette (BBL Culture Swab) (T092)

Container/Tube: Sterile container with transport media

Specimen Volume: Entire collection

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosal surface around buccal cavity (the space near the upper rear molars between the cheek and the teeth) to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4, M5, Barthels FlexTrans Media or Jiangsu Transport Media)

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

MUMPU 617824 Mumps Virus, Molecular Detection, PCR, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	7 days

CPT Code Information:

87798

MBX 70594 Muscle Pathology Consultation

Specimen Requirements:

Preferred:

Specimen Type: Frozen muscle biopsy tissue

Supplies: Muscle Biopsy Kit (T541)

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Additional Information:

1. All specimens and additional paperwork must be labeled with:

-Two patient identifiers (first and last name, date of birth, or medical record number)

-Pathology accession/case number

-Anatomic site

2. Contact the Mayo Clinic Muscle Laboratory at 800-533-1710 for special problems to maximize benefit of the muscle biopsy.

Acceptable:

Specimen Type: Stained muscle biopsy slides

Collection Instructions:

1. Submit all stains performed on the case.

2. **All specimens must be labeled with anatomic site**.

Additional Information: Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: 1.5 cm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	
	Ambient	

CPT Code Information:

88342 (if appropriate)

88341 (if appropriate)

88346 (if appropriate)

88350 (if appropriate)

88305 (if appropriate)

88313 (if appropriate)

88319 (if appropriate)

88314 (if appropriate)

88321 (if appropriate)

88323 (if appropriate)

88323-26 (if appropriate)

88325 (if appropriate)

FCDU9 Muscle Relaxants, Umbilical Cord Tissue (Non-Orderable)

75784

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

MUSK
64277

Muscle-Specific Kinase (MuSK) Autoantibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86366

FMUSG
57659

Mushroom IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MUSH
82626

Mushroom, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMTFG
57679

Mustard Food IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MSTD
82801

Mustard, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MUXF3
609437

MUXF3 (Cross-reactive Carbohydrate Determinant), IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86008

FBMO
75510

MVista Blastomyces Quantitative Antigen, Fluid

Specimen Requirements:

Specimen Type: Spinal fluid (CSF) or bronchoalveolar fluid

Container/Tube: Sterile, leak-proof screw cap container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 2 mL of CSF or bronchoalveolar lavage fluid.

2. Submit specimen in a sterile, leak-proof screw cap container labeled with specimen type.

3. Send refrigerate.

Additional Information:

1. Specimen type is required.

2. A separate order is required for each specimen.

3. Sputolysin, sodium hydroxide, and potassium hydroxide treatment degrade the analyte detected in the assay.

Specimen Minimum Volume: Spinal fluid: 0.8 mL; Bronchoalveolar fluid: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

87449

FMVCO 57122 MVista Coccidioides Quantitative Antigen EIA

Specimen Requirements:

Specimen Type: Urine

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 2 mL

Collection Instructions: Collect 2 mL random urine specimen. Ship specimen refrigerated in a plastic, preservative-free urine container.

Note: Sputolysin and Sodium Hydroxide are interfering substances.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	48 hours
	Frozen	

CPT Code Information:

87449

FHST 91957 MVista Histoplasma Ag Quantitative EIA

Specimen Requirements:

Specimen Type: Bronchial washing or body fluid

Container/Tube: Sterile, leak-proof screw cap container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 2 mL of bronchial washing or body fluid in a sterile, leak-proof screw cap container.

2. **Label with specimen type (required)**.

3. Send refrigerate.

Note:

1. MiraVista will test most body fluids with the following disclaimer: The reference range and other method performance specifications have not been established for this test in this type of body fluid. The test results should be integrated into the clinical context for interpretation.

2. New York approved for Urine, Serum, Plasma, spinal fluid (CSF), and bronchoalveolar lavage (BAL) specimens.

Specimen Minimum Volume: Bronchial washing/Body fluid: 0.5 mL (Note: Minimum volume does not allow for repeats.)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

87385

FHSAG 90017 MVista Histoplasma Ag Quantitative, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal Fluid

Sources: CSF

Container/Tube: Sterile container

Specimen Volume: 0.8 mL

Collection Instructions: Collect 0.8 mL of spinal fluid (CSF). Ship refrigerated, 0.8 mL of spinal fluid. Send specimen in a plastic, screw-capped vial refrigerated.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

87385

MGMR 608980 Myasthenia Gravis Evaluation with Muscle-Specific Kinase (MuSK) Reflex, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

	Ambient	72 hours

CPT Code Information:

86041

86043 (if appropriate)

86366 (if appropriate)

MGLE 608979 Myasthenia Gravis/Lambert-Eaton Myasthenic Syndrome Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86041

86596

86043 (if appropriate)

86366 (if appropriate)

SGTF 35860 MYB (6q23) Rearrangement FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods will be attempted but are less favorable for successful results by fluorescence in situ hybridization testing.

2. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Collection Instructions: Submit 2 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

MYC
70515

MYC Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CFCTB 623130 Mycobacteria and Nocardia Culture, Cystic Fibrosis, Varies

Specimen Requirements:

Specimen Type: Respiratory fluid

Sources: Bronchoalveolar lavage fluid, bronchial washing, sputum (saliva is **not acceptable**)

Container/Tube: Sterile container

Specimen Volume: 3 to 5 mL

Collection Instructions:

1. Collect 3 separate respiratory specimens for acid-fast smears and culture in patients with clinical and chest X-ray findings compatible with tuberculosis.

2. These 3 specimens should be collected at 8 to 24-hour intervals (24 hours when possible) and should include at least 1 first-morning specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87116-Mycobacterial Culture

87015-Mycobacteria Culture, Concentration (if appropriate)

87118-Id MALDI-TOF Mass Spec AFB (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87176-Tissue Processing (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

CTB 8205 Mycobacteria and Nocardia Culture, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Bone marrow

Container/Tube: Sterile container, or green top (lithium or sodium heparin)

Specimen Volume: Entire collection

Specimen Type: Gastric washing

Container/Tube: Sterile container

Specimen Volume: 10 mL

Collection Instructions: Neutralize specimen within 4 hours of collection with 100 mg of sodium carbonate per 5 to 10 mL of gastric wash.

Specimen Type: Respiratory

Sources: Bronchoalveolar lavage fluid, bronchial washing, sputum

Container/Tube: Sterile container

Specimen Volume: 3 mL

Collection Instructions:

1. Collect 3 respiratory specimens for acid-fast smears and culture in patients with clinical and chest X-ray findings compatible with tuberculosis.

2. These 3 specimens should be collected at 8 to 24-hour intervals (24 hours when possible) and should include at least 1 **first-morning** specimen.

Specimen Type: Stool

Supplies: Stool Collection Kit, Random (T635)

Container/Tube: Sterile container

Specimen Volume: 5 to 10 g

Specimen Type: Tissue

Container/Tube: Sterile container

Specimen Volume: 5 to 10 mm

Collection Instructions: Collect a fresh tissue specimen.

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 20 to 50 mL

Collection Instructions: Collect a random urine specimen.

Fresh tissue or body fluid are preferred over a swab specimen. Recovery of mycobacteria from swabs is generally very low yield. Only submit a swab specimen if tissue biopsy or fluid aspiration is not possible.

Specimen Type: Swab

Supplies: BD E-swab (T853)

Sources: Surgically collected wound, tissue, or body fluid

Container/Tube: Sterile, screw-capped tube containing Liquid Amies Medium with flocked swab (eg, E-swab)

Specimen Volume: Flocked swab in 1 mL of Liquid Amies Medium

Collection Instructions:

1. Before collecting specimen, wipe away any excessive amount of secretion and discharge, if appropriate.

2. Obtain secretions or fluid from source with sterile flocked swab. **Paranasal sinus**

collections must use a nasopharyngeal flocked swab.

3. Place flocked swab in sterile, screw-capped tube containing 1 mL of Liquid Amies Medium.

4. If smear and culture are requested or both a bacterial culture and mycobacterial culture are requested, collect a second swab to maximize test sensitivity. Submit each swab in a separate sterile, screw-capped tube with 1 mL of Liquid Amies Medium.

Additional Information: Swabs from the following sources are **not acceptable**: respiratory fluid (eg, sputum), nasal, sinus, outer ear, mouth, throat, or scalp. Surgically collected inner ear swabs **are** acceptable.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87116-Mycobacterial Culture

87015-Mycobacteria Culture, Concentration (if appropriate)

87118-Id MALDI-TOF Mass Spec AFB (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87176-Tissue Processing (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

CTBBL
82443

Mycobacterial Culture, Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: SPS (sodium polyanethol sulfonate)

Specimen Volume: 8 to 10 mL per culture

Collection Instructions:

1. Send whole blood specimen in original tube.

2. SPS tubes are acceptable, but not preferred.

3. Note: SPS tubes must be clearly labeled as SPS. If label is obscured, sample may be canceled, as ACD (yellow top) is not an acceptable tube type.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

87116-Mycobacterial Culture

87118-Id MALDI-TOF Mass Spec AFB (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87153-Mtb PZA Confirmation, pcnA sequence (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

DMAID 619537 Mycobacterium abscessus Complex Subspecies Identification, Next-Generation Sequencing

Specimen Requirements:

Supplies: Infectious Container, Large (T146)

Specimen Type: *Mycobacterium abscessus* isolate growing in pure culture

Container/Tube:

Preferred: Solid slant medium (eg. Middlebrook 7H10, 7H11 or Lowenstein Jensen agar)

Acceptable: Broth medium (eg. Mycobacteria Growth Indicator Tube, 7H9 broth, BACT/ALERT MP or VersaTREK bottle). **Note:** Broth specimens will require subculture which will delay results**.**

Specimen Volume:

Solid slant medium: Isolate with visible growth

Broth medium: Greater than or equal to 3 mL of broth culture

Collection Instructions:

1. Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

2. Place specimen in a large infectious container and label as an etiologic agent/infectious substance.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87153

TBSP 607364 Mycobacterium tuberculosis Complex Species Identification, PCR, Varies

Specimen Requirements:

Specimen Type: *Mycobacterium tuberculosis* complex isolate growing in pure culture

Supplies: Infectious Container, Large (T146)

Container/Tube: Growth on solid slant media, eg, Middlebrook 7H10, 7H11 and Lowenstein Jensen; growth in broth medium, eg, Mycobacteria Growth Indicator Tube, 7H9 broth BACT/ALERT MP or VersaTREK

Specimen Volume: Isolate with visible growth on solid media; if broth is sent, 3 mL or more of broth culture

Collection Instructions:

1. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures**.

2. Place specimen in a large infectious container and label as an etiologic agent/infectious substance.

Additional Information: If subculture to Middlebrook agar medium is needed to ensure purity, turnaround time for results may be delayed.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87150

MTBXS 619564 Mycobacterium tuberculosis Complex, Molecular Detection and Rifampin Resistance, PCR, Sputum

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycobacterium tuberculosis* DNA is unlikely.

Specimen Type: Sputum (undigested)

Container/Tube: Sterile container

Specimen Volume: 3 mL

Specimen Stability Information: Refrigerated (preferred) 7 days/Ambient 72 hours

Additional Information:

1. If a single specimen is being shared between mycobacterial culture, acid-fast smear, and/or *M tuberculosis* PCR, a minimum volume of 3 mL for the respiratory specimen is required. Specimen volumes less than indicated may decrease sensitivity of testing.

2. If insufficient volume is submitted, testing will be canceled.

Specimen Type: N-acetyl-L-cysteine/sodium hydroxide (NALC/NaOH)-digested sputum

Container/Tube: Sterile container

Specimen Volume: 3 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Stability Information: Refrigerated 7 days

Additional Information:

1. If a single specimen is being shared between mycobacterial culture, acid-fast smear, and/or *M tuberculosis* PCR, a minimum volume of 3 mL for the respiratory specimen is required. Specimen volumes less than indicated may decrease sensitivity of testing.

2. If insufficient volume is submitted, testing will be canceled.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Sputum	Refrigerated (preferred)	7 days
	Ambient	72 hours

CPT Code Information:

87564

DMTBR 622531 Mycobacterium tuberculosis Complex, Molecular Detection of Drug Resistance Markers, Next-Generation Sequencing

Specimen Requirements:

Supplies: Infectious Container, Large (T146)

Specimen Type: *Mycobacterium tuberculosis* complex isolate growing in pure culture

Container/Tube:

Preferred: Solid slant medium (eg. Middlebrook 7H10, 7H11 or Lowenstein Jensen agar)

Acceptable: Broth medium (eg. Mycobacteria Growth Indicator Tube, 7H9 broth, BACT/ALERT MP or VersaTREK bottle). **Note:** Broth specimens will require subculture which will delay results.

Specimen Volume:

Solid slant medium: Isolate with visible growth

Broth medium: Greater than or equal to 3 mL of broth culture

Collection Instructions:

1. Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

2. Place specimen in a large infectious container and label as an etiologic agent/infectious substance.

3. Turnaround time for results may be delayed if subculture is needed to ensure purity.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87153 - *M tuberculosis* drug resistance, NGS

MTBT 62203 Mycobacterium tuberculosis Complex, Molecular Detection, PCR, Paraffin, Tissue

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycobacterium tuberculosis* DNA is unlikely.

Preferred:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded (FFPE) tissue block

Sources: Body tissue

Container/Tube: Tissue block

Collection Instructions: Submit a FFPE tissue block to be cut and returned.

Acceptable:

Specimen Type: Formalin-fixed, paraffin-embedded (FFPE) tissue sections (scrolls)

Sources: Body tissue

Container/Tube: Sterile containers, one for each individual cut section (scroll).

Collection Instructions:

1. Perform microtomy and prepare five separate 10-micron sections.

2. **Place each section (scroll) in a separate sterile container for submission.**

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87556

MTBRP 88807 Mycobacterium tuberculosis Complex, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit a raw clinical sample (not a culture isolate) for testing.

Fresh, undigested specimens are preferred. If no fresh specimen is available, digested respiratory specimens treated with N-acetyl-L-cysteine/sodium hydroxide (NALC/NaOH) are acceptable for some specimen types. For detailed information, see the **Specimen Type** information below.

The high sensitivity of amplification by polymerase chain reaction (PCR) requires the specimen to be processed in an environment in which contamination of the specimen by *Mycobacterium tuberculosis* DNA is unlikely.

If a single specimen is being shared between mycobacteria culture, acid-fast smear, and/or *M tuberculosis* PCR, a minimum volume of 2 mL for body fluid, 3 mL for respiratory specimen, or a pea-sized piece of tissue should be obtained. Specimen

volumes less than indicated may decrease sensitivity of testing. If insufficient volume is submitted, test or tests will be canceled.

Submit only 1 of the following specimens:

Preferred Specimen Types

Specimen Type: Body fluid

Sources: Body, bone marrow aspirate, ocular, or CSF

Container/Tube: Sterile container

Specimen Volume: 1 mL

Additional Information: Only fresh, non-NALC/NaOH-digested body fluids are acceptable.

Specimen Type: Gastric washing

Container/Tube: Sterile container

Specimen Volume: 2 mL

Specimen Type: Respiratory

Sources: BAL, bronchial washing, tracheal secretion, or sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL if only PCR ordered or 3 mL if PCR ordered with smear and culture

Additional Information: Before collecting sputum specimens, see Ordering Guidance.

Specimen Type: Feces

Container/Tube: Sterile container

Specimen Volume: 5 to 10 g

Additional Information: Only fresh, non-NALC/NaOH-digested fecal specimens are acceptable.

Specimen Type: Tissue

Sources: Fresh tissue, bone, or bone marrow biopsy

Container/Tube: Sterile container

Specimen Volume: 5 to 10 mm

Collection Instructions: Keep moist with sterile water or sterile saline

Additional Information: Only fresh, non-NALC/NaOH-digested tissue is acceptable.

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Collect a random urine specimen.

Acceptable Specimen Types

Specimen Type: NALC/NaOH-digested respiratory specimens

Sources: Lavage fluid, bronchial washing, gastric washing, respiratory fluid, sputum, or tracheal secretion

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Minimum Volume: Body fluid, nondigested respiratory specimen, urine: 0.5 mL; Fresh tissue or bone: 5 mm; NALC-NaOH-digested specimen, gastric washing: 1 mL; Stool: 5 g

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87556-*Mycobacterium tuberculosis*, complex, molecular detection, PCR

87015-Mycobacteria culture, concentration (if appropriate)

MTBPZ 56099 Mycobacterium tuberculosis Complex, Pyrazinamide Resistance by pncA DNA Sequencing, Varies

Specimen Requirements:

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook 7H10 agar slant

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87153-Mtb PZA Confirmation, pncA Sequence

MPA 81563 Mycophenolic Acid, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Draw blood immediately before the next scheduled dose (trough).

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

80180

RPMPM 610248 Mycoplasma (Mycoplasmoides) pneumoniae Macrolide (Azithromycin) Resistance Prediction, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycoplasma (Mycoplasmoides) pneumoniae* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-swab (T853)

-Culture Swab-Liquid Stuarts/Single Swab (NP Swab) (T515)

-M4-RT (T605)

Sources: Throat, nasal, or nasopharyngeal

Container/Tube:

Preferred: Culture swab transport system (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Acceptable: Culture transport swab (Stuart's media) or place swab in M4, M4-RT, M5, M6, universal transport media, or ESwab

Specimen Volume: Swab

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Specimen Type: Fluid

Sources: Pleural, pericardial, cerebrospinal

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, tracheal secretions, sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: Respiratory: 0.5 mL
Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

Mycoplasma (Mycoplasmoides) pneumoniae with Macrolide Resistance Reflex, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycoplasma (Mycoplasmoides) pneumoniae* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-Culture Swab-Liquid Stuarts/Single Swab (NP Swab) (T515)

-M4-RT (T605)

Sources: Throat, nasal, or nasopharyngeal

Container/Tube:

Preferred: Culture swab transport system (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Acceptable: Culture transport swab (Stuart's media) or place swab in M4, M4-RT, M5, M6, universal transport media, or ESwab

Specimen Volume: Swab

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Specimen Type: Fluid

Sources: Pleural, pericardial, cerebrospinal

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, tracheal secretions, sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: Respiratory specimen: 0.5 mL
Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87581

MMGEN 620733 Mycoplasma genitalium, Transcription-Mediated Amplification, Post-Prostatic Massage Fluid/Urine or Peritoneal Fluids

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Post-prostatic massage fluid/urine (VBIII)

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima Urine Specimen Transport Tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should void a small amount of urine prior to prostatic massage. Pre-massage urine can be discarded.

3. Patient should cease voiding and a prostatic massage should be performed by urologist or other healthcare professional.

4. Collect post-massage urine into a sterile, plastic, preservative-free container.

5. Transfer 2 mL of urine into the Aptima Urine Specimen Transport Tube using the disposable pipette provided within 24 hours of collection. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

6. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

7. Transport and store post-prostatic massage fluid/urine (VBIII) specimen transport container at 2 to 30 degrees C (refrigerate is preferred temperature) within 30 days of collection. **Specimen cannot be sent frozen**.

Specimen Type: Peritoneal fluid (pelvic wash, cul-de-sac fluid)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima Specimen Transfer Tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen by paracentesis using needle and syringe using routine procedures. Refer to the references below.

2. Thoroughly vortex specimen upon receipt to the laboratory.

3. Transfer 1 mL of peritoneal fluid into the Aptima Specimen Transfer kit using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug) within 24 hours of collection.

4. Recap the specimen transfer tube tightly.

5. Transport and store specimen container at 2 to 30 degrees C (refrigerate is preferred temperature) within 30 days of collection. **Specimen cannot be sent frozen**.

Specimen Minimum Volume: Peritoneal fluid: 1 mL; Post-prostatic massage fluid/urine (VBIII): See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		APTIMA VIAL

CPT Code Information:

87563

AMGEN 616513 Mycoplasma genitalium, Transcription-Mediated Amplification, Urine or Urogenital Swab

Specimen Requirements:

Submit only 1 of the following specimens:

Swab specimen must be collected using an Aptima Collection Unisex Swab or Aptima Collection Multitest Swab based on the corresponding source. These swabs are contained in the Aptima Collection Kit.

Specimen Type: Endocervix/cervix

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Unisex Swab Collection Kit**

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

3. Discard the cleaning swab.

4. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

5. Place the blue shaft swab into transport tube provided in collection kit.

6. Snap off swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Multitest Swab Collection Kit.**

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate is preferred temperature) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Type: Urethra (Male patients only)

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Unisex Swab Collection Kit.**

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate swab gently at least 1 full rotation using sufficient pressure to ensure swab comes in contact with all urethral surfaces. Allow swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate is preferred temperature) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima Urine Specimen Transport Tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection, transfer 2 mL of urine into the Aptima Urine Specimen Transport Tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport container at 2 to 30 degrees C (refrigerate is preferred temperature) within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C up to an additional 90 days.

Specimen Type: Penile/meatal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Multitest Swab Collection Kit.**

2. Roll the swab (pink shaft) just at the tip or outside the opening to the penis. Be sure to roll the swab completely around the opening to get the best sample.

3. Immediately place the pink swab into the collection tube.

4. Carefully break the pink swab shaft at the score line and recap the tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate is preferred temperature) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Minimum Volume: Urine-first void: 2 mL; Swabs (endocervical, urethral, vaginal, penile/meatal): See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		APTIMA VIAL

CPT Code Information:

87563

MYCO
48394

Mycoplasma pneumoniae Antibodies, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86738 x 2-*Mycoplasma pneumoniae* by EIA

86738-*Mycoplasma pneumoniae* by indirect IFA (if appropriate)

MYCOG 48317 Mycoplasma pneumoniae Antibodies, IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MYCO / *Mycoplasma pneumoniae* Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86738

MYCOM 48318 Mycoplasma pneumoniae Antibodies, IgM, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MYCO / *Mycoplasma*

pneumoniae Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86738

MYCON 48319 Mycoplasma pneumoniae Antibody Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MYCO / *Mycoplasma pneumoniae* Antibodies, IgG and IgM, Serum.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CXCFX 601508 MYD88 Reflex to CXCR4 Mutation Detection, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see LPLFX / Lymphoplasmacytic Lymphoma/Waldenstrom Macroglobulinemia, *MYD88* L265P with Reflex to *CXCR4*, Varies.

Specimen Minimum Volume: Whole blood, Bone marrow: 1 mL; Extracted DNA: 20 mcL with a concentration of at least 10 nanograms per mcL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81479

MYD88 62927 MYD88, L265P, Somatic Gene Mutation, DNA Allele-Specific PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection

2. Indicate specimen source.

Specimen Stability Information: Frozen

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 50 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81305

MOGFS 65563 Myelin Oligodendrocyte Glycoprotein (MOG-IgG1) Fluorescence-Activated Cell Sorting (FACS) Assay, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection should occur prior to initiation of immunosuppressant medication.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86363

86363-titer (if appropriate)

MAGES 607034 Myelin-Associated Glycoprotein Autoantibodies, IgM, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83520

MDSDF 614276 Myelodysplastic Syndrome (MDS), Diagnostic FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 12, 88275 x 6, 88291-FISH Probe, Analysis, Interpretation; 6 probe sets

88271 x 2, 88275-FISH Probe, Analysis; each additional probe set (if appropriate)

MDSMF 614287 Myelodysplastic Syndrome (MDS), Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x2, 88275 x1, 88291 x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271 x2, 88275 x1 – FISH Probe, Analysis; each additional probe set (if appropriate)

MYEFL 63414 Myelodysplastic Syndrome, Flow Cytometry, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA), green top (heparin)

Specimen Volume: 2-5 mL

Slides: Include 1 to 2 unstained bone marrow smears labeled with 2 unique identifiers.

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each) x18

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

MSTF
35844

Myeloid Sarcoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be

attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. If individual probe sets are chosen: For each probe set ordered, 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If a complete myeloid sarcoma FISH panel is ordered, include 10 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88291

88271 x 2 (if appropriate)

88271 x 2 (if appropriate)

88271 (if appropriate)

88271 x 2 (if appropriate)

88271 x 3 (if appropriate)

88274 w/modifier 52 (if appropriate)

88274 (if appropriate)

88275 (if appropriate)

MSMRD 616030 Myeloma Stratification and Risk-Adapted Therapy with Reflex to Measurable/Minimal Residual Disease, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 4 mL

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88182-Flow cytometry, cell cycle or DNA analysis

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow cytometry interpretation, 2 to 8 Markers

MFCDF 614298 Myeloma, High Risk with Reflex Probes, Diagnostic FISH

Evaluation, Fixed Cell Pellet

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 Fixed cell pellet

Collection Instructions: Place specimen in a sterile container with a 3:1 methanol:glacial acetic acid (or similar) fixative.

Transport Temperature:

Specimen Type	Temperature	Time
Fixed Cell Pellet Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x6, 88275x3, 88291 x1-FISH Probe, Analysis, Interpretation; 3 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

MAS1 605125 Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 15

86053

86363

86341

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN2TS (if appropriate)

86256 AN3TS (if appropriate)

86255 APBCS (if appropriate)

86256 APBTS (if appropriate)

86256 CRMTS (if appropriate)
84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86363 MOGTS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86053 NMOTS (if appropriate)

84182 PC1BS (if appropriate)

86256 PC1TS (if appropriate)

86256 PC2TS (if appropriate)

86255 SP7CS (if appropriate)

86256 SP7TS (if appropriate)

86255 T46CS (if appropriate)

86256 T46TS (if appropriate)

MAC1
605126

Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 17

86053

86341

84182 AGNBC (if appropriate)

86256 AGNTC (if appropriate)

86255 AINCC (if appropriate)

86256 APHTC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 AN2TC (if appropriate)

86256 AN3TC (if appropriate)

86255 APBCC (if appropriate)

86256 APBTC (if appropriate)

86256 CRMTC (if appropriate)
84182 CRMWC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86255 NFHCC (if appropriate)

86256 NIFTC (if appropriate)

86255 NFLCC (if appropriate)

86053 NMOTC (if appropriate)

84182 PC1BC (if appropriate)

86256 PC1TC (if appropriate)

86256 PC2TC (if appropriate)

86255 SP7CC (if appropriate)

86256 SP7TC (if appropriate)

86255 T46CC (if appropriate)

86256 T46TC (if appropriate)

MYPO 70511 Myeloperoxidase (MPO) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MPO 80389 Myeloperoxidase Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

MPNCM 65115 Myeloproliferative Neoplasm, CALR with Reflex to MPL, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5 to 2 mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow: 0.5 mL; Extracted DNA: 50 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81219-CALR (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9

81339 -MPL (MPL proto-oncogene, thrombopoietin receptor) (eg, myeloproliferative disorder) gene analysis; sequence analysis, exon 10 (if appropriate)

MPNJM 606805 Myeloproliferative Neoplasm, JAK2 V617F with Reflex to CALR and MPL, Bone Marrow

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Note: Extracted DNA from bone marrow is **not** acceptable.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81270-JAK2 (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

81219-CALR (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9 (if appropriate)

81339-MPL (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence (if appropriate)

MPNR 63031 Myeloproliferative Neoplasm, JAK2 V617F with Reflex to CALR and MPL, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not** aliquot.

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Blood, bone marrow: 0.5 mL; Extracted DNA: 50 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81270-JAK2 (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant
81219-CALR (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9 (if appropriate)
81339-MPL (MPL proto-oncogene, thrombopoietin receptor) (eg, myeloproliferative disorder) gene analysis; sequence analysis, exon 10 (if appropriate)

GNMY9 619299 MYH9-Related Disorders, MYH9 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

MEF2B 619494 Myocyte Enhancer Factor 2 (MEF2B) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYODT 619685 MYOD1 Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88381-Microdissection, manual

81479

MYOD1 70518 Myogenic Differentiation Antigen 1 (MYOD1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYOGE 70516 Myogenin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYOGL Myoglobin Immunostain, Technical Component Only
70517

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYGLS 621090 Myoglobin, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

83874

FMMPP MyoMarker 3 Plus Profile
75594

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Separate serum from cells within 1 hour of collection and send 5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 4 mL (volume does NOT allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516 x 10

86235 x 7

FMYO3 MyoMarker 3 Profile
75595

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is

acceptable. Separate serum from cells within 1 hour of collection and send 5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 4 mL (volume does NOT allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516 x 9

86235 x 7

MSAES 622116 Myositis Specific Antibody Evaluation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86255 x10

82397

86235

84182-SRPBS (if appropriate)

86256-SRPTS (if appropriate)

DDITF 35265 Myxoid/Round Cell Liposarcoma, 12q13 (DDIT3 or CHOP) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291 DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

NAACD
621423

N-Acetylaspartic Acid, Canavan Disease, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Pediatric: If insufficient collection volume, submit as much specimen as possible in a single container; the laboratory will determine if volume is sufficient for testing.

Collection Instructions:

1. Collect a random urine specimen (first morning void preferred).

2. No preservative.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	90 days
	Refrigerated	90 days
	Ambient	14 days

CPT Code Information:

83921

NAT2Q
610617

N-Acetyltransferase 2 (NAT2) Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant or liver transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell or liver transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479-Unlisted molecular pathology procedure

NMH24 N-Methylhistamine, 24 Hour, Urine

605135

Specimen Requirements:

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Aliquot into plastic tube and send at refrigerate temperature.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

NMH1D 605159 N-Methylhistamine, 24 Hour, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NMH24 / *N*-Methylhistamine, 24 Hour, Urine.

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Aliquot into plastic tube and send at refrigerate temperature.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

NMHR
604981

N-Methylhistamine, Random, Urine

Specimen Requirements:

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

82570

NMHR1
605015

N-Methylhistamine, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NMHR / *N*-Methylhistamine, Random, Urine.

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

FINA
91447

NAbFeron (IFNB-1) Neutralizing Antibody Test

Specimen Requirements:

Patient Preparation: Specimen should be collected either before treatment with interferon or more than 24 hours following the most recent dose. Patient **should not** be on steroid therapy for at least two weeks prior to testing.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Within 48 hours of collection, centrifuge and aliquot 2 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

86382

FNAD
80761

Nadolol, Serum/Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Submit only 1 of the following specimens:

Specimen Type: Serum

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

3. Send refrigerated.

Specimen Type: Plasma

Collection Container/Tube: Lavender top or pink top (EDTA) (plasma gel/PST are **not acceptable**)

Specimen volume: 1 mL Plasma

Collection Instructions:

1. Promptly centrifuge and aliquot plasma into a plastic vial.

2. Label specimen as plasma.

3. Send refrigerated.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80299

FNALO 91784 Naloxone - Total (Conjugated/Unconjugated), Screen, Urine

Specimen Requirements:

****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect 1 mL random urine specimen in a plastic preservative-free urine container.

2. Send refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

80307

80362 if appropriate

G0480 if appropriate

NAPSN 70519 Napsin A Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NARC 82026 Narcolepsy-Associated Antigen, HLA-DQB1 Typing, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A), lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original vial. **Do not aliquot**.

Additional Information: Specimen acceptability is based on extracted DNA concentration and not sample age.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

81376-HLA Class II typing, low resolution (eg, antigen equivalents); one locus (eg, HLA-DRB1/3/4/5, -DQB1, -DQA1, -DPB1, or -DPA1), each

QNKS 60616 Natural Killer (NK)/Natural Killer T-Cell Subsets, Quantitative, Blood

Specimen Requirements:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	30 hours	GREEN TOP/HEP

CPT Code Information:

86356 x3

86359

86357

NMS1 603542 Necrotizing Myopathy Evaluation, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge within 2 hours of collection and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

82397

86256 (if appropriate)

84182 (if appropriate)

FNECT 57941 Nectarine (Prunus spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

NECT4 622259 Nectin-4, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

NEGCT Negative Control, Technical Component Only

70410

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

GCCON 623156 Neisseria gonorrhoeae Confirmation, Nucleic Acid Amplification, Varies

Specimen Requirements:

If testing is being performed on a previously tested specimen that resulted as positive, submit that specimen.

If testing is being performed on a newly collected specimen, submit only 1 of the following specimens:

Specimen Type: Ocular (corneal/conjunctiva)

Supplies:

-Aptima Unisex Swab Collection Kit (T583)

-Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab or Aptima Collection Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using Aptima Multitest Swab or Aptima Collection Unisex Swab. Specimens must be collected using either of these options.

Note: The white swab provided within the collection kit is a cleaning swab and should not be used for collection. Discard the white cleaning swab.

2. Place collection swab in transport tube provided in collection kit.

3. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimen must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection. transfer 2 mL of urine into the urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport tube at 2 to 30 degrees C (refrigerated temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Oropharynx/Pharynx/Throat

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Collection Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Swab site using Aptima Collection Multitest Swab (pink shaft).

3. Place pink swab in transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Rectal/Anal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab into rectum about 3 to 5 cm past anal margin and gently rotate swab for 10 seconds.

3. Place collection swab in transport tube provided in collection kit.

4. Snap off swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is

preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87591

MGRNA 61646 Neisseria gonorrhoeae, Miscellaneous Sites, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Ocular (corneal/conjunctiva)

Supplies:

-Aptima Unisex Swab Collection Kit (T583)

-Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab or Aptima Collection Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using Aptima Multitest Swab or Aptima Collection Unisex Swab. Specimens must be collected using either of these options.

Note: The white swab provided within the collection kit is a cleaning swab and should not be used for collection. Discard the white cleaning swab.

2. Place collection swab in transport tube provided in collection kit.

3. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and

time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 60 days of collection.

Specimen Type: Peritoneal fluid (pelvic wash, cul-de-sac fluid)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Transfer 1 mL of peritoneal fluid directly into the Aptima specimen transfer tube within 24 hours of collection.

2. Cap tube securely and label tube with patient's entire name and collection date and time.

3. Maintain Aptima specimen transfer tube at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 30 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87591

GCRNA 61552 Neisseria gonorrhoeae, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Endocervix/Cervix

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Unisex Swab Collection Kit.

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

3. Discard the cleaning swab.

4. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

5. Place blue swab into transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimen must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urethra (Males Only)

Supplies: Swab, Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimen must be collected using the Aptima Unisex Swab Collection Kit.

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate blue swab gently at least 1 full rotation using sufficient pressure to ensure swab comes into contact with all urethral surfaces. Allow blue swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection. transfer 2 mL of urine into the urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport

tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: ThinPrep Specimen (Endocervix/Cervix)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect ThinPrep sample as per normal collection process.

2. **ThinPrep specimen must be aliquoted (as outlined below) before it is processed for Papanicolaou testing (Pap smear).**

3. Vortex ThinPrep/PreservCyt vial 3 to 10 seconds.

4. Within 1 minute of vortexing:

a. Transfer 1 mL of specimen into the Aptima specimen transport tube using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug).

b. Process only 1 ThinPrep and transfer tube set at a time.

c. Recap Aptima Specimen Transfer Tube tightly and gently invert 3 times to mix.

5. Label Aptima specimen transfer tube with appropriate label.

6. Use remainder of ThinPrep specimen for Pap testing.

7. Maintain specimen transport tube at 2 to 8 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection, or within 14 days if stored at 15 to 30 degrees C. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Oropharynx/Pharynx/Throat

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Collection Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Swab site using Aptima Collection Multitest Swab (pink shaft).

3. Place pink swab in transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Type: Rectal/Anal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab into rectum about 3 to 5 cm past anal margin and gently rotate swab for 10 seconds.

3. Place collection swab in transport tube provided in collection kit.

4. Snap off swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL

	Ambient	APTIMA VIAL
	Frozen	APTIMA VIAL

CPT Code Information:

87591

SCGCV 621939 Neisseria gonorrhoeae, Self-Collect, Amplified RNA, Vaginal (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SCCGV / *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Vaginal.

Specimen Type: Vaginal

Supplies: Aptima Vaginal Swab Self-Collection Kit (T1001)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1001 or available separately)**.**

2. Provide patient with the Aptima Vaginal Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for up to 12 months.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL

	Ambient	APTIMA VIAL
	Frozen	APTIMA VIAL

CPT Code Information:

SCGCR 621936 Neisseria gonorrhoeae, Self-Collect, Nucleic Acid Amplification, Rectal (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SCCGR / *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Rectal

Specimen Type: Rectal/anal

Supplies: Aptima Rectal Swab Self-Collection Kit (T1000)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1000 or available separately)**.**

2. Provide patient with the Aptima Rectal Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

SCGCT 621933

Neisseria gonorrhoeae, Self-Collect, Nucleic Acid Amplification, Throat (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see: SCCGT/ *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Throat

Specimen Type: Oropharynx/pharynx/throat

Supplies: Aptima Throat Swab Self-Collection Kit (T1002)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. **Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab** (provided in T1002 or available separately)**.**

2. Provide patient with the Aptima Throat Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

FNMEN Neisseria Meningitidis IgG Vaccine Response

91669

Specimen Requirements:

Container/Tube:

Preferred: Red top tube

Acceptable: Serum gel tube

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Note: Serum gel tube is acceptable, but must pour off into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86317/x4

FNNAT 75929 Neonatal Alloimmune Thrombocytopenia (NAIT)-Serial Monitoring Maternal only

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instruction:

1.Centrifuge and aliquot 5 mL serum into a plastic vial.

2.Send refrigerated.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

86022

86023 x 2

NBILI
82133

Neonatal Bilirubin, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: 2 Serum gel Microtainers

Acceptable: 2 Red top Microtainers

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel microtainers should be centrifuged within 2 hours of collection.

2. Red-top microtainers should be centrifuged, and the serum aliquoted into an amber vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	24 hours	LIGHT PROTECTED
	Frozen	30 days	LIGHT PROTECTED
	Ambient	6 hours	LIGHT PROTECTED

CPT Code Information:

82247

82248

FNEOS 75451 Neopterin

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 0.8 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.8 mL of serum into an amber vial to protect from light.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL; Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Frozen (preferred)	180 days	LIGHT PROTECTED
	Refrigerated	72 hours	LIGHT PROTECTED

CPT Code Information:

83520

RSCGP 618128 Nephrocalcinosis, Nephrolithiasis, and Renal Electrolyte Imbalance Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404 x 4

81405 x 2

81406 x 8

81407 x 2

81479

81479 (if appropriate for government payers)

NETT
82734

Nettle, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

NEUN 70631 Neu-N Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NONCP 603047 Neuro-Oncology Expanded Gene Panel with Rearrangement, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 72 mm(2)

-If ordered in conjunction with CMAPT / Chromosomal Microarray, Tumor, Formalin-Fixed Paraffin-Embedded, the preferred amount of tissue is 430 mm(2), the minimum amount is 180 mm(2).

-These amounts are cumulative over up to 15 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 6 mm x 6 mm x 10 slides as preferred: approximate/equivalent to 360 mm(2) and the minimum as 4 mm x 4 mm x 10 slides: approximate/equivalent to 144 mm(2).

Preferred: Submit 2, if available, of the following specimens.

Acceptable: Submit **at least one** of the following specimens. Tissue blocks are preferred over tissue slides.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 15 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

15 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 15

slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81455

81456

NONCM 622295 Neuro-Oncology Gene Panel, Mutations Only, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 288 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-If ordered in conjunction with CMAPT / Chromosomal Microarray, Tumor, Formalin-Fixed Paraffin-Embedded, the preferred amount of tissue is 430 mm(2), the minimum amount is 180 mm(2).

-These amounts are cumulative over up to 15 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

- For this test, at least 6 mm x 6 mm areas on 8 unstained slides is preferred: this is approximately equivalent to 288 mm(2). The minimum acceptable area is 6 mm x 6 mm on 1 unstained slides: approximately equivalent to 36 mm(2). For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing.

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Acceptable:

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 15 unstained

Collection Instructions:

Submit the following slides:

1 Slide stained with hematoxylin and eosin

AND

15 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 15 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81457

Tumor

Specimen Requirements:

This assay requires at least 60% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 144 mm(2)

-These amounts are cumulative over up to 20 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 6 mm x 6 mm x 10 slides as preferred: approximate/equivalent to 360 mm(2) and the minimum as 4 mm x 4 mm x 10 slides: approximate/equivalent to 144 mm(2).

-If ordered in conjunction with other tests, please refer to specimen requirements for the specific tests.

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with an acceptable amount of tumor tissue

Acceptable:

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 20 unstained

Collection Instructions:

Submit the following slides:

1 Slide stained with hematoxylin and eosin

AND

20 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue

Note: The total amount of required tumor nuclei can be obtained by scraping up to 20 slides from the same block.

Additional Information:

1. Unused unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81524

NCDCS 615865 Neurochondrin Antibody, Cell-Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

NCDCC 615864 Neurochondrin Antibody, Cell-Binding Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

NCDTS 616111 Neurochondrin Antibody, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

NCDTC 616112 Neurochondrin Antibody, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

NF4FS 614591 Neurofascin-155 IgG4, Flow Cytometry, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-CIDP / Chronic Inflammatory Demyelinating Polyradiculoneuropathy/Nodopathy Evaluation, Serum

-DMNES / Peripheral Nervous System Demyelinating Neuropathy, Autoimmune Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

NF1Z 614585 Neurofibromatosis Type 1, NF1, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes

prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal

testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81408

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

NF2F 70520 Neurofilament (2F11) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NFSMI
71354

Neurofilament (SMI31) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NFLP
622765

Neurofilament Light Chain, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: None

Submission Container/Tube: Plastic screw-top vial

Specimen Volume: 0.6 mL

Collection Information: Centrifuge and aliquot plasma into a plastic vial. **Do not submit specimen in original tube.**

Specimen Minimum Volume: 0.50 mL

Transport Temperature:

Specimen Type	Temperature	Time
EDTA Plasma	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

83884

PNEFS Neuroimmunology Antibody Follow-up, Serum

84300

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 AAGFS (if appropriate)

86043 ACMFS (if appropriate)

84182 AGNBS (if appropriate)

86255 AK5CS (if appropriate)

86255 AK5IS (if appropriate)

86255 AINCS (if appropriate)

86255 AMPCS (if appropriate)

86255 AMPHS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

84182 AN2BS (if appropriate)

86255 AGN1S (if appropriate)

86255 ANN1S (if appropriate)

86255 ANN2S (if appropriate)

86255 ANN3S (if appropriate)

86255 APBCS (if appropriate)

86255 APBIS (if appropriate)

86255 CS2CS (if appropriate)

86255 CRMS (if appropriate)

86255 DPPCS (if appropriate)

86255 DPPIS (if appropriate)

86255 GABCS (if appropriate)

86255 GFACS (if appropriate)

86255 GFAIS (if appropriate)

86255 GRFCS (if appropriate)

86255 GRFIS (if appropriate)

83516 IGG_D (if appropriate)

86255 IG5CS (if appropriate)

86255 IG5IS (if appropriate)

83516 IGM_D (if appropriate)

83516 IGM_M (if appropriate)

86255 ITPCS (if appropriate)

86255 ITPIS (if appropriate)

86255 LG1CS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86255 GL1IS (if appropriate)

86255 GL5CS (if appropriate)

86255 GL5IS (if appropriate)

86255 NCDCS (if appropriate)

86255 NCDIS (if appropriate)

86255 NFHCS (if appropriate)

86255 NIFIS (if appropriate)

86255 NFLCS (if appropriate)

86255 NMDCS (if appropriate)

83519 CCPQ (if appropriate)

84182 PC1BS (if appropriate)

84182 PCTBS (if appropriate)

86255 PCABP (if appropriate)

86255 PCAB2 (if appropriate)

86255 PCATR (if appropriate)

86255 PDEIS (if appropriate)

86255 SP5CS (if appropriate)

86255 SP5IS (if appropriate)

86255 SP7CS (if appropriate)

86255 SP7IS (if appropriate)

86255 SRPIS (if appropriate)

84182 SRPBS (if appropriate)

86255 T46CS (if appropriate)

86255 T46IS (if appropriate

PNEFC
84299

Neuroimmunology Antibody Follow-up, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Vial number 1

Acceptable: Any vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 AK5CC (if appropriate)

86255 AK5IC (if appropriate)

84182 AGNBC (if appropriate)

86255 AINCC (if appropriate)

86255 AMPCC (if appropriate)

86255 AMPHC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

84182 AN2BC (if appropriate)

86255 AGN1C (if appropriate)

86255 ANN1C (if appropriate)

86255 ANN2C (if appropriate)

86255 ANN3C (if appropriate)

86255 APBCC (if appropriate)

86255 APBIC (if appropriate)

86255 CS2CC (if appropriate)

84182 CRMWC (if appropriate)

86255 CRMC (if appropriate)

86255 DPPCC (if appropriate)

86255 DPPIC (if appropriate)

86255 GABCC (if appropriate)

86255 GFACC (if appropriate)

86255 GFAIC (if appropriate)

86255 GRFCC (if appropriate)

86255 GRFIC (if appropriate)

86255 IG5CC (if appropriate)

86255 IG5IC (if appropriate)

86255 ITPCC (if appropriate)

86255 ITPIC (if appropriate)

86255 LG1CC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86255 GL1IC (if appropriate)

86255 GL5CC (if appropriate)

86255 GL5IC (if appropriate)

86255 NCDCC (if appropriate)

86255 NCDIC (if appropriate)

86255 NFHCC (if appropriate)

86255 NIFIC (if appropriate)

86255 NFLCC (if appropriate)

86255 NMDCC (if appropriate)

84182 PC1BC (if appropriate)

84182 PCTBC (if appropriate)

86255 PCTRC (if appropriate)

86255 PCA1C (if appropriate)

86255 PCA2C (if appropriate)

86255 PDEIC (if appropriate)

86255 SP5CC (if appropriate)

86255 SP5IC (if appropriate)

86255 SP7CC (if appropriate)

86255 SP7IC (if appropriate)

86255 T46CC (if appropriate)

86255 T46IC (if appropriate)

NMOFS 38324 Neuromyelitis Optica (NMO)/Aquaporin-4-IgG Fluorescence-Activated Cell Sorting (FACS) Assay, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86053

86053-titer (if appropriate)

NMOFC 38325 Neuromyelitis Optica (NMO)/Aquaporin-4-IgG Fluorescence-Activated Cell Sorting (FACS) Assay, Spinal Fluid

Specimen Requirements:

Collection Container/Tube: Sterile vial

Specimen Volume: 3 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86053

86053-titer (if appropriate)

NSESF
81796

Neuron-Specific Enolase (NSE), Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	15 days
	Ambient	72 hours

CPT Code Information:

83520

NSEI
70630

Neuron-Specific Enolase Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NSE
80913

Neuron-Specific Enolase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. **Specimens should not be transported by tube system prior to centrifugation**.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Ambient	5 days

CPT Code Information:

83520

NCLGP
608014

Neuronal Ceroid Lipofuscinosis (Batten Disease) Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

NCLBS 616838 Neuronal Ceroid Lipofuscinosis, Two-Enzyme Panel, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete, ie, unpunched.

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood Spots: 1; Whole Blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

NCLW 616839 Neuronal Ceroid Lipofuscinosis, Two-Enzyme Panel, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

NEUD1 622267 Neuronal Differentiation 1 (NeuroD1), Technical Component only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

NSAIP 621136 Neurosyphilis IgG Antibody Index with VDRL, Serum and Spinal Fluid

Specimen Requirements:

Both spinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within a maximum of 24 hours of each other.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. The spinal fluid (CSF) specimen **must be** collected within 24 hours of the serum specimen, preferably at the same time.

2. The CSF aliquot should be from the second, third, or fourth CSF vial collected during the lumbar puncture. **Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection**.

3. Label vial as spinal fluid or CSF.

4. Band CSF specimen together with the serum sample.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL Serum

Collection Instructions:

1. Within 24 hours of collection of the spinal fluid specimen, a serum specimen **must**

also be collected, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label tube as serum.

4. Band serum specimen together with the CSF sample.

Specimen Minimum Volume: Spinal fluid/Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	10 days
	Frozen	10 days
Serum	Refrigerated (preferred)	10 days
	Frozen	10 days

CPT Code Information:

86780

82784 x 2-(if appropriate)

82040-(if appropriate)

86780 x 2-(if appropriate)

82042-(if appropriate))

86593-(if appropriate)

NSAI 621141 Neurosyphilis IgG, Antibody Index, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NSAIP / Neurosyphilis IgG Antibody Index with VDRL, Serum and Spinal Fluid.

Both spinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within a maximum of 24 hours of each other.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. The CSF specimen **must be** collected within 24 hours of the serum specimen, preferably at the same time.

2. The CSF aliquot should be from the second, third, or fourth CSF vial collected during the lumbar puncture. **Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection**.

3. Label vial as spinal fluid or CSF.

4. Band CSF specimen together with the serum sample.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL Serum

Collection Instructions:

1. Within 24 hours of collecting the spinal fluid specimen, a serum specimen **must also be** collected, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label tube as serum.

4. Band serum specimen together with the CSF sample.

Specimen Minimum Volume: Spinal fluid/Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	10 days
	Frozen	10 days

CPT Code Information:

82784 x 2

82040

86780 x 2

82042

FNEU2 Neurotensin
75905

Specimen Requirements:

Patient preparation:

1. Fasting: 10 to 12 hours

2. Patient should not be on any antacid medication or medications that affect gastroentero-intestinal function, if possible, for at least 48 hours prior to collection.

Supplies: Z-Tube Preservative Inter Science Institute (T701)

Collection Container/Tube: Z-tube

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL plasma

Collection Instructions:

1. Draw 10 mL of blood directly into special Z-tube

2. As soon as possible, centrifuge and aliquot plasma into a plastic vial.

3. Freeze immediately after separation.

4. Ship frozen.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Z-Tube Plasma	Frozen	180 days

CPT Code Information:

83519

NADF
35312

Newborn Aneuploidy Detection, FISH, Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot**.

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

4. Cord blood is acceptable.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

NIU
8626

Nickel, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic aliquot container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 0.55 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83885

NIS
8622

Nickel, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

Metal Free Specimen Vial (T173)

Collection Container/Tube: Royal blue-top (metal-free, no additive)

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, vial

Specimen Volume: 0.55 mL Serum

Collection Instructions:

1. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a Mayo metal-free, screw-capped vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	7 days	METAL FREE
	Ambient	7 days	METAL FREE
	Frozen	7 days	METAL FREE

CPT Code Information:

83885

NIUCR
614552

Nickel/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 2.55 mL

Collection Instructions:

1. Collect a random urine specimen

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83885

82570

NIUC 614553 Nickel/Creatinine Ratio, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NIUCR / Nickel/Creatinine Ratio, Random, Urine.

Patient Preparation:

High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 2.55 mL

Collection Instructions:

1. Collect a random urine specimen

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

NICOU
82510

Nicotine and Metabolites, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

G0480

80323 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

NICOS
82509

Nicotine and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

G0480

80323 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

NITU
607705

Nitrogen, Total, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative. Refrigerated is the preferred preservation method.

Specimen Stability Information: Frozen 3 years

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	

CPT Code Information:

84999

NITF
607704

Nitrogen, Total, Feces

Specimen Requirements:

Patient Preparation: Laxatives and enemas should not be used during collection as barium and boric acid interfere with test procedure.

Supplies: Stool Containers - 24, 48, 72 Hour Kit (T291)

Container/Tube: Stool container; complies with shipping requirements, do not use other containers.

Specimen Volume: Entire collection (24, 48, 72, or 96 hours)

Collection Instructions:

1. All containers must be sent together.

2. Entire collection must contain at least 5 g of feces.

3. The number of containers sent should be indicated on the labels (ie, 1 of 4)

Specimen Stability Information: Frozen 3 years

Additional Information: Patient can store sample at refrigerate temperature during collection period.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84999

NKX3 606692 NKX3.1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SSF1 87294 Nocardia Stain, Varies

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 0.2 mL

Collection Instructions: Collect a raw specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87206

87176-Tissue processing (if appropriate)

NDSPC 113339 Non-Gynecologic Direct Smear (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88104

NTPPC 113337 Non-Gynecologic, ThinPrep (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88112

NSIP
31769

Non-Seasonal Inhalant Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 10

NSFIB
604200

Nonalcoholic Steatohepatitis (NASH)-FibroTest, Serum and Plasma

Specimen Requirements:

Both serum and plasma are required for this test.

Patient Preparation:

Fasting:12 hours, required

Specimen Type: Serum

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 4 mL

Collection Instructions:

1. Centrifuge and aliquot serum into an amber vial within 2 hours of collection.

2. Centrifuged serum must be light protected within 4 hours of collection. It is acceptable to draw the blood and then protect it from light after centrifugation as long as it's within 4 hours of collection.

3. Label specimen as serum.

Specimen Type: Plasma

Collection Container/Tube: Gray top (potassium oxalate/sodium fluoride)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot plasma into plastic vial.

2. Label specimen as plasma.

Specimen Minimum Volume: Serum: 2 mL
Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma NaFl-KOx	Refrigerated (preferred)	7 days	
	Frozen	14 days	

	Ambient	7 days	
Serum	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	14 days	LIGHT PROTECTED
	Ambient	24 hours	LIGHT PROTECTED

CPT Code Information:

0003M

NSRGG 617393 Noonan Syndrome and Related Conditions Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to

determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81442

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

LNORO 65170 Norovirus PCR, Molecular Detection, Feces

Specimen Requirements:

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium - either Cary-Blair, Para-Pak Culture and Sensitivity Media)

Specimen Volume: Representative portion of diarrheal fecal sample, 1 gram or 5 mL

Collection Instructions:

1. Collect fresh feces and place in preservative within 1 hour of collection.

2. Visibly formed feces are not consistent with Norovirus gastrointestinal disease and should not be submitted for testing.

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient	7 days

CPT Code Information:

87798 x 2

NEREG
31767

Northeast Regional Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 10

NOTRP
37119

Nortriptyline, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

NTC3Z 616563 NOTCH3 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese

Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FCDU4 75779 Novel Psychoactives, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

NR4A3
64940

NR4A3 (9q22.33) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be

rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-NA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

PBNP1 NT-Pro B-Type Natriuretic Peptide, Serum
615897

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes, aliquot the serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	7 days

CPT Code Information:

83880

NTRKM 619703 NTRK Genes Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81479

NMRLP 603839 Nuclear Magnetic Resonance Lipoprotein Profile, Serum

Specimen Requirements:

Patient Preparation:

1. **Fasting: 12 hours**, **required**; On night before examination, evening meal should be eaten before 6 p.m. and should contain no fatty foods.

2. Patient must not consume any alcohol for 24 hours before the specimen is collected.

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Allow isopropyl alcohol (from phlebotomy site prep) to dry thoroughly before venipuncture.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83704

NMITO 617090 Nuclear Mitochondrial Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will

be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or

equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81440

NPMFM 620654 Nucleophosmin (NPM1) Mutation Analysis, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Refrigerated (preferred)	72 hours
	Ambient	72 hours

CPT Code Information:

81310

NPM1Q Nucleophosmin (NPM1) Mutation Analysis, Varies

604418

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix whole blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	72 hours

CPT Code Information:

81310-*NPM1* (nucleophosmin) (eg, acute myeloid leukemia) gene analysis; exon 12 variants

NUT
70521

NUT Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NUT1F
63431

NUTM1 (15q14) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Container/Tube: Formalin-fixed, paraffin-embedded tumor tissue block

Specimen Type: Slides

Specimen Volume: 4 Consecutive, unstained, 5 micron-thick sections placed on positively charged slides and 1 hematoxylin and eosin-stained slide

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: 2 consecutive, unstained, 5 micron- thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274-w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

NMEG
82497

Nutmeg, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

NUTSP
31771

Nuts Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

FNGPG
57930

Nuts and Grains Panel IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 17

FOAKE
57999

Oak Live (Quercus virginiana) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FROE
57907

Oak Red (Quercus rubra) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

OAK
82673

Oak, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOATG
57576

Oat IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

OATS
82688

Oat, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OCT2
70522

OCT-2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OCT4
70523

OCT3/4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OCTO
82820

Octopus, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLNZ
91129

Olanzapine (Zyprexa)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.50 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

OLIG2 OLIG2 Immunostain, Technical Component Only
71357

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OLIGS Oligoclonal Banding, Serum (Non-Orderable)
2783

Specimen Requirements:

Only orderable as part of a profile. For more information see:

OLIG / Oligoclonal Banding, Serum and Spinal Fluid

MSP3 / Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83916

OLIG
8017

Oligoclonal Banding, Serum and Spinal Fluid

Specimen Requirements:

Both serum and spinal fluid are required. Spinal fluid must be obtained within 7 days of serum collection.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Label specimen as spinal fluid.

Specimen Minimum Volume: Serum, Spinal fluid: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83916 x 2

OLIGC
3484

Oligoclonal Banding, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information, see:

OLIG / Oligoclonal Banding, Serum and Spinal Fluid

MSP3 / Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83916

OLIGU
64889

Oligosaccharide Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 8 mL

Pediatric Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No added preservative.

3. Immediately freeze specimen.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	365 days
	Refrigerated	15 days
	Ambient	7 days

CPT Code Information:

84377

FOLBG
57671

Olive Black IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days

	Ambient	7 days

CPT Code Information:

86001

OLIV
82733

Olive Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OLIVF
86306

Olive-Food, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FONG
57636

Onion IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

ONIN
82806

Onion, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OPTMX
62736

Opiate Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain of Custody Meconium Kit (T653) includes the specimen containers, seals, and documentation required.

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Additional Information:

1. Specimen that arrives with a broken seal does not meet the chain of custody requirements.

2. The laboratory recommends sending chain-of-custody specimens by overnight shipment.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATM
84326

Opiate Confirmation, Meconium

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATS
623435

Opiate Panel, Free (Unconjugated), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

80356 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATX 62735 Opiates Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 5 mL

Collection Instructions: Collect urine specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 5 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

80362 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATU 8473 Opiates Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

80362 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FCDU1 Opiates, Umbilical Cord Tissue (Non-Orderable)
75776

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

FORNG Orange IgG
57632

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

ORNG Orange, IgE, Serum
82740

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ORCH
82907

Orchard Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OREG
82496

Oregano, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ORXNA
604230

Orexin-A/Hypocretin-1, Spinal Fluid

Specimen Requirements:

Patient Preparation: Patient **should not** have recently received radioisotopes, either therapeutically or diagnostically, due to potential assay interference.

Collection Container/Tube: Sterile vial

Submission Container/Tube: Plain vial with no additives

Specimen Volume: 1.5 mL

Pediatric Volume: 0.5 mL

Collection Instructions:

1. Obtain aliquot from second collection vial (preferred, not required).

2. Hemolyzed specimens will give false-positive results. Specimens should be centrifuged to remove any red cells prior to shipping.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	120 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

83519

OAU
80619

Organic Acids Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Pediatric: If insufficient collection volume, submit as much specimen as possible in a single container; the laboratory will determine if volume is sufficient for testing.

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	416 days
	Refrigerated	14 days

CPT Code Information:

83919

OAUS
610707

Organic Acids Screen, Urine Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Collection Container:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell filter paper

Specimen Volume: 1 Filter paper card soaked with urine, typically requires 2 to 3 mL of urine

Collection Instructions

1. Soak a filter paper card with urine, approximately 22 cm(2) in area (typically requires 2 to 3 mL of urine; exact measurement is not important as urine volume will be normalized to creatinine).

Note: Filter paper sample may be collected by dipping the card into a collection cup of urine. Avoid dilute urine if possible.

2. Let urine dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen (preferred)	28 days	FILTER PAPER
	Ambient	14 days	FILTER PAPER

	Refrigerated	14 days	FILTER PAPER

CPT Code Information:

83919

IDENT 9221 Organism Referred for Identification, Aerobic Bacteria

Specimen Requirements:

Preferred:

Specimen Type: Bacterial isolate swab

Supplies:

E-swab (T853)

Infectious Container, Large (T146)

Container/Tube:

Preferred: E-Swab collection and transport system

Acceptable: Flocked swab and 1-mL liquid Amies transport medium in 12 x 80 mm tube

Collection Instruction:

1. Perform isolation of infecting bacteria.

2. Utilize the flocked swab to obtain an adequate sample of pure cultured isolate. **Do not submit mixed cultures.**

3. Place swab into the transport system containing 1-mL liquid Amies transport medium.

4. If needed, break off end of swab and close the transport tube.

5. Place the transport system into the secondary infectious container for shipment.

6. Each isolate must be submitted under a separate order.

Note: For the following organisms, submit an agar slant or other appropriate media to ensure viability upon arrival to the laboratory; *Neisseria gonorrhoeae*, *Campylobacter* sp., *Helicobacter pylori* and any other fastidious organism.

Acceptable:

Specimen Type: Pure culture of organism from source cultured

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Collection Instructions:

1. Perform isolation of infecting bacteria.

2. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures.**

3. Place the agar slant or other appropriate media into the secondary infectious container for shipment.

4. **Each isolate must be submitted under a separate order.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87077-Organism Referred for Identification, Aerobic Bacteria

87077-Identification Commercial Kit (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147 x 3-Serologic Agglut Method 1 Ident (if appropriate)

87147-Serologic Agglut Method 2 Ident (if appropriate)

87147 x 4-Serologic Agglut Method 3 Ident (if appropriate)

87147 x 2-6 - Serologic Agglut Method 4 Ident (if appropriate)

87077-Identification Staphylococcus (if appropriate)

87077-Identification Streptococcus (if appropriate)

87798-Identification by PCR (if appropriate)

ANIDE
8114

Organism Referred for Identification, Anaerobic Bacteria

Specimen Requirements:

Specimen Type: Pure culture of organism from a source not normally colonized by anaerobes

Acceptable Sources: Abscesses, percutaneous transtracheal aspirates, sterile body fluids, suprapubic aspirations, or wounds

Supplies:

-Anaerobic Transport Tube (T588)

-Infectious Container, Large (T146)

Container/Tube:

Preferred: Anaerobic transport tube

Acceptable: Thioglycollate broth or any other suitable anaerobic transport system

Collection Instructions:

1. Perform isolation of infecting bacteria.

2. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87076-Organism ref for ID, anaerobic bact

87076-Id MALDI-TOF mass spec anaerobe (if appropriate)

87076-Anaerobe Ident (if appropriate)

87153-Anaerobe ident by sequencing (if appropriate)

87077-Identification commercial kit (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria identification (if appropriate)

87077-Additional identification procedure (if appropriate)

87077-Identification Staphylococcus (if appropriate)

87077-Identification Streptococcus (if appropriate)

87147 x 3-Serologic agglut method 1 ident (if appropriate)

87147-Serologic agglut method 2 ident (if appropriate)

87147 x 4-Serologic agglut method 3 ident (if appropriate)

87147 x 2-6-Serologic Agglut Method 4 Ident (if appropriate)

87153-Aerobe ident by sequencing (if appropriate)

87798-Identification by PCR (if appropriate)

OROT 8905 Orotic Acid, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative needed.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	90 days

CPT Code Information:

83921

Orris Root (Iris florentina) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

OTP
620710

Orthopedia Homeobox Protein (OTP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OPTU
614360

Orthostatic Protein, Timed Collection, Urine

Specimen Requirements:

Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Supplies: 2 Sarstedt 5 mL Aliquot Tube (T914)

Daytime Collection

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a 16-hour (daytime) urine specimen.

2. No preservative.

3. Invert well before taking 4-mL aliquot.

4. Do not over fill aliquot tube 4 mL at most.

5. Collect specimen per instructions in Orthostatic Protein Measurement 24-Hour Urine:

Collection Site Instructions (T546) in Special Instructions.

Nighttime (Supine) Collection

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect an 8-hour (nighttime) urine specimen.

2. No preservative.

3. Invert well before taking 4-mL aliquot at most.

Specimen Minimum Volume: 1 mL from 16-hour (daytime) urine collection/1 mL from 8-hour (nighttime) urine collection

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156 x 2

ORTHP 617766 Orthostatic Proteinuria, Random, Urine

Specimen Requirements:

Patient Preparation: Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. This is a nighttime (supine) collection.

2. Void immediately prior to lying down to sleep.

3. Collect an 8-hour (nighttime) urine specimen.

4. Upon awaking in the morning, the patient should immediately void into the specimen cup.

4. No preservative.

5. Invert well before taking 4-mL (maximum) aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

CRETR- 82570

PRCON- 84156

OSM24 Osmolality, 24 Hour, Urine

622326

Specimen Requirements:

Supplies:

-Diazolidinyl Urea (Germall), 5.0 mL (T822)

-Urine Tubes, 10 mL (T068)

Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours

2. Add 5 mL of diazolidinyl urea as preservative at start of collection or refrigerate specimen during and after collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83935

UOSMU Osmolality, Random, Urine
606520

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions: Collect a random urine specimen

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83935

UOSMS Osmolality, Serum
614057

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	24 hours

CPT Code Information:

83930

FRAG 9064 Osmotic Fragility, Erythrocytes

Specimen Requirements:

Both patient and shipping control whole blood specimens are required. For complete instructions, refer to Specimen Collection and Labeling Instructions for Osmotic Fragility Testing of Erythrocytes.

Shipping Control Specimen

Specimen Type: Whole blood (non-patient)

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a shipping control specimen from a healthy person unrelated to the patient at the same time (or within 4 hours) as the patient specimen.

Note: The shipping control specimen can be collected from a phlebotomist, volunteer or another heathy patient.

2. Clearly handwrite "CONTROL" on the outermost label of the tube.

3. Immediately refrigerate (or place on cold gel pack/small amount of wet ice) after collection.

4. Send shipping control in the original tube. **Do not aliquot**.

5. Keep the shipping control and the patient specimens **together**, either rubber banded or in a bag.

Additional Information:

1. **The shipping control and patient specimens must be handled identically** from the time of collection through receipt in the testing laboratory.

2. **If the shipping control is not sent with the patient specimen, test cancellation is likely.**

3. The shipping control specimen evaluates whether the patient result has been compromised by handling conditions such as temperature, motion, or other transportation interferences, as temperature and handling extremes can adversely impact the integrity of the specimen.

Patient Specimen

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect and label patient specimen.

2. Immediately refrigerate (or place on cold gel pack/small amount of wet ice) specimen after collection.

3. Send whole blood specimen in the original tube. **Do not aliquot**.

4. Keep the shipping control and the patient specimens together, either rubber banded or in a bag.

Additional Information: The shipping control and patient specimens must be handled identically from the time of collection through receipt in the testing laboratory.

Specimen Minimum Volume: Patient whole blood, shipping control: 2 mL each

Transport Temperature:

Specimen Type	Temperature	Time	
Control	Refrigerated	72 hours	PURPLE OR PINK TOP/EDTA
Whole Blood EDTA	Refrigerated	72 hours	

CPT Code Information:

85557

OSCAL 80579 Osteocalcin, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required

2. For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

83937

OIBFG 617407 Osteogenesis Imperfecta and Bone Fragility Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional information:

1. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated

Additional Information:

1. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Molecular Testing, Chorionic Villi/Products of Conception.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information: All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 x2

81408 x2

81479

81479 (if appropriate for government payers)

Ova and Parasite, Microscopy, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Bile

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) and/or slides

Specimen Volume: 4 mL EDTA and/or 1 slide (unstained preferred)

Collection Instructions:

1. Bone marrow and/or slides will be accepted for this test.

2. If submitting slides with EDTA tube, label and bag specimens together. Send refrigerate as one collection.

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Colonic washing

Container/Tube:

Preferred: ECOFIX preservative

Acceptable: 10% Buffered formalin AND zinc polyvinyl alcohol (Zn PVA) (one vial of each)

Specimen Volume: at least 2 mL

Collection Instructions:

1. Place specimen into preservative within 30 minutes of collection.

2. Follow instructions on the container as follows:

a. Colonic washings should be placed into ECOFIX preservative vial (or 10% formalin

and Zn-PVA vials) in a ratio of 1 part preservative:1 part specimen. Some preservative may need to be removed from the vial to maintain this ratio.

b. Place washings into vial, twist the cap tightly closed, and shake vigorously until the contents are well mixed.

c. Do not fill above the fill line indicated on the preservative container.

Specimen Stability Information: Ambient (preferred)/Refrigerate 5 days

Specimen Type: Duodenal aspirate

Preferred: ECOFIX preservative

Acceptable: 10% Buffered formalin AND Zn PVA (one vial of each)

Specimen Volume: at least 2 mL

Collection Instructions:

1. Place specimen into preservative within 30 minutes of collection.

2. Follow instructions on the container as follows:

a. Duodenal aspirate should be placed into ECOFIX preservative vial (or 10% formalin and Zn-PVA vials) in a ratio of 1 part preservative:1 part specimen. Some preservative may need to be removed from the vial to maintain this ratio.

b. Place washings into vial, twist the cap tightly closed, and shake vigorously until the contents are well mixed.

c. Do not fill above the fill line indicated on the preservative container.

Specimen Stability Information: Ambient (preferred)/Refrigerate 5 days

Specimen Type: Spinal fluid

Container/Tube: Sterile container

Specimen Volume: At least 1 mL

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Fluid, abscess, drainage material

Sources: Abdominal, ascites, brain, cyst (must specify anatomical location of cyst), liver, lymphatic, peritoneal, splenic

Container/Tube: Sterile container AND either ECOFIX preservative vial or one zinc (Zn)-PVA AND one 10% formalin vial

Specimen Volume: 15 mL

Collection Instructions:

1. Place half of collection into preservative (either ECOFIX or PVA and 10 % Formalin combo) in a ratio of 1 part preservative: 1 part specimen. Some preservative may need to be removed from the vial to maintain this ratio.

2. Label the preservative type on the side of each vial.

3. Place other half of collection in a sterile container.

3. Label both specimens, bag together, and send refrigerate as one collection.

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Respiratory

Source: Bronchial washing, bronchoalveolar lavage, sputum

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Tissue

Sources: Bladder, brain, colon, intestine, liver, lymph node, lung, muscle, rectal, spleen (must specify anatomical location of tissue)

Container/Tube: Sterile container

Specimen Volume: 5 to10 mm

Collection Instructions: Place specimen in sterile container with 1 to 2 drops of sterile saline to keep tissue moist.

Specimen Stability Information: Refrigerate 5 days

Specimen Minimum Volume: Respiratory specimens, spinal fluid, abscess, or drainage material: 0.5 mL
Tissue: 3 mm
Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	5 days

CPT Code Information:

87015-Concentration (any type), for infectious agents (if appropriate)

87209-Smear, primary source with interpretation; complex special stain (eg, trichrome, iron hematoxylin) for ova and parasites (If appropriate)

87210-Wet mount for infectious agents (if appropriate)

87207-Smear, primary source, with interpretation; special stain for inclusion bodies or intracellular parasites (if appropriate)

OPE 619736 Ova and Parasite, Travel History or Immunocompromised, Feces

Specimen Requirements:

Patient Preparation: Specimen collection should be delayed for 7 to 10 days after administration of barium, bismuth, kaolin, magnesia, castor oil or mineral oil, and 2 to 3 weeks after antibiotics have been given since these may interfere with identification of protozoa.

Specimen Type: Preserved feces (Unpreserved specimens are **not acceptable)**

Supplies: ECOFIX Stool Transport Vial (Kit) (T790)

Container/Tube:

Preferred: ECOFIX preservative

Acceptable: 10% Buffered formalin AND zinc polyvinyl alcohol (Zn PVA) (one vial of each)

Specimen Volume: Portion of stool: 2 g or 2 mL

Collection Instructions:

1. Follow instructions on the container as follows:

a. Place specimen into Ecofix preservative vial or one 10% formalin AND one Zn PVA preservative vial within 30 minutes of passage or collection. Refer to the fill line on the preservative vial. Do not fill above the line indicated on the container.

b. Mix the contents of the tube with the spoon, twist the cap to close tightly, and shake vigorously until the contents are well mixed.

Additional Information:

1. Stool placed in 10% buffered formalin can be accepted if accompanied by Zn PVA-preserved specimen; **10% buffered formalin-preserved specimens submitted without an accompanying Zn PVA-preserved specimen will be canceled**.

2. Specimen submitted in low viscosity (LV)-PVA, mercury PVA or copper (Cu)-PVA **will be rejected**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	21 days
	Refrigerated	21 days

CPT Code Information:

87177-Concentration (any type), for infectious agents

87209-Smear, primary source with interpretation; complex special stain (eg, trichrome, iron hematoxylin) for ova and parasites

OVAL
82826

Ovalbumin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

FOVAS 57836 Ovarian Antibody Screen with Reflex to Titer, IFA

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86255

86256 (if appropriate)

OVAPA 623946 Ovarian Malignancy Risk Assessment, Ova1Plus, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Information:

1. Allow the specimen to clot for at least 30 minutes.

2. Within 2 hours of collection, centrifuge the specimen.

3. For serum gel tubes, aliquot serum into a plastic vial or send centrifuged serum gel tube.

4. For red top tubes, aliquot serum into a plastic vial within 10 minutes of centrifugation.

Specimen Minimum Volume: Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	8 days
	Frozen	90 days

CPT Code Information:
81503

0003U

OVAWA 623945 Ovarian Mass Risk Monitoring, OvaWatch, Serum

Specimen Requirements:

Collection Container/Tube: Sarstedt Aliquot Tube, 5 mL (T914)

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Information:

1. Allow the specimen to clot for at least 30 minutes.

2. Within 2 hours of collection, centrifuge the specimen.

3. For serum gel tubes, aliquot serum into a plastic vial or send centrifuged serum gel tube.

4. For red top tubes, aliquot serum into a plastic vial within 10 minutes of centrifugation.

Specimen Minimum Volume: Serum 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	8 days

	Frozen	90 days

CPT Code Information:

0375U

OVMU
82825

Ovomucoid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

OXI
82679

Ox-Eye Daisy, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DOXA1 Oxalate Analysis, Hemodialysate
606473

Specimen Requirements:

Specimen Type: Dialysate fluid

Patient Preparation: Patient should avoid taking vitamin C supplements for 24 hours prior to dialysis

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions: Adjust the pH of the specimen to 2.5 to 3.0 with 6M Hydrochloric Acid

Additional Information: Nonacidified frozen hemodialysate delivered to the laboratory within 3 days from collection will be accepted and the following comment will be added to the result: **In nonacidified hemodialysate stored frozen, oxalate values may increase spontaneously**.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Dialysate Fluid	Frozen	14 days

CPT Code Information:

83945

OXU
606737

Oxalate, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: For 24 hours before, as well as during the collection process, patient should not take large doses (>2 g orally/24 hours) of vitamin C.

Supplies:

-Diazolidinyl Urea (Germall) 5.0 mL (T822)

-Sarstedt Aliquot Tube, 5mL (T914)

Container/Tube: Plastic tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Add 5 mL of diazolidinyl urea (Germall) as a preservative at start of collection or refrigerate specimen during and after collection.

2. Collect urine for 24 hours.

3. Mix container thoroughly and aliquot urine into plastic vial.

4. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 indicate bacterial contamination, and testing will be canceled. **Do not** attempt to adjust pH, as it will adversely affect results.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83945

POXA1
606472

Oxalate, Plasma

Specimen Requirements:

Any client who has never collected a specimen for this test should call 800-533-1710 or 507-266-5700 and ask for the Clinical Specialty Laboratory for more detailed instructions.

Patient Preparation:

1. Fasting: 12 hours, preferred but not required

2. For 24 hours before specimen collection, the patient should not take vitamin C supplements.

Specimen Type: Acidified plasma

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Place specimen on wet ice immediately.

2. Within 1 hour of collection, centrifuge for 10 minutes at 3500 rpm. Use of a 4 degrees C refrigerated centrifuge is optimal but not required.

3. Aliquot plasma into a plastic vial.

4. Adjust the pH of the plasma specimen to a pH of 2.3-2.7 with approximately 10 mcL concentrated (12M) hydrochloric acid (or 20 mcL of 6M HCl) per 1 mL plasma.

Additional Information: Nonacidified specimens can be accepted if the heparinized plasma is properly frozen. A disclaimer will be added in nonacidified plasma, "Sample was received nonacidified and frozen. In nonacidified samples oxalate values may increase spontaneously." Although there can be a small absolute increase of plasma oxalate in nonacidified specimens, this will not change the clinical interpretation.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Frozen	30 days

CPT Code Information:

83945

ROXUR Oxalate, Random, Urine
606747

Specimen Requirements:

Patient Preparation: Avoid taking large doses (>2 g orally/24 hours) of vitamin C prior to specimen collection.

Supplies: Urine Tubes, 10 mL tube (T068)

Container/Tube: 10-mL plastic tube or a clean, plastic container with no metal cap

Specimen Volume: 7 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 may indicate bacterial contamination, and testing will be cancelled. **Do not** attempt to adjust pH as it will adversely affect results.

Specimen Minimum Volume: 6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83945

82570

OXCO1 Oxalate, Random, Urine (Non-Orderable)
606748

Specimen Requirements:

Only orderable as part of a profile. For more information see ROXUR / Oxalate, Random, Urine.

Patient Preparation: Avoid taking large doses (>2 g orally/24 hours) of vitamin C prior to specimen collection.

Supplies: Urine Tubes, 10 mL tube (T068)

Container/Tube: 10-mL plastic tube or a clean, plastic container with no metal cap

Specimen Volume: 7 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 may indicate bacterial contamination, and testing will be cancelled. Do not attempt to adjust pH as it will adversely affect results.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83945

RAT11 Oxalate/Creatinine Ratio, Urine (Non-Orderable)

606751

Specimen Requirements:

Only orderable as part of a profile. For more information see ROXUR / Oxalate, Random, Urine.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

OXAZS Oxazepam, Serum

623016

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

OMHC 81030 Oxcarbazepine Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST is **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80183

OXYDS
623456

Oxycodone and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYSX
61727

Oxycodone Screen, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 20 mL

Collection Instructions: Collect a random specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

OXYSU
62623

Oxycodone Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5 mL tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

OXYCX 61728 Oxycodone with Metabolite Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 5 mL

Collection Instructions: Collect urine specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain of custody testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYCU 62616 Oxycodone with Metabolite Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYMS 623460 Oxymorphone and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYMU 62622 Oxymorphone Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYWB 113429 Oxysterols, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin) or yellow top (ACD B)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	72 hours
	Ambient	48 hours

CPT Code Information:

82542

OXYBS 63147 Oxysterols, Blood Spot

Specimen Requirements:

Supplies:

-Card-Blood Spot Collection (Filter Paper) (T493)

-Card-Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Blood Spot Collection Card (Filter Paper)

Acceptable: Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper,

Munktell filter paper, Postmortem Screening Card or collected with EDTA, sodium heparin, lithium heparin, or ACD B-containing devices

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Let blood dry completely on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

2. At least 1 spot should be complete, (ie, unpunched).

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	10 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Ambient	10 days	FILTER PAPER

CPT Code Information:

82542

OXNP 62988 Oxysterols, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin or lithium heparin), yellow top (ACD B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions:

1. Centrifuge at 4 degrees C.

2. Aliquot plasma into plastic vial. **Do not disturb the buffy coat layer**.

3. Send frozen.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	65 days

CPT Code Information:

82542

OYST
82883

Oyster, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

CCPQ 81185 P/Q-Type Calcium Channel Antibody, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MGLE / Myasthenia Gravis/Lambert-Eaton Myasthenic Syndrome Evaluation, Serum

PNEFS / Neuroimmunology Antibody Follow-Up, Serum

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86596

P16 70524 p16 (INK4a/CDKN2A) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

P40NA 70526 p40 + Napsin A Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

P40 p40 Immunostain, Technical Component Only
70527

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P504S P504S Immunostain, Technical Component Only
610458

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P53
70528

p53 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections

4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigered	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P57I 70529 p57 (KIP2/CDKN1C) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P62 70629 p62 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC,If additional IHC

P63 70530 p63 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

SQUI
82821

Pacific Squid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PANOX 62737 Pain Clinic Survey 10, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 30 mL

Collection Instructions: Collect a random specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 30 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 20 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

PN10X 62911 Pain Clinic Survey 10, Chain of Custody, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PANOX / Pain Clinic Survey 10, Chain of Custody, Random Urine.

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 30 mL

Collection Instructions: Collect a random specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 30 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 20 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

FPALI
75392

Paliperidone, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Serum

Collection Container/Tube: Red

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain, red-top tube(s). (Serum gel tube is not acceptable.) Spin down and send 1 mL of serum refrigerate in a preservative-free plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

80342

FPAN2 Pancreastatin, Plasma
75907

Specimen Requirements:

Patient Preparation:

1. Fasting: 10 hours, recommended; water, but no other liquid, may be taken as needed

2. Patient should not be on any medications that may influence insulin levels, if possible, for at least 48 hours prior to collection.

Collection Container/Tube: Z tube (T701)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL plasma

Collection Instructions:

1. Draw 10 mL of blood in a pre-chilled Z-tube.

2. Immediately after collection, centrifuge in a refrigerated centrifuge and aliquot 3 mL of plasma into a plastic vial.

3. Freeze immediately.

4. Send frozen.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Z-Tube Plasma	Frozen	60 days

CPT Code Information:

83519

ELASF
609492

Pancreatic Elastase, Feces

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a fresh random fecal specimen, no preservatives.

2. If specimen is sent refrigerate, send immediately after collection.

3. If specimen cannot be sent immediately, freeze specimen and send frozen.

Additional Information:

1. **Separate specimens must be submitted when multiple tests are ordered**, with the exception of CALPR / Calprotectin, Feces. **If only a single specimen is collected, it must be split prior to transport**.

2. Testing cannot be added on to a previously collected specimen.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	28 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

82653

HPP
8014

Pancreatic Polypeptide, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, required

2. The patient **should not** have recently received radioisotopes, either therapeutically or

diagnostically, due to potential assay interference. See Ordering Guidance.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Plasma

Collection Instructions:

1. Place specimen on wet ice and keep cold at all times following collection.

2. Centrifuge (refrigerated centrifuge is not required) and aliquot plasma into a plastic vial. Freeze immediately.

Specimen Minimum Volume: Plasma: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	90 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

83519

WESPR 621326 Panel to Whole Exome Sequencing Reflex Test, Varies

Specimen Requirements:

For most patients, a new specimen submission will not be required. Testing can be performed using stored DNA from the original whole exome sequencing test. To order testing on the stored specimen, see Additional Testing Requirements.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report.

Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81417-Patient only

81417, 81416-Patient and one family member comparator sample (duo) (as appropriate)

81417, 81416 x 2-Patient and two family member comparator samples (trio or non-traditional trio) (as appropriate)

81417, 81416 x 3-Patient and three family member comparator samples (quad) (as appropriate)

PAPY
82356

Papaya, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PAPR
82810

Paprika, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PFIB
601950

Parafibromin, Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PNPAB
61881

Paraneoplastic Pemphigus Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

86255

PVLE
607409

Paraneoplastic Vision Loss Evaluation, Serum

Specimen Requirements:

Patient Preparation:

1. For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication.

2. This test should not be requested for patients who have recently received radioisotopes, therapeutically or diagnostically, because of potential assay interference. The specific waiting period before specimen collection will depend on the isotope administered, the dose given, and the clearance rate in the individual patient. Specimens will be screened for radioactivity prior to analysis. Radioactive specimens received in the laboratory will be held 1 week and assayed if sufficiently decayed or canceled if radioactivity remains.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x1

84182 x1

84182 CRMWS (if appropriate)

86256 CRMTS (if appropriate)

PARID
9202

Parasite Identification, Varies

Specimen Requirements:

Specimen Type: Parasitic worms, insects, or mites

Container/Tube: Sterile container (10% formalin or 70% alcohol may be added if appropriate specimen type)

Specimen Volume: Entire specimen

Collection Instructions:

1. For scabies, submit skin scrapings on glass microscope slide. Cover with a clean slide and use a rubber band to hold the 2 slides together (do not tape). Place the slides in a clean, dry container for transport. If no slides are available at time of collection, the dry skin scraping can be submitted in a sterile container and will be placed on slide at time of examination. Skin scraping must be visible with the naked eye to be of a quantity sufficient for testing.

2. Submit whole worms and worm segments in 70% alcohol or formalin. Worms must be

clean of stool to be suitable for examination.

3. Submit arthropods (ticks, lice, nits, bed bugs, etc) in a clean, dry container. Do not wrap in tape, gauze, or tissue that might obscure them during examination.

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87168-Arthropod (if appropriate)

87169-Parasite (if appropriate)

87210 - Arthropod or Parasite (if appropriate)

PTH
70544

Parathyroid Hormone (PTH) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PTHFN 61526 Parathyroid Hormone, Fine-Needle Aspiration Biopsy (FNAB)-Needle Wash

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. Needle wash specimens for analysis should be collected in conjunction with cytology specimens.

2. Have saline available prior to start of procedure. Saline is the only acceptable solution for needle washings.

3. After each fine-needle aspiration biopsy (FNAB) has been collected and the material in the needle has been expelled onto a slide for cytologic analysis, attach the used FNAB needle to an empty syringe.

4. Draw between 0.10 mL and 0.25 mL of saline up through the needle until the saline starts to fill the hub of the needle or end of the syringe.

5. Expel this fluid back through the needle into a separate plastic aliquot tube. This is the needle washing used for analysis.

6. Repeat steps 2 through 4 for each needle pass of the same biopsied site and empty into the same tube, accumulating a total of 0.5 mL to 1.5 mL of fluid to send to the laboratory. (If more than 1 site is biopsied, see Additional Information)

7. Inspect specimen for visible blood or tissue contamination:

a. If bloody, centrifuge specimen and transfer supernatant to a new plastic aliquot tube (5-mL standard tube) to send to laboratory. The supernatant, not the cellular material, is used for analysis.

b. If specimen is clear, centrifugation is not necessary.

8. Freeze within 2 to 4 hours of collection.

Additional Information:

1. If more than 1 site is biopsied, each washing material should be submitted on a separate tube and under a different order number.

2. A minimum of 0.5 mL is required for testing; however, the total collection volume should not exceed 1.5 mL. Sample volumes outside these parameters may be rejected.

3. Do not send saline control. This test has been validated to rule-out saline matrix effect.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fine Needle Wash	Frozen (preferred)	30 days
	Refrigerated	4 hours

CPT Code Information:

83970

PTH2 Parathyroid Hormone, Serum
28379

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, preferred but not required

2. **For 12 hours before specimen collection,** patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

83970

PTHRP
81774

Parathyroid Hormone-Related Peptide, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Ice-cooled, lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Plasma

Collection Instructions:

1. Centrifuge specimen in a refrigerated centrifuge or in chilled centrifuge cups.

2. Aliquot plasma into plastic vial and freeze.

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	30 days

CPT Code Information:

82397

PPAP
52964

Parental Sample Prep for Prenatal Microarray Testing, Blood

Specimen Requirements:

This test requires 2 blood specimens: 1 sodium heparin and 1 EDTA

Specimen Type: Whole blood

Container/Tube: Green top (sodium heparin) **and** lavender top (EDTA)

Specimen Volume: 3 mL EDTA **and** 4 mL sodium heparin

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimens in original tubes. **Do not aliquot.**

Specimen Minimum Volume: 2 mL EDTA and 2 mL sodium heparin whole blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

This test ID contains no charge and serves as a way to correlate proband parental specimens. If additional testing is warranted, the appropriate tests will be added.

PCAB
83728

Parietal Cell Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

PJUD
82877

Parietaria judaica, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

POFF
82549

Parietaria officinalis, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PLINK 62139 Paroxysmal Nocturnal Hemoglobinuria, PI-Linked Antigen, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 2.6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow cytometry, RBC x 1

88184-Flow cytometry, WBC x 1

88185-Flow cytometry, additional marker (each), RBC x 1

88185-Flow cytometry, additional marker (each), WBC x 6

88188-Flow Cytometry Interpretation, 9-15 Markers x 1

FPRTF 57967 Parrot Australian (Budgerigar) Feathers IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FPARG 57686 Parsley IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PSLY
82765

Parsley, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PARVS
48395

Parvovirus B19 Antibodies, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86747 x 2

PARVG 48320 Parvovirus B19 Antibodies, IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PARVS / Parvovirus B19 Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86747

PARVM 48321 Parvovirus B19 Antibody, IgM, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PARVS / Parvovirus B19 Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86747

PARVN 48322 Parvovirus B19 Antibody, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PARVS / Parvovirus B19 Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

PARVP Parvovirus B19, Molecular Detection, PCR, Plasma

86337

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PARVO
83151

Parvovirus B19, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 0.5 mL

Collection Instructions:

1. Do not centrifuge.

2. Label specimen as amniotic fluid.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Do not centrifuge.

2. Label specimen as spinal fluid.

Specimen Type: Synovial fluid

Container/Tube: Sterile vial or lavender top (EDTA)

Specimen Volume: 0.5 mL

Collection Instructions: Label specimen as synovial fluid.

Alternate:

Specimen Type: Bone marrow

Container/Tube: Sterile container or lavender top (EDTA)

Specimen Volume: 0.5 mL

Collection Instructions: Label specimen as bone marrow.

Specimen Minimum Volume: Amniotic Fluid, Bone Marrow, and Spinal Fluid 0.3 mL; Synovial fluid 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PARVI
70532

Parvovirus Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PFRUT
82355

Passion Fruit, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CONFL
620379

Pathology Consultation

Specimen Requirements:

Additional Information:

All specimens and additional paperwork must be labeled with:

-Two patient identifiers (first and last name, date of birth, or medical record number)

-Specimen type and anatomic site

Supplies: MCF Pathology Packaging Kit (T994)

Specimen Type: Tissue

Container/Tube: Formalin fixed paraffin-embedded blocks, stained slides and/or unstained slides

Note: Submit only pertinent hematoxylin and eosin (H and E) and all special stains relevant to the diagnostic questions for the case. Include positively charged unstained slides and/or a formalin-fixed, paraffin-embedded tissue block if it is anticipated that additional stains or ancillary testing may be necessary.

Bone Marrow Case Specimen Requirements

For information on collecting specimens, see Bone Marrow Core Biopsy, Clot, and Aspirate Collection Guideline.

Specimen Type: Bone marrow aspirate

Slides: 5

Preferred: Freshly prepared slides made at the time of specimen collection

Acceptable: Slides made from EDTA specimen, within 2 hours of collection

Submission Container/Tube: Plastic slide holder

Specimen Type: Bone marrow aspirate clot

Container/Tube: Bone marrow aspirate clot embedded in paraffin block

Collection Instructions: Process and embed clot in paraffin

Specimen Type: Bone marrow core biopsy

Container/Tube: Fixed biopsy core embedded in paraffin block

Collection Instructions: Process and embed core in paraffin.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
MMLDRY	Ambient	

CPT Code Information:

88321 (if appropriate)

88323 (if appropriate)

88325 (if appropriate)

PATHC 70317 Pathology Consultation

Specimen Requirements:

Additional Information:

All specimens and additional paperwork must be labeled with:

-Two patient identifiers (first and last name, date of birth, or medical record number)

-Specimen type and anatomic site

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Tissue

Container/Tube: Formalin-fixed paraffin-embedded blocks, stained slides, and/or unstained slides

Note: Submit only pertinent hematoxylin and eosin (H and E) and all special stains relevant to the diagnostic questions for the case. Include positively charged unstained slides and/or a formalin-fixed, paraffin-embedded tissue block if it is anticipated that additional stains or ancillary testing may be necessary.

Bone Marrow Case Specimen Requirements

For information on collecting specimens, see Bone Marrow Core Biopsy, Clot, and Aspirate Collection Guideline.

Specimen Type: Bone marrow aspirate

Slides: 5

Preferred: Freshly prepared slides made at the time of specimen collection

Acceptable: Slides made from EDTA specimen, within 2 hours of collection

Submission Container/Tube: Plastic slide holder

Specimen Type: Bone marrow aspirate clot

Container/Tube: Bone marrow aspirate clot embedded in paraffin block

Collection Instructions: Process and embed clot in paraffin

Specimen Type: Bone marrow core biopsy

Container/Tube: Fixed biopsy core embedded in paraffin block

Collection Instructions: Process and embed core in paraffin.

Specimen Type: Blood

Slides: 2

Preferred: 2 Freshly prepared fingerstick slides

Acceptable: 2 Slides made from whole blood in EDTA, made within 8 hours of collection

Submission Container/Tube: Plastic slide holder

Transport Temperature:

Specimen Type	Temperature	Time
MMLDRY	Ambient	

CPT Code Information:

88321 (if appropriate)

88323 (if appropriate)

88325 (if appropriate)

PAX5 PAX-5 Immunostain, Technical Component Only
70533

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PAX2 PAX2 Immunostain, Technical Component Only
607795

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PAX8
70534

PAX8 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

PDGF 58102 PDGFB (22q13), Dermatofibrosarcoma Protuberans/Giant Cell Fibroblastoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods will be attempted but are less favorable for successful results

2. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Collection Instructions: Submit 2 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

FBEP
57935

Pea Black-Eyed/Cow Pea (Vigna sinensis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FPGNG 57654 Pea Green IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPEAC 57666 Peach IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PECH
82816

Peach, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPNTG 57537 Peanut IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPNG4 57571 Peanut IgG4

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PEANT 64756 Peanut, IgE with Reflex to Peanut Components, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86003

PEAN 82888 Peanut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86003

PEAR
82807

Pear, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPCFG 57688 Pecan Food IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PCANH 62600 Pecan Hickory, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

PEC
82880

Pecan-Food, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AMLFP
622390

Pediatric Acute Myeloid Leukemia Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x22, 88275x11, 88291 x1-FISH Probe, Analysis, Interpretation; 11 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

PAS38 Pediatric Allergy Screen 3 to 8 Years, Serum

83346

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.55 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 6

PAS3 Pediatric Allergy Screen

83345

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

PAS8
83347

Pediatric Allergy Screen >8 Years, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

PCDES
605129

Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86341

86363

86053

86255 x 12

86256 AMPIS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86255 GL5CS (if appropriate)

86256 GL5TS (if appropriate)

86363 MOGTS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86256 NMDIS (if appropriate)

86053 NMOTS (if appropriate)

84182 PCTBS (if appropriate)

86256 PCTTS (if appropriate)

PCDEC 605130 Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 13

86341

86053

86256 AMPIC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 GL5CC (if appropriate)

86256 GL5TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86256 NMDIC (if appropriate)

86053 NMOTC (if appropriate)

84182 PCTBC (if appropriate)

86256 PCTTC (if appropriate)

BALFP 622400 Pediatric B-Lymphoblastic Leukemia/Lymphoma Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x19, 88275 x9, 88291 x1-FISH Probe, Analysis, Interpretation; 9 probe sets

88271 x2, 88275 x1-FISH Probe, Analysis; each additional probe set (if appropriate)

TALFP 622420 Pediatric T-Lymphoblastic Leukemia/Lymphoma Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x14, 88275x7, 88291x1- FISH Probe, Analysis, Interpretation; 7 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

PBPO
82660

Penicillin G, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PENIV
82656

Penicillin V, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PENL
82913

Penicillium chrysogenum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PENTS
8239

Pentobarbital, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before the next scheduled dose.

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FPBPG
57657

Pepper Bell/Paprika IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPBLG
57645

Pepper Black IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPCYE
57538

Pepper Cayenne (Capsicum frutescens) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days

	Frozen	28 days

CPT Code Information:

86003

FPEPI
75916

Pepsinogen I (PG I)

Specimen Requirements:

Patient preparation:

1. Fasting: 10 to 12 hours, recommended

2. Antacids or other medications affecting stomach acidity or gastrointestinal motility should be discontinued, if possible, for at least 48 hours prior to collection.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. Centrifuge and aliquot 3 mL serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

83520

PERAM
609438

Perampanel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top (serum gel/SST is **not acceptable**)

Acceptable: None

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. For sustained-release formulations ONLY, draw blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

SAT 2503 Percent Saturation (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see SFEC / Iron and Total Iron-Binding Capacity, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	365 days

Perch Ocean IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SMPB
37406

Peripheral Blood Smear Review (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood Slide	Refrigerated		CARTRIDGE

CPT Code Information:

85060

PNBX
70598

Peripheral Nerve Pathology Consultation

Specimen Requirements:

Specimen Type: Nerve biopsy tissue, slides, or block

Supplies: Nerve Biopsy Collection Kit (T457)

Container/Tube: Nerve Biopsy Collection Kit

Collection Instructions: Prepare and transport specimen per Nerve Biopsy Specimen Preparation Instructions (T580).

Specimen Minimum Volume: 4.5 cm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Frozen	

CPT Code Information:

88305 (if appropriate)

88313 (if appropriate)

88321 (if appropriate)

88323 (if appropriate)

88323-26 (if appropriate)

88325 (if appropriate)

88362 (if appropriate)

88348 (if appropriate)

88342 (if appropriate)

88341 (if appropriate)

DMNES 621103 Peripheral Nervous System Demyelinating Neuropathy, Autoimmune Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended to occur before starting immunosuppressant medication or intravenous immunoglobulin (IVIg) treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516 x5

83520

86255 x2

83520 x4 (if appropriate)

ACASM 83632 Pernicious Anemia Cascade, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required

2. For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

3. For 1 week before specimen collection, if medically feasible, patient **should not** take proton pump inhibitors (omeprazole, lansoprazole, dexlansoprazole, esomeprazole, pantoprazole, and rabeprazole).

4. For at least 2 weeks before specimen collection, patient **should not** take or receive drugs that interfere with gastrointestinal motility (eg, opioids).

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. Centrifuge, divide specimen into 3 plastic vials:

Vial 1 (B12PA): 1 mL of serum

Vial 2 (PAMMA): 1.5 mL of serum

Vial 3 (PAGAS): 0.5 mL of serum

2. Band specimens together and send frozen.

Specimen Minimum Volume: Serum: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	24 hours

CPT Code Information:

82607

82941-if appropriate

83921-if appropriate

86340-if appropriate

PDGP
608013

Peroxisomal Disorder Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable,

specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24

hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

PNZN 9789 Perphenazine, Serum or Plasma

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Plasma

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Green-top (sodium heparin) (Plasma gel/PST are **not** acceptable)

Submission Container/Tube: Amber plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a green-top (sodium heparin) tube.

2. Centrifuge and aliquot 3 mL plasma into an amber vial to protect from light and refrigerate immediately.

3. Ship refrigerated.

Specimen Type: Serum

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Red top (Serum gel/SST are **no**t acceptable)

Submission Container/Tube: Amber Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a plain red-top tube.

2. Centrifuge and aliquot 3 mL of serum into an amber vial to protect from light and refrigerate immediately.

3. Ship refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	180 days	LIGHT PROTECTED
	Ambient	72 hours	LIGHT PROTECTED

CPT Code Information:

80299

PERS 82353 Persimmon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

STK1Z 614588 Peutz-Jeghers Syndrome, STK11, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole Blood 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

UPH24 pH, 24 Hour, Urine
606521

Specimen Requirements:

Supplies:

Diazolidinyl Urea (Germall), 5.0 mL (T822)

Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours

2. Add 5 mL of diazolidinyl urea as preservative at start of collection **or** refrigerate specimen during and after collection.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH >8 indicate bacterial contamination, and testing will be cancelled. Do not attempt to adjust pH as it will adversely affect results.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83986

UPHB
606522

pH, Body Fluid

Specimen Requirements:

Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	24 hours

CPT Code Information:

83986

FPHFL
57309

pH, Fecal

Specimen Requirements:

Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a random unpreserved fecal specimen

2. Send frozen

Note: **Separate specimens must be submitted when multiple tests are ordered**.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

83986

PHU_ 606510 pH, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collections Instructions: Collect a random urine specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83986

FPHAS 57580 Phadiatop (Allergy Screen)

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

86005

PHCGD 621575 Phagocytic Disorders and Chronic Granulomatous Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions.

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81223

81249

81479

81479 (if appropriate for government payers)

PCPMX 62740 Phencyclidine (PCP) Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain of Custody Meconium Kit (T653)

Container/Tube: **Chain of custody kit containing the specimen containers, seals, and documentation required**

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen in the container provided. Seal and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Specimen that arrives with a broken seal does not meet the chain-of-custody requirements.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPMC 89069 Phencyclidine (PCP) Confirmation, Meconium

Specimen Requirements:

Supplies: Stool container. Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium (newborn's first bowel movements) specimen.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPX 62739 Phencyclidine Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and required documentation

Specimen Volume: 10 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPU
80371

Phencyclidine Confirmation, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container Tube: Plastic, 10 mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not** accepted for this test.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPS
624109

Phencyclidine Confirmation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FCDUA
75785

Phencyclidine, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

PBR
37049

Phenobarbital, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80184

FPGT
91757

Phenosense Combination HIV Drug Resistance Assay

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volume: 3 mL

Collection Instructions: Draw blood in either PPT (pearl top) or lavender-top (EDTA) tubes. Centrifuge specimen within six hours of collection. Remove plasma from cells immediately, and transfer specimen to a screw-capped, plastic vial. Freeze 3 mL of PPT plasma or EDTA plasma immediately, send specimen frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load.

2. Viral load collection date.

NOTE: 1. Intended for use only for patients with viral loads greater than or equal to 500 copies/mL. For best results, viral loads should be confirmed within 2 weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be cancelled.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87900/Infectious agent drug susceptibility phenotype prediction

87901/Infectious agent genotype analysis by nucleic acid; reverse transcriptase and protease

87903/Infectious agent phenotype analysis by nucleic acid with drug resistance tissue culture analysis; first through 10 drugs tested

87904/x12 Each additional drug tested

FPFUZ Phenosense Entry HIV Drug Resistance Assay

91755

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volume: 3 mL

Collection Instructions: Draw blood into **two** 5-mL PPT (pearl top) or EDTA (lavender top) tube. Centrifuge specimen within six hours of collection. Transfer plasma to one or more polypropylene screw-capped tube(s) and freeze. Freeze immediately. Send 3 mL plasma in a screw-cap vial frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load

2. Viral load collection date

NOTE:

1. Intended for use only for patients with viral loads greater than or equal to 500 copies/mL. For best results, viral loads should be confirmed within two weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be canceled.

Specimen Minimum Volume: 1.0 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87903

FPHIV 91756 Phenosense HIV Drug Resistance Replication Capacity

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volumes: 3 mL

Collection Instructions: Draw blood into **two** 5-mL PPT (pearl top) or EDTA (lavender top) tubes. Centrifuge specimen within 6 hours of collection. Transfer plasma to one or more polypropylene screw-capped tube(s) and freeze. Freeze immediately. Send 3 mL plasma in a screw-cap vial frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load.

2. Viral load collection date.

NOTE:

1. Intended for use only for patients with viral loads greater than or equal to 500 copies/mL. For best results, viral loads should be confirmed within two weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be canceled.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87903

87904 x12

PKUBS 65593 Phenylalanine and Tyrosine, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 90 days/Refrigerated 90 days/Frozen 90 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate 6 days

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

84030

84510

82542 (if appropriate for government payers)

PKU
8380

Phenylalanine and Tyrosine, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required; Infants should have specimen collected before next feeding

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Green top (lithium heparin), lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge and aliquot plasma into a plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

84030 Phenylalanine

84510 Tyrosine

82542 (if appropriate for government payers)

PKUSC
610508

Phenylalanine and Tyrosine, Self-Collect, Blood Spot

Specimen Requirements:

Container/Tube: Blood Spot Card

Specimen Volume: 2 Blood spots

Additional Information:

1. Order test each time the patient is to collect a dried blood specimen at home and mail the specimen directly to Mayo Clinic Laboratories.

2. Order should be placed a minimum of 3 days prior to desired date of collection.

3. Enter patient's address information for each order created, including street address (**post office [PO] boxes are not acceptable delivery locations**), city, state abbreviation, zip code, country, and home phone number.

4. For each order, the Blood Spot Collection-Self Collect kit will be mailed directly to the patient for self-collection (**delivery to a PO box will not occur**).

5. For more information on how to collect blood spots, see the following:

-How to Collect a Dried Blood Spot Sample via fingerstick.

-Blood Spot Collection Instructions-Fingerstick

-Blood Spot Collection Instructions-Fingerstick-Spanish

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	90 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Refrigerated	90 days	FILTER PAPER

CPT Code Information:

0382U

PHEGP 608032 Phenylalanine Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test

(including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81406 x 2

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

PNYF
37052

Phenytoin, Free, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days

	Ambient	7 days

CPT Code Information:

80186

PNTFT 37051 Phenytoin, Total and Free, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

Phenytoin, total-80185

Phenytoin, free-80186

PNYG 37050 Phenytoin, Total and Phenobarbital Group, Serum

Specimen Requirements:

One serum specimen (0.5 mL of serum) may be sent if using a red top tube.

Serum for Phenytoin:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Serum for Phenobarbital:

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL for 2 specimens; 0.25 mL for 1 serum red top

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

80184-Phenobarbital

80185-Phenytoin, total

PNYA
37048

Phenytoin, Total, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

80185

PHMA
82736

Phoma betae, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PTEN 614126 Phosphatase and Tensin Homolog (PTEN) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PETH 617480 Phosphatidylethanol Confirmation, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Do not use alcohol to clean arm. Use alternative (such as Betadine, also known as Povidone Iodine) to cleanse site before specimen collection.

2. Draw blood in tubes smaller than 10 mL if possible.

3. Do not centrifuge.

4. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Frozen (preferred)	28 days
	Refrigerated	14 days

CPT Code Information:

G0480

80321 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PSPTG
62578

Phosphatidylserine/Prothrombin Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86148

PT217 621635 Phospho-Tau 217, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic screw-top vial

Specimen Volume: 0.6 mL plasma

Collection Information: Centrifuge and aliquot plasma into plastic vial. **Do not submit in original tube.**

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
EDTA Plasma	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

84393

PDETS 620070 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PDETC 620069 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PDEIS 620068 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

PDEIC 620067 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

PETNU 623429 Phosphoethanolamine, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Clean plastic urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	70 days
	Refrigerated	14 days

CPT Code Information:

82131

PFK1
607456

Phosphofructokinase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA) or yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood ACD-B	Refrigerated	11 days

CPT Code Information:

82657

PFKC 608422 Phosphofructokinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	11 days

CPT Code Information:

82657

PGK1 607457 Phosphoglycerate Kinase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA) or yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

PGKC 608423 Phosphoglycerate Kinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

PLAIF 70592 Phospholipase A2 Receptor (PLA2R), Renal Biopsy

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Unstained slides (unfixed)

Source: Kidney tissue

Slides: 2 Slides

Collection Instructions: 2 frozen tissue unstained positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick, **centered on the slide,** and

submitted on dry ice.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Acceptable:

Specimen Type: Unfixed tissue block (O.C.T)

Source: Kidney tissue

Specimen Volume: Entire specimen

Collection Instructions:

1. Embed in O.C.T. compound.

2. Freeze specimen and ship on dry ice.

Acceptable:

Specimen Type: Wet tissue

Source: Kidney tissue

Supplies: Renal Biopsy Kit (T231)

Container/Tube: Renal Biopsy Kit, Zeus/Michel's

Specimen Volume: Entire specimen

Collection Instructions: Collect specimens according to the instructions in Renal Biopsy Preparation Instructions.

Additional Information: If standard immunoglobulin and complement immunofluorescence have already been performed, submit the residual frozen tissue (must contain glomeruli) on dry ice.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Frozen	

CPT Code Information:

88346-primary IF

EURO 606475 Phospholipase A2 Receptor, Enzyme Linked Immunosorbent Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PMND1 / Primary Membranous Nephropathy Diagnostic Cascade, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial within 2 hours of collection

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83520

PLA2I 607367 Phospholipase A2 Receptor, Immunofluorescence, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

86255

PLA2M
607366

Phospholipase A2 Receptor, Monitoring, Enzyme-Linked Immunosorbent Assay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83520

ACLIP
86179

Phospholipid (Cardiolipin) Antibodies, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147

CLPMG
82976

Phospholipid (Cardiolipin) Antibodies, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147 x 2

GCLIP 80993 Phospholipid (Cardiolipin) Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147

MCLIP 81900 Phospholipid (Cardiolipin) Antibodies, IgM, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147

PMMIL 89656 Phosphomannomutase and Phosphomannose Isomerase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

82657

PHOS
8408

Phosphorus (Inorganic), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	7 days

CPT Code Information:

84100

PHOSS
622796

Phosphorus (Inorganic), Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

-RTRP2 / Tubular Reabsorption of Phosphorus, Random Urine and Serum

-TRPP / Tubular Reabsorption of Phosphorus, Serum/Plasma/Urine

Patient Preparation: Fasting 12 hours, required

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

POU
610832

Phosphorus, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-Hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84105

RPHOC
610829

Phosphorus, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84105

STAT3 622778 Phosphorylated STAT3 at Tyrosine 705 (pSTAT3-Y705), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PHTDP 71482 Phosphorylated TDP43 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

PAHD
82786

Phthalic Anhydride, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CVSPC
72163

Physician Interpretation Conventional, Varies

Specimen Requirements:

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		SLIDE
	Refrigerated		SLIDE

CPT Code Information:

88141

TPSPC
72162

Physician Interpretation Screen, Varies

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		THIN PREP
	Refrigerated		THIN PREP

CPT Code Information:

88141

TPDPC
72129

Physician Interpretation, Diagnostic, Varies

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		THIN PREP
	Refrigerated		THIN PREP

CPT Code Information:

88141

PIGE
82781

Pig Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPIGF
75555

Pigeon Feathers IgG

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PIGF
82145

Pigeon Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPDD
75548

Pigeon/Dove Droppings Gel Diffusion

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86331

PINE
82381

Pine Nut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPINP
75410

Pine Ponderosa IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FPIAP
57670

Pineapple IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PNAP
82815

Pineapple, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PINW
9204

Pinworm Exam, Perianal

Specimen Requirements:

Specimen Type: Perianal

Container/Tube:

Preferred: ParaClick Pinworm Paddle

Acceptable: SWUBE disposable paddle or similar method of collection

Specimen Volume: Entire specimen

Collection Instructions: See Pinworm Collection Instructions

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

87172

PIPU
81248

Pipecolic Acid, Random, Urine

Specimen Requirements:

Supplies: Urine tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	94 days
	Refrigerated	14 days

CPT Code Information:

82542

PIPA
81326

Pipecolic Acid, Serum

Specimen Requirements:

Patient Preparation:

Fasting 12 hours, required; Infants and small children should have specimen collected before next feeding/meal

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	94 days
	Refrigerated	14 days

CPT Code Information:

82542

PISTA
82808

Pistachio, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PIT1
72124

PIT-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PKLRZ
610058

PKLR Full Gene Analysis, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

PLAP
70539

Placental Alkaline Phosphatase (PLAP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PLAI
82837

Plaice, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PBLI
9302

Plasma Cell Assessment, Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA)

Specimen Volume: 10 mL

Collection Instructions:

1. **Do not centrifuge.**

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow cytometry, cell surface, cytoplasmic

88185 x 5-Each additional marker

88187-Flow cytometry, interpretation; 2 to 8 markers

PCPRO 61654 Plasma Cell DNA Content and Proliferation, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 4 mL

Specimen Stability Information: 3 days

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88182-Flow cytometry, cell cycle or DNA analysis

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow cytometry interpretation, 2 to 8 Markers (added as FCINT)

CSPMM 618626 Plasma Cell Myeloma Pre-Analysis Cell Sorting, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see NGPCM / MayoComplete Plasma Cell Myeloma, Next-Generation Sequencing, Varies.

Container/Tube: Lavender or pink top (EDTA) or yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Minimum plasma cell percentage is 1%.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

4. Label specimen as bone marrow.

5. **Fresh specimen is required for this test**, as testing is performed on sorted cells.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient	4 days

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow Cytometry, additional cell surface, cytoplasmic or nuclear marker (each)

PLASF 35293 Plasma Cell Proliferative Disorder, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions: Submit 10 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 7 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

PCPDS 606079 Plasma Cell Proliferative Disorder, High Risk with Reflex Probes, Diagnostic FISH Evaluation, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88271 x 2, 88274, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271 x 2, 88274-FISH Probe, Analysis; each additional probe set (if appropriate)

CSPCF 607625 Plasma Cell Proliferative Disorder, Pre-Analysis Cell Sorting, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see PCPDS / Plasma Cell Proliferative Disorder, High Risk with Reflex Probes, Diagnostic FISH Evaluation, Bone Marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow Cytometry, additional cell surface, cytoplasmic or nuclear marker (each)

PLHBB 9096 Plasma Free Hemoglobin, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

2. Results could be falsely elevated due to artifactual red blood cell lysis if not centrifuged within 2 hours of collection.

Specimen Minimum Volume: Plasma: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	20 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

83051

PGRBC
609675

Plasmalogens, Blood

Specimen Requirements:

Patient Preparation: Specimen must be collected either prior to or 6 weeks after a blood transfusion.

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin), yellow top (ACD solution A or B)

Specimen Volume: 5 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Ambient	14 days

CPT Code Information:

82542

PGDBS 609664 Plasmalogens, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Patient Preparation: Specimen must be collected either prior to or 6 weeks after a blood transfusion.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood spot collection card

Acceptable: Local newborn screening card, PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, or blood collected in tubes containing ACD, EDTA, or heparin spotted and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 90 days/Refrigerated 90 days/Frozen 90 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Acceptable

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin), yellow top (ACD solution A or B)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not** aliquot.

Specimen Stability Information: Refrigerate (preferred) 14 days/Ambient 11 days

Specimen Minimum Volume: Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82542

PAI1
86083

Plasminogen Activator Inhibitor Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, at-40 degrees C or below.

Specimen Stability Information: Frozen 2 years

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85415

PSGN 9079 Plasminogen Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and

Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85420

PLABN 35794 Platelet Antibody Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Additional Information: For testing neonates, draw blood **from the mother** of the neonate. Then follow the Collection Instructions prior to sending.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	365 days
	Refrigerated	48 hours

CPT Code Information:

86022

GNPLT 619285 Platelet Disorders, Comprehensive Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Blood: 1 mL; Cultured fibroblasts/skin biopsy: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

GNPFD 619355 Platelet Function Defect Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

Platelet Neutralization Procedure, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Specimen Stability Information: Frozen 2 years

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85597

GNSPD 619327 Platelet Storage Pool Deficiency Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710..

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

PLAFL 64278 Platelet Surface Glycoprotein, Flow Cytometry, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Collection Container/Tube: ACD solution A or B

Specimen Volume: 6 mL

Pediatric Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not** aliquot.

Specimen Minimum Volume: Adult: 1 mL; Pediatric: 200 mcL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Ambient	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each) X5

88187-Flow cytometry interpretation, 2 to 8 markers

PTEM 63682 Platelet Transmission Electron Microscopic Study, Whole Blood

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Collection Container/Tube:

Preferred: Yellow top (ACD, solution B)

Acceptable: Yellow top (ACD, solution A)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Ambient	72 hours

CPT Code Information:

85390

88348

FPLAT 75450 Platinum, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Draw blood in a plain, royal blue top tube(s). **(Serum gel tube is not acceptable.)** Spin down and send 1 mL of serum refrigerated in an acid washed (MCL Supply T619) or

trace metal-free plastic container.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	60 days
	Ambient	60 days
	Frozen	60 days

CPT Code Information:

83018

PLUM
82809

Plum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PMLR
84114

PML::RARA Quantitative, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top **(**EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Blood: 8 mL; Bone Marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA

	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81315-PML/RAR-alpha (t(15;17)), (PML-RARA regulated adaptor molecule 1) (eg, promyelocytic leukemia) translocation analysis; all breakpoints (eg, intron 3, intron 6 and variable in exon 6), qualitative or quantitative

PMPDD 66569 PMP22 Gene, Large Deletion/Duplication Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81324

PMS2 70540 PMS2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FPA15 Pneumococcal Antibody Panel - PCV15 (15-serotype)
75952

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Allow blood to clot for 30 to 60 minutes.

2. Centrifuge at ambient temperature, 1000 x g for 15 minutes.

3. Aliquot 1 mL of serum into a plastic vial.

4. Send refrigerated.

Specimen Minimum Volume: Serum: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	42 days
	Ambient	42 days
	Frozen	

CPT Code Information:

86581

PNRP 81698 Pneumocystis jiroveci, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Pneumocystis* species DNA is unlikely.

Preferred Specimens: Pleural, respiratory (eg, bronchoalveolar lavage [BAL], bronchial washing, sputum), or fresh tissue

Acceptable Specimens: If no fresh specimen is available, digested respiratory specimens treated with N-acetyl-L-cysteine-sodium hydroxide (NALC/NaOH) are acceptable (eg, BAL, bronchial washing, respiratory fluid, sputum, or tracheal secretion)

Submit only 1 of the following specimens:

Preferred

Specimen Type: Body fluid

Sources: Pleural

Container/Tube: Sterile container

Specimen Volume: 1 mL

Additional Information: Only fresh, non-NALC/NaOH-digested body fluid is acceptable.

Specimen Type: Respiratory

Sources: BAL, bronchial washing, tracheal secretions, or sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL if only PCR ordered or 3 mL if PCR ordered with smear and culture

Specimen Type: Tissue

Sources: Respiratory

Container/Tube: Sterile container

Specimen Volume: 5-10 mm

Collection Instructions:

1. Submit fresh tissue.

2. Keep tissue moist with sterile water or sterile saline

Acceptable

Specimen Type: NALC/NaOH-digested respiratory specimens

Sources: BAL, bronchial washing, respiratory fluid, sputum, or tracheal secretion

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Minimum Volume: Body fluid or nondigested respiratory specimen: 0.5 mL; Fresh tissue: 5 mm; NALC-NaOH-digested specimen: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87594

D240 70416 Podoplanin (D2-D40) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

POLET 619676 POLE Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81407

FPOLO 75165 Poliovirus (Types 1, 3) Antibodies, Neutralization

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Ship refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	5 days

CPT Code Information:

86382 x 2

FPOLE 57942 Pollock White (Pollachius virens) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

PVJAK 65116 Polycythemia Vera, JAK2 V617F with Reflex to JAK2 Exon 12-15, Sequencing for Erythrocytosis, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	5 days

CPT Code Information:

81270-JAK2 V617

0027U (if appropriate)

TALDO
61843

Polyols, Quantitative, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82542

FPOM
57918

Pomegranate (Punica granatum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

PD2T
65296

Pompe Disease Second-Tier Newborn Screening, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA and then spotted and dried on filter paper.

Specimen Volume: 3 Blood spots

Collection Instructions:

1. Completely fill at least 3 circles on the filter paper card (approximately 100 microliters blood per circle).

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	28 days

CPT Code Information:

83789

PDBS
602280

Pompe Disease, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA, and dried on filter paper

Specimen Volume: 3 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 3 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry completely on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	28 days

CPT Code Information:

83789

GAAN 622582 Pompe Disease, GAA Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. Do not aliquot.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a

minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen can only tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order

MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FPOPW 57557 Poplar White (Populus alba) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

POPSD
82632

Poppy Seed, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL; For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPORG
57627

Pork IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPRK4 Pork IgG4

57564

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PORK
82700

Pork, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PBALP
64661

Porphobilinogen and Aminolevulinic Acid, Plasma

Specimen Requirements:

Patient Preparation: Patient **must not** consume any alcohol for at least 24 hours prior to specimen collection.

Supplies: Amber Frosted Tube, 5mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Green top (lithium heparin), lavender top (EDTA), yellow top (ACD A or B)

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. It is recommended that specimen collection occur during the acute phase. Porphobilinogen and aminolevulinic acid may be normal when the patient is not exhibiting symptoms.

2. Centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma	Frozen (preferred)	21 days	LIGHT PROTECTED
	Refrigerated	7 days	LIGHT PROTECTED

CPT Code Information:

82542

82135

PBGDW 31894 Porphobilinogen Deaminase, Washed Erythrocytes

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as ethanol induces porphobilinogen deaminase activity, which may lead to a false-normal result.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. Record volume of packed cells and the total volume of the specimen.

4. Discard supernatant plasma.

5. Wash packed erythrocytes 2 times by resuspension of at least an equal amount of cold 0.9% saline, mix, and centrifuge for 5 minutes at 2000 rpm, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	14 days
	Refrigerated	14 days
	Ambient	48 hours

CPT Code Information:

82657

PBGD_ 88925 Porphobilinogen Deaminase, Whole Blood

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as ethanol induces porphobilinogen deaminase activity, which may lead to a false-normal result.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Specimen Volume: 4 mL; full tube

Collection Instructions: Refrigerate specimen as soon as possible.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	8 days
	Ambient	7 days

CPT Code Information:

82657

PBGU
82068

Porphobilinogen, Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: Patient **must not** consume any alcohol for at least 24 hours before specimen collection.

Supplies: Urine Container-Amber, 60 mL (T596)

Specimen Volume: 20 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative necessary but pH must be above 5.0. If pH is below 5.0, specimen will be rejected.

3. Specimens should be protected from light and frozen immediately following collection.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen (preferred)	7 days	LIGHT PROTECTED
	Refrigerated	7 days	LIGHT PROTECTED

CPT Code Information:

84110

PCGP
608023

Porphyria Comprehensive Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710**.**

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days /Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent Cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81406 x 2

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

PEWE
31893

Porphyrins Evaluation, Washed Erythrocytes

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Dark blue top (metal free heparin), green top (lithium heparin), or lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. Record volume of packed cells and the total volume of the specimen.

4. Discard supernatant plasma.

5. Wash packed erythrocytes 2 times by resuspension of at least an equal amount of cold 0.9% saline, mix, and centrifuge for 5 minutes at 2000 rpm, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

84311-Spectrophotometry, analyte not elsewhere specified

82542-Chromatography (if appropriate)

PEE
88886

Porphyrins Evaluation, Whole Blood

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA), dark blue top (metal free heparin), royal blue top (metal free EDTA), green top (lithium heparin)

Specimen Volume: 4 mL

Collection Instructions: Immediately place specimen on wet ice.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

84311

82542-if appropriate

FQPPS
81652

Porphyrins, Feces

Specimen Requirements:

Patient Preparation:

1. For 3 days before, as well as during, specimen collection, patient **must not eat red meat or take any aspirin-containing medications.**

2. For 24 hours before, as well as during, specimen collection, patient **must not use** barium, laxatives, or enemas.

Supplies: Stool Containers - 24, 48, 72 Hour Kit (T291)

Container/Tube: Stool containers

Specimen Volume: Entire collection (48, 72, or 96 hours). 24-Hour collection is adequate if the collection volume is at least 100 g

Collection Instructions:

1. Collect all stool specimens within a 24, 48, 72, or 96-hour timeframe.

2. Do not add preservative.

3. Send entire collection.

Additional Information: Specimens smaller than 100 g may not provide accurate results.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

84126

PQNU 8562 Porphyrins, Quantitative, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: For the 24 hours before, as well as during, specimen collection, patient **must not** consume any alcohol.

Supplies:

-Urine Container - Amber, 60-mL (T596)

-Sodium Carbonate, 5 gram (T272)

Container/Tube: 60-mL Amber urine container

Specimen Volume: 20 to 50 mL

Collection Instructions:

1. Add 5 g of sodium carbonate as preservative at start of collection. This preservative is intended to achieve a pH above 7. **Do not substitute sodium bicarbonate for sodium carbonate.**

2. Collect a 24-hour urine specimen.

3. The container should be refrigerated and protected from light as much as possible during collection.

4. Record volume and duration.

5. Aliquot 20 to 50 mL into an amber container and freeze.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen	7 days	LIGHT PROTECTED

CPT Code Information:

84110-Porphobilinogen, quantitative

84120-Porphyrins, quantitation and fractionation

PQNRU 60597 Porphyrins, Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: For at least 24 hours before specimen collection, patient **must not** consume any alcohol.

Supplies: Urine Container - Amber, 60 mL (T596)

Container/Tube: 60-mL Amber urine container

Specimen Volume: 20 to 50 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative3. Specimens should be protected from light and frozen immediately following collection.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen	72 hours	LIGHT PROTECTED

CPT Code Information:

84110-Porphobilinogen, quantitative

84120-Porphyrins, quantitation and fractionation

82135- ALA Delta Random Urine

PTP 8731 Porphyrins, Total, Plasma

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Lavender top (EDTA)

Submission Container/Tube: Amber vial

Specimen Volume: 3 mL Plasma

Collection Instructions: Centrifuge specimen and aliquot plasma into an amber vial.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma	Frozen (preferred)	14 days	LIGHT PROTECTED
	Refrigerated	14 days	LIGHT PROTECTED

CPT Code Information:

84311-Porphyrins, total

82542-Porphyrins, fractionation (if appropriate)

FPPOS 75889 Posaconazole Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Complete and submit with specimen:

1. Fungus Test Lab Request Form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999 - Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 - mould - MIC microdilution or agar dilution (if appropriate)

87186 - yeast - MIC microdilution or agar dilution (if appropriate)

POSA
89591

Posaconazole, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge, and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80187

POSV
9205

Post Vasectomy Check, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Container/Tube: Semen Analysis Kit (T178)

Specimen Volume: Total ejaculate

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89321

PMAOG
620568

Postmortem Aortopathy Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81410

PMARG 620582 Postmortem Arrhythmia Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81439

PMCAG 620596 Postmortem Cardiomyopathy and Arrhythmia Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81439

PMCMG 620610 Postmortem Cardiomyopathy Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81439

PCMSP 620638 Postmortem Inherited Congenital Myasthenia Syndrome Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block.

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81443

PMHLH 620624 Postmortem Primary Hemophagocytic Lymphohistiocytosis (HLH) Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81443

PMSBB 81931 Postmortem Screening, Bile and Blood Spot

Specimen Requirements:

Both bile and blood spots are required.

Supplies: Card-Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Card-Postmortem Screening Card (Filter Paper)

Acceptable: Whatman Protein Saver 903 paper or local newborn screening card

Specimen Volume: Properly completed screening card

Collection Instructions:

1. Collect blood in a heparin-containing tube and drop 25 mcL of blood onto the 2 circles labeled Blood.

2. Collect bile by direct puncture of the gallbladder and drop 25 mcL of bile onto the 2 circles labeled Bile.

3. Allow to dry at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Fill out information on page 2 of collection card.

5. Do not expose specimen to heat or direct sunlight.

6. Do not stack wet specimens.

7. Keep specimen dry.

Specimen Minimum Volume: Bile spot: 1; Blood spot: 1

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)		FILTER PAPER
	Frozen		FILTER PAPER
	Refrigerated		FILTER PAPER

CPT Code Information:

83789

KUR
614060

Potassium, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-Hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-Hour collection.

Additional Information:

See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84133

RKUR
610696

Potassium, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84133

KS
602352

Potassium, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	48 hours

CPT Code Information:

84132

FPTWG
57539

Potato White IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

POU2F POU2F3 Immunostain, Technical Component Only
618483

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FMPG Poultry and Meat Panel IgG
57931

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 7

POWV
616098

Powassan Virus, IgM, Enzyme-Linked Immunosorbent Assay, Serum

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

86790

POWVB 617484 Powassan Virus, RNA, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

POWVU 617487 Powassan Virus, RNA, Molecular Detection, PCR, Random, Urine

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect random urine in a sterile container.

2. Label specimen as urine.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

POWVS 617486 Powassan Virus, RNA, Molecular Detection, PCR, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 2 hours of collection centrifuge and aliquot serum into a sterile container.

2. Serum specimens not aliquoted from the serum gel collection tube into a sterile container will be rejected.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

POWVC 617485 Powassan Virus, RNA, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Send specimen from collection vial 2.

2. Do not centrifuge or heat inactivate.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PWAS 35535 Prader-Willi/Angelman Syndrome, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks is required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81331-*SNRPN/UBE3A*, (small nuclear ribonucleoprotein polypeptide Nand ubiquitin protein ligase E3A) (eg, Prader-Willi syndrome and/or Angelman syndrome), methylation analysis

88235-Tissue culture for amniotic fluid (if appropriate)

88240-Cryopreservation (if appropriate)

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells (if appropriate)

PRAME PRAME Immunostain, Technical Component Only

615794

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PALB Prealbumin, Serum

9005

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

84134

C2NAD
621651

PrecivityAD, Plasma

Specimen Requirements:

Supplies: Screw cap micro tube, 2 mL, PCR Performance Tested, Low protein-binding (T983)

Collection Container/Tube: 10 mL Purple top (K EDTA)

Submission Container/Tube: 2-mL screw cap micro tubes

Specimen Volume: 1.5 mL

Collection Instructions:

1. Centrifuge within two hours of collection.

2. Aliquot plasma into a 2 mL micro tube.

3. Freeze plasma (no longer than 2 hours after collection) at -20 degrees C or below.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	

CPT Code Information:

0412U

C2AD2 PrecivityAD2, Plasma
621652

Specimen Requirements:

Supplies: Screw cap micro tube, 2 mL, PCR Performance Tested, Low protein-binding (T983)

Collection Container/Tube: 10 mL Purple top (K EDTA)

Submission Container/Tube: Two 2-mL screw cap micro tubes

Specimen Volume: 3 mL in 2 tubes, each containing 1.5 mL

Collection Instructions:

1. Centrifuge within two hours of collection.

2. Label two 2-mL screw-cap micro tubes.

3. Aliquot 1.5 mL of plasma into each labeled micro tube.

4. Freeze plasma (no longer than 2 hours after collection) at or below -20 degrees C.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	

CPT Code Information:

0503U

AD2AR PrecivityAD2, Reflex to Apolipoprotein E, Plasma
621654

Specimen Requirements:

Supplies: Screw cap micro tube, 2 mL, PCR Performance Tested, Low protein-binding

(T983)

Collection Container/Tube: 10 mL Purple top (K EDTA)

Submission Container/Tube: Two 2-mL screw cap micro tubes

Specimen Volume: 3 mL in 2 tubes, each containing 1.5 mL

Collection Instructions:

1. Centrifuge within two hours of collection.

2. Label two 2-mL screw-cap micro tubes.

3. Aliquot 1.5 mL of plasma into each labeled micro tube.

4. Freeze plasma (no longer than 2 hours after collection) at or below -20 degrees C.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	

CPT Code Information:

0503U

PERA
621166

Preeclampsia sFlt-1/PlGF (Soluble fms-Like Tyrosine Kinase 1/ Placental Growth Factor) Ratio, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, the patient **should not** receive intravenous heparin.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	24 hours

CPT Code Information:

0482U

PGN
65119

Pregabalin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

17PRN
88646

Pregnenolone and 17-Hydroxypregnenolone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	28 days

CPT Code Information:

84140-Pregnenolone

84143-17-Hydroxypregnenolone

PREGN
88645

Pregnenolone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen and immediately aliquot serum into a plastic vial.

2. Freeze immediately.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	28 days

CPT Code Information:

84140

PADF
35313

Prenatal Aneuploidy Detection, FISH

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 to 25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

3. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 12 mL.

4. If ordering with CHRAF / Chromosome Analysis, Amniotic Fluid, submit a minimum of 12 mL.

5. If ordering with both CMAP and CHRAF, then submit a minimum of 26 mL.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

Acceptable:

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 30 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium (Such as CVS Media (RPMI) and Small Dish).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

4. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 12 mg.

5. If ordering with CHRCV / Chromosome Analysis, Chorionic Villus Sampling, submit a minimum of 12 mg.

6. If ordering with both CMAP and CHRCV, then submit a minimum of 26 mg.

Specimen Minimum Volume: Amniotic fluid: 2 mL; Chorionic villi: 2 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		
	Ambient		

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

PHSP 5566 Prenatal Hepatitis Evaluation, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial and ship frozen (preferred).

Specimen Minimum Volume: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Frozen (preferred)	84 days	
	Refrigerated	6 days	

CPT Code Information:

87340

86803

G0472 (if appropriate for government payers)

87522 (if appropriate)

86707 (if appropriate)

87341 (if appropriate)

87350 (if appropriate)

PHEPU 620973 Previous Viral Hepatitis (Unknown Type), Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 2.6 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot 1.8 mL serum into a plastic vial and ship frozen (preferred).

Specimen Minimum Volume: 1.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

86704

86706

86708

86803

87340

87341 (if appropriate)

87522 (if appropriate)

PBCPN 620737 Primary Biliary Cholangitis Antibody Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86039

83516 x2

86381

PCDGG 617421 Primary Ciliary Dyskinesia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

HLHGP 619830 Primary Hemophagocytic Lymphohistiocytosis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When

applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

PMND1 609778 Primary Membranous Nephropathy Diagnostic Cascade, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83520

86255 (x1 or x2, if appropriate)

PRMB 37053 Primidone and Phenobarbital, Serum

Specimen Requirements:

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

PRIMD-80188

PBR-80184

PTRE
82784

Privet Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PRKSG 617435 PRKAR1A Full Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination

studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

GAL2 606833 Probability of Hepatocellular Carcinoma, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see HCCGS / Hepatocellular Carcinoma Risk Panel with GALAD Score, Serum.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	5 days

PA 8683 Procainamide and N-Acetylprocainamide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, plasma may sit on gel refrigerated or be aliquoted into a plastic vial.

3. For red top tubes, aliquot serum immediately into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

80192

PRCAL 602598 Procalcitonin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	48 hours
	Ambient	24 hours

CPT Code Information:

84145

PINP
61695

Procollagen I Intact N-Terminal, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

83519

LDAI
621645

Progentec aiSLE DX Lupus Disease Activity Index, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Yellow top (ACD solution A)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Centrifuge at 1500 x *g* for 20 minutes.

2. Aliquot plasma into plastic vial.

3. Freeze specimen within 2 hours of collection.

Note: Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	74 days

CPT Code Information:

0446U

LFRI
621644

Progentec aiSLE DX Lupus Flare Risk Index, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Yellow top (ACD solution A)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Centrifuge at 1500 x *g* for 20 minutes.

2. Aliquot plasma into plastic vial.

3. Freeze specimen within 2 hours of collection.

Note: Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	74 days

CPT Code Information:

0447U

PROG 70542 Progesterone Receptor (PR) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

PGSN 8141 Progesterone, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection and serum aliquoted into a plastic vial.

2. Red-top tubes should be centrifuged, and serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

84144

22C3 603762 Programmed Death-Ligand 1 (PD-L1) (22C3), Semi-

Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

288PD 609994 Programmed Death-Ligand 1 (PD-L1) (28-8), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1.One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

SP142 603769 Programmed Death-Ligand 1 (PD-L1) (SP142), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

SP263 603755 Programmed Death-Ligand 1 (PD-L1) (SP263), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

PINS
80908

Proinsulin, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required; Infants younger than 2 years should fast a maximum of 6 hours

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Ice-cooled, lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Immediately after draw, place the whole blood on ice for at least 10 minutes.

2. Centrifuge at refrigerated temperature, aliquot plasma into a plastic vial, and freeze specimen.

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	30 days

CPT Code Information:

84206

PRLI
70541

Prolactin (PRL) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PLPMA
35090

Prolactin, Pituitary Macroadenoma, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

84146

PRL
85670

Prolactin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

84146

APRI 603183 Prolonged Clot Time Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

APROL 603308 Prolonged Clot Time Profile, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 5 mL Platelet-poor plasma in five (5) 13-mm x 75-mm, 5-mL,

polypropylene, plastic vials , each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 5 separate 13-mm x 75-mm, 5-mL, polypropylene, plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: Four (4) 13-mm x 75-mm, 5-mL, polypropylene, plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85379-DIMER

85384-CLFIB

85390-26-APRI

85610-PTSC

85613-DRV1

85670-TTSC

85730-APTSC

85130-Chromogenic FVIII (if appropriate)

85130-Chromogenic FIX (if appropriate)

85210-Factor II (if appropriate)

85220-Factor V (if appropriate)

85230-Factor VII (if appropriate)

85240-Factor VIII (if appropriate)

85250-Factor IX (if appropriate)

85260-Factor X (if appropriate)

85270-Factor XI (if appropriate)

85280-Factor XII (if appropriate)

85335-Bethesda titer (if appropriate)

85335-Factor V inhibitor screen (if appropriate)

85335-Factor VIII inhibitor screen (if appropriate)

85335-Factor IX inhibitor screen (if appropriate)

85366-Soluble fibrin monomer (if appropriate)

85385-PT-Fibrinogen (if appropriate)

85597-Platelet neutralization for lupus inhibitor (if appropriate)

85598-Hex LA (if appropriate)

85611-PT mix 1:1 (if appropriate)

85613-DRVVT mix (if appropriate)

85613-DRVVT confirm (if appropriate)

85635-Reptilase time (if appropriate)

85732-APTT mix 1:1 (if appropriate)

FPHEG Promethazine (Phenergan)

90101

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

FPIBD
75953

PROMETHEUS IBD Precis

Specimen Requirements:

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 2 mL serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Refrigerated (preferred)	14 days

	Frozen	30 days
	Ambient	7 days

CPT Code Information:

81599

PFN
80295

Propafenone, Serum

Specimen Requirements:

Patient Preparation: Specimens should only be collected after patient has been receiving propafenone for at least 3 days. Trough concentrations should be collected just before administration of the next dose.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FPROS
75902

Prostaglandin D2 (PG D2), Random Urine

Specimen Requirements:

Patient Preparation: Patient **should not** be on aspirin, indomethacin, or anti-

inflammatory medications, if possible, for at least 48 hours prior to collection of specimen.

Submission Container/Tube: Plastic vial

Specimen Volume: 10 mL

Collection Instructions:

1. Collect 10 mL random urine (24-hour urine collection is **not acceptable**)

2. No preservative.

2. Freeze immediately and send specimen frozen in a plastic vial.

3. Specimen must remain frozen prior to and during shipping.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	60 days

CPT Code Information:

84150

FPDPG 75900 Prostaglandin D2 (PGD2)

Specimen Requirements:

Patient preparation: Patient **should not** be on aspirin, indomethacin, or anti-inflammatory medications, if possible, for at least 48 hours prior to collection of specimen.

Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge as soon as possible, aliquot 3 mL of serum into a plastic vial.

2.Freeze specimen immediately after separation.

3. Send frozen.

Specimen Minimum Volume: Serum:1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	60 days

CPT Code Information:

84150

PHI11 Prostate Health Index Reflex, Serum
113000

Specimen Requirements:

Patient Preparation:

1. Specimens for testing should be collected prior to prostate manipulations such as digital rectal examination, prostatic massage, transrectal ultrasound, and prostatic biopsy.

2. A 6-week waiting period between needle biopsy and specimen collection is recommended.

3. Specimens should not be collected from patients receiving therapy with high biotin (vitamin B7) doses (ie, >5 mg/day) until at least 8 hours following the last biotin administration.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 3 hours of collection, centrifuge, aliquot serum into a plastic vial, and refrigerate. Freeze sample within 24 hours of collection and send frozen.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	150 days

	Ambient	24 hours
	Refrigerated	24 hours

CPT Code Information:

84153 - Total PSA

84154 - Free PSA (if appropriate)

86316 - [-2]ProPSA (if appropriate)

PSAIM 70543 Prostate Specific Antigen (PSA) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PTRIP
610456

Prostate Triple (P63/KRT/P504S), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

P2PHI
608462

Prostate-Specific Antigen (PSA) and [-2]ProPSA with Prostate Health Index (PHI), Serum

Specimen Requirements:

Patient Preparation:

1. Specimens for testing should be collected prior to prostate manipulations such as digital rectal examination (DRE), prostatic massage, transrectal ultrasound (TRUS), and prostatic biopsy.

2. A 6-week waiting period between needle biopsy and specimen collection is recommended.

3. Specimens should not be collected from patients receiving therapy with high biotin (vitamin B7) doses (ie, >5 mg/day) until at least 8 hours following the last biotin administration.

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 3 hours of collection, centrifuge, aliquot serum into a plastic vial, and refrigerate. Freeze sample within 24 hours of collection and send frozen.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	150 days
	Ambient	24 hours
	Refrigerated	24 hours

CPT Code Information:

84153-Total PSA

84154-Free PSA

86316-[-2]ProPSA

PSA
9284

Prostate-Specific Antigen (PSA) Diagnostic, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions:

1. For serum gel tubes, centrifuge within 2 hours of collection.

2. For red top tubes, centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Additional Information: Free prostate-specific antigen (PSA) can only be added on within 72 hours of performing total PSA. Specimen must have been shipped frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	5 days

CPT Code Information:

84153

SPSA
82023

Prostate-Specific Antigen (PSA) Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Red-top tube must be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

2. Serum gel tube must be centrifuged within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	5 days

CPT Code Information:

84153

G0103 (if appropriate)

PSAU 64061 Prostate-Specific Antigen (PSA) Ultrasensitive, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes, aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

84153

PSAFT 81944 Prostate-Specific Antigen (PSA), Total and Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Within 3 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	72 hours

CPT Code Information:

84153

84154

PACPI 70531 Prostatic Acid Phosphatase (PACP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PACP 8019 Prostatic Acid Phosphatase, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days

CPT Code Information:

84066

CFX 9339 Protein C Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85303

PCAG 9127 Protein C Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Plasma

Specimen Stability Information: Frozen 2 years

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or ideally, at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85302

GNPRC 619173 Protein C Deficiency, PROC Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

SFX
617792

Protein S Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Patient Preparation: Patient must not be receiving Coumadin.

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze specimen immediately (no longer than 4 hours after collection) below -40 degrees C.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85306

PSF
80338

Protein S Antigen, Free, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile, see PSTF / Protein S Antigen, Plasma.

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). Coumadin will lower protein S. If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been

transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma in 2 plastic vials, each containing 0.5 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

5. Send specimens in the same shipping container.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85306

PSTF
83049

Protein S Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). Warfarin will lower protein S. If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma in 2 plastic vials, each containing 0.5 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot 0.5 mL of plasma into 2 plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

5. Send specimens in the same shipping container.

Additional Information: A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85306-Free

85305-Total (if appropriate)

PST 80994 Protein S Antigen, Total, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PSTF / Protein S Antigen, Plasma.

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). Warfarin will lower protein S. If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma in 2 plastic vials, each containing 0.5 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

5. Send specimens in the same shipping container.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85305

GNPRS 619187 Protein S Deficiency, PROS1 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

12PU1 Protein, Total, 12 Hour, Urine

614042

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5 mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a 12-hour urine specimen. Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

2. No preservative.

3. Invert well before taking 4-mL aliquot.

4. Do not overfill aliquot tube, 4 mL at most.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

Protein, Total, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours. Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

2. No preservative.

3. Invert well before taking 4-mL aliquot.

4. Do not over fill aliquot tube 4 mL at most.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

TPBF
606619

Protein, Total, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

-Pericardial

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84157

PRCON 617782 Protein, Total, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ORTHP / Orthostatic Proteinuria, Random, Urine

Patient Preparation: Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. This is a nighttime (supine) collection.

2. Void immediately prior to lying down.

3. Collect an 8-hour (nighttime) urine specimen.

4. Upon awaking in the morning, the patient should immediately void into the specimen cup.

4. No preservative.

5. Invert well before taking 4-mL (maximum) aliquot.

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

TP
8520

Protein, Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days

CPT Code Information:

84155

TPSF
872

Protein, Total, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge specimen to remove any cellular material.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	72 hours
	Frozen	180 days

CPT Code Information:

84157

RPTU1
614004

Protein/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Invert well before taking 4 mL aliquot.

4. Do not over fill aliquot tube, 4 mL at most.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

82570

RATO2
617783

Protein/Creatinine, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ORTHP / Orthostatic Proteinuria, Random, Urine.

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

PR3
82965

Proteinase 3 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

FPT12 Prothrombin Fragment 1+2

75957

Specimen Requirements:

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Do not draw from an arm with a heparin lock or heparinized catheter.

2. Draw blood in a blue-top tube containing 3.2% buffered sodium citrate.

3. Mix immediately by gently inverting at least 6 times.

4. Centrifuge and aliquot 1 mL citrated plasma into a plastic vial.

5. Send frozen.

Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Plasma: 0.5 mL; Note: This volume does not allow for

repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	28 days

CPT Code Information:

83520

PTNT 81742 Prothrombin G20210A Mutation, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD solution B), light-blue top (sodium citrate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not** aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

81240-F2 (prothrombin, coagulation factor II) (eg, hereditary hypercoagulability) gene analysis, 20210G>A variant

PTSC 602171 Prothrombin Time (PT), Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile or as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85610

PTMSC 602183 Prothrombin Time Mix 1:1, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85611

PTTP
40934

Prothrombin Time, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL platelet-poor plasma

Collection Instructions:

1. For complete instructions see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, remove plasma, and centrifuge plasma again.

3. Aliquot plasma into plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

Additional Information: A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen (preferred)	30 days
	Ambient	24 hours

CPT Code Information:

85610

PPFWE
31891

Protoporphyrins, Fractionation, Washed Erythrocytes

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Dark blue top (metal free heparin), green top (lithium heparin), or lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. Record volume of packed cells and the total volume of the specimen.

4. Discard supernatant plasma.

5. Wash packed erythrocytes 2 times by resuspension of at least an equal amount of cold 0.9% saline, mix, and centrifuge for 5 minutes at 2000 rpm, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

82542

PPFE
8739

Protoporphyrins, Fractionation, Whole Blood

Specimen Requirements:

Patient Preparation: Patient **must not** consume any alcohol for 24 hours before specimen collection.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA), dark blue top (metal free heparin), royal blue top (metal free EDTA), green top (lithium heparin)

Specimen Volume: 4 mL

Collection Instructions: Refrigerate specimen as soon as possible.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

82542

PROTR 9797 Protriptyline (Vivactyl)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable**. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80335

PRSSZ 35532 PRSS1 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81404-PRSS1 (protease, serine, 1 [trypsin 1]) (eg, hereditary pancreatitis), full gene sequence

PCHE1 606604 Pseudocholinesterase, Total, Serum

Specimen Requirements:

Patient Preparation: For patients with prolonged apnea following surgery, wait at least 24 hours before obtaining specimen.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	24 hours

CPT Code Information:

82480

PSY
62235

Psychosine, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Card-Blood Spot Collection (Filter Paper)

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing EDTA (preferred) or heparin and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	96 days	FILTER PAPER
	Frozen	96 days	FILTER PAPER
	Refrigerated	96 days	FILTER PAPER

CPT Code Information:

82542

PSYCF 606146 Psychosine, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial.

Specimen Volume: 0.15 mL

Collection Instructions: Do not aliquot.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen	7 days

CPT Code Information:

82542

PSYR 606145 Psychosine, Whole Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

82542

PSYQP 610060 Psychotropic Pharmacogenomics Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days (preferred)/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81418

PTFIB
65820

PT-Fibrinogen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85385

PTNZ
614586

PTEN Hamartoma Tumor Syndrome, PTEN Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81321

FPTH
90182

PTH Antibody

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube: Red or SST

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and send 1 mL of serum ambient in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Ambient (preferred)	14 days
	Frozen	28 days
	Refrigerated	14 days

CPT Code Information:

83519

PU1
70545

PU.1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PUSE
82362

Pumpkin Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PUPYP
65151

Purines and Pyrimidines Panel, Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Centrifuge at 4 degrees C and aliquot plasma into plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: Plasma: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	90 days

CPT Code Information:

82542

PUPYU
41977

Purines and Pyrimidines Panel, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative needed.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	90 days

CPT Code Information:

82542

PC1TS
43437

Purkinje Cell Cytoplasmic Antibody Type 1 (PCA-1) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For further information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PC1TC
43446

Purkinje Cell Cytoplasmic Antibody Type 1 (PCA-1) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PC2TS 43438 Purkinje Cell Cytoplasmic Antibody Type 2 (PCA-2) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-PAVAL / Paraneoplastic, Autoantibody Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PC2TC
43447

Purkinje Cell Cytoplasmic Antibody Type 2 (PCA-2) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PCTTS
43439

Purkinje Cell Cytoplasmic Antibody Type Tr (PCA-Tr) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-PAVAL / Paraneoplastic Autoantibody Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PCTTC 43448 Purkinje Cell Cytoplasmic Antibody, Type Tr (PCA-Tr) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time

CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

FPYRE
57540

Pyrethrum IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

PLP
42359

Pyridoxal 5-Phosphate, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required; Infants should have specimen collected before next feeding, water may be taken as needed

2. For 24 hours before specimen collection, patient **should not take multivitamins or vitamin supplements**.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin) or Plasma gel separator (PST)

Acceptable: None

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge at 4 degrees C.

2. Aliquot all plasma into amber vial and freeze immediately.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Frozen	29 days	LIGHT PROTECTED

CPT Code Information:

84207

B6PA 42361 Pyridoxic Acid, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see B6PRO / Vitamin B6 Profile (Pyridoxal 5-Phosphate and Pyridoxic Acid), Plasma.

Patient Preparation:

1. **Fasting: 12 hours, required**; Infants-should have specimen collected before next feeding, water can be taken as needed

2. For 24 hours before specimen collection, patient **must not** take multivitamins or

vitamin supplements.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Green top (sodium or lithium heparin) or plasma gel separator (PST)

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge at 4 degrees C.

2. Aliquot all plasma into amber vial and freeze immediately.

Specimen Minimum Volume: Plasma: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Frozen	29 days	LIGHT PROTECTED

CPT Code Information:

82542

P5NT
80650

Pyrimidine 5' Nucleotidase, Blood

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

83915

PK1 607459 Pyruvate Kinase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

84220

PKC 608418 Pyruvate Kinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

HAEV1 / Hemolytic Anemia Evaluation, Blood

EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

84220

PYRC
83356

Pyruvate, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.6 mL

Collection Instructions: Send specimen from vial 4.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

84210

PYR
8657

Pyruvic Acid, Blood

Specimen Requirements:

Call 800-533-1710 or 507-266-5700 to order special collection tube.

Patient Preparation:

Fasting: 4 hours, required

Supplies: Perchloric Acid-Pyruvate Tube (T012)

Container/Tube: Special collection tube containing 2.5 mL of 6% perchloric acid

Specimen Volume: Exactly 1 mL

Collection Instructions:

1. Special collection tube must be prechilled prior to collection.

2. Draw enough blood directly into syringe to add exactly 1 mL of blood to the prechilled special collection tube.

3. Taking care to not spill any of the preservative, cautiously remove the cap from the tube.

4. Immediately transfer blood, once drawn, to the prechilled, special collection tube, recap, and shake vigorously to mix.

Additional Information:

1. **Check expiration date before using.** Supplied collection tube expires 14 months after preparation.

2. If perchloric acid (preservative) spills, obtain a new, prechilled tube.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated	15 days	PYRUVATE

CPT Code Information:

84210

QFEVR 616832 Q Fever Antibody Screen with Titer Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

86638

86638 x4 (if appropriate)

QFP
83149

Q Fever IgM and IgG, Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see QFEVR / Q Fever Antibody Screen with Titer Reflex, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

86638 x 4

QUAD1
113145

Quad Screen (Second Trimester) Maternal, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Do not collect specimen after amniocentesis as this could affect results.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Additional Information:

1. For an assessment that includes neural tube defect results, gestational age must be between 15 weeks, 0 days and 22 weeks, 6 days.

2. Assessments for trisomy 21 (Down syndrome) and trisomy 18 (Edwards syndrome) only are available between 14 weeks, 0 days and 22 weeks, 6 days.

3. Initial or repeat testing is determined in the laboratory at the time of report and will be reported accordingly. To be considered a repeat test for the patient, the testing must be within the same pregnancy and trimester, with interpretable results for the same tests, and both tests are performed at Mayo Clinic.

4. Maternal Serum Screening patient education brochure (T522) is available upon request.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

81511

QFT4 113563 QuantiFERON-TB Gold Plus, Blood

Specimen Requirements:

Supplies:

-QuantiFERON-TB Gold Plus Collection Kit (T794)

-QuantiFERON-TB Gold Plus HIGH ALTITUDE Collection Kit (T795)

Collection Instructions:

1. Special collection, incubation, and centrifugation procedures must be followed.

2. For blood collection options (1-tube collection or 4-tube collection) and specimen transport instructions, see Mycobacterium tuberculosis Infection Determination by Quanti-FERON-TB Gold Plus Collection and Processing Instructions (T688).

Specimen Minimum Volume: 4 mL: 1 mL per tube (4 tubes)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated	28 days	QTBKIT

CPT Code Information:

86480

TBBS 9336 Quantitative Lymphocyte Subsets: T, B, and Natural Killer (NK) Cells, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	52 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86355

86357

86359

86360

QPALM Queen Palm, IgE, Serum
82863

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FQUET Quetiapine (Seroquel)
91727

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube not acceptable. Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

FQUIN 57922 Quinoa (Chenopodium quinoa) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

REPII Rabbit Epithelium, IgE, Serum
82782

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RAMB Rabbit Meat, IgE, Serum
82860

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RSER
82544

Rabbit Serum Proteins, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RUPR
82148

Rabbit Urine Proteins, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FRFIT
90330

Rabies Antibody Endpoint

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Note: Collection date is required.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

86382

FRAD 57933 Radish (Raphanus sativus) IgE

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

RASE 82366 Rape Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RWEED Rape Weed, IgE, Serum

82616

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BRTP 619957 Rapid Hereditary Breast Cancer Treatment Decision Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81432

MAL 9240 Rapid Malaria/Babesia Smear, Varies

Specimen Requirements:

Both blood and slides are required.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Type: Blood films

Slides: 2 Thin blood films and 2 thick blood films

Container/Tube: Plastic slide container

Collection Instructions:

1. For complete instructions, see Thick and Thin Blood Slide Preparation Instructions.

2. Slides must be clean and grease-free.

3. Blood films should be made from fresh blood using fingerstick or drops of blood from needle following venipuncture. However, EDTA anticoagulated blood is also acceptable.

4. Prepare thin blood films as follows:

a. Prepare 2 thin smears with a slide prep machine. OR

b. Prepare a thin film with a "feathered edge" that is no more than a single cell thick.

c. Allow the film to thoroughly air dry and then fix by briefly immersing in either absolute or 95% methyl alcohol.

d. Allow to air dry after fixation.

5. Prepare thick blood films as follows:

a. Place a large drop of blood (approximately the size of a dime and preferably from a fingerstick) on a slide.

b. Using a corner of a second slide spread the drop in a circular motion while applying firm pressure to literally scratch the blood onto the carrier slide. This technique allows the blood to dry quickly and adhere well to the slide. Use approximately 20 circular sweeps with the second slide. The drop of blood should be about the size of a quarter when finished.

c. **Do not fix.** Air dry thoroughly (approximately 45 minutes) before placing in transport container.

Specimen Minimum Volume: Blood: 0.5 mL
Slides: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

87207

RPRT1 616863 Rapid Plasma Reagin (RPR) Screen with Reflex to Titer, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86592

86593-Rapid Plasma Reagin Titer (if appropriate)

RPRT3 616970 Rapid Plasma Reagin (RPR) with Reflex to Treponema pallidum Particle Agglutination, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86592

86593- (if appropriate)

86780- (if appropriate)

RRPRS 616866 Rapid Plasma Reagin Screen with Reflex to Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see SYPH1 / Syphilis IgG with Reflex, Enzyme Immunoassay, Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86592-Rapid Plasma Reagin Screen

86593-Rapid Plasma Reagin Titer (if appropriate)

86780-Syphilis Antibody by TP-PA (if appropriate)

RPRS 603261 Rapid Plasma Reagin Screen with Reflex, Serum

Specimen Requirements:

Only available as a reflex test. For more information see SYPHT / Syphilis Total Antibody with Reflex, Serum.

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86593-Rapid Plasma Reagin Titer (if appropriate)

86780-Syphilis Antibody by TP-PA (if appropriate)

RPRT2 616865 Rapid Plasma Reagin Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see RPRT1 / Rapid Plasma Reagin (RPR) Screen with Reflex to Titer, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86593

RPDEI 620377 Rapidly Progressive Dementia Evaluation Interpretation, Spinal Fluid (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Frozen (preferred)	28 days	BlueTop SARSTEDT
	Refrigerated	14 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

RPDE 620376 Rapidly Progressive Dementia Evaluation, Spinal Fluid

Specimen Requirements:

Supplies: CJD/RPD Evaluation Kit (T966)

Container/Tube:

Preferred: 2 Sarstedt CSF False Bottom Tubes 63.614.625 (2.5 mL)

Acceptable: Sarstedt 72.703.600 (1.5 mL) or Sarstedt 72.694.600 (2 mL)

Specimen Volume: 2 tubes; each containing 1.5 to 2.5 mL

Collection Instructions:

1. Perform lumbar puncture and discard the first 1 to 2 mL of cerebrospinal fluid (CSF).

2. Collect two tubes of CSF directly into an acceptable collection tube until the tube is at least 50% full.

3. Send CSF specimen in original collection tube. **Do not aliquot.**

Note: Polystyrene collection tubes are not acceptable. Exposure of CSF to polystyrene tubes may result in falsely low Abeta42 concentrations.

The Alzheimer's Association consensus protocol for handling of CSF for clinical measurements of Abeta42 and tau recommends using the drip method for CSF collection and directly collecting into a low-bind polypropylene tube. Although some clinicians prefer the syringe pull method due to speed of collection, the drip method reduces the risk of Abeta42 binding to the plastic of any syringe used.

4. Collection instructions can also be found on Spinal Fluid Specimen Collection Instructions for Creutzfeldt-Jakob Disease and Rapidly Progressive Dementia Evaluations (T974).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Frozen (preferred)	28 days	BlueTop SARSTEDT
	Refrigerated	14 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

CPT Code Information:

0584U

82234

84393

84394

RSBV 621403 Rare Subepithelial Autoimmune Blistering Disease Variants, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

86255

FRASP 57665 Raspberry IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

RASP
86305

Raspberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RAT
82725

Rat Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RTSP
82793

Rat Serum Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RTUP
82794

Rat Urine Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RAVMP
618645

Ravulizumab Monitoring Panel, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. Suggest discontinuing natalizumab at least 4 weeks before specimen collection for ravulizumab testing. Patient should consult the healthcare professional who prescribed this drug to determine if discontinuation is an option. If not, it is okay to proceed with testing while taking natalizumab.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: 2 Plastic vials

Specimen Volume: 2 mL Serum in **2 separate plastic vials** (see Collection Instructions)

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

3. Centrifuge at 4 degrees C.

4. Aliquot serum into two separate plastic vials, each containing 1 mL.

5. Within 30 minutes of centrifugation, freeze specimen. Serum must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 1 mL in 2 separate plastic vials, each containing 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days
Serum Red	Frozen	14 days

CPT Code Information:

80299

86161

RAVU
609420

Ravulizumab, Serum

Specimen Requirements:

Patient Preparation: Consider discontinuing natalizumab at least 4 weeks prior to specimen collection. Patient should consult the healthcare provider who prescribed this medication to determine if discontinuation is an option. If not, ok to proceed with testing while taking natalizumab.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Immediately after specimen collection, place the tube on wet ice.

3. After specimen has clotted on wet ice, centrifuge at 4 degrees C and aliquot serum into a plastic vial.

4. Freeze specimen within 30 minutes of centrifugation. Specimen must be placed on dry ice if not frozen immediately.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

80299

NGSFX 65718 Reanalysis of Acute Myeloid Leukemia 4- or 11- Gene Panels, Additional Genes (Non-Orderable)

Specimen Requirements:

No additional specimen is required. This is a bioinformatics review of additional gene regions not analyzed in the previously ordered NGAMT / MayoComplete Acute Myeloid Leukemia, Therapeutic Gene Mutation Panel (*FLT3*, *IDH1*, *IDH2*, *TP53*), Next-Generation Sequencing, Varies or NGAML / MayoComplete Acute Myeloid Leukemia, 11-Gene Panel, Varies. Call 800-533-1710 for assistance with ordering.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

NCYB 618977 Recessive Congenital Methemoglobinemia, CYB5 and CYB5 Reductase Genetic Analysis, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

RCVBS 610009 Recoverin-IgG Antibody, Immunoblot, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

EEEV1 Red Blood Cell (RBC) Enzyme Evaluation, Blood
607493

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 12 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	11 days

CPT Code Information:

82955-G6PD Enzyme Activity

84087-Glucose Phosphate Isomerase

84220-Pyruvate Kinase Enzyme Activity

82657-Hexokinase

82657-Adenylate Kinase

82657-Phosphofructokinase

82657-Phosphoglycerate Kinase

82657-Triosephosphate Isomerase

82978-Glutathione

83915-Pyrimidine 5' Nucleotidase

EEEVI 608087 Red Blood Cell (RBC) Enzyme Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	11 days

IDCOR 622374 Red Blood Cell Antigen Genotyping, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Pink top (EDTA)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Pediatric Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Label specimen as blood.

3. Send whole blood specimen in the original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

0084U

NENZ
619047

Red Blood Cell Enzyme Disorders Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81443

NMEM 619061 Red Blood Cell Membrane Disorders Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81479

81479 (if appropriate for government payers)

RBCME 64897 Red Blood Cell Membrane Evaluation, Blood

Specimen Requirements:

Both patient and shipping control specimens are required.

For complete instructions, refer to Specimen Collection and Labeling Instructions for Osmotic Fragility Testing of Erythrocytes.

Shipping Control Specimen

Specimen Type: Whole blood (non-patient)

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a shipping control specimen from a healthy person unrelated to the patient at the same time (or within 4 hours) as the patient specimen.

Note: The shipping control specimen can be collected from a phlebotomist, volunteer or another heathy patient.

2. Clearly handwrite "CONTROL" on the outermost label of the tube.

3. Refrigerate (or place on cold gel pack/small amount of wet ice) immediately after collection.

4. Send shipping control in the original tube. **Do not aliquot**.

5. Keep the shipping control and the patient specimens **together**, either rubber banded or in a bag.

Additional Information:

1. **The shipping control and patient specimens must be handled identically** from the time of collection through receipt in the testing laboratory.

2. **If the shipping control is not sent with the patient specimen, test cancellation is likely.**

3. The shipping control specimen evaluates whether the patient result has been compromised by handling conditions such as temperature, motion, or other transportation interferences, as temperature and handling extremes can adversely impact the integrity of the specimen.

Patient Specimen

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect and label patient specimen.

2. Refrigerate (or place on cold gel pack/small amount of wet ice) specimen immediately after collection.

3. Send whole blood specimen in the original tube. **Do not aliquot**.

4. Keep the shipping control and the patient specimens together, either rubber banded or in a bag.

Additional Information: The shipping control and patient specimens must be handled identically from the time of collection through receipt in the testing laboratory.

Patient Specimen

Specimen Type: Slides

Container/Tube: Blood smears

Specimen Volume: 2 Well-made peripheral blood smears

Collection Instructions:

1. Prepare 2 peripheral blood smears from the EDTA tube collected from the patient.

2. Either stain the smear with Wright stain or fix the smear with absolute methanol prior to shipping.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Whole blood patient/Shipping control: 2 mL; Slides: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Control	Refrigerated	72 hours	PURPLE OR PINK TOP/EDTA
Whole Blood EDTA	Refrigerated	72 hours	
Whole Blood Slide	Refrigerated		CARTRIDGE

CPT Code Information:

85557-Osmotic fragility

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

85060-Morphology review

FRSE 57939 Red Snapper (Lutjanus spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SORR 82737

Red Sorrel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

REDT
82901

Red Top, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

UREDF
607696

Reducing Substance, Feces

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz Random (T288)

Container/Tube: Fecal container

Specimen Volume: 3 g

Collection Instructions:

1. Collect a loose, unpreserved, random fecal specimen.

2. Freeze immediately.

Additional Information: If additional tests are ordered, aliquot and separate sample prior to freezing to allow 1 container per test.

Specimen Minimum Volume: 2 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen	7 days

CPT Code Information:

84376

RBCS
36440

Relative B-Cell Subset Analysis Percentage, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume:

< or =14 years: 4 mL

>14 years: 10 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as blood for RBCS.

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: < or =14 years: 3 mL; >14 years: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Refrigerated	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x7

TFEBF 64973 Renal Cell Carcinoma, 6p21.1 (TFEB) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded (FFPE) tumor tissue block. Blocks prepared with alternative fixation methods will be attempted but are less favorable for successful results by FISH testing.

2. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

RFAMA 113634 Renal Function Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	24 hours

CPT Code Information:

KS: 84132

NAS: 84295

CL: 82435

HCO3: 82374

BUN: 84520

CRTS1: 82565

CA: 82310

GLURA: 82947

ALB: 82040

PHOS: 84100

RPCWT 70591 Renal Pathology Consultation, Wet Tissue

Specimen Requirements:

Specimen Type: Kidney biopsy

Supplies: Renal Biopsy Kit (T231)

Specimen Volume: Entire specimen

Collection Instructions: Collect and prepare biopsy specimens per instructions in Renal Biopsy Preparation Instructions.

Additional Information: On Saturdays and holidays, RUSH (same day as receipt) interpretation is available for clinically emergent cases (eg, acute kidney failure, rapidly progressive glomerulonephritis, acute kidney allograft dysfunction) but requires advanced notification and approval by a Mayo Clinic Renal Pathologist (507-284-5677).

To request RUSH service outside of regular business hours, contact Mayo Clinic Laboratories at 800-533-1710.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Kidney Biopsy	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305 (If appropriate)

88348 (If appropriate)

88313 (If appropriate)

88346 (If appropriate)

88350 (If appropriate)

RFCWT 620415 Renal Pathology Tissue Consultation, Wet Tissue

Specimen Requirements:

Specimen Type: Tissue

Supplies: Renal Biopsy Kit (T968)

Source: Kidney

Specimen Volume: Entire specimen

Collection Instructions: Collect specimens according to the instructions in Renal Biopsy Preparation Instructions.

Additional Information:

Rush (same day as receipt) interpretation is available on Saturday and holidays for clinically emergent cases (eg, acute kidney failure, rapidly progressive glomerulonephritis, acute kidney allograft dysfunction) but requires advanced notification and approval by a Mayo Clinic Renal Pathologist 904-956-3318.

To request RUSH service outside regular business hours, call 800-533-1710.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Kidney Biopsy	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305 (If appropriate)

88348 (If appropriate)

88313 (If appropriate)

88346 (If appropriate)

88350 (If appropriate)

PRA
8060

Renin Activity, Plasma

Specimen Requirements:

Patient Preparation: For 4 to 6 weeks before specimen collection, spironolactone (Aldactone) should be discontinued, as plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Chilled, lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions:

1. Draw blood in a chilled syringe from a patient in a seated position; place specimen in a chilled, 3-mL lavender-top (EDTA) tube and mix well. Alternatively, draw blood directly in a chilled, EDTA tube.

2. Immediately place EDTA tube into an ice-water bath until thoroughly cooled.

3. Immediately centrifuge using a refrigerated centrifuge and aliquot plasma into a plastic vial.

Note: If a refrigerated centrifuge is unavailable, chill the centrifuge carriers before use. Centrifuge specimen for 5 minutes or less, then promptly aliquot plasma into a plastic vial.

4. Immediately freeze plasma.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

84244

RTSCJ
622836

Reptilase Time, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. For complete instruction, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally below -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85635

RTSC
602185

Reptilase Time, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Platelet poor plasma

Collection Instructions:

1. For complete instruction, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85635

RP
609409

Respiratory Panel, PCR, Nasopharyngeal

Specimen Requirements:

Specimen Type: Nasopharyngeal swab

Supplies:

-Culture Swab - Liquid Stuarts/Single Swab (NP Swab) (T515)

-M4-RT (T605)

Collection Container/Tube: Swab. See Additional Information for acceptable swab.

Submission Container/Tube: Transport medium. See Additional Information for acceptable media.

Specimen Volume: Nasopharyngeal swab in minimum volume of 1 mL of transport media

Collection Instructions: Nasopharyngeal swab specimens should be collected according to standard technique and immediately placed into transport media and submitted for testing.

Additional Information:

If any nasopharyngeal swab or transport media not listed below is utilized, testing may be canceled.

-Acceptable nasopharyngeal (NP) swabs are Copan Rayon Swabs, Copan Nylon Flocked Swabs, Copan Polyester Swabs, Puritan Calcium Alginate Swabs, SteriFlock NP Swab

-Acceptable transport media are Remel M4, Remel M4-RT, Remel M5, Remel M6, BD Universal Viral Transport Media (VTM), Copan Universal Transport Media (UTM), PrimeStore Molecular Transport Medium (MTM)

-Acceptable collection and transport system is Sigma-Virocult Viral Collection and Transport System (Swab and transport medium)

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	72 hours
	Frozen	30 days
	Ambient	4 hours

CPT Code Information:

0202U

RPB 616389 Respiratory Panel, PCR, Varies

Specimen Requirements:

Specimen Type: Respiratory fluid

Sources: Bronchoalveolar lavage (BAL) or bronchial washing

Collection Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	72 hours
	Frozen	30 days
	Ambient	4 hours

CPT Code Information:

87633

RPR1 62046 Respiratory Profile, Region 1, North Atlantic (CT, MA, ME, NJ, NH, NY, PA, RI, VT), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.8 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785

86003 x 25

RPR10 62056 Respiratory Profile, Region 10, Southwestern Grasslands (OK, TX), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.55 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 25 Each individual allergen

RPR11 62057 Respiratory Profile, Region 11, Rocky Mountain (AZ [Mt]; CO; ID [Mt]; NM, UT [Mt]; WY), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 24-Each individual allergen

RPR12 62058 Respiratory Profile, Region 12, Arid Southwest (Southern AZ Desert, Southern CA Desert), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR13 62059 Respiratory Profile, Region 13, Southern Coastal California, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 24-Each individual allergen

RPR14 62060 Respiratory Profile, Region 14, Central California, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR15 62061 Respiratory Profile, Region 15, Intermountain West (Southern ID, NV), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR16 62062 Respiratory Profile, Region 16, Inland Northwest (OR, Central and Eastern WA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 21-Each individual allergen

RPR17 62063 Respiratory Profile, Region 17, Pacific Northwest (Northwestern CA, Western OR, WA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR18 62064 Respiratory Profile, Region 18, Alaska, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 15-Each individual allergen

RPR19 Respiratory Profile, Region 19, Puerto Rico, Serum

62065

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 21-Each individual allergen

RPR2 62047 Respiratory Profile, Region 2, Mid-Atlantic (DC, DE, MD, NC, VA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR3 62048 Respiratory Profile, Region 3, South Atlantic (GA, N.FA, SC), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR4 62049 Respiratory Profile, Region 4, Subtropic Florida (Florida S. of Orlando), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR5 62050 Respiratory Profile, Region 5, Ohio Valley (IN, KY, OH, TN, WV), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 26-Each individual allergen

RPR6 62051 Respiratory Profile, Region 6, South Central (AL, AR, LA, MS), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR7 62052 Respiratory Profile, Region 7, Northern Midwest (MI, MN, WI), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR8 62053 Respiratory Profile, Region 8, Central Midwest (IA, IL, MO), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.55 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 25-Each individual allergen

RPR9 62054 Respiratory Profile, Region 9, Great Plains (KS, ND, NE, SD), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RSVAB 601948 Respiratory Syncytial Virus (RSV) In Situ Hybridization, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-4 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-TC, Primary

88364-TC, if additional ISH

FRSVQ 75669 Respiratory Syncytial Virus (RSV) RNA, Qualitative Real-Time PCR

Specimen Requirements:

Specimen Type: Bronchoalveolar lavage or bronchial wash

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

Collect in a sterile leak-proof container (no media or preservative). Ship refrigerated.

Note: Specimen type is required.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	30 days

	Ambient	7 days

CPT Code Information:

87634

RETB
616856

Reticulocyte Profile, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	48 hours
	Ambient	24 hours

CPT Code Information:

85046

RB1
604028

Retinoblastoma Protein (Rb) Immunostain, Tech Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained

slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FRBP
75570

Retinol Binding Protein

Specimen Requirements:

Specimen Type: Serum

Patient Preparation:

Fasting: 12 hours, required

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 2 mL serum

Collection Instructions:

1. Centrifuge and aliquot 2 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	20 days
	Frozen	180 days

CPT Code Information:

83883

RB24
609449

Retinol-Binding Protein, 24 Hour, Urine

Specimen Requirements:

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Mix well before taking 5-mL aliquot.

Additional Information:

See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83883

RBR
610010

Retinol-Binding Protein, Random, Urine

Specimen Requirements:

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83883-Retinol -Binding Protein, Random, U

82570-Creatinine, Random, U

RAPAN
616785

Rheumatoid Arthritis Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86200

86431

RFPN
621421

Rheumatoid Factor Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

86431 x2

RHUT
603415

Rheumatoid Factor, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

86431

RHNI
82856

Rhizopus nigricans, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FRDG
57959

Rhodotorula IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FRUB 57920 Rhubarb (Rheum rhaponticum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

VITB2
42363

Riboflavin (Vitamin B2), Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required; Infants should have specimen collected before next feeding

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Light-green top (sodium or lithium heparin plasma gel)

Submission Container/Tube: Amber vial

Specimen Volume: 2 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a amber vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Refrigerated (preferred)	28 days	LIGHT PROTECTED
	Frozen	28 days	LIGHT PROTECTED
	Ambient	72 hours	LIGHT PROTECTED

CPT Code Information:

84252

RIB
87837

Ribosome P Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

FRICE
57633

Rice IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

RICE
82709

Rice, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMTAG
57260

Rickettsia typhi (murine typhus) Antibodies, IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	5 days

CPT Code Information:

86757

FRIFA
75628

Rifampin Level

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Allow serum to clot for 30 minutes.

2. Immediately centrifuge and aliquot 2 mL of serum into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	365 days

CPT Code Information:

80299

RISAP
621813

Risankizumab Quantitation with Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80299

82397

RISA
621304

Risankizumab, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before the next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

ROMA2
46917

Risk Score, if Postmenopausal, Serum (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	84 days

	Refrigerated	48 hours

ROMA1 Risk Score, if Premenopausal, Serum (Non-Orderable)

46916

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	84 days
	Refrigerated	48 hours

FRISP Risperidone (Risperdal) and 9-Hydroxyrisperidone

91105

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Specimen Type: Plasma (Preferred)

Container/Tube: Green-top (sodium heparin) tube(s).

Specimen volume: 3 mL

Collection Instructions: Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Specimen Type: Serum

Container/Tube: Red-top tube, serum gel is not acceptable.

Specimen volume: 3 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

RIVAR
65847

Rivaroxaban, Anti-Xa, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen should be collected 2 to 4 hours (peak) after a dose or just prior (trough) to the next dose for rivaroxaban concentrations.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	42 days

CPT Code Information: 80299

RNAP
83397

RNA Polymerase III Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

RNP
81357

RNP Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

ROPAN
615248

Ro52 and Ro60 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days

CPT Code Information:

86235 x 2

RO52
615246

Ro52 Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days

CPT Code Information:

86235

RO60
615247

Ro60 Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days

CPT Code Information:

86235

ROMA
62661

ROMA Score (Ovarian Malignancy Risk Algorithm), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	84 days
	Refrigerated	48 hours

CPT Code Information:

86305-HE4, S

86304-Cancer Ag 125 (CA 125), S

FROPI
57171

Ropivacaine, Serum/Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Submit only 1 of the following specimens:

Serum

Draw blood in a plain, red-top tube(s). (Serum gel tube is not acceptable.) Spin down and send 1 mL serum refrigerate in a plastic vial.

Plasma

Draw blood in a lavender-top (EDTA) or pink top tube(s). (Plasma gel tube is not acceptable.) Spin down and send 1 mL plasma refrigerate in plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	30 days
	Frozen	90 days
	Ambient	30 days

CPT Code Information:

80299

ROTA
8886

Rotavirus Antigen, Feces

Specimen Requirements:

Specimen Type: Fresh unpreserved feces

Supplies: Stool Collection Kit, Random (T635)

Container/Tube:

Preferred: Sterile fecal container

Acceptable: Swab

Specimen Volume: 5 to 10 grams

Collection Instructions: Place specimen in a tightly sealed plastic bag.

Specimen Minimum Volume: 1 gram

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	7 days
	Refrigerated	72 hours

CPT Code Information:

87425

MARS
82701

Rough Marsh Elder, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RRRP
82723

Rough Pigweed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RBPG
34938

Rubella Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Instructions: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86762

ROC
5194

Rubeola (Measles) Antibodies, IgG and IgM, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.25 mL

Collection Instructions: Submit specimen from collection vial number 2 (preferred), 3, or 4.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765 x 2

RUFI
63030

Rufinamide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect blood immediately before next scheduled dose.

2. For sustained-release formulations ONLY, collect blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80210

T821Q 615905 RUNX1::RUNX1T1 Translocation (8;21), Minimal Residual Disease Monitoring, Quantitative, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: Peripheral blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	72 hours

CPT Code Information:

81401

RUSS
82681

Russian Thistle, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FRFYG
57579

Rye Food IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

RYEG
82908

Rye Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RYE
82689

Rye, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

2SC
610031

S-(2-Succinyl)-Cysteine (2SC) Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Preferred: 2 unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick.

Acceptable: Formalin-fixed, paraffin-embedded (FFPE) tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

S100
70547

S-100 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

F100B
57349

S-100B Protein, Serum

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL serum

Collection Instructions:

1. Allow specimen to clot completely at ambient temperature.

2. Within 2 hours of collection, centrifuge and aliquot 2 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

86316

SSCTU 607001 S-Sulfocysteine Panel, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative needed.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	90 days

CPT Code Information:

82542

SCERA 610002 Saccharomyces cerevisiae Antibody, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86671

SCERG
610003

Saccharomyces cerevisiae Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86671

FSFLE
57541

Safflower (Carthamus tinctorius) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FSAG 57957 Sage (Artemisia spp.) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SALCA Salicylate, Serum
37061

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80179

SALL4 SALL4 Immunostain, Technical Component Only
71534

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FSALG
57631

Salmon IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.
2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.
3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SALM
82754

Salmon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SALMC
606220

Salmonella Culture, Feces

Specimen Requirements:

Patient Preparation: Patient **should not** use barium or bismuth for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-Salmonella Culture, Stool-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147 x 1-3-Serologic Agglut Method 1 Ident (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

HEXBZ
608026

Sandhoff Disease, HEXB Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes

prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SARD
82818

Sardine (Pilchard), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SATB2
607600

SATB2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

C5B9
616921

SC5b-9 Level Terminal Complement Complex, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. **Do not** collect specimens for at least 48 hours following plasma exchange.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (K2 EDTA)

Acceptable: Lavender top (K3 EDTA), light-blue top (sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Centrifuge between 1000 and 2000 x *g* for 10 minutes at 4 degrees C.

3. Aliquot plasma into a plastic vial.

4. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the plasma is aliquoted and frozen.

Specimen Minimum Volume: Plasma: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

86160

SCLE
82716

Scale, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SCALS
82259

Scallop, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SHUR
60451

Schistosoma Exam, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen. Preferred time of collection between the hours of

12 noon and 3 p.m. but not required. A 24-hour urine collection is also acceptable.

2. No preservative.

Additional Information: A specimen volume of at least 10 mL is recommended for optimal recovery of *Schistosoma* species ova.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	7 days

CPT Code Information:

87210

87015

BILHA
65019

Schistosoma species Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86682

SCL70
80178

Scl 70 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

FSCPR
75625

Scleroderma Comprehensive Profile

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL (volume does not allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

86038

83516

86235 x 5

86256

FSCN4
75263

SCN4A (Myotonia) DNA Sequencing Test

Specimen Requirements:

Note: Collection date and informed consent are required.

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 8 mL

Pediatric (0-3 years) **Volume**: 2 mL

Collection Instructions:

1. Invert several times to mix.

2. Send ambient.

Additional Information: Higher blood volumes ensure adequate DNA quantity, which varies with WBC, specimen condition, and need for confirmatory testing. Patients aged 0 to 3 years have higher WBC, yielding more DNA per mL of blood.

Specimen Minimum Volume: 6 mL; Pediatric (0-3 years): 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

81406

SDHMP 619712 SDH Genes Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

15 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81404

81405 x2

81406

81479 (if appropriate for government payers)

SDHB
70550

SDHB Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SEAFP
31770

Seafood Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

SEAS
31766

Seasonal Inhalants Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 10

SECOS
8243

Secobarbital, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Collect specimen immediately before the next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FSHPU
58038

Sedative Hypnotic Panel, Urine

Specimen Requirements:

Collect 10 mL random urine without preservatives. Ship refrigerated in a plastic container.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307

SEWB
65600

Selenium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours.**

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 0.8 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

84255

SES 9765 Selenium, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium, iodine, and barium are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium-, iodine-, and barium containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: 6-mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.8 mL serum

Collection Instructions:

1. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a metal-free, polypropylene vial, avoid transferring the cellular components of blood. **Do not** insert a pipette into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE

	Frozen	28 days	METAL FREE

CPT Code Information:

84255

SEMA3 617055 Semaphorin 3B (SEMA3B) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SEMB 60556 Semen Analysis with Strict Morphology, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Supplies: Semen Analysis Kit - Dilution Media (T178)

Specimen Volume: Total ejaculate

Collection Instructions:

See Semen Collection Specimen Preparation and Packaging Instructions included with the kit

1. Prior to use, allow dilution medium to warm to room temperature for 45 to 60 minutes.

2. Allow semen to liquefy at room temperature for up to 30 minutes.

3. Use sterile volumetric pipet or tube for volume measurement.

4. Pour liquefied semen into 50-mL dilution medium container within 60 minutes of semen collection time, cap tightly, but do not overtighten, and gently mix.

Note: Proper temperature maintenance of specimen throughout processing and shipping is critical. All materials the specimen is exposed to should be at room temperature (20–28 degrees C).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	36 hours

CPT Code Information:

89310-Semen Analysis

89398-Strict Criteria Sperm Morphology

If both components performed,

89322-Semen Analysis with Strict Morphology

FER
81641

Semen Analysis, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Supplies: Semen Analysis Kit - Dilution Media (T178)

Specimen Volume: Total ejaculate

Collection Instructions:

See complete Semen Collection instructions included with the kit.

1. Prior to use, allow dilution medium to warm to room temperature for 45 to 60 minutes.

2. Allow semen to liquefy at room temperature for up to 30 minutes.

3. Use sterile volumetric pipet or tube for volume measurement.

4. Pour liquefied semen into 50-mL dilution medium container within 60 minutes of semen collection time, cap tightly, but do not overtighten, and gently mix.

Note: Proper temperature maintenance of specimen throughout processing and shipping is critical. All materials the specimen is exposed to should be at room temperature (20-28 degrees C).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89310

SMFL
82858

Seminal Fluid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL

More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SP5CS 615869 Septin-5 Antibody, Cell Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP5CC 615868 Septin-5 Antibody, Cell-Binding Assay, Spinal Fluid (Non-

Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid.

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP5TS 616113 Septin-5 Antibody, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SP5TC
616114

Septin-5 Antibody, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid.

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SP7CS
615873

Septin-7 Antibody, Cell-Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP7CC 615872 Septin-7 Antibody, Cell-Binding Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENC2 / Encephalopathy, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP7TS 616115 Septin-7 Antibody, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SP7TC 616116 Septin-7 Antibody, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENC2 / Encephalopathy, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Autoimmune Myelopathy /Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SEP9Z
617714

SEPTIN9 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

FSERO
75621

SeroNeg RAdx3 Profile

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 3 mL (volume does NOT allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83520

83516 x2

SRAU
616216

Serotonin Release Assay, Unfractionated Heparin, Mass Spectrometry, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required; Collect during an episode of suspected heparin-induced thrombocytopenia

2. Patient **should not** be on ticagrelor (Brilinta) as this may interfere with the assay, yielding a false-negative result.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Serum

Collection Instructions:

1. After collection, allow the specimen to sit for a minimum of 1 hour at ambient temperature to ensure it clots completely.

2. Within 48 hours of collection, centrifuge and aliquot serum into a 13-mm x 75-mm, 5-mL, polypropylene vial

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	
	Refrigerated	7 days

CPT Code Information:

82542

SERU
87834

Serotonin, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before and during specimen collection, patient should **not** eat avocados, bananas, butternuts, cantaloupe, dates, eggplant, grapefruit, hickory nuts, honeydew melon, kiwifruit, melon, nuts, pecans, pineapple, plantains, plums, tomatoes, or walnuts, which are high in serotonin or tryptophan.

2. Patient **should not** take medications that may elevate serotonin levels, including lithium, monoamine oxidase inhibitors, methyldopa, morphine, and reserpine, or selective serotonin reuptake inhibitors (SSRI, eg, PROZAC) which can lead to depletion of platelet serotonin levels and result in false-negative serotonin results for a minimum of 72 hours before specimen collection. Some drugs with longer half-lives (i.e. fluoxetine) can require months after discontinuation for serotonin levels to return to baseline.

3. Patient should avoid heavy nicotine consumption during the 24-hour collection period.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative at start of collection.

2. Collect urine for a full 24 hours (required) and record the total volume.

3. Refrigerate specimen during collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

84260

SERWB Serotonin, Blood

84373

Specimen Requirements:

Patient Preparation: Patient **should not** take medications that may elevate serotonin levels, including lithium, monoamine oxidase inhibitors, methyldopa, morphine, and reserpine, or selective serotonin reuptake inhibitors (SSRI, eg, PROZAC) which can lead to depletion of platelet serotonin levels and result in false-negative serotonin results for a minimum of 72 hours before specimen collection. Some drugs with longer half-lives (i.e. fluoxetine) can require months after discontinuation for serotonin levels to return to baseline.

Supplies: Serotonin Tube (T259)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Serotonin tube containing ascorbic acid

Specimen Volume: 2.5 mL

Collection Instructions:

1. Immediately after collection, transfer approximately 2.5 mL of whole blood to the serotonin tube and mix well (any volume of whole blood from 1.5-3 mL is acceptable).

2. Immediately freeze specimen (necessary to lyse the red blood cells).

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Frozen	90 days	SEROTONIN TUBE

CPT Code Information:

84260

SER
84395

Serotonin, Serum

Specimen Requirements:

Patient Preparation: Patient **should not** take medications that may elevate serotonin levels, including lithium, monoamine oxidase inhibitors, methyldopa, morphine, and reserpine, or selective serotonin reuptake inhibitors (SSRI, eg, PROZAC) which can lead to depletion of platelet serotonin levels and result in false-negative serotonin results for a minimum of 72 hours before specimen collection. Some drugs with longer half-lives (i.e. fluoxetine) can require months after discontinuation for serotonin levels to return to baseline.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge as soon as blood has clotted and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	4 days

CPT Code Information:

84260

SERPZ
63128

SERPINA1 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days /Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

FSERT
91345

Sertraline (Zoloft) and Desmethylsertraline

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Specimen Type: Plasma

Container/Tube: Green Top

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a green-top (sodium heparin) tube, **plasma gel tube is not acceptable.** Spin down and send 2 mL of sodium heparin plasma refrigerated in a plastic vial.

Serum

Specimen Type: Serum

Container/Tube: Red

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a plain, red-top tube, **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80332

FSESG 57682 Sesame Seed IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SESA 82728 Sesame Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SESPF 621157 Sesame Seed, IgE, with Reflex to Sesame Seed Component, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SCOF3 622027 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS CoV-2) and Influenza Virus Type A and Type B RNA, Molecular Detection, PCR, Varies

Specimen Requirements:

Preferred:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with viral transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4 or M5), saline, or phosphate buffered saline (PBS). Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchoalveolar lavage fluid

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Acceptable:

Specimen Types: Oropharyngeal (throat) swab, nasal mid-turbinate, or nares/nasal swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-Swab, Sterile Polyester (T507)

Collection Container/Tube:

Preferred: BBL Culture Swab, COPAN Mid-turbinate Swab

Acceptable: Dacron-tipped swab with plastic handle

Submission Container/Tube: Sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions: Swab must be placed into viral transport media (eg, M4-RT, M4, or M5), saline, or PBS. Media **should not** contain guanidine thiocyanate (GTC).

Specimen Types: Bronchial washings, endotracheal aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Specimen Minimum Volume: Upper respiratory tract swab: See Specimen Required; lower respiratory specimens: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	14 days
	Refrigerated	72 hours

CPT Code Information:

87636

SFRB 617492 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS CoV-2), Influenza A and B, and Respiratory Syncytial Virus, Molecular Detection, Bronchoalveolar Lavage

Specimen Requirements:

Source: Bronchoalveolar lavage (BAL)

Container/Tube: Sterile container

Specimen Volume: 1 mL

Additional Information: Do not aliquot into viral transport medium.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Bronchoalveolar Lavage	Refrigerated (preferred)	72 hours
	Ambient	24 hours

CPT Code Information:

87637

SCOF4
622031

Severe Acute Respiratory Syndrome Coronavirus 2 (SARS CoV-2), Influenza Virus Type A and Type B RNA, and Respiratory Syncytial Virus (RSV) Molecular Detection, PCR, Varies

Specimen Requirements:

Preferred:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4 or M5), saline, or phosphate buffered saline (PBS). Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchoalveolar lavage fluid

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Acceptable:

Specimen Type: Oropharyngeal (throat) swab, nasal mid-turbinate, or nares/nasal swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-Swab, Sterile Polyester (T507)

Collection Container/Tube:

Preferred: BBL Culture Swab, COPAN Mid-turbinate Swab

Acceptable: Dacron-tipped swab with plastic handle

Submission Container/Tube: Sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions: Swab must be placed into viral transport media (eg, M4-RT, M4, or M5), saline, or PBS. Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchial washings, endotracheal aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Specimen Minimum Volume: Upper respiratory tract swab: See Specimen Required; lower respiratory specimens: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	14 days
	Refrigerated	72 hours

CPT Code Information:

87637

CoV-2), Molecular Detection, Varies

Specimen Requirements:

Preferred:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with viral transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4 or M5), saline, or phosphate buffered saline (PBS). Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchoalveolar lavage fluid

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: **Do not aliquot into viral transport media.**

Acceptable:

Specimen Type: Oropharyngeal (throat) swab, nasal mid-turbinate, or nares/nasal swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-Swab, Sterile Polyester (T507)

Collection Container/Tube:

Preferred: BBL Culture Swab or COPAN Mid-turbinate Swab

Acceptable: Dacron-tipped swab with plastic handle

Submission Container/Tube: Swab in sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of

transport media.

Collection Instructions: Swab must be placed into viral transport media (eg, M4-RT, M4, or M5), saline, or PBS. Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchial washings, endotracheal aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Specimen Minimum Volume: Upper respiratory tract swab: See Specimen Required; lower respiratory specimens: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	14 days
	Refrigerated	72 hours

CPT Code Information:

87635

COVTA 622465 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), Nucleocapsid, Total Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

86769

COVSQ 622462 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), Spike Antibody, Semi-Quantitative, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86769

SCIDP 620134 Severe Combined Immunodeficiency (SCID) Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

SCCNP 619872 Severe Congenital and Cyclic Neutropenia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Blood: 1 mL; Skin biopsy or cultured fibroblasts: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

SCTF 35843 Sex Chromosome Determination, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

SHBG1 Sex Hormone-Binding Globulin, Serum

608102

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

84270

SRYF
35301

Sex-Determining Region Y, Yp11.3 Deletion, FISH

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 to 25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

3. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic

Fluid/Chorionic Villus Sampling, submit a minimum of 12 mL.

4. If ordering with CHRAF / Chromosome Analysis, Amniotic Fluid, submit a minimum of 12 mL.

5. If ordering with both CMAP and CHRAF, submit a minimum of 26 mL.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

4. Specimen cannot be frozen.

Specimen Type: Autopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 4 mm diameter

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. Do not use alcohol or iodine preparations.

4. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Type: Whole blood

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

4. Cord blood is acceptable

Additional Information:

1. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

2. Specimen cannot be frozen.

Specimen Type: Chorionic villus

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 25 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium.

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Type: Fixed cell pellet

Container/Tube: Sterile container with a 3:1 fixative (methanol:glacial acetic acid)

Specimen Volume: Entire specimen

Specimen Type: Products of conception or stillbirth

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 1 cm(3) of placenta (including 20-mg of chorionic villi) **and** a 1-cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions: If a fetus cannot be specifically identified, collect villus material or tissue that appears to be of fetal origin.

Additional Information: Do not send entire fetus.

Specimen Type: Skin biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 4 mm diameter

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. **Do not** use alcohol or iodine preparations.

4. A local anesthetic may be used.

5. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Minimum Volume: Amniotic fluid: 5 mL; Autopsy, skin biopsy: 4 mm; Whole blood: 2 mL; Chorionic villi: 5 mg; Fixed cell pellet: 1 pellet

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88273 w/modifier 52-Chromosomal in situ hybridization, less than 10 cells (if appropriate)

88273-Chromosomal in situ hybridization, 10-30 cells (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

SZDIA
64750

Sezary Diagnostic Flow Cytometry, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA), green top (sodium heparin)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88188-Flow Cytometry Interpretation, 9 to15 markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or more markers (if appropriate)

SZMON
64749

Sezary Monitoring Flow Cytometry, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA), green top (sodium heparin)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88188-Flow Cytometry Interpretation, 9 to15 markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or more markers (if appropriate)

SF1
72121

SF-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SHWL
82747

Sheep Wool, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

STFRP
35148

Shiga Toxin, Molecular Detection, PCR, Feces

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen

to be processed in an environment in which contamination of the specimen by shiga toxin DNA is unlikely.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Preserved feces

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of nonnutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of feces; 5 mL

Collection Instructions:

1. Collect fresh fecal specimen and submit in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

Specimen Stability Information: Ambient (preferred) 7 days/Refrigerated 7 days

Acceptable:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: Representative portion of feces

Collection Instructions: Collect fresh fecal specimen and submit representative sample in fecal container.

Specimen Stability Information: Refrigerated (preferred) 7 days/Frozen 7 days

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies	7 days

CPT Code Information:

87798

SHIGC
606221

Shigella Culture, Feces

Specimen Requirements:

Patient Preparation: The patient **should not** use antacids, barium, bismuth, antidiarrheal medication, or oily laxatives for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-Shigella Culture, Stool-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147 x 4-Serologic Agglut Method 3 Ident (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

SQTSG 617463 Short QT Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies.

Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to

DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81406 x 2

81479

81479 (if appropriate for government payers)

SRW
82667

Short Ragweed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PSRCH
620808

Short Renal Clearance, Iohexol, Preparation

CPT Code Information:

36415 x2-Blood

81050 x3-Urine

96372-Injection

Q9961-Contrast dye

FSHRG
57542

Shrimp IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SHRI
82677

Shrimp, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SAU
620916

Sialic Acid, Free and Total, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Pediatric Volume: 0.5 mL

Collection Instructions: Collect a random urine specimen **(early morning preferred).**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	90 days
	Frozen	90 days

CPT Code Information:

84275

SDEX 9180 Sickle Solubility, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD solution B), green top (heparin)

Specimen Volume: 1 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	14 days

CPT Code Information:

85660

STAT6 70554 Signal Transducer and Activator of Transcription 6 (STAT6), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FSILS
75690

Silicon, Serum

Specimen Requirements:

*****Must submit one specimen per orders. Specimens cannot be shared between multiple orders.*****

Supplies: NMS Acid Washed "Precleaned", 10 mL (T619)

Collection Container/Tube: Royal blue top (no additive, metal-free)(serum gel/SST are **not acceptable**)

Submission Container/Tube: Trace metal-free plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a trace metal-free plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	21 days	Acid Washed Plastic (MML Supply T619)
	Ambient	14 days	Acid Washed Plastic (MML Supply T619)
	Frozen	14 days	Acid Washed Plastic (MML Supply T619)

CPT Code Information:

84285

BIR
82674

Silver Birch, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FSINS
75400

Sinemet, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a plastic vial.

2. **Immediately freeze specimen on dry ice or results will be compromised**.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	6 days

CPT Code Information:

80299

SIIRO 35144 Sirolimus, Whole Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood immediately before a scheduled dose.

2. Do not centrifuge.

3. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Therapeutic range applies to trough specimen collected immediately prior to a.m. dose.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80195

SMAD4 SMAD4 Immunostain, Technical Component Only

613980

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SDLDL
619974

Small Dense Low Density Lipoprotein Cholesterol, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours; required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

83722

SLL
65884

Small Lymphocytic Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. If individual probe sets are chosen: For each probe set ordered, 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If a complete small lymphocytic lymphoma (SLL) panel is ordered, submit 14 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88377-if 1 probe set

88377 x 2-if 2 probe sets

88377 x 3-if 3 probe sets

88377 x 4-if 4 probe sets

88377 x 5-if 5 probe sets

88377 x 6-if 6 probe sets

88377 x 7-if 7 probe sets

88377 x 8-if 8 probe sets

SM
81358

Smith (Sm) Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

DHCRZ
608025

Smith Lemli Opitz Syndrome, DHCR7 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit of only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood**

specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SLO 81595 Smith-Lemli-Opitz Screen, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Lavender top (EDTA), pearl white top (EDTA plasma gel), yellow top (ACD solution A or B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL plasma

Collection Instructions:

1. Centrifuge and aliquot plasma into plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: Plasma: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

82542

SMN1Z SMN1 Gene, Full Gene Analysis, Varies

65941

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2**.** A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81336

88233-Tissue culture, skin, or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SMAS
609515

Smooth Muscle Antibody Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86015

SMAT
608956

Smooth Muscle Antibody Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a reflex. For more information see SMAS / Smooth Muscle Antibody Screen, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86015

SMOTH
70552

Smoothelin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

FCRNS 57961 Smut Corn (Ustilago maydis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SNAIL
82344

Snail, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SOD1Z
617740

SOD1 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have

received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin

and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81404

NAU
610734

Sodium, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84300

RNAUR
610785

Sodium, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84300

NAS
602353

Sodium, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Ambient	14 days

CPT Code Information:

84295

SOLEF
86310

Sole, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FSFM
58015

Soluble Fibrin Monomer

Specimen Requirements:

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Platelet-poor plasma

Collection Instructions:

1. Centrifuge at 1700 x *g* for 15 minutes.

2. Aliquot 1.5 platelet-poor plasma into plastic vial making sure not to disturb the buffy coat.

3. Send frozen.

Note: **Critical frozen**. **Separate specimens must be submitted** when multiple tests are ordered.

Specimen Minimum Volume: Platelet-poor plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85366

SOLFM 602175 Soluble Fibrin Monomer, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85366

SLA 624092

Soluble Liver Antigen (SLA) IgG Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

SLC1Q
610061 Solute Carrier Organic Anion Transporter Family Member 1B1 (SLCO1B1) Genotype, Statin, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant or liver transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell or liver transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81328

FSOM1 Somatostatin (Somatotropin Release-Inhibiting Factor, SRIF)
75906

Specimen Requirements:

Patient Preparation:

1. Fasting: 10 to 12 hours

2. Patient **should not** be on any medications that affect insulin secretion or intestinal motility, if possible, for at least 48 hours prior to collection.

Supplies: GI Preservative (T125)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: GI preservative tube

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL plasma

Collection Instructions:

1. Centrifuge as soon as possible and aliquot 3 mL of plasma into a plastic vial.

2. Freeze plasma immediately after separation

3. Send frozen.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	180 days
	Refrigerated	7 days

CPT Code Information:

84307

SOMAT 70553 Somatostatin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SSTR2 113597 Somatostatin Receptor 2 (SSTR2), Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SORBU 614934 Sorbitol and Mannitol, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82542

SXDBS 623688 Sorbitol and Xylitol, Quantitative, Blood Spot

Specimen Requirements:

Fasting: 8 hours, required. Infants **should not fast** overnight; collect just before next feeding.

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Card-Blood Spot Collection (Filter Paper)

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing EDTA (preferred) or sodium heparin and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Blood Spot	Ambient (preferred)	31 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Refrigerated	90 days	FILTER PAPER

CPT Code Information:

82542

SORD
620920

Sorbitol and Xylitol, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Refrigerate immediately.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82542

SORDB
623501

Sorbitol and Xylitol, Quantitative, Whole Blood

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin)

Specimen Volume: 1 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Freeze whole blood specimens in the original tube. Frozen aliquots from well-mixed specimens are also acceptable.

3. Send frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Frozen (preferred)	90 days
	Refrigerated	31 days

CPT Code Information:

82542

FSOTA
91123

Sotalol (Betapace)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days

	Ambient	72 hours

CPT Code Information:

80299

SOX10 SOX10 Immunostain, Technical Component Only
70555

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SOX11 SOX11 Immunostain, Technical Component Only
70556

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SOX9
623382

SOX9 (SRY-Box Transcription Factor) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

FSOYG 57551 Soybean IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FSYG4 Soybean IgG4
57574

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SOY Soybean, IgE, Serum
82886

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PBC2
620736

SP100 and GP210 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.0 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516 x 2

SP100
620725

SP100 Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

SPAGR
113392

Special Red Cell Antigen Typing, Whole Blood

Specimen Requirements:

Container/Tube:

Preferred: Pink top (EDTA)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Pediatric Volume: 3 mL Blood in pink-top or lavender-top (EDTA) tube

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	72 hours

CPT Code Information:

86905-Each red cell antigen typing (if more than one ordered)

SGUR
606565

Specific Gravity, Random, Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 20 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

81005

FSPNG
57678

Spinach IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SPIN
86312

Spinach, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

SMNCS 65574 Spinal Muscular Atrophy Carrier Screening, Deletion/Duplication Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For more information about testing patients who have received a hematopoietic stem cell, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the

cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) Ambient/Refrigerated <24hours

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81329

88233 (if appropriate)

88240 (if appropriate)

SMNDX 65575 Spinal Muscular Atrophy Diagnostic Assay, Deletion/Duplication Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Direct testing of an uncultured specimen may be attempted for this test. Contact the laboratory at 800-533-1710 if direct testing is desired.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81329

88235 (if appropriate)

88240 (if appropriate)

88233 (if appropriate)

88240 (if appropriate)

81265 (if appropriate)

SBULB 35542 Spinobulbar Muscular Atrophy (Kennedy Disease), Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone

marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic,

Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81204-*AR (*androgen receptor)(eg, spinal and bulba muscular atrophy, Kennedy disease, X chromosome inactivation) gene analysis; characterization of alleles (eg, expanded size or methylation status)

SCAP 609505 Spinocerebellar Ataxia Repeat Expansion Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4

days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81178

81179

81180

81181

81184

81479 (if appropriate for government payers)

SCARA 609698 Spinocerebellar Ataxia Type 1, 2, 3, 6, or 7, Repeat Expansion Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal

testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional information:

1. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

2. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated

Additional Information:

1. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

2. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information: All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Amniotic fluid: 10 mL
Blood: 0.5 mL
Chorionic villi: 5 mg

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

88233-Fibroblast Culture (if appropriate)

88235-Amniotic Fluid Culture (if appropriate)

88240-Cryopreservation (if appropriate)

81265-Maternal Cell Contamination (if appropriate)

81178 (if appropriate)

81179 (if appropriate)

81180 (if appropriate)

81181 (if appropriate)

81184 (if appropriate)

SFGP 83679 Spotted Fever Group Antibody, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86757 x 2

SPRU
82394

Spruce, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SCCA
610033

Squamous Cell Carcinoma Antigen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial. **Do not submit in original tube**.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86316

SQUA
82797

Squash, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SQUID
82631

Squid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SSAB
82403

SS-A and SS-B Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235 x 2

SSA
81360

SS-A/Ro Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

SSB
81359

SS-B/La Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

STLPC 83916 St. Louis Encephalitis Antibody Panel, IgG and IgM, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ABOPC / Arbovirus Antibody Panel, IgG and IgM, Spinal Fluid.

Container/Tube: Sterile vial

Preferred: Vial number 2

Acceptable: Any vial

Specimen Volume: 0.7 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86653 x 2

FSTAB
57891

Stachybotrys chartarum/atra IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

STRC
83316

Standard Renal Clearance, Plasma and Random Urine

Specimen Requirements:

Contact MLI for approval prior to ordering this test.

Both plasma and urine are required.

Specimen Type: Plasma

Container/Tube: 6-mL Green top (sodium heparin)

Specimen Volume: 3-4 mL

Collection Instructions: Label specimen as plasma.

Specimen Type: Urine

Container/Tube: Sarstedt 5 mL Aliquot Tube (T914)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine collection.

2. No preservative.

3. Label specimen as urine.

Specimen Minimum Volume: 2 mL plasma/5 mL urine

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Refrigerated (preferred)	7 days
	Frozen	7 days
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

82542 x 7

STEM
82696

Stemphyllium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

STER
82079

Sterols, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Lavender top (EDTA), pearl white top (EDTA plasma gel), yellow top (ACD solution A or B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL plasma

Collection Instructions:

1. Centrifuge specimen and aliquot plasma into plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: Plasma: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

82542

STFR1 sTfR (Soluble Transferrin Receptor), Serum
623578

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes, aliquot serum into a plastic vial prior to sending.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

84238

SPPS Stiff-Person Spectrum Disorders Evaluation, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Serum
614599

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 2

86341

0431U

84182 (if appropriate)

86256 (if appropriate)

86256 (if appropriate)

SPPC
614600

Stiff-Person Spectrum Disorders Evaluation, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume:

2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 2

86341

0431U

84182 (if appropriate)
86256 (if appropriate)

86256 (if appropriate)

SPPCI
614602

Stiff-Person Spectrum Disorders including Progressive Encephalomyelitis with Rigidity and Myoclonus, Interpretation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

SPPSI
614601

Stiff-Person Spectrum Disorders, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Interpretation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

INSEC
31765

Stinging Insects Allergen Profile, Serum

Specimen Requirements:

Supplies:

Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

FSTBG 57656 Strawberry IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

STBY
82676

Strawberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SABP
86537

Streptococcal Antibodies Profile, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

86060

86215

GAPCR 610254 Streptococcus Group A, Molecular Detection, PCR, Throat

Specimen Requirements:

Patient Preparation: Do not collect throat swab specimens immediately after use of antiseptic mouthwash.

Specimen Type: Throat swab

Supplies: BD E-Swab (T853)

Container/Tube:

Preferred: BD Liquid Amies Elution Swab (ESwab)

Acceptable: Copan Liquid Amies Elution Swab (ESwab)

Specimen Volume: Entire collection/1 ESwab

Collection Instructions:

1. **Must be collected and transported using the ESwab collection kit.**

2. Using a tongue blade to hold the tongue down, firmly rub the swab over the tonsils, posterior pharynx, and any areas of inflammation with exudate. Ensure that areas with exudate or inflammation are swabbed.

3. Avoid the tongue, teeth, and gums as the swab is withdrawn.

4. Insert the swab back into the ESwab transport tube and break the swab at the indicated score line.

5. Label specimen according to policy.

6. Send refrigerated. **Do not freeze**.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	6 days
	Ambient	48 hours

CPT Code Information:

87651

SPNEU Streptococcus pneumoniae Antigen, Random, Urine

83150

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Excessively bloody or very turbid specimens containing protein, cells, or particulates will be canceled, as they can inhibit the function of the test.

4. **Do not centrifuge to remove particulates**.

5. Specimens with any dyes or unnatural color are not acceptable and will be canceled.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	24 hours

CPT Code Information:

87899

SPNC
89971

Streptococcus pneumoniae Antigen, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen collected in vial 2, if possible. If not possible, note the vial from which the aliquot was obtained.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899

PN23M
619921

Streptococcus pneumoniae IgG Antibodies, 23 Serotypes, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86581

PNTO
608969

Streptococcus pneumoniae IgG Antibodies, Total, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86317

PNTOR 608970 Streptococcus pneumoniae IgG Antibodies, Total, with Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86317

86317 x 23 (if appropriate)

MSTC 80749 Strict Criteria Sperm Morphology for Infertility Diagnosis and Treatment, Semen

Specimen Requirements:

Container/Tube: Semen Analysis Kit (T178)

Collection Instructions: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Specimen Volume: Total ejaculate

Specimen Minimum Volume: A minimum count is needed. Lab will determine.

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89398

MSTC1 35184 Strict Criteria Sperm Morphology for Infertility Diagnosis and Treatment, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Container/Tube: Slides

Specimen Volume: 2 slides-10 microL of liquefied semen on each slide

Collection Instructions:

1. If sperm concentration is <10 million/mL, centrifuge the specimen at 300 x G for 10 minutes before making slides.

2 Label 2 frosted slides **in pencil** with the patient's first and last name and the date of specimen collection. No adhesive labels.

3. Allow the semen to liquefy for 30 minutes.

4. Place 10 mcL of liquefied semen on the label end of each slide, and evenly smear the specimen using a plain slide (this process is the same as making a blood smear).

5. Allow the smears to air dry for 15 minutes before placing both slides into 1 slide holder for shipment.

Specimen Minimum Volume: A minimum count of sperm is needed; lab will determine

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89398

STRNG
63866

Strongyloides Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86682

FSTYR
91094

Styrene, Occupational Exposure, Blood

Specimen Requirements:

Collect 2 tubes green-top (sodium heparin) whole blood. Send 20 mL sodium heparin whole blood refrigerated. Collect specimen at end of shift or prior to next shift. Tubes should be filled to prevent loss of volatile compound into headspace.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
WB Sodium Heparin	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

84600

SUAC 83635 Succinylacetone, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood Spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tube containing EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 1 spot should be complete (ie, unpunched).

3. Let blood dry on the Blood Spot Collection Card (T493) at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 7 days/Refrigerated 14 days/Frozen 90 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate 6 days

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

84510

82542

82542 (if appropriate for government payers)

SUDC
606930

Sudden Cardiac Death Pathology Consultation

Specimen Requirements:

1. The heart specimen is required for testing.

2. Send whole blood in addition to heart specimen, if available.

Specimen Type: Heart specimen

Supplies:

-Pathology Packaging Kit (T554)

-Ambient Shipping Box, 5 lb (T970)

Specimen Volume: Entire specimen

Collection Instructions:

1. Fix entire specimen in formalin for a minimum of 24 hours.

2. Following fixation, lightly wrap specimen in formalin-soaked paper towel, place in a leak-proof sealable bag with a small amount of formalin, seal, and ship ambient in a sturdy shipping container.

Additional Information: Paraffin block may be accepted.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen.

Specimen Minimum Volume: Heart specimen: See Specimen Required; Blood: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
SUDC Study Specimen	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

88037

FSCNE 57543 Sugar Cane (Saccharum officinarum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SBSE
82382

Sugarbeet Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SFZ
8238

Sulfamethoxazole, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (gel tubes/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum for a peak level should be collected 60 minutes after dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

SULFU
606479

Sulfate, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL urine tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Collection instructions with **no preservative:**

-Specimen must be kept refrigerated during and after collection.

3. Collection instructions **with preservative:**

a. If preservative is **not** added **before** collection, the specimen **must be kept refrigerated until preservative is added**.

b. Preservative must be added **within 4 hours** of the end of collection.

4. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH >8 indicate bacterial contamination, and testing will be cancelled. Do not attempt to adjust pH as it will adversely affect results.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

84392

FSLFU 57710 Sulfonylurea Screen, Urine

Specimen Requirements:

Collection Container: Plastic urine container

Specimen Volume: 5 mL

Collection Instructions:

1. Collect 5 mL random urine without preservative.
2. Ship refrigerated in a plastic container.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80377

FSUNG
57681

Sunflower Seed IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SUNFS
82813

Sunflower Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SUNF
82615

Sunflower, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SUP24 Supersaturation Profile, 24 Hour, Urine

616180

Specimen Requirements:

Patient Preparation:

X-ray dyes and contrast media will affect uric acid test results.

-If a kidney X-ray with dye or computerized tomography (CT) scan with contrast has been performed, patient should wait a minimum of 1 day before starting collection.

-If a cholangiography (bile duct X-ray) has performed, patient should wait 7 days before starting collection.

-Urine must be collected before tablets have been taken for gallbladder X-ray, otherwise patient should wait 7 days before starting collection.

Supplies: Diazolidinyl Urea (Germall) 5.0 mL (T822)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 35 mL

Collection Instructions:

1. Add 5 mL of diazolidinyl urea as preservative at start of collection or refrigerate specimen during and after collection.

2. Collect urine for 24 hours.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if diazolidinyl urea is added at the beginning of the collection or kept refrigerated during the entirety of the collection. Specimens with pH above 8 indicate bacterial contamination, and testing will be canceled. Do not attempt to adjust pH as it will adversely affect results.

4. **If not using Germall, the specimen must be kept refrigerated** during the entirety of the collection and sent frozen.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 16 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82340-Calcium

82436-Chloride

82507-Citrate excretion

82570-Creatinine

83735-Magnesium

83935-Osmolality

83945-Oxalate

83986-pH

84105-Phosphorus

84133-Potassium

84300-Sodium

84392-Sulfate

84560-Uric acid

82140-Ammonium

84540-Urea Nitrogen

SUPRA 616375 Supersaturation Profile, Random, Urine

Specimen Requirements:

Supplies:

-Urine Tubes, 10 mL (T068)

-Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: 90 mL urine container

Submission Container/Tube: 2 x 10 mL plastic urine tubes and 5 x 5 mL plastic urine tubes

Specimen Volume: 40 mL

Collection Instructions:

1. Collect a random urine specimen, mix collection container thoroughly and divide the urine into 7 plastic tubes (2 x 10 mL and 5 x 5 mL).

2. Refrigerate specimen after collection. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 indicate bacterial contamination, and testing will be canceled. **Do not** attempt to adjust pH as it will adversely affect results.

Specimen Minimum Volume: 16 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82310-Calcium

82436-Chloride

82507-Citrate excretion

82570-Creatinine

83735-Magnesium

83935-Osmolality

83945-Oxalate

83986-pH

84105-Phosphorus

84133-Potassium

84300-Sodium

84392-Sulfate

84560-Uric acid

82140-Ammonium

SSH24 Supersaturation Profile, Self-Collect, 24 Hour, Urine

616228

Specimen Requirements:

Patient Preparation:

X-ray dyes and contrast media will affect uric acid test results.

-If a kidney X-ray with dye or computerized tomography (CT) scan with contrast has been performed, the patient should wait at least 24 hours before starting collection.

-If a cholangiography (bile duct X-ray) has been performed, the patient should wait 7 days before starting collection.

-If patient has taken tablets for a gallbladder X-ray, the patient should wait 7 days before starting collection.

Supplies: SSH24 Home Collection Kit (T921)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 35 mL

Collection Instructions:

1. Order test each time the patient is to collect a 24-hour urine specimen at home and mail the specimen directly to Mayo Clinic Laboratories.

2. Order should be placed a minimum of 3 days prior to desired date of collection.

3. Enter patient's address information for each order created, including street address, city, state abbreviation, zip code, country, and home phone number.

4. For each order, the SSH24 Home Collection kit will be mailed directly to the patient for self-collection. This kit contains instructions for how to properly collect and transport the specimen after collection.

Specimen Minimum Volume: 16 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82340-Calcium

82436-Chloride

82507-Citrate excretion

82570-Creatinine

83735-Magnesium

83935-Osmolality

83945-Oxalate

83986-pH

84105-Phosphorus

84133-Potassium

84300-Sodium

84392-Sulfate

84560-Uric acid

82140-Ammonium

84540-Urea Nitrogen

SNS
82594

Supplemental Newborn Screen, Blood Spot

Specimen Requirements:

Patient must be older than 24 hours and less than 1 week of age.

Supplies: Card-Blood Spot Collection Filter Paper (T493)

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, Munktell, PerkinElmer 226 filter paper, or local newborn screening card

Specimen Volume: 3 Blood spots

Collection Instructions:

1. Do not use device or capillary tube containing EDTA to collect specimen.

2. Completely fill at least 3 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the Blood Spot Collection Card at ambient temperature in a horizontal position for 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)		FILTER PAPER
	Frozen		FILTER PAPER
	Refrigerated		FILTER PAPER

CPT Code Information:

83789

TB2LN 34550 Susceptibility, Mycobacterium tuberculosis Complex, Second Line, Varies

Specimen Requirements:

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook 7H10 agar slant

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87186-Susceptibility, Mtb Cx, 2nd Line

SV40
618827

SV40 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SGUM
82483

Sweet Gum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SPOT
82799

Sweet Potato, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

VERG
82909

Sweet Vernal Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HBRM
622927

SWI/SNF Chromatin Remodeling Complex Subunit ATPase 2 (SMARCA2 or BRM) Immunostain, Technical Component

Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

SWORD
82346

Swordfish, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SYNAP
70557

Synaptophysin (SYNAPTO) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SS18F 35303 Synovial Sarcoma (SS), 18q11.2 (SS18 or SYT) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

SSX18 Synovial Sarcoma Panel, Technical Component Only

610609

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FSCBB 75967 Synthetic Cannabinoid Metabolites Confirmation (Qualitative)-Expanded, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see FSCBA / Synthetic Cannabinoid Metabolites Screen-Expanded, Urine

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80352

FSCBA 75966 Synthetic Cannabinoid Metabolites Screen-Expanded, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic, preservative-free urine container

Specimen Volume: 3 mL

Collection Instructions:

1. Collect 3 mL random urine specimen without preservative.

2. Send refrigerated in a plastic preservative-free urine container.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80307

SGSU
81035

Synthetic Glucocorticoid Screen, Random, Urine

Specimen Requirements:

Supplies: Urine tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	14 days

CPT Code Information:

80299

SGSS
81031

Synthetic Glucocorticoid Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

80299

TPPA 61480 Syphilis Antibody, Treponema pallidum Particle Agglutination, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86780

NSYPH 616862 Syphilis IgG Enzyme Immunoassay, Serum

Specimen Requirements:

Collection Container/Tube: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86780

SYPH1 616860 Syphilis IgG with Reflex, Enzyme Immunoassay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86780-Syphilis IgG Screen

86592-Syphilis Rapid Plasma Reagin Screen (if appropriate)

86593-Rapid Plasma Reagin Titer (if appropriate)

86780-Syphilis Antibody by TP-PA (if appropriate)

SYPHT
603264

Syphilis Total Antibody with Reflex, Serum

Specimen Requirements:

Collection Container/Tube: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86780

86780-Syphilis Antibody by TP-PA (if appropriate)

86593-Rapid Plasma Reagin Titer (if appropriate)

SSCP
622520

Systemic Sclerosis Criteria Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86039

86235

83516

TBNY
82589

T, B and NK Lymphocyte Quantitation, New York, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	52 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86355

86357

86359

86360

TBET
70559

T-Box Expressed in T Cells (TBET) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

COGTF 113529 T-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), Children's Oncology Group Enrollment Testing, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x2, 88275 x1, 88291 x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271 x2, 88275x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

TALAF 609557 T-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), FISH, Adult, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x20, 88275x10, 88291x1-FISH Probe, Analysis, Interpretation; 10 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

TALMF 614323 T-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275x1, 88291x1-FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

BTIA1
605160

T-Cell Intracellular Antigen 1 (TIA-1) Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TIA1
70566

T-Cell Intracellular Antigen 1 (TIA-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TCL1A
70561

T-Cell Leukemia/Lymphoma Protein 1A (TCL1A) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

TLBLF 65413 T-Cell Lymphoblastic Leukemia/Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. If individual probe sets are chosen: For each probe set ordered, submit 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If a complete T-lymphoblastic leukemia/lymphoma (TLBL) panel is ordered: submit 16 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

TLPFD 622430 T-Cell Lymphoma Panel, Diagnostic, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2-3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x8, 88275 x4, 88291 x1- FISH Probe, Analysis, Interpretation; 4 probe set

88271 x2, 88275 x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

TLYM 65911 T-Cell Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. For each probe set ordered, 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If ordering TCL1A or TRA, 4 unstained slides are necessary; the break-apart TCL1A and TRA break-apart probe sets are performed simultaneously.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377 (if 1 probe set)

88377 x 2 (if 2 probe sets)

88377 x 3 (if 3 probe sets)

88377 x 4 (if 4 probe sets)

88377 x 5 (if 5 probe sets)

88377 x 6 (if 6 probe sets)

88377 x 7 (if 7 probe sets)

88377 x 8 (if 8 probe sets)

TLPMF
614346

T-Cell Lymphoma, Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole Blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275 x1, 88291x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275 x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

TCRF1 70560 T-Cell Receptor Beta (TCR Beta F1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRBC3 622845 T-Cell Receptor Constant Beta Chain (TRBC1) +CD3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

TRBC1 622852 T-Cell Receptor Constant Beta Chain (TRBC1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TCRGD 70562 T-Cell Receptor Delta Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRECS 610068 T-Cell Receptor Excision Circles Analysis, Blood

Specimen Requirements:

For serial monitoring, it is recommended to perform specimen collection at the same time of day, if possible.

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Lavender top (EDTA)

Specimen Volume:

Adults: 10 mL

Pediatrics

-Preferred volume for >1 year: 5 mL

-Preferred volume for < or =1 year old: 3 mL

Collection Instructions:

1. Do not collect specimen using a butterfly needle.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Adults: 10 mL
Pediatrics: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81479-Unlisted molecular pathology procedure

TCGR
83122

T-Cell Receptor Gene Rearrangement, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCGBM T-Cell Receptor Gene Rearrangement, PCR, Bone Marrow

31139

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCGET
802124

T-Cell Receptor Gene Rearrangement, PCR, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue slides

Slides: 4 unstained

Collection Instructions: Submit 4 consecutive, positively charged, unbaked, unstained slides with 10-micron thick sections

Acceptable

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block containing neoplastic cells

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCGRV
31140

T-Cell Receptor Gene Rearrangement, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

3. Include pathology report.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 Unstained slides

Container/Tube: Transport in plastic slide holders

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

4. Include pathology report.

Specimen Stability Information: Ambient

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Sources: Pleural, peritoneal, vitreous and spinal fluid

Container/Tube: Sterile container

Specimen Volume: At least 5 mL

Collection Instructions:

1. If the volume is large, pellet cells prior to sending.

2. Send less volume at ambient temperature or as a frozen cell pellet.

3. Specify the type of fluid being submitted.

Specimen Stability Information:

Body fluid: Ambient 4 days/Refrigerated/Frozen

Cell pellet: Frozen

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection.

2. Indicate specimen source.

Specimen Stability Information: Frozen

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods

are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Body fluid: 1 mL; Frozen tissue: 50 mg; Extracted DNA: 50 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s), using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCRB 610067 T-Cell Receptor V-Beta Repertoire Analysis, Spectratyping, Blood

Specimen Requirements:

For serial monitoring, it is recommended to perform specimen collection at the same time of day, if possible.

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Specimen Type: Blood

Container/Tube: Lavender top (EDTA)

Specimen Volume:

Adults: 10 mL

Pediatrics:

-Preferred volume for >1 year: 3 mL

-Preferred volume for < or =1 year: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Adults: 5 mL

Pediatrics: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81340-TRG (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, polymerase chain reaction)

TCP
89319

T-Cell Subsets, Naive, Memory, and Activated, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x 7

TREGS
89318

T-Cell Subsets, Regulatory (Tregs), Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86359

86361

CYTH1 T-Helper Cell Type 1 Cytokine Panel, Plasma

622036

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Within 2 hours of collection, centrifuge at 4 degrees C, 1500 x *g* for 10 minutes and aliquot plasma into a plastic vial

3. Freeze immediately.

Additional Information:

1. Plasma for this test cannot be shared with other tests. Submit specimen for this test in its own vial.

2. This test cannot be added on to a previously collected specimen.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

83520 x5

CYTH2 T-Helper Cell Type 2 Cytokine Panel, Plasma
622929

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Within 2 hours of collection, centrifuge at 4 degrees C, 1500 x *g* for 10 minutes and aliquot plasma into a plastic vial.

3. Freeze immediately.

Additional Information:

1. **Plasma for this test cannot be shared with other tests**. Submit specimen for this test in its own vial.

2. **This test cannot be added onto a previously collected specimen**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

83520 x4

TPIT T-PIT Immunostain, Technical Component Only
607889

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

T3FR
621321

T3 (Triiodothyronine), Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

84481

RT3
9405

T3 (Triiodothyronine), Reverse, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

84482

T3
8613

T3 (Triiodothyronine), Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

84480

FRT3S
75752

T3 Reverse, LC/MS/MS

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Allow blood to clot at room temperature for 30 minutes.

2. Within 24 hours of collection, centrifuge and aliquot serum immediately into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84482

FRT4D
8859

T4 (Thyroxine), Free, Dialysis, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.6 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	21 days
	Ambient	7 days

CPT Code Information:

84439

FRT4
8725

T4 (Thyroxine), Free, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84439

T4FT4
36108

T4 (Thyroxine), Total and Free, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.625 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84436

84439

T4
8724

T4 (Thyroxine), Total Only, Serum

Specimen Requirements:

Patient Preparation: For 4 to 6 weeks before specimen collection, patient should not be taking lipid-lowering medications containing dextrothyroxine.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

84436

FFT4F
91548

T4, Free, Direct Dialysis

Specimen Requirements:

Patient Preparation:

Fasting: 8-16 hours, preferred but not required

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collections Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	35 days
	Ambient	4 days

CPT Code Information:

84439

TAKRO Tacrolimus, Blood
35145

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood immediately before a scheduled dose.

2. Do not centrifuge.

3. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Therapeutic range applies to trough specimens collected

immediately prior to a.m. dose.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80197

TACPK 88157 Tacrolimus, Peak, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Do not centrifuge.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80197

TAPEN 62594 Tapentadol and Metabolite, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

G0480

80372 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TAPNS 624021 Tapentadol and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.0 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80372 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FIOCA
57944

Tapioca IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days

	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

TBSU
604262

Targeted Benzodiazepine Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

G0480

80347 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TOSU
604261

Targeted Opioid Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

G0481
80364 (if appropriate for select payers)

TSPU
610272

Targeted Stimulant Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	14 days
	Frozen	28 days

CPT Code Information:

G0480

80326 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TARR
82486

Tarragon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TRAP
70570

Tartrate-Resistant Acid Phosphatase (TRAP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TAU3
70628

TAU 3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TAU4
70627

TAU 4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TAUI
70558

TAU Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HEXAN
622610

Tay-Sachs Disease, HEXA Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic

donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FGTEA
57684

Tea IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

TEA
82625

Tea, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TELDP
619886

Telomere Biology Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing, Chorionic Villi/Products of Conception/Tissue. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. **Cultured cells from a prenatal specimen will not be accepted**.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing, Chorionic Villi/Products of Conception/Tissue. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Whole blood: 1 mL; Skin biopsy or cultured fibroblasts: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

TEMZS
623017

Temazepam and Oxazepam, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

TDT
70563

Terminal Deoxynucleotidyl Transferase (TdT) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TERTD
618208

TERT Promoter Mutation Analysis, Droplet Digital PCR, Tumor

Specimen Requirements:

This assay requires at least 5% nuclei of tumor cells/cells of interest.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue144 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent nuclei of tumor cells/cells of interest.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-Cytology fixatives: Cytology smears fixed in alcohol and thin preps fixed with CytoLyt.

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81345

88381-Microdissection, manual

FFTFT 75751 Testosterone, Free (Dialysis) and Total, MS

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 1.8 mL

Collection Instructions:

1. Allow blood to clot at room temperature. Centrifuge and immediately aliquot 1.8 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Stability Information: Refrigerated (preferred) 21 days/Ambient 7 days/Frozen 2 years

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	21 days
	Ambient	7 days
	Frozen	

CPT Code Information:

84402

84403

FFTST 75642 Testosterone, Free and Weakly Bound, With Total Testosterone, LC/MS-MS

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Within one hour of collection, centrifuge and aliquot 2 mL serum into a plastic vial.

2. Ship refrigerated.

Specimen Minimum Volume: 1.4 mL (does not allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

84410

FFBTT
75934

Testosterone, Free, Bioavailable, and Total, MS

Specimen Requirements:

Caution: Administration of STRENSIQ may interfere in certain assays and may falsely elevate values. For patients receiving STRENSIQ, consideration should be given to using alternate methods.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.8 mL Serum

Collections Instructions:

1. Allow blood to clot at room temperature.

2. Centrifuge and aliquot 2.8 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: Serum: 1.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82040

84270

84403

TTBS
80065

Testosterone, Total and Bioavailable, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84403

84410

TGRP
8508

Testosterone, Total and Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84402

84403

TTFB
83686

Testosterone, Total, Bioavailable, and Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84402

84403

84410

TTST
8533

Testosterone, Total, Mass Spectrometry, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.215 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84403

TTIGS
36667

Tetanus Toxoid IgG Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86317

TGOGF 621841 TGFBR3 (1p22), OGA (10q24) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 6 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 6 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291 DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

THEV1 608085 Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

Specimen Requirements:

Blood and serum are required.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 15 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Type: Serum

Patient Preparation: For 12 hour before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Within 2 hours of collection, serum gel tubes should be centrifuged.

2. Within 2 hours of collection, red-top tubes should be centrifuged and the serum aliquoted into a plastic vial.

3. Label specimen as serum.

Specimen Minimum Volume: Blood: 2.5 mL
Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	7 days
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

83020-26-Hemoglobinopathy Interpretation

83020-Hb Variant, A2 and F Quantitation

83021

82728

82664 (if appropriate)

83068 (if appropriate)

83789 (if appropriate)

88184 (if appropriate)

THEV0 608092 Thalassemia Summary Interpretation, Blood (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	

TLU 8603 Thallium, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to

potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium or iodine containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. Aliquot 10 mL into a plastic 10-mL urine tube (T068) or a clean, plastic aliquot container with no metal cap or glued insert.

4. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: For multiple collections, see Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

TLB
8149

Thallium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based

metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD Vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

TLUCR 615255 Thallium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap

or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 7 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.3mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83018

82570

FHCCF
75831

THC Confirmation, MS, SP

Specimen Requirements:

Submit only 1 of the following specimens:

Serum:

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 5mL of serum refrigerated in plastic vial.

Plasma:

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable**. Spin down and send 5 mL sodium heparin plasma refrigerated in plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

80349

G0480 (if appropriate)

FMARX THC, MS, WB/SP Rfx (Non-Orderable)
75832

Specimen Requirements:

Only orderable as a reflex from FMARI

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

THEO Theophylline, Serum
8661

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, serum may sit on gel refrigerated or aliquoted into a plastic vial.

3. For red top tubes, aliquot serum immediately into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80198

TDP 42356 Thiamine (Vitamin B1), Whole Blood

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required; Infants should have specimen collected before next feeding, water can be taken as needed

2. For 12 hours before specimen collection, patient **should not** take vitamin supplements.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Amber vial

Specimen Volume: 4 mL

Collection Instructions:

1. Invert 8 to 10 times to mix whole blood.

2. Transfer whole blood into amber vial or tube and freeze within 24 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Frozen	28 days	LIGHT PROTECTED

CPT Code Information:

84425

DTHPV 70336 ThinPrep Diagnostic with Human Papillomavirus (HPV) Reflex, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume=4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

88141(if appropriate)

TPRPD 70333 ThinPrep Diagnostic, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Pap smears should be collected near the middle of the menstrual cycle. Avoid douching, lubricant use, or sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in on direction. **Do not over-rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

88141 (if appropriate)

STHPV 70335 ThinPrep Screen with Human Papillomavirus (HPV) Reflex, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Supplies: Thin Prep Media with Broom Kit (T056)

Specimen Type: Cervical

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Specimen Type: Ectocervix and endocervix

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one quarter or one half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

G0123

88142

88141 (if appropriate)

TPRPS ThinPrep Screen, Varies
70332

Specimen Requirements:

Patient Preparation: For optimal interpretation, Pap smears should be collected near the middle of the menstrual cycle. Avoid douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm to warm water and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm to warm water and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in one direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

G0123 (Government Payers)

88142

88141-TPSPC (if appropriate)

DTPCO 70338 ThinPrep with Human Papillomavirus (HPV) Co-Test-Diagnostic, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: ThinPrep Media with Broom Kit (T056)

Collection Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: ThinPrep Media with Spatula and Brush Kit (T434)

Collection Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

G0123 (for government payers)

87626 (if appropriate)

87626 (if appropriate)

88141 (if appropriate)

CTPCO 616073 ThinPrep with Human Papillomavirus (HPV) Co-Test-Screen with p16/Ki67 Dual Stain Reflex, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou (Pap) smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, or sexual intercourse for 24 hours before specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep/PreservCyt vial

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the broom collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they tend to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep/PreservCyt vial

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire ectocervix while maintaining tight contact with ectocervical surface.

2. Rinse spatula as quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over-rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they tend to leak during transport.

9. Place labels on the vial and bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

G0123 (Government payers)

G0124 (Government payers, if appropriate)

88141 -TPSPC (if appropriate)

88344 -CINTC/CINPC (if appropriate)

STPCO 70337 ThinPrep with Human Papillomavirus (HPV) Co-Test-Screen, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

G0123 (for government payers)

87626 (if appropriate)

87626 (if appropriate)

88141 (if appropriate)

FFTIO
57708

Thiocyanate, Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 1 mL sodium heparin plasma refrigerated in plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 1 mL of serum refrigerated in plastic vial.

Specimen Minimum Volume: 0.50 mL Does not allow for repeat testing

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

84430

THIO
65381

Thiopurine Metabolites, Whole Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not** aliquot, centrifuge, or freeze.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	8 days
	Ambient	24 hours

CPT Code Information: 80299

TPNUQ
610062 Thiopurine Methyltransferase (TPMT) and Nudix Hydrolase (NUDT15) Genotyping, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

0034U

TPMT3 65188 Thiopurine Methyltransferase Activity Profile, Erythrocytes

Specimen Requirements:

Patient Preparation: Thiopurine methyltransferase (TPMT) enzyme activity can be inhibited by several drugs and may contribute to falsely low results. Patients should abstain from the following drugs for at least 48 hours prior to TPMT testing: naproxen (Aleve), ibuprofen (Advil, Motrin), ketoprofen (Orudis), furosemide (Lasix), sulfasalazine (Azulfidine), mesalamine (Asacol), olsalazine (Dipentum), mefenamic acid (Ponstel), trimethoprim (Proloprim), methotrexate, thiazide diuretics, and benzoic acid inhibitors.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin), dark blue top (metal free sodium heparin), or non-centrifuged plasma gel tubes

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen. **Do not centrifuge**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

84433

FSULU
75890

Thiosulfate, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 4 mL

Collection Instructions:

1. Collect 4 mL random urine specimen in a plastic preservative-free urine container.

2. Send refrigerated.

Specimen Minimum Volume: 1.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days

	Frozen	14 days

CPT Code Information:

82542

82570

81002 (if appropriate)

TTSC 602184 Thrombin Time (Bovine), Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85670

FATCX
75958

Thrombin-Antithrombin Complex

Specimen Requirements:

Collection Container/Tube: Blue top (3.2% buffered sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Do not draw from an arm with a heparin lock or heparinized catheter.

2. Draw blood in a blue-top tube containing 3.2% buffered sodium citrate.

3. Mix immediately by gently inverting at least 6 times.

4. Centrifuge, aliquot plasma into a plastic vial, being careful not to disturb the cells, and freeze immediately.

5. Send frozen.

Note: **Critical frozen. Separate specimens must be submitted when multiple tests are ordered.**

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	28 days

CPT Code Information:

83520

AATHI
603184

Thrombophilia Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

AATHR
603304

Thrombophilia Profile, Plasma and Whole Blood

Specimen Requirements:

Both blood and plasma are required.

Patient Preparation:

1. Specimen **must be** collected prior to initiation of anticoagulants and thrombolytic therapy.

2. Patient **should not** be receiving warfarin, heparin, direct thrombin inhibitors (argatroban, dabigatran), or direct factor Xa inhibitors (apixaban, rivaroxaban, and edoxaban).

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), light-blue top (3.2% sodium citrate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 5 mL Platelet-poor plasma in 5 plastic vials; each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 5 separate plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information: Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: Platelet-poor plasma: 5 mL total, 5 plastic vials each containing 1 mL, Whole blood: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days
Whole blood	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

81240-F2 PTNT

85300-ATTF

85303-CFX

85306-PSF

85307-APCRV

85379-DIMER

85384-CLFIB

85390-26-AATHI

85610-PTSC

85613-DRV1

85670-TTSC

85730-APTSC

81241-F5 (coagulation factor V) (eg, hereditary hypercoagulability) gene analysis, Leiden variant (if appropriate)

85210-Factor II (if appropriate)

85220-Factor V (if appropriate)

85230-Factor VII (if appropriate)

85240-Factor VIII (if appropriate)

85250-Factor IX (if appropriate)

85260-Factor X (if appropriate)

85270-Factor XI (if appropriate)

85280-Factor XII (if appropriate)

85301-Antithrombin antigen (if appropriate)

85302-Protein C antigen (if appropriate)

85305-Protein S antigen, total (if appropriate)

85306-Protein S activity (if appropriate)

85366-Soluble fibrin monomer (if appropriate)

85385-PT-Fibrinogen (if appropriate)

85597-Platelet neutralization for lupus inhibitor (if appropriate)

85598-Hex LA (if appropriate)

85611-PT mix 1:1 (if appropriate)

85613-DRVVT mix (if appropriate)

85613-DRVVT confirmation (if appropriate)

85635-Reptilase (if appropriate)

85732 - APTT Mix 1:1 (if appropriate)

FFTPO 57822 Thrombopoietin (TPO)

Specimen Requirements:

Specimen Type: Serum
Container/Tube: Red or SST
Specimen Volume: 1 mL
Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	30 days

CPT Code Information:

83520

GNTHR 619271 Thrombosis Disorders, Comprehensive Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

THSIF 605244 Thrombospondin Type 1 Domain Containing 7A (THSD7A), Immunofluorescence

Specimen Requirements:

Specimen Type: Kidney tissue

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; paraffin sections 3 to 4-microns thick

Acceptable: Formalin-fixed, paraffin-embedded (FFPE) kidney tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

88346-Primary IF

88350-If additional IF

THSD7 607079 Thrombospondin Type-1 Domain-Containing 7A Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 1 mL

Collection Information: Centrifuge and aliquot serum into plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

86255

THYM 82606 Thyme, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TGAB
84382

Thyroglobulin Antibody, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Red top (gel tubes/**SST are not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86800

TGABI
610071

Thyroglobulin Antibody, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see IETG / Interference Evaluation Heterophile, Thyroglobulin Tumor Marker, Serum.

Patient Preparation: For 12 hours before specimen collection, patient should not take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: None (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86800

THYR
70565

Thyroglobulin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TGMS
62749

Thyroglobulin Mass Spectrometry, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.25 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	416 days
	Ambient	72 hours

CPT Code Information:

84432

TGII 610072 Thyroglobulin, Interference Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

HTGR 62936 Thyroglobulin, Tumor Marker Reflex, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: None (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

86800

84432 (if appropriate)

TFNAB 607714 Thyroglobulin, Tumor Marker, Fine-Needle Aspiration Biopsy Needle Wash

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. Needle wash specimens for analysis should be collected in conjunction with cytology specimens.

2. Have saline available prior to start of procedure. **Saline is the only acceptable solution for needle washings.**

3. After each fine-needle aspiration biopsy (FNAB) has been collected and the material in the needle has been expelled onto a slide for cytologic analysis, attach the used FNAB needle to an empty syringe.

4. Withdraw between 0.10 mL and 0.25 mL of saline up through the needle until the saline starts to fill the hub of the needle or end of the syringe.

5. Expel this fluid back through the needle into a separate plastic screw-top tube. This is the needle washing used for analysis.

6. Repeat steps 2 through 4 for each needle pass of the same biopsied site and empty into the same tube, accumulating a total of 0.5 mL to 1.5 mL of fluid to send to the laboratory. (If more than 1 site is biopsied, see Additional Information)

7. Inspect specimen for visible blood or tissue contamination:

a. If bloody, centrifuge specimen and transfer supernatant to a new plastic aliquot tube (5-mL standard tube) prior to sending to laboratory. The supernatant, not the cellular material, is used for analysis.

b. If specimen is clear, centrifugation is not necessary.

Additional Information:

1. If more than 1 site is biopsied, each washing material should be submitted on a separate tube and under a different order number.

2. A minimum of 0.5 mL is required for testing; however, the total collection volume should not exceed 1.5 mL. Specimen volumes outside these parameters may be rejected.

3. Do not send a saline control. This test has been validated to rule-out saline matrix effect.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fine Needle Wash	Frozen (preferred)	90 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

84432

HTG2 Thyroglobulin, Tumor Marker, Serum

62800

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84432

86800

TGQN
610070

Thyroglobulin, Tumor Marker, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see IETG / Interference Evaluation Heterophile, Thyroglobulin Tumor Marker, Serum.

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: None (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	90 days

	Ambient	7 days

CPT Code Information:

84432

TAB 82041 Thyroid Autoantibodies Profile, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86376-Thyroperoxidase antibody

86800-Thyroglobulin antibody

THCSM 622850 Thyroid Function Cascade, Monitor (without Thyroid Peroxidase Antibody), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84443

84439-(if appropriate)

84480-(if appropriate)

THSCM
83633

Thyroid Function Cascade, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84443

84439 (if appropriate)

84480 (if appropriate)

86376 (if appropriate)

TTF8G 70575 Thyroid Transcription Factor (8G7G3/1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TTFSP 70576 Thyroid Transcription Factor (SPT24) Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostains Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick.

Acceptable: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TTFK5 70489 Thyroid Transcription Factor 1 (TTF1) (SPT24) and Keratin 5 (KRT5) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

TSH
70574

Thyroid-Stimulating Hormone (TSH), Beta Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

STSH 8939 Thyroid-Stimulating Hormone-Sensitive (s-TSH), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84443

FFTSI
75917

Thyroid-stimulating Immunoglobulin (TSI)

Specimen Requirements:

Collection Container/Tube:

Preferred: Red-top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.3 mL Note: This volume does not allow for repeat testing

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

84445

TSI
8634

Thyroid-Stimulating Immunoglobulin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	60 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

84445

TPO
81765

Thyroperoxidase Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86376

THYRO
81797

Thyrotropin Receptor Antibody, Serum

Specimen Requirements:

Patient Preparation:

1. For 12 hours before specimen collection. patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

2. Patient **should not** be receiving heparin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

83520

TBGI
9263

Thyroxine-Binding Globulin (TBG), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

84442

FGTIA 75019 Tiagabine (Gabitril), Serum

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80199

STICK 602733 Tick-Borne Antibodies, Modified 2-Tier, ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.35 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	14 days

CPT Code Information:

86618

86666 x 2

86753

86617 x 2 (if appropriate)

TICKS
83265

Tick-Borne Disease Antibodies Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	14 days

CPT Code Information:

86618

86666 x 2

86753

86617 x 2-Lyme disease Western blot (if appropriate)

TIKLB
618311

Tick-Borne Panel, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred).

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798 x4

87469

87468

87484

87478

87999 (if appropriate for government payers)

BRBST 620148 Tickborne Bacterial, PCR and Sequencing, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: If not submitting in original vial, mix well before transferring to a sterile vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87801-Broad Range Bacterial PCR and Sequencing

87798-Specimen Identification by PCR (if appropriate)

TILAP 619510 Tilapia, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TIMG
82891

Timothy Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FFTIB
91137

Tin, Blood

Specimen Requirements:

Draw blood in a royal blue-top (metal free EDTA) tube. Send 2 mL metal free EDTA whole blood refrigerated.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA - Metal Free (ERB)	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

83018

FFTIN
91101

Tin, Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Serum

Draw blood in a metal free tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of metal free serum refrigerated in a plastic vial.

Plasma

Draw blood in an EDTA metal-free royal blue-top tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL metal-free EDTA plasma refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days

	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

83018

TSTGP
83671

Tissue Transglutaminase Antibodies, IgA and IgG Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86364 x 2

TTGA
82587

Tissue Transglutaminase Antibody, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86364

TTGG
83660

Tissue Transglutaminase Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86364

TIWB
614612

Titanium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Collection Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381)

Specimen Volume: 0.55 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original collection tube. **Do not aliquot.**

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

TIS
89367

Titanium, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.55 mL Serum

Collection Instructions:

1. Allow specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Frozen	28 days	METAL FREE
	Ambient	7 days	METAL FREE

CPT Code Information:

83018

TOBPA Tobramycin, Peak, Serum
37063

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80200

TOBRA
37065

Tobramycin, Random, Serum

Specimen Requirements:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80200

TOBTA
37064

Tobramycin, Trough, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood no more than 30 minutes before next scheduled dose.

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80200

FHIPP
91121

Toluene as Hippuric Acid, Occupational Exposure, Urine

Specimen Requirements:

10 mL aliquot of random or spot urine collected at end of shift. Send specimen refrigerated. Tubes should be filled to prevent loss of volatile compound into headspace.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

83921

82570

FFTLB
91141

Toluene, Occupational Exposure, Blood

Specimen Requirements:

Draw blood in 2 green top (sodium heparin) tubes. Tubes should be filled to prevent loss of volatile compound into headspace. Send 20 mL of sodium heparin whole blood refrigerated.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
WB Sodium Heparin	Refrigerated (preferred)	14 days
	Frozen	365 days

CPT Code Information:

84600

FMATG
57628

Tomato IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

TOMA
82695

Tomato, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TOPI
81546

Topiramate, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80201

TRCHG 800270 ToRCH Profile IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86644-CMV

86695-Herpes simplex, type 1

86696-Herpes simplex, type 2

86762-Rubella

86777-*Toxoplasma*

TIBC
2501

Total Iron Binding Capacity, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see SFEC / Iron and Total Iron Binding Capacity, Serum

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. For 24 hours before collection, patient **should not** take iron-containing supplements.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood before noon (preferred).

2. Within 2 hours of collection, serum gel tubes should be centrifuged.

3. Within 2 hours of collection, red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	365 days

CPT Code Information:

83550

TOXCG
616734

Toxocara Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	5 days

CPT Code Information:

86682

FGGMC
75519

Toxoplasma gondii Antibodies (IgG, IgM), ELISA, CSF

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Submit 1 mL of spinal fluid (CSF) in a sterile, leak-proof container.

2. Send refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86777-IgG

86778-IgM

TOXGP
800077

Toxoplasma gondii Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86777

TXPG
621838

Toxoplasma gondii Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

86777

TXMGP
802127

Toxoplasma gondii Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Aliquot tube

Specimen Volume: 1.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86778-Toxoplasma IgM

86777-Toxoplasma IgG

TXPAB
621836

Toxoplasma gondii Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

86778-Toxoplasma IgM

86777-Toxoplasma IgG

TXM
802126

Toxoplasma gondii Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86778

TXPM
621837

Toxoplasma gondii Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

86778

TOXB 62977 Toxoplasma gondii, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PTOX 81795 Toxoplasma gondii, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Spinal fluid

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: 12 x 75-mm screw cap vial

Acceptable: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Fresh tissue

Supplies: M4-RT (T605)

Container/Tube:

Preferred: Multi-microbe medium (eg, M4-RT)

Acceptable: Sterile container with 1 to 2 mL of sterile saline

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue in a sterile container containing 1 mL to 2 mL of sterile saline or multi-microbe medium (M4-RT, M4, or M5)

Specimen Type: Ocular fluid

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container: 12 x 75-mm screw cap vial **(swabs are NOT acceptable)**

Specimen Volume: 0.3 mL

Collection Instructions:

1. Aliquot collected fluid into screw-cap vial. **Do not submit ocular fluid in syringe.**

2. **Do not centrifuge or dilute the specimen.**

Specimen Minimum Volume: Amniotic fluid, Ocular fluid, Spinal fluid: 0.3 mL; Tissue: 2 x 2 mm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

TOXO
70569

Toxoplasma Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CSP53 607599 TP53 Gene Somatic Mutation Pre-Analysis Cell Sorting, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see P53CA / Hematologic Neoplasms, *TP53* Somatic Mutation, DNA Sequencing Exons 4-9, Varies.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

88184-Flow cytometry, first cell surface, cytoplasmic or nuclear marker

88185 x 4-Each additional marker

TP53 TP53 Mutation Analysis, Next-Generation Sequencing, Tumor

616496

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81351

TRAM 62595 Tramadol and Metabolite, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

G0480

80373 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TRAMS
623268

Tramadol and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

G0480

80373 (if appropriate for select payers)

TFE3I
70564

Transcription Factor E3 (TFE3) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TRSF
34623

Transferrin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, aliquot serum into a plastic vial prior to shipment.

3. For red-top tubes, aliquot the serum into a plastic vial immediately after centrifuging.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	7 days

CPT Code Information:

84466

FFTRZ
75024

Trazodone (Desyrel)

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80338

FHEAV
57949

Tree of Heaven (Ailanthus spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

TREE1 Tree Panel #1, Serum
81886

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TREE3 Tree Panel #3, Serum
81704

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TREE4
81705

Tree Panel #4, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TRCNG
616733

Trichinella Antibody, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	5 days

CPT Code Information:

86784

FFTRU 91099 Trichloroacetic Acid, Urine

Specimen Requirements:

Submit a 10 mL aliquot from a random or spot urine collected at end of shift, end of exposure, or end of workweek. Send specimen refrigerated.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

83921-Organic acid, single, quantitative

82570-Creatine, other source

TRVI 82853 Trichoderma viride, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MTRNA 61756 Trichomonas vaginalis, Nucleic Acid Amplification, Varies

Specimen Requirements:

This test should only be performed in men.

Submit only 1 of the following specimens:

Specimen Type: Post-prostatic massage fluid (VBIII)

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima Urine Specimen Transport Tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should void a small amount of urine prior to prostatic massage. Pre-massage urine can be discarded or submitted for other testing as applicable.

3. Patient then ceases voiding and a prostatic massage is performed by the urologist or other health care professional.

4. Collect post-massage fluid into a sterile, plastic, preservative-free container.

5. Transfer 2 mL of post-massage fluid specimen into the Aptima urine specimen transport tube using the disposable pipette provided within 24 hours of collection. The correct volume of fluid has been added when the fluid level is between the black fill lines on the Aptima urine transport tube.

Specimen Type: Urethral (male only)

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: Swab

Collection Instructions:

1. **Urethral specimens must be collected using an Aptima Unisex Swab Collection kit**.

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate blue swab gently at least 1 full rotation using sufficient pressure to ensure swab comes into contact with all urethral surfaces. Allow blue swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in the Aptima transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely and label tube with patient's entire name and collection date and time.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	30 days	APTIMA VIAL
	Ambient	30 days	APTIMA VIAL

CPT Code Information:

87661

TVRNA 61755 Trichomonas vaginalis, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Endocervix/Cervix

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Unisex Swab Collection Kit**.

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

Discard the cleaning swab.

3. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal, and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

4. Place blue swab into Aptima transport tube provided in collection kit.

5. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 60 days of collection.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Multitest Swab Collection Kit**.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into Aptima transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 60 days of collection.

Specimen Type: ThinPrep Specimen (Endocervix/Cervix)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect ThinPrep sample as per normal collection process.

2. **ThinPrep specimen must be aliquoted (as outlined below) before it is processed/tested for Papanicolaou testing (Pap smear)**.

3. Vortex ThinPrep/PreservCyt vial 3 to 10 seconds.

4. Within 1 minute of vortexing:

a. Transfer 1 mL of specimen into the Aptima specimen transfer tube using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug).

b. Process only 1 ThinPrep and transfer tube set at a time.

c. Recap Aptima specimen transfer tube tightly and gently invert 3 times to mix.

4. Label Aptima transfer tube with appropriate label.

5. Use remainder of ThinPrep specimen for Pap testing.

6. Maintain specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 30 days of collection when stored at 2 to 14 degrees C or within 14 days when stored at 15 to 30 degrees C.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection, transfer 2 mL of urine into the Aptima urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the Aptima urine transport tube.

4. Maintain Aptima urine specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 30 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Frozen	180 days	APTIMA VIAL
	Ambient		APTIMA VIAL

CPT Code Information:

87661

FFTMV 75554 Trichophyton Mentagrophytes (var interdigitale) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

TCPT
82720

Trichophyton rubrum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TRPS1
616890

Trichorhinophalangeal syndrome type 1 (TRPS1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TGLBF
606918

Triglycerides, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial Fluid

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

84478

TRIGC
606879

Triglycerides, CDC, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see LMPP / Lipoprotein Metabolism Profile, Serum.

Patient Preparation:

1. **Fasting: 12 hours, required**

2. Patient **must not** consume any alcohol for 24 hours before specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

84478

TRIG1
616665

Triglycerides, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Container/Tube: Plastic vial

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

84478

TMP
80146

Trimethoprim, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (gel tubes/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum for a peak level should be collected at least 60 minutes after a dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

TRMP
64269

Trimipramine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

TPI1
607458

Triosephosphate Isomerase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not** aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

TPIC
608424

Triosephosphate Isomerase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

T46CS
616449

Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Cell Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

T46CC 616450 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Cell Binding Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

T46TS 616447 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

T46TC 616448 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

T46IS 616445 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation,

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

T46IC 616446 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

FFTRP 91774 Trofile Co-Receptor Tropism Assay

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volume: 3 mL

Collection Instructions: Draw blood in either PPT (pearl top) or lavender-top (EDTA) tubes.

Centrifuge specimen within six hours of collection. Remove plasma from cells immediately, and transfer specimen to a screw-capped, plastic vial. Freeze 3 mL of PPT plasma or EDTA plasma immediately, send specimen frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load

2. Viral load collection date

Note: 1. Intended to use only for patients with viral loads greater than or equal to 1000 copies/mL. For best results, viral loads should be confirmed within two weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be cancelled.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87999

FFTRO 57159 Trofile DNA Co-Receptor Tropism Assay

Specimen Requirements:

Specimen Type: Whole Blood

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 4 mL

Collection Instructions: Draw 4 mL blood in a lavender-top (EDTA) tube(s), (Do not centrifuge.) Freeze and ship frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

Note: Trofile DNA is recommended for patients with undetectable viral loads.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Frozen	14 days

CPT Code Information:

87999

WHIPB 87974 Tropheryma whipplei, Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Tropheryma whipplei* DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred)

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

87798

TWRP
80909

Tropheryma whipplei, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Tropheryma whipplei* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Fresh tissue or biopsy

Sources: Small intestine (duodenum, ileum, or jejunum), lymph node, bone, joint, synovial, liver, pancreas, spleen, lung, heart valve (and other heart tissues), or brain

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3) - approximately the size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue only, do not add fluid to tissue.

3. Refrigerate or freeze specimen.

Specimen Stability Information: Refrigerated (preferred) 7 days /Frozen 7 days

Preferred Paraffin-embedded tissue block:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Small intestine (duodenum, ileum, or jejunum), lymph node, bone, joint, synovial, liver, pancreas, spleen, lung, heart valve (and other heart tissues), or brain

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Small intestine (duodenum, ileum, or jejunum), lymph node, bone, joint, synovial, liver, pancreas, spleen, lung, heart valve (and other heart tissues), or brain

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Fluid

Sources: Cerebrospinal or ocular (eg, vitreous humor)

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Specimen Stability Information: Refrigerated (preferred) 7 days/Frozen 7 days

Specimen Type: Synovial fluid

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red clot tube (no anticoagulant), or sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send fluid specimen in original tube (preferred).

Specimen Stability Information: Refrigerated (preferred) 7 days /Frozen 7 days

Specimen Minimum Volume: Fresh tissue or biopsy: 5 mm(3); Paraffin-embedded tissue block: Two 10-micron sections; All other specimens: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87798

TROP2
621064

Trophoblast Antigen 2 (TROP2) Immunostain, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

TRK
603300

Tropomyosin Receptor Kinase (TRK) Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostains Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick.

Acceptable: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HSTNI
614422

Troponin I, High Sensitivity, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Mint green top (lithium heparin gel)

Acceptable: Green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Plasma gel tube should be centrifuged within 2 hours of collection.

2. Green-top tube should be centrifuged, and the plasma aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	180 days
	Refrigerated	48 hours
	Ambient	4 hours

CPT Code Information:

84484

TRPS
65832

Troponin T, 5th Generation, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Light-green top (lithium heparin gel)

Acceptable: Green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Within 2 hours of collection, lithium heparin gel tubes should be centrifuged.

2. Within 2 hours of collection, lithium heparin tubes should be centrifuged, and the plasma aliquoted into a plastic vial .

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Li Heparin	Frozen (preferred)	365 days
	Ambient	24 hours
	Refrigerated	24 hours

CPT Code Information:

84484

TROT
82788

Trout, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CRUZI
621062

Trypanosoma cruzi (Chagas) Antibody Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86753 x2

CHAGL 622447 Trypanosoma cruzi IgG, Lateral Flow Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CRUZI / *Trypanosoma cruzi* (Chagas) Antibody Panel, Serum.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86753

CHAGS 615244 Trypanosoma cruzi Total Antibody, Enzyme-Linked Immunosorbent Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CRUZI / *Trypanosoma cruzi* (Chagas) Antibody Panel, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86753

CHAGI 622448

Trypanosoma cruzi, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CRUZI / *Trypanosoma cruzi* (Chagas) Antibody Panel, Serum.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

TRYPN
70572

Trypsin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRPTS 70573 Tryptase Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRYPA 32283 Tryptase, Autopsy, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Additional Information: Tryptase degenerates very quickly when left in the presence of red blood cells.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	7 days

CPT Code Information:

83520

TRYPT
81608

Tryptase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Additional Information: Tryptase degenerates very quickly when left in the presence of red blood cells.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Frozen (preferred)	14 days
	Refrigerated	7 days

CPT Code Information:

83520

TRYPP Tryptophan, Plasma

82955

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, required; Infants should have specimen collected before next feeding (2-3 hours without total parenteral nutrition if possible)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect specimen and place on wet ice. Note: Thrombin-activated tubes **should not be used** for collection.

2. Centrifuge immediately or within 4 hours of collection if the specimen is kept at refrigerated temperature.

3. Being careful to ensure that no buffy coat is transferred, aliquot plasma into a plastic vial and freeze.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	14 days

CPT Code Information:

82131

TRYPU Tryptophan, Random, Urine

83823

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions: Collect a random urine specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	70 days
	Refrigerated	14 days

CPT Code Information:

82131

TTF40
602647

TTF41 (SPT24) + p40 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

TTRZ
617753

TTR Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin):

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1year /Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood**

specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

TSCP 616537 Tuberous Sclerosis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions T800)

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

81407

81265-Maternal cell contamination (if appropriate)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88235-Amniotic Fluid culture (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

RTRP2 614047 Tubular Reabsorption of Phosphorus, Random Urine and Serum

Specimen Requirements:

Both serum and urine are required.

Patient Preparation:

Fasting: 12 hours, required

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

Specimen Type: Urine

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Label specimen as urine.

Specimen Minimum Volume: Urine: 1 mL; Serum: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

82565

82570

84100

84105

TUNA
82547

Tuna, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TURKF
82824

Turkey Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FGORG
57641

Turkey IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

TURK 82702 Turkey, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FCTUR 57544 Turmeric (Curcuma longa) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

TRYPI
82848

Tyrophagus putrescentiae, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL

For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TYROS 70577 Tyrosinase (TYROS) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TYRGP 608033 Tyrosine Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies

on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

TYRBS
607550

Tyrosinemia Follow Up Panel, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing EDTA and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 7 days/Refrigerated 14 days/Frozen 90 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate 6 days

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

84510

84030

82542

80299

82131

82542 only (if appropriate for government payers)

TYRSC 610495 Tyrosinemia Follow-Up Panel, Self-Collect, Blood Spot

Specimen Requirements:

Container/Tube: Blood Spot Card

Specimen Volume: 2 Blood spots

Additional Information:

1. Order test each time the patient is to collect a dried blood specimen at home and mail the specimen directly to Mayo Clinic Laboratories.

2. Order should be placed a minimum of 3 days prior to desired date of collection.

3. Enter patient's address information for each order created, including street address (**post-office [PO] boxes are not acceptable delivery locations**), city, state abbreviation, zip code, country, and home phone number.

4. For each order, the Blood Spot Collection-Self Collect kit will be mailed directly to the patient for self-collection (**delivery to a PO box will not occur**).

5. For more information on how to collect blood spots, see the following

-How to Collect a Dried Blood Spot Sample via fingerstick

-Blood Spot Collection Instructions-Fingerstick

-Blood Spot Collection Instructions-Fingerstick-Spanish

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	7 days	FILTER PAPER

	Frozen	90 days	FILTER PAPER
	Refrigerated	14 days	FILTER PAPER

CPT Code Information:

0383U

UBA1Q 620900 UBA1 Mutation Quantitative Detection, VEXAS syndrome, Droplet Digital PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD-B) or green top (heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability: Refrigerated (preferred) 7 days/Ambient 7 days

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD-B) or green top (heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Stability: Refrigerated (preferred) 7 days/Ambient 7 days

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

2. Provide volume and concentration of DNA. The minimum required volume 50 mcL at a concentration of 50 ng/mcL

Specimen Stability: Frozen (preferred) 1 year/Refrigerate

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood: 4 mL; Bone marrow: 2 mL; Extracted DNA: 50 mcL at 50 ng/mcL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

81403

UBIQ
70578

Ubiquitin (UBIQ) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

UGTFZ 610064 UDP-Glucuronosyltransferase 1A1 (UGT1A1), Full Gene Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor or a liver transplant will interfere with testing. For information about testing patients who have received a hematopoietic stem cell or liver transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

ULCH 82546 Ulocladium chartarum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

UNIPD
35566

Uniparental Disomy, Varies

Specimen Requirements:

Specimens from both parents and the child or fetus are recommended for optimal interpretation of results. Each specimen must have a separate order for this test. Only the proband specimen will be charged.

Testing can be performed if only one parent specimen is submitted, however, biparental inheritance and some types of uniparental disomy (UPD) cannot be definitively established in the absence of one parent. Additionally, there is a higher likelihood for uninformative or inconclusive results.

If all required specimens are not received within one month of ordering, testing will be canceled.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or

equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL Tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81402 – proband

81402 (if appropriate for parental specimen)

FURA
90316

Uranium, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Patient Preparation: For 48 hours before specimen collection, patient should avoid exposure to gadolinium-based contrast media.

Supplies: NMS Acid Washed "Precleaned", 10 mL (T619)

Container/Tube: Acid washed or trace metal free plastic container

Specimen Volume: 1 mL

Collection Instructions:

1**.** Collect urine in acid washed or trace metal free plastic container.

2. Submit 1 mL of urine refrigerated.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Refrigerated (preferred)	16 days	Acid Washed Plastic (MML Supply T619)
	Frozen	30 days	Acid Washed Plastic (MML Supply T619)
	Ambient	9 days	Acid Washed Plastic (MML Supply T619)

CPT Code Information:

83018

UCDP
608020

Urea Cycle Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume

must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that

all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order

MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

UEBF
606598

Urea Nitrogen, Body Fluid

Specimen Requirements:

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Peritoneal dialysate (dialysis fluid)

-Pericardial

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84520

URAU
607234

Urea, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, urine tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Mix well before taking aliquot.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84540

URCON
614061

Urea, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84540

URBRP 65133 Ureaplasma species, Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Ureaplasma* DNA is not likely.

Container/Tube:

Preferred Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred)

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798 x 2

87999 (if appropriate for government payers)

URPRP 65135 Ureaplasma species, Molecular Detection, PCR, Plasma

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Ureaplasma* DNA is unlikely.

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Submission Container/Tube: Screw-capped, sterile container

Specimen Volume: 1 mL Plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798 x 2

87999 (if appropriate for government payers)

URRP 60758 Ureaplasma species, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Ureaplasma* DNA is not likely.

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-M4-RT (T605)

Sources: Vaginal, cervix, urethra, urogenital, chest/mediastinal; bronchus or lung (donor swab), or upper respiratory sources

Sources for infants younger than 3 months: Nasopharynx, nose, throat

Container/Tube:

Preferred: Culture swab transport system (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Acceptable: Swab in transport media: M4, M4-RT, M5, M6, universal transport media, or ESwab

Specimen Volume: 1 Swab

Collection Instructions:

Vaginal:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Urethra or cervical:

1. Collect specimen by inserting swab 1 to 3 cm and rotating 360 degrees.

2. Place swab back into swab cylinder.

Wound:

1. Collect specimen by swabbing back and forth over wound surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Specimen Type: Fluid

Sources: Pelvic, peritoneal, amniotic, prostatic secretions, semen, reproductive drainage or fluid, pleural/chest, chest tube, pericardial

Container/Tube: Sterile container

Specimen Volume: 1 to 2 mL

Specimen Type: Respiratory

Sources: Sputum, tracheal secretions, bronchial washings, bronchoalveolar lavage, lung fluid

Sources for infants younger than 3 months: Nasal washings (not acceptable for older patients))

Container/Tube: Sterile container

Specimen Volume: 1 to 2 mL

Specimen Type: Synovial fluid

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red top (no anticoagulant), or sterile container

Specimen Volume: 1 mL

Collection Instructions: Send fluid specimen in original tube.

Specimen Type: Urine-first void, kidney/bladder stone, or ureter

Container/Tube: Sterile container

Specimen Volume: 10 mL or entire specimen

Collection instructions:

Urine first void:

1. Specimen can be collected at any time during the day. The patient should not have urinated for at least 1 hour prior to specimen collection.

2. The first voided portion is the initial 20 to 30 mL of the urine stream obtained **without** cleaning the external urethra.

Specimen Type: Tissue

Sources: Placenta, products of conception, urogenital, respiratory, bronchus, chest/mediastinal, bone, spine, or joint

Container/Tube: Sterile container

Specimen Volume: 5 mm(3)

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit fresh tissue only, **do not add fluid to tissue**

3. Refrigerate or freeze specimen.

Specimen Minimum Volume: Urine-first void: 2 mL; All other specimens: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798 x 2

87999 (if appropriate for government payers)

URCU 614044 Uric Acid, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-Hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84560

FUABF
75099

Uric Acid, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Sources: Drain, peritoneal/ascites, pleural or synovial fluid

Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove cellular material.

2. Aliquot 1 mL body fluid into a plastic vial.

3. Send frozen.

4. **Specimen source must be provided**.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen (preferred)	180 days
	Refrigerated	5 days
	Ambient	24 hours

CPT Code Information:

84560

URIC
8440

Uric Acid, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Submission Container/Tube: Plastic vial

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days

CPT Code Information:

84550

RURC1 621395 Uric Acid/Creatinine Ratio, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84560

82570

U1A1Q 610063 Uridine Diphosphate (UDP) Glucuronosyltransferase 1A1 TA Repeat Genotype, UGT1A1, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have

received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81350-*UGT1A1* (UDP glucuronosyltransferase 1 family, polypeptide AI) (eg, irinotecan metabolism), gene analysis, common variants (eg, *28, *36, *37)

GALE
64372

Uridine Diphosphate-Galactose 4' Epimerase, Blood

Specimen Requirements:

Multiple whole blood tests for galactosemia can be performed on 1 specimen. Prioritize order of testing when submitting specimens. For a list of tests that can be ordered together, see Galactosemia-Related Test List.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin) or yellow top (ACD)

Specimen Volume: 5 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Ambient	6 days

CPT Code Information:

82542

UPII
610490

Uroplakin II Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

UPGDW Uroporphyrinogen Decarboxylase, Washed Erythrocytes
31892

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as alcohol suppresses enzyme activity for 24 hours after ingestion.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. Record volume of packed cells and the total volume of the specimen.

4. Discard supernatant plasma.

5. Wash erythrocytes 2 times by resuspension with 5 mL of cold 0.9% saline, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	14 days
	Refrigerated	14 days
	Ambient	4 days

CPT Code Information:

82657

UPGD
8599

Uroporphyrinogen Decarboxylase, Whole Blood

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as alcohol suppresses enzyme activity for 24 hours after ingestion.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Specimen Volume: 4 mL; full tube

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Ambient	7 days

CPT Code Information:

82657

FUROC 35328 UroVysion for Detection of Bladder Cancer, Urine

Specimen Requirements:

Specimen Type: Urine

Sources: Voided urine, catheterized urine, bladder washings, stoma collections, ureteral brushings or washings, renal pelvic brushings or washings

Supplies: FISH for Urothelial Carcinoma Urine Collection Kit (T509)

Container/Tube:

Preferred: FISH for Urothelial Carcinoma in Urocyte Urine Collection Kit

Acceptable: 70% ethanol, PreservCyt, CytoLyt, ThinPrep UroCyte (UroCyte PreservCyt Solution)

Specimen Volume: 30 mL

Collection Instructions:

1. Follow instructions included with Urocyte Urine Collection Kit.

2. If kit is not used, submit a random urine specimen with an equal volume of 70% ethanol, PreservCyt, or CytoLyt.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88120

USPF 58104 USP6 (17p13), Aneurysmal Bone Cyst and Nodular Fasciitis, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 3 unstained

Collection Instructions: Submit 3 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

USTEK 609476 Ustekinumab Quantitation with Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before the next dose of drug administration (trough level).

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

80299

83520

USNU
82388

Ustilago nuda, Mold Grain Rust, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FNSVG
75140

Vaginitis (VG), NuSwab

Specimen Requirements:

Specimen Type: Vagina swab

Container/Tube: Aptima Multitest Swab Specimen Collection kit; Aptima Unisex Swab Specimen Collection Kit

Specimen Volume: 1 Swab

Collection Instructions:

1. Collect vaginal fluid sample using the Aptima swab by contacting the swab to the lower third of the vaginal wall and rotating the swab for 10 to 30 seconds to absorb fluid.

2. Immediately place the swab into the transport tube and carefully break the swab shaft

against the side of the tube.

3. Tightly screw on the cap.

4. Submit one vaginal Aptima Multitest or Unisex swab.

5. Send refrigerate.

Specimen Minimum Volume: One swab

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Refrigerated (preferred)	30 days
	Ambient	30 days

CPT Code Information:

87801

87798 x 3

87661

VALPG 37067 Valproic Acid, Free and Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

VALPA-80164

VALPF-80165

VALPF 37068 Valproic Acid, Free, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80165

VALPA 37066 Valproic Acid, Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80164

VANPA 37069 Vancomycin, Peak, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum for a peak level should be drawn 1 hour after completion of dose.

2. Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	48 hours

CPT Code Information:

80202

VANRA Vancomycin, Random, Serum
37071

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	48 hours

CPT Code Information:

80202

VANTA Vancomycin, Trough, Serum
37070

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw specimen immediately prior to the next dose (within 30 minutes).

2. Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	48 hours

CPT Code Information:

80202

VRERP 84406 Vancomycin-Resistant Enterococcus, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by vancomycin-resistant *Enterococcus* DNA is unlikely.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Perianal, perirectal, rectal, anal

Supplies: Culturette (BBL Culture Swab) (T092)

Container/Tube: Culture transport swab (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Specimen Volume: Swab

Specimen Stability Information: Refrigerated (preferred)/Frozen

Acceptable:

Specimen Type: Preserved feces

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair, Para-Pak C and S vial)

Specimen Volume: Representative portion of feces

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

Specimen Stability Information: Ambient (preferred)/ Refrigerated

Specimen Type: Unpreserved feces

Supplies:

-Stool Container, Small (Random), 4 oz Random (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: Representative portion of feces

Collection Instructions: Collect fresh fecal specimen and submit representative sample in fecal container.

Specimen Stability Information: Refrigerated (preferred)/Frozen

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87500

Vanilla IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

VANIL
82621

Vanilla, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

VH 9254 Vanillylmandelic Acid and Homovanillic Acid, Random, Urine

Specimen Requirements:

Patient Preparation: Administration of L-dopa may falsely increase homovanillic acid and vanillylmandelic acid results. For 24 hours prior to specimen collection, the patient should **not** take L-dopa.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Adjust the urine pH to a level between 1 and 5 by adding 50% acetic acid or hydrochloric acid dropwise and checking the pH.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	180 days

CPT Code Information:

83150-HVA

84585-VMA

VMA
9454

Vanillylmandelic Acid, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Administration of L-dopa may falsely increase vanillylmandelic acid results. For 24 hours prior to specimen collection, the patient should **not** take L-dopa.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative at the start of collection. Use 15 mL of 50% acetic acid for children younger than 5 years. If specimen is refrigerated during collection, preservative may be added up to 12 hours after collection. This preservative is intended to achieve a pH of between approximately 1 and 5.

2. Collect a 24-hour urine specimen.

3. If necessary, adjust urine pH to a level between 1 and 5 by adding 50% acetic acid or hydrochloric acid dropwise and checking the pH.

Additional Information: For multiple collections see Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	180 days

CPT Code Information:

84585

VMAR
60274

Vanillylmandelic Acid, Random, Urine

Specimen Requirements:

Patient Preparation: Administration of L-dopa may falsely increase vanillylmandelic acid results. For 24 hours prior to specimen collection, the patient should **not** take L-dopa.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Adjust the random urine pH to a level between 1 and 5 by adding 50% acetic acid or hydrochloric acid dropwise and checking the pH.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	180 days

CPT Code Information:

84585

VZV
70581

Varicella Zoster Virus (VZV) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

VZPG
34944

Varicella-Zoster Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86787

VZGM
61856

Varicella-Zoster Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86787 x2

VZM
80964

Varicella-Zoster Virus (VZV) Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86787

FVZGC
58045

Varicella-Zoster Virus Antibody, IgG, CSF

Specimen Requirements:

Specimen Type: Spinal fluid (CSF)

Container/Tube: Sterile plastic container

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect 0.5 mL CSF in sterile plastic container.

2. Ship refrigerated.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	365 days

CPT Code Information:

86787

LVZV
800168

Varicella-Zoster Virus, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Sources: Spinal, pleural, peritoneal, ascites, pericardial, amniotic, or ocular

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Swab

Sources: Miscellaneous; dermal, eye, nasal, or throat

Supplies:

-Culturette (BBL Culture Swab) (T092)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT) and ESwabs

Collection Instructions: Place swab back into multimicrobe media (M4-RT, M4, or M5).

Specimen Type: Genital swab

Sources: Cervix, vagina, urethra, anal/rectal, or other genital sources

Supplies:

-Culturette (BBL Culture Swab) (T092)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT) (T605) and ESwabs

Collection Instructions: Place swab back into multimicrobe media (M4-RT, M4, or M5).

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or washing, sputum, or tracheal aspirate

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Tissue

Sources: Brain, colon, kidney, liver, lung, etc.

Supplies: M4-RT (T605)

Container/Tube:

Preferred: Multimicrobe media (M4-RT)

Acceptable: Sterile container with 1 to 2 mL of sterile saline

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue in a sterile container containing 1 mL to 2 mL of sterile saline or multimicrobe medium (M4-RT, M4, or M5)

Specimen Minimum Volume: Body Fluid, Ocular Fluid, or Spinal Fluid: 0.3 mL
Respiratory Specimens: 1 mL
Tissue: 2 x 2-mm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

VZVPV
618309

Varicella-Zoster Virus, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Spinal fluid

Collection Container/Tube:

Preferred: Vial number 2

Acceptable: Any vial number

Submission Container/Tube: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Specimen Type: Body fluid

Sources: Pleural, peritoneal, ascites, pericardial, amniotic, or ocular

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Swab

Sources: Miscellaneous; dermal, eye, nasal, throat, or genital

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-M4-RT (T605)

Container/Tube: Multimicrobe media (M4-RT, M4, M5, Bartels, or Jiangsu) and E-Swab or Culturette

Collection Instructions: Place swab back into multimicrobe media.

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, nasopharyngeal aspirate or washing, sputum, or tracheal aspirate

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Fresh tissue

Supplies: M4-RT (T605)

Container/Tube:

Preferred: Multimicrobe media (M4-RT, M4, M5, Bartels, or Jiangsu)

Acceptable: Sterile container with 1 to 2 mL of sterile saline

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue. Fixed tissue is **not** acceptable.

Specimen Minimum Volume: Ocular Fluid and Spinal Fluid: 0.3 mL
Body Fluid (pleural, peritoneal, ascites, and pericardial): See Specimen Required
Respiratory Specimens: 1 mL
Tissue: 2 x 2 mm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

VEGF
63019

Vascular Endothelial Growth Factor, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender-top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Centrifuge at 4 degrees C, 1500 x *g* for 10 minutes.

3. Aliquot plasma into plastic vial.

4. Within 2 hours of collection, freeze specimen.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	21 days
	Refrigerated	24 hours

CPT Code Information:

83520

VIP
8150

Vasoactive Intestinal Polypeptide, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions: Centrifuge, aliquot plasma into a plastic vial, and freeze immediately.

Specimen Minimum Volume: Plasma: 0.55 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	90 days

CPT Code Information:

84586

VIPI
70580

Vasoactive Intestine Polypeptide (VIP), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

VDSFT
621912

VDRL Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see NSAIP / Neurosyphilis IgG Antibody Index with VDRL, Serum and Spinal Fluid.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. The spinal fluid (CSF) specimen **must be** collected within 24 hours of the serum specimen, preferably at the same time.

2. The CSF aliquot should be from the second, third, or fourth CSF vial collected during the lumbar puncture. **Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection**.

3. Label vial as spinal fluid or CSF.

4. Band CSF specimen together with the serum sample.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86593

VDSFQ
65036

VDRL Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see VDSF / VDRL, Spinal Fluid.

Collection Container/Tube: Sterile vial

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Submit specimen collected in vial 2 if possible. If not, note from which vial the aliquot was obtained.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	14 days

	Refrigerated	14 days

CPT Code Information:

86593

VDSF 9028 VDRL, Spinal Fluid

Specimen Requirements:

Collection Container/Tube: Sterile vial

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Submit specimen collected in vial 2, if possible. If not, note from which vial the aliquot was obtained.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

86592

VEDOZ 603025 Vedolizumab Quantitation with Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80280

82397

VEDOL
602807

Vedolizumab Quantitation with Reflex to Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80280

82397 (if appropriate)

VELV
82917

Velvet Leaf, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FVENE
75577

Venison IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FBMBL 57975 Venom Bumble Bee (Bombus terrestrus) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FHOBG 57714 Venom Honey Bee IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FWFHG 57799 Venom W-F Hornet IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

ACADV 622680 Very Long-Chain Acyl-CoA Dehydrogenase (VLCAD) Deficiency, ACADVL Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4

days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

VIBC Vibrio Culture, Feces

89658

Specimen Requirements:

Patient Preparation: Patient should **not use** barium or bismuth for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-*Vibrio* culture, stool

87077-Identification by MALDI-TOF mass spec (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

VIGA
91089

Vigabatrin (Sabril)

Specimen Requirements:

Submit only 1 of the following specimens

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 1 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80339

VIM
70579

Vimentin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

VIRID 619900 Viral Susceptibility, Defects in Intrinsic and Innate Immunity, Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested

volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant

risk for test failure due to insufficient DNA.

2. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SVISC
610406

Viscosity, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. Use a warm pack or heat block to keep specimen at 37 degrees C until after centrifugation.

2. Centrifuge at 37 degrees C and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2.0 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

85810

VITAE
605267

Vitamin A and Vitamin E, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required; infants should have specimen collected before next feeding.

2. Patient **must not** consume any alcohol for 24 hours before specimen collection.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a light protected plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	LIGHT PROTECTED
	Frozen	28 days	LIGHT PROTECTED
	Ambient	7 days	LIGHT PROTECTED

CPT Code Information:

84446

84590

VITA
42357

Vitamin A, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required; infants should have specimen collected before next feeding

2. Patient **must not** consume any alcohol for 24 hours before specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

84590

VITAP
605124

Vitamin A, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see VITAE / Vitamin A and Vitamin E, Serum.

Patient Preparation:

1. Fasting: 12 hours, required; Infants should have specimen collected before next feeding

2. Patient **must not** consume any alcohol for 24 hours before specimen collection.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into light protected plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	LIGHT PROTECTED
	Frozen	28 days	LIGHT PROTECTED
	Ambient	7 days	LIGHT PROTECTED

CPT Code Information:

84590

FB12
9156

Vitamin B12 and Folate, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, required

2. Do not order on patients who have recently received methotrexate or other folic acid antagonists.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

82607-Vitamin B12

82746-Folate

B12
9154

Vitamin B12 Assay, Serum

Specimen Requirements:

Patient Preparation: This test **should not** be performed on patients who have received a vitamin B12 injection or radiolabeled vitamin B12 injection within the previous 2 weeks.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes aliquot the serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

82607

VITB3
604987

Vitamin B3 and Metabolites, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (K2 EDTA)

Acceptable: K3 EDTA

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

84591

FPAB
57394

Vitamin B5 (Pantothenic Acid) Bioassay

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Allow specimen to clot for 30 minutes at room temperature.

2. Centrifuge for 15 minutes at 1000 x g.

3. Aliquot 0.5 mL serum into a plastic vial.

4. Send frozen.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	21 days
	Refrigerated	7 days

CPT Code Information:

84591

B6PRO 42360 Vitamin B6 Profile (Pyridoxal 5-Phosphate and Pyridoxic Acid), Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required; Infants should have specimen collected before next feeding, water may be taken as needed

2. For 24 hours before specimen collection, **patient must not take multivitamins or vitamin supplements.**

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin) or plasma gel separator (PST)

Acceptable: None

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge at 4 degrees C and aliquot into an amber vial.

2. Freeze immediately.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Frozen	29 days	LIGHT PROTECTED

CPT Code Information:

82542

84207

FBIOT
91902

Vitamin B7, H (Biotin)

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot 2 mL of serum into an amber vial to protect from light.

2. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Frozen (preferred)	14 days	LIGHT PROTECTED
	Refrigerated	7 days	LIGHT PROTECTED

CPT Code Information:

84591

VITE
42358

Vitamin E, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required; infants should have specimen collected before next feeding

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a light protected plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	44 days	LIGHT PROTECTED
	Frozen	44 days	LIGHT PROTECTED
	Ambient	7 days	LIGHT PROTECTED

CPT Code Information:

84446

VITK1
42364

Vitamin K1, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required; Infants should have specimen collected before next feeding

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes:
Preferred: Immediately aliquot serum into an amber or light-protected plastic vial.

Acceptable: Serum may be on whole blood cells a maximum of 2 hours before aliquoting. Protect the collection tube from light.

3. For serum-gel tubes:

Preferred: Immediately aliquot serum into an amber or light-protected plastic vial.

Acceptable: Serum may be aliquoted into an amber or light-protected plastic vial within 24 hours of collection when kept at ambient temperature and protected from light or within 7 days of collection if kept refrigerated and protected from light.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84597

VLTB
89190

Volatile Screen, Blood

Specimen Requirements:

Container/Tube:

Preferred: Gray top (potassium oxalate/sodium fluoride)

Acceptable: Lavendar top (EDTA) or green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. **Do not use alcohol to clean arm.** Use alternatives such as Betadine to cleanse arm before collecting any specimen for volatile testing.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL or amount to fill 1 tube

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80320 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

VLTBX
62745

Volatile Screen, Chain of Custody, Blood

Specimen Requirements:

Supplies: Chain-of-Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen container seals and documentation required.

Preferred: Gray top (potassium oxalate/sodium fluoride)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. **Do not use alcohol to clean arm.** Use alternative such as Betadine to cleanse arm before collecting any specimen for volatile testing.

2. Specimen must be sent in original tube. Collect specimen, seal, and submit with the associated documentation to satisfy the legal requirements for chain of custody testing.

Specimen Minimum Volume: 0.5 mL or amount to fill 1 tube

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80320 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

VLTUX 62746 Volatile Screen, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain-of-Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation required.

Specimen Volume: 10 mL

Collection Instructions: Collect a random urine specimen using the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain of custody testing.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80320 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

VLTU 8826 Volatile Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative is required.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80320 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

VLTS
8632

Volatile Screen, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: Full tube

Collection Instructions:

1. Do not use alcohol to clean arm. Use alternative such as Betadine to cleanse arm before collecting any specimen for volatile testing.

2. Avoid exposure of specimen to atmosphere.

3. Within 2 hours of collection, centrifuge the specimen.

4. For serum gel tubes, **do not aliquot**. Send centrifuged specimen in original tube.

5. For red-top tubes, aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80320 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

VHLZZ 614589 Von Hippel Lindau Syndrome, VHL, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be

attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

VWD8B 605011 von Willebrand Disease 2N (Subtype Normandy), Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, transfer all plasma into a vial, and centrifuge plasma again.

3. Aliquot plasma into a separate tube leaving 0.25 mL in the bottom of the centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, less than or equal to -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	56 days

CPT Code Information:

85246

AVWPQ 603186 von Willebrand Disease Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

AVWPI 603551 von Willebrand Disease Profile Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see AVWPR / von Willebrand Disease Profile, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

AVWPR 603550 von Willebrand Disease Profile, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been

transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 3 mL Platelet-poor plasma in three (3) 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred), each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Within 4 hours of collection, centrifuge and transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 3 separate 13-mm x 75-mm, 5-mL, polypropylene vials , leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately (within 2 hours of processing) freeze plasma at -20 degrees C or, ideally, -40 degrees C or below.

Specimen Minimum Volume: Platelet-poor plasma: 2 13-mm x 75-mm, 5-mL, Polypropylene vials , each containing 1 mL platelet-poor plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85240-Coagulation factor VIII assay

85246-von Willebrand factor antigen

85397-von Willebrand factor activity

85390-Technical interpretation

85245-von Willebrand factor ristocetin cofactor activity (if appropriate)

85247-von Willebrand factor multimer (if appropriate)

85335-Bethesda titer (if appropriate)

85335-Coagulation factor VIII inhibitor screen (if appropriate)

85390-26-Special coagulation interpretation (if appropriate)

GNVWD 619201 von Willebrand Disease, VWF and GP1BA Genes, Next-Generation Sequencing, Varies

Specimen Requirements:

A previous bone marrow transplant from an allogenic donor will interfere with whole blood testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred) <24 hours/Ambient <24 hours

Additional information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Amniotic fluid: 10 mL; Other specimen types: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81408

81479

81479 (if appropriate for government payers)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

VWACT 602170 von Willebrand Factor Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 2 mL Platelet-poor plasma in 2 vials, each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a polypropylene vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 2 separate polypropylene vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, < or =-40 degrees C.

6. Send specimens in the same shipping container.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85397

VWAG 9051 von Willebrand Factor Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, less than or equal to -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85246

VWFMS 603851 von Willebrand Factor Multimer Analysis, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. Specimen should be collected prior to coagulation factor replacement therapy.

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	42 days

CPT Code Information:

85247

VWFMP 604411 von Willebrand Factor Multimer Analysis, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AVWPR / von Willebrand Disease Profile, Plasma

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	42 days

CPT Code Information:

85247

FVOST
75888

Voriconazole Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen volume: Organism in pure culture

Complete and submit with specimen:

1.Fungus Testing Lab request form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999-Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 x 4 for mould MIC microdilution or agar dilution (if appropriate)

87186 x 4 for yeast MIC microdilution or agar dilution (if appropriate)

VORI
88698

Voriconazole, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80285

FWALP 57561 Wall Eyed Pike (Sander vitreus)(Stizostedium vitreum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FWCR1 Walnut Component rJug r 1
75573

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

86008

FFWNC Walnut Component rJug r 3
75584

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	84 days
	Ambient	28 days

CPT Code Information:

86008

FWALG 57640 Walnut Food (Juglans spp) IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

WALN 82732

Walnut Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BLWX 618850 Walnut-Food Components, IgE, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see BLWRF / Walnut-Food, IgE, with Reflex to Walnut-Food Components, IgE, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008 x2

BLW
82898

Walnut-Food, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BLWRF
618849

Walnut-Food, IgE, with Reflex to Walnut-Food Components, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WARSQ Warfarin Response Genotype, Varies
610065

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant or liver transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell or liver transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

0030U

FWARP 75517 Warfarin, Plasma

Specimen Requirements:

****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Lavender top or pink top (EDTA) (Plasma gel tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen volume: 1 mL Plasma

Collection instructions:

1. Promptly centrifuge and aliquot 1 mL of plasma into a preservative-free plastic vial.

2. Send refrigerated.

Specimen Stability Information: Refrigerated (preferred) 30 days/Ambient 30 days/Frozen 15 months

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	

CPT Code Information:

80375

WSPV
82659

Wasp Venom, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL

For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FWATG
57677

Watermelon IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

WMEL
86304

Watermelon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WEED1 Weed Panel # 1, Serum
81882

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

WEED2 Weed Panel # 2, Serum
81883

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WEED3 Weed Panel # 3, Serum
81884

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WEED4 Weed Panel # 4, Serum
81885

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WNVCI West Nile CSF Interpretation (Non-Orderable)
36779

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	30 days

WNVSI
36778

West Nile Serum Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

WNS
36769

West Nile Virus Antibody, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

IgG-86789

IgM-86788

WNC
36772

West Nile Virus Antibody, IgG and IgM, Spinal Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Sterile vial

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen from collection vial 2, 3, or 4

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

86789

86788

WNGS
36771

West Nile Virus Antibody, IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see WNS / West Nile Virus Antibody, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86789

WNGC
36774

West Nile Virus Antibody, IgG, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see WNC / West Nile Virus Antibody, IgG and IgM, Spinal Fluid.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Sterile vial

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen from collection vial 2, 3, or 4

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

86789

WNMS 36770 West Nile Virus Antibody, IgM, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see WNS / West Nile Virus Antibody, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86788

WNMC 36773 West Nile Virus Antibody, IgM, Spinal Fluid(Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see WNC / West Nile Virus Antibody, IgG and IgM, Spinal Fluid.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Sterile vial

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen from collection vial 2, 3, or 4

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

86788

WNVBL 608438 West Nile Virus, RNA, PCR, Molecular Detection, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

WNVUR 608437

West Nile Virus, RNA, PCR, Molecular Detection, Random, Urine

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect random urine in a sterile container.

2. No preservative should be added.

3. Label specimen as urine.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

WNVS
608436

West Nile Virus, RNA, PCR, Molecular Detection, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Sterile container

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a sterile container.

Additional Information: Serum specimens not aliquoted from the serum gel collection tube into a sterile container **will be rejected**.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

WNCSF 608435 West Nile Virus, RNA, PCR, Molecular Detection, Spinal Fluid

Specimen Requirements:

Collection Container/Tube:

Preferred: Vial number 2

Acceptable: Any vial number

Submission Container/Tube: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge or heat inactivate.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

FONS 75448 Western blot for anti-optic nerve autoantibodies in the serum

Specimen Requirements:

Complete and submit with specimen:

-Completed OHSU Ocular Requisition Form

-Clinical history

-Healthcare professional information (name and phone number)

NOTE: Without this information, testing cannot be completed.

Submit only 1 of the following specimens:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

3. Send refrigerated.

Specimen Type: Plasma

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen volume: 5 mL Plasma

Collection Instructions:

1. Centrifuge and aliquot plasma into a plastic vial.

2. Label specimen as plasma.

3. Send refrigerated.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	7 days

CPT Code Information:

84182

WRW
82666

Western Ragweed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FWHTG
57553

Wheat IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FWHG4 Wheat IgG4
57570

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

WHT
82686

Wheat, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FWHGY
57577

Whey IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

WHEY
82622

Whey, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

ASHW White Ash, IgE, Serum
82730

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BENW White Bean, IgE, Serum
82726

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WFHV
82658

White Faced Hornet Venom, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WHIC
82719

White Hickory, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WPIN
82729

White Pine, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

POTA
82710

White Potato, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FWHFE
57545

Whitefish IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SC018 Whole Blood

619408

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD) or light-blue top (3.2% sodium citrate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not** aliquot.

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

WEGG

610357

Whole Egg, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WESMT 616787 Whole Exome and Mitochondrial Genome Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80 mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient <24 hours/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81415-Patient only

81415, 81416-Patient and one family member comparator sample (duo) (as appropriate)

81415, 81416 x 2-Patient and two family member comparator samples (trio or non-traditional trio) (as appropriate)

81415, 81416 x 3-Patient and three family member comparator samples (quad) (as appropriate)

81460-Whole Mitochondrial Genome

81465-Whole Mitochondrial Genome Large Deletion Analysis

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

WESDX 616068 Whole Exome Sequencing for Hereditary Disorders, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred)/ 4 days Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81415-Patient only

81415, 81416-Patient and one family member comparator sample (duo) (as appropriate)

81415, 81416 x 2-Patient and two family member comparator samples (trio or non-traditional trio) (as appropriate)

81415, 81416 x 3-Patient and three family member comparator samples (quad) (as appropriate)

WESR 616070 Whole Exome Sequencing Reanalysis, Varies

Specimen Requirements:

For most patients, a new specimen submission will not be required. Testing can be performed using stored DNA from the original whole exome sequencing test. To order testing on the stored specimen, see Additional Testing Requirements.

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filtration Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 (formerly Ahlstrom 226) filter paper or blood spot collection card

Specimen Volume: 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

2. For collection instructions, see Blood Spot Collection Instructions

3. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

4. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	

	Refrigerated

CPT Code Information:

The first reanalysis: No charge

For all subsequent reanalysis requests: 81417

WGSDX 614363 Whole Genome Sequencing for Hereditary Disorders, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens for each family member.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81425-Patient only

81425, 81426-Patient and one family member comparator sample (duo) (as appropriate)

81425, 81426 x 2-Patient and two family member comparator samples (trio or non-traditional trio) (as appropriate)

81425, 81426 x 3-Patient and three family member comparator samples (quad) (as appropriate)

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

Specimen Requirements:

For most patients, a new specimen submission will not be required. Testing can be performed using stored DNA from the original whole genome sequencing test. To order testing on the stored specimen, see Additional Testing Requirements.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens for each family member.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the

cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80 mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: Whole blood: 1 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

First reanalysis: No charge

81427-For all subsequent reanalysis requests

WRGR
82830

Wild Rye Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WILL
82731

Willow, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

WT1I
70582

Wilms Tumor (WT-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

WNDZ 619409 Wilson Disease, ATP7B Full Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese

Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic

fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture

fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

FWING 57955 Wingscale (Atriplex Canescens) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

WORM
82680

Wormwood, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

ABCD1 X-Linked Adrenoleukodystrophy (XALD), ABCD1 Gene Sequencing with Deletion/Duplication, Varies

622694

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A

maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;**

order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen can only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405-*ABCD1* (ATP-binding cassette, sub-family D [ALD] member 1) (eg, adrenoleukodystrophy) full gene sequence

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

XHIM
82964

X-Linked Hyper IgM Syndrome, Blood

Specimen Requirements:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 4 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	72 hours	GREEN TOP/HEP

CPT Code Information:

88184-Flow cytometry, cell surface, cytoplasmic

88185 x 6-Each additional marker

FCDX3 75843 Xylazine Confirmation (Qualitative), Umbilical Cord, Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80375

YMCRO 35576 Y Chromosome Microdeletions, Molecular Detection, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81403-DAZ/SRY (deleted in azoospermia and sex determining region Y) (eg, male infertility), common deletions (eg, AZFa, AZFb, AZFc, AZFd)

YFHV
82657

Yellow Faced Hornet Venom, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

YJV
82661

Yellow Jacket Venom, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

YERSC
606222

Yersinia Culture, Feces

Specimen Requirements:

Patient Preparation: Patient **should not** use barium or bismuth for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-Yersinia Culture, Stool-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

YAP1 70583 Yes-Associated Protein (YAP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FYOG 57915 Yogurt (Lactobacillus bulgaricus) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

BTB46 ZBTB46 Immunostain, Technical Component Only
605254

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick.

Acceptable: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

VZIKM Zika Virus, IgM Antibody Capture ELISA, Serum
618559

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Allow blood to clot at ambient temperature (20-25 degrees C) for 30 to 60 minutes, then centrifuge and aliquot serum into plastic vial.

2. Send serum specimen frozen.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	30 days

CPT Code Information:

86794

VZIKU
619431

Zika Virus, PCR, Molecular Detection, Random, Urine

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect random urine in a sterile container.

2. Label specimen as urine.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87662

VZIKS
619432

Zika Virus, PCR, Molecular Detection, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Sterile container

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot the serum into a sterile container.

Additional Information: Serum specimens not aliquoted from the serum gel collection tube into a sterile container **will be rejected**.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87662

ZNU
8591

Zinc, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of barium are known to interfere with this test. If barium-containing contrast media has been administered**, the specimen should not be collected for at least 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

84630

ZN_S
7735

Zinc, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium, iodine, and barium are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium-, iodine-, and barium containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: 6-mL Plain, royal blue-top Vacutainer plastic trace element

blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.8 mL serum

Collection Instructions:

1. **This specimen must always be drawn first**.

2. **Do not collect specimen from a line.** Avoid hemolysis.

3. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

4. Within 4 hours of specimen collection, remove serum from cellular fraction. Remove the stopper. Carefully pour specimen into a Mayo metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

5. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

84630

ZNUCR 615259 Zinc/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of barium are known to interfere with this test. If barium-containing contrast media has been administered, **the specimen should not be collected for at least 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap

or glued insert

Submission Container/Tube: Plastic urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

84630

82570

ZNCU
615260

Zinc/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ZNUCR / Zinc/Creatinine Ratio, Random, Urine.

Patient Preparation: High concentrations of barium are known to interfere with most metal tests. If barium-containing contrast media has been administered, the specimen should not be collected for at least 96 hours.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

84630

FZIP 57107 Ziprasidone (Geodone, Zeldox)

Specimen Requirements:

Submit only 1 of the following specimens

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

FZNT8 ZnT8 Antibodies
75969

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 45 minutes of collection, centrifuge and aliquot serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.2 mL; Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	64 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86341

FZOLP Zolpidem (Ambien), serum or plasma
57738

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80368

FCDUD 75787 Zolpidem, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

ZONI 83685 Zonisamide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST is **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80203

FZCCE
57562

Zucchini (Cucurbita spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

MULT
35577

Zygosity Testing (Multiple Births), Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Cord blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send cord blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. While a properly collected cord blood sample may not be at risk for maternal cell contamination, unanticipated complications may occur during collection. Therefore, maternal cell contamination studies are recommended to ensure the test results reflect that of the patient tested and are available at an additional charge. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88235-Tissue culture for amniotic fluid (if appropriate)

88240-Cryopreservation (if appropriate)

81266-Each additional specimen (eg additional cord blood donor, additional fetal samples from different cultures, or additional zygosity in multiple birth pregnancies) (as needed)